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Chapter

NON-RELATIVISTIC PARTICLE DYNAMICS

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4.1 INTRODUCTION

According to Einstein, “The purpose of mechanics is to describe how bodies change their position in space with time” Dynamics is the study of motion of bodies and the relationship of this motion with the force producing it. It is thus an important branch of mechanics and starts with the simplest of bodies, *viz.*, particles, where particle means a quantity of matter having negligible dimensions.

Classical or Newtonian Mechanics

In dynamics, since the time of Newton, the calculations has been based on ‘*Newtons laws of motion*’ in which we use *invariance of space* and time as well as *mass*. In other words, the assumption that the distance between two points, the interval between two events in space and the mass of a body at any point in space are *the same for all observers, irrespective of their state of rest or motion*. This system is referred as *classical* or *Newtonian mechanics*. This invariance of space, time and mass was straight way accepted and hence enjoyed complete sway right up to the end of 19th century and no doubts were raised or questions asked as to its validity and accuracy.

Relativistic or Einsteinian Mechanics

However, the concepts have been completely changed due to Einstein’s revolutionary theory popularly known as ‘*Special theory of relativity*’. This was the year 1905, the new era in science and technology, from which the *direction of thinking* has turned. Einstein (1879-1955) in this theory catagorically stated that the speed of light (c) is constant irrespective of any relative motion between the source of light and the observer. In addition to this postulate of constancy of speed of light (c), he denied the Newton’s invariance of both space and times. According to Einstein, the mass (m) of a body increases with its velocity and is γ times its *rest mass* (m_0) at velocity v , where $\gamma = 1/\sqrt{1 - (v^2/c^2)}$. The Newtonian mass or momentum of a fast moving body must be multiplied by the factor γ to obtain its correct (relativistic) mass or momentum at its given speed. The system of calculations dealing with bodies moving with high velocities, approaching λ , thus came to be known as the Relativistic or Einsteinian mechanics.

Quantum or Wave Mechanics

Again, in the year 1924, the French Physicist, *Louis de Broglie*, put forward the view that ‘light behaves like a wave on a macroscopic scale and as a particle (*viz.* a *photon*) on the *microscopic scale* (*i.e.*, in the absorption or emission of light). This is known as wave-matter *dual nature*, *i.e.* light behaves as corpuscles or particles as well as *waves*.

This is applicable to small and moving particles like electrons, protons, atoms, molecules and all fundamental particles, in general as well as to photons. They exhibit a similar duality of nature in the reverse sense, *i.e.*, behave as corpuscles or particles on a macroscopic scale and as waves on a microscopic scale, with a frequency and a wavelength, called *de Broglie wavelength* associated with them – this wavelength being given by $\lambda = h/mc$ for a photon (its velocity being c) and $\lambda = h/mv$ for a particle, in general, where h is the well known Plank’s constant, equal to 6.62×10^{-27} erg-sec. or 6.62×10^{-34} J-sec.

Thus, with the wave-like nature of tiny particles, their behaviour cannot possibly be predicted by means of either the Classical or the Relativistic mechanics. Indeed, *Werner Heinsberg* enunciated in the year 1927 his famous **principle of uncertainty**, according to which *the position and the momentum of a particle cannot both be measured simultaneously with an equal degree of accuracy*, meaning thereby that if the position of the particle be measured accurately, the inaccuracy in the measurement of its momentum increases correspondingly and vice versa, such that *the product of the uncertainty Δr in the position of the particle at any given instant and the uncertainty Δp in its momentum at the same instant is equal, or greater than, the Plank’s constant h , i.e.,*

$$\Delta r \Delta p \geq h?, \quad [\text{where } h = 6.62 \times 10^{-27} \text{ erg-sec.}]$$

This uncertainty is by no means due to any want of perfection in the measuring instruments or the experiment but is *inherent* in the very nature of things,—the measurement of one (position or momentum) disturbing the other by an unknown amount.

This strange behaviour of matter on the atomic scale lies in the domain of *Quantum mechanics* (or *wave mechanics*), due to painstaking mathematical investigations of *Schrodinger*, *Heinsberg*, *Bohr* and *Dirac*, the contribution of *Dirac* being, in main, the incorporation into it of the relativistic effects in conformity with the Special Theory of Relativity.

On account of the very small value of h , however, the uncertainty in the measurement of position and momentum of a particle is appreciable only if it be of atomic size and is quite negligible for larger particles. So that, in the case of relatively larger bodies, quantum mechanics too gets reduced, for all intents and purposes, to classical mechanics.

Thus, except in the case of bodies moving very fast ($v \rightarrow c$) or having extremely small or atomic size, the classical mechanics gives results quite in conformity with experimental facts. And, since in everyday life we mostly come across bodies of relatively large size, moving with speeds far less than that of light (c), we shall concern ourselves, in general, with only the classical mechanics except in cases where it happens to be patently inapplicable.

I – DYNAMICS OF UNCHARGED PARTICLES

4.2 EQUATIONS OF MOTION OF AN UNCHARGED PARTICLE

As mentioned already, the classical mechanics is based on Newton's law of motion and the invariance of space, time and mass. So that, starting with the 2nd law, which is the *law* of motion including, as it does, both the first and the third laws, we have

force equal to rate of change of momentum, i.e.

$$F = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) \quad \left[\because \vec{p} = m\vec{v} \right]$$

$$\text{Since mass remains constant, we have } \vec{F} = m \frac{d\vec{v}}{dt} = m \frac{d^2\vec{r}}{dt^2} = m\vec{a} \quad \dots(A)$$

where $d\vec{v}/dt = d^2\vec{r}/dt^2 = \vec{a}$ the acceleration of the particle.

(i) Now, putting relation (A) above as $d^2\vec{r}/dt^2 = \vec{F}/m = \vec{a}$, we have on integrating with respect to t ,

velocity of the particle at instant t , i.e., $d\vec{r}/dt = \vec{a}t + C_1$ where C_1 is the constant of integration.

Since at $t = 0$, $d\vec{r}/dt = C_1$, we have x initial velocity of the particle (*i.e.*, its velocity just before the force is applied to it) equal to C_1 .

Representing the *initial velocity* by the usual notation $\vec{u}$ and the final velocity at instant t by $\vec{v}$, we therefore, have

$$\frac{d\vec{r}}{dt} = \vec{v} = \vec{u} + \vec{a}t \quad \dots(i)$$

(ii) Integrating equation (i) again with respect to t , we have *position of the particle at instant t* given by $\vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2 + C_2$

where C_2 is another constant of integration.

Again, at $t = 0$, $\vec{r} = C_2$ where C_2 represents the *initial position vector* $\vec{r}_0$ of the particle.

So that, $\vec{r} = \vec{u}t + \frac{1}{2}\vec{a}t^2 + \vec{r}_0$ whence, $(\vec{r} - \vec{r}_0) = \vec{u}t + \frac{1}{2}\vec{a}t^2$

Clearly, $(\vec{r} - \vec{r}_0) = \vec{S}$ the *displacement vector* of the particle at instant t .

We, therefore, have $\vec{S} = \vec{u}t + \frac{1}{2}\vec{a}t^2$... (ii)

(iii) We have *acceleration of the particle*, $\vec{a} = d\vec{v}/dt$.

Taking its *dot product* with the velocity $\vec{v}$ of the particle at instant t ,

we have, $\vec{v} \cdot \vec{a} = \vec{v} \cdot d\vec{v}/dt$

or, since $\vec{v} = d\vec{r}/dt$ we have $\frac{d\vec{r}}{dt} \cdot \vec{a} = \vec{v} \cdot \frac{d\vec{v}}{dt}$, integrating which with respect to t , we have

$$\int \frac{d\vec{r}}{dt} \cdot \vec{a} dt = \int \vec{v} \cdot \frac{d\vec{v}}{dt} dt \text{ or, } \vec{r} \cdot \vec{a} = \frac{1}{2} \vec{v} \cdot \vec{v} + C_3 \text{ or, } \vec{r} \cdot \vec{a} = \frac{1}{2} v^2 + C_3$$

where C_3 is yet another constant of integration.

At $t = 0$, clearly, $\vec{v} = \vec{u}$ and $\vec{r} = \vec{r}_0$ So that, $\vec{r} \cdot \vec{a} = \frac{1}{2} u^2 + C_3$

or, $C_3 = \vec{a} \cdot \vec{r}_0 - \frac{1}{2} u^2$

Substituting this value of C_3 in the expression above, we therefore have

$$\vec{r} \cdot \vec{a} = \frac{1}{2} v^2 + \vec{a} \cdot \vec{r}_0 - \frac{1}{2} u^2 \text{ or, } \frac{1}{2} (v^2 - u^2) = \vec{a} \cdot (\vec{r} - \vec{r}_0)$$

or, $v^2 - u^2 = 2aS$ or, $v^2 = u^2 + 2aS$... (iii)

These relations, (i), (ii) and (iii), are the three *general equations of motion of a particle*.

Let us now consider *two particular cases*.

(a) **Case of a body falling freely under the action of gravity.** If we consider a body acted upon only by the gravitational force of attraction due to earth, the acceleration on it (called the *acceleration due to gravity*) acts *vertically downwards*, so that, taking the origin at the surface of the earth and the x -axis *vertically upwards* we have *equation of motion* of the body or the particle, $md^2x/dt^2 = -mg$.

Or, *acceleration of the particle*, $d^2x/dt^2 = -g$.

This expression, on integration with respect to t , gives the velocity v of the particle at a given instant t , i.e., $dx/dt = v = -gt + C_1$, where C_1 is the *constant of integration*.

Since at $t = 0$, $v = u$, the initial velocity of the particle, we have $C_1 = u$, and, therefore,

$$dx/dt = v = -gt + u. \text{ Or, } v = u - gt. \text{ ... (iv)}$$

Again, integrating the above expression with respect to t , we have *position of the body or the particle at instant t* given by

$$x = ut - \frac{1}{2}gt^2 + C_2 \text{ where } C_2 \text{ is another constant of integration}$$

Again, at, $t = 0$, $x = x_0$, so that, $C_2 = x_0$.

And therefore, $x = ut - \frac{1}{2}gt^2 + x_0$ whence, $(x_0 - x) = -ut + \frac{1}{2}gt^2$

Since $(x_0 - x) = h$, the *height through which the body has fallen*, we have

$$h = -ut + \frac{1}{2}gt^2 \text{ ... (v)}$$

Finally, we have $g = -dv/dt$.

Taking the dot product of this expression with v , we have

$$vg = -v \cdot dv/dt \text{ or, } (dx/dt) \cdot g = -v \cdot dv/dt, \quad [\because v = dx/dt.]$$

Integrating which with respect to t , we have

$$\int \frac{dx}{dt} \cdot g \cdot dt = \int -v \cdot \frac{dv}{dt} dt$$

or, $x \cdot g = -v^2/2 = C_3$, where C_3 is a constant of integration.

Since at $t = 0$, $x = x_0$ and $v = u$, we have $x_0 g = -u^2/2 + C_3$, whence, $C_3 = u^2/2 + x_0 g$.

$\therefore xg = -v^2/2 + u^2/2 + x_0 g$. or, since $(x_0 - x) = h$, we have

$$v^2 = u^2 + 2gh, \quad \dots(vi)$$

whence, the *velocity acquired by a body starting from rest and falling freely through height h* is clearly, $v = \sqrt{2gh}$

For a body thrown vertically upwards, we simply put $-h$ in place of h in the expressions deduced above, when we obtain the following corresponding relations:

$$v = u - gt \quad \dots(vii),$$

$$h = ut - \frac{1}{2}gt^2 \quad \dots(viii)$$

$$\text{and} \quad v^2 = u^2 - 2gh \quad \dots(ix)$$

So that, from relation (ix), the *maximum height attained by the body* (when $v = 0$) is $h = u^2/2g$ and, from relation (vii), the *time taken to do so* is $t = u/g$.

(b) **The projectile.** In the cases of motion considered above, the motion of the body or the particle takes place *along the direction of the applied force* and the problem is thus *one-dimensional*. Let us now consider a two-dimensional problem, viz., the motion of a *projectile*.

A *projectile* is a body or a particle, projected at an angle to the horizontal, this angle being referred to as its *angle of elevation or projection* and the path taken by it as its *trajectory*.

Thus, suppose a particle is projected with velocity u from a point O at an angle α with the horizontal OX , (Fig. 4.1). Let OY be an up ward vertical through O and let the position of the particle at time t after projection be $P(x, y)$, as shown.

Then, assuming no resistance due to air, the particle is throughout its motion, under the influence of the uniform acceleration due to gravity alone, acting vertically downwards. Its equations of motion along the horizontal and the vertical are, therefore, $md^2x/dt^2 = 0$ and $md^2y/dt^2 = -mg$ respectively, i.e. $d^2x/dt^2 = 0$ and $d^2y/dt^2 = -g$.

Since u is the velocity of projection of the particle at an angle α with the horizontal, its components *along the horizontal and the vertical* are respectively.

$$u_x = u \cos \alpha \quad \text{and} \quad u_y = u \sin \alpha.$$

Then, since the particle has no acceleration along the horizontal, its position after time t along this direction is given by

$$x = u_x t = u \cos \alpha t \quad \dots(x)$$

And since there is an acceleration $-g$ acting on it along the vertical, its position after time t along this direction is, in accordance with relation (viii) above, given by

$$y = u_y t - \frac{1}{2}gt^2 = u \sin \alpha t - \frac{1}{2}gt^2 \quad \dots(xi)$$

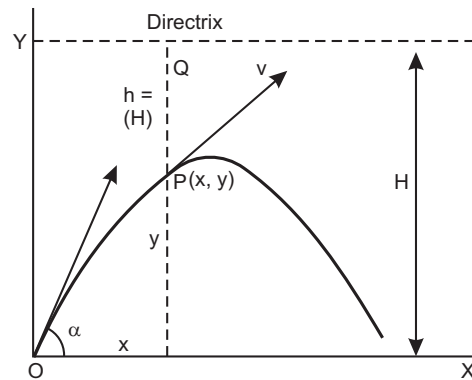


Fig. 4.1

From relation (x), $t = x/u \cos \alpha$. Substituting this value of t in relation (xi), we, therefore, have

$$y = u \sin \alpha \frac{x}{u \cos \alpha} - \frac{1}{2} g \left(\frac{x}{u \cos \alpha} \right)^2$$

$$\text{or,} \quad y = x \tan \alpha - \frac{gx^2}{2u^2 \cos^2 \alpha} \quad \dots (xii)$$

indicating that *the trajectory of the projectile is parabolic in shape.*

Now, relation (xii) may be put as

$$\frac{g - x^2}{2u^2 \cos^2 \alpha} - x \frac{\sin \alpha}{\cos \alpha} + y = 0$$

$$\text{or,} \quad x^2 - x \frac{\sin \alpha}{\cos \alpha} \cdot \frac{2u^2 \cos^2 \alpha}{g} + y \frac{2u^2 \cos^2 \alpha}{g} = 0$$

$$\text{or,} \quad x^2 - \frac{2u^2 x \sin \alpha \cos \alpha}{g} + \frac{2u^2 \cos^2 \alpha \cdot y}{g} = 0$$

$$\text{or,} \quad x^2 - \frac{2u^2 x \sin \alpha \cos \alpha}{g} + \frac{u^4 \sin^2 \alpha \cos^2 \alpha}{g^2} = \frac{u^4 \sin^2 \alpha \cos^2 \alpha}{g^2} - \frac{2u^2 \cos^2 \alpha y}{g}$$

$$\text{or,} \quad \left(\frac{x - u^2 \sin \alpha \cos \alpha}{g} \right)^2 = - \frac{2u^2 \cos^2 \alpha}{g} \left(y - \frac{u^2 \sin^2 \alpha}{2g} \right)$$

showing that *the vertex of the parabola is at the point $u^2 \sin \alpha \cos \alpha / g$, $u^2 \sin^2 \alpha / 2g$, and that the length of the latus rectum is $2u^2 \cos^2 \alpha / g$. The directrix is thus horizontal and at a height $u^2 \cos^2 \alpha / 2g$ above the vertex or $u^2 / 2g$ above the horizontal line OX .*

The velocity of the projectile at any instant t is given by $v^2 = v_x^2 + v_y^2$

where v_x and v_y are the velocities of the particle at time t along the axes of x and y respectively.

There being no acceleration along the x -axis, $v_x = u_x = u \cos \alpha$. And the acceleration along the axis of y being $-g$, $v_y = u_y - gt = u \sin \alpha - gt$, in accordance with relation (vii) above. So that,

$$\begin{aligned} v^2 &= v_x^2 + v_y^2 = (u \cos \alpha)^2 + (u \sin \alpha - gt)^2 \\ &= u^2 \cos^2 \alpha + u^2 \sin^2 \alpha - 2u \sin \alpha gt + g^2 t^2 \\ &= u^2 - 2u g \sin \alpha t + g^2 t^2 = u^2 - 2g u \sin \alpha t - \frac{1}{2} g t^2. \end{aligned}$$

$$\text{or,} \quad v^2 = u^2 - 2g y,$$

$$\text{whence,} \quad v = \sqrt{u^2 - 2gy}$$

This velocity v is the same as that the particle would acquire in falling freely to P from a point on the directrix vertically above P . (See worked example 4).

Range of the projectile. Putting $y = 0$ in relation (xii) above,

$$\text{we have} \quad \tan \alpha = gx / 2u^2 \cos^2 \alpha.$$

$$\text{or,} \quad x = \frac{\sin \alpha}{\cos \alpha} \left(\frac{2u^2 \cos^2 \alpha}{g} \right) = \frac{\sin \alpha \cdot 2u^2 \cos \alpha}{g}$$

$$\text{or,} \quad x = \frac{u^2 \sin 2\alpha}{g}$$

This gives the *maximum horizontal distance covered by the projectile* or the *range of the projectile on the horizontal plane*. Denoting it by R , we therefore, have

$$R = u^2 \sin 2\alpha / g.$$

For a given range R and for a given velocity of projection, u , there are two possible angles of elevation or projection, viz., α and $(\pi/2 - \alpha)$ and, clearly, the range (R) has its maximum value u^2/g , when $\sin 2\alpha = 1$ i.e., when, $2\alpha = 90^\circ$ or $\alpha = 45^\circ$ or $\pi/4$.

Example 1. A force $\vec{F} = 25\hat{i} + 6\hat{j}$ newtons acts on a particle of mass 2.5 kg for 5 sec. If the initial position of the particle is $\vec{r}_0 = 6\hat{j} + 8\hat{k}$ metre and the initial velocity $\vec{u} = 2.5\hat{i} + 3\hat{k}$ ms⁻¹ calculate (i) the final velocity of the particle, (ii) the final position of the particle, (iii) the work done by the force on the particle. (Agra,)

Solution. (i) Since the initial velocity of the particle, $\vec{u} = 2.5\hat{i} + 3\hat{k}$ m/sec and its acceleration $\vec{a} = \vec{F}/m = (2.5\hat{i} + 6\hat{j})/2.5$ m/sec², we have

final velocity of the particle after time $t = 5$ sec, i.e., $\vec{v} = \vec{u} + \vec{a}t$

$$\text{or, } \vec{v} = 2.5\hat{i} + 3\hat{k} + \left(\frac{25\hat{i} + 6\hat{j}}{2.5}\right)5 = 2.5\hat{i} + 3\hat{k} + 50\hat{i} + 12\hat{j} = 52.5\hat{i} + 12\hat{j} + 3\hat{k} \text{ m/sec}$$

(ii) The final position of the particle is given by $\vec{r} = \vec{r}_0 + \vec{u}t + \frac{1}{2}\vec{a}t^2$

$$= (6\hat{j} + 8\hat{k}) + (2.5\hat{i} + 3\hat{k})5 + \frac{1}{2}\left(\frac{25\hat{i} + 6\hat{j}}{2.5}\right)25 = 6\hat{j} + 8\hat{k} + 12.5\hat{i} + 15\hat{k} + 125\hat{i} + 30\hat{j}$$

$$\text{or, } \vec{r} = 137.5\hat{i} + 36\hat{j} + 23\hat{k}$$

(iii) Clearly, work done by the force on the particle, i.e., $W = (\text{force} \times \text{distance through which the force is applied}) \vec{F} = (\vec{r} - \vec{r}_0)$

$$= (25\hat{i} + 6\hat{j})[(137.5\hat{i} + 36\hat{j} + 23\hat{k}) - (6\hat{j} + 8\hat{k})]$$

$$\text{or, } W = (25\hat{i} + 6\hat{j})(137.5\hat{i} + 30\hat{j} + 15\hat{k}) = (3437.5 + 180) = 3617.5 \text{ joules.}$$

Example 2. The initial positions of two particles are $-2, 0$ and $0, -2$ and they start simultaneously along the axes of x and y with uniform velocities $3\hat{i}$ cm/sec and $4\hat{j}$ cm/sec respectively. (i) Obtain the vector representing the position of the second particle with respect to the first as a function of time. (ii) When and where will the two particles be closest to one another?

Solution. Let the velocities of the two particles be $\vec{u}_1$ and $\vec{u}_2$ respectively. Then, we have $\vec{u}_1 = 3\hat{i}$ cm/sec and $\vec{u}_2 = 4\hat{j}$. And, therefore,

(i) Position of the first particle after time t is given by

$$\vec{r}' = \vec{r}'_0 + \vec{u}_1 t = -2\hat{i} + (3\hat{i})(t) = (3t - 2)\hat{i}$$

and position of the second particle after the same time t is given by

$$\vec{r}'' = \vec{r}''_0 + \vec{u}_2 t = -2\hat{j} + (4\hat{j})(t) = (4t - 2)\hat{j}$$

$\therefore$ Position vector of the second particle with respect to the first is given by

$$\vec{r} = \vec{r}'' - \vec{r}' = (4t - 2)\hat{j} - (3t - 2)\hat{i} \text{ cm}$$

(ii) Obviously, the two particles will be closest together when $\vec{r}$ has the minimum value, i.e.,

$$\text{when } \frac{d}{dt}(r^2) = 0 \text{ or, } \frac{d}{dt}[(4t - 2)^2 + (3t - 2)^2] = 0$$

i.e., when $2(4t - 2)(4) + 2(3t - 2)(3) = 0$, or, when $t = 28/50 = 0.56$ sec.

At this time, the position of the second particle with respect to the first will be given by

$$\vec{r} = (4 \times 0.56 - 2)\hat{j} - (3 \times 0.56 - 2)\hat{i} = (0.32\hat{i} + 0.24\hat{j}) \text{ cm.}$$

Thus, the two particles will be the closest to one another after 0.56 sec when the position vector of the second particle with respect to the first is given by

$$\vec{r} = (0.32\hat{i} + 0.24\hat{j}) \text{ cm.}$$

II – DYNAMICS OF CHARGED PARTICLES

4.3 SOME DEFINITIONS AND PRELIMINARY RELATIONS

Before proceeding with the study of the behaviour of a charged particle in an electric or a magnetic field, let us refresh our memory as to the meanings of some of the terms we shall have occasion to use so frequently in our discussion, as also recapitulate some simple preliminary relations.

(i) **Electric field and intensity of the field.** The area round about an electric charge q , within which its influence is perceptible, is referred to as its *electric field*, and the force experienced by a unit positive charge placed at any point in the field is called the *strength* or the *intensity* of the field at that point, or, usually, simply the *field* at that point and is denoted by the symbol E . Having both magnitude and direction, it is a *vector quantity* and hence the intensity at a point due to a number of charges is equal to the *vector sum* of the intensities there due to the individual charges.

Further, if the intensity of the field be the same at all points in it, it is said to be a *uniform electric field*.

It follows, therefore, that the force experienced by a charge q , when placed at a point where the intensity of the field is E , is equal to qE .

It may be as well be mentioned here that in the *electrostatic system of units*, q is measured in *esu* or *statcoulombs*. The *practical unit of charge* in this system is the *coulomb* $= 3 \times 10^9$ *esu* or *statcoulombs*. In the *Rationalised MKS* (or the *SI*) system, which is now increasingly coming into vogue, the unit of charge is the *coulomb* the same as the practical unit in the electrostatic system of units.

Since the intensity of the electric field E is the force on unit charge, its unit in the *esu* system is *dynes/esu*. And since *field intensity is also defined as rate of change of potential with distance* and the unit of potential in the electrostatic system is the *statvolt*, the unit of intensity in this system is also 1 *statvolt/cm*. The *practical unit of E* in this system is *volt/cm*, where 1 volt is 1/300th of the *esu*, i.e., $= 1/300$ *statvolt*.

In the *RMKS* (or the *SI*) system, the unit of electric intensity is *Newton/coulomb* (i.e., N/C) or 1 *volt/meter*, where 1 *newton* $= 10^5$ *dynes*. We, thus have $1 N/C = 1 \text{ volt}/1\text{meter} = 10^5 \text{ dynes}/3 \times 10^9 \text{ stat coulombs} = 1/3 \times 10^4 \text{ statvolts/cm}$.

(ii) **Kinetic energy of a charged particle in an electric field.** Since the potential (V) between two points in an electric field is defined as the work done in moving a unit positive charge from one point to the other (quite independently of the path taken), it is clear that the work done in taking a charge q from one point to the other (along any path) will be qV .

This work appears in the form of increased *K.E.* of the particle. So that, if m be the *mass* of the particle carrying the charge q and v , the velocity acquired by it, we have

$$\frac{1}{2}mv^2 = qV = \text{charge} \times \text{potential difference.}$$

Thus, a particle carrying a positive charge *gains* or *loses* energy according as it moves from a higher to a lower potential or from a lower to a higher one. Naturally, the reverse will be the case if the charge carried by the particle be negative.

In the electrostatic system, this *K.E.* is measured in *ergs* and in the *RMKS* (or the *S.I.*) system, in *joules*. More often than not, however, it is measured in *electron volts (eV)** where $1 \text{ eV} = 1.60 \times 10^{-12} \text{ ergs} = 1.60 \times 10^{-19} \text{ joules}$.

(iii) **Magnetic field and strength of a magnetic field—Flux density.** The region in which magnetic effects can be detected is called a *magnetic field* and the strength of a magnetic field at any point in it is measured by the *magnetic flux density* B at that point, where flux density is defined as *the magnetic flux per unit area perpendicular to the direction of the flux*. In the *emu* or the *Gaussian system*, its unit in the *gauss* and in the *RMKS* (or the *SI*) system, it is *weber/metre² (Wb/m²)* of *tesla (T)*, where $1 \text{ Wb/m}^2 \text{ (or T)} = 10^4 \text{ gauss}$.

The flux density (B) is a *vector quantity*, having *magnitude* as well as *direction* (which is the same as that of the lines of magnetic flux).

(iv) **Force on a moving charge in a magnetic field.** We know that the force acting on a conductor of length δl carrying a current I in a magnetic field of strength B is given by $F_{\text{mag}} = BI \delta l \sin \theta$, where θ is the angle that the conductor makes with the direction B . This force is perpendicular to both the field (B) and the current (I) and its direction is given by the familiar *Fleming's left hand rule*. Putting it in vector notation, we have

$$\vec{F}_{\text{mag}} = I \delta \vec{l} \times \vec{B} = \delta l \vec{I} \times \vec{B}$$

where $\delta \vec{l}$ and $\vec{I}$ are obviously *parallel vectors*.

If the quantities δl , I and B be measured in *emu*, F_{mag} is in dynes, but if they be measured in *RMKS* (or *SI*) units F_{mag} is in newtons (where $1 \text{ N} = 10^5 \text{ dynes}$).

Now, as we know, *a moving charge constitutes a current* and must, therefore, experience a similar force in a magnetic field.

Lorentz force

Thus, if $\vec{v}$ be the velocity of a charge q moving in a direction making an angle θ with the field of ($\vec{B}$), as shown in Fig. 4.2, the distance covered by it in time $\delta t = \vec{v} \cdot \delta t$. This is equivalent to a conductor of length $\delta l = \vec{v} \cdot \delta t$ inclined to the direction of $\vec{B}$ at an angle θ and carrying a current $I = q/\delta t$ ($\because$ current is the rate of flow of charge).

$\therefore$ Force acting on the charge, i.e.,

$$F_{\text{mag}} = \frac{q}{\delta t} \vec{v} \delta t \times \vec{B} \sin \theta = qvB \sin \theta$$

or, in vector notation, $\vec{F}_{\text{mag}} = q \vec{v} \times \vec{B}$

This force is called *Lorentz force* and its direction, as given by the *right-handed screw rule*, is perpendicular to both $\vec{v}$ and $\vec{B}$.

If the quantities q , $\vec{v}$ and $\vec{B}$ be measured in *emu*, $\vec{F}_{\text{mag}}$ is in dynes, but if measured *RMKS* (or *SI*) units, it is in *newtons*.

Quite often, however, q and $\vec{B}$ are measured in the *Gaussian system of units*, i.e., q in *esu* and B in *emu* (or *gauss*). We have, therefore, to convert q into *emu* by dividing it by c (the speed of light in free space), because $1 \text{ esu of charge} = 1/c \text{ of emu of charge}$. In such a case, then, the expression for F_{mag} becomes.

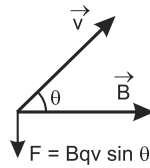


Fig. 4.2

* An *electron-volt* is the energy acquired by an electron (carrying a charge $e = 4.8 \times 10^{-19} \text{ esu}$) on moving through a potential difference of 1 *volt* (equal to $1/300 \text{ esu}$) and is thus equal to $qV = eV = 4.8 \times 10^{-10} \times 1/300 = 1.60 \times 10^{-12} \text{ ergs}$ or $1.60 \times 10^{-12}/10^7 = 1.60 \times 10^{-19} \text{ joules}$.

$$\vec{F}_{mag} = \frac{q}{c} \vec{v} \times \vec{B}$$

It will be noted that *for a force to act on a charged particle in a magnetic field,*

(i) *the particle must be in motion, i.e., $v \neq 0$, because a stationary charge does not constitute a current.*

(ii) *the velocity $\vec{v}$ of the particle must not be in the same direction as $\vec{B}$, or else, under the rule of vector product, $\vec{v} \times \vec{B}$ would be equal to zero.*

Further, since the force ($\vec{F}_{mag}$) acts in a direction perpendicular to $\vec{v}$ the velocity of the particle, *there is no change in the magnitude of the velocity but only a continuous change in its direction.*

This means, in other words, that *there is no change in the kinetic energy of a charged particle during its motion in a magnetic field.*

It may as well as be mentioned here that in case the charged particle is also simultaneously subjected to an electric field $\vec{E}$, the force on it due to the latter is $\vec{F}_{el} = q\vec{E}$. And therefore, *total force acting on the charged particle, i.e.,*

$$\vec{F} = \vec{F}_{el} + \vec{F}_{mag} = q\vec{E} + \frac{q}{c} \vec{v} \times \vec{B}$$

This force $\vec{F}$, due to the two fields together, is also called *Lorentz force*.

4.4 CHARGED PARTICLE IN A UNIFORM AND CONSTANT ELECTRIC FIELD—

We have seen under §4.2 above how, in accordance with Newton's second law of motion, the force acting on a particle is given by $\vec{F} = d^2 \vec{r} / dt^2$, where m is the *mass* of the particle (remaining constant at non-relativistic speeds) and $d^2 \vec{r} / dt^2$ its acceleration.

Now, imagine a particle of mass m and carrying a charge q to be placed in a uniform and constant electric field $\vec{E}$. The force acting on it will obviously be $q\vec{E}$ (see §4.3 (i) above). If the acceleration acquired by $t^{1/1}$ particle under the action of this force be $\vec{a} = d^2 \vec{r} / dt^2$,

we clearly have

$$m\vec{a} = md^2 \vec{r} / dt^2 = q\vec{E} \text{ whence, } d^2 \vec{r} / dt^2 = \frac{q}{m} \vec{E}.$$

Integrating expression (i) with respect to t , we have *velocity of the particle*, $\vec{v} = \frac{d\vec{r}}{dt} = \frac{q\vec{E}}{m} t + \vec{C}_1$ where C_1 is the constant of integration.

Clearly, at $t = 0$, $\vec{v} = \vec{u}$, the *initial velocity* of the particle. So that $C_1 = \vec{u}$. We therefore, have

$$\frac{d\vec{r}}{dt} = \vec{v} = \frac{q\vec{E}}{m} t + \vec{u}$$

As will be easily seen, this is a relation similar to $\vec{v} = \vec{u} + \vec{a}t$ for an uncharged particle, since $q\vec{E}/m$ here is equal to $\vec{a}$.

Again integrating relation (ii), we have

$$\vec{r} = \frac{1}{2} \frac{q\vec{E}}{m} t^2 + \vec{u}t + \vec{C}_2$$

where C_2 is another constant of integration.

Again, at $t = 0$, $\vec{r} = \vec{r}_0$ the initial position vector of the particle, and we have $\vec{r}_0 = \vec{C}_2$

$$\text{So that, } \vec{r} = \frac{1}{2} \frac{q\vec{E}}{m} t^2 + \vec{u}t + \vec{r}_0 = \frac{1}{2} \vec{a} t^2 + \vec{u}t + \vec{r}_0 \quad \dots (iii)$$

The vector diagram corresponding to it is as shown in Fig. 4.3, where $\vec{OA}$ represents the *initial position vector* $\vec{r}_0$ at $t=0$, $\vec{AB}$ represents $\vec{u}t$, the distance covered in time t due to the initial velocity $\vec{u}$ and $\vec{BC}$ represents $\frac{1}{2}\vec{a}t^2$ or $\frac{1}{2}\frac{q\vec{E}}{m}t^2$, the distance covered on account of the acceleration acquired due to field $\vec{E}$, directed along BC . The resultant displacement $\vec{r}$ of the particle is the vector sum of all the three and is represented by OC , in accordance with the polygon law of addition of vectors.

Two particular cases. Having discussed above the general case of the motion or displacement of a charged particle when placed in an electric field, let us consider two particular cases, viz., (i) when the electric field acts along the direction of motion of the particle and (ii) when it acts in a direction perpendicular to it.

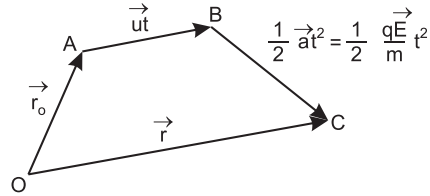


Fig. 4.3

(i) It will be easily seen that in the case of both $\vec{u}$ and $\vec{r}_0$ being directed along the electric field $\vec{E}$, the acceleration produced in the particle will also be directed along $\vec{E}$. The electric field is, in this case called a **longitudinal field** and the acceleration acquired by the particle, a **longitudinal acceleration**. The resultant position vector $\vec{r}$ will now also be directed along $\vec{E}$, thus making the whole problem a simple *one-dimensional* one.

Also, if $\vec{r}_0 = 0$ and $\vec{u} = 0$, i.e., if the particle is initially at the origin and starts from rest, we shall again have a one-dimensional problem with

$$\vec{r} = \frac{1}{2}\vec{a}t^2 = \frac{1}{2}\frac{q}{m}\vec{E}t^2 \text{ in the direction of } \vec{E},$$

exactly in the manner that the distance covered by an uncharged particle, starting from rest, is given by $\vec{S} = \frac{1}{2}\vec{a}t^2$

So that, in either case, we may replace the vectors $\vec{r}_0$, $\vec{r}$, $\vec{u}$, $\vec{v}$ and $\vec{E}$ by their respective scalar magnitudes.

(ii) Let us now discuss the case **when the electric field is a transverse one**, i.e., in a direction perpendicular to the initial velocity of the particle. In this case, the acceleration acquired by the particle is also referred as the **transverse acceleration**.

Thus, imagine a beam of charged particles, each of mass m and carrying a charge q , to be travelling along the axis of x with velocity v and passing in between two metallic plates P_1 and P_2 (Fig. 4.4), maintained at a constant potential difference V , with the upper plate positive with respect to the lower one.

Then, if d be the distance between the two plates, the electric field set up between them is V/d and since it is directed from the upper to the lower plate, or along the axis of y , let it be denoted by E_y .

As soon as the beam of charged particles enters in between the plates at the origin O , it is subjected to this transverse electric field E_y . Since this field

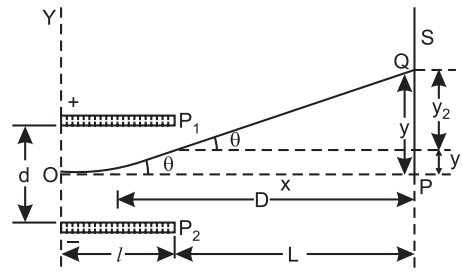


Fig. 4.4

is perpendicular to the velocity v_x of beam, the latter remains unaffected and the beam, therefore, continues to cover the horizontal distance between the plates with the same velocity v_x . If l be the length of the plates or that of the electric field set up between them, the time taken by the beam to cover this distance is l/v_x .

During this time, obviously, each particle of the beam acquires a *transverse acceleration* $a = qE_y/m$ and hence a transverse velocity $v_y = 0 + at$.

$$\text{or} \quad v_y = 0 + q \cdot \frac{E_y}{m} t = \frac{qE_y}{m} \cdot \frac{1}{v_x}$$

[$\because$ the initial transverse velocity is *zero*.]

$\therefore$ *Transverse displacement of the beam* (upwards or downwards, according as the charge on the particles is *negative* or *positive*) during the time $t = l/v_x$ that it takes to cover the entire electric

field of length l is say, $y_1 = 0 + \frac{1}{2} at^2$

$$\text{i.e.} \quad y_1 = \frac{1}{2} \frac{qE_y}{m} \cdot l^2 = \frac{qE_y}{2mv_x^2} \cdot l^2$$

This is clearly the equation to a parabola ($qE_y/2mv_x^2$ being a constant quantity). So that, the beam of charged particles takes a parabolic path in between the two plates, as shown.

It will be seen that the problem here is a *two-dimensional* one, similar to that of a particle thrown horizontally and subjected to the downward acceleration due to gravity (*i.e.*, similar to that of a projectile). The important difference, however, is that, unlike the gravitational force acting on the uncharged particle, we can here alter the electric field and hence the force acting on the particle, as desired. Also, the electric field is effective only within a l , the length of the two plates, beyond which it suddenly drops to *zero* and the beam, therefore, moves straight along the direction in which it emerges from the electric field between the plates.

If there be a screen S at a distance L from the ends of the plates, the beam will strike it at, say, Q , making an angle θ with the x -axis, instead of at P , along it, as it would if it were undeflected. The *total vertical displacement* of the beam is thus $PQ = y$, which is clearly made up of two parts: y_1 within the plates (or the electric field) and y_2 , on emergence from the plates (or the electric field).

$$\text{As we have seen above,} \quad y = \frac{qE_y}{2mv_x^2} \cdot l^2$$

$$\text{And} \quad y = L \tan \theta,$$

$$\text{where} \quad \tan \theta = \frac{v_y}{v_x} = \frac{qE_y / mv_x}{v_x} = \frac{qE_y l}{mv_x^2}$$

$$\text{So that,} \quad y_2 = \frac{LqE_y l}{mv_x^2}$$

And, therefore, *total displacement* of the beam,

$$y = y_1 + y_2 = \frac{qE_y l^2}{2mv_x^2} + \frac{LqE_y l}{mv_x^2}$$

$$\text{or,} \quad y = \frac{qE_y l}{mv_x^2} \left(L + \frac{l}{2} \right)$$

Since $qE_y l / mv_x^2 = \tan \theta$ and $(L + l/2) =$ distance D of the screen from the centre C of the electric field between the plates, we have

$$y = D \tan \theta$$

This is the *principle* underlying the *cathode ray oscillograph*, *television picture tubes* and a host of other experiments, like determination of e/m (i.e., *charge to mass ratio*) for an electron etc.

In all these cases, an electron beam is obtained from a hot cathode or filament, surrounded by a hollow metallic cylinder, kept at a negative potential with respect to the cathode and called the *modulator*. This helps to make the electrons emitted by the cathode into a compact beam. The electron beam is then subjected to a high accelerating potential V' , say, between the cathode and a pair of anodes, so that the energy acquired by each electron is eV' , where e is the *charge* on an electron. If m be the *mass* of an electron and v_x , the velocity acquired by it along the axis of x , clearly,

$$\frac{1}{2}mv_x^2 = eV', \text{ whence, } v_x = \sqrt{2eV'/m}.$$

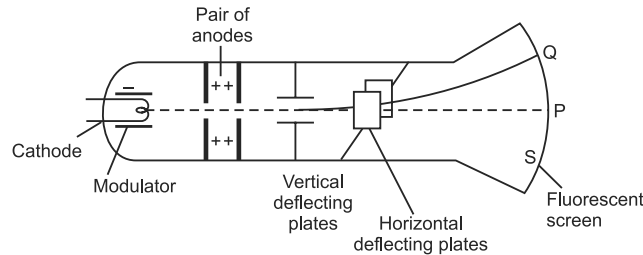


Fig. 4.5

In the case of the *cathode ray oscillograph*, (Fig. 4.5), this high velocity electron beam is passed through two pairs of plates which, when a potential difference is applied to them, have a vertical and a horizontal electric field set up in between them and thus deflect the electron beam passing through them vertically and horizontally respectively. Thus, activating any one pair of plates at a time, the beam may be deflected vertically or horizontally, as desired. The deflection of the beam is noted on fluorescent screen from the position of the bright spot produced first by the undeflected beam at P and then by the deflected beam at Q , say in the case of vertical deflection.

The cathode ray oscillograph is put to a number of uses, one of which is to demonstrate in a spectacular manner the formation of Lissajous' figures when a particle is subjected simultaneously to simple harmonic motions at right angle to each other.

Example 3. At time $t = 0$, the velocity of an electron is $10^6 \hat{i}$ cm/sec and the initial position vector $\vec{r}_0 = 100\hat{j}$ cm. (i) Obtain the position vector after 0.1 sec. (ii) If the electric field by $10^{-2} \hat{i}$ statvolt/cm, obtain the position and velocity vectors after 10^{-8} sec.

Solution. (i) The position vector after t sec will be $\vec{r} = \vec{r}_0 + \vec{u}t$ where $\vec{u}$ is the *velocity* of the electron (supposed uniform). So that,

$$\vec{r} = 100\hat{j} + 10^6 \hat{i} (0.1) = 10^5 \hat{i} + 10^2 \hat{j} \text{ cm}$$

(ii) In this case, the electron will have an acceleration $\vec{a} = (\text{force acting on it/its mass}) = e\vec{E}/m$ where e is the *charge on the electron* $= -4.8 \times 10^{-10}$ esu and $\vec{E}$, the electric field $= 10^{-2} \hat{i}$ esu and m , the mass of the electron $= 0.91 \times 10^{-27}$ gm.

$\therefore$ Velocity of the electron after $t = 10^{-8}$ sec, i.e., $\vec{v} = \vec{u} + \vec{a}t$

$$= 10^6 \hat{i} - \frac{4.8 \times 10^{-10} (10^{-2} \hat{i})}{0.91 \times 10^{-27}} \times 10^{-8}$$

Or,
$$\vec{v} = \left(10^{-1} - \frac{4.8}{0.91} \right) 10^7 \hat{i} = -5.2 \times 10^7 \hat{i}$$

And position vector after $t = 10^{-8}$ sec, i.e., $\vec{r} = \vec{r}_0 + \vec{u}t + \frac{1}{2} \vec{a}t^2$

$$= \vec{r}_0 + \vec{u}t + \frac{1}{2} \frac{e\vec{E}}{m} t^2$$

Or,
$$\vec{r} = 100 \hat{j} + 10^5 \hat{i} (10^{-8}) - \frac{1}{2} \frac{4.8 \times 10^{-10} (10^{-2} \hat{i})}{0.91 \times 10^{-27}} \times (10^{-8})^2$$

$$= 10^2 \hat{j} + 10^{-2} \hat{i} \frac{1}{2} \frac{4.8}{0.91} \times 10^{-1} \hat{i}$$

$$= 10^2 \hat{j} + 10^{-2} \hat{i} \left(1 - \frac{24}{0.91} \right) = 10^2 \hat{j} + 10^{-2} \hat{i} \left(-\frac{23.09}{0.91} \right)$$

$$= 10^2 \hat{j} - 25 \times 10^{-2} \hat{i}$$

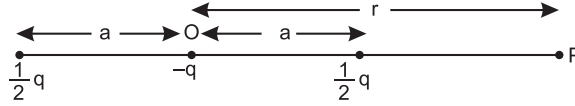
Or,
$$r = 10^2 \hat{j} - 0.25 \times 10^{-2} \hat{i} \text{ cm}$$

Example 4. The adjoining figure shows a particular charge distribution $\frac{1}{2}q, -q, \frac{1}{2}q$. Show that for $r \gg a$, the electric field at P is given by $3Q/r^4$, where $Q = qa^2$.

Solution. Here, clearly, electric field at P due to the charge $\frac{1}{2}q$ to the left of the point O is

$$= \frac{\frac{1}{2}q}{(r+a)^2} \vec{r} = \frac{q}{2(r+a)^2} \vec{r} \text{ directed outwards,}$$

where r is the unit vector along OP .



Electric field at P due to charge $\frac{1}{2}q$ to the right of the point O

$$= \frac{q}{2(r-a)^2} \vec{r} \text{ also directed outwards,}$$

and electric field at P due to the charge $-q$ at $O = \frac{-q}{r^2} \vec{r}$ directed inwards.

$\therefore$ Resultant electric field at P due to the three charges, say $\vec{E}$

$$= \left(\frac{q}{2(r+a)^2} + \frac{q}{2(r-a)^2} - \frac{q}{r^2} \right) \vec{r}.$$

$$= \frac{q}{2r^2} \left[\left(1 + \frac{a}{2} \right)^{-2} + \left(1 - \frac{a}{2} \right)^{-2} - 2 \right] \hat{r}$$

Or, expanding by Binomial Theorem and remembering that $(a/r) \ll 1$ or $a \ll r$, or what is the same thing, $r \gg a$, we have

$$\begin{aligned}\vec{E} &= \frac{q}{2r^2} \left[\left(1 - \frac{2a}{r} + \frac{(-2)(-3)}{2!} \cdot \frac{a^2}{r^2} \right) + \left(1 + \frac{2a}{r} + \frac{(-2)(-3)}{2!} \cdot \frac{a^2}{r^2} \right) \right] \vec{r} \\ &= \frac{q}{2r^2} \left(\frac{3a^2}{r^2} + \frac{3a^2}{r^2} \right) \vec{r} = \frac{3qa^2}{r^4} \vec{r} \text{ directed outwards.}\end{aligned}$$

Or, magnitude of the electric field at P due to the three charges

$$= 3qa^3/r^4 = 3Q/r^4 \text{ directed outwards.} \quad [\because qa^2 = Q]$$

Example 5. A cathode ray oscillograph has deflecting plates of length 2 cm and separation 0.5 cm. Calculate the potential difference (in volts) between the plates which will cause angular deviation of 0.04 radian in an electron beam of speed 8.0×10^3 cm/sec.

$[e/m = 5 \times 10^{17}$ esu/gm]

(Agra 2003)

Solution. Let the potential difference required to be applied to the plates be V volts = $V/300$ stat volts or esu.

$\therefore$ Electric field in between the plates

$$= \frac{\text{p.d.}}{\text{distance between the plates}} = \frac{V}{300 \times 0.5} = \frac{V}{150} \text{ statvolts/cm}$$

Hence acceleration produced in an electron in between the plates

$$= \frac{\text{force acting on the electron}}{\text{mass of the electron}} = \frac{Ve}{150 \times m} \text{ cm/sec}^2,$$

where e is the charge on the electron and m , its mass.

$$= \frac{V}{150} \times 5 \times 10^{17} = \frac{V \times 10^{17}}{30} \text{ cm/sec}^2.$$

Now, the electron remain in between the plates for an interval of time

$$t = \frac{\text{length of the plates}}{\text{horizontal velocity of the electron } (v_x)}$$

Here, length of the plates = 2 cm/sec and horizontal velocity of the electron

$$= 8 \times 10^8 \text{ cm/sec}$$

So, that

$$t = \frac{2}{8 \times 10^8} = 0.25 \times 10^{-6} = 2.5 \times 10^{-7} \text{ sec.}$$

$\therefore$ vertical velocity acquired by the electron in time, t say, v_y

$$= 0 + at = 0 + \frac{V \times 10^{17}}{30} \times 2.5 \times 10^{-7} = \frac{5V}{6} \times 10^7 \text{ cm/sec}$$

$\therefore$ angular deviation of the electron, $\theta = \frac{v_y}{v_x}$ radiation = 0.04 radian (given)

$$\text{or } \frac{5V \times 10^7}{6 \times 8 \times 10^8} = 0.04, \text{ whence, } V = \frac{6 \times 8 \times 10^8 \times 0.04}{5 \times 10^7} = \frac{19.2}{5} = 3.84 \text{ volts.}$$

Thus, the potential difference that must be applied to the plates is 3.84 volts.

Example 6. (a). Calculate the intensity of the electric field which would produce in an electron an acceleration equal to that due to gravity. (Mass of an electron = 0.91×10^{-27} gm., charge on it, $e = 4.80 \times 10^{-10}$ esu and $g = 980$ cm/sec²).

(b) Obtain the ratio of the electrostatic force between two electrons distant r apart and the gravitational force between them. ($G = 6.67 \times 10^{-8}$ dynes-cm²/gm²).

Solution. (a) Let $\vec{E}$ be the required intensity of the field. Then, clearly, force acting on an electron placed in this field $= \vec{E}e$ and, therefore, acceleration produced in the electron, $\vec{a} = \text{force/mass} = \vec{E}e/m$ cm/sec², i.e.,

$$\vec{a} = \frac{\vec{E} (4.8 \times 10^{-10})}{0.91 \times 10^{-27}} \text{ cm/sec}^2.$$

This must be equal to the acceleration due to gravity, i.e., 980 cm/sec².

$$\text{We, therefore, have } \frac{\vec{E} (4.8 \times 10^{-10})}{0.91 \times 10^{-27}} = 980, \text{ whence, } E = \frac{980 \times 0.91 \times 10^{-27}}{4.8 \times 10^{-10}}$$

$$= \frac{980 \times 0.91}{4.8} \times 10^{-17} \text{ statvolts/cm.}$$

$$\text{or, } E = \frac{980 \times 0.91}{4.8} \times 10^{-17} \times 300 = \frac{98 \times 0.91 \times 10^{-14}}{1.6} = 5.57 \times 10^{-13} \text{ volts/cm.}$$

(b) The electrostatic force between the two electrons, say, $F_E = \frac{e \cdot e}{r^2} = \frac{e^2}{r^2}$, and gravitational force between them, say, $F_G = \frac{m \times m}{r^2} G = \frac{m^2}{r^2} G$

$\therefore$ Ratio of electrostatic force and gravitational force,

$$\text{i.e., } F_E/F_G = \frac{(4.8 \times 10^{-10})^2}{(0.91 \times 10^{-27})^2 \times 6.67 \times 10^{-8}} = 4.17 \times 10^{42}$$

Example 7. A particle of charge q and mass m enters an electric field $-E\hat{j}$ between two plates with an initial velocity $v_0\hat{i}$ (i) Calculate the forces acting on it in the any y directions. (ii) Will a force in the y -direction influence the x -component of the velocity? (iii) Solve for v_x and v_y as functions of time and write the complete vector equation for the velocity $\vec{v}$ acquired after time t . (iv) Taking the origin at the point of entry, write the complete vector equation for the position of the particle as a function of time while the particle is between the plates.

Solution. (i) The x -direction being perpendicular to the direction of the field which is along the y -direction, there will be no force acting on the particle along the x -direction, i.e., $F_x = 0$. In the y -direction, naturally, the force acting on the particle $= \text{charge on the particle} \times \text{intensity of the field}$. So that, $F_y = -qE\hat{j}$.

(ii) A force in the y -direction will have no component in the x -direction and can, therefore, have no effect on the velocity of the particle in this direction.

(iii) The initial velocity of the particle is given to be $v_0\hat{i}$ along the x -direction. Since there is no force acting on the particle in this direction, its velocity in this direction remains unaltered at $v_0\hat{i}$. And, since a force $-qE\hat{j}$ acts on the particle along the y -direction, its acceleration along this direction is $a = -qE/m$. The velocity acquired by the particle along this direction in time t is, therefore, $v = 0 + at$ (the initial velocity in this direction being zero), i.e., $v = -\frac{qE}{m}t$.

The complete vector equation for the velocity of the particle is thus $\vec{v} = v_0\hat{i} - \frac{qE}{m}t\hat{j}$

(iv) The distance covered by the particle along the x -direction in time t is v_0t and the distance covered by it along the y -direction is $ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}\frac{qE}{m}t^2$

If therefore, $\vec{r}$ be the position vector of the particle after time t , we have

$$\vec{r} = \vec{r}_0 + v_0 t \hat{i} - \frac{qE}{m} t^2 \hat{j}$$

Since the origin is taken at the point of entry, $r_0 = 0$. We, therefore, have *vector equation for the position of the particle*, $\vec{r} = v_0 t \hat{i} - \frac{qE}{m} t^2 \hat{j}$.

Example 8. In example 7 above, if the charged particle be an electron, having an initial energy 10^{-10} ergs, if the electric field strength be 0.01 statvolt/cm and if the length of the plates be 2 cm, find (a) the velocity vector as it leaves the region between the plates, (b) the angle (v, i) for the particle as it leaves the plates, (c) the displacement of the electron and (d) the point of intersection of the x -axis with its direction as it leaves the field.

Solution. (a) Here, if v_0 be the initial velocity of the electron, we have $\frac{1}{2} m v_0^2 = 10^{-10}$ ergs

where m is the mass of electron, so that $v_0 = \sqrt{2 \times 10^{-10}} \text{ per } m = \sqrt{2 \times 10^{-10} / 9.1 \times 10^{-28}} \text{ cm/sec}$ along the axis of x , i.e., equal to v_x .

Since this velocity remains unaffected by the electric field, the time taken by the electron to cross the field-length l between the plates is given by $t = l/v_0 = 2/v_0 \text{ sec}$.

$\therefore$ Velocity acquired by the electron in time $t \text{ sec}$, i.e.,

$\vec{v} = v_0 \hat{i} + \frac{eE}{m} \left(\frac{l}{v_0} \right) \hat{j} = v_0 \hat{i} + \frac{eE}{m} \left(\frac{l^2}{v_0^2} \right)^{\frac{1}{2}} \hat{j}$ where eE/m is the acceleration acquired by the electron.

$$\begin{aligned} \text{or,} \quad \vec{v} &= \left(\frac{2 \times 10^{-10}}{9.1 \times 10^{-28}} \right)^{\frac{1}{2}} \hat{i} + \frac{4.8 \times 10^{-10} (10^{-2})}{9.1 \times 10^{-28}} \left(\frac{4 \times 9.1 \times 10^{-28}}{2 \times 10^{-10}} \right)^{\frac{1}{2}} \hat{j} \\ &= \left(\frac{2 \times 10^{18}}{9.1} \right)^{\frac{1}{2}} \hat{i} + \frac{4.8 \times 10^{16}}{9.1} (2 \times 9.1 \times 10^{-18})^{\frac{1}{2}} \hat{j} \end{aligned}$$

$$\text{or,} \quad \vec{v} = 4.7 \times 10^2 \hat{i} + 2.25 \times 10^7 \hat{j} \text{ cm/sec.}$$

(b) The angle θ that the electron makes with the x -axis as it leaves the plates is given by the relation $\tan \theta = v_y/v_x = 2.25 \times 10^7 / 4.7 \times 10^2 = 0.0475$, whence $\theta = 2^\circ 44'$.

(c) The displacement of the electron (see Fig. 4.4) is the distance covered by it along the y -direction in time t , i.e., equal to $y = 0 + \frac{1}{2} a t^2$, because the initial velocity along the y -direction is zero.

$$\begin{aligned} \text{or, displacement,} \quad y &= \frac{1}{2} \frac{E}{m} t^2 = \frac{1}{2} \frac{eE}{m} \frac{l^2}{v^2} = \frac{1}{2} \frac{4.8 \times 10^{-10} \times 10}{9.1 \times 10^{-28}} \left(\frac{4 \times 9.1 \times 10^{-28}}{2 \times 10^{-10}} \right) \\ &= \frac{1}{2} \left(\frac{4.8 \times 10^{16}}{9.1} \right) (2 \times 9.1 \times 10^{-18}) = 4.8 \times 10^2 \text{ cm.} \end{aligned}$$

(d) If x be the distance of the point of intersection of the direction of the electron, as it emerges from the field, with the axis of x , we have (see Fig. 4.4)

$$\tan \theta = \frac{y}{l-x}, \text{ whence, } l-x = \frac{y}{\tan \theta}$$

$$\text{or,} \quad x = l - \frac{y}{\tan \theta} = 2 - \frac{4.8 \times 10^{-2}}{4.78 \times 10^{-2}} = 2 - 1 = 1 \text{ cm}$$

Thus, the point of intersection of the direction in which the electron emerges from the field and the axis of x lies at a distance of 1 cm from the origin or the point of entry of the electron into the field.

Example 9. An electron emitted with zero velocity from the hot cathode in a vacuum tube is accelerated by the electric field towards the anode. If the anode is at positive potential of 1000 V with respect to the cathode, find the velocity acquired by the electron as it reaches the anode. ($e = 1.6 \times 10^{-19}$ C, $m = 9.11 \times 10^{-31}$ kg) (Nagpur Uni. 2007)

Solution. Given: $V_a = 1000$ V, $e = 1.6 \times 10^{-19}$ C, $m = 9.11 \times 10^{-31}$ kg

$$\begin{aligned}\frac{1}{2}mv^2 &= eV_a \\ v &= \sqrt{\frac{2eV_a}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 1000}{9.11 \times 10^{-31}}} \\ &= \sqrt{\frac{2 \times 1.6 \times 10^{15}}{9.11}} = 10^7 \sqrt{\frac{2 \times 16}{9.11}} \\ &= 1.874 \times 10^7 \text{ m/s}\end{aligned}$$

Example 10. A beam of electrons moving horizontally with a velocity of 10^7 m/s enters midway a uniform electric field between two horizontal parallel plates 5 cm long and 1.8 cm apart. On emerging from the plates the beam just grazes the edge of the positive plate. Calculate the potential difference V applied between them (Given $e/m = 1.8 \times 10^{11}$ C/kg). (Nagpur Uni. 2007)

Solution. $v = 10^7$ m/s At $x = 5 \times 10^{-2}$ m, $y = \frac{1.8 \times 10^{-2}}{2} = 9 \times 10^{-3}$ m

and $d = 1.8 \times 10^{-2}$ m

The equation of the path between the plates is

$$y = \frac{eE}{2mv^2} x^2 = \frac{1}{2} \left(\frac{e}{m} \right) E \left(\frac{x}{v} \right)^2$$

$$\therefore E = 2 \left(\frac{m}{e} \right) \left(\frac{v}{x} \right)^2 y \quad \dots(1)$$

The p.d. is given by

$$\begin{aligned}V &= Ed \\ &= 2 \left(\frac{m}{e} \right) \left(\frac{v}{x} \right)^2 yd \\ &= 2 \left(\frac{1}{1.8 \times 10^{11}} \right) \left(\frac{10^7}{5 \times 10^{-2}} \right)^2 9 \times 10^{-3} \times 1.8 \times 10^{-2} \\ &= \frac{2 \times 9 \times 10^2}{25} = 72 \text{ V}\end{aligned}$$

Example 11. A beam of positive ions moving along the x -axis enters a region of uniform electric field of intensity 3 KV/m along the y -axis and magnetic field of 1 Kilo Gauss along the z -axis. Calculate the speed of those ions which pass undeviated. What will happen to those ions which are moving (i) faster, (ii) slower than these ions? (Nagpur Uni. 2008)

Solution. Given $E = 10^3$ V/m, $B = 10^3$ Gauss, 10^4 Gauss = 1 Wb/m²

$$\therefore B = \frac{10^3}{10^4} = 0.1 \text{ Wb/m}^2 = 0.1 \text{ Tesla}$$

For positive ions which pass undeviated
downward magnetic force = upward electric force

$$qvB = qE$$

$$\therefore v = \frac{E}{B} = \frac{10^3}{0.1} = 10^4 \text{ m/s}$$

- (i) The ions which move with speed *more* than this value will be deflected *downward*, and
(ii) the ions which move with speed less than this value will be deflected *upward*.

Example 12(a). If deflection sensitivity of a CRO is 2mm/V, calculate the deflection of the spot when a voltage of 15 V applied. (Nagpur Uni. 2006)

Solution. For 1 volt, the deflection produced is 2 mm.

$\therefore$ 15 V will produce deflection = $2 \times 15 = 30 \text{ mm} = 3 \text{ cm}$.

Example 12(b). Determine the deflection sensitivity of a signal for a CRT in which $l = 2 \text{ cm}$, $L = 30 \text{ cm}$, $d = 0.5 \text{ cm}$ and $V_a = 2000 \text{ V}$. (Pbi. U. 2001)

Solution. For a CRT, Electrostatic deflection sensitivity

$$S = \frac{IL}{2dV_0} \text{ m/volt}$$

Here

$$l = 2 \text{ cm} = 0.02 \text{ m}; L = 30 \text{ cm} = 0.3 \text{ m}$$

$$d = 0.5 \text{ cm} = 0.005 \text{ m}; V_a = 2000 \text{ V}$$

$$\therefore \text{Deflection sensitivity} = \frac{0.02 \times 0.3}{2 \times 0.005 \times 2000} \\ = 3 \times 10^{-4} \text{ m/V} = 0.3 \text{ mm/V}$$

Example 13. The length of deflecting plates in a C.R.O. is 5 cm. They are separated by 4 mm. The distance of the fluorescent screen from the nearest edge of the deflecting plates is 15 cm. A d.c voltage of 25 V is applied to the deflecting plates. If the accelerating potential difference is 1000 volt find the displacement of the spot on the screen. (Bangalore. U. 2001)

Solution. For a CRT, total deflection due to electrostatic field is given by

$$\text{Total deflection} = \frac{IL}{2d} \frac{V_d}{V_a}$$

and the displacement of the spot on the screen is equal to total deflection. Referring to Fig. 9.10, we have

$$l = 5 \text{ cm} = 0.05 \text{ m}; d = 4 \text{ mm} = 0.004 \text{ m}; D = 15 \text{ cm}$$

$$L = \left[\frac{l}{2d} + D \right] = 2.5 + 15 = 17.5 \text{ cm} = 0.175 \text{ m}$$

$$V_d = 25 \text{ V}; V_a = 1000 \text{ V}.$$

$$\therefore \text{Total deflection} = \frac{0.05 \times 0.175 \times 25}{2 \times 0.004 \times 1000} \\ = 27.34 \times 10^{-3} \text{ m} = 27.34 \text{ mm}$$

4.5 CHARGED PARTICLE IN AN ALTERNATING ELECTRIC FIELD

Suppose we have a particle of mass m and carrying a charge q placed in an alternating electric field whose intensity E at any instant t is given by $E = E_0 \sin \omega t$, where E_0 is the maximum or the peak value of E , i.e., the amplitude of the electric field vector. and $2\pi/\omega$, its time-period.

If, therefore, the acceleration produced in the particle under the action of this field be d^2r/dt^2 , we have

$$m \frac{d^2r}{dt^2} = qE = qE_0 \sin \omega t, \text{ whence, } \frac{d^2r}{dt^2} = \frac{qE_0}{m} \sin \omega t \quad \dots(i)$$

indicating that *the acceleration of the particle too, like the applied electric field, varies sinusoidally with time*, as shown in Fig. 4.6, having an amplitude (or peak value) qE_0/m , and the same time-period $2\pi/\omega$ as that of the applied field.

Integrating relation I with respect to time, we have

$$\frac{dr}{dt} = v = \frac{qE_0}{m\omega} \cos \omega t + C_1$$

where C_1 is the constant of integration to be determined from the initial or boundary conditions.

Initially, i.e., at $t = 0$, when the electric field (E) is zero, the initial velocity of the particle too is zero and we have

$$0 = -\frac{qE_0}{m\omega} + C_1 \quad \text{Or, } C_1 = qE_0/m\omega.$$

So that,

$$\frac{dr}{dt} = v = \frac{-qE_0}{m\omega} \cos \omega t + \frac{qE_0}{m\omega}$$

or,

$$\frac{dr}{dt} = v = \frac{-qE_0}{m\omega} (1 - \cos \omega t) + \frac{2qE_0}{m\omega} \sin^2 \omega t \quad \dots(ii)$$

indicating that *the velocity of the particle at any given instant is always positive, in the direction of E_0 , and hence unidirectional, though it varies after the manner of a $\sin^2$ -curve*, as shown in Fig. 4.7, and has an amplitude or peak value $2qE_0/m\omega$.

Again, integrating relation II, with respect to t , we have

$$r = \left(\frac{qE_0}{m\omega} \right) t - \left(\frac{qE_0}{m\omega} \right) \sin \omega t + C_2 \quad \dots(iii)$$

where C_2 is another constant of integration.

Since at $t = 0$, $r = r_0$, the initial position vector, we have

$$r = \left(\frac{qE_0}{m\omega} \right) t - \left(\frac{qE_0}{m\omega} \right) \sin \omega t + r_0 \quad \dots(iii)$$

where r_0 may be zero or have a finite value, depending upon the choice of the origin.

However, if $r = 0$ at $t = 0$, we have $r_0 = 0$. In that case, therefore,

$$r = \left(\frac{qE_0}{m\omega} \right) t - \left(\frac{qE_0}{m\omega^2} \right) \sin \omega t \quad \dots(iv)$$

It will thus be seen that under the initial or boundary condition chosen here (viz., $r = 0$ at $t = 0$), the displacement vector (r) is the sum of two vectors, with the value of one varying linearly, and that of the other, sinusoidally, with time. In consequence, the resulting motion of the particle is a simple harmonic oscillation of time-period $2\pi/\omega$ (the same as that of the applied electric field) and amplitude $qE_0/m\omega^2$, superimposed on a constant drift velocity $qE_0/m\omega$, as shown in Fig. 4.8. This is obvious from the fact that in this particular

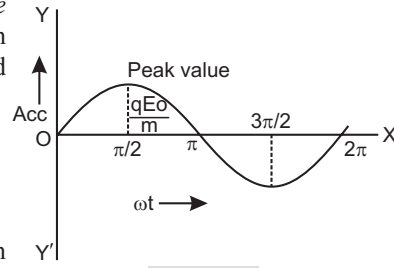


Fig. 4.6

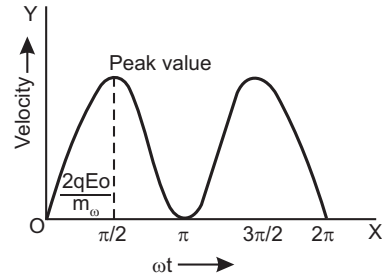


Fig. 4.7

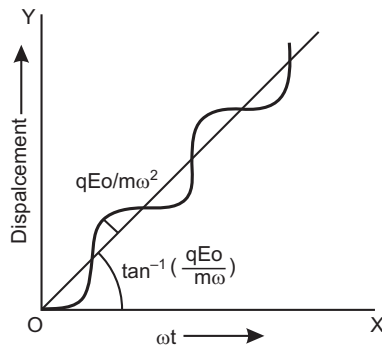


Fig. 4.8

case ($r = 0$ at $t = 0$) the expression for the velocity has no sine or cosine term attached to it, indicating that the velocity does not get reversed at any time (i.e., remains unidirectional throughout) and the particle continues to advance in one and the same direction

Example 14. Alpha particles of speed $v_x = 2.0 \times 10^9$ cm/sec pass through 10 cm length of a field where the alternating electric field $E_x = 5 \times 10^3 \sin(\omega t + \phi)$ volts/cm exists. Calculate the difference of speed between the fastest and slowest particle emerging from the field.

Solution. Here, clearly, the *amplitude* or the *peak value* of the field, i.e., $E_0 = 5 \times 10^3$ volts per cm or $5 \times 10^3/300$ statvolt/cm.

With the high speed of the particles, the value of the field may be assumed to remain the same during the small fraction of time they take to cross just 10 cm length of it. So that, an α -particle (with its positive charge) gains energy when the field is positive and loses an equal amount of energy when the field is negative, the maximum gain or loss obviously occurring when the field has its peak value.

$\therefore$ Assuming the field to be as its peak positive value, we have

force acting on the α -particle $= q \times E_0$ and hence

energy gained by the particle in crossing 10 cm of the field $= q \times E_0 \times 10$,

The particle will, therefore, have the maximum speed.

Similarly, when the field is at its peak negative value, the *energy lost by the particle in traversing 10 cm of the field* $= q \times E_0 \times 10$. So that, the particle will now have the minimum speed.

The difference in the energies of the fastest and the slowest particles will thus be $2q \times E_0 \times 10$.

Now, an α -particle carries a positive charge of magnitude $2e$, where e is the charge on a proton (equal to 4.8×10^{-10} esu). So that, *difference in the energies of the fastest and the slowest particles*, i.e.,

$$\Delta E = 2 \times 2e \times E_0 \times 10 = 2 \times 2 \times 4.8 \times 10^{-10} \times (5 \times 10^3/300) \times 10 = 3.2 \times 10^7 \text{ ergs.}$$

Since energy $E = \frac{1}{2} mv^2$, where m is the mass and v , the *velocity* of the α -particle, we have $\Delta E = mv\Delta v$, whence, $\Delta v = \Delta E/mv$.

$\therefore$ Substituting the values of ΔE , m (equal to 4 times the mass of a proton) and v (given), we have

difference in the velocities of the fastest and the slowest particles, i.e.

$$\Delta v = \frac{3.2 \times 10^7}{4 \times 1.6725 \times 10^{-24} \times 2.0 \times 10^9} = \frac{3.2 \times 10^8}{4 \times 1.6725 \times 2} = \frac{0.4 \times 10^8}{1.6725} = 2.39 \times 10^7 \text{ cm/sec}$$

Example 15. A stream of protons is first accelerated through a potential difference V until it acquires a constant velocity v and then passed through two parallel plates to which an alternating potential $v_0 \sin \omega t$ is applied. If the frequency of the alternating potential be not very large and the time of transit of the stream through the plates comparatively small, show

that the stream emerges from the plates with a velocity equal to $\left[\frac{2e}{m} (V + V_0 \sin \omega t) \right]^{\frac{1}{2}}$, where e is the charge on a proton and m , its mass.

Solution. Since v is the velocity acquired by a proton on being accelerated through a potential difference V , we have $\frac{1}{2} mv^2 = eV$.

On being subjected to the alternating potential $V_0 \sin \omega t$, the proton will acquire an additional energy $eV_0 \sin \omega t$. If, therefore, v' be its velocity *now*, clearly, *increase in its energy* $= \frac{1}{2} mv'^2 - \frac{1}{2} mv^2 = \frac{1}{2} m(v'^2 - v^2)$.

$$\begin{aligned}\text{We, therefore, have } \frac{1}{2} m(v'^2 - v^2) &= eV_0 \sin \omega t. \text{ Or, } mv'^2 = 2eV_0 \sin \omega t + mv^2 \\ &= 2eV_0 \sin \omega t + 2eV,\end{aligned}$$

$$\text{whence, } v'^2 = \frac{2eV_0 \sin \omega t}{m} + \frac{2eV}{m} = \frac{2e}{m} (V + V_0 \sin \omega t)$$

Or, velocity of the proton stream on emerging through the plates, i.e.,

$$v' = \left[\frac{2e}{m} (V + V_0 \sin \omega t) \right]^{1/2}$$

4.6 LINEAR ACCELERATOR

For the study of nuclear reactions, charged particles having energies of many million electron-volts (MeV) are required. It is difficult to generate direct voltages of the order of 10 million volts mainly due to insulation difficulties. To obtain linear acceleration of a charged particle in excess of 10 MeV, some indirect methods are used. The first type is known as '*Drift tube accelerator*' and the second type as '*Wave guide accelerator*'. Let us discuss first type i.e., as '*Wave guide accelerator*'. Let us discuss first type i.e., a linear accelerator.

A linear accelerator (or a *Linac*) is a device which accelerates charged particles in a straight line by means of oscillating electric field that provides either a series of steady accelerating steps in correct phase at a series of gaps between electrodes or accompanies the charged particles as a travelling wave.

In a linear accelerator, a moderate accelerating potential is applied a number of times so that the charged particles are accelerated along a straight line. A simple form of the linear accelerator is shown in Fig. 4.9. The charged particles or ions travel through an aperture A and move along the axis of a series of coaxial cylindrical electrode 1, 2, 3, 4 etc. These cylindrical electrodes are known as *drift tubes*. The drift tubes are connected to an A.C. source of very high frequency say a high frequency (H.F.) oscillator so that alternate tubes have potentials of opposite sign. Thus, in *one-half* cycle if tubes 1 and 3 are positive, 2 and 4 will be negative. After half a cycle the polarities are reversed i.e., 1 and 3 will be negative and 2 and 4 positive.

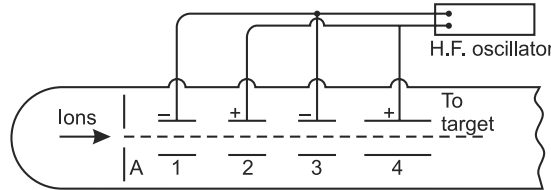


Fig. 4.9

Suppose a positive ion leaves A and is accelerated during the half cycle when the drift tube No. 1 is negative with respect to A. If V is the potential of drift tube 1 with respect to A, then velocity v_1 of the ion on reaching the drift tube is given by

$$\frac{1}{2} mv_1^2 = Ve$$

Or

$$v_1 = \sqrt{\frac{2Ve}{m}}$$

where e is the charge and m the mass of the ion. It is supposed that v_1 is small as compared to c the velocity of light so that the change in mass due to relativity effect is negligible. The ions are accelerated in the gap between the tubes but travel with constant velocity in the *field free space* within the tubes themselves. The length of the tube 1 is so adjusted that as the positive ions come

out of it, the tube has a positive potential and the next tube No. 2 has a negative potential, *i.e.*, the potential change sign. The positive ion is again accelerated in the space between the tubes 1 and 2 and on reaching the tube 2 its velocity v_2 is given by

$$\frac{1}{2}mv_2^2 = 2Ve$$

Or
$$v_2 = \sqrt{2} \sqrt{\frac{2Ve}{m}} = \sqrt{2} v_1$$

This shows that v_2 is $\sqrt{2}$ times v_1 . In order that this ion on coming out of tube 2 may find tube 3 just negative and the tube 2 positive, it must take the same time to travel through the tube 2. As the velocity is $\sqrt{2}v_1$ the length of tube 2 must be $\sqrt{2}$ times the length of tube 1. For successive accelerations in successive gaps the tubes 1, 2, 3, 4, etc., must have lengths proportional to 1, $\sqrt{2}$, $\sqrt{3}$, $\sqrt{4}$ etc., to a first approximation.

Energy of the ion. If n is the number of gaps that the ion travels in the accelerator and v_n is the final velocity acquired by it, then

Velocity of the ion as it emerges out of the n th tube

$$v_n = \sqrt{n} \sqrt{\frac{2Ve}{m}}$$

$\therefore$ Kinetic energy of the ion $\frac{1}{2}mv_n^2 = nVe = nVe$

The final energy of the ions when they strike the target depends upon the overall length of the accelerator *i.e.*, the total number of gaps and on the energy gained in each gap. The beam striking the target consists of pulses of particles. The number of pulses per second is equal to the frequency of the alternating voltage applied to the drift tubes.

Length of the cylinder. As the ion is accelerated in the gap between two cylindrical electrodes, the time taken by the ion to travel through the cylinder should be equal to half the time period of the high frequency voltage so that each time the ion comes out of the cylinder the polarity changes.

If v_n is the velocity of the ion, the time of passage through the n th cylinder of length L_n

$$t = \frac{L_n}{v_n} = \frac{T}{2} = \frac{1}{2f}$$

where f is the frequency of the oscillating electric field.

$$L_n = \frac{v_n}{2f} = \left(\frac{2nVe}{m} \right)^{1/2} \frac{1}{2f}$$

This equation shows that the length of the successive cylinders has to be increased in order to get a resonance acceleration of the ion at each gap and the length $L_n \propto n^{1/2}$

Limitations:

The limitations of linear accelerator are:

- (i) The length of the accelerator becomes inconveniently large and it is difficult to maintain vacuum in such a large chamber.
- (ii) The ion current is available in the form of pulses of short duration.

Circular Accelerator

A *circular accelerator* is another example of drift tube accelerator. Circular accelerator is a device which can accelerate charged particles by passing them again and again in a radio-frequency (*r.f.*) electric field along a closed path. Familiar examples are (i) **cyclotrons** which accelerates protons, deuterons and α -particles and (ii) **Betatrions** which accelerates electrons.

Wave Guide Accelerator

Wave Guide Accelerator makes use of electromagnetic radiations (waves) travelling in a wave guide to accelerate charged particles. A wave guide is a hollow pipe (cylindrical or rectangular) of conducting material in which oscillating electro-magnetic fields of the order of 10^5 MHz can be established. If the electro-magnetic wave is fed at one end of the wave guide and there is an electrical conductor at the other end, the advancing and the reflected waves superimpose to form *standing* or *stationary* wave. But if the wave gets completely absorbed at the other end the result is a *travelling* wave. For this purpose, the wave guide is terminated at its end by an impedance equal to its characteristic impedance so that the travelling wave is maintained from one end to the other.

To accelerate the charged particle, the speed of the *e.m. wave is synchronised with that of the particle*. The particle, therefore, gains energy from the (axial) *electric field component* of the *e.m. wave* but the *magnetic field component* remains ineffective being at right angles to the path of the particle.

The *R.F.* signal is obtained from a standard master oscillator and is amplified at each feeding station using *Klystron amplifier*.

Acceleration of electrons. To accelerate the electrons a *disc loaded wave guide system* is used. The beam of electrons is injected along the axis of the metallic tube of the wave guide and the *phase velocity* of the *e.m. waves* is *matched* with that of electrons.

After the electrons have attained their maximum kinetic energy, they automatically emerge out of the accelerator in the form of a fine collimated beam.

Acceleration of protons. Wave guide accelerators can be used for protons. But the design of such accelerator is very different. The electrons being light particles travel most of the time with a velocity very close to the velocity of light. On the other hand, protons gain speed along its total path. As the protons have a *much lower injection velocity*, disc loaded wave guides are not found practicable for phase velocities as low as $0.1c$ but in its place *long cylindrical resonant cavity* has to be used.

Example 19. Protons are accelerated in a cyclotron in which the magnetic field strength is 1 Wb/m^2 . What must be the frequency of the oscillator supplying power to the dees?

(Given: mass of proton = $1.67 \times 10^{-27} \text{ kg}$, electronic charge = $1.6 \times 10^{-19} \text{ C}$) (Nagpur U. 2009)

Solution.

$$B = 1 \text{ Wb/m}^2, m_p = 1.672 \times 10^{-27} \text{ kg}$$

$$q = e = 1.6 \times 10^{-19} \text{ C}$$

$$T = \frac{2\pi m}{qB}$$

$$\begin{aligned} f &= \frac{1}{T} = \frac{qB}{2\pi m_p} \\ &= \frac{1.6 \times 10^{-19} \times 1}{2 \times 3.14 \times 1.672 \times 10^{-27}} = \frac{1.6 \times 10^8}{2 \times 3.14 \times 1.672} \\ &= 0.1524 \times 10^8 \text{ Hz} \\ &= 15.24 \times 10^6 \text{ Hz} = \mathbf{15.24 \text{ MHz}} \end{aligned}$$

Example 20. Deutrons are to be accelerated with a cyclotron. If its magnets produce a flux density of 2.475 T , what must be the frequency of the oscillating potential applied to the Dee's? Mass of ${}_1\text{H}^2 = 2 \text{ a.m.u.}$ (P.U. 2001; Nagpur U. 2009)

Solution. Mass of ${}_1\text{H}^2$ atom = $2 \text{ a.m.u} = 2 \times 1.66 \times 10^{-27} \text{ kg} = 3.32 \times 10^{-27} \text{ kg}$

$$e = 1.6 \times 10^{-19} \text{ C}, B = 2.475 \text{ T}$$

$$\begin{aligned}\text{Frequency of the applied field (or oscillating potential)} f &= \frac{Be}{2\pi m} \\ &= \frac{2.475 \times 1.6 \times 10^{-19}}{2\pi \times 3.32 \times 10^{-27}} = 18.98 \times 10^6 \text{ Hz} = \mathbf{18.98 \text{ MHz}}\end{aligned}$$

Example 21. A cyclotron with dees of radius 90 cm has a transverse magnetic field of 0.8 Tesla. Calculate the energies to which (i) a proton and (ii) deuteron are accelerated. (Given mass of the proton = 1.67×10^{-27} kg, Mass of the deuteron = 3.34×10^{-27} kg.)

Solution. (i) Proton.

$$\text{Mass of the proton } m = 1.67 \times 10^{-27} \text{ kg}$$

$$\text{Charge on the proton } e = 1.6 \times 10^{-19} \text{ C.}$$

$$\begin{aligned}\text{Energy of the emergent proton } E &= \frac{1}{2} \frac{B^2 e^2 r^3}{m} \\ &= \frac{0.8 \times 0.8 \times 1.6 \times 10^{-19} \times 1.6 \times 10^{-19} \times 0.9 \times 0.9}{2 \times 1.67 \times 10^{-27}} \\ &= 0.3973 \times 10^{-11} \text{ J} = \frac{0.3973 \times 10^{-11}}{1.6 \times 10^{-13}} \text{ MeV} \\ &= \mathbf{24.8 \text{ MeV.}}\end{aligned}$$

(ii) Deuteron.

$$\text{Mass of the deuteron } m = 3.34 \times 10^{-27} \text{ kg}$$

$$\text{Charge on the deuteron } e = 1.6 \times 10^{-19} \text{ C.}$$

$$\begin{aligned}\therefore \text{Energy of the emergent deuteron} &= \frac{0.8 \times 0.8 \times 1.6 \times 10^{-19} \times 1.6 \times 10^{-19} \times 0.9 \times 0.9}{2 \times 3.34 \times 10^{-27}} \\ &= 0.1986 \times 10^{-11} \text{ J} \\ &= \mathbf{12.4 \text{ MeV.}}\end{aligned}$$

Example 22. A cyclotron has a magnetic field of 10^4 Gauss and radius of 80 cm. Calculate the frequency of the alternating electric field that must be applied and to what energy deuterons can be accelerated? (Mass of deuteron = 2 a.m.u.) (P.U. 2000)

Solution.

$$B = 10^4 \text{ Gauss} = 1 \text{ T}; r = 80 \text{ cm} = 0.8 \text{ m}$$

$$m = 2 \text{ a.m.u.} = 2 \times 1.66 \times 10^{-27} \text{ kg} = 3.32 \times 10^{-27} \text{ kg};$$

$$e = 1.6 \times 10^{-19} \text{ C.}$$

$\therefore$ Frequency of alternating electric field

$$\begin{aligned}f &= \frac{Be}{2\pi m} = \frac{1 \times 1.6 \times 10^{-19}}{2\pi \times 3.32 \times 10^{-27}} \\ &= 7.7 \times 10^6 = \mathbf{7.7 \text{ MHz}}\end{aligned}$$

$$\begin{aligned}\text{Energy } E &= \frac{1}{2} \frac{B^2 e^2 r^2}{m} \\ &= \frac{1 \times 1 \times 1.6 \times 10^{-19} \times 1.6 \times 10^{-19} \times 0.8 \times 0.8}{2 \times 3.32 \times 10^{-27}} = 24.67 \times 10^{-13} \text{ J} \\ &= \mathbf{15.4 \text{ MeV}}\end{aligned}$$

Example 23. Between the Dee's of a cyclotron 1.5 metre in diameter an alternating potential difference of 15 mega cycles is applied. Calculate the energy in MeV of the protons issuing out of the cyclotron. (Mass of proton = 1.672×10^{-27} kg.)

Solution. In terms of frequency of the applied field and radius of cyclotron Dee's the energy is given by

$$E = 2\pi^2 m f^2 r_{\max}^2$$

Mass of proton = 1.672×10^{-27} and $f = 15 \times 10^6$

$$\therefore E \text{ (in MeV)} = \frac{2\pi^2 \times 1.672 \times 10^{-27} \times 15 \times 10^6 \times 15 \times 10^6 \times .75 \times .75}{1.6 \times 10^{-13}}$$

$$= 26.11 \text{ MeV}$$

Example 24. A cyclotron with Dee's diameter 1.8 m has a magnetic field of 0.8 tesla. Calculate the energy to which the doubly ionised helium ion He^{++} can be accelerated. Also calculate the number of revolutions the particle makes in attaining this energy.

(Mass of $\text{He}^{++} = 6.68 \times 10^{-27} \text{ kg}$.)

Solution. Mass of the α -particle (He^{++})

$$m = 6.68 \times 10^{-27} \text{ kg}$$

Charge on He^{++} ion, $e = 2 \times 1.6 \times 10^{-19} \text{ C}$

Now
$$E = \frac{B^2 e^2 r^2}{2m}$$

$$B = 0.8 \text{ Tesla; } r = 0.9 \text{ m}$$

$$\therefore E = \frac{0.8 \times 0.8 \times 2 \times 1.6 \times 10^{-19} \times 2 \times 1.6 \times 10^{-19} \times 0.9 \times 0.9}{2 \times 6.68 \times 10^{-27}}$$

$$= 0.397 \times 10^{-11} \text{ J} = \frac{0.397 \times 10^{-11}}{1.6 \times 10^{-13}} = 24.8 \text{ MeV.}$$

The frequency of the alternating electric field is given by

$$f = \frac{Be}{2\pi m} = \frac{0.8 \times 2 \times 1.6 \times 10^{-19}}{2\pi \times 6.68 \times 10^{-27}} = 0.061 \times 10^8 = 6.1 \times 10^6 \text{ s}^{-1}$$

This gives the number of times the He^{++} ion comes out of the gap between the dees each time undertaking a semi-circular path.

$\therefore$ Number of complete revolution made by He^{++} ion in attaining the above energy

$$= \frac{f}{2} = \frac{1}{2} \times 6.1 \times 10^5 \times 6.1 \times 10^6 = 3.05 \times 10^6 \text{ s}^{-1}$$

Example 25. Deutrons in a cyclotron describe a circle of radius 0.32 m just before emerging out of the Dee's. The frequency of the applied e.m.f. is 10 MHz. Find the flux density of the magnetic field and the velocity of the deuterons emerging out of the cyclotron. (Mass of deuteron is $3.32 \times 10^{-27} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ C}$.) (Bang. U. 2000)

Solution. The frequency of the applied electric field is given by

$$f = \frac{Be}{2\pi m} \therefore B = \frac{2\pi m f}{e}$$

Here

$$m = 3.32 \times 10^{-27} \text{ kg; } f = 10 \text{ MHz} = 10 \times 10^6 \text{ Hz;}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$B = \frac{2\pi \times 3.32 \times 10^{-27} \times 10^7}{1.6 \times 10^{-19}} = 1.304 \text{ Tesla}$$

Radius of the circle just before deuterons emerge $r_{\max} = 0.32 \text{ m}$

Now $Bev = \frac{mv^2}{r_{\max}}$

$\therefore v = \frac{Ber_{\max}}{m} = \frac{1.304 \times 1.6 \times 10^{-19} \times 0.32}{3.32 \times 10^{-27}}$

$= 2.01 \times 10^7 \text{ ms}^{-1}$

Example 26. A cyclotron of radius 0.462 m is used to accelerate deuterons. The oscillator frequency is 25 MHz . Find the magnetic flux density needed and also energy acquired by the deuteron. (Given $m = 3.32 \times 10^{-27} \text{ kg}$) (Bang. U. 2004)

Solution.

$$m = 3.32 \times 10^{-27} \text{ kg}, f = 25 \text{ MHz} = 25 \times 10^6 \text{ Hz}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$B = \frac{2\pi mf}{e} = \frac{2\pi \times 3.32 \times 10^{-27} \times 25 \times 10^6}{1.6 \times 10^{-19}} = 3.26 \text{ T}$$

$$E = 2\pi^2 mf^2 r_{\max}^2$$

$$= 2\pi^2 \times 3.32 \times 10^{-27} \times 25 \times 10^6 \times 25 \times 10^6 \times 0.462 \times 0.462$$

$$= 87.4 \times 10^{-13} \text{ J} = \frac{87.4 \times 10^{-13}}{1.6 \times 10^{-13}} = 54.6 \text{ MeV.}$$

Example 27. A cyclotron oscillator frequency 1 MHz is used to accelerate protons. If the radius of the dees is 60 cm , find the magnetic field in Tesla. (Nagpur Uni. 2008; H.P.U. 2002)

Solution.

$$m = 1.67 \times 10^{-27} \text{ kg}, f = 1 \text{ MHz} = 10^6 \text{ Hz}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$B = \frac{2\pi mf}{e} = \frac{2 \times 3.142 \times 1.67 \times 10^{-27} \times 10^6}{1.6 \times 10^{-19}} = 6.56 \text{ T}$$

Example 28. The dees of a cyclotron, used for accelerating protons, have radii of 50 cm , each and an alternating potential difference of frequency $10 \text{ megacycles per second}$ and maximum value 10^4 volts is applied to them. Obtain (i) the strength of the magnetic field of the cyclotron, (ii) the velocity and the kinetic energy acquired by a proton, (iii) the maximum number of revolutions made by the proton and (iv) the time spent by the proton inside the dees.

Solution (i). We have cyclotron frequency, $n = qB/2\pi mc$, where q is in *esu* and B in *emu* or *gauss*. So that,

$$\text{Magnetic field } B = \frac{mc}{q} 2\pi n = \frac{1.67 \times 10^{-24} \times 3 \times 10^{10} \times 2\pi \times 10 \times 10^7}{4.8 \times 10^{-10}}$$

$$= \frac{1.67 \times 3 \times 2\pi}{4.8} \times 10^3 = 6557 \text{ Gauss.}$$

(ii) We know that the maximum velocity of the proton (corresponding to the radius of its orbit, i.e., the outer radius of the dees) is given by $v = qBr/mc$ (using *Gaussian units*, as here). So that,

$$\text{velocity acquired by the proton, } v = \frac{4.8 \times 10^{-10} \times 6557 \times 50}{1.67 \times 10^{-24} \times 3 \times 10^{10}} = \frac{8 \times 6557}{1.67} \times 10^5$$

$$= 3.142 \times 10^9 \approx 3 \times 10^9 \text{ cm/sec.}$$

And $\therefore$ Energy acquired by the proton, $E = \frac{1}{2}mv^2 = \frac{1}{2}(1.67 \times 10^{-24})(3 \times 10^9)^2 \text{ mv}^2$

$$\begin{aligned}
 &= \frac{15.03}{2} \times 10^{-6} = 7.5 \times 10^{-6} \text{ ergs.} \\
 &= \frac{7.5 \times 10^{-6}}{1.6 \times 10^{-12}} = 4.7 \times 10^6 = 4.7 \times 10^6 \text{ eV.}
 \end{aligned}$$

(iii) If N be the number of rotations made by the proton inside the dees before emerging out, we have *energy gained by it* $= 2NeV$ ergs, because in one rotation it crosses the gap between the dees two times and each time acquires an energy eV ergs.

Since $2NeV$ ergs $= 2NV$ electron volts, we have $2NV = 4.7 \times 10^6$,

$$\text{whence, } N = \frac{4.7 \times 10^6}{2V} = 2.35 \times 10^2 = 235.$$

Thus, the *number of revolutions made by the proton inside the dees* $= 235$.

(iv) The frequency of the alternating potential difference applied to the two dees being 10 megacycles per second, the time spent by the proton inside the dees during one rotation $= 1/10 \times 10^6$ sec.

Hence, *time spent by the proton inside the dees in 235 rotations*

$$= 235/10 \times 10^6 = 2.35 \times 10^{-5} \text{ sec}$$

4.7 CHARGED PARTICLE IN A UNIFORM AND CONSTANT MAGNETIC FIELD

We have seen under § 4.3 (iii) how when a particle carrying a charge q moves with velocity $\vec{v}$ in a magnetic field $\vec{B}$, it is acted upon by a *Lorentz force* $\vec{F} = q\vec{v} \times \vec{B}$ in *emu*, *RMKS* or *SI* units, or equal to $\frac{q}{c} \vec{v} \times \vec{B}$ in *Gaussian units* (i.e., with q in *esu* and B in *emu*).

In the absence of any other force, therefore, (the force due to gravity being negligible compared with the Lorentz force), if m be the *mass* of the particle, and $d^2\vec{r}/dt^2$, its *acceleration* we have

$$m \frac{d^2\vec{r}}{dt^2} = m \frac{d\vec{v}}{dt} = q\vec{v} \times \vec{B} \text{ in } emu, RMKS \text{ or } SI \text{ units} \quad \dots(i)$$

$$\text{Or} \quad = (q\vec{v}/c) \times \vec{B} \text{ in } Gaussian \text{ units.}$$

Now, if the velocity of the particle, $\vec{v}$ be perpendicular to the magnetic field the *Lorentz force* $\vec{F} = q\vec{v} \times \vec{B}$ in *emu*, *RMKS* or *SI* units, or, equal to $(qv/c) \times \vec{B}$ in *Gaussian units*, acts upon it in a direction perpendicular to $\vec{v}$; so that, although it does not change the magnitude of the velocity, it continuously changes its direction, resulting in the particle moving in a circular path with the force $\vec{F}$ directed towards its centre, as shown in Fig. 4.10 (a). If r be the *radius* of this circular path, clearly, the balancing *centrifugal force*, acting on the particle *outwards*, (away from the centre)

$$= mv^2/r.$$

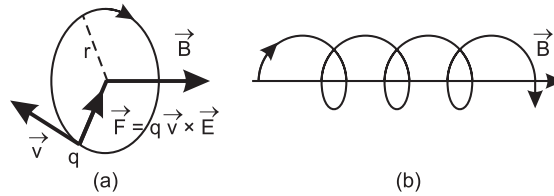


Fig. 4.10

$$\text{So that,} \quad mv^2/r = qv \times B$$

whence, $r = m\vec{v}/q\vec{B}$, if *emu*, *RMKS* or *SI* units be used or $r = m\vec{v}c/q\vec{B}$ if *Gaussian units* be used. ...(ii)

If, however, the velocity $\vec{v}$ of the particle be inclined to the magnetic field at an angle other than 90° (i.e., be not normal to the field), it may be resolved into a component $\vec{v}_1$, parallel to the field, and a component $\vec{v}_2$, at right angles to it. The Lorentz force acting on the particle will then be $q\vec{v}_2 \times \vec{B}$ in *emu*, **RMKS** or **SI** units, or $(q\vec{v}_2/c) \times \vec{B}$ in Gaussian units, making the particle move in a circular path of radius $\vec{r} = m\vec{v}_2/q\vec{B}$, if *emu*, **RMKS** or **SI** units be used or, $\vec{r} = m\vec{v}_2 c/q\vec{B}$ if Gaussian units be used.

And, since the component $\vec{v}_1$ of the velocity, parallel to the field B makes the particle move in the direction of the field, the combined effect of the circular and the linear horizontal motion is to produce a *helical motion* (or a *moving circle*), as shown in Fig. 4.10 (b) the direction of the magnetic field ($\vec{B}$) as axis. ... (iii)

The radius r of the circular, or the helical path, described by a charged particle in a uniform magnetic field is sometimes called **gyro radius** or **cyclotron radius**.

N.B. It will be noted that in expressions (ii) and (iii) above, for r , the terms mv and mv_2 denote the momentum of the particle in the plane perpendicular to B . The relations would, therefore, hold good even in the case of relativistic velocities if we use the appropriate value of the momentum (p) in place of mv or mv_2 . These relations may, therefore, be used to obtain the value of p for high energy particles.

If ω be the angular velocity of the particle in circular or helical motion in the magnetic field, we have $\omega = v/r$ (or v_2/r) = qB/m (or qB/mc if Gaussian units be used).

Therefore, *time-period of the particles* (or *the time taken by it to describe one full circle*) is given by $T = 2\pi/\omega = 2\pi m/qB$ (or $2\pi mc/qB$ in Gaussian system of units).

The distance through which the particle advances forward as it completes one full rotation (in case of helical motion) or the linear distance it covers in one full time-period T is called the *pitch of the helix* and is clearly equal to $v_1 T = 2\pi m/qBv_1$ (or $2\pi mc/qBv_1$ in Gaussian system of units)

And, the *frequency of the particle* or the number of rotations made by it per second, here called **gyro frequency** or **cyclotron frequency**, $n = 1/T = qB/2\pi m$ [or $qB/2\pi mc$ in Gaussian system of units].

As will be readily seen, *the gyro frequency or cyclotron frequency, of a particle is quite independent of the velocity, and hence also of the energy, of the particle* so long as the velocity lies in the non-relativistic region, i.e., so long as $v \ll c$. This, as shall see in the next article, is the principle underlying the cyclotron.

This expression for frequency no longer holds good as $v \rightarrow c$, for then the mass of the particle varies in accordance with the relation $m = m_0/\sqrt{1 - v^2/c^2}$, where m_0 is the *rest mass* of the particle.

Alternative method. we could tackle the problem analytically as follows:

Let the magnetic field be directed along the axis of x , so that $\vec{B} = B\hat{i}$. Then, expressing $\vec{v}$ in terms of its components along the three coordinate axes, relation (i) above may be put in the form

$$m \frac{d}{dt} (v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) = q(v_x \hat{i} + v_y \hat{j} + v_z \hat{k}) \times (B\hat{i})$$

$$\text{or, } \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k} = \frac{qB}{m} (0 + v_z \hat{j} - v_y \hat{k}) \quad \dots (iv)$$

Equating the coefficients of $\hat{i}$ on either side of the equation, we have

$$\frac{dv_x}{dt} = 0 \quad \dots (v)$$

indicating that *the particle does not accelerate in the direction of the field or that v_x , the velocity of the particle in this direction (i.e., along the field) remains constant.*

This velocity, in our earlier discussion above, we have denoted by v_1 . So that,

$$v_x = v_1.$$

Similarly, equating the coefficients of $\mathbf{j}$ and $\mathbf{k}$ on either side, we have

$$\frac{dv_y}{dt} = \frac{qB}{m} v_z \dots (vi) \quad \text{and} \quad \frac{dv_z}{dt} = -\frac{qB}{m} v_y \dots (vii)$$

Differentiating relation VI with respect to t , we have

$$\frac{d^2 v_y}{dt^2} = \frac{qB}{m} \frac{dv_z}{dt}$$

$\therefore$ Substituting the value of dv_z/dt from relation (vii) above, we have,

$$\frac{d^2 v_y}{dt^2} = -\left(\frac{qB}{m}\right)^2 v_y$$

Or, putting $qB/m = \omega$, we have $\frac{d^2 v_y}{dt^2} = -\omega^2 v_y$

As will be readily seen, this an equation similar to that for the *simple harmonic motion* of a particle expect that in place of the *displacement y of the particle, we have here its velocity v_y* . On that analogy, therefore, that solution of this equation is

$$v_y = A \sin(\omega t + \phi)^\dagger \dots (ix)$$

where the values of the constant A and ϕ can be obtained directly from the initial or boundary conditions.

From relation (vi), therefore, we have

$$\frac{d}{dt} [A \sin(\omega t + \phi)] = \frac{qB}{m} v_z = \omega v_z$$

i.e., $A \omega \cos(\omega t + \phi) = \omega v_z$.

Or, $v_z = A \cos(\omega t + \phi) \dots (x)$

Squaring and adding relations ix and x, we have

$$v_y^2 + v_z^2 = A^2 \dots (xi)$$

But we have, in our earlier discussion, taken the velocity of the particles in the Y-Z plane to be v_2 . So that $A^2 = v_2^2$ Or, $A = v_2$.

Since $v_y = dy/dt$ and $A = v_2$, we have from relation (ix),

$$dv_y/dt = v_2 \sin(\omega t + \phi) \dots (xii)$$

Equation (xi) and (xii), on integration, respectively yield

† In case Gaussian units be used, i.e., with q in esu and B in emu, q must every where be replaced by q/c , since q esu of charge = q/c cmu of charge.

† Or, this equation may be obtained thus:

Multiplying both sides of equation viii by $2d/dt$ and integrating, we have

$$\left(\frac{dv_y}{dt}\right)^2 = (\omega v_y)^2 + C = C - (\omega v_y)^2 \quad \text{where, } C \text{ is a constant of integration. So, that}$$

$$\frac{dv_y}{dt} = \sqrt{C - (\omega v_y)^2} \quad \text{or} \quad \frac{dv_y}{\sqrt{C - (\omega v_y)^2}} = dt \quad \text{which, on integration, gives,}$$

$$\frac{1}{\omega} \sin^{-1} \left(\frac{\omega v_y}{\sqrt{C}} \right) = t + C' \quad \text{where } C' \text{ is another constant of integrator}$$

$$\text{i.e. } v_y = \frac{\sqrt{C}}{\omega} \sin(\omega t + \omega C') \quad \text{Or putting } \sqrt{C}/\omega = A \text{ and } \omega C' = \phi \text{ we have}$$

$$v_y = A \sin(\omega t + \phi)$$

$$y = -\frac{v_2}{\omega} \cos(\omega t + \phi) \text{ and } z = \frac{v_2}{\omega} \sin(\omega t + \phi)$$

squaring and adding which, we have

$$y^2 + z^2 = \frac{v_2^2}{\omega^2} = r^2$$

This is the equation to a circle of radius r in the $Y-Z$ plane,

$$\text{where } r = v_2 / \omega = v_2 / \frac{qB}{m} = mv_2 / qB$$

which is thus the value of *gyro radius* or the *cyclotron radius*. Since the particle has also a constant horizontal velocity v_1 along the axis of x (or the direction of the field), the resultant motion of the particle is *helical*, with the axis of the helix lying along the direction of the field (B).

Note. It may be mentioned again that if we use *Gaussian units*, i.e., if we take q in *esu* and B in *emu* (or gauss), we must use q/c instead of q in all the expression obtained above, since q *esu* of charge = q/c *emu* of charge.

Example 16. Calculate radius of path of an electron in a magnetic field of induction 10^{-4} wb/m^2 perpendicular to its path. (Velocity of the electron = $1.9 \times 10^5 \text{ m/s}$, $m = 9.1 \times 10^{-31} \text{ kg}$, $e = 1.6 \times 10^{-19} \text{ C}$) (Nagpur Uni. 2008)

Solution.

$$\begin{aligned} R &= \frac{mv}{eB} = \frac{9.1 \times 10^{-31} \times 1.9 \times 10^5}{1.6 \times 10^{-19} \times 10^{-4}} \\ &= \frac{9.1 \times 1.9}{1.6} \\ &= 10.81 \text{ m} \end{aligned}$$

Example 17. An electron having kinetic energy $2 \times 10^5 \text{ eV}$ enters a uniform magnetic field of induction $3 \times 10^{-3} \text{ Wb/m}^2$ in a direction perpendicular to the field. Find the radius of the electron path in the field.

Solution. Given

$$\begin{aligned} m &= 9 \times 10^{-31} \text{ kg}, e = 1.6 \times 10^{-19} \text{ C}, 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}, \\ B &= 3 \times 10^{-3} \text{ Wb/m}^2, \text{K.E.} = 2 \times 10^5 \times 1.6 \times 10^{-19} \text{ J} \end{aligned}$$

$$\frac{1}{2}mv^2 = \text{Kinetic energy } k \text{ say}$$

$$\therefore v = \sqrt{\frac{2k}{m}}$$

Radius of the path is given by

$$\begin{aligned} R &= \frac{mv}{eB} = \frac{m}{eB} \sqrt{\frac{2k}{m}} \\ &= \sqrt{\frac{2km}{eB}} \\ &= \frac{\sqrt{2 \times 2 \times 10^5 \times 1.6 \times 10^{-19} \times 9 \times 10^{-31}}}{1.6 \times 10^{-19} \times 3 \times 10^{-3}} \\ &= \frac{\sqrt{4 \times 1.6 \times 9 \times 10^{-45}}}{1.6 \times 3 \times 10^{-22}} = \frac{\sqrt{4 \times 16 \times 9 \times 10^{-46}}}{48 \times 10^{-23}} \\ &= \frac{24}{48} = 0.5 \text{ m} \end{aligned}$$

Example 18. What would be the length of the last drift tube in a linear accelerator which produces 120 MeV C^{12} ions, using frequency of 70 MHz?

(1 eV = 1.6×10^{-19} J, and 1 amu = 1.66×10^{-27} kg)

(Nagpur U. 2009)

Solution.

$$\begin{aligned} E_n &= 120 \text{ MeV} = 120 \times 1.6 \times 10^{-19} \times 10^6 \text{ J} \\ &= 1.2 \times 1.6 \times 10^{-11} \text{ J} \quad m = 12 \times 1.66 \times 10^{-27} \text{ kg} \\ f &= 70 \text{ MHz} = 70 \times 10^6 \text{ Hz} = 7 \times 10^7 \text{ Hz} \end{aligned}$$

Let l_n be the length of the last tube.

We have,

$$\frac{1}{2}mv_n^2 = E_n$$

$$\therefore v_n = \sqrt{\frac{2E_n}{m}}$$

According to the condition of synchronous acceleration

$$\frac{l_n}{v_n} = \frac{T}{2} = \frac{1}{2f}$$

$$\begin{aligned} l_n &= \frac{v_n}{2f} = \frac{1}{2f} \sqrt{\frac{2E_n}{m}} \\ &= \frac{1}{2 \times 7 \times 10^7} \sqrt{\frac{2 \times 1.2 \times 1.6 \times 10^{-11}}{12 \times 1.66 \times 10^{-27}}} \\ &= \frac{1}{14 \times 10^7} \sqrt{\frac{2 \times 1.2 \times 1.6 \times 10^{-11}}{12 \times 1.66 \times 10^{-27}}} \\ &= \frac{1}{14 \times 10^7} \sqrt{\frac{2 \times 1.6 \times 10^{15}}{1.66}} \\ &= \frac{10^7}{14 \times 10^7} \sqrt{\frac{2 \times 1.6 \times 10}{1.66}} \\ &= \frac{1}{14} \times 4.391 = 0.3136 \text{ m} \\ &= \mathbf{31.36 \text{ cm}} \end{aligned}$$

4.8 THE CYCLOTRON

The α and β - particles given out by natural radioactive substances neither possess sufficiently large speeds nor their speeds are under control. It was, therefore, felt necessary to accelerate charged particles to very high velocities by the application of electric and magnetic fields. Cockroft and Walton produced fast moving protons by electronic voltage multiplication device. The best arrangement was, however, made by Lawrence and Livingstone in 1935 and called a *cyclotron*. This arrangement won the Nobel prize for Lawrence.

Construction. The cyclotron essentially consists of two semicircular hollow cylindrical boxes, D_1 and D_2 usually of copper and called the dees on account of their shape like the letter D . These are placed in an evacuated cylindrical steel tank, with their edges parallel but with a small diametrical gap G between them as shown in fig. 4.11 (a)

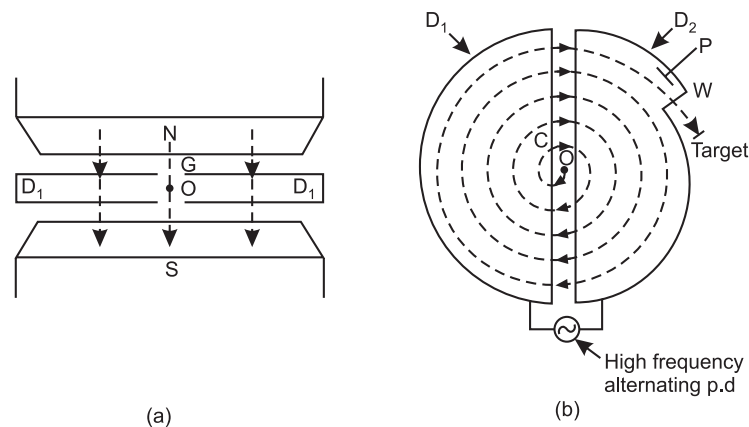


Fig. 4.11

Role of Electric and Magnetic field. Simultaneous application of electric field and magnetic field plays an important role in the working of cyclotron.

Magnetic field. The basic function of the magnetic field is to move the charged particle into a *semi-circular path with the dees*.

The whole assembly is arranged to lie in between the pole-pieces of large and powerful electromagnet NS, of about the same diameter as that of the two dees; so that, magnetic field is parallel to the axis of the tank and perpendicular to its base and hence also to the plane of the dees, with a flux of the order of $1.5 \text{ weber-metre}^2$ or 15000 gauss.

Electric field. The primary function of the electric field is to provide a potential difference between the dees of the cyclotron to *accelerate the charged particles*. Thus, an alternating electric field having a frequency such that the time taken by the particle to travel through the semi-circular path within the dees is equal to half the time period of the alternating electric field is required.

The dees, also referred to as the *electrodes*, are connected to a radio frequency oscillator in order to produce a high alternating potential difference across them, of the order of 10^4 volts , or more and a very high frequency in the neighbourhood of 10 MHz.

The positive ions to be accelerated may be obtained in one of the following ways:

(i) The required gas, (for example, *hydrogen* for obtaining *protons*, *heavy hydrogen* for obtaining *deutrons* and *helium* for obtaining α -particles) is filled into the steel tank at a low pressure of about 10^{-3} mm of mercury and a filament, at or near the centre of the tank, *just outside the dees*, is heated. With a small potential difference applied between the filament and the tank, the electrons emitted by the hot filament acquire sufficient energy to be able to ionise some of the atoms of the gas and thus produce a vertical column of positive ions. Some of these positive ions then find their way into the gap between the two dees.

(ii) Alternatively, what is known as a *capillary ion source* is used. The ions here too are produced in the same manner as indicated under (i) above, but *in a separate or an auxiliary chamber* and led from there into the steel tank through a narrow opening or a capillary tube which is so directed that there is again a narrow vertical column of positive ions near the centre O of the gap G between the two dees.

Both the methods have their advantages and disadvantages in to which, however, we need not go here.

Principle. The cyclotron is a *magnetic resonance type positive ion* accelerator. The charged particle to be accelerated placed at a centre point O of the Dee, rapidly passes through an alternating electric field along a closed path, its energy being increased each time. A strong magnetic field

is used to control the motion of the particles and to return them periodically to the region of the acceleration electric field. The particle passes definite points of the alternating electric field almost in union when the field is in the same phase *i.e.* in *resonance*.

The resonance condition is discussed below along with the cyclotron resonance equation.

Working. On account of the high frequency alternating potential difference applied to the dees D_1 and D_2 , they alternately acquire positive and negative potentials about 10^7 time in a second. So that, if at a given instant D_1 happens to be negative, some of the positive ions, say *protons*, in the gap G between the dees, get attracted towards it and enter into the hollow space within it. Now, although the electric field ceases to be effective inside D_1 (there being no electric field inside a hollow charged conductor), the magnetic field acts in a direction perpendicular to their plane of motion and the positive ions, or the protons, are thus compelled to take a circular path inside D_1 .

The frequency of the alternating potential difference applied to the dees is so adjusted that not only does the potential changes direction but has also its peak or maximum value V when, after covering their semi-circular path through D_1 , the positive ions or the protons, emerge out into the diametrical gap G ; so that, they now find D_2 to be negative and D_1 positive, and therefore cross over from D_1 to D_2 *each acquiring an amount of energy qV in the process* (where q is the charge on it). Hence, as under the action of the magnetic field perpendicular to them, they again describe a semi-circular path inside D_2 they emerge out into the gap G with a higher velocity.

If this frequency of the alternating potential difference be further equal to the gyro frequency or the cyclotron frequency $qB/2\pi m$ (by suitable adjustment of the magnetic field B), the frequency of rotation of the ions or the protons then being quite independent of their velocity inside the dees, they always emerge out into the gap from either dee to find that the dees have, in the meanwhile, changed their signs of potential. The motion of the ions through the dees is thus in resonance with the alternating potential applied. Every time, therefore, the ions gain an amount of energy qV each as they cross over from one dee to the other, with the result that they continue to move in circular paths of continually increasing radii inside the two dees, thus describing an ever-expanding spiral, as shown in Fig. 4.11 (a), until the radius of their path becomes nearly the same as the outermost radius of either dee.

By this time, the ion beam acquires a sufficient high velocity and is brought out of dee D_2 through a thinly covered window W by means of a deflecting plate P which is maintained at negative potential. To make sure that this negative potential is with respect to the earth, it is *so arranged that the window W lies at right angles to the diametrical gap G between the dees, *i.e.*, OW is perpendicular to the gap G , because then, the instantaneous potential difference across the gap G is *zero* at the time that the ions or the protons emerge out at W . (This is so because the time taken by the ions to cover the distance from the gap to W , inside D_2 , is one-quarter of their time-period and since the potential difference across the gap had its peak or maximum value when the ions crossed the gap from D_1 to D_2 it must, after a quarter of a time-period, have an instantaneous value *zero*). Thus, deflected by the negative plate P the high energy beam of positive ions (or protons) is allowed to fall on the target, placed in an evacuated side-tube attached to the steel tank.*

Resonance condition: As we have discussed above, cyclotron is a resonance device. In a cyclotron the value of the magnetic field strength depends upon the frequency of the oscillating electric field applied between the dees. It is so chosen as to give *resonance* between the arrival of the charged particle in the gap and reversal of the voltage between the dees. This is done by adjusting the time taken by the charged particle to describe a semi-circular path equal to half the time period of oscillation of the applied high frequency electric field *i.e.*

$$\frac{\pi}{\omega} = \frac{T}{2}$$

$$f = \frac{1}{T} = \frac{\omega}{2\pi} = \frac{Be}{2\pi m}$$

The frequency

This gives the *cyclotron resonance condition* for charged particles of a given value of $\frac{e}{m}$. If this condition is not satisfied there will be no resonance between the arrival of the charged particle in the gap and the reversal of the voltage between the dees. The particle will go out of step and will not be accelerated. This is why it is said that a cyclotron is a *resonance device*.

Theory. If a positive ion is generated at a point O, as shown in Fig. 4.11(b), within the gap at a time when D_2 is at positive potential and D_1 at a negative potential, it will be accelerated across the gap to D_1 and enter the hollow segment D_1 with velocity v given by

$$Ve = \frac{1}{2}mv^2$$

where, V is the applied voltage and e and m are charge and mass of the particle respectively. When it is inside the conductor, it will not be acted upon by the electric field but under the influence of the applied magnetic field having a flux density B , it will travel along a circular path, the radius of which is given by

$$\frac{mv^2}{r} = Bev$$

and finally emerges at C in the direction indicated.

The time taken by the positive ion to travel the semicircular path

$$t = \frac{\pi}{\omega} = \frac{\pi r}{v} = \frac{\pi m}{Be}$$

where ω is the angular velocity of the ion in the circular path and

$$\omega = \frac{Be}{m}$$

The value of t is a constant being independent of the velocity of the ion and the radius in which it travels. If the frequency of the applied voltage is adjusted in such a manner that it is reversed as soon as the particle comes out of D_1 , the particle at C will be accelerated across the gap to D_2 and will describe a further circular path in D_2 . The radius of the semi-circle as well as speed of the particle will, now, be greater than that in the first case, but as proved above, the time taken by the particle to travel the semi-circular path in D_2 will be the same. Everytime the particle emerges out of the dees, the direction of the voltage is reversed and the particle is accelerated across the gap. The path of the particle will be a spiral and it will finally come out of the Dees in the direction indicated, through the window W .

Maximum kinetic energy of the particle. The final energy E of the charged particle is given by

$$E = \frac{1}{2}mv_{\max}^2$$

where $v_{\max}$ is the maximum velocity gained by the charged particle in its final orbit of radius $r_{\max}$.
Now

$$\frac{mv_{\max}^2}{r_{\max}} = Bev_{\max}$$

$$r_{\max} = \frac{Be}{m}r_{\max}$$

Hence,

$$E = \frac{1}{2}mv_{\max}^2 = \frac{1}{2}m \frac{B^2 e^2}{m^2} r_{\max}^2$$

$$= \frac{1}{2} \frac{B^2 e^2 r_{\max}^2}{m}$$

This relation gives the maximum kinetic energy of the charged particle in terms of applied magnetic field and dee radius.

The condition for optimal acceleration of the ion in the inter dee gap is that the time taken by the ion to travel the semi-circular path (t) is equal to half the time period (T) of oscillation of the applied high frequency electric field

$$\begin{aligned} \text{i.e.,} \quad t &= \frac{T}{2} \\ \text{or} \quad \frac{\pi m}{Be} &= \frac{T}{2} \\ \text{or} \quad T &= \frac{2\pi m}{Be} \end{aligned}$$

If f is the frequency of the oscillating electric field, then

$$f = \frac{1}{T} = \frac{Be}{2\pi m}$$

This is the basic cyclotron resonance equation.

Hence, in terms of f the maximum energy of the charged particle

$$\begin{aligned} E_{\max} &= \frac{1}{2} \frac{B^2 e^2 r_{\max}^2}{m} \\ &= \frac{1}{2} 4\pi^2 m \frac{B^2 e^2}{4\pi^2 m^2} r_{\max}^2 \\ &= 2 \pi^2 m f^2 r_{\max}^2 \end{aligned}$$

The particles are ejected out of the cyclotron as pulse streams and not continuously.

With a comparatively small potential difference of the order of 50,000 volts, very fast moving particles can be produced. For example, if the particle makes 200 revolutions before emerging out it will gain a velocity equivalent to a total fall through a potential of

$$2 \times 50000 \times 200 = 20 \text{ million volts}$$

If heavy hydrogen is used instead of ordinary hydrogen, a beam of high energy deuterons is obtained. As their mass is double they possess greater energy and are more useful as atomic projectiles.

Maximum radius of curvature. If V is the average voltage applied between the dees of a cyclotron and the charged particle crosses the gaps between the dees n times to reach the orbit of maximum radius, then energy acquired by the ion having a charge e is given by

$$\begin{aligned} E_{\max} &= neV \\ \text{But} \quad E_{\max} &= \frac{1}{2} \frac{B^2 e^2 r_{\max}^2}{m} \\ \therefore \quad \frac{B^2 e^2 r_{\max}^2}{2m} &= neV \\ \text{or} \quad r_{\max} &= \frac{1}{B} \left(\frac{2mV}{e} \right)^{1/2} n^{1/2} \end{aligned}$$

If the values of B and V are kept constant, $r_{\max}$ is directly proportional to the square root of the number of time the particles crosses the gap between the dees.

Limitations of the cyclotron

(i) The energy to which a particle can be accelerated in a cyclotron is limited due to change in mass with velocity. The mass of a particle, when moving with a velocity v is given by

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

where m_0 is the rest mass and c the velocity of light. As already proved, the time taken by a particle

to travel the semi-circular path is $\frac{\pi m}{Be} = \frac{T}{2}$

$$\therefore \text{Frequency} \quad f = \frac{1}{T} = \frac{Be}{2\pi m}$$

$$= \frac{Be\sqrt{1 - \frac{v^2}{c^2}}}{2\pi m_0}$$

Hence the frequency of rotation of the charged particle decreases as the velocity increases. As a result it takes a longer time to complete its semi-circular path and the particle continuously goes on lagging the applied alternating potential difference till a stage is reached when it can no longer be accelerated further. This discrepancy is optimised by the following two methods:

(i) **Field variation.** The frequency of the ion can be kept constant by taking $B\sqrt{1 - \frac{v^2}{c^2}}$ constant. For this purpose the value of the magnetic field B should increase as the velocity of the ion increases so that the product remains unchanged.

(ii) **Frequency modulation.** In the alternative method, the frequency of the applied $A.C$ is varied so that it is always equal to the frequency of rotation of the ion. The machine in which the frequency of electric field is kept constant and magnetic field is varied is called **Synchrotron**, whereas a machine in which magnetic field is kept constant and the frequency of the applied electric field is varied is known as a **Frequency modulated cyclotron** or **Synchro-cyclotron**.

(2) A cyclotron cannot be used to produce high energy electron beams. The reason for the same is that there is an appreciable increase in the mass of the electron at fairly low energies. For example, a 10% increase in the rest mass of the electron take place at an energy of 50 KeV only. Electron being a very light particle there is an appreciable increase in its velocity at low energies which is not the case with massive particles like the proton or the α -particle.

(3) Neutron can not be accelerated by a cyclotron. A neutron carries no charge; it can not be accelerated by the electric field between the two Dee's. It can also be not acted upon by the magnetic field so that its path within the dees cannot be regulated.

4.9 MAGNETIC FIELD FOCUSSED

An appropriate configuration of non-uniform electric field, is used for focussing of electrons. An electron beam can also be focussed with the help of magnetic field. Depending upon the nature of the magnetic field and the direction of its application, two different types of focussing are obtained.

- (i) Longitudinal magnetic field focussing, and
- (ii) Transverse magnetic field focussing.

4.9.1 Longitudinal Magnetic Field Focussing

Charged particles like electrons can be focussed at a point by employing a uniform magnetic field acting along the direction of motion of beam of charged particles; as shown in fig. 4.12. The path of an electron.

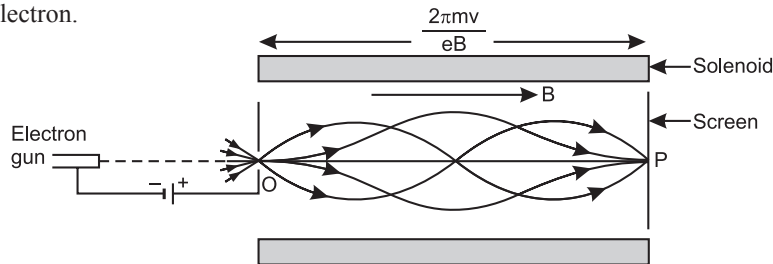


Fig. 4.12 Focusing of charged particles in a longitudinal magnetic field.

in a uniform magnetic field would be a helix if electrons enter at an angle ϕ , the path (p) of the helix is given by

$$p = \frac{2\pi mv}{eB} \cos \phi \quad \dots(i)$$

Suppose a beam of electrons enters at point O with a small angle ϕ and with a velocity v . If ϕ is taken too small ($\phi \leq 10^\circ$), $\cos \phi$ may be taken as unity. In time T , the charged particles come to focus at point P , the length $OP = l$ is given by

$$l = \frac{2\pi mv}{eB} \quad \dots(ii)$$

Therefore, the beam of electron entering the field B making a small angle of divergence ϕ come to focus at a distance l or any point which is located at a distance of integral multiple of l i.e. nl .

4.9.2 Transverse Magnetic Field Focussing (180°)

Particle carrying the *same charge* (q) and having the *same momentum* (p), even if moving in different directions, can all be brought to focus at very nearly the same point on a screen by means of a suitably applied magnetic field. This is called *magnetic focussing*, and since the particles come to common focus after describing an angle of 180° from their point of entry into the magnetic field, we qualify it as **180° magnetic focussing**.

If, on the other hand, the particles carry the same charge but have different momenta, they are brought to focus at different points on the screen.

To take the second case first, if a beam of particles, each carrying a charge q , enters normally, through a slit S in a screen, [Fig. 4.13 (a)], into a magnetic field of flux density B perpendicular to the beam (i.e., perpendicular to the plane of the paper in the case shown), a particle of the beam is acted upon by a force equal to qvB , where v is the velocity of the particle. Under the action of this force, the particle takes a

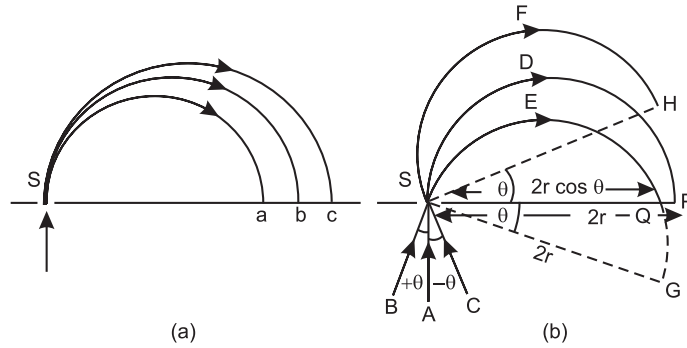


Fig. 4.13

circular path, such that if m be the mass of the particle and r , the radius of its circular path, we have

$$mv^2/r = qvB, \text{ whence, } r = mv^2/qvB = mv/qB. \quad \dots(i)$$

Now, q and B being constants, i.e., the same for each particle, r varies with the momentum $p = mv$ of the particles. So that particles having different momenta come to focus at different points a, b, c on the screen. The device thus acts as a **momentum selector**.

Obviously, even if the velocities of the particles in the beam be arranged to be the same, their momenta will still be different on account of their different masses. So that, particles of different masses will then come to focus at different points. This is the principle underlying the mass spectrographs, like the *Bainbridge mass spectrograph*, for the separation of isotopes*, i.e., elements having the same atomic number and chemical properties but different atomic weights.

* **Isotopes:** Isotopes are nuclei having the same atomic (proton) number Z but different atomic mass A . The isotopes of an element contain the same number of protons but different number of neutrons. For example: Oxygen has two isotopes $^{16}\text{O}_8$ and $^{17}\text{O}_8$. Hydrogen has three isotopes: (i) Hydrogen $^1\text{H}_1$ (ii) Deuterium $^2\text{H}_1$ and (iii) Tritium $^3\text{H}_1$. Neon has two isotopes $^{20}\text{Ne}_{10}$, $^{22}\text{Ne}_{10}$. Chlorine also has two isotopes $^{35}\text{Cl}_{17}$ and $^{37}\text{Cl}_{17}$. At present about 297 different isotopes are known.

Let us now consider the first case, viz., the case of particles carrying the same charge and having the same momentum.

Thus, suppose we have a conical beam of particles, all carrying the same charge q and having the same momentum $p = mv$, entering the magnetic field through a slit S as shown in Fig. 4.13(b). Then the particles entering the slit normally along AS describe, under the action of the magnetic field, the trajectory SDP , thus covering an angle of 180° from S and coming to focus on the screen at P , such that the diameter of the circular path taken is $SP = 2r$.

The particles entering the slit along BS at an angle $+\theta$ with AS describe the trajectory SEQ and come to focus at Q , such that the diameter SG of their circular path is also inclined to SP at an angle θ , and therefore, the chord $SQ = 2r \cos \theta$.

Similarly, the particles entering the slit along CS at an angle $-\theta$ with AS describe the trajectory SFQ and also come to focus on the screen at Q , with the diameter SH of their circular path inclined to SP at an angle θ .

Thus, the particles of the conical beam are all spread over the small distance QP on the screen, where

$$QP = SP - SQ = 2r - 2r \cos \theta = 2r(1 - \cos \theta)$$

Or, Since $\cos \theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} \dots = 1 - \frac{\theta^2}{2}$, neglecting the higher power of θ , we have

$$QP = 2r[1 - (1 - \theta^2/2)] = 2r \cdot \theta^2/2 = r \theta^2.$$

θ^2 being negligible for a small value of θ , points Q and P will be practically coincident or the same.

In other words, *all particles having the same charges and momentum came to focus at the same point P, after covering an angle of 180° from their point of entry (S) into the magnetic field.*

If the screen be a photographic plate and the slit (S) parallel to the magnetic field, we shall have line images of the slit corresponding to particles of different momenta at point like a, b, c in the first case [Fig. 4.13 (a)] and a single thin line image of the slit at P in the second case, [Fig. 4.13(b)], the line images corresponding to all particles being coincident there.

Determination of relative abundance of isotopes in an element. Accelerating the ions of a given through slit S. [Fig 4.13 (b)] by means of an adjustable potential difference and using another slit at P , we can estimate the relative abundance of the different isotopes in the element. For, obviously only those ions will pass through the slit at P for which the ratio *momentum to charge*, i.e., mv/q is *constant* $= Br$ (see relation I, above). These ions are allowed to send a current through a sensitive galvanometer and the deflection noted. The accelerating potential is now altered so that another set of ions, with a slightly different value of m/q now enter the slit S and emerge through the slit of P , producing a slightly different deflection in the galvanometer. These deflections in the galvanometer being proportional to the number of ions present in the beam and responsible for the current through it, we can easily estimate from them the relative abundance of the different ions present in the beam and hence that of the different isotopes present in the given element.

Example 29. An electron describes a helix of radius 10 cm and pitch 3 cm in a magnetic field of 50 gauss. Calculate the components of its velocity along and perpendicular to the field. (Take mass of electron $= 9 \times 10^{-28}$ gm)

Solution. Let the components of the velocity of the electron along and perpendicular to the magnetic field be v_1 and v_2 respectively, the former responsible for its forward motion along the direction of the field and the latter for its rotation, making its resultant path a helix.

The radius of the helix, therefore, is given by the relation $r = mv_2 c/qB$ (see §4.7).

$$\text{whence, } v_2 \left(\frac{qB}{mc} \right) r = \left(\frac{4.80 \times 10^{-10} \times 50}{9 \times 10^{-28} \times 3 \times 10^{10}} \right) 10 = (8.9 \times 10)^8 10 = 8.9 \times 10^9 \text{ cm/sec.}$$

And, since $\text{pitch of the helix} = \frac{2\pi mc}{qB} v_1 = 3$, we have

$$v_1 = \frac{3}{2\pi} \left(\frac{qB}{mc} \right) = \frac{3}{2\pi} (8.9 \times 10^8) = 4.25 \times 10^8 \text{ cm/sec.}$$

Thus, the components of the velocity of the electron along and perpendicular to the magnetic field are respectively $4.25 \times 10^8 \text{ cm/sec}$ and $8.9 \times 10^9 \text{ cm/sec}$.

Example 30. An electron of energy 20 electron volts moving in a direction perpendicular to a magnetic field $B = 10^2 \text{ Wb/m}^2$ describes a circle of radius r . Calculate the value of r , ($1 \text{ ev} = 1.6 \times 10^{-19}$) joules, mass of electron may be taken to be $9 \times 10^{-31} \text{ kg}$.

Solution. Let v be the velocity of the electron perpendicular to the magnetic field. Then if m be its mass, we have

$$\frac{1}{2}mv^2 = 20 \text{ eV} = 20 \times 1.6 \times 10^{-19} \text{ joules}$$

whence
$$v = \left(\frac{2 \times 20 \times 1.6 \times 10^{-19}}{m} \right)^{\frac{1}{2}} \left(\frac{64 \times 10^{-19}}{m} \right)^{\frac{1}{2}} \text{ m/sec.}$$

Now, as we know, the radius of the circle described by a charged particles in a uniform magnetic field (§4.7) or the gyro (or cyclotron) radius, as it is called, is given by $r = mv/qB^\dagger$. So, that, we have

$$\begin{aligned} r &= \frac{m(64 \times 10^{-19})^{\frac{1}{2}}}{\sqrt{m} \times q \times B} = \frac{\sqrt{m} (64 \times 10^{-19})^{\frac{1}{2}}}{q \times B} = \frac{(9 \times 10^{-31} \times 64 \times 10^{-19})^{\frac{1}{2}}}{1.6 \times 10^{-19} \times 10^2} \\ &= \frac{3 \times 8}{1.6} \times 10^{-8} = 15.0 \times 10^{-8} \text{ metre.} \end{aligned}$$

Thus, the radius of the circular path described by the electron = $15.0 \times 10^{-8} \text{ metre}$.

Example 31. An electron of velocity $\vec{v} = (2\hat{i} + 3\hat{j})10^8 \text{ cm/sec}$ enters a region of uniform magnitude field $B = 500\hat{i} \text{ gauss}$, so that its path become helical.

(i) In what direction does the axis of the helix lie? (ii) Calculate the radius of the helix. (iii) Calculate the number of rotations as the electron advance 10 cm along the axis of the helix.

[Agra (supp)]

Solution. Here, the velocity v of the electron may be resolved into two rectangular components, $\vec{v}_1 = 2 \times 10^8 \hat{i} \text{ cm/sec}$ along the direction of the magnetic field $\vec{B} = 500\hat{i} \text{ gauss}$ and $\vec{v}_2 = 3 \times 10^8 \hat{j} \text{ cm/sec}$, perpendicular to it, the former component being responsible for the forward motion of the electron along the field and the latter for circular motion (see § 4.7). So that,

(i) the axis of the helix lies along the direction of the magnetic field, i.e., along the axis of x :

(ii) the radius of the helix is given by $r = mv_2c/qB$. Taking $m = 9 \times 10^{-28} \text{ gm}$, $q = 4.80 \times 10^{-10} \text{ esu}$, $v_2 = 3 \times 10^8 \text{ cm/sec}$ and $B = 500 \text{ gauss}$ (given)

we have

$$r = \frac{9 \times 10^{-28} \times 3 \times 10^8 \times 3 \times 10^{10}}{4.80 \times 10^{-10} \times 500} = \frac{81}{4.80 \times 500} = 0.03375 \approx 0.034 \text{ cm}$$

(iii) The forward distance covered by the electron in one full rotation, i.e. pitch of the helical path $\frac{2\pi mc}{qB} v_1 = \frac{2\pi \times 9 \times 10^{-28} \times 3 \times 10^{10}}{4.80 \times 10^{-10} \times 500} \times 2 \times 10^8$

[†] $r = mvc/qB$ if Gaussian units be used, i.e., if q be in esu and B in emu or gauss.

$\therefore$ number of rotations made by the electron as it advance through a distance of 10 cm

$$= \frac{10}{\text{pitch of the helix}} = \frac{10 \times 4.80 \times 10^{-10} \times 500}{2\pi \times 9 \times 10^{-28} \times 3 \times 10^{10} \times 2 \times 10^8}$$

$$= \frac{4.8 \times 5}{2\pi \times 54} \times 10^3 = 70.74.$$

Example 32. For a proton initially travelling with velocity $v_x = 5 \times 10^8$ cm/sec, calculate the following: (i) Force experienced in a magnetic field $\vec{B} = 2000\hat{j} + 4000\hat{k}$ gauss. (ii) Acceleration acquired in an electric field $\vec{E} = 200\hat{i} + 100\hat{j}$ volts/cm (iii) Transverse deflection in travelling a length $L_z = 10$ cm in an electric field $E_y = 200$ volts/cm. (Agra 2002)

Solution. (i) We know that force acting on a charge q moving perpendicular to a magnetic field $\vec{B}$ with velocity $\vec{v}$ is given by, $\vec{F} = \frac{q}{c}(\vec{v} \times \vec{B})$ using Gaussian units. So that,

$$\vec{F} = \frac{q}{c}(5 \times 10^8 \hat{i})(2000\hat{j} + 4000\hat{k})$$

Here $q = e = 4.8 \times 10^{-10}$ esu and $v = v_x = 5 \times 10^8$ cm/sec. We, therefore, have

$$\vec{F} = \frac{4.8 \times 10^{-10}}{3 \times 10^{10}} \times 5 \times 10^8 \times 2000(\hat{k} - 2\hat{j}) = 1.6 \times 10^8 (\hat{k} - 2\hat{j}) \text{ dyne}$$

or magnitude of the force, $F = 1.6 \times 10^8 \sqrt{5} = 3.578 \times 10^8$ dynes

and its direction cosines, clearly $0, -2\sqrt{5}$ and $1\sqrt{5}$

(ii) We know that the force experienced by a particles carrying a charge q in an electric field E is equal to qE . Therefore, if m be the mass of the particle and a , the acceleration produced in it, we have

$$m\vec{a} = q\vec{E} \quad \text{whence,} \quad \vec{a} = q\vec{E}/m$$

So that acceleration of the proton in the electric field. i.e.,

$$\vec{a} = \frac{4.8 \times 10^{-10}}{1.67 \times 10^{-24}} \left(\frac{200\hat{i} + 100\hat{j}}{300} \right) = \frac{4.8 \times 10^{-10}}{1.67 \times 10^{-24}} \left(\frac{2\hat{i} + \hat{j}}{3} \right)$$

$$\therefore \text{ magnitude of the acceleration } a = \frac{4.8 \times 10^{-10}}{1.67 \times 10^{-24} \times 3} \sqrt{5} = \frac{1.6 \times \sqrt{5}}{1.67} \times 10^{14}$$

$$= 2.142 \times 10^{14} \text{ cm/sec}^2,$$

and its direction cosines, $2\sqrt{5}, 1\sqrt{5}$ and 0 .

(iii) The transverse displacement of the particle will clearly be y , where acceleration

$$a = (\text{charge} \times \text{electric field})/\text{mass} = qE_y/m \text{ and } t = L_x/v_x.$$

$$\text{So that, } y = \frac{1}{2} \frac{4.8 \times 10^{-10}}{1.67 \times 10^{-24}} \times \frac{200}{300} \times \left(\frac{10}{5 \times 10^8} \right)^2 = \frac{1.6}{1.67 \times 25} = 0.0383 \text{ cm}$$

Example 33. (a) A stream of protons and deuterons in a vacuum chamber enters a uniform magnetic field. Both protons and deuterons have been subjected to the same accelerating potential; hence kinetic energies of the particles are the same if the ion stream is perpendicular to the magnetic field and the protons move in a circular path of radius 15 cm., find the radius of the path traversed by the deuterons. Given that the mass of a duetron is twice that of a proton. (Agra)

(b) Calculate the Lorentz force on a proton moving with velocity $(2\hat{i} + 3\hat{j})10^8$ cm/sec in an electric field of intensity $(\hat{i} + 2\hat{j} + 3\hat{k})$ statvolts/cm and a magnetic field of $(300\hat{j} + 300\hat{k})$ gauss.

Solution. (a) We know that the radius of the path traversed by a charged particle in a magnetic field is given by $\vec{r} = mvc/q\vec{B}$ using Gaussian units.

So, that, radius of the path of proton $\vec{r}_p = m_p v_p c / q \vec{B}$ and radius of the path of the deuteron $\vec{r}_D = m_D v_D c / q \vec{B}$.

$$\therefore \frac{\vec{r}_D}{\vec{r}_p} = \frac{m_D v_D c}{q \vec{B}} \times \frac{q \vec{B}}{m_p v_p c} = \frac{m_D v_D}{m_p v_p}$$

Now, we are given that $m_D = 2m_p$. And, since the kinetic energies of the two particles are the same, their velocities must be inversely proportional to the square roots of their masses. For,

$$\frac{1}{2} m_D v_D^2 = \frac{1}{2} m_p v_p^2 \quad \therefore \quad v_D/v_p = \sqrt{m_p/2m_p} = 1/\sqrt{2}. \text{ or } v_D = v_p/\sqrt{2}$$

$$\therefore \quad r_D/r_p = 2m_p \times v_p \sqrt{2} / m_p \times v_p = \sqrt{2}$$

Since $r_p = 15$ cm (given), we have $r_D = r_p \sqrt{2} = 15\sqrt{2} = 21.21$ cm.

Thus, the radius of the circular path of the deuteron = 21.21 cm.

(b) As we know, Lorentz force $\vec{F} = q\vec{E} + \frac{q}{c} \vec{v} \times \vec{B}$. Substituting the given values, therefore, we have

$$\begin{aligned} \vec{F} &= 4.8 \times 10^{-10} (i + 2k + 3k) + \frac{4.8 \times 10^{-10}}{3 \times 10^{10}} (2i + 3j) 10^2 \times (300j + 300k) \\ &= 4.8 \times 10^{-10} [\hat{i} + 2\hat{j} + 3\hat{k} + 2\hat{k} - 2\hat{j} + 3\hat{i}] = 4.8 \times 10^{-10} (4\hat{i} + 5\hat{k}) \end{aligned}$$

$\therefore$ magnitude of the force, $F = 4.8 \times 10^{-10} \sqrt{16 + 25} = 4.8 \times 10^{-10} \sqrt{41} = 3.07 \times 10^{-9}$ dyne.

and the direction cosines = $4/\sqrt{41}, 0, 5/\sqrt{41}$.

Example 34. An electron enters a transverse magnetic field B at a point P with velocity v along the axis of x and describing a curved path of radius of curvature r under the action of the field, emerges from it at a point Q where the distance between the two points is x . Show that

- (i) its displacement along the y -axis as it emerges from the field is $y = r \left[1 - \sqrt{1 - \left(\frac{x}{2r}\right)^2} \right]$ and
(ii) its angular deflection $\theta = qBx/mvc$.

Solution. In Fig. 4.13 (a) let the dotted circle represents the region of the magnetic field B perpendicular to the plane of the paper and directed downwards, and let P be the point of entry of the electron into the field (in a direction perpendicular to it), PQ , the curved path of radius of curvature r described by it in the field in a direction tangential to the curved path at Q , such that horizontal distance between P and Q is x .

Then, clearly, displacement suffered by the electron in the magnetic field is y and the angular displacement suffered by it, θ , where θ is the angle that curved path subtends at its centre of curvature O .

(i) Drop QS perpendicular on to OP . Then, $OP = r$, $SP = y$ and $OS = (r - y)$.

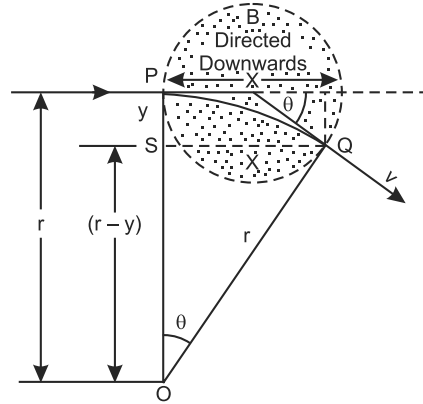


Fig. 4.13 (a)

Now, in the right angled triangle OSQ , we have $OQ^2 = OS^2 + SQ^2$.
Or, $r^2 = (r - y)^2 + x^2$,

whence, $r - y = \sqrt{r^2 - x^2}$ or $y = r - \sqrt{r^2 - x^2}$

$$\text{Or, displacement } y = r \left(1 - \sqrt{1 - \frac{x^2}{r^2}} \right) = r \left[1 - \sqrt{1 - \left(\frac{x}{r} \right)^2} \right]$$

(ii) Again, from $\triangle OSQ$, we have $x = r \sin \theta$, or $\sin \theta$ is always small, we have $x = r\theta$, whence, $\theta = x/r$.

But r , as we know, is given by the relation $r = mvc/qB$. So that, angular displacement suffered by the electron, $\theta = qBx/mvc$.

Alternatively, we could obtain θ as follows: The angular velocity of the electron inside the magnetic field being $\omega = qB/mc$, we have $\theta = \omega t$ where t is the time taken by the electron to cover the distance x in the magnetic field and, therefore, equal to x/v . So that, $\theta = \omega x/v = qBx/mvc$.

Example 35. A beam of charged particles (having same charge and momentum) enters a magnetic field and is deflected by 180° in a circular path of radius 20 cm. Find the width of the focal line if beam is in the form of a cone of 6° .

$$\text{Solution. } R = 20 \text{ cm} = 0.2 \text{ m}, \phi = \left(\frac{6}{2} \right)^\circ = 3^\circ = \frac{3 \times \pi}{180} \text{ radian}$$

$$\begin{aligned} \text{width of the focal line} &= R \phi^2 \\ &= 0.2 \left(\frac{3 \times \pi}{180} \right)^2 \\ &= 5.48 \times 10^{-4} \text{ m} \end{aligned}$$

Example 36. An electron moving in a horizontal direction with a speed of 5.0×10^7 m/sec enters a region where there is a uniform electric field of 2000 V/m directed upwards in the plane of its motion. Find the electrons coordinates referred to the point of entry and the direction of its motion 4×10^{-8} second later.

$$\text{Solution. } z = \frac{1}{2} \left(\frac{qE}{m} \right) t^2 \quad \text{Here } q = e = 1.6 \times 10^{-19} \text{ C}$$

$$y = \frac{1}{2} \left[\frac{2000 \times 1.6 \times 10^{-19}}{9.1 \times 10^{-31}} \right] (4 \times 10^{-8})^2 = 0.28 \text{ m}$$

$$\text{Along x-axis. } x = v \times t = 5 \times 10^7 \times 4 \times 10^{-8} = 2.0 \text{ m}$$

$$\tan \phi = \frac{y}{x} = \frac{0.28}{2.0} = 0.14$$

$$\phi = \tan^{-1}(0.14) = 8^\circ \text{ w.r. to horizontal axis.}$$

Example 37. Find the distance separating singly charged ions, of potassium isotopes of atomic weight 39 and 40 and the same kinetic energy, at the focus of 180° mass spectrometer if the former ions follow a path of 50 cm. radius of curvature. What is the maximum permissible spread θ on each side of the initial ion beam so that the two focal lines may not overlap?

Solution. Let v_1 and v_2 be the velocity of the two types of ions and m_1 and m_2 their atomic weight respectively. Then, since their kinetic energy is the same, we have $v_2/v_1 = \sqrt{m_1/m_2}$

$$\text{Now, } r = \frac{cmv}{qB} \quad \therefore \quad r_1 = \frac{cm_1 v_1}{qB} \quad \text{and} \quad r_2 = \frac{cm_2 v_2}{qB}$$

Or
$$\frac{r_2}{r_1} = \frac{r_2}{50} = \frac{m_2}{m_1} \frac{v_2}{v_1} = \frac{m_2}{m_1} \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{40}{39}},$$

whence,
$$r_2 = 50 \left(1 + \frac{1}{39}\right)^{\frac{1}{2}} = 50 \left(1 + \frac{1}{2} \cdot \frac{1}{39} + \dots\right) = \left(50 + \frac{50}{78}\right) \text{ cm}$$

$\therefore$ Separation of the two focal lines $= 2(r_2 - r_1) = 2\left(50 + \frac{50}{78} - 50\right)$

$$= 2 \times \frac{50}{78} = 1.28 \text{ cm}$$

Now, in order that the two may not overlap we must have width of the first focal line $r_1 \theta^2 = 1.28$. Or, $50 \theta^2 = 1.28$

where θ is the semi-angle of the cone of the ions as they enter the magnetic field. So that

$$\theta = \sqrt{1.28/50} = 0.16 \text{ radian} \quad \text{or} \quad \frac{0.16 \times 180}{\pi} = 9.17^\circ.$$

Example 38. The Uranium isotopes of atomic weight 235 and 238 may be separated by 180° magnetic focussing. Calculate the separation of their beams at the focus, brought in a magnetic field of 10^4 gauss if the radius of curvature of the heavier isotope be 100 cm when (i) the velocities (ii) the energies of the beams are equal.

Solution. The separation of the two beams at the focus is clearly equal to the difference between the diameters of their semicircular paths in the magnetic field, i.e., equal to $2(r_2 - r_1)$, where r_1 and r_2 are the radii of the two semicircular paths respectively. Now,

(i) in case the respective velocities of the two beams be v_1 and v_2 , we have

$$r_1 = m_1 v_1 c / qB \text{ and } r_2 = m_2 v_2 c / qB,$$

whence, $r_1/r_2 = m_1 v_1 / m_2 v_2$. Or since the velocities of the two beams are equal,

we have $v_1 = v_2$ and $\therefore r_1/r_2 = m_2/m_1 = 235/238$.

Or
$$\frac{r_2 - r_1}{r_2} = \frac{238 - 235}{238} = \frac{3}{238} \quad \text{Or, } r_2 - r_1 = \frac{3}{238} r_2 = \frac{3}{238} \times 100$$

Hence, separation of the two beams $= 2(r_2 - r_1) = 2 \times \frac{3}{238} \times 100 = 2.522 \text{ cm}$.

(ii) And, when the energies of the two beams are equal, we have

$$\frac{1}{2} m_1 v_1^2 = \frac{1}{2} m_2 v_2^2 \quad \text{i.e., } v_1/v_2 = \sqrt{m_2/m_1}$$

But, as seen under (i) above, $r_1/r_2 = m_1 v_1 / m_2 v_2$. So that,

$$\frac{r_1}{r_2} = \frac{m_1}{m_2} \sqrt{\frac{m_2}{m_1}} = \sqrt{\frac{m_1}{m_2}} = \sqrt{\frac{235}{238}} = \left(1 - \frac{3}{238}\right)^{\frac{1}{2}} = \left(1 - \frac{3}{2 \times 238}\right)$$

$\therefore$
$$r_2 - r_1 = \frac{3}{2 \times 238} r_2 = \frac{3}{2 \times 238} \times 100$$

Hence separation of the two beams $= \frac{2 \times 3}{2 \times 238} \times 100 = 1.261 \text{ cm}$

4.10 CHARGED PARTICLE IN A COMBINED ELECTRIC AND MAGNETIC FIELD—

Having studied the behaviour of a charged particle in an electric and a magnetic field separately, let us now see how it will behave in a combined electric and magnetic field, (i) when the two fields are crossed, i.e., at right angles to one another, and (ii) when the two field are parallel (or antiparallel), to each other. Each is discussed below:

4.11 ELECTRIC AND MAGNETIC FIELDS IN CROSSED CONFIGURATION (i.e., MUTUALLY PERPENDICULAR)

In atomic physics, quite often it becomes necessary to select only those particles (*viz.* electron, protons etc.) that have the same velocity, out of the bunch of particles having a wide range of velocities. This is the basis of *velocity selector* or *velocity filter* and achieved by using electric and magnetic fields in crossed configuration. When uniform electric and magnetic fields act perpendicular simultaneously over the same region, the fields are said to be in **crossed configuration**.

When electrons pass through such a region, they are deflected simultaneously by both the fields. Two metal plates separated by a small distance, when charged set up a uniform electric field E in the y -direction and a uniform magnetic field B is also set up in the same region between the plates in z -direction. Let the magnetic field B is acting into the page. Both these fields are manually perpendicular to the direction of motion of electrons, *i.e.*, x -direction. Let the stream of electrons enter the region of crossed configuration with a velocity v [Fig. 4.14].

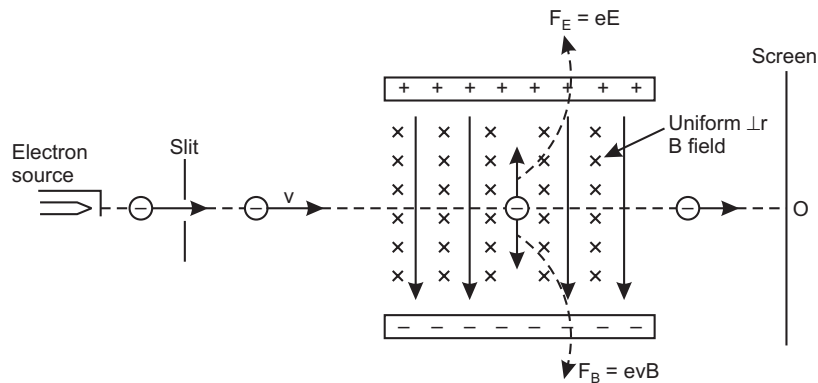


Fig. 4.14 Crossed E and B fields

The electric field E deflects the *electrons upward* with a force

$$F_E = eE \quad \dots(i)$$

and magnetic field B deflects them *downwards* with a force

$$F_B = evB \quad \dots(ii)$$

If the magnitudes of these two fields are adjusted such that the force acting on the electrons are equal, so that the electrons will not experience any net force. Under this condition

$$eE = evB$$

$$\therefore v = \frac{E}{B}$$

The electrons experience a zero net force as the above two forces are equal and opposite thereby balance each other. Therefore, they will not deviate from their original straight line path, *i.e.*, x -axis and travel without any change in their velocity v . By knowing the value of E and B , the velocity of electrons and hence their kinetic energy can be determined. In 1897, J.J. Thomson used this method for the determination of electron beam velocity.

4.12 VELOCITY SELECTOR

Velocity selector also known as velocity filter is an electro-optic device which uses the combined effect of two mutual perpendicular fields E and B acting simultaneously over the same region. This crossed-configuration of E and B selects a stream of charged particles of single velocity from a beam of charged particles having a wide range of velocities. [Fig 4.15].

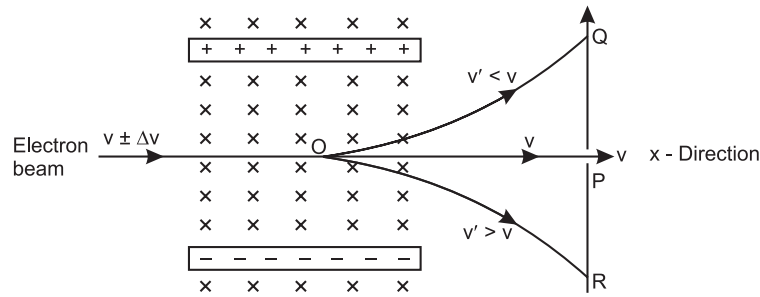


Fig. 4.15

Here the electric field E is produced in vertically downward direction ($-y$ axis) by a set of charged particles plates and a uniform magnetic field B is applied perpendicular to it, acting into the paper. If the two fields E and B are adjusted such the electric field force balances the magnetic force B acting on the electrons moving with velocities $v + \Delta v$, then those electrons are not deflected and continue to travel along the straight line, having velocity v , subsequently, they pass through the slit at P . Those electrons moving with velocity lesser than v ($v' < v$) will get deflected upward along OQ and those moving with greater velocity than v ($v' > v$) will get deflected downward along OR . Thus, a strictly homogenous single velocity electron beam travelling along OP is obtained. This arrangement is therefore, known as a **velocity selector** or a **velocity filter**. This forms an essential component in Bainbridge mass spectrograph.

4.13 MASS SPECTROGRAPHS

Mass spectrograph is an instrument employed to separate different isotopes from a stream of positive ions of an element and measures their individual masses as well as their relative abundances, by using electric and magnetic fields over the same region since the instrument is used to separate ions of different masses, is called as mass spectrograph.

Bainbridge's Mass Spectrograph. A schematic diagram showing the construction of a Bainbridge mass spectrograph is shown in Fig. 4.16.

A beam of *positive ions* produced in a discharged tube is collimated into a narrow beam by the two slits S_1 and S_2 . After emerging from the slit S_2 , the positive ions enter a *velocity selector*.

Velocity selector. It consists of two plates P_1 and P_2 between which a steady electric field E is maintained in a direction at right angles to the ion beam. A magnetic field B_1 is produced by an electromagnet (represented by the dotted circle). The magnetic field B_1 is at right angles to the electric field E as well as the ion beam.

The electric field and the magnetic field of the *velocity selector* are so adjusted that the deflection produced by one is exactly equal and opposite to the deflection produced by the other so that, there is no net deflection for ions having a particular velocity v .

If E and B_1 are the electric field intensity and magnetic induction, then

$$eE = B_1 ev$$

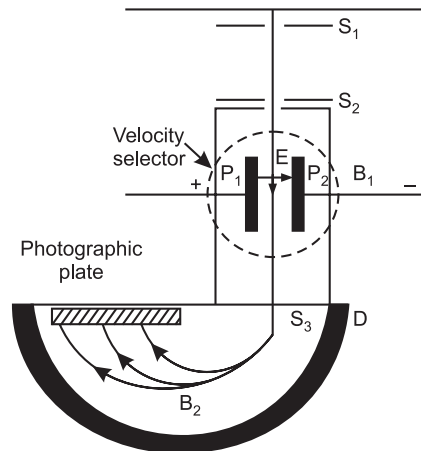


Fig. 4.16. Bainbridge's mass spectrograph

or
$$v = \frac{E}{B_1}$$

Only those ions having this velocity v alone pass through the entry slit S_3 and enter the evacuated chamber D . Thus, all ions entering the chamber have the same velocity.

Working. The positive ions which enter the chamber D are subjected to a strong magnetic field of intensity B_2 , perpendicular to the path. The force acting on each ion is B_2ev . As the force acts at right angles to the direction of motion of the ion, it moves in a circular path of radius R given by

$$B_2ev = \frac{Mv^2}{R}$$

or
$$R = \frac{Mv}{B_2e}$$

$$\therefore \frac{e}{M} = \frac{V}{B_2R} = \frac{E}{B_1B_2R} \quad \left[\because V = \frac{E}{B_1} \right]$$

As E , B_1 and B_2 are constant

$$\frac{e}{M} \propto \frac{1}{R}$$

After describing the semicircular path the ions strike a photographic plate.

Determination of mass. If e is the same for all ions, $M \propto R$.

Thus, we get a linear mass scale on the photograph plate. The ion of different mass will traverse paths of different radii and strike the photographic plates at different points, thereby giving a typical mass spectrum.

The method is very accurate as the mass scale is linear.

No change in kinetic energy of a charged particle in magnetic field. When a charged particle having a charge q , and velocity v (kinetic energy = $\frac{1}{2}mv^2$) enters a magnetic field of intensity B perpendicular to its path the force acting on the charge particles and the direction of the magnetic field. The charged particle will therefore move in a circular path of radius r , given by

$$Bqv = \frac{mv^2}{r}$$

or
$$r = \frac{mv}{Bq}$$

As the force due to the magnetic field acts in direction at right angles to the direction of motion, the force will only change the direction but will not be able to bring about a change in the magnitude of the velocity. *The kinetic energy of the particle will, therefore, remain unchanged.* The force due to magnetic field will only make the charged particle move along a circular path with the same velocity and, therefore, the same kinetic energy.

Example 39. In a Bainbridge mass spectograph radius of curvature of the path of Ne^{20} ions is 0.2 m. Calculate that for Ne^{22} ions. Assume single ionization of both the ions.

(Nagpur Uni 2007, 2008, 2005)

Solution. Given $m_1 \propto 20$, $m_2 \propto 22$, $R_1 = 0.26$ m

$$R_1 = km_1$$

$$R_2 = km_2$$

$$\therefore \frac{R_1}{R_2} = \frac{m_1}{m_2}$$

$$\text{or } R_2 = \frac{m_2}{m_1} \times R_1 = \frac{22 \times 0.26}{20} = 0.286 \text{ m}$$

Example 40. A positive ion beam moving in the x -direction enters a region in which there is an electric field $E_y = 3000$ volts/cm and a magnetic field $B_z = 300$ gauss. Deduce the speed of those ions which may pass undeflected through the region. What will happen to ions which are (i) faster, (ii) slower than these. [Agra (Supp),]

Solution. As we have seen in § 4.11, the velocity v_x of a charged particles, to remain undeflected in the crossed electric and magnetic fields E and B along the y -axis and z -axis respectively, must be given by the relation $v_x = E/B$. Or, if the Gaussian units be used (*i.e.*, E in stavolts/cm and B in gauss), we must have

$$v_x = \frac{cE}{B}$$

Here, $E = E_x = 3000$ volts/cm = 3000/300 or 10 stavolts/cm, $B = B_z = 3000$ gauss and $c = 3 \times 10^{10}$ cm/sec. So that,

$$v_x = 3 \times 10^{10} \times 10/300 = 10^2 \text{ cm/sec.}$$

Thus, the speed of the ions must be 10^2 cm/sec.

At this speed, obviously, the deflection of the $+ve$ ions in the downward direction in the plane of paper *i.e.* in the $-y$ direction, due to the magnetic field is just balanced by their deflection in the $+y$ direction due to the electric field in that direction.

(i) If, therefore, the speed v_x of the ion-beam be higher than 10^2 cm/sec, its deflection in the $-y$ direction due to the magnetic field will exceed that in the $+y$ direction due to the electric field and there will thus be a resultant deflection in the $-y$ direction.

(ii) If, on the other hand, the speed v_x of the ion-beam be lower than 10^2 cm/sec, the deflection in the $-y$ direction due to the magnetic field will be less than that in the $+y$ direction due to the electric field and there will, therefore, be a resultant deflection of the beam in the $+y$ direction.

4.14 WHEN ELECTRIC AND MAGNETIC FIELDS IN PARALLEL CONFIGURATION.

(*i.e.*, E AND B ARE IN PARALLEL) – THOMSON PARABOLAS

When electricity neutral atom losses one or more electrons, it becomes a positively charged atom and is called a positive ion.

Suppose a beam of such positive ions, each of mass m and charge q moving with a velocity v enters a region in which a uniform electric field of intensity E and a uniform magnetic field of induction B are present. Both the fields E and B are parallel to each other and are at right angle to the direction of motion of the beam (fig. 4.17).

In the region of the fields, an ion experience the electric force qE in the direction of E and at the same time also experiences a magnetic force qvB in the direction perpendicular to both v and B . In the figure qvB is directed out of the plane of the paper. Therefore, during its passage through the fields, the direction of the ion due to E in the y -direction and the deflection due to B is in the x -direction.

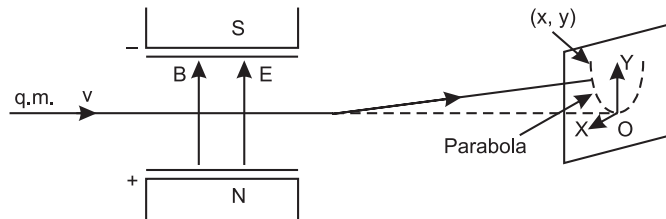


Fig. 4.17

The deflection on the screen are given by

$$S_E = \frac{qEL}{mv^2}$$

and

$$S_B = \frac{qBIL}{mv}$$

where l is the length of the horizontal path through the fields and L is the distance of the screen from the centre of the fields. If x and y are the coordinates of the point on the screen where the deflected beam meets the screen, then

$$y = \frac{qElL}{mv^2} \quad \dots(i)$$

$$x = \frac{qBIL}{mv} \quad \dots(ii)$$

To eliminate v , we square equation (ii) and then dividing by equation (i), we get

$$\begin{aligned} \frac{x^2}{y} &= \frac{q^2 B^2 (lL)^2}{m^2 v^2} \cdot \frac{mv^2}{qElL} \\ &= \left(\frac{lLB^2}{E} \right) \frac{q}{m} \end{aligned}$$

$$\text{Therefore,} \quad x^2 = \left(\frac{lLB^2}{E} \right) \frac{q}{m} y \quad \dots(iii)$$

This is the equation of the parabola with its vertex at the origin O of the axes on the screen. The parabola is symmetrical about the y -axis. Thus, for constant values of l , L , B and E positive ions having the same charge to mass ratio (q/m), but moving with different velocities, will lie on the parabola represented by equation. (iii).

If each ion has a positive charge equal in magnitude to the electronic charge e , then equation (iii) is written as

$$x^2 = \left(\frac{lLB^2}{E} \right) \frac{e}{m} y \quad \dots(iv)$$

or

$$x^2 = C \frac{e}{m} y$$

where

$$C = lLB^2/E$$

equation (iv) represents again a parabola, *i.e.* the path traced by a positive charge has parabolic nature. This principle was used by J.J. Thomson and he determined $\frac{e}{m}$ by using his very famous Thomson's parabola method.

4.15 DETERMINATION OF E/M FOR POSITIVE RAYS

It was first shown by *Goldstein* in the year 1886, that if a perforated cathode be used in a discharge tube in which the pressure is a little higher than in the usual ones for the production of cathode rays (so that there is an appreciable amount of residual gas in the tube), luminous streamers are seen travelling into the back region of the cathode (Fig. 4.18), away from the anode, *i.e.*, in the opposite direction of the cathode rays (which travel towards the anode), the colour of the streamers depending upon the nature of the residual gas inside the tube.

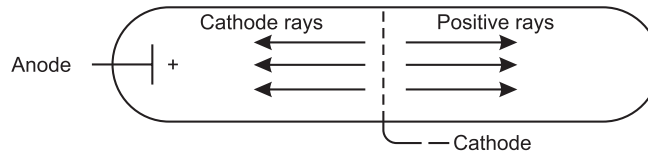


Fig. 4.18

These came to be known as *canal rays*, since they appeared to issue from the cylindrical chamels or cannels in the cathode, or *positive rays*, since they carried a positive charge on them, as evidence by their travelling away from the anode as also by the direction of their deflection by a magnetic or an electric field.

These rays, as we know, are just *stream of positive ions* produced because the fast moving electrons, rushing from the cathode towards the anode, *ionising the gas*, i.e., knocking out electrons from some of the gas atoms they collide with. These positive ions carry a charge equal in magnitude to e (the charge on an electron) or a multiple of e , according as one or more electron have been knocked out of an atom.

J.J Thomson was the first to have devised a method, in the year 1911, to determine their charge to mass ratio e/m by the application of a combination of a magnetic and an electric field, parallel to each other. This is explained as under.

Example 41. A stream of ions, of charge to mass ratio 1.44×10^{14} esu/gm, travelling horizontally in vacuum, encounters vertical electric and magnetic field in the same region, on passing through which it falls on the fluorescent screen, showing a displacement of 2 cm both horizontally and vertically. What will be the horizontal displacements for a stream of ions of twice the above charge to mass ratio if the vertical displacement for it be 9 cm.

Would it be possible to distinguish between streams of deuterons and α -particles in such an experiment?

Solution. Here, again, $\frac{x^2}{y} = \frac{q}{m}, \frac{B^2}{E} k$,

For the stream of ions for which $q/m = 1.44 \times 10^{14}$ esu/gm and $x = y = 2$ cm, we, therefore have,

$$\frac{(2)^2}{2} = 2 = (1.44 \times 10^{14}) \frac{B^2}{E} k, \quad \text{whence} \quad \frac{B^2 k}{E} = \frac{2}{1.44 \times 10^{14}}$$

$\therefore$ For ions for which $q/m = 2(1.44 \times 10^{14})$ esu/gm and for which $y = 9$ cm, we have

$$\frac{x^2}{9} = 2(1.44 \times 10^{14}) \frac{B^2}{E} k = 2(1.44 \times 10^{14}) \frac{2}{(1.44 \times 10^{14})} = 4$$

or $x^2 = 36$, whence, $x = 6$ cm.

Thus, the horizontal displacement of this stream of ions = 6 cm.

Since, the charge to mass ratio (q/m) is the same for both a deuteron and an α -particle, they will both suffer identical horizontal and vertical displacements in such an experiment and will thus be indistinguishable from the parabolas traced by them on the fluorescent screen.

Example 42. An electric field, 10^3 volt/meter and a magnetic field 3.2×10^{-3} weber/m² are parallel to each other along the z -direction. In the region of this combination of fields are introduced hydrogen ions of energy 200 electron volts along the axis of x . How long would it takes the hydrogen ions to hit the fluorescent screen placed at a distance of 0.8 metres from the origin perpendicular to their direction of motion and what will be the coordinates of the point where the screen is hit? (Mass of proton may be taken to be 1.6×10^{-27} kg and the charge on it equal to 1.6×10^{-19} coulombs).

Solution. Let v be the velocity of the hydrogen ions along the axis of x . Then $\frac{1}{2}mv^2 = 200$ eV

$$\text{whence,} \quad v = \sqrt{\frac{2 \times 200 \times 1.6 \times 10^{-19}}{1.6 \times 10^{-27}}} = 2 \times 10^5 \text{ m/sec.}$$

(Because 1 eV = 1.6×10^{-19} joules).

Since no force is acting along this axis, the velocity remains unaffected and hence *time taken to cover the distance $x = 0.8$ meter to the screen* $= 0.8/2 \times 10^5 = 4 \times 10^{-6}$ sec.

Now a moving charge constitutes an electric current and, therefore, the force acting on the hydrogen ions due to the magnetic field B is equal to qvB^* in a direction perpendicular to both B and v and, therefore, in the y -direction (in according with Fleming's left hand rule). So that, *acceleration of the ions in this direction* $= qvB/m$.

$\therefore$ Distance covered by the ions in this direction in time $t = 4 \times 10^{-6}$ sec is given by

$$y = \frac{1}{2} \frac{qvB}{m} t^2 = \frac{1}{2} \frac{1.6 \times 10^{-19} \times 2 \times 10^5 \times 3.2 \times 10^{-3}}{1.6 \times 10^{-27}} \times (4 \times 10^{-10})^2 = 0.512 \text{ m}$$

because $u = 0$

Similarly, distance covered by the ions in the z -direction in time $t = 4 \times 10^{-6}$ sec is given by

$$z = \frac{1}{2} \frac{qE}{m} t^2 = \frac{1}{2} \frac{1.6 \times 10^{-19} \times 10^3}{1.6 \times 10^{-27}} \times (4 \times 10^{-6})^2 = 0.8 \text{ m}$$

Thus, *coordinates of the point where the screen is hit* are 0.8, 0.512 and 0.8 m.

4.16 THOMSON'S PARABOLA METHOD

(Determination of q/m of positive ions)

J.J. Thomson, in 1911, developed a method of measuring the relative masses of different atoms by employing a parallel configuration of electric field E and magnetic field B simultaneously. The experimental arrangement is shown in fig. 4.19.

F is a glass flask in which gaseous discharge is produced. The anode A is in a side tube. The cathode C is made of a tube of soft iron faced with aluminium. This tube is fixed to the neck of the flask. Along the axial hole in this tube there is a copper tube of fine bore of about 0.1 mm diameter through which positive ions produced in the flask can pass. I is the inlet for an experimental gas. An electromagnet NS with soft iron pole pieces P and Q provides a strong magnetic field in the y -direction. The pole pieces P and Q are insulated from N and S by thin micasheet GG . The electric field is set up in the same direction by applying a p.d. between P and Q . Thus, parallel electric and magnetic fields are produced at right angles to the axis of the copper tube. A vertical photographic plate V is at the end of the conical vessel V .

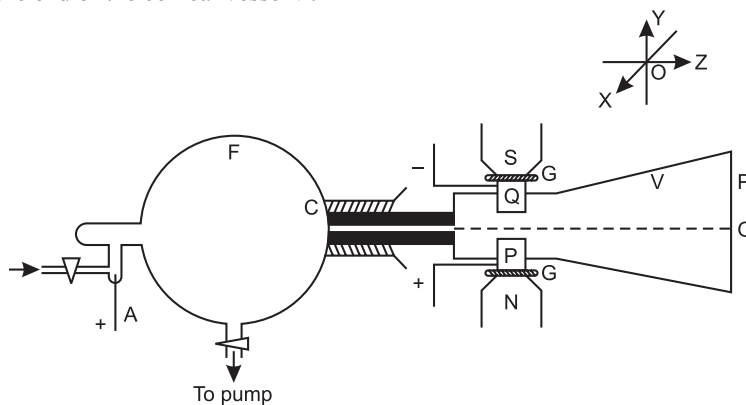


Fig. 4.19

Procedure for q/m of the isotopes of an element in gaseous state:

The flask F is evacuated of air to pressure of about 0.001 mm of Hg by means of a vacuum pump. The experimental gas at about the same pressure is now introduced into the flask through the inlet I . A high voltage of about 20000V is applied between the anode A and cathode C . Cathode

* Since the charge, the magnetic and electric field are all expressed here in RMKS (or SI) units, we need not divide the charge q by c as we do when using Gaussian units, i.e., the charge in esu and the magnetic field in emu or gauss.

rays or electrons then flow from C and A and positive ions produced by ionisation of the gas-atoms flow to C . After emerging from C they pass through the region of parallel electric and magnetic field. Then the deflected ions are incident on the photographic plate P where, they produce traces of parabolas on one side of the y -axis. The fields are then reversed to obtain the parabolic traces on the other side of the y -axis.

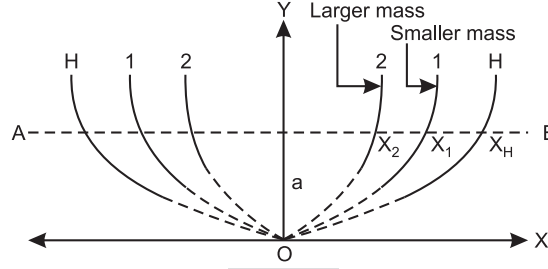


Fig. 4.20

After development of the plate parabolic traces are found on it (Fig. 4.20).

The outermost parabola H is produced by hydrogen ions because they have greatest value of q/m . (Hydrogen ions are usually present due to the residual air in the flask and the hydrogen parabola is taken as the standard). The parabolas (1) and (2) are produced by the ions of the isotopes of the gas introduced. To find q/m of the ions of the isotopes in terms of q/m of hydrogen ions a straight line $AB : y = a$; parallel to the x -axis is drawn and the x -coordinates x_H, x_1, x_2 of its point of intersection with the parabolic traces are found (Fig. 4.20). Then for hydrogen using equation (iv) of article 4.14 as

$$x^2 = C \frac{q}{m} y$$

we have

$$x_H^2 = C \frac{q}{m_H} a$$

$$\text{Therefore, } \frac{q/m_H}{x_H^2} = \frac{1}{Ca} \quad \dots(i)$$

Similarly, for the isotopes

$$\frac{q/m_1}{x_1^2} = \frac{1}{Ca} \quad \dots(ii)$$

$$\frac{q/m_2}{x_2^2} = \frac{1}{Ca} \quad \dots(iii)$$

where m_H, m_1 and m_2 are the corresponding atomic masses. From equations (i), (ii) and (iii), we have

$$\frac{q/m_1}{x_1^2} = \frac{q/m_2}{x_2^2} = \frac{q/m_H}{x_H^2}$$

$$\text{Hence, } \frac{q}{m_1} = \frac{x_1^2}{x_H^2} \left(\frac{q}{m_H} \right) \quad \dots(iv)$$

$$\text{and } \frac{q}{m_2} = \frac{x_2^2}{x_H^2} \left(\frac{q}{m_H} \right) \quad \dots(v)$$

Thus the value of q/m of the ions of the isotopes in terms of that of hydrogen ions are found.

Atomic Masses:

If the ionic charge q is the same in each case, then from Eq. (iv) and (v), we have

$$m_1 = \frac{x_H^2}{x_1^2} m_H \quad \dots(vi)$$

and
$$m_2 = \frac{x_H^2}{x_2^2} m_H \quad \dots(vii)$$

This is how, we determine the mass of the isotopes of the desired element.

Discovery of Isotopes

In 1912, Thomson attempted to determine the atomic weight of neon using parabola method. He discovered two parabolas, a strong one corresponding to a mass 20 and a weaker one corresponding to mass 22. Thus, for the first time, Thomson found two kinds of neon atoms, identical in chemical nature and having the same optical spectra but different in mass. In fact, it was shown latter by Thomson's co-worker F.W. Aston that there are three kinds of neon atoms. 90.5% have a mass value of 20, 9.2% have mass 22 and 0.3% have a mass 21. Thus, neon has three isotopes. Thomson method is an effective experimental tool which provides the firm experiment of the existence of stable isotopes of any desired element.

4.17 MAGNETIC LENS

An axially symmetric magnetic fields have a focusing property when an electron beam passes through them. Such fields are generated by using short solenoids. By encasing the coils in hollow iron shields the magnetic fields are concentrated and improved focusing action is obtain. Such Solenoids are, therefore, called as *thin magnetic lenses*.

As we know, that an electron travelling in a non-magnetic field describes a helical path of pitch.

$$p = \frac{2\pi m v}{eB}$$

The radius of the helical loop goes on progressively decreasing as the electron approaches to stronger and stronger magnetic field B . Thus, the radii of progressive solenoidal loops goes on shrinking into tighter loops and ultimately $r \rightarrow 0$ and the electron beam comes to focus at a point (Fig. 4.21). In other words, magnetic lens converges the electron beam. However, diverging action is impossible in magnetic lenses.

By adjusting the current through the solenoid and the initial accelerating voltage, the focal distance of magnetic lens can be monitored.

Uses: Magnetic lenses are widely used in electron microscopes and in instruments where such an action is desired.

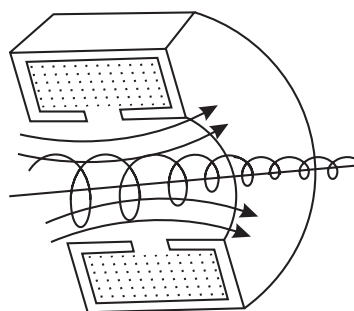


Fig. 4.21

EXERCISE

Long Answer Questions

- (a) Describe the motion of an electron subjected to a uniform electric field acting normal to the electron velocity.

(b) Show that an electron moving with uniform velocity follows a parabolic path in a transverse uniform electric field. *(Nagpur Uni. 2009)*
- Show that the velocity acquired by an electron in a uniform electrostatic field varies as the square root of potential difference through which it is accelerated. *(Nagpur Uni. 2001)*
- Prove that an electron moves along a parabolic path, when it enters in a uniform electric field applied perpendicular to its motion. What will happen if the electric field is not uniform?
- (a) Describe the path of electron of mass ' m ' and charge ' e ' moving with velocity v in (i) uniform magnetic field, and (ii) uniform electrostatic field. Both the fields are applied perpendicular to the direction of motion of the particle.

- (b) Does kinetic energy of a charged particles change when it enters a magnetic field.
5. (a) Draw the block diagram of a cathode ray oscillograph. Describe its construction and explain its working. Give uses of CRO.
(Pbi. U. 2004, 2002, 2000; G.N.D.U. 2004; Pat. U. 2004; Nag. U. 2002; Gharwal U. 2000)
- (b) Why is a sweep circuit necessary in a CRO? What will happen if fly back time is comparable to sweep time in a sawtooth voltage?
(Nag. U. 2000)
6. (a) Explain the necessity of time base circuit.
(b) Write the application and uses of C.R.O.
(Pbi U. 2003; MDU 2001; Pat. U. 2004; Garhwal U. 2000)
- (c) Describe the working and function of electron gun in a CRO. Which type of emission is employed is it.
(MDU 2003)
- (d) Describe the role of aquadag in cathod ray oscilloscope.
(P.U. 2001)
7. (a) Explain the principle of electrostatic focussing in a CRO.
(Nagpur Uni 2007)
- (b) What is the difference between CRT and CRO?
(Nagpur Uni 2007)
- (c) The willemite is used on the screen of a CRO. We get (i) blue colour (ii) green colour (iii) red colour (iv) white colour.
8. (a) Describe dual beam cathode ray oscilloscope.
(G.N.D.U. 2002, 2001)
- (b) Who Willemite is most suitable coating material for screen of C.R.T.
(H.P.U. 2003)
9. Explain carefully the principle of Linear accelerator
Deduce the expression for the energy of the particle and length of cylinders in terms of the constants of the apparatus.
(Nagpur Uni 2009, 2007, 2006, GNDU 2004, 2003; H.P.U. 2001; Guahati U. 2003; P.U. 2004; K.U. 2000)
10. (a) Describe the principle construction and working of a cyclotron. Derive expression for the maximum kinetic energy achieved by a particle of mass m in terms of the applied magnetic field and dee radius. Also state the relation in terms of the frequency of the applied electric field. Discuss its limitations.
(M.D.U. 2000, KU. 2002, 2004; Nag. Uni 2010, 2007, 2006; G.N.D.U. 2002, Pbi. U. 2003; Bang. U. 2001; Kerala U. 2001; Cal. U. 2003; P.U. 2002, 2000)
- (b) Can we accelerate neutrons by a cyclotron?
11. (a) Show that the maximum radius of curvature of the path of a particle inside of the dees of a cyclotron is proportional to the square root of the number of times it crosses the gap between the dees.
(b) What are the primary function of (i) electric field and (ii) magnetic field in a cyclotron?
12. (a) A Cyclotron is a called resonance device. Explain.
(b) Can a cyclotron be used to accelerate electrons? If not way?
(KU 2002; Pbi,U. 2001,2000; HPU 2002, 2001)
13. Give theory of Thomson's parabola method to determine the specific charge of positive ions. Give the necessary theory.
(Nagpur Uni 2007)
14. Describe the principle, construction and working of Bainbridge's mass spectrograph. How is the nuclear mass determined using the spectrograph?
(KU. 2002, 2000, MDU. 2002)
15. Draw figure showing the parabolic traces obtained on the photographic place in thomson's method due to two isotopes of atomic masses m_1 and m_2 where $m_1 < m_2$. Explain how the traces are analysed to determine the atomic masses.
16. Draw a neat diagram of Thomson's apparatus for determining charges to mass ratio of positive ions. Give a brief description of the apparatus. Describe Thomson's parabola method to determine e/m of the isotope of an element in gaseous state.
17. Draw a neat diagram of cyclotron and obtain resonance condition.
(Nagpur Uni 2002)
18. (a) Explain the principle and working of cyclotron. Show that the time spent by the charged particle in the Dec of the cyclotron is independent of its velocity.
(b) Mention the limitations on the energy achieved by a particles in the cyclotron.
(Nagpur Uni 2007)

Short Answer Questions

1. Show that a charged particle does not change its energy when fired in a magnetic field.
2. Magnetic field changes the velocity of a charged particle without changing its speed. Explain.
(Nagpur Uni. 2001)
3. Show that the radius of orbit of a charged particle moving at right angles to magnetic field is proportional to its momentum.
(Nagpur Uni. 2000)
4. Consider a particles of mass m and charge q moving with velocity v . The particle enters a region where a perpendicular uniform magnetic field B acts. Show that in the region the kinetic energy of the particle is proportional to the square of the radius of its orbit.
5. Explain the working of an electrostatic electron lens.
6. What is deflection sensitivity of a cathode ray tube? Obtain expression for (i) electrostatic deflection sensitivity and (ii) magnetic deflection sensitivity. Give their importance.
(Nagpur U. 2009; Bang. U. 2004)
7. Explain how a CRO may be used to measure voltage.
(Bang U. 2000)
8. Discuss how a cathode ray oscilloscope may be used to measure the phase difference between two a.c. signals of the same frequency and amplitude.
(Kolkata U. 2001)
9. Explain how a CRO may be used to measure frequency.
(Nag. U. 2002; Kolkata U. 2001; Bang. U. 2000)
10. What is the difference between linear accelerator and circular accelerator?
(P.U. 2001)
11. Which accelerator makes use of electromagnetic radiations for accelerating particles?
(G.N.D. U. 2000)
12. What is cyclotron? Show that the maximum kinetic energy gained by a particle in a cyclotron is
$$K.E. = 2\pi^2 m f^2 r_{\max}^2$$

symbols have their usual meaning.
13. What is velocity selector? Explain the function of velocity selector in Bainbridge mass spectrograph in determining the isotopes of various elements.
(Nagpur Uni. 2008)
14. Explain with neat diagram the working of magnetic lens. Where it is used.
15. An electric field E is applied between the vertical deflecting plates of a CRO. Obtain an expression for the displacement of an electron entering the field.
(Nagpur Uni 2007)
16. Obtain an expression for the deflection produced on the screen at a distance L from the centre of the plates.
(Nagpur Uni 2007)
17. Obtain expression for maximum K.E. of particles coming out of a cyclotron.
(Nagpur Uni 2007)
18. State any two limitations of a linear accelerator.
(Nagpur Uni 2008)
19. What is the path traced out by a beam of electrons when subjected to parallel E and B fields?
(Nagpur Uni 2007)
20. What is discharge tube? Describe its construction.
(Nagpur Uni 2008)
21. Explain the principle of 180° magnetic focusing.
(Agra U. 2005; Nagpur U. 2008, 2006)
22. If a charged particle passes through, mutually perpendicular electric and magnetic field, then prove that its path will be cycloid.
(Agra U 2004)
23. Explain the principle of velocity selector.
(Agra U 2004)
24. Show that in Brainbridge mass spectrograph, the radii of curvature of path of positive ions is proportional to masses of ions.
(Nagpur U. 2009)

I. Dynamics of uncharged particles

1. The initial position of a particle of mass 100 gm is $\vec{r}_0 = 5\hat{j} + 7\hat{k}$ cm and its initial velocity $\vec{u} = 30\hat{i} + 10\hat{k}$ cm/sec. If force $\vec{F} = 100\hat{i} + 75\hat{j}$ dynes acts upon it for 4 sec, obtain its final velocity and final position. Also calculate the work done by the force.
[Ans. $34\hat{i} + 3\hat{j} + 10\hat{k}$; $128\hat{i} + 11\hat{j} + 47\hat{k}$; 13250 ergs]

2. A particle is projected upward in a direction inclined at 60° to the horizontal. Show that its velocity when at its greatest height is half its initial velocity. (Neglect the resistance of the air)
3. Find the greatest distance that a stone can be thrown inside a horizontal tunnel 10 ft high with a velocity of projection 80 ft/sec. Find also the corresponding time of flight. [Ans. 120 ft; 1.58 sec.]
4. A particles P is projected with initial velocity $5\hat{i} + 6\hat{j} - 7\hat{k}$ m/sec. and it experience a uniforms acceleration $6\hat{j}$ m/sec². Another particles Q is projected from the same position 20 sec. later with initial velocity $2\hat{i} - 3\hat{j} + 4\hat{k}$ m/sec. and it experiences a uniform acceleration $-3\hat{j}$ m/sec². Deduce the position of Q relative to P , 50 sec. after P is projected. (Agra supp)

[Ans. $19\hat{i} - 127.5\hat{j} + 47\hat{k}$]

II. Dynamics of charged particles

1. Charged particles in a uniform electric field

5. Show that when an electric field is applied at right angles to the direction of motion of a charged particle, it describes a path similar to that described by a particles thrown horizontally in the Earth's gravitational field.
6. A potential difference of 650 volts retards the motion of an electron proceeding normally from one electrode to another, parallel to it. If the initial energy possessed by the electron by 10^9 ergs, (i) will it be able to reach the second electrode? (ii) What should be the value of the retarding potential so that the electron may just reach the second electrode? [Ans. (i) No.; (ii) 625 volts]
7. Calculate (i) the force on a proton in an electric field of 100 statvolts/cm, (ii) the intensity of the electric field that can just support the weight of an electron. [Ans. (i) 4.80×10^{-8} dynes; (ii) 1.86×10^{-15} statvolts/cm]
8. Calculate the intensity of the electric field which will produce in an alpha particles an acceleration equal to that of g (i.e. 980 cm/sec²). (The mass of an alpha particle is four times that of a proton and the charge on it twice that on a proton). [Ans. 2.05×10^{-2} volts/cm]
9. A proton is accelerated through a potential difference V and then allowed to cross a field-free evacuated space. How long would it takes to cover the distance between two given points in this space, distance r apart. [Ans. $r / \sqrt{2eV/m}$]
10. In an electric field $E = 10^{-2}\hat{i}$ statvolts/cm, the velocity of an electron is $v_0 = 10^5\hat{i}$ cm/sec and its position vectors at $\vec{r}_0 = 10^2 - \hat{j}$ cm at $t = 0$. Obtain its velocity and position vectors at $t = 10^{-9}$ sec. [Ans. $\vec{v} = 5.4 \times 10^7\hat{i}$ cm/sec; $\vec{r} = 10^2\hat{j} - 0.25\hat{i}$ cm]
11. An electron beam is accelerated through apd. of 1000 volts, and then passed through a pair of deflecting plates 2 cm long and 0.5 cm apart. What should be the p.d. between the plates in order that the electron beam may be deflected through 5° ? [Ans. 4.7×10^2 volts]
12. The accelerating voltage in a cathode ray oscilloscope is 2400 volts, the effective length of the deflecting plates is 3 cm, their distance apart 0.5 cm and the distance of the screen from their mid-point 30 cm. If the p.d. across the deflecting plates be 20 volts, obtain, (i) the velocity of an electron as it enters the deflecting field (ii) the time it takes to traverse this field, (iii) the angle of deflection due to the field and its displacement on the screen. (e/m for an electron = 5.2×10^{17} esu/ gm. [Ans. (i) 2.88×10^9 cm/se, (ii) 1.04×10^{-9} sec, (iii) $1^\circ 26'$, (iv) 0.75 cm]
13. What is the energy in Joules of an electron accelerated through a potential of 1000 kV. (Nagpur U. 2009)

[Ans. 1.6×10^{-13} J]

2. Charged particle in an alternating electric field - Linear accelerator

14. An alternating potential difference $V = 2 \sin(2\pi \times 10^8) t$ is applied to two plane and parallel plates A and B arranged 0.5 cm apart along the axis of x . An electron is emitted from plate A with zero velocity just when plate B is acquiring positive potential. What will be the maximum speed acquiring by the electron? [Ans. 6.7×10^8 cm/sec.]
- [Hint. Here, obviously $V_0 = 2$, statvolts and $\omega = 2\pi \times 10^8$. So that, peak value of the electric field, i.e.,

$E_0 = V_0/0.5 = 2/0.5 = 4$ stat volt/cm. Therefore, $m d^2x/dt^2 = -eE_0 \sin \omega t$ $d^2x/dt^2 = -(e/m)E_0 \sin \omega t$. Hence, $v = dx/dt = eE_0(1 - \cos \omega t)/m\omega$. The value of v will be the maximum when $\cos \omega t = -1$. Thus, $v_{\max} = 2eE_0/m\omega = 2 \times 4.8 \times 10^{-10} \times 4/9.1 \times 10^{-28} \times 2\pi \times 10^8 = 6.7 \times 10^9$ cms/sec.]

15. An alternating electric field of peak value 3 volts/cm and frequency 1 MHz. (i.e., 10^8 c.p.s.) is established between two plane and parallel plates. Show that the electrons starting at different times with zero initial velocities in this field will acquire maximum velocities of $\pm 8.4 \times 10^8$ cm/sec.

[Hint. Proceed exactly as in problem 13 above and obtain $v_{\max}$]

16. The energy of a C^{12} ion when it emergence from the last drift tube of a linear accelerator is 120 MeV. What would be the energy of a O^{16} ion from the accelerator? (Amaravati U. 2008) [Ans. 160 MeV]

3. Charged particle in a magnetic field - Cyclotron

17. Show that the path of a charged particle in a uniform magnetic field is, in general, a helix. Under what condition is this path reduced to a circle?
18. A charged particles of mass M and charge q is moving in a constant magnetic field B . Show that (i) the component of the velocity of the particle along the axis of the magnetic field is constant, (ii) the magnetic field does not change the kinetic energy of the particle, (iii) for a uniform circular motion in a plane perpendicular to the axis of the magnetic field, the angular velocity of the particle, ω , is given by $\omega = qB/Mc$, where c is the speed of light in free space. (Agra)
19. Calculate the gyroradius and gyrofrequency of an electron moving in a uniform magnetic field of 10^2 gauss with a velocity 3×10^9 cm/sec in a direction perpendicular to the field.

[Ans. 0.17 cm, $8.8 \times 10^9 \approx 9 \times 10^9$ hertz or c.p.s.]

20. An electron describes a helix of radius 6.0 cm and pitch 2.0 cm in a magnetic field of intensity 30 gauss. Obtain the components of its velocity along and perpendicular to the magnetic field.

[Ans. 1.68×10^8 cm/sec, 3.39×10^9 cm/sec.]

21. Particles of charge q and mass m are projected with the velocity $a\hat{i} + b\hat{j}$ in a magnetic field $B\hat{j}$ of units.

Show that they follow a helical path. Deduce an expression for the pitch of the helix. What is the significance of the fact that this is independent of the component of velocity? (Agra)

[Ans. Pitch of helix = $\frac{2\pi mc}{qB} b$]

22. An electron with a velocity 10^7 m/sec enters a region of uniform magnetic flux density of 0.10 weber metre⁻², the angle between the direction of the field and the initial path of the electron being 25° . Find the pitch of its helical path. (e/m for an electron = 1.8×10^{11} coulomb/kg) [Ans. 0.32 cm.]
23. An electron, moving with a velocity equal to $4/5$ th of that of light in free space, enters a magnetic field of 300 gauss in a direction perpendicular to that of the field. Calculate the radius of the circular path described by the electron inside the field on non-relativistic as well as relativistic considerations.

[Ans. 4.55 cm; 7.58 cm,]

[Hint. Relativistically, the mass of the electron is given by $m = m_0/\sqrt{1 - v^2/c^2}$ where $m_0 = 9.1 \times 10^{-28}$ gm]

24. An alpha particle moves with velocity 10^8 cm/sec at an angle of 30° with a magnetic field of 10^2 gauss. Obtain the values of (i) the Lorentz force acting on it, (ii) the radius and the pitch of the helix described by it and (iii) its frequency of rotation. (Take mass of α -particle = $4 \times 1.6 \times 10^{-24}$ gm and the charge on it = $2 \times 4.8 \times 10^{-10}$ esu).

[Ans. (i) 16×10^{-16} dynes, (ii) 10 cm, 108.7 cm, (iii) $7.96 \times 10^5 \approx 8 \times 10^5$ c.p.s.]

[Hint. Lorentz force. $\vec{F} = \frac{q}{c}(\vec{v} \times \vec{B})$ So that, its magnitude $\frac{q}{c}vB \cos \theta$]

25. Explain clearly the principle underlying the working of a cyclotron. What are its drawbacks and how have they been remedied. Which is preferable, a lower or a higher radiofrequency voltage, to be applied to the dees?

26. What will be the energy of (i) a proton, (ii) an electron as it emerges from a cyclotron after making 100 revolution inside it, if the p.d. across the dees be 10^2 volts? Which of these will emerge with a higher velocity?

[Ans. (i) 2×10^5 eV, (ii) 2×10^5 eV; The electron]

27. Calculate, ignoring relativistic effects, the amount of work done on an electron in a cyclotron if, starting from rest, it finally emerges with a velocity of 4×10^9 cm/sec.
(Take mass of electron = 9×10^{-28} gm) [Ans. 4.5×10^2 eV]
28. (a) Show that the cyclotron frequency of a given kind of ions is independent of the energy. State the limiting condition, if any. (Agra)
(b) Explain the principle of magnetic focussing. (Agra)
29. The radius of the dees of a cyclotron used accelerate protons is 50 cm and they are connected to a radiofrequency supply of 10 megacycles per second at maximum value 6000 volts. Calculate (i) the strength of the magnetic field applied, (ii) the velocity and the energy of the emerging protons, (iii) the minimum number of rotations made by them and (iv) the time spent by them inside the dees.
[Ans. (i) 6.557×10^3 gauss, (ii) 3.142×10^9 cm/sec, 5.15×10^8 eV, (iii) 430, (iv) 4.3×10^{-5} sec.]
30. A particle of charged q esu and mass m gm enters a magnetic field of $2000 \hat{i}$ gauss with a velocity $(2\hat{i} + 3\hat{j}) 10^8$ cm/sec. Deduce (i) radius of the helical path, (ii) number of rotations made by the particle while travelling forward 10 cm along the axis of the helix. (Agra (supp))
[Ans. (i) $4.5 \times 10^{15} \left(\frac{m}{q}\right)$ cm, (ii) $\left(\frac{q}{m}\right)(5.3 \times 10^{-16})$]
31. The pole pieces of a cyclotron are 1.2 m in diameter and provide a magnetic field of 1.6 Wb m^{-2} . What will be the energy of the alphas, deuteron and protons in such a machine? What should be the range of oscillator frequency to cover the acceleration of the above particles? Mass of proton = 1.67×10^{-27} kg and charge 1.6×10^{-19} C. (Calicut Uni. 2003) [Ans. 12.20 to 24.40 MHz]
32. In a cyclotron the magnetic flux density is 1.5 m weber/m^2 . A proton of mass 1.67×10^{-19} C is accelerated. The radius at which the proton leaves the system is 0.5 m. Find its maximum K.E. and the time of flight through one dee. (Nagpur U 2010, 2004) [Ans. 2.18×10^{-8} S]
33. A cyclotron oscillator frequency is 10 MHz. What would be the operating magnetic field for accelerating proton? If the radius of its dees is 60 cm, what is the kinetic energy of proton beam produced by the accelerator? Given: $e = 1.6 \times 10^{-19}$ C. and $m = 1.67 \times 10^{-27}$ kg.
(Nagpur U 2008) [Ans. 0.656 Wb/m^2 , 7.4 MeV]

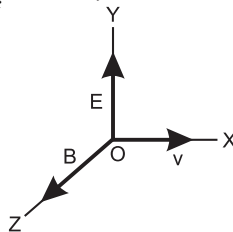
III. Magnetic focusing

34. Calculate the width of the focal line obtained by 180° -magnetic focussing if the radius of curvature of the earth of the ion beam by 50 cm and the semi angle of the cone of the beam, as it enters the magnetic field, be 5° . [Ans. 0.38.07 cm]
35. The width of the focal line of an ion beam obtained by 180° -magnetic focussing is 0.3656 cm. If the angle of the cone of the ion beam entering the magnetic field be 4° , calculate the radius of curvature of the path taken by the beam in the magnetic field. [Ans. 30 cm]

IV. Charged particle in crossed electric and magnetic fields

36. A charged particle moves in the x -direction through a region in which there is an electric field E_y and a perpendicular magnetic field B_z . What is the condition necessary to ensure that the net force on the particle will be zero? Show the v , E and B vectors on a diagram. What is the condition of v_x if $E_y = 10$ statvolts and $B_z = 300$ gauss?

[Ans. $F_{\text{elec}} + F_{\text{mag}} = 0$ i.e., $q\vec{E} + \frac{q}{c}\vec{v} \times \vec{B} = 0$, which leads to $v_x = cE_y/B_z$; $v_x = 1 \times 10^9$ cm/sec]



Vector diagram

37. A stream of charged particles, moving with a velocity of 6×10^9 cm/sec passes between two parallel plates 1 cm apart and having a p.d. of 27 volts across them. What should be the magnitude and direction of the magnetic field that should be applied in the region of the electric field in order that the stream of particles may remain undeflected?
[Ans. 0.45 gauss, perpendicular to both E and v]
38. What should be the velocity of a stream of (i) protons, (ii) deuterons in order that they may remain undeflected when passing through crossed electric and magnetic fields, $E = 900$ volts/cm and $B = 600$ gauss?
[Ans. (i) 1.5×10^8 cm/sec; (ii) 1.5×10^8 cm/sec.]
39. Explain the principle of a velocity selector for charged particles, using crossed electric and magnetic fields. (Agra)
40. If the electric field in the velocity selector at a Bainbridge mass spectrograph is 10^4 V/m and magnetic flux density is 0.2 Wb/m², find the speed of an ion which will go undeflected through the crossed-field.
(Nagpur 2010, 2004) [Ans. 5×10^4 m/s]

V. Charged particle in parallel electric and magnetic fields

41. A beam containing ions of various charges q , masses m and speeds v (along the x -axis) enters a region of uniform electric field E and magnetic field B (both along the y -axis). The ions are received on a screen after travelling a distance L_x . Show that (i) all ions of a given q/m value fall along a parabola irrespective of their speeds, (ii) all ions of a given speed fall along a straight line irrespective of their q/m value. (Delhi U. 2003)
42. Ions with charge to mass ratio 2.9×10^{14} esu/gm travelling horizontally pass through a region of combined and parallel electric and magnetic fields along the vertical so as to hit a fluorescent screen on which they show horizontal and vertical displacements of 4 cm each. What will be the charge to mass ratio for charged particles which show horizontal and vertical displacements of 1 cm and 8 cm respectively? [Ans. 9.1×10^{12} esu/gm]
43. A beam of protons having a velocity of 2×10^7 cm/sec along the axis of x is introduced into a combination of parallel electric and magnetic fields of 100.2 volts/cm and 3.34×10^2 gauss respectively along the z -direction. Obtain the coordinates of the points where the beam will hit a photographic plate placed perpendicular to the direction of the beam at a distance of 6 cm from the region.
[Ans. 6, 2.88, 4.32 cm]
44. In Thomson's parabola method, parabolic traces were obtained on the photographic plate for Neon and Hydrogen ions. X -coordinates of the points of intersection of a straight line, parallel to the X -axis, with the parabolic traces were 4.77, 5, 22.4 mm. Find the atomic masses of the isotopes of neon.
(Atomic mass of $H = 1$ amu) (Nagpur U. 2007)
45. Two positive ray parabolas are obtained in a particular experiment for H_2 and a gas X . At a particular y value the x values are in the ratio 4 : 1. Determine the isotopic mass of element X and identify it. Given: $m_H = 1$ amu. (Nagpur U. 2009)
[Ans. 16 amu]

5

Chapter

CONSERVATION LAWS

PART I: LAW OF CONSERVATION OF ENERGY

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INTRODUCTION

In Physics we often deal with conservation principles this is because most physical properties abide by conservation laws. In general, a conservation law states that a ***particular measurable property of an isolated physical system does not change as the system evolves***. For example, the conservation of energy follows from the time-invariance of physical systems. At one time, scientists thought that the law of conservation of mass and the law of conservation of energy were two distinct laws. In the early part of the twentieth century, however, German-born American physicist Albert Einstein (1879-1955) demonstrated that matter and energy are two forms of the same thing. He showed that matter can change into energy and that energy can change into matter. In some instances, a tiny bit of matter can be created or destroyed in a change. But the total amount of matter PLUS energy before and after a change still remains constant. This statement is now accepted as the law of conservation of mass and energy.

Examples of the law of conservation of mass and energy are common in every day life. The manufacturer of an electric heater can tell consumers how much heat will be produced by a given model of heater. The amount of heat produced is determined by the amount of electrical current ($H = I^2R$) that goes into the heater. Similarly that amount of gasoline that can be formed in the breakdown of petroleum can be calculated by the amount of petroleum used in the process. And the amount of nuclear energy produced by a nuclear power plant can be calculated by the amount of Uranium-235 used in the plant. Thus, the energy is conserved. It may take various forms, such as electrical energy, heat, light, magnetism or kinetic energy (due to its motion), sound, etc., but the relationship is always the same. The amount of energy used to initiate a change is the same as the amount of energy detected at the end of the change. In other words, energy can not be created or destroyed in a physical or chemical change, only changes the form of its manifestation. This statement summarizes the law of conservation of energy.

A partial listing of conservation laws that are said to be exact laws, or more precisely have never been shown to be violated.

- Conservation of mass-energy
- Conservation of linear momentum
- Conservation of angular momentum
- Conservation of electric charge
- Conservation of parity*

The most important consequence of the law of conservation of energy is that *perpetual motion* machines can only work perpetually if they deliver no energy to their surroundings.

5.1 SIGNIFICANCE OF CONSERVATION LAWS

In Physics, we come across various conservation laws, as mentioned above, and even though all of them may not be equally exact or accurate, they nevertheless prove helpful in many ways. Thus, for instance,

- (i) without going into details of the trajectories or the forces involved in any particular case,

* Conservation laws are now widely regarded as some of the most fundamental laws in all of nature. It was a great Shock, therefore, when two American physicists, Val Lodgson Fitch (1923) and James Watson Cronin (1921) discovered mid-1960s that certain subatomic particles known as K-mesons appear to violate a conservation law. That law is known as the conservation of parity; which defines the basic symmetry of nature: that an object and its mirror image will behave the same way. Scientists have not yet fully explained this unexpected experimental result.

they give us a broad and generalised picture of the significant facts that emerge in consequence of the equations of motion:

(ii) even in cases where the nature of the forces involved is not clearly known, the conservation laws have been successfully invoked, particularly in the realm of what are called fundamental or elementary particles and have, indeed, helped predict the existence of quite a few more of them *viz.*, conservation of parity.

(iii) they forewarn us of the impossibility of the occurrence of certain types of phenomena (like, for example, a perpetual motion machine) and thus prevent wastage of time and effort that we might otherwise feel tempted to devote in tackling such problems;

(iv) they seem to have an intimate relationship with the concept of invariance and we may often use them, with success, in exploring and unravelling new and hitherto not well understood phenomena. Thus, as we have seen in chapter II, the principle of conservation of linear momentum has been obtained more or less as a direct consequence of Galilean invariance.

With these few examples regarding the importance of conservation laws, let us now proceed to deal with three important ones of them *viz.*, those of energy, linear and angular momentum, in proper detail.

I – LAW OF CONSERVATION OF ENERGY

5.2. CONCEPTS OF WORK, POWER AND ENERGY

Before dealing with the law of conservation of energy itself, it is better to be quite clear about the concepts of work, power and energy involved in the law.

1. Work. Suppose a particle of mass m is acted upon by a force $\vec{F}$, (Fig. 5.1), under the influence of which it is set into motion, not necessarily in the direction of the force, its actual path depending upon its initial velocity and the magnitude and direction of the force applied. Let it be along a curve C , joining the points A and B . It may be noted that the magnitude and direction of the force may vary from point to point along the curve.

To determine the work done in displacing the particle from A to B , let P and P' be two neighbouring points such that their position vectors are $\vec{r}$ and $\vec{r} + \delta\vec{r}$ and, therefore,

$\vec{PP'} = \delta\vec{r}$ (or $\delta\vec{s}$ if we prefer to work in terms of the length of curve C).

Then, for a displacement of the particle (or the point of application of the force) from P to P' , we have

$$\text{Work done by the force, } dW = \vec{F} \cdot d\vec{s}$$

$\therefore$ Total work done by the force in displacing the particle along the whole length of the curve from A to B , i.e. W , is given by the relation

$$W = \int_A^B \vec{F} \cdot d\vec{r} = \int_A^B \vec{F} \cdot d\vec{s} = \int_A^B F \cos \theta ds, \quad \dots(i)$$

where θ is the angle that the direction of the force makes with the tangent to the curve at the point P and, therefore, $F \cos \theta$, the tangential component of the force along the curve at the point P .

Such an integral evaluated along the path of the particle is called the **line integral of the force** $\vec{F}$.

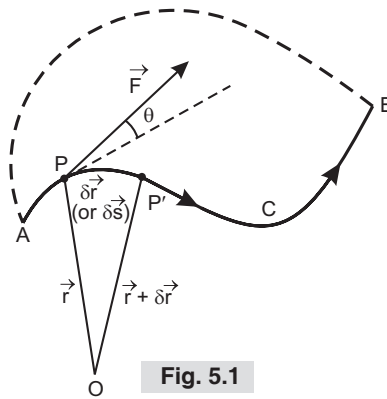


Fig. 5.1

We may, therefore, define work done by the force on the particles as the *line integral of the tangential component of the force evaluated along the actual path of the particle*.

If the Cartesian components of $\vec{F}$ and $d\vec{r}$ (or $d\vec{s}$) be respectively $F_x\hat{i}, F_y\hat{j}, F_z\hat{k}$ and $d_x\hat{i}, d_y\hat{j}$ and $d_z\hat{k}$, we have

$$\begin{aligned} W &= \int_A^B (F_x\hat{i} + F_y\hat{j} + F_z\hat{k})(d_x\hat{i} + d_y\hat{j} + d_z\hat{k}) \\ &= \int_A^B F_x d_x + \int_A^B F_y d_y + \int_A^B F_z d_z. \end{aligned} \quad \dots(ii)$$

Further, if there be a number of forces, $\vec{F}_1, \vec{F}_2$ etc., acting simultaneously on the particle, such that their resultant $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$, we have work done on the particle given by

$$\begin{aligned} W &= \int_A^B \vec{F} \cdot d\vec{r} = \int_A^B (\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots) d\vec{r} \\ &= \int_A^B \vec{F}_1 \cdot d\vec{r} + \int_A^B \vec{F}_2 \cdot d\vec{r} + \dots, \end{aligned} \quad \dots(iii)$$

i.e., equal to the sum of the work done by each individual force.

In case force $\vec{F}$ be a *constant or a nonvariable* one and the displacement of the particle takes place *along a straight line*, we have

$$W = \int_A^B \vec{F} \cdot d\vec{r} \left(= \int_A^B \vec{F} \cdot d\vec{s} \right) = \vec{F} \cdot \vec{r} (= \vec{F} \cdot \vec{s}) = F \cos \theta(r) = F \cos \theta(s), \quad \dots(iv)$$

where $\vec{r}$ (or $\vec{s}$) is the displacement of the particle along the straight line.

This is in accordance with the elementary definition of work with which we are already familiar from our junior classes, viz., that work done is given by

$$W = \text{Component of the force along the displacement} \times \text{The displacement.}$$

Units of work (i) In the *C.G.S. system*, the unit of work is the **erg**, which is the work done by a force of 1 *dyne* acting through a distance of 1 *cm*, i.e., 1 *erg* = 1 *dyne* $\times$ 1 *cm*.

(ii) In the *FPS system*, the unit of work is the **foot-poundal**, which is the work done by a force of 1 *poundal* acting through a distance of 1 *ft*, i.e., 1 *ft-poundal* = 1 *poundal* $\times$ 1 *foot*.

(iii) In the *RMKS (or SI) system*, the unit of work is the **joule**, which is the work done by a force of 1 *newton* acting through a distance of 1 *meter*, i.e., 1 *joule* = 1 *newton* $\times$ 1 *metre*.

(Since 1 *newton* = 10^5 *dynes* and 1 *metre* = 10^2 *cm*, 1 *joule* = $10^5 \times 10^2 = 10^7$ *ergs*).

These are called the *absolute units*. The corresponding *gravitational units* are *g times the absolute units*. Thus, the gravitational units of work in the three systems are (i) the *gm wt-cm* = 981 *ergs*, (ii) the *foot-pound* = 32 *foot-poundals* and (iii) the *kilogramme-meter* = 9.81 *joules*.

2. Power of Activity. It is defined as the *rate of doing work* and is denoted by the latter *P* or *A*.

Thus, in Fig. 5.1, since $\vec{F} \cdot d\vec{r}$ is the work done by the force in moving the particle from *P* to *P'* in time *dt*, the rate of working of the force or its *power on activity* $P = \vec{F} \cdot d\vec{r} / dt = \vec{F} \cdot \vec{V}$... (v) where $\vec{V}$ is the *instantaneous value* of the velocity acquired by the particle.

Relation (v) thus gives the *instantaneous value of the power of the force*. The *average value of power* is obtained by dividing the total work done (*W*) by the total time taken (*t*), i.e., *average power* = *W/t*.

It will be easily seen from the above that the work done is given by

$$W = \int_A^B P \cdot dt.$$

Units of power. (i) In the *C.G.S. system*, the unit of power is 1 *erg/sec*. (ii) In the *F.P.S. system*, it is 1 *foot-poundal/sec*. The unit commonly used in this system is the *horse power (H.P.)*, equal to 550 *ft-pounds/sec* or 550×60 or 33,000 *ft-pounds/min*. (iii) In the *RMKS* or (*SI*) system, the unit of power is 1 *joule/sec*, called the **watt**, usually used to measure *electrical power*.

3. Energy. We shall deal with it later in § 5.4.

5.3 CONSERVATIVE FORCES

In the type of cases discussed under § 5.2 above, the force $\vec{F}$ generally varies from point to point and it may, therefore, appear that the work done by it (W) in moving the particle from point A to point B will depend upon the path taken between A and B . In actual fact it may or may not, depending upon the type of force we are dealing with. For, forces may, in general be divided into two broad categories; **conservative** and **non-conservative**.

The work done by a conservative force in displacing a particle from one point to another depends only on the position of the two given points and is quite independent of the actual path taken between them. On the other hand, the work done by a non-conservative force does depend upon the actual path taken.

To the conservative class of forces belong most of what are called **central forces**, i.e., forces which are directed towards, or away from, a fixed centre, as in the case of gravitational and electrostatic forces respectively. Such forces are *position-dependent* forces, so to speak, since they depend only upon the instantaneous position of the particle with respect to the fixed centre and on nothing else.

Forces which are *velocity-dependent* i.e., whose value depends upon the magnitude and direction of velocity, like frictional and viscous forces, are in general, *non-conservative*, though, quite a few among them too may turn out to be conservative. Thus, for example, *Lorentz force* $\frac{q}{c}(\vec{v} \times \vec{B})$, although velocity-dependent, is conservative because, acting as it does, in a direction perpendicular to both the magnetic field B and the velocity of the particle v , the work done by it, irrespective of the path taken, is always zero and thus independent of the path.

A region of space in which a particle experiences a conservative force at every point is referred to as a *conservative-force field* or a *conservative field of force*. In case the conservative force in the field happens to be a central force, it may be specified as a *central-force field*.

That a central force or a position-dependant force is a conservative force may be easily shown as follows:

Taking the origin as the fixed centre, a central force may be expressed as $\vec{F} = F(r)\hat{r}$ where $F(r)$ is a function of r only and $\hat{r}$, the unit vector along $\vec{r}$.

Clearly, work done by such a force in displacing a particle from a point A to a point B is given by

$$W_{AB} = \int_A^B \vec{F} \cdot d\vec{r} = \int_A^B F(r)\hat{r} \cdot d\vec{r}$$

As we know, $r^2 = r^2$, so that $2\vec{r} \cdot d\vec{r} = 2rdr$ and $\therefore d\vec{r} = \hat{r}dr$.

Hence
$$W_{AB} = \int_A^B F(r)dr$$

an expression dependent only on the functional form of $F(r)$ and the two end-points A and B .

The force in question, viz, $\vec{F}$ is, therefore, a *conservative force*.

It is clear from the above that the work done by the force F in moving the particle from B to A along any path will be given by

$$W_{BA} = \int_B^A \vec{F} \cdot d\vec{r} = - \int_A^B \vec{F} \cdot d\vec{r} = -W_{AB}.$$

It follows, therefore, that the total work done in displacing a particle from A to B and back to A along whatever path (as shown dotted in Fig. 5.1) is $W_{BA} - W_{AB} = 0$, indicating that the *work done by a conservative force along a closed path, i.e., along a closed curve, is zero*. This is denoted symbolically as $\oint \vec{F} \cdot d\vec{r} = 0$, (the symbol $\oint$ standing for the *integral* over a closed path).

This is a *characteristic property of a conservative force*. In fact, we may perhaps best define a conservative force as one by which the work done in a round trip is zero and, therefore, a non-conservative force as one by which the work done in a round trip is not zero. In the first case, when the particle (or the body) comes back to its starting point, it has the same kinetic energy with which it started, so that its ability to do work is *conserved*. In the second case, the particle, as it comes back to its starting point, has less or more kinetic energy than the one with which it started, so that its capacity to do work is *not conserved* (i.e., it is either less or more than before).

5.4 ENERGY

The energy of a particle is its *capacity to do work* and is measured by the amount of work it is capable of doing in virtue of (i) its motion or (ii) position or condition. The former is called the *kinetic energy* and the latter, the *potential energy* of the particle, represented by the symbols K (or now, more usually, T) and U (or V) respectively.

Since energy is really *work*, its units are the same as those of work, viz., the **erg**, the **foot-poundal** (or the **foot-pound**) and the **joule** in the three systems respectively.

We shall now proceed to consider the two types of energy separately and in necessary detail.

(i) **Kinetic energy–Work-energy principle.** Since kinetic energy (T) of a particle is due to its motion, let us obtain a proper expression for it for a particle moving with velocity v .

Let a force $\vec{F}$ be applied to the particle in a direction opposite to that of its motion. Then, clearly, *work done by the particle against the applied force, i.e.,* $W = - \int \vec{F} \cdot d\vec{r}$. Since, in accordance with Newton's second law of motion,

$$\vec{F} = m d\vec{v}/dt,$$

Where m is mass of the particle, we have

$$W = -m \int \frac{d\vec{v}}{dt} \cdot d\vec{r} = -m \int \frac{dr}{dt} dv = -m \int_v^0 f \cdot dv = \frac{1}{2} mv^2.$$

Since energy of the particle is its capacity to do work, we have

$$\text{Kinetic energy of the particle, } T = \frac{1}{2} mv^2,$$

It may as well be noted that $dT/dt = d\left(\frac{1}{2} mv^2\right)/dt = mv(dv/dt) = F \cdot v = P$, i.e., the *rate of change of energy is equal to the power or activity of the force*.

Again, invoking Newton's second law of motion, we have

$$mdv/dt = F.$$

Dot multiplying both sides by v or dr/dt and integrating with respect to t between the limits A and B (Fig. 5.1), we have

$$\int_A^B m \frac{d\vec{v}}{dt} \cdot \vec{v} dt = \int_A^B \vec{F} \cdot \frac{d\vec{r}}{dt} dt.$$

Or,
$$\left[\frac{1}{2} m \vec{v} \cdot \vec{v} \right]_A^B = \int_A^B \vec{F} \cdot d\vec{r}.$$

Since $\vec{v} \cdot \vec{v} = v^2$ and v_A and v_B are the values of the initial and final velocities of the particle at points A and B respectively, we have

$$\frac{1}{2} m v_B^2 - \frac{1}{2} m v_A^2 = \int_A^B \vec{F} \cdot d\vec{r} = W$$

where W is the work done by the force in displacing the particle from point A to point B .

Since $\frac{1}{2} m v_B^2$ is the kinetic energy of the particle at point B and $\frac{1}{2} m v_A^2$, its kinetic energy at point A , we have

$$\begin{aligned} \text{work done in displacing the particle from } A \text{ to } B &= \text{its K.E. at } B - \text{its K.E. at } A \\ &= \text{change in energy in proceeding from } A \text{ to } B. \end{aligned}$$

i.e. *Work done by a force on a particle is equal to the change in the kinetic energy of the particle.*

This is spoken of as the **work-energy principle** or the **work-energy theorem**.

As will be readily seen, the kinetic energy of the particle increases if work be done *on it* (by the force) and decreases if work be done *by it* (against the force).

(ii) **Potential energy.** We have already defined the potential energy of a particle or a body as its capacity to do work in virtue of its position (or condition). It is measured by the amount of work done by the force to restore the particle or the body from its present position (or condition) to a standard position (or condition) and is denoted by the symbol U or V . Thus, if the present or the existing position of the particle be specified by a vector $\vec{r}$ and its standard position by a vector r_0 , we have

$$U = \int_r^{r_0} \vec{F} \cdot d\vec{r}.$$

It may be emphasised here that *this relationship exists only in the case of a conservative field of force*, for the work done by a conservative force alone depends on the position of the particle, irrespective of the path taken. In the case of a non-conservative force, the work done cannot be expressed in the form of potential energy since it depends on the directions of motion as also on the shape of the path taken and even, at times, on the magnitude of the velocity.

The standard position (or condition) chosen is the one in which the force acting on the particle or the body is zero, so that its *P.E.* in this position (or condition) may also be taken to be zero. The difference of *P.E.* in this position (or condition) and in the existing position (or condition) of the particle then gives its *P.E.* in the latter position.

Thus, for example, the standard condition in the case of a spring is taken to be its normal, unstretched (or uncompressed) state. And, although for convenience we take the electrostatic and gravitational potential energies to be zero on the surface of the earth, the electrostatic and gravitational forces involved are actually governed by the inverse square law, so that the electrostatic force due to a charge or the gravitational force due to a material body is zero only at an infinite distance from it and the standard or the zero potential energy position is thus infinity. In such cases, therefore, $r_0 = \infty$ and we have

$$U = \int_r^\infty \vec{F} \cdot d\vec{r} \quad \text{or,} \quad U = - \int_\infty^r \vec{F} \cdot d\vec{r},$$

i.e., the potential energy of a particle at a point $\vec{r}$ is given by the amount of work done in moving it from ∞ to that point.

In calculating the work we may, of course, use any convenient path since, as we know, the work done does not depend on the actual path taken but only on the positions of the end-points.

5.5 CONSERVATIVE FORCE AS NEGATIVE GRADIENT OF

POTENTIAL ENERGY –CURL $\vec{F} = 0$

We have just seen in § 5.4 above how

$$U = - \int_\infty^r \vec{F} d\vec{r} \quad \text{where } \vec{F} \text{ is a conservative force.}$$

Expressing $\vec{F}$ and $\vec{r}$ in terms of their components along the three coordinate axes, we have

$$\vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k}, \quad \vec{r} = x\hat{i} + y\hat{j} + z\hat{k} \quad \text{and} \quad d\vec{r} = d_x \hat{i} + d_y \hat{j} + d_z \hat{k}.$$

So that,

$$\begin{aligned} U &= - \int_\infty^r (F_x \hat{i} + F_y \hat{j} + F_z \hat{k})(d_x \hat{i} + d_y \hat{j} + d_z \hat{k}) \\ &= - \int_\infty^r (F_x dx + F_y dy + F_z dz) \end{aligned}$$

Which, on partial differentiation with respect to x, y, z , gives

$$F_x = -\partial U / \partial x, \quad F_y = -\partial U / \partial y, \quad F_z = -\partial U / \partial z,$$

$$\text{and } \therefore \quad \vec{F} = F_x \hat{i} + F_y \hat{j} + F_z \hat{k} = - \left(\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k} \right) = - \text{grad } U$$

$$\text{or} \quad -\vec{\nabla} U$$

Thus, a conservative force F is equal to the negative gradient of potential energy U .

If instead of the three dimensions, we restrict ourselves to one dimension only, we have $F = -dU/dx$, showing that a conservative force is the negative of the space rate of change of potential energy.

$$\text{Now, we have } \text{Curl } \vec{F} = \vec{\nabla} \times \vec{F} = -\vec{\nabla} \times \vec{\nabla} U$$

$$\begin{aligned} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial U}{\partial x} & \frac{\partial U}{\partial y} & \frac{\partial U}{\partial z} \end{vmatrix} \\ &= - \left[\hat{i} \left(\frac{\partial^2 U}{\partial y \partial z} - \frac{\partial^2 U}{\partial z \partial y} \right) + \hat{j} \left(\frac{\partial^2 U}{\partial z \partial x} - \frac{\partial^2 U}{\partial x \partial z} \right) + \hat{k} \left(\frac{\partial^2 U}{\partial x \partial y} - \frac{\partial^2 U}{\partial y \partial x} \right) \right] \end{aligned}$$

Since, with U taken as a perfect differential, $\partial^2 U / \partial y \partial z = \partial^2 U / \partial z \partial y$, $\partial^2 U / \partial z \partial x = \partial^2 U / \partial x \partial z$ and $\partial^2 U / \partial y \partial x = \partial^2 U / \partial x \partial y$ we have

$$\text{Curl } \vec{F} = 0.$$

This condition ($\text{Curl } \vec{F} = 0$) implies the existence of a potential function U such that $\vec{F} = - \text{grad } U$. It thus follows that $\text{curl } \vec{F} = 0$ is a necessary and a sufficient condition for a conservative field of force $\vec{F}$.

5.6. LAW OF CONSERVATION OF MECHANICAL ENERGY

It follows from § 5.5 above that

$$\vec{F} \cdot d\vec{r} = -(\partial U/\partial x)\delta x - (\partial U/\partial y)\delta y - (\partial U/\partial z)\delta z = -\delta U.$$

∴ work done in displacing a particle from A to B is given by

$$W = \int_A^B \vec{F} \cdot d\vec{r} = -\int_A^B dU = U_A - U_B.$$

But, as shown under § 5.4 (i)

$$W = \int_A^B \vec{F} \cdot d\vec{r} = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2$$

Obviously, therefore, $U_A - U_B = \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2$ indicating that *what the particle loses in P.E.*

it gains in K.E.

So that, $\frac{1}{2}mv_A^2 + U_A = \frac{1}{2}mv_B^2 + U_B$.

Or, putting T_A and T_B respectively for $\frac{1}{2}mv_A^2$ and $\frac{1}{2}mv_B^2$, we have

$$T_A + U_A = T_B + U_B, \text{ or, } T + U = E, \text{ a constant,}$$

showing that *the sum total (E) of the kinetic and potential energies of a particle remains unaltered as the particle moves from one point A to another point B in a conservative force field. In other words, the mechanical energy of a particle remains constant in such a field.* This is referred to as the **law of conservation of mechanical energy for conservative forces**. We shall see later that it may be easily extended to include also other forms of energy when it is called the *general law of conservation of energy*.

This is perhaps by far the most accurate of all the conservation laws and there has been found not one single exception to it so far.

It is easy enough to see that *the law holds good also for a system of particles moving in a conservative field of force* for the simple reason that it holds good for each individual particle.

The quantity $E = (T + U)$ is often spoken of as the energy function of the particle and is obviously *invariant to both changes of time and position of the particle*. This may be seen from a simple illustration of the law in the case of a falling particle.

Let the particle of mass m be at rest at a height h above the ground, so that it has only potential energy $U = mgh$, its kinetic energy T being zero. Its *total energy* is thus $(T + U) = (0 + mgh) = mgh$.

Let it now be allowed to fall under the action of gravity, so that as it gathers speed, it acquires K.E. at the expense of its P.E.

When it has fallen to a height z above the ground, *i.e.*, through a distance $(h - z)$, acquiring a velocity v , we have $v^2 - 0 = 2g(h - z)$ and the K.E. gained by it is, therefore,

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m[2g(h - z)] = mg(h - z).$$

And since its height from the ground is now z , the P.E. lost by it is $\int_h^z -mgdr = -mg(h - z)$ and ∴ its P.E. at this height (z) above the ground, *i.e.*, $U = mgh - [-mg(h - z)] = mgz$.

∴ *The sum total of its energy, $E = T + U = mg(h - z) + mgz = mgh$, the same as at its initial height h .*

And, when the particle has reached the ground, *i.e.*, when $z = 0$, its, $P.E.$, $U = mgh = 0$ and its $K.E.$, $T = mg(h - z) = mgh$.

And $\therefore$ its total energy $E = T + U = mgh + 0 = mgh$, the same as to start with.

Thus, the total energy of the particle remains the same mgh , in accordance with the law of conservation at any height from the ground between 0 and h , indicating that the energy function $E = T + U$ is independent of both time and the position of the particle.

Example 1. The position of a moving particle is at any instant given by $\vec{r} = A \cos \theta \hat{i} + A \sin \theta \hat{j}$. Show that the force acting on it is a conservative one. Also calculate the total energy of the particle.

Solution. The particle is obviously moving along a circular path of radius A . It must, therefore, be acted upon by a force directed towards the centre of its circular path, *i.e.*, the force acting upon it is a radial or a central one and hence necessarily *conservative*. However, this may also be show as follows:

The position vector of the particle at an instant t is

$$\vec{r} = A \cos \theta \hat{i} + A \sin \theta \hat{j} = A(\cos \omega t \hat{i} + \sin \omega t \hat{j})$$

Where ω is the angular velocity of the particle and $\omega t = \theta$.

The *linear velocity* of the particle at the given instant is

$$\vec{v} = d\vec{r}/dt = A\omega(-\sin \omega t \hat{i} + \cos \omega t \hat{j})$$

and hence its *acceleration* $\vec{a} = d^2\vec{r}/dt^2 = -A\omega^2(\cos \omega t \hat{i} + \sin \omega t \hat{j}) = -\omega^2\vec{r}$.

$\therefore$ If m be the *mass* of the particle, the force acting on it is $\vec{F} = -m\omega^2\vec{r}$.

Now, if this force ($\vec{F}$) be a conservative one, we should have $\text{curl } \vec{F} = 0$.

Let us see if this is so.

We have $\text{Curl } \vec{F} = \nabla \times \vec{F}$

$$\text{And, } \nabla \times \vec{F} = -m\omega^2 \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & 0 \end{vmatrix}, \text{ which is clearly equal to 0.}$$

The force acting on the particle is, therefore, a conservative one.

Now, since $\vec{F}$ is a conservative force, we have $\vec{F} = -\text{grad } U$, where U is the *P.E.* of the particle at the given instant.

We, therefore, have $\vec{F} = -\text{grad } U = -\left(\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right) = -\frac{dU}{dr}$

$$\therefore U = -\int \vec{F} \cdot d\vec{r} = -\int -m\omega^2\vec{r} \cdot d\vec{r} = \frac{1}{2}m\omega^2 r^2.$$

And if $\vec{v}$ be the linear velocity of the particle at the given instant, we have

$$\text{K.E. of the particle, } T = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 r^2 \quad [\because v = r\omega.]$$

$$\therefore \text{ Total energy of the particle, } E = T + U = \frac{1}{2}m\omega^2 r^2 + \frac{1}{2}m\omega^2 r^2 = m\omega^2 r^2.$$

Or,

$$E = m\omega^2(A^2\cos^2\omega t + A^2\sin^2\omega t) = m\omega^2A.$$

Thus, the *total energy of the particle* = $m\omega^2A$.

Example 2. Show that the following two forces are conservative:

(i) $\vec{F} = (y^2 - x^2)\hat{i} + 2xy\hat{j}$ and (ii) $F = (2xy + z^2)\hat{i} + x^2\hat{j} + 2xz\hat{k}$.

Also calculate the work done on a particle in case (ii) in moving it from the position 0, 1, 2 to the position 5, 6, 8.

Solution. Clearly, if the curl of each of the two forces be zero, they will be conservative.

Now, in case (i),

$$\begin{aligned} \text{curl } \vec{F} &= \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (y^2 - x^2) & 2xy & 0 \end{vmatrix} \\ &= \left[0 - \frac{\partial}{\partial z}(2xy) \right] \hat{i} + \left[\frac{\partial}{\partial z}(y^2 + x^2) - 0 \right] \hat{j} + \left[\frac{\partial}{\partial x} 2xy - \frac{\partial}{\partial y}(y^2 - x^2) \right] \hat{k} \\ &= -\frac{\partial}{\partial z}(2xy)\hat{i} + \frac{\partial}{\partial z}(y^2 + x^2)\hat{j} + \left[\frac{\partial}{\partial x}(2xy) - \frac{\partial}{\partial y}(y^2 - x^2) \right] \hat{k} \\ &= 0 + 0 + (2y - 2y)\hat{k} = 0. \end{aligned}$$

Thus, $\text{curl } \vec{F}$ being zero, force $\vec{F}$ is a conservative force.

In case (ii), again, $\text{curl } \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xy + z^2 & x^2 & 2xz \end{vmatrix}$

$$\begin{aligned} &= \left[\frac{\partial}{\partial y}(2xz) - \frac{\partial}{\partial z}(x^2) \right] \hat{i} + \left[\frac{\partial}{\partial z}(2xy + z^2) - \frac{\partial}{\partial x}(2xz) \right] \hat{j} \\ &\quad + \left[\frac{\partial}{\partial x}(x^2) - \frac{\partial}{\partial y}(2xy + z^2) \right] \hat{k} \\ &= 0 + (2z - 2z)\hat{j} + (2x - 2x)\hat{i} = 0 \end{aligned}$$

Again, $\text{curl } \vec{F}$ being zero, the force is a conservative one.

Now, work done by force $\vec{F}$ in case (ii) in displacing a particle from A to B is given by

$$\begin{aligned} W &= \int_B^A \vec{F} \cdot d\vec{r} = \int_A^B (F_x dx + F_y dy + F_z dz) \\ &= \int_A^B [(2xy + z^2)dx + x^2 dy + 2xz dz] \\ &= \int_A^B (2xy dx + x^2 dy) + (z^2 dx + 2xz dz) = \int_A^B d(x^2 y) + d(z^2 x) \\ &= \int_A^B d(x^2 y + z^2 x) = [x^2 y + z^2 x]_{0,1,2}^{5,6,8} = 5 \times 5 \times 6 + 8 \times 8 \times 5 - 0 - 0 \end{aligned}$$

or,

$$W = 150 + 320 = 470.$$

Thus, work done by the force = 470 units.

Example 3. Show that the force $\vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$ is conservative force. (Pbi. U. 2003)

Solution. A force is said to be conservative if the curl of the force is zero.

$$\text{Or, } \vec{\nabla} \times \vec{F} = 0$$

$$\text{Given, } \vec{F} = yz\hat{i} + zx\hat{j} + xy\hat{k}$$

$$\begin{aligned} \therefore \vec{\nabla} \times \vec{F} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & zx & xy \end{vmatrix} \\ &= \hat{i} \left[\frac{\partial}{\partial y}(xy) - \frac{\partial}{\partial y}(zx) \right] + \hat{j} \left[\frac{\partial}{\partial z}(yz) - \frac{\partial}{\partial x}(xy) \right] + \hat{k} \left[\frac{\partial}{\partial x}(zx) - \frac{\partial}{\partial y}(yz) \right] \\ &= \hat{i}(x - x) + \hat{j}(y - y) + \hat{k}(z - z) = 0 \end{aligned}$$

Since, $\vec{\nabla} \times \vec{F} = 0$, the force $\vec{F}$ is conservative.

Example 4. A particle lying in the $x - y$ plane is acted upon by a force directed towards the origin whose magnitude is kr where r (equal to $\sqrt{x^2 + y^2}$) is the distance from the particle to the origin. (a) Find the work that must be done to move the particle from the origin to the point $x = 1, y = 1$ along the radius vector to that point. (b) Find the work that must be done if the particle is first moved to $x = 1, y = 0$ and from there to $x = 1, y = 1$. (c) Is this particular force conservative or non-conservative?

Solution. (a) Clearly, work done in moving the particle from 0 to r is given by

$$W = \int_A^B \vec{F} \cdot d\vec{r} = \int_A^B \vec{F} \cos \theta \cdot dr = \int_0^r F dr$$

$$\left[\begin{array}{l} \because \text{Tangential component of } F \\ F \cos \theta = F, \cos \theta \text{ being equal to } 1. \end{array} \right]$$

$$\text{Since } F = kr, \text{ we have } W = \int_0^r kr \cdot dr = \left[\frac{kr^2}{2} \right]_0^r = \frac{kr^2}{2}.$$

Now, $r = \sqrt{x^2 + y^2} = \sqrt{1+1} = \sqrt{2}$, We, therefore, have

$$W = \frac{k(\sqrt{2})^2}{2} = k.$$

Thus, the work done = k units.

(b) In this case, the components of force $\vec{F}$ along the x and y -axis are F_x and F_y respectively

[$\because r = \sqrt{x^2 + y^2}$]. And, clearly, no work is done against the component F_y in taking the particle from O to A [Fig. 5.2], along the axis of x , (where $OA = x = 1$), and similarly, no work is done against the component F_x in taking the particle from A to P along the axis of y , (where $AP = y = 1$).

$\therefore$ Work done in taking the particle from the origin O to A ($x = 1$)

$$= \int_0^1 kx \cdot dx = \frac{kx^2}{2} = \frac{k}{2}.$$

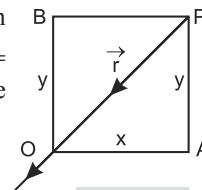


Fig. 5.2

And, work done in taking the particle from A to P ($y = 1$)

$$= \int_0^1 ky \cdot dy = \frac{ky^2}{2} = \frac{k}{2}.$$

$\therefore$ total work done in taking the particle from the origin O to A ($x = 1, y = 0$) and from A to P ($x = 1, y = 1$) $= k/2 + k/2 = k$, the same as in case (i).

(c) It will be easily seen that if we take the particle to P by first moving it from the origin (O) to B ($y = 1, x = 0$) and from there to P ($y = 1, x = 1$), we shall again have to do the same work

$$\int_0^1 ky \, dy + \int_0^1 kx \, dx = \frac{k}{2} + \frac{k}{2} = k \text{ as in cases (i) and (ii).}$$

Thus, the work done in taking the particle from the origin to P is the same (k) whatever the path taken, i.e., it depends only upon the position of the point P and not upon the actual path taken. The force on the particle is, therefore, a *conservative force*.

Example 5. A point-mass m starts from the point A of a curved track of the form shown in Fig. 5.3 with a horizontal velocity $\vec{v}_0$. The track is frictionless except for the small horizontal portion DE of length L . (i) Calculate the velocity of the mass at points B and C . (ii) What constant deceleration is required to stop the mass at E if the retarding frictional force starts operating at point D ?

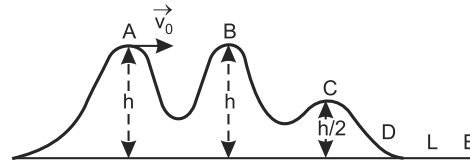


Fig. 5.3

Solution. (i) Clearly, the total energy possessed by the mass at the point $A = T + U = \frac{1}{2}mv_0^2 + mgh$. Since point B is also at the same height h from the horizontal as the point A , the kinetic and potential energies of the mass there too are the same as at A , viz., $\frac{1}{2}mv_0^2$ and mgh respectively.

At the point C , however, which is at a height $h/2$ from the horizontal, the *P.E.* of the mass is reduced to $\frac{1}{2}mgh$. Since the total energy of the mass must remain the same as before, it follows that half the *P.E.* of the mass, i.e., $\frac{1}{2}mgh$ is converted here into *K.E.* So that,

$$K.E. \text{ of the mass at } C = \frac{1}{2}mv_0^2 + \frac{1}{2}mgh$$

But if v_c be the velocity of the mass at C , its *K. E.* $= \frac{1}{2}mv_c^2$.

We $\therefore$ have $\frac{1}{2}mv_c^2 = \frac{1}{2}mv_0^2 + \frac{1}{2}mgh$, whence, $v_c = \sqrt{v_0^2 + gh}$.

Thus, the *velocity of the mass at the point C* $= \sqrt{v_0^2 + gh}$.

(ii) By the time the mass reaches the point D , its *P.E.* is reduced to zero, so that the whole of its energy, $\frac{1}{2}mv_0^2 + mgh$ is here converted into kinetic energy.

If therefore v_D be the velocity of the mass at D , we have

$$\frac{1}{2}mv_D^2 = \frac{1}{2}mv_0^2 + mgh, \text{ whence, } v_D = \sqrt{v_0^2 + 2gh}.$$

Now, if $-a$ be the constant acceleration of the mass between the points D and E which brings it to rest at E , we have

$$0 - v_D^2 = 2(-a)(L). \text{ Or, } -a = -v_D^2/2L = -\sqrt{v_0^2 + 2gh} / 2L.$$

Thus, the constant deceleration required to stop the mass at $E = \sqrt{v_0^2 + 2gh} / 2L$.

Example 6. A particle of mass m slides along a track with elevated ends and a flat central part, as shown in Fig. 5.4. The flat part has a length equal to 2 meters. The curved portions of the track are frictionless. For the flat part the coefficient of kinematics friction is $\mu = 0.20$. The particle is released at point A which is at a height $h = 1.0$ metre above the flat part of the track, where does the particle finally come to rest?

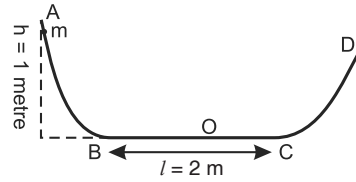


Fig. 5.4

Solution. Here, clearly, the only energy that the particle possesses at the point A is potential energy $= mgh$, relative to the flat or the horizontal part BC. As it is released at A, this P.E. naturally starts converting itself into K.E. until at B, the vertical distance being reduced to zero, the whole of it is present in the kinetic form. So that, if v be velocity acquired by the particle on reaching B,

$$\text{We have } \frac{1}{2}mv^2 = mgh.$$

As the particle now starts moving along the horizontal portion BC, it has to overcome the frictional force F and when the whole of its kinetic energy is used up in doing work against this frictional force over a distance S , say it will come to rest. We, therefore, have

$$\frac{1}{2}mv^2 = Fs.$$

Now, as we know, $F/R = \mu$ or $F = \mu R$, where R is the normal reaction of the horizontal surface over which the particle moves and is equal to mg , and μ , the coefficient of kinematic friction between the horizontal surface and the particle.

So that, $F = \mu R = \mu mg$.

$$\therefore \frac{1}{2}mv^2 = \mu mg.s. \text{ Or, } mgh = \mu mgs, \text{ whence, } s = h/\mu.$$

Since $h = 1$ metre and $\mu = 0.2$, we have $s = 1/0.2 = 5$ metres i.e., the particle will come to rest after transversing a 5 metre length of the flat surface.

The length of the flat surface being only 2 metres, the particle will still have some K.E. after travelling from B to C. It will, therefore, go up the frictionless, curved part CD until this K.E. is converted into P.E. It will then turn back, its P.E. will get converted into K.E. until at C, it will again have the same K.E. as before when it went up along CD. It will, therefore, again cover a distance of 2 metres on the flat surface from C to B, with some more part of its K.E. used up in doing work against the frictional force encountered. It will, however, still have some K.E. left and will, therefore, go up the frictionless part BA until this K.E. is converted into P.E. It will then turn back and its P.E. converted into K.E. until at B its K.E. will again be the same as that when it went up along BA. It will now, however, be able to cover a distance of only 1 metre on the flat surface, from B to O, the mid-point of the surface, for by then it will have covered a full 5 metre length of the flat surface that it could in virtue of its initial K.E.

The particle will thus finally come to rest at the mid-point O of the flat surface.

Example 7. The bob of a simple pendulum, initially held in the horizontal position at A, as shown in Fig 5.5, is released such that it goes in a circle (shown dotted) round a point P at a

vertical distance d below the point of suspension S of the pendulum. Show that the distance d should at least be $0.6l$, where l is the length of the pendulum.

Solution: Clearly, $K.E.$ gained by the bob on reaching point $B = mgl$, where m is the mass of the bob.

If, therefore, v be the velocity of the bob at B , we have

$$\frac{1}{2}mv^2 = mgl, \text{ whence, } v^2 = 2gl.$$

In order that the bob should go in a circle round the point P , as shown dotted, its velocity v' at the uppermost point C of the circle should be such that the centrifugal force mv'^2/r just supports the weight mg of the bob, where r is the radius of the circle, equal to $(l - d)$, i.e.,

$$mv'^2/r = mg. \text{ Or, } v'^2 = rg.$$

Thus, $K.E. \text{ of the bob at } C = \frac{1}{2}mv'^2.$

$$\therefore \text{ Loss of } K.E. \text{ in proceeding from } B \text{ to } C = \frac{1}{2}mv^2 - \frac{1}{2}mv'^2.$$

This must clearly be equal to the gain in $P.E.$ of the bob in rising through a vertical distance BC (or $2r$), i.e. equal to $mg(2r)$.

$$\therefore \frac{1}{2}mv^2 - \frac{1}{2}mv'^2 = 2mgr. \text{ Or, } v^2 - v'^2 = 4gr.$$

Or, substituting the values of v^2 and v'^2 , we have

$$2gl - rg = 4gr. \text{ Or, } 2l = 5r = 5(l - d) = 5l - 5d.$$

$$\text{Or, } 5d = 3l, \text{ whence, } d = \frac{3}{5}l = 0.6l$$

Thus, the least distance of point P below the point of suspension S of the pendulum must be $0.6l$.

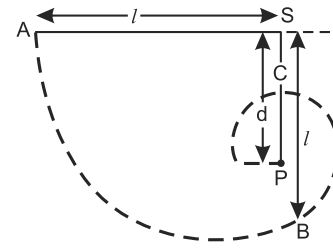


Fig. 5.5

5.7. POTENTIAL ENERGY IN AN ELECTRIC FIELD-ELECTRIC POTENTIAL

We know that the intensity $\vec{E}$ of an electric field at a point is defined as the force experienced by a unit positive charge when placed at that point in the field. It follows, therefore, that the force experienced by a charge q placed at the point will be $q\vec{E}$.

Again, the potential (V) at a point in an electric field is defined as the amount of work done in moving a unit positive charge from ∞ (where the potential is assumed to be zero) to that point. It is thus, in other words, the electrostatic potential energy of a unit positive charge placed at that point. So that, if the potential energy of a charge of $+q$ units be U at a point having position vector $\vec{r}$ in an electric field, we have

$$V = U/q = -\int_{\infty}^{\vec{r}} \vec{E} \cdot d\vec{r}$$

where $\vec{E}$ is the intensity of the field at the point.

Hence, the potential difference between two points $\vec{r}_1$ and $\vec{r}_2$ is

$$\text{given by } V_2 - V_1 = \int_{\vec{r}_1}^{\vec{r}_2} \vec{E} \cdot d\vec{r}$$

where V_1 and V_2 are the potentials at the two points respectively.

This is obviously equal to the change in the electrostatic potential energy of a unit positive charge when moved from one point to the other. Clearly, therefore, if a charge q be moved between the two given points, the *change in electrostatic potential energy* will be

$$U_2 - U_1 = q(V_2 - V_1),$$

Where U_1 and U_2 are the electrostatic potential energies of the charge q at the two points respectively.

Now, it may be noted that the expression for potential, $V = -\int_{\infty}^r \vec{E} \cdot d\vec{r}$ is similar to that for the potential energy of a particle, viz., $U = -\int_{\infty}^r \vec{F} \cdot d\vec{r}$, (§ 5.4). So that, taking

$$\vec{E} = (E_x\hat{i} + E_y\hat{j} + E_z\hat{k}) \text{ and } d\vec{r} = (dx\hat{i} + dy\hat{j} + dz\hat{k})$$

and proceeding exactly as in §5.5, we obtain

$$\vec{E} = \left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k} \right) = -\text{grad } V \text{ or } -V\vec{\nabla}$$

and in one dimension,

$$E = -dV/dx \text{ (only magnitude).}$$

This enables us to define *electric potential* in two ways, viz.,

(i) *as the scalar quantity whose gradient with the negative sign gives the intensity of the electric field at the point in question.* Or,

(ii) *as the scalar quantity whose negative space rate of change in any given direction gives the component of the field intensity in that direction.*

Now, in the electrostatic system of units, we have electric intensity at a point distant r from a charge q in a medium of dielectric constant K given by $\vec{E} = q\hat{r}/Kr^2$, where $\hat{r}$ is the unit vector in the direction of $\vec{r}$.

$$\therefore V = -\int \vec{E} \cdot d\vec{r} = -\int \frac{q\hat{r}}{Kr^2} \cdot d\vec{r} = -\int \frac{qdr}{Kr^2} = \frac{q}{Kr} + C,$$

where C is a constant integration.

Since at $r = \infty$, $V = 0$, we have $C = 0$. And, therefore,

$$V = q/Kr \text{ esu or statvolt.}$$

In the RMKS (or SI) system, the unit of potential is the volt, where $1 \text{ volt} = 1 \text{ joule}/1 \text{ coulomb} = 10^7/3 \times 10^9 \text{ esu}$ or $\text{statcoulomb} = 1/300 \text{ esu}$.

Based on the volt, we have a very convenient unit for the measurement of energy, namely, the **electron volt** which is equal to the work done by an electron moving through a potential difference of 1 volt. Thus, $1 \text{ eV} = 4.80 \times 10^{-10} \text{ esu} \times 1/300 \text{ esu} = 1.6 \times 10^{-12} \text{ ergs} = 1.6 \times 10^{-19} \text{ joules}$.

Example 8. Calculate the distance of closest approach of (i) two protons, each of energy 500 MeV, approaching each other from opposite directions, (ii) a proton of energy V electron volts approaching a nucleus of atomic number Z .

(i) As the two protons approach each other, they have to oppose the electrostatic force of repulsion between them and their kinetic energy is gradually converted into potential energy. Their distance of closest approach will, therefore, be that at which the whole of their kinetic energy is so converted into potential energy. Let it be $r \text{ cm}$.

Then, since e is the charge on each proton, the potential energy of the two protons at distance r will be $(e)(e)/r = e^2/r$. This must be equal to their total kinetic energy,

$$\begin{aligned} 2 \times 500 \text{ MeV} &= 2 \times 500 \times 10^6 \text{ eV} = 2 \times 500 \times 10^6 \times 1.6 \times 10^{-12} \\ &= 1.6 \times 10^{-3} \text{ ergs.} \end{aligned}$$

$$\begin{aligned} \text{Thus, } e^2/r &= 1.6 \times 10^{-3}, \text{ whence, } r = e^2/1.6 \times 10^{-3} = (4.8 \times 10^{-10})^2 / 1.6 \times 10^{-3} \\ &= 1.44 \times 10^{-16} \text{ cm.} \end{aligned}$$

Or, distance of closest approach of the two protons = 1.44×10^{-16} cm.

(ii) Here, too, the proton will approach the comparatively heavier nucleus until the whole of its *K.E.* is converted into *P.E.*

Now, charge on the proton is e and that on the nucleus, Ze . So that, if r cm be the distance of closest approach, the *P.E.* of the proton and the nucleus = $(Ze)(e)/r = Ze^2/r$. This must be equal to the *K.E.* of the proton, given to be V electron volts = $V \times 1.6 \times 10^{-12}$ ergs, i.e.,

$$Ze^2/r = V \times 1.60 \times 10^{-12}, \text{ whence, } r = Ze^2/V \times 1.60 \times 10^{-12}.$$

$$\begin{aligned} \text{Or, distance of closest approach, } r &= \frac{Z(4.8 \times 10^{-10})^2}{V \times 1.60 \times 10^{-12}} = \frac{14.4Z}{V} \times 10^{-3} \text{ cm} \\ &= \frac{14.4Z}{V} \text{ A.U.} \end{aligned}$$

Example 9. An electron is moving in a circular orbit of radius 2×10^{-8} cm about a proton. Supposed to be at rest. Calculate the velocity of the electron as also its kinetic and potential energies.

How much energy is required to ionize the system (i.e., to remove the electron to an infinite distance from the proton, so as to have zero kinetic energy)? What is the value of the ionization potential?

Solution. For the electron to be moving in the circular orbit around and proton, the centrifugal force acting on it must *just* be balanced by the force of attraction between them. We, therefore, have $mv^2/r = (e)(e)/r^2$, whence, $v = \sqrt{e^2/mr}$, where m is the mass of the electron; v , its velocity; r the radius of the orbit and e , the charge on both the electron and the proton.

Substituting the values of e , m and r , therefore, we have

$$\begin{aligned} \text{Velocity of the electron, } v &= \sqrt{\frac{(4.8 \times 10^{-10})^2}{9.1 \times 10^{-28} \times 2 \times 10^{-8}}} = \frac{4.8 \times 10^{-10}}{(18.2 \times 10^{-36})^{\frac{1}{2}}} \\ &= 11.26 \times 10^7 \text{ cm/sec.} \end{aligned}$$

$$\begin{aligned} \therefore \text{ Kinetic energy of the electron} &= \frac{1}{2}mv^2 = \frac{1}{2} \times 9.1 \times 10^{-28} \times (11.26 \times 10^7)^2 \\ &= 5.76 \times 10^{-12} \text{ ergs} = \frac{5.76 \times 10^{-12}}{1.6 \times 10^{-12}} = 3.6 \text{ eV.} \end{aligned}$$

$$\begin{aligned} \text{And, potential energy of the electron} &= \frac{(-e)(e)}{r} = \frac{-e^2}{r} = -\frac{(4.8 \times 10^{-10})^2}{2 \times 10^{-8}} \\ &= -11.5 \times 10^{-12} \text{ ergs} = -\frac{11.5 \times 10^{-12}}{1.6 \times 10^{-12}} = -7.2 \text{ eV.} \end{aligned}$$

Clearly, work done in removing the electron to an infinite distance from the proton, i.e., its ionization energy = total energy of the electron at infinity – its total energy at the given distance r from the proton, i.e., **ionisation energy** = $0 - (\text{K.E. of electron at distance } r \text{ from proton} + \text{P.E. of electron at distance } r \text{ from proton}) = 0 - (3.6 - 7.2) = 3.6 \text{ eV} = 3.6 \times 1.6 \times 10^{-12} = 5.76 \times 10^{-12} \text{ ergs}$.

Now, *ionization potential* is the potential through which the electron must be accelerated to acquire its ionization energy.

Since the ionization energy of the electron here is 3.6 *electron volts*, it must have been accelerated through a potential of $3.6/1 = 3.6$ volts.

$$\therefore \text{ Ionisation potential of the electron} = 3.6 \text{ volts.}$$

5.8 LINEAR RESTORING FORCE

As its very name indicates, a linear restoring force is the force which is proportional to the linear displacement of a particle from a fixed point – its mean position of rest or equilibrium—and is always directed towards that position (*i.e.*, opposite to that of its displacement), tending to restore the particle back to its original position. This is known as **Hooke's law**.

Thus, taking the mean position of the particle as the origin, if its displacement at a given instant be $\vec{r}$ we have

$$\vec{F} \propto (-\vec{r}) \text{ Or, } \vec{F} = -C\vec{r},$$

where C is a positive constant, called the *force constant*.

Or, since a small linear restoring force is often provided by a stretched or a compressed spring, C is also sometimes called the *spring factor*.

The negative sign of the expression for $\vec{F}$ indicates that *the force is directed oppositely to the displacement $\vec{r}$ from the mean position*. A graph between $\vec{F}$ and $\vec{r}$ will thus be a straight line passing through the origin and having a slope $-C$.

Obviously, work has to be done against this restoring force in displacing the particle from its mean position and this work is stored up in the particle in the form of potential energy.

To obtain the value (U) of this potential energy, we note that the linear restoring force is a conservative force as may be easily established by showing that $\text{curl } \vec{F} = 0$, this being an essential condition for a conservative force (see §5.5). We thus have

$\text{curl } \vec{F} = \vec{\nabla} \times (-C\vec{r})$ where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$. So that,

$$\text{curl } \vec{F} = -C \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = 0,$$

because the partial differential coefficients ($\partial z/\partial y$ etc.) are all equal to zero.

And since it is a conservative force, it may be expressed as the negative of the gradient of potential energy U to the particle, acquired on displacement r , *i.e.* $\vec{F} = \text{grad } U$, and, therefore,

$$\begin{aligned} U &= -\int (F_x dx + F_y dy + F_z dz) \\ &= \frac{C}{2}(x^2 + y^2 + z^2) + A = \frac{Cr^2}{2} + A, \end{aligned}$$

where A is a constant of integration.

Since, with displacement zero, *i.e.* $r=0$, the P.E., U is also zero, we have $A=0$. And, therefore,

$$U = Cr^2/2.$$

Thus, if the displacement be along the x direction, *i.e.*, if $\vec{r} = x\hat{i}$,

where $\hat{i}$ is the unit vector along the axis of x , we have $U = Cx^2/2$. So that, plotting P.E., (U), against displacement x , we obtain a parabola, as shown in Fig. 5.6, clearly bringing out the fact that *the work done in displacing a particle against a linear restoring force, and hence the potential energy acquired by it, is proportional to the square of its displacement from its mean or equilibrium position*.

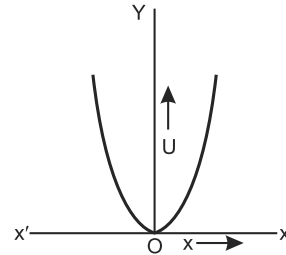


Fig. 5.6

If the *maximum displacement* of the particle be a , the *maximum potential energy* acquired by it $= \frac{1}{2}ma^2$.

Let us imagine the particle to be initially at its position of maximum displacement a , where the whole of its energy is present in the potential form and is equal to $\frac{1}{2}ma^2$. On releasing it, it will start moving towards its mean or equilibrium position under the action of the restoring force, thus acquiring kinetic energy at the expense of its potential energy. And since the restoring force, is as we have seen, a conservative force, the sum total of its energy E , at any point of its displacement, will remain constant. So that, at displacement x , if its velocity be v , we have its *kinetic energy*, $T = \frac{1}{2}mv^2$ and its *potential energy*, $U = \frac{1}{2}Cx^2$.

$$\therefore \text{its total energy, } E = T + U = \frac{1}{2}mv^2 + \frac{1}{2}Cx^2 = \frac{1}{2}Ca^2,$$

$$\text{whence, } \frac{1}{2}mv^2 = \frac{1}{2}C(a^2 - x^2) \text{ and } \therefore v = \pm \sqrt{\frac{C}{m}(a^2 - x^2)}.$$

Again, as the particle reaches its mean or equilibrium position at $x = 0$, its $P.E = 0$ and its kinetic energy, the maximum, equal to $\frac{1}{2}Ca^2$. Its velocity here will therefore be the maximum, say, $v_{\max}$.

$$\text{We, therefore, have } \frac{1}{2}mv_{\max}^2 = \frac{1}{2}Ca^2, \text{ whence, } v_{\max} = \pm \sqrt{\frac{c}{m}}a.$$

Example 10. A car carries a framework $ABCD$, shown below (Fig. 5.7) in which a 200 gm mass P is supported between two springs of force constant 5000 dyne/cm each. Side AB is kept horizontal and along the length of the car and the pointer attached to P reads zero when the car is at rest. What will the pointer read when the car has (i) uniform speed 20 m/sec on a straight road

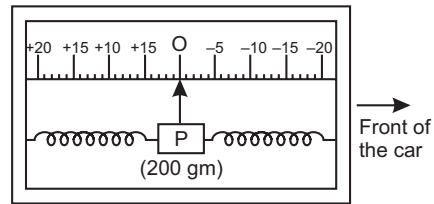


Fig. 5.7

(ii) Uniform speed 10 m/sec on a circular road of radius 20 m, (iii) uniform acceleration of 0.5 m/sec^2 on a straight road, (iv) uniform acceleration 1.0 m/sec^2 on a straight road? (Guwahati 2002)

Solution. (i) In this case, the motion of the car being unaccelerated, there is no force acting on mass P and hence the pointer attached to it continues to read zero on the scale.

(ii) Here, since the car is moving with a uniform speed along a circular path, the centripetal force (mv^2/r) will act in a direction perpendicular to the length of the two springs, thus producing neither extension nor contraction in them. The pointer will, therefore, again continue to read zero on the scale.

(iii) The forward acceleration of the car being 0.5 m/sec^2 or 50 cm/sec^2 , the force of reaction acting on the mass $P = 200 \times 50$ dynes in the backward direction. The first spring (to the left of P) will thus get compressed and the second, stretched through a distance x say. We shall, therefore, have

$$200 \times 50 = -(C + C)x = -2Cx. \text{ Where } C \text{ is the force constant of each spring.}$$

$$\text{Or, } x = -\frac{200 \times 50}{2C} = -\frac{200 \times 50}{2 \times 5000} = -1 \text{ cm} = -10 \text{ mm}$$

Thus, the mass will move 10 mm to the left and the pointer will read + 10 mm on the scale, to the left of the zero mark.

(iv) Here, the backward acceleration of the car being 1.0 m/sec^2 or 100 cm/sec^2 , the force of reaction 200×100 dynes will act on the mass in the forward direction. As a result, the first spring (to the left of P) will get extended and the second, compressed by the same amount x' , say. We, therefore, have

$$200 \times 100 = 2 \times 5000 x', \text{ whence, } x' = 200 \times 100 / 2 \times 5000 = +2 \text{ cm or } +20 \text{ mm.}$$

The mass will thus move forward through 20 mm and the pointer will thus read -20 mm on the scale, to the right of the zero mark.

Example 11. An ideal massless spring S can be compressed 1.0 metre by a force of 100 newtons. This same spring is placed at the bottom of a frictionless inclined plane which makes an angle $\theta = 30^\circ$ with the horizontal (Fig. 5.8). A 10 kg mass m is released from rest at the top of the incline and is brought to rest momentarily after compressing the spring 2.0 metres. (a) Through what distance does the mass slide before coming to rest? (b) What is the speed of the mass just before it reaches the spring?

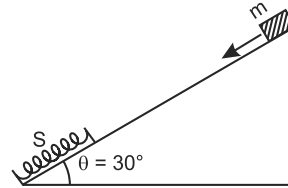


Fig. 5.8

Solution. (a) Let the mass m slide a total distance s along the inclined plane before coming to rest on compressing the spring through 2 metres. Then, clearly,

$$\text{Gravitational potential energy lost by the mass} = mg \sin \theta \times s.$$

This must obviously be equal to the work done in compressing the spring through a distance of 2 metres, i.e. equal to $\frac{1}{2} Cx^2 = \frac{1}{2} \times 100(2)^2 = 200$ joules, because $C = 100 \text{ N/m}$ (given)

$$\text{We, therefore, have } mg \sin \theta \times s = 200. \text{ Or, } 10 \times 9.8 \times \frac{1}{2} \times s = 200,$$

$$\text{i.e., } 49s = 200. \text{ Or, } s = 200/49 = 4.08 \text{ m, or } 4.0 \text{ metres, say.}$$

Thus, the distance through which the mass slides before coming to rest = 4 metres.

(b) Since this total distance of 4.08 metres also includes the 2 metres through which the spring is compressed, the distance through which the mass slides from the top of the plane until it reaches the spring is $4.08 - 2 = 2.08$ metres.

And, since its initial velocity $u = 0$ ($\because$ it starts from rest), its velocity v on covering a distance of 2.08 metres along the incline is given by the relation $v^2 - u^2 = 2as$. i.e., $v^2 = 2 \times g \sin \theta \times 2.08$
 $= 2 \times 9.8 \times \frac{1}{2} \times 2.08 = 20.38$, whence,
 $v = \sqrt{20.38} = 4.514 \text{ m/sec or } 4.5 \text{ metres per sec.}$

Thus, the speed of the mass just before it reaches the spring = 4.5 metres/sec.

Example 12. Obtain an expression for restoring force as a function of position for a particle moving in a potential energy field $U = A - Bx + Cx^2$. At what point does the force vanish? Is this a point of stable equilibrium? If so, obtain the value of the force constant.

Solution. Clearly, linear restoring force $F = -dU/dx = B - 2Cx$.

This force will vanish when $dU/dx = 0$, or, $B - 2Cx = 0$, i.e., when $x = B/2C$.

Now, for stable equilibrium of the particle d^2U/dx^2 , i.e., $2C$ must be positive.

Hence if C be a positive quantity, we shall have a minimum potential energy position of the particle at $x = B/2C$ and this will, therefore, be a point of stable equilibrium.

From the relation $F = B - 2Cx$, it is clear that the force constant = $2C$.

Example 13. Two ideal and identical springs S_1 and S_2 , each of normal length l and force constant C are fixed at points $(-l, o)$ and $(+l, o)$ and connected together at the other end, as shown in Fig. 5.9 (a).

(a) Show that for the displacement, x, y , of their joined ends,

$$U = \frac{C}{2} \left[\{(x+l)^2 + y^2\}^{\frac{1}{2}} - l \right]^2 + \frac{C}{2} \left[\{(l-x)^2 + y^2\}^{\frac{1}{2}} - l \right]^2$$

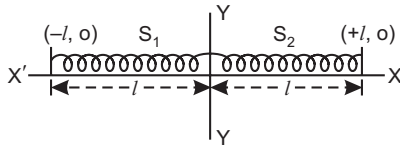


Fig. 5.9 (a)

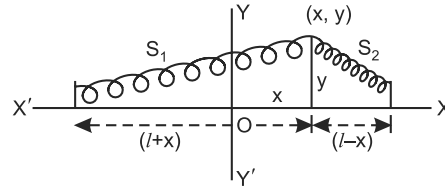


Fig. 5.9 (b)

(b) Obtain the values of the functions $U(x, 0)$ and $U(0, y)$ and show that $U(x, 0)$ assumes the shape of a parabola for all values of x and that if $y \gg l$, $U(0, y)$ too assumes a shape which is a near parabola.

(c) Calculate the force components F_x for $y = 0$ and F_y for $x = 0$.

Solution. (a) With the joined ends of the springs given the displacement (x, y) , as shown in Fig. 5.9 (b), we find that spring S_1 gets stretched and springs S_2 , compressed, their increased and decreased lengths now being,

$$[(l+x)^2 + y^2]^{\frac{1}{2}} \text{ and } [(l-x)^2 + y^2]^{\frac{1}{2}} \text{ respectively. So that,}$$

$$\text{increase in length of spring } S_1 = [(l+x)^2 + y^2]^{\frac{1}{2}} - l$$

$$\text{and decrease in length of spring } S_2 = [(l-x)^2 + y^2]^{\frac{1}{2}} - l$$

$$\therefore U(x, y) = \frac{1}{2} C [(l+x)^2 + y^2]^{\frac{1}{2}} - l]^2 + \frac{1}{2} C [(l-x)^2 + y^2]^{\frac{1}{2}} - l]^2.$$

$$\begin{aligned} \text{(b) Obviously, } U(x, 0) &= \frac{1}{2} C [(l+x)^2]^{\frac{1}{2}} - l]^2 + \frac{1}{2} C [(l-x)^2]^{\frac{1}{2}} - l]^2 \\ &= \frac{1}{2} C (l+x-l)^2 + \frac{1}{2} C (l-x-l)^2 = Cx^2, \end{aligned}$$

indicating that $U(x, 0)$ assumes the shape of a parabola for all values of x .

$$\begin{aligned} \text{And } U(0, y) &= \frac{1}{2} C [(l^2 + y^2)^{\frac{1}{2}} - l]^2 + \frac{1}{2} C [(l^2 + y^2)^{\frac{1}{2}} - l]^2 \\ &= C [(y^2 + l^2)^{\frac{1}{2}} - l]^2 \end{aligned}$$

$$\text{Now, if } y \gg l, \text{ we have } (y^2 + l^2)^{\frac{1}{2}} = y + \frac{1}{2} y^{-1} l^2 \approx y + l^2/2y$$

$$\therefore U(0, y) \approx C \left(y + \frac{l^2}{2y} - l \right)^2 \approx C \left[y \left(1 + \frac{l^2}{2y^2} \right) - l \right]^2$$

Since $y \gg l$, we have $l^2/2y^2 \ll 1$ and $\therefore U(0, y) \approx C(y-l)^2$

Again, y being very much greater than l , $(y-l) \approx y$. So that,

$$U(0, y) \approx Cy^2.$$

Indicating that $U(0, y)$ too takes the shape of a near parabola.

C (or G) corresponding to minimum potential energy, any slight displacement of the particle, either way, results in a force acting on it, tending to bring it back into its original position C (or G).

These points (like C or G) of minimum potential energy are, therefore, called **positions of stable equilibrium**.

In case the curve has also a horizontal portion, such as shown dotted in the figure, signifying a region of *constant potential energy*, the slope of the curve, and hence the force acting on the particle at any point on it, remains zero even if slightly displaced from its initial position, so that it stays in its new position, without either tending to go further away from it or to come back to its initial position. Such a region is, therefore, called a **region neutral equilibrium**.

Bounded region or the Potential well. Let us consider again the particle at the minimum potential energy or the stable equilibrium position, C , with its total energy $E(=T + U)$ represented by the horizontal line shown, cutting the energy curve at B and D . Since the system is conservative and the energy is conserved, it follows that the *P.E.* of the particle at C being the minimum, its *K.E.* there must be the maximum, so that it moves towards either B or D . As it proceeds towards D , say, it comes under the action of the restoring force $F = -dU/dx$, so that its *K.E.* decreases and its *P.E.* increases (the sum total remaining the same, E), until when it reaches the point D , the whole of its energy is in the potential form, equal to E , its kinetic energy, and hence its velocity, now being reduced to *zero*. Here, therefore, the particle remains momentarily at rest and then start moving back under the action of the restoring force and again reaches C where its potential energy is reduced to the minimum and its kinetic energy becomes the maximum. It, therefore, continues to move towards B , its potential energy now increasing and its kinetic energy correspondingly decreasing (the sum total remaining constant at E) until at B , the whole of its energy is again in the potential form and its kinetic energy, and hence its velocity, *zero*. Again, therefore, it is momentarily at rest at B and then starts moving back towards C under the action of the restoring force.

Thus, the particle remains confined to the region BCD , oscillating (simple harmonically) between the *turning points* B and D . Such a region (between the turning points B and D) is appropriately called a bounded region or a potential well (*or potential valley*) and *always exists about a point of minimum potential energy or stable equilibrium*.

The difference of energy between the top and the bottom of the well is called the *binding energy* for well and the particle confined to the well is said to be in the *bound state* or to perform a *bound motion* and, as we have seen, if its motion be one-dimensional, it is necessarily *periodic* and keeps on repeating itself after regular intervals, called its *time-period*.

It will be readily seen that

(i) If the total energy of the particle be E_1 , the particle will have four turning points, J, L, F and H and it can oscillate in either of the two potential wells JCL and $F GH$;

(ii) If its total energy be E_2 , it will obviously have only one turning point at M and will always turn to the right from here, its speed increasing and decreasing according as its potential energy (U) decreases or increases.

(iii) And, if its energy be greater than E_3 , there will be no turning point at all, *i.e.*, the particle will not change direction and only its speed will vary as in case (ii).

Now, the potential energy function $U = f(x)$ may, in general, be expanded about the position x_0 of stable equilibrium by means of *Taylor's theorem*. Thus,

$$U = U_{x_0} + \left(\frac{dU}{dx}\right)_{x_0} (x - x_0) + \left(\frac{d^2U}{dx^2}\right)_{x_0} \left(\frac{x - x_0}{2}\right)^2 + \dots,$$

where $(dU/dx)_{x_0}$, $(d^2U/dx^2)_{x_0}$ etc. are the differentials at the stable equilibrium position (x_0) and, as such, are constants.

If we measure displacement from the position x_0 , i.e., if we take x_0 as the origin, so that $(x - x_0) = (x - 0) = x$, we have

$$U = U_0 + \left(\frac{dU}{dx}\right)_0 x + \left(\frac{d^2U}{dx^2}\right)_0 \frac{x^2}{2!} + \dots$$

Since x_0 is a point of stable equilibrium, and hence of minimum potential energy, we have $(dU/dx)_0 = 0$ and $(d^2U/dx^2)_0$, a positive quantity, say C . Then, denoting $(d^3U/dx^3)_0, \dots$ etc. by $C_1, C_2, \dots$, we have

$$U = U_0 + \frac{Cx^2}{2!} + \frac{C_1x^3}{3!} + \dots,$$

and, therefore, force acting on the particle i.e., $F = -\frac{dU}{dx}$

Or,

$$F = -Cx - \frac{C_1}{2}x^2 - \frac{C_2}{6}x^3 \dots$$

In case the displacements be small, the higher powers of x become negligible and we, therefore, have

$$F = -Cx \text{ and } U = U_0 + \frac{1}{2}Cx^2,$$

indicating that force F is a *linear restoring force*, proportional to displacement and directed towards the position of equilibrium. The positive constant C is thus the *force constant*.

The potential energy curve (or the graph between U and x) in this region of small displacements, x_1 and x_2 , say, about the position of stable equilibrium (i.e., within the potential well) thus comes out to be parabolic in form (as shown in Fig. 5.11), with its vertex at x_0, U_0 . In case the minimum potential energy (U_0) be assumed to be zero, the apex of the parabola will touch the x -axis at the origin O .

The particle thus oscillates inside the potential well about the stable equilibrium (or minimum potential energy) position between the limits x_1 and x_2 , so that its total energy $E = T + U$. It will be easily seen that the particle cannot cross the limits x_1 and x_2 for, within this region (x_1, x_2), the sum total of its energy is $E = T + U$. And at x_1 and x_2 , $T = 0$ and, therefore $U = E$, i.e., the entire energy is present in the potential form. Beyond x_1 and x_2 , the potential energy U would be greater than this total energy E , implying thereby that, in order that $T + U$ may be equal to E , the kinetic energy T must be *negative*, which is simply absurd.

The motion of the particle thus remains restricted between x_1 and x_2 on either side of O , so long as the total energy is E and there are no non-conservative forces (like friction etc.) operating. In case the value of E be different, the values of x_1 and x_2 too will naturally be correspondingly different. This is in fact the case of a simple harmonic oscillator which we shall study in some detail later in chapter 8.

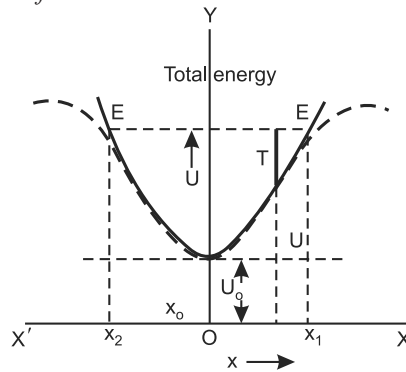


Fig. 5.11

Example 14. Fig. 5.12 shows the potential energy curve of a system in arbitrary units for U and r . State the possible r -values about which the system can oscillate, also the maximum energy of oscillations in each case. If the system just escapes from the potential well at smaller r , with what kinetic energy does it go to $r \rightarrow \infty$? (Agra 2001)

If 0.1 unit be the mass of the system, calculate its velocity when it escapes from the potential well at smaller r .

Solution. As we know, the system can oscillate only about its position of stable equilibrium of minimum potential energy. As is clear from the graph, these positions occur at A and B , where the values of r are **2 units** and **4.5 units** respectively.

The *maximum energies of oscillation* at these positions are, obviously,

$1 - (-2.5)$ or **3.5 units** and $0.5 - (-1)$ or **1.5 units**, respectively.

And, as the system just escapes at C from the potential well at smaller $r = 2$ units, its *K.E.* will obviously be **1 unit**.

If v be the velocity of the system as it escapes from the potential well at smaller r , we have its $K.E. = \frac{1}{2}mv^2 = 1$ unit. So that,

$$\frac{1}{2}(0.1)v^2 = 1. \text{ Or, } v = \sqrt{20} = 4.47 \text{ units.}$$

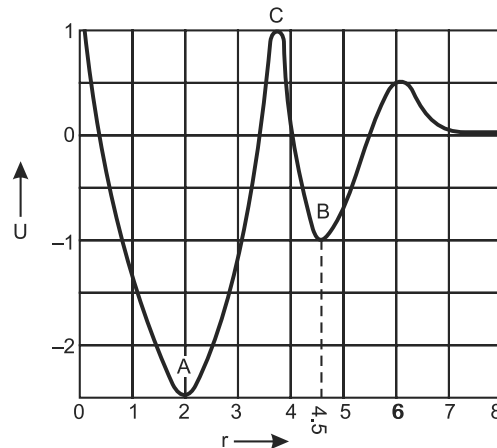


Fig. 5.12

Example 15. The figure below (Fig. 5.13) shows the potential energy diagram of a system like a diatomic molecule, potential energy U being plotted against inter-nuclear distance r .

Deduce values of the following:

- (i) Equilibrium inter-nuclear distance r_0 .
- (ii) Binding energy E_b .
- (iii) Force constant near the bottom of the well.

(Agra 2003)

Solution. (i) At the equilibrium inter-nuclear distance, the potential energy must be the minimum. As we can see from the graph supplied (Fig 5.13), this happens at the point A which corresponds to $r = 2$ Angstrom units. This, therefore, gives the value of r_0 . Thus,

$$\text{Equilibrium inter-nuclear distance, } r_0 = 2 \text{ A.U.} = 2 \times 10^{-8} \text{ cm.}$$

(ii) Since *binding energy* is the difference between the potential energies at the top and the bottom of the potential concerned, we have

$$\text{binding energy, here } E_b = 0 - (-4) = 4\text{eV} = 4 \times 1.6 \times 10^{-12} = 6.4 \times 10^{-12} \text{ ergs.}$$

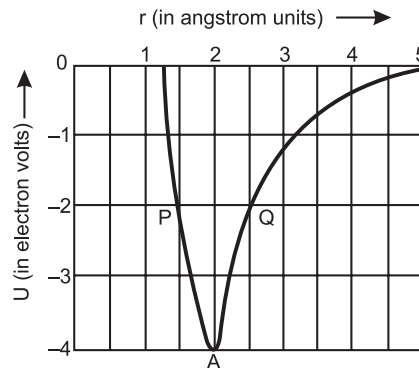


Fig. 5.13

(iii) The potential energy curve near the bottom of the potential well, i.e., its portion PAQ is parabolic in shape and, therefore,

$$U = U_A + \frac{1}{2}(r - r_0)^2, \text{ whence the linear restoring force } F = -dU/dr = -C(r - r_0)$$

So that,

$$C = \frac{dU/dr}{(r - r_0)} = \frac{dU}{dr(r - r_0)}$$

Considering the displacement from A to Q , therefore, we have

$$dU = -2 - (-4) = 2eV = 2 \times 1.60 \times 10^{-12} \text{ ergs}, dr = 0.5 \text{ A.U.} = 0.5 \times 10^{-8} \text{ cm and } (r - r_0) = (2.5 - 2.0) = 0.5 \text{ A.U.} = 0.5 \times 10^{-8} \text{ cm.}$$

$$\text{Hence, force constant } C = \frac{2 \times 1.60 \times 10^{-12}}{(0.5 \times 10^{-8})(0.5 \times 10^{-8})} = \frac{3.20 \times 10^{-12}}{0.25 \times 10^{-16}}$$

$$= 12.8 \times 10^4 \text{ dyne/cm.}$$

Example 16. The potential energy function for the force between two atoms in a diatomic molecule can approximately be expressed as $U_{(x)} = \frac{a}{x^{12}} - \frac{b}{x^8}$,

where a and b are positive constants and x is the distance between the atoms.

(i) For what values of x , is $U_{(x)}$ equal to zero? (ii) For what values of x , is $U_{(x)}$ minimum? (iii) Calculate the force between the two atoms and plot $F_{(x)}$ against x . Show that the two atoms repel each other for x less than x_0 and attract each other for x greater than x_0 . What is the value of x_0 ? (iv) Assuming that one of the atoms remains stationary and the other moves along the x -axis, describe the possible motions. (v) Calculate the dissociation energy (the energy required to break the molecule into atoms) of the molecule. (Agra 2005)

Solution. (i) From the relation $U_{(x)} = \frac{a}{x^{12}} - \frac{b}{x^6}$, it is clear that for $U_{(x)}$ to be zero, we should have either (i) $a/x^{12} - b/x^6 = 0$, whence, $x^6 = a/b$ and, therefore,

$$x = (a/b)^{1/6}, \text{ or (ii) } x = \infty$$

Thus, the two values of x for which $U_{(x)}$ will be zero are $x = (a/b)^{1/6}$ and $x = \infty$.

(ii) For $U_{(x)}$ to be a minimum, we should have $dU_{(x)}/dx = 0$,

$$\text{i.e., } \frac{d}{dx} \left(\frac{a}{x^{12}} - \frac{b}{x^6} \right) = 0$$

$$\text{Or, } -\frac{12a}{x^{13}} + \frac{6b}{x^7} = 0 \text{ whence, } x^6 = \frac{2a}{b} \text{ and, therefore, } x =$$

$$(2a/b)^{1/6}$$

Substituting this value of x in the expression for $U_{(x)}$ above, we have

$$U_{(x)\min} = b^2/4a.$$

Thus, $U_{(x)}$ will have the minimum value $-b^2/4a$ when $x = (2a/b)^{1/6}$.

If, therefore, we plot a graph between x , the distance between the two atoms, and the potential energy $U_{(x)}$, we

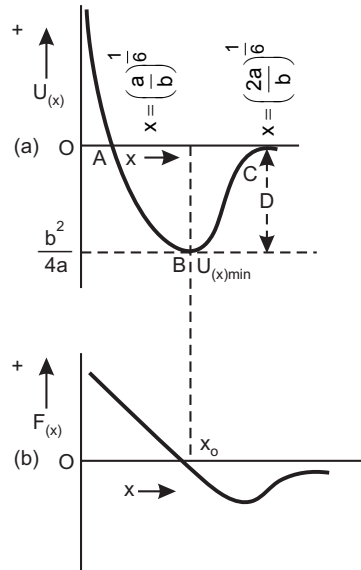


Fig. 5.14

obtain a curve of the form shown in Fig. 5.14(a), which shows that $U_{(x)} = 0$ both at $x = (a/b)^{\frac{1}{6}}$ and at $x = \infty$ and that $U_{(x)}$ is a minimum at $x = (2a/b)^{\frac{1}{6}}$.

(iii) As we know, the force between the two atoms is given by $F = -\frac{d}{dx}U_{(x)}$ which gives $F(x) = \frac{12a}{x^{13}} - \frac{6b}{x^7}$, whence,

$$F_{(x)} = 0, \text{ when } x = (2a/b)^{\frac{1}{6}}$$

This is, therefore, the point of *stable equilibrium* and we denote this value of x by x_0 .

At values of x less than x_0 , i.e., between $x = 0$ and $x_0 = (2a/b)^{\frac{1}{6}}$, F will be *positive*, indicating that the force is one of repulsion or that the atoms repel each other, and that at values of x higher than x_0 , i.e., between $x_0 = (2a/b)^{\frac{1}{6}}$ and ∞ , F will be *negative*, indicating that the force is one of attraction or that the atoms now attract each other.

A graph plotted between $F_{(x)}$ and x comes out to be of the form shown in Fig. 5.14 (b), showing $F_{(x)} = 0$ at $x_0 = (2a/b)^{\frac{1}{6}}$, positive between $x = 0$ and $x = x_0$ and negative between $x = x_0$ and $x = \infty$.

(iv) If one of the atoms remains stationary and the other moves along the x -axis, it will have its minimum potential energy, or stable equilibrium, position at $x_0 = (2a/b)^{\frac{1}{6}}$ from the stationary atom and will therefore oscillate simple harmonically about this position or within the potential well ABC (as explained under §5.9).

(v) Once one of the atoms of the molecule acquires sufficient energy to get over the potential well, it will cease to be bound to the other atom and the molecule will thus get broken or dissociated. This energy to be able to get over the potential well thus gives the *dissociation energy* D of the molecule.

The value of D is thus the difference between the potential energies at the top and the bottom of the potential well ABC , i.e., between 0 and the minimum energy $-b^2/4a$ or equal to $0 - (-b^2/4a) = b^2/4a$.

Thus, *the dissociation energy of the molecule* $= b^2/4a$, meaning thereby that if the value of the kinetic energy at the equilibrium position be $b^2/4a$ (or more), the diatomic molecule will break up into two separate atoms.

5.10 NON-CONSERVATIVE FORCES—GENERAL LAW OF CONSERVATION OF ENERGY

We have seen in §5.6 how when conservative forces alone are concerned, the mechanical energy of a particle, or a system of particles, remains conserved, i.e., the sum of its kinetic energy (T) and potential energy (U) gives its total energy E which remains constant or that the change in its total energy (ΔE) is *zero* i.e.,

$$T + U = E, \text{ a constant. Or, } \Delta T + \Delta U = \Delta E = 0.$$

Since the work done on a particle by a conservative force remains stored up in it in the form of potential energy, it follows that this work done by a conservative force, say, $W_{(c)}$ is equal to the increase in its potential energy, with the sign reversed, i.e., $W_{(c)} = -\Delta U$.

Let us now see whether energy is conserved even if a non-conservative force acts on a particle,—a non-conservative force being one, the work done by which in displacing a particle from one point to another depends upon the actual path traversed between them. Frictional and viscous forces are obvious examples of such forces and, as we know, the longer the path traversed between two

given points, the greater the amount of work done by the frictional force. So that, unlike the case of conservative forces, the work done here in taking a particle from a point A to another point B is not equal and opposite to the work done in taking it from B to A and the total work done from A to B and back is therefore not, zero. In fact, there is loss of kinetic energy either way, *i.e.*, in proceeding from A to B as also in proceeding back from B to A , so that the resultant kinetic energy of the particle is less than its initial value by an amount equal to the work done in performing the round trip from A back to A .

Thus, if we have both, a conservative and a non-conservative, force acting on a particle and if the work done by the two be $W_{(c)}$ and $W_{(N)}$ respectively, then if ΔT be the loss suffered by the particle in its kinetic energy, we have $W_{(c)} + W_{(N)} = \Delta T$. Or, putting $-\Delta U$ for $W_{(c)}$, we have $\Delta T + \Delta U = W_{(N)}$. Obviously, $\Delta T + \Delta U = \Delta E$, the *change in the total energy* of the particle. And, therefore, $\Delta E = W_{(N)}$.

Thus, the change in the total energy of the particle is no longer zero but $W_{(N)}$, the work done by the non-conservative force.

If the non-conservation force be a *frictional force*, the work done by it [$W_{(N)}$] appears in the form of equivalent heat energy, say, equal to Q , where, clearly, $W_{(N)} = -Q$. We, therefore, have

$$\Delta E = -Q \text{ or } \Delta E + Q = 0,$$

i.e., the change in the total energy of the particle is zero or its total energy remains conserved.

Similarly, in the case of other non-conservative forces, the work done by them appears in some other forms of energy, like sound, light etc., but in *all cases, without exception*, the total energy is conserved. *The general law of conservation of energy thus holds good in the case of both conservative and non-conservative forces.*

EXERCISE

- (a) What is meant by a conservative force? Show that if the force between two bodies is of a central kind, it is also conservative.

(b) Show that in the case of a conservative force, the work done around a closed path is zero.

(c) Show that *Lorentz force* is a conservative force.
- Show that for a conservative force field $\vec{F}(r)$ we can define a scalar function $U(r)$ such that $\vec{F}(r) = -\vec{\nabla}U(r)$. Also show that $\text{curl } \vec{F}(r) = 0$.
- State and explain the work-energy principle and write a short note on the utility of *conservation laws*, in general.
- Give a homely example or two of a conservative force and show that the force acting on a particle whose position at any instant t is given by $\vec{r} = a \cos \omega t \hat{i} + b \sin \omega t \hat{j}$ is a conservative force.
- (a) A centripetal force of 18 N is used to keep a 2 kg ball in a uniform circular motion at the end of a string 1 m long. How much work does the force do in each revolution of the ball?

(b) What is the work done by the earth's gravitational force of attraction on a satellite moving in a circular orbit around it? [Ans. (a) 0; (b) 0]
- (a) A particle at a point $(2, 1, 3)$ is acted upon by a force $\vec{F} = 3\hat{i} + 2xy\hat{j} + xz^2\hat{k}$. Calculate the work done in moving it to a point $(4, 1, 3)$.

(b) A particle moves along one quarter of the circumference of a circle of radius 0.5 m . If the force applied be 0.2 N , inclined at an angle of 60° with the tangent to the circle at the point, calculate the work done. [Ans. (a) 12 units; (b) $7.852 \times 10^{-2} \text{ joule}$]
- Investigate in two different ways whether the force $\vec{F} = (y^2 - x^2)\hat{i} + 3xy\hat{j}$ is a conservative one. [Ans. Non-conservative]

[Hint. (i) Show that $\text{curl } \vec{F} \neq 0$. (ii) Evaluate the line integral from the point $(0, 0)$ to the point

(x_0, y_0) along one path, made up of two straight sections $(0, 0)$ to $(x_0, 0)$ and $(x_0, 0)$ to (x_0, y_0) , as shown in Fig. 5.15, and then along the other path made up of two similar straight sections $(0, 0)$ to $(0, y_0)$ and $(0, y_0)$ to (x_0, y_0) , as shown dotted in the Figure. Show that work done along the two paths is not the same.]

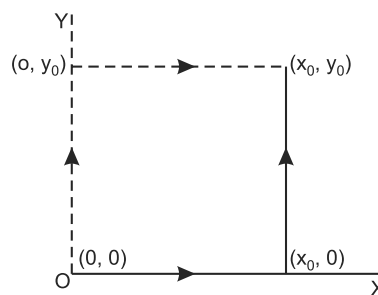


Fig. 5.15

8. (a) If the magnitude of the force of attraction between two particles of masses m_1 and m_2 respectively, and distance x apart, be given by $F = k m_1 m_2 / x^2$, where k is a constant, find (i) the potential energy function and (ii) work required to be done to increase the separation of the masses from $x = x_1$ to $x = x_1 + d$.

[Ans. (i) $U(x) = -k m_1 m_2 / x$, if $V(\infty) = 0$ (ii) $k m_1 m_2 d / x_1 (d + x_1)$]

- (b) Enunciate the principle of conservation of mechanical energy and illustrate by means of a homely example.

Fall under the action of gravity

9. A particle of mass m is attached to a fixed point by means of a string of length l and hangs freely. Show that if it is projected horizontally with a velocity greater than $\sqrt{5gl}$ it will completely describe a vertical circle.
10. Show that for a given initial speed u , a projectile will have the same speed at the same elevation irrespective of its angle of projection.
11. A small block of mass m slides along the frictionless loop-the-loop track shown in Fig. 5.16. Calculate the height h at which the mass just be released on the track to be able to go round the loop of radius R . [Ans. $h = 5R/2$]

[Hint. If v be the velocity acquired by the block in falling through height h , we have $v^2 - 0 = 2gh$. Or, $v^2 = 2gh$. As the mass reaches the top of the loop, let its velocity be v' , so that centrifugal force exerted by the mass on the track upwards $= mv'^2/R$. In order that the block may be able to go round the loop, this must balance the weight mg of the block. So that, $mv'^2/R = mg$, whence, $v'^2 = gR$.

Then, $\frac{1}{2}mv^2 - \frac{1}{2}mv'^2 = \text{work done by the block in covering a vertical distance } 2R \text{ against gravity from } A \text{ to } B$, i.e., $\frac{1}{2}m(v^2 - v'^2) = 2mgR$.

Or, $v^2 - v'^2 = 4gR$.

Substituting the values of v and v' , we have $2gh - gR = 4gR$.

Or, $2h = 5R$, whence, $h = 5R/2$.]

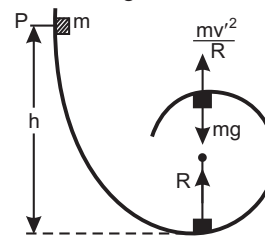


Fig. 5.16

12. A light rigid rod of length l has a mass attached to its end, forming a simple pendulum. It is inverted and then released. What is its speed v at the lowest point and what is the tension T in the suspension at that instant? [Ans. $2\sqrt{gl}; 5mg$]

II-Potential energy in an electric field

13. What is the energy (in electron volts and in ergs) acquired by an alpha particle accelerated through a potential difference of 10^3 volts and what is then its velocity? (Charge on an alpha particle $= +2e$ and its mass $= 4$ times the mass of a proton). [Ans. 2000 eV or 3.2×10^{-9} ergs; 3.09×10^7 cm/sec.]
14. (a) What is the electrostatic potential in volts due to an atomic nucleus of atomic number 10 at a distance of 2 Angstrom units from it? (b) What will be the energy of (i) an electron (ii) a proton, placed at that point? (Charge on the nucleus $= +10e$; 1 Angstrom unit $= 10^{-8}$ cm.) [Ans. (a) 72 volts; (b) (i) -72 eV (ii) +72 eV.]
15. (a) Calculate (i) the intensity of the electric field (in volts/cm), (ii) the potential (in volts) at a distance of 1 A.U. from a proton. (Charge on a proton $= +e = 4.80 \times 10^{-10}$ esu; 1 esu of potential $= 300$ volts; 1 A.U. $= 10^{-8}$ cm.)

- (b) A proton is released from rest at a distance of 1 A.U. from another proton. What is the kinetic energy when the protons have moved infinitely apart from each other? If one proton be kept at rest, what is the terminal velocity acquired by the other? (Mass of proton = $1.67 \times 10^{-24} \text{ gm}$).

[Ans. (a) (i) $1.44 \times 10^9 \text{ volt/cm}$, (ii) $4.8 \times 10^{-2} \text{ esu}$ or 14.4 volts .

(b) $23 \times 10^{-12} \text{ erg}$; $5.25 \times 10^6 \text{ cm/sec}$.]

16. (a) Obtain the distance of closest approach of a particle of mass m and charge q , approaching with velocity v a massive particle carrying a charge Q and supposed to remain stationary.

[Ans. $2Qq/mv^2$]

- (b) Alpha particles of speed $2.0 \times 10^9 \text{ cm/sec}$ are shot into a thin film of gold ($Z = 79$). Deduce the distance of nearest approach between the α -particles and nuclei of gold atoms, assuming Coulomb's law to hold.

[Ans. $2.84 \times 10^{-12} \text{ cm}$]

III—Linear restoring force

17. A mass of 1 kg is supported by a spring of force constant 10^5 dynes/cm . It is pulled down through a distance of 20 cm and then released. Calculate its highest velocity.

[Ans. 200 cm/sec]

18. A spring which does not conform to Hooke's law obeys the force law $F = -Kx^3$. Calculate the potential energy at x , referred to $U = 0$ at $x = 0$. Also calculate the work done in stretching the spring from $x = x_1$ to $x = x_2$.

[Ans. $\frac{1}{4}Kx^4$; $\frac{1}{2}K(x_2^4 - x_1^4)$]

19. An object is attached to a vertical spring and slowly lowered to its equilibrium position. This stretches the spring by an amount d . If the same object is attached to the same vertical spring but permitted to fall instead, through what distance does it stretch the spring?

[Ans. $2d$]

20. Show that a mass M falling freely from a vertical height h on a vertical spring, of force constant C , would produce a maximum compression y_0 in the spring, where $y_0 = \frac{mg}{C} + \frac{1}{C}\sqrt{m^2g^2 + 2mghC}$.

IV—Potential energy curve—potential well

21. What is the potential energy curve of a particle? What significant information does it give about the behaviour of the particle?

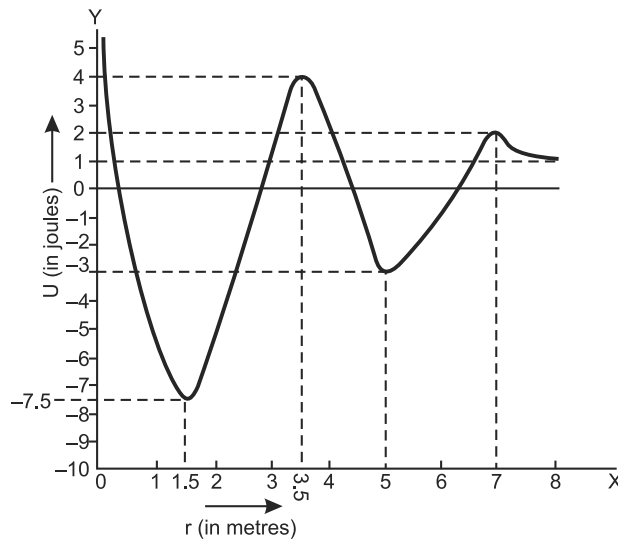
22. What is meant by a potential well (or a potential valley)? Show that for small oscillations of a particle, the well (or the valley) is parabolic in shape. What special significance attaches to this particular shape of the well (or the valley)?

23. The energy of a particle is given by $U = A - \frac{B}{x} + \frac{C}{x^2}$, where A , B and C are all positive constants.

What is the position of stable equilibrium of the particle? What is the *force constant* for small oscillations of the particle about this position? For what values of the total energy can the motion of the particle be bounded (*i.e.*, for what values of the total energy will it be inside a potential well)?

[Ans. $2C/B$; $B^4/8C^3$; for total energy $E < A$]

24. The accompanying figure (Fig. 5.17) shows the potential energy curve of a mass of 0.25 kg , with r in metres and U in joules. Supply the following information from the curve:

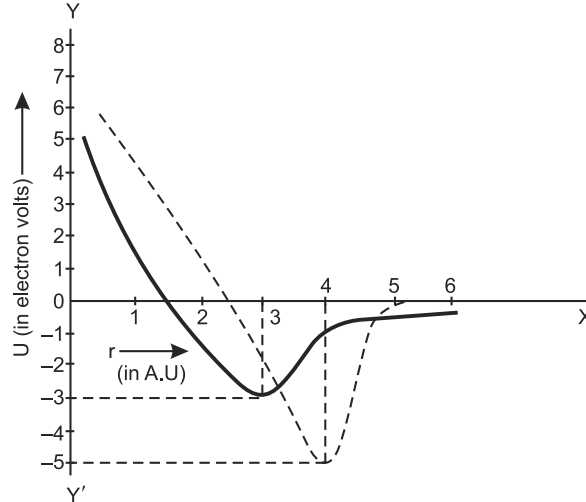
**Fig. 5.17**

- (i) The positions of stable equilibrium of the particle about which it can oscillate.
- (ii) The maximum energy of the oscillation in each case.
- (iii) The kinetic energy with which the particle goes to $r \rightarrow \infty$ on just escaping from the potential well at the larger value of r and its velocity then.
- (iv) The maximum total energy that would make the motion of the particle unbounded.

[Ans. (i) $r = 1.5$ m and $r = 50$ m; (ii) 11.5 joules and 7 joules;

(iii) 2 joules; 4 m/sec; (iv) 4 joules]

25. In Fig. 5.18, we have two potential energy curves of two diatomic molecules, with the interatomic distance (r) in *Angstrom units* and potential energy U in *electron units*. What are the intra-atomic distances (r) in *Angstrom units* and potential energy U in *electron volts*. What are the intra-atomic distances for equilibrium and the binding energies in the two cases?



What does the narrower potential well in the second case signify?

[Ans. 3 A.U. or 3×10^{-8} cm and 4 A.U. or 4×10^{-8} cm; 3eV and 5eV. The narrower potential well in the second case indicates a more rapid change of gradient dU/dr and hence a larger value of the force constant]

6

Chapter

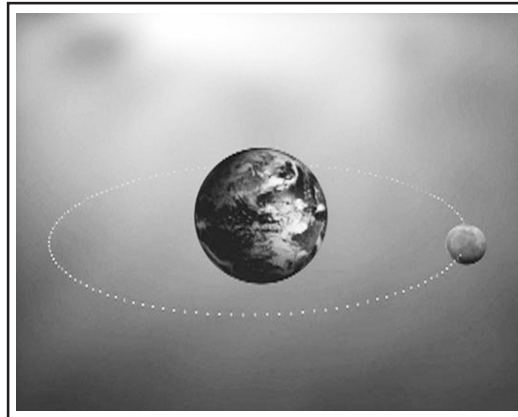
CONSERVATION LAWS

PART II: CONSERVATION OF LINEAR AND ANGULAR MOMENTUM

Brief Contents

Introduction

- 6.1 Conservation of Linear Momentum
- 6.2 Centre of mass
- 6.3 Motion of Velocity of Centre of Mass
- 6.4 Total linear momentum about the centre of mass
- 6.5 System of two particles
- 6.6 Equation of motion of centre of mass
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- 6.9 Collision
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- 6.11 Value of the scattering angle
- 6.12 Systems of variable mass—The Rocket
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- 6.14 Conservation of angular momentum
- 6.15 Motion of a planet in an elliptical orbit around the sun
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- 6.17 Some illustrative examples of conservation of angular momentum
- 6.18 Angular momentum of elementary particles—its quantisation



INTRODUCTION

In the previous chapter we have seen that the law of conservation of energy is an *empirical law* of physics. It concluded that the total amount of energy in an *isolated system* remains constant and is said to be conserved, over time. The consequence of this law is that energy can neither be created nor destroyed: it can only be transformed from one state into the other. The only thing that can happen to energy in a *closed system* is that it can change form: for instance, *Chemical energy* can become *Kinetic energy*.

However, the law of conservation of energy fails to provide complete or satisfactory solutions to a host of problems involving interaction of bodies. Let us take the most familiar example of bullet fired from a gun (or rifle). The law of conservation of energy tells us that the kinetic energies of the forward moving bullet and the backward moving or recoil of gun, along with the heat and sound energies liberated, must together be equal to the chemical energy of the explosive detonated in the gun. It does not tell us, how this total energy is distributed among the bullet, the gun and the atmosphere. And further, *energy* being a *scalar quantity*, the fact of its remaining conserved does not even suggest that the bullet and the gun must move in opposite direction. In dynamics there are many more problems in which we have no knowledge of the active forces actually involved, some additional principle of a vector nature is, therefore, obviously needed. It is the law of *conservation of momentum*, (being vector quantity) that gives the solution to above problems.

Linear and Angular Momenta

Every particle or a body can possess two types of motions, either linear or angular (*i.e.* rotational) motion. Depending upon the type of motion, a particle can have either linear momentum or angular momentum. If a particle of mass m is moving with a linear velocity $\vec{v}$, then it possesses linear momentum, $\vec{p}$ which is expressed as the product of its mass and velocity. Hence linear momentum, $\vec{p} = m \vec{v}$. The another type is angular momentum, denoted by $\vec{J}$ and is defined as moment of its linear momentum. Hence, $\vec{J} = \vec{r} \times \vec{p}$. We will discuss angular momentum, and its conservation at the end of this chapter in detail. Let us study the conservation of linear momentum and its applications.

6.1 CONSERVATION OF LINEAR MOMENTUM

Linear Momentum. The linear momentum of a body is defined as the product of its mass and linear velocity. If m is the mass of the body and $\vec{v}$ its velocity, then

$$\text{Linear momentum, } \vec{p} = m \vec{v}$$

It is a vector quantity. Its units are kg ms^{-1} .

Principle of conservation of Linear momentum. It states “The total linear momentum of a system of particles free from the action of external forces and subjected only to their mutual interaction, remains constant, no matter how complicated the forces are.”

Mathematically, according to Newton’s second law of motion, $\vec{F} = \frac{d\vec{p}}{dt}$ where $\vec{F}$ is the applied external force acting on the system. If for an *isolated system*, the external force is absent, $\vec{F} = 0$

$$\therefore \frac{d\vec{p}}{dt} = 0 \quad \text{or} \quad \vec{p} = \text{a constant}$$

i.e., the total linear momentum of the system remains constant.

Example 1. When a bullet is fired from a gun, the bullet of mass m moves forward with a velocity $\vec{v}$ and the gun of mass M kicks backward (*i.e.* recoils) with a velocity $\vec{V}$.

Solution. Momentum of the bullet in the forward direction, $\vec{p}_1 = m \vec{v}$

Momentum of the gun in the backward direction, $\vec{p}_2 = -M \vec{V}$

Thus, the total linear momentum, after the bullet is fired

$$m \vec{v} - M \vec{V} = 0$$

Also, the linear momentum before the bullet is fired is zero. Thus, the total linear momentum of the bullet and the gun which was zero before the bullet was fired remains the same (*i.e.* zero) even after firing the bullet, *i.e.* the linear momentum is conserved.

Example 2. Imagine a rocket ship about to be fired into space (Fig. 6.1 a). If the rocket ship is initially at rest, its speed is 0, so its linear momentum must be zero. No matter what its mass is, the linear momentum of the rocket = mass $\times$ 0 km/hr = 0.

Solution. When the rocket is fired, then hot gases escape from the rear of the rocket ship, as shown in Fig. 6.1 (b). The momentum of those gases is equal to product of the total mass of the gases (m_g) and their velocity ($\vec{V}_g$) *i.e.*, $-m_g \cdot \vec{V}_g$. The negative sign indicates that the gases are escaping backward, or to the left. The law of conservation of linear momentum says, then, that the rocket has to move in the opposite direction, to the right or the +ve direction with a momentum $M_r \vec{V}_r$. That must be true because, then $-m_g \vec{V}_g$ (from the gases) plus $(+M_r \vec{V}_r)$ (from the rocket) = 0. If you know the mass of the rocket (M_r), its velocity ($\vec{V}_r$) towards right can be calculated.

Impulse. The law of conservation of linear momentum is also valid in case of an impulse, a large force acting during a small time interval. The linear momentum, $\vec{p} = \text{mass} \times \text{velocity} = m \vec{v}$.

Since in accordance with Newton's second law,

$$\vec{F} = d\vec{p}/dt = d(m\vec{v})/dt,$$

we have

$$\vec{F} dt = d(m\vec{v}).$$

integrating which with respect to t , we have $\int_{t_1}^{t_2} \vec{F} dt = \int_{m\vec{v}_1}^{m\vec{v}_2} d(m\vec{v}) = m\vec{v}_2 - m\vec{v}_1$, where $(m\vec{v}_2 - m\vec{v}_1)$

is the change in the momentum of the particle in the interval of time from t_1 to t_2 and $\int_{t_1}^{t_2} \vec{F} dt$, the *impulse* of the force during the *same* interval (the integral being taken because the force may vary during the time interval). So that, we have

Impulse = change in momentum.

Now, in case the external force applied to a particle (or a body) be *zero*, we have

$$\vec{F} = d\vec{p}/dt = d(m\vec{v})/dt = 0, \text{ whence, } \vec{p} = m\vec{v} = \text{a constant},$$

showing that *in the absence of an external force, the momentum of the particle (or the body) remains constant.*

We have already discussed under § 2.10 (ii) the *law of conservation of linear momentum* in the case of a system of two mutually interacting particles, so that, in the absence of an external force on the system,

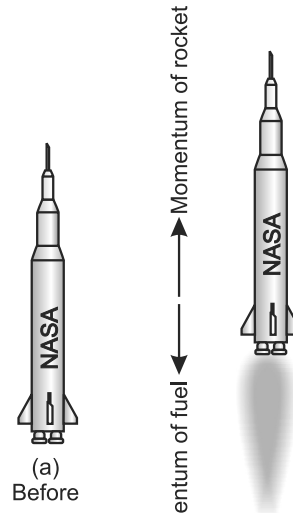


Fig. 6.1

$$\vec{p}_1 + \vec{p}_2 \text{ or } m_1 \vec{v}_1 + m_2 \vec{v}_2 = \text{constant},$$

i.e., the total momentum of the particles remains conserved.

Example 3. It would be interesting to give here an example to show that the law of conservation of momentum enables us to solve a problem which cannot be solved by a straight application of the relation $\vec{F} = m \vec{a}$.

Solution. Suppose a particle of mass m , initially at rest, suddenly explodes into two fragments of masses m_1 and m_2 which fly apart with velocities v_1 and v_2 respectively. Obviously, the forces resulting in the explosion of the particle must be internal forces since no external force has been applied. In the absence of the external force, therefore, the momentum must remain conserved and we should have $m\vec{v} = m_1\vec{v}_1 + m_2\vec{v}_2$.

Since the particle was initially at rest, $\vec{v} = 0$ and therefore, also $m_1\vec{v}_1 + m_2\vec{v}_2 = 0$, whence,

$$v_1/v_2 = -m_2/m_1.$$

showing at once that the velocities of the two fragments must be inversely proportional to their masses and in opposite directions along the same line. This result could not possibly be arrived at from the relation $F = ma$, since we know nothing about the forces that were acting during the explosion. Nor could we derive it from the law of conservation of energy, as already pointed out.

Principle applied to a system of particles

The law may be easily extended to any number of particles, either in fixed relative positions so as to form a rigid body or in the form of a loose conglomeration and thus being free to move into all sorts of different positions.

Let $m_1, m_2, \dots, m_n$ be the masses of the different particles and, in addition to their own interaction with each other, let there also be external forces acting on them, so that they acquire velocities $\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n$ respectively. Then, clearly, their total momentum $\vec{p}$, say, is the vector sum of their individual momenta $\vec{p}_1, \vec{p}_2, \vec{p}_3, \dots$ etc., i.e.,

$$\vec{P} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots \vec{p}_n = m_1 \vec{v}_1 + m_2 \vec{v}_2 + m_3 \vec{v}_3 + \dots m_n \vec{v}_n.$$

Differentiating with respect to t , we have

$$\frac{d\vec{p}}{dt} = \frac{d\vec{p}_1}{dt} + \frac{d\vec{p}_2}{dt} + \dots \frac{d\vec{p}_n}{dt} = \frac{d}{dt}(\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots \vec{p}_n),$$

$$\text{i.e.,} \quad \vec{F} = \vec{F}_1 + \vec{F}_2 + \dots \vec{F}_n,$$

where $\vec{F}_1, \vec{F}_2, \vec{F}_3, \dots, \vec{F}_n$ are the forces acting on the particles of masses $m_1, m_2, m_3, \dots, m_n$ respectively.

Now, the internal forces alone cannot bring about any changes in the momentum of the body since, forming pairs of equal and opposite forces, they give rise to equal and opposite changes of momentum which cancel out. The above forces $\vec{F}_1, \vec{F}_2, \dots, \vec{F}_n$ thus actually represent only the external forces acting on the particles and $\vec{F}$, their resultant. So that if $\vec{F} = 0$, we have $d\vec{p}/dt = 0$ and, therefore $\vec{p} = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots \vec{p}_n$, a constant, implying that even though the individual values of $\vec{p}_1, \vec{p}_2$ etc. may change, their total sum $\vec{p}$ remains unaltered. This is the law of conservation of linear momentum and may formally be stated thus:

Statement:

When the vector sum of the external forces acting upon a system of particles equals zero, the total linear momentum of the system remains constant.

The conservation of linear momentum of a system is a *direct consequence* of the *translational invariance of the potential energy of the system* which, as we know, is one of the results of Galilean invariance.

Thus, considering, for the sake of simplicity, the case of two particles in one dimension, say, along the axis of x , such that their coordinates are x_1 and x_2 , we have their potential energy $U(x_1, x_2)$, depending upon their positions.

Now, Galilean invariance demands that this *P.E.* should remain constant even if each particle is displaced through the *same* distance d , i.e., the potential energy $U[(x_1 + d), (x_2 + d)]$ after the displacement must be the same as before the displacement, viz., $U(x_1, x_2)$. This means that if $U(x_1, x_2) = (x_1 - x_2)^2$, we should also have $U[(x_1 + d), (x_2 + d)]$ equal to $(x_1 - x_2)^2$. Let us see if this is so.

We have $U[(x_1 + d), (x_2 + d)] = [(x_1 + d) - (x_2 + d)]^2 = (x_1 - x_2)^2 = U(x_1, x_2)$, clearly showing that *the potential energy is independent of the displacement d suffered by each particle.*

In general therefore, if the potential energy $U(\vec{r}_1, \vec{r}_2)$ is a function of only $(\vec{r}_1 - \vec{r}_2)$, it is *translationally invariant*.

Since the force on a particle is the negative of $\partial u / \partial x$, we have

$$F_1 = -\partial u / \partial x_1 \text{ and } F_2 = -\partial u / \partial x_2.$$

But U being a function of $(x_1 - x_2)$ only, we have

$$\frac{\partial U}{\partial x_1} = \frac{dU}{dx} \frac{dx}{dx_1} = \frac{dU}{dx} \text{ and } \frac{\partial U}{\partial x_2} = \frac{dU}{dx} \frac{dx}{dx_2} = -\frac{dU}{dx}.$$

And, therefore, $\frac{\partial U}{\partial x_1} = -\frac{\partial U}{\partial x_2}$ or, $F_1 = -F_2$.

So that, the total force acting on the particles which interact with each other is $\vec{F} = \vec{F}_1 + \vec{F}_2 = 0$.

The total force acting on the system of the two particles thus being *zero*, we have, by Newton's second law, in vector form,

$$\frac{d}{dt}(m_1 \vec{v}_1 + m_2 \vec{v}_2) = 0. \text{ Or, } m_1 \vec{v}_1 + m_2 \vec{v}_2 = \text{constant},$$

i.e., *the total momentum of the two particles remains conserved.*

The same argument can be extended to any number of particles, the one and only condition being that the potential energy of the system must be translationally invariant.

The law of conservation of linear momentum is one of the fundamental conservation laws and is also applicable to atomic and nuclear physics even though Newtonian mechanics is not. It is applicable even at relativistic velocities, provided we take the relativistic value of the mass of the particles, given by $m = m_0 / \sqrt{1 - v^2/c^2}$.

6.2. CENTRE OF MASS

When we consider the motion of a system consisting of a large number of particles, say, N , there is one point in it which behaves as though the entire mass of the system (i.e., the sum of the masses of all the N individual particles) were concentrated there and its motion is the same as would ensue if the external forces acting on the system were applied directly to it. This point is called the *centre of mass* of the system. As we shall see as we proceed along, this concept of centre of mass has proved most useful in tackling many a problem and, in particular, those concerned with collision of particles (§ 6.6).

Taking first the simple case of a system of just two particles of masses m_1 and m_2 and of a total mass M , with $\vec{r}_1$ and $\vec{r}_2$ as their position vectors with respect to some origin, if $\vec{R}$ be the position vector of the centre of mass, we have

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{M}$$

Or,
$$M \vec{R} = m_1 \vec{r}_1 + m_2 \vec{r}_2,$$

i.e., the product of the total mass of the system and the position vector of the centre of mass (c.m.) is equal to the sum of the products of the individual masses and their respective position vectors.

The position of the c.m. of the system may thus be easily obtained.

To take a general case, if we have a system consisting of N particles, of masses $m_1, m_2, \dots, m_n$, with $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_n$ as their position vectors at a given instant, the position vector $\vec{R}$ of the centre of mass of the system Fig. 6.2 at that instant is given by the relation $M \vec{R} = m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_n \vec{r}_n$, where M is the total mass of the system.

Clearly, $M = m_1 + m_2 + \dots + m_n = \sum_{k=1}^N m_k$, where m_k is the mass of the k_{th} particle. If, therefore, $\vec{r}_k$ be the position vector of the k_{th} particle, we have

$$\begin{aligned} \vec{R} &= \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_N \vec{r}_N}{m_1 + m_2 + \dots + m_N} \\ &= \frac{\sum_{k=1}^N m_k \vec{r}_k}{\sum_{k=1}^N m_k} = \frac{\sum_{k=1}^N m_k \vec{r}_k}{M}, \end{aligned}$$

where $\sum_{k=1}^N m_k \vec{r}_k$ is called the *first moment of* Z

mass for the system.

Now, $\vec{r}_k = r_k \hat{i} + y_k \hat{j} + z_k \hat{k}$ and $\vec{R} = x \hat{i} + y \hat{j} + z \hat{k}$. So that, if X, Y and Z be the Cartesian coordinates of the centre of mass, we have

$$X = \frac{m_1 x_1 + m_2 x_2 + \dots + m_N x_N}{m_1 + m_2 + \dots + m_N} = \frac{\sum_{k=1}^N m_k x_k}{\sum_{k=1}^N m_k} = \frac{\sum_{k=1}^N m_k x_k}{M}.$$

Or,
$$X = \frac{1}{M} \sum_{k=1}^N m_k x_k,$$

$$Y = \frac{m_1 y_1 + m_2 y_2 + \dots + m_N y_N}{m_1 + m_2 + \dots + m_N} = \frac{\sum_{k=1}^N m_k y_k}{\sum_{k=1}^N m_k} = \frac{\sum_{k=1}^N m_k y_k}{M}.$$

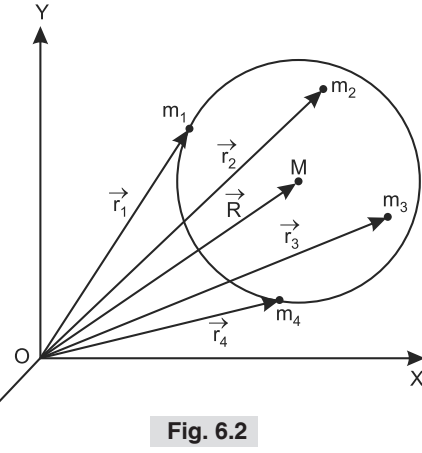


Fig. 6.2

Or,

$$Y = \frac{1}{M} \sum_{k=1}^N m_k y_k,$$

and

$$Z = \frac{m_1 z_1 + m_2 z_2 + \dots + m_N z_N}{m_1 + m_2 + \dots + m_N} = \frac{\sum_{k=1}^N m_k z_k}{\sum_{k=1}^N m_k} = \frac{\sum_{k=1}^N m_k z_k}{M}.$$

Or,

$$Z = \frac{1}{M} \sum_{k=1}^N m_k z_k,$$

Since a rigid body consists of a large number of particles, compactly packed together, it must also have a *centre of mass*. In view, however, of the large number of particles constituting it and the extremely small spacing between them, the distribution of mass may be deemed to be continuous throughout the body. The sign of summation used above is thus replaced by that of integration taken over the whole volume of the body.

If ρ be the density of the body at a point whose position vector is $\vec{r}$, the mass of an element of volume dV there is clearly $dm = \rho dV$. So that, proceeding as above, if $\vec{R}$ be the position vector of the centre of mass of the body, we have

$$\vec{R} = \frac{1}{M} \int_V \vec{r} dm = \frac{\int_V \vec{r} dm}{\int_V dm}$$

and its cartesian coordinates,

$$X = \frac{\int_V x dm}{\int_V dm} = \frac{1}{M} \int_V x dm = \frac{\int_V x \rho dv}{\int_V \rho dv},$$

$$Y = \frac{\int_V y dm}{\int_V dm} = \frac{1}{M} \int_V y dm = \frac{\int_V y \rho dv}{\int_V \rho dv},$$

and

$$Z = \frac{\int_V z dm}{\int_V dm} = \frac{1}{M} \int_V z dm = \frac{\int_V z \rho dv}{\int_V \rho dv},$$

For homogenous bodies of regular geometrical shapes, the centre of mass obviously lies on the point or the line of symmetry as the case may be, as, for example, at the centre in the case of a sphere and on the axis in the case of a cone. In fact, it is located at a point for which $\int dm \vec{r} = 0$ and consequently $\vec{R} = 0$. This integral also, like the summation $\sum m k \vec{r}_k$ (mentioned earlier) is called the *first moment of mass for the system*.

6.3 MOTION OF VELOCITY OF CENTRE OF MASS

Velocity of Centre of Mass. Let us consider the motion of the system consisting of n particles and total mass M assuming that the mass of the system remains constant *i.e.*, no mass enters or

leaves the system.

$$\therefore M \vec{R} = \sum m_i \vec{r}_i = m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_i \vec{r}_i + \dots + m_n \vec{r}_n$$

Differentiating with respect to time, we get

$$M \frac{d\vec{R}}{dt} = m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} + \dots + m_i \frac{d\vec{r}_i}{dt} + \dots + m_n \frac{d\vec{r}_n}{dt}$$

But $\frac{d\vec{R}}{dt} = \vec{V}$ the velocity of centre of mass

and $\frac{d\vec{r}_1}{dt} = \vec{v}_1, \frac{d\vec{r}_2}{dt} = \vec{v}_2 \dots \frac{d\vec{r}_i}{dt} = \vec{v}_i$ and $\frac{d\vec{r}_n}{dt} = \vec{v}_n$

which represent the velocities of individual particles.

$$\therefore M \vec{V} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots + m_i \vec{v}_i + \dots + m_n \vec{v}_n = \sum m_i \vec{v}_i \quad \dots(i)$$

The velocity of centre of mass is given by

$$\vec{V} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots + m_i \vec{v}_i + \dots + m_n \vec{v}_n}{M} = \frac{\sum m_i \vec{v}_i}{M} \quad \dots(ii)$$

From relation (i) we also find the *vector sum of the linear momenta of the individual particles i.e., the total linear momentum of the system is required to the product of the total mass of the system and the velocity of centre of mass.*

In the absence of any external force, the total linear momentum of the system $\vec{P}$ is conserved

$$\therefore M \vec{V} = \sum m_i \vec{v}_i = \vec{P} = \text{a constant}$$

or $\vec{V} = \text{a constant vector}$

Hence the velocity of the centre of mass of a system remains constant if no external force is applied to it.

Acceleration. Differentiating equation (i) with respect to time t , we have

$$M \frac{d\vec{V}}{dt} = m_1 \frac{d\vec{v}_1}{dt} + m_2 \frac{d\vec{v}_2}{dt} + \dots + m_i \frac{d\vec{v}_i}{dt} + \dots + m_n \frac{d\vec{v}_n}{dt}$$

or $M \vec{a} = m_1 \vec{a}_1 + m_2 \vec{a}_2 + \dots + m_i \vec{a}_i + \dots + m_n \vec{a}_n$

where $\vec{a}$ is the acceleration of the centre of mass and $\vec{a}_1, \vec{a}_2$ etc. are the accelerations of individual particles.

The acceleration of the centre of mass is given by

$$\vec{a} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2 + \dots + m_i \vec{a}_i + \dots + m_n \vec{a}_n}{M} = \frac{\sum m_i \vec{a}_i}{M} \quad \dots(iii)$$

According to Newton's second law of motion

$$m_1 \vec{a}_1 = \vec{F}_1, m_2 \vec{a}_2 = \vec{F}_2 \dots m_i \vec{a}_i = \vec{F}_i \dots m_n \vec{a}_n = \vec{F}_n$$

The external forces acting on different particles

$$\therefore M \vec{a} = \vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_i + \dots + \vec{F}_n$$

Thus the product of the total mass of a system and the vector acceleration of the centre of mass is equal to the vector sum of all the external forces acting on the individual particles of the system.

When no external force is acting

$$\therefore \vec{F}_1 + \vec{F}_2 + \dots + \vec{F}_i + \dots + \vec{F}_n = \sum \vec{F}_i = 0 \quad \therefore M \vec{a} = 0 \text{ or } \vec{a} = 0 \text{ vector}$$

Hence in the absence of an external force, the acceleration of the centre of mass is zero and therefore the velocity is a constant vector.

6.4 TOTAL LINEAR MOMENTUM ABOUT THE CENTRE OF MASS

Let C be the centre of mass of a number of particles of mass $m_1, m_2 \dots m_i \dots m_n$ and $\vec{r}_1, \vec{r}_2 \dots \vec{r}_i \dots \vec{r}_n$ their position vectors with respect to the origin O , then

Position vector of centre of mass

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + \dots + m_i \vec{r}_i + \dots + m_n \vec{r}_n}{m_1 + m_2 + \dots + m_i + \dots + m_n} = \frac{\sum m_i \vec{r}_i}{\sum m_i} = \frac{\sum m_i \vec{r}_i}{M} \quad \dots(i)$$

where

$$M = m_1 + m_2 + \dots + m_i + \dots + m_n = \sum m_i$$

Consider one of the particles of mass m_i having position vector $\vec{r}_i$ then its position with respect to the centre of mass C , $\vec{r}_{ci}$ is given by $\vec{R} + \vec{r}_{ci} = \vec{r}_i$

The position of all other particles with respect to the centre of mass C is also given by similar relations.

Substituting $\vec{r}_i = \vec{R} + \vec{r}_{ci}$ in Equation (i), we get

$$\begin{aligned} M \vec{R} &= \sum m_i (\vec{R} + \vec{r}_{ci}) \\ &= \sum m_i \vec{R} + \sum m_i \vec{r}_{ci} = M \vec{R} + \sum m_i \vec{r}_{ci} \end{aligned}$$

$$\text{or } \sum m_i \vec{r}_{ci} = 0 \quad \dots(ii)$$

i.e., the sum of the products of mass and position vector of all the particle about the centre of mass is zero.

Differentiating relation (ii) with respect to t we get $\sum m_i \frac{d\vec{r}_{ci}}{dt} = \sum m_i \vec{v}_{ci} = 0$

because $\frac{d\vec{r}_{ci}}{dt} = \vec{v}_{ci}$ = velocity of the particle of mass m_i relative to centre of mass. $\sum m_i \vec{v}_{ci}$, therefore gives the total linear momentum of all the particles about the centre of mass.

Thus the total linear momentum of the system of particles about the centre of mass is zero.

Note. The centre of mass frame is therefore, sometimes called zero momentum frame.

6.5 SYSTEM OF TWO PARTICLES

(i) **Position vector of centre of mass.** The position of centre of mass for a number of particles is mathematically given by

$$\vec{R} = \frac{\sum m_i \vec{r}_i}{\sum m_i}$$

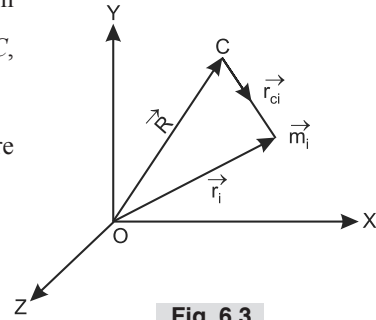


Fig. 6.3

For two mass points m_1 and m_2 , $\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} \dots (i)$

where $\vec{r}_1$ and $\vec{r}_2$ are the vector distances of the particles of mass m_1 and m_2 respectively from the origin O .

If, however the centre of mass lies at the origin of the co-ordinate system as in Fig. 6.5.

$$\vec{R} = 0$$

$$\therefore m_1 \vec{r}_1 + m_2 \vec{r}_2 = 0 \text{ or } \frac{m_1}{m_2} = -\frac{\vec{r}_2}{\vec{r}_1}$$

Thus the centre of mass divides the line joining the two masses in the inverse ratio of the masses i.e., the heavier mass lies nearer the centre of mass of the two particle system. It should be noted that if O is the origin and $\vec{r}_2$ is positive, $\vec{r}_1$ must be negative.

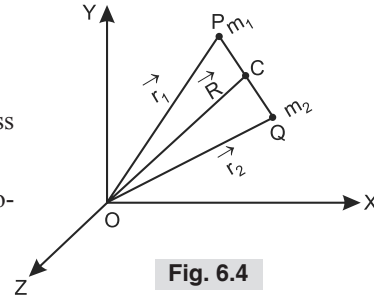


Fig. 6.4

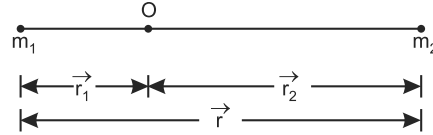


Fig. 6.5

(ii) Velocity of centre of mass. Rewriting equation (i) we have $m_1 \vec{r}_1 + m_2 \vec{r}_2 = (m_1 + m_2) \vec{R}$

Differentiating with respect to time we have $m_1 \dot{\vec{r}}_1 + m_2 \dot{\vec{r}}_2 = (m_1 + m_2) \dot{\vec{R}}$

or Velocity of centre of mass $\vec{R} = \frac{m_1 \dot{\vec{r}}_1 + m_2 \dot{\vec{r}}_2}{m_1 + m_2} \dots (ii)$

or $\vec{V} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} \dots (iii)$

where $\vec{V} = \dot{\vec{R}}$, $\vec{v}_1 = \dot{\vec{r}}_1$ and $\vec{v}_2 = \dot{\vec{r}}_2$.

(iii) Acceleration of centre of mass. Differentiating equation (ii) again with respect to time, we have the acceleration of the centre of mass.

$$\vec{R} = \frac{m_1 \ddot{\vec{r}}_1 + m_2 \ddot{\vec{r}}_2}{m_1 + m_2} \text{ or } \vec{a} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2}{M} \dots (iv)$$

where $\vec{a} = \ddot{\vec{R}}$, $\vec{a}_1 = \ddot{\vec{r}}_1$, $\vec{a}_2 = \ddot{\vec{r}}_2$, and $M = m_1 + m_2$

(iv) Linear momentum of centre of mass. From equation (iii) we have

$$(m_1 + m_2) \vec{V} = m_1 \vec{v}_1 + m_2 \vec{v}_2$$

or $M \vec{V} = m_1 \vec{v}_1 + m_2 \vec{v}_2$ [where $M = m_1 + m_2$]

or $\vec{P} = \vec{p}_1 + \vec{p}_2$

where $\vec{P}$ is the linear momentum of the centre of mass and $m_1 \vec{v}_1$ and $m_2 \vec{v}_2$ the linear momentum of masses m_1 and m_2 respectively.

Hence the linear momentum of a system of two particles is equal to linear momentum of the centre of mass.

6.6 EQUATION OF MOTION OF CENTRE OF MASS

Consider two mass points m_1 and m_2 as shown in Fig. 6.6. If c.m. is the centre of mass for these two mass points and $\vec{R}$ the radius vector for the centre of mass, then

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

where $\vec{r}_1$ is the radius vector for mass point m_1 and $\vec{r}_2$ that for mass point m_2 .

Equivalent one body problem. Suppose there is no external force acting on the system and the only forces are those of mutual interaction, then the velocity of the centre of mass is a *constant*. As the centre of mass must be on the line joining m_1 and m_2 , the force on m_1 due to m_2 as well as the force on m_2 due to m_1 are both directed towards the centre of mass. Hence these forces are *central forces*.

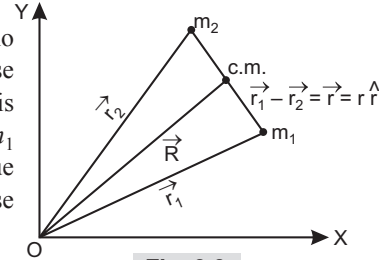


Fig. 6.6

If we denote the force on m_1 as $\vec{F}_{12} = F(r) \hat{r}$ the force on m_2

being equal and opposite will be denoted by $\vec{F}_{21} = -F(r) \hat{r}$.

$$m_1 \frac{d^2 \vec{r}_1}{dt^2} = \vec{F}_{12} = F(r) \hat{r} \text{ or } \frac{d^2 \vec{r}_1}{dt^2} = \frac{1}{m_1} F(r) \hat{r} \quad \dots(i)$$

and

$$m_2 \frac{d^2 \vec{r}_2}{dt^2} = \vec{F}_{21} = -F(r) \hat{r} \text{ or } \frac{d^2 \vec{r}_2}{dt^2} = -\frac{1}{m_2} F(r) \hat{r} \quad \dots(ii)$$

Subtracting (ii) and (i) we have $\frac{d^2 \vec{r}_1}{dt^2} - \frac{d^2 \vec{r}_2}{dt^2} = \left(\frac{1}{m_1} + \frac{1}{m_2} \right) F(r) \hat{r}$

Now

$$\frac{d^2 \vec{r}_1}{dt^2} - \frac{d^2 \vec{r}_2}{dt^2} = \frac{d^2}{dt^2} (\vec{r}_1 - \vec{r}_2) = \frac{d^2 \vec{r}}{dt^2} \quad [\because \vec{r} = \vec{r}_1 - \vec{r}_2]$$

If we put $\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2} = \frac{m_1 + m_2}{m_1 m_2}$ or $\mu = \frac{m_1 m_2}{m_1 + m_2}$

then,

$$\frac{d^2 \vec{r}}{dt^2} = \left(\frac{1}{m_1} + \frac{1}{m_2} \right) F(r) \hat{r} = \frac{1}{\mu} F(r) \hat{r}$$

or

$$\mu \vec{\ddot{r}} = F(r) \hat{r} \quad \dots(iii)$$

i.e. the system behaves as a single particle of mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$. Here, μ is known as the **reduced**

mass of the system and acts at the point known as **centre of mass**. The relation given by Eq. (iii) is known as *equation of motion* of the centre of mass of a particle having mass equal to the reduced mass (μ) at a vector distance $\vec{r}$ from one of the particles to the other and shows that two separate equations of motion (i) and (ii) have been reduced to a single equation (iii) involving reduced mass (μ) concept.

We have thus, reduced the two body problem to a one body problem.

6.7 MOTION OF REDUCED MASS UNDER INVERSE SQUARE FORCE

The most familiar example of inverse square force is the gravitational force. For forces of gravitational attraction between the two point masses m_1 and m_2 .

$$\vec{F}(r) = -\frac{Gm_1m_2}{r^2}\hat{r} \text{ or } \mu \frac{d^2\vec{r}}{dt^2} = \mu \vec{\ddot{r}} = -\frac{Gm_1m_2}{r^2}\hat{r}$$

$$\text{or, } \frac{m_1m_2}{m_1+m_2} \vec{\ddot{r}} = -\frac{Gm_1m_2}{r^2}\hat{r}$$

$$\therefore \vec{\ddot{r}} = -\frac{GM}{r^2}\hat{r} \text{ where } M = m_1 + m_2$$

This is clearly the equation of motion of a particle of unit mass at a vector distance $\vec{r}$ (equal to the distance between the two particles) from a fixed mass $M = (m_1 + m_2)$ exerting a force of attraction on it. Further, the acceleration of one mass with respect to the other mass will appear to be the same but in opposite direction whether the observer is at mass m_1 or m_2 .

6.8 LABORATORY CENTRE-OF-MASS AND FRAMES OF REFERENCE

A frame of reference carried by the centre of mass of an *isolated system* of particles (*i.e.*, a system not subjected to any external forces) is called the *centre-of-mass* or the *C-frame of reference*.

In this frame, obviously, $\vec{R}$ (the position vector of the centre of mass) is equal to zero and hence the velocity of the centre of mass, *i.e.*, $\vec{V} = d\vec{R}/dt$ also equal to zero. In consequence, the linear momentum $\vec{p}$ of the system too, given by $\vec{p} = M\vec{V} = \sum m_k \vec{v}_k$ is zero. The C-frame of reference is, therefore, also called the *zero-momentum frame*. And, since in the absence of external forces, the centre of mass, and hence the C-frame, moves with constant velocity, it is also an *inertial frame*. This makes it particularly useful in solving many a problem which are difficult to solve in the laboratory frame of reference, as we shall presently see. Indeed, it has become almost customary to deal with all collisions in Nuclear Physics in this frame of reference alone.

In the study of collisions between two particles we shall come across the laboratory system or laboratory frame of references and the centre of mass system or centre of mass frame of reference.

A reference frame is the space determined by a rigid body regarded as the base. The rigid body is supposed to extend in all directions as far as necessary. A point in space is located by the three coordinates taken with respect to the origin of the reference system.

If the origin of the reference system is a point rigidly fixed to the laboratory it is known as the laboratory frame.

The laboratory frame is inertial so long as earth is taken to be an inertial frame.

Centre of mass system (Frame of reference). *If the origin of the reference system is a point rigidly fixed to the centre of mass of a system of particles on which no external force is acting it is known as the centre of mass frame of reference.*

In the centre of mass reference frame the position vector of the centre of mass $\vec{R} = 0$ as the centre of mass is itself the origin of the reference system.

$\therefore$ The velocity of centre mass $\vec{V} = \frac{d\vec{R}}{dt} = 0$ and the linear momentum $\vec{p} = M\vec{V}$ of the system is also = 0. Hence it is known as a *zero momentum frame*.

Advantages of studying collision process in centre of mass system. (i) In the absence of any external force the velocity of the centre of mass is a *constant*. In other words, the centre of mass

reference frame moves with a constant velocity with respect to the laboratory frame. *Hence the centre of mass frame is also an inertial frame.*

Various physical quantities measured in the two systems are related to each other by Galilean transformations provided the velocity of centre of mass is small as compared to the velocity of light.

(ii) A system of two particles requires six co-ordinates to describe the motion in the laboratory system. Three co-ordinates are required to describe the motion of centre of mass and three more co-ordinates are required to describe the relative motion. But in the centre of mass frame we require only three co-ordinates as the centre of mass is itself at rest in this frame.

The discussion of a collision process, therefore, becomes much simpler in the centre of mass frame of reference than in the laboratory frame.

6.9 COLLISION

When two bodies are approaching each other, a force comes into play between them for a finite time and brings about a measurable change in their velocities, momenta and energy according to the respective laws of conservation, a collision is said to have taken place.

Contrary to the meaning of the term ‘collision’ in our everyday parlance, in Physics it does not necessarily mean one particle (or body) ‘striking’ against another. Indeed, two particles may not even ‘touch’ each other* and may still be said to collide. All that is implied is that as the particles approach each other, (i) *a large force for a relatively short time (i.e., an impulse) acts on each colliding particle*, (ii) *the motion of a particles (or, at least, of one of the particles) is changed rather abruptly* and (iii) *the total momentum (as also the total energy) of the particles remains conserved.*

The essence of a collision is, in fact, *a redistribution of the total momentum of the particles.* This is the reason why the law of conservation of momentum is simply indispensable in dealing with the phenomenon of collision between particles.

During collision, the force of some interaction comes into play between the two colliding particles or systems for a finite small time which brings about a change in their relative motions. If the collision acts for a time t_0 to $t_0 + \Delta t$, then a time t less than t_0 is known as *time before collision* and a time t greater than $t_0 + \Delta t$, is called *time after collision*.

The collision is termed **scattering** if the nature of particles does not change after collision. Familiar examples of collision or scattering are the deflection of a comet as it passes near the solar system and the deflection of an α -particle by an atomic nucleus. The study of collision is of particular importance in atomic and nuclear physics. The bodies involved may be atoms, nuclei or various elementary particles such as electrons, protons, *etc.*

There are two types of collisions:

(1) Elastic Collision, and (2) Inelastic Collision

(1) Elastic scattering (or Elastic collision). A collision (scattering) is said to be an elastic collision if (i) the final particles after collision are the same as the initial particles before collision, (ii) the sum of the kinetic energies of the particles after collision is the same as the sum of the kinetic energies of the particle before collision.

(2) Inelastic scattering (or Inelastic collision). A collision is said to be an inelastic collision if

- (i) The final particles after collision are the same as the initial particles before collision.
- (ii) The sum of the kinetic energies of the particles after collision is either more or less than the sum of the kinetic energies of the particles before collision.

Some of the collisions between atomic, nuclear and fundamental particles belong to this class.

* As for example, in the case of an alpha particle and a nucleus of gold—due, no doubt, to the electrostatic force of repulsion between them.

In such a collision between two particles, therefore, we have $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$ as also $\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$, where m_1 and m_2 are the respective masses of the two particles and u_1, u_2 and v_1, v_2 their velocities *before* and *after* the collision, *i.e.*, initially before the interacting forces between the particles have yet started acting and finally, after these forces have again been reduced effectively to zero.

A perfectly inelastic collision, in which the particles stick permanently together on impact and the loss of kinetic energy is the maximum consistent with the requirements of conservation of momentum. A bullet remaining embedded in a target is one such example.

In either case, however, since the interacting forces after the collision become effectively zero, the potential energy of the system remains the same both before and after the collision.

In the case of inelastic collision between larger particles, the loss of kinetic energy occurs mostly in the form of heat energy due to the increased vibrations of the constituent atoms of the particles. On the other hand, in the case of inelastic atomic collisions, an atom may absorb some energy and get into what is called an 'excited state', with one or more of its electrons moving into a higher orbit or a higher energy level, the energy so absorbed (and remaining in the potential form) being referred to as 'excitation energy' (ξ). The final kinetic energy of the system is thus less by this much amount than the initial one, *i.e.*, *final kinetic energy of the system* + ξ = *initial kinetic energy of the system* or, $\frac{1}{2}m_1\vec{v}_1^2 + \frac{1}{2}m_2\vec{v}_2^2 + \xi = \frac{1}{2}m_1\vec{u}_1^2 + \frac{1}{2}m_2\vec{u}_2^2$.

In case of the atom be already in an excited state before collision, it may come into its normal state on collision, *i.e.*, the electron (or electrons) may fall back to the original orbit, thereby releasing the potential energy ξ absorbed during excitation, resulting in an increase in the kinetic energy of the system after collision. Thus, in the case, we have *final K.E. of the system* - ξ = *initial kinetic energy of the system*. Or, $\frac{1}{2}m_1\vec{v}_1^2 + \frac{1}{2}m_2\vec{v}_2^2 - \xi = \frac{1}{2}m_1\vec{u}_1^2 + \frac{1}{2}m_2\vec{u}_2^2$.

It is to be carefully noted here that in the case of inelastic collision and reaction, only the sum of the kinetic energies after collision is different from the sum of the kinetic energies before collision but the total energy of the system, including kinetic energy, potential energy and any other form of energy remains conserved.

6.10 CALCULATION OF FINAL VELOCITIES OF COLLIDING PARTICLES

The velocities of the particles after a collision can be easily calculated by solving their equations of motion if we know their velocities before collision and the forces acting during the collision. Unfortunately, these interacting forces are not always known to us. Even so, we can obtain the velocities of the particles after collision from those before collision in case (i) *the collision is perfectly in-elastic* and (ii) *the collision is elastic but one-dimensional, i.e.*, in which the relative motion of the particles before and after the collision is along the *same* line.

In case of the elastic collision be two or three-dimensional, the velocities of the particles after collision can be determined only if we also know the direction of motion of the particles after the collision. (See case (iii) below).

(I) Inelastic collision. In the case of a perfectly inelastic collision, if m_1 and m_2 be the masses of the two colliding particles and $\vec{u}_1$ and $\vec{u}_2$ their velocities before collision, we have

$$\text{momentum of the particles before collision} = m_1\vec{u}_1 + m_2\vec{u}_2.$$

After the collision, the two particles stick together and move with a velocity $\vec{v}$, say. So that,

$$\text{momentum of the particles after collision} = (m_1 + m_2)\vec{v}.$$

Since momentum is always conserved in a collision, we have

$$(m_1 + m_2) \vec{v} = m_1 \vec{u}_1 + m_2 \vec{u}_2, \text{ whence } \vec{v} = \frac{m_1 \vec{u}_1 + m_2 \vec{u}_2}{m_1 + m_2}.$$

Thus, knowing the masses of the two particles and their initial velocities, we can easily obtain their common velocity ($\vec{v}$) after the collision.

In case the second particle be at rest, $\vec{u}_2 = 0$. So that, we shall have $(m_1 + m_2) \vec{v} = m_1 \vec{u}_1$, whence, $\vec{v} = \frac{m_1}{m_1 + m_2} \vec{u}_1$

That there will be a loss of *K.E.* may be easily seen from the fact that *K.E. of the particles before collision* = $\frac{1}{2} m_1 u_1^2$ [the second particle being at rest, $\vec{u}_2 = 0$].

$$\text{and } K.E. \text{ of the particles after collision} = \frac{1}{2} (m_1 + m_2) v^2,$$

$$\begin{aligned} \therefore \frac{K.E. \text{ after collision}}{K.E. \text{ before collision}} &= \frac{\frac{1}{2} (m_1 + m_2) v^2}{\frac{1}{2} m_1 u_1^2} \\ &= \frac{1}{2} (m_1 + m_2) \left(\frac{m_1 u_1}{m_1 + m_2} \right)^2 \frac{1}{\frac{1}{2} m_1 u_1^2} \\ &= \frac{m_1}{m_1 + m_2}, \text{ i.e., } < 1 \end{aligned}$$

showing that *K.E. after collision* < *K.E. before collision*.

As already pointed out, this loss of *K.E.* on collision manifests itself in the form of heat energy in the system due to increased vibrations of the atoms of the two particles.

Let us now consider the problem if the two particles be in the *centre-of-mass frame of reference* in which the centre of mass remains at rest and hence the momentum both before and after the collision is *zero*.

Since the velocity of the centre of mass in the laboratory frame of reference is $\vec{v} = m_1 \vec{u}_1 / (m_1 + m_2)$ in accordance with relation (iii), § 6.3, it follows that the centre of mass frame moves this velocity ($\vec{V}$) relative to the laboratory frame. The initial velocities of the two particles in this frame. The initial velocities of the two particles in this frame are, therefore,

$$\vec{u}'_1 = \vec{u}_1 - \vec{V} = \vec{u}_1 - m_1 \vec{u}_1 / (m_1 + m_2) = m_2 \vec{u}_1 / (m_1 + m_2)$$

$$\text{and } \vec{u}'_2 = \vec{u}_2 - \vec{V} = 0 - \vec{V} = -m_1 \vec{u}_1 / (m_1 + m_2).$$

After the collision, the two particles stick together, their combined mass being $(m_1 + m_2)$ but they must be at rest since their momentum in this frame must remain *zero*. Relative to the laboratory frame, however the, velocity of the combined mass in the centre-of-mass frame must naturally be the same as that of the frame itself, viz., $\vec{V} = m_1 \vec{u}_1 / (m_1 + m_2)$ which, it will be readily seen, is the same as the velocity $\vec{v}$ of the combined mass in the laboratory frame.

(II) Elastic one-dimensional collision. Again, if m_1 and m_2 be the masses of two particles, $\vec{u}_1, \vec{u}_2$ and $\vec{v}_1, \vec{v}_2$, their respective velocities before and after a *head-on collision along the line joining their centres*, we have,

$$\overset{\rightarrow}{m_1 u_1} + \overset{\rightarrow}{m_2 u_2} = \overset{\rightarrow}{m_1 v_1} + \overset{\rightarrow}{m_2 v_2} \quad \dots(i)$$

$$\text{and K.E. equation, } \frac{1}{2} m_1 \vec{u}_1^2 + \frac{1}{2} m_2 \vec{u}_2^2 = \frac{1}{2} m_1 \vec{v}_1^2 + \frac{1}{2} m_2 \vec{v}_2^2 \quad \dots(ii)$$

Or, writing these two equations as

$$m_1(u_1 - v_1) = m_2(v_2 - u_2) \quad \dots(iii)$$

$$\text{and} \quad m_1(u_1^2 - v_1^2) = m_2(v_2^2 - u_2^2) \quad \dots(iv)$$

and dividing the latter by the former, we have

$$u_1 + v_1 = v_2 + u_2. \quad \text{Or, } u_1 - u_2 = v_2 - v_1, \quad \dots(v)$$

showing that *in an elastic one-dimensional collision, the relative velocity with which the two particles approach each other before collision is equal to the relative velocity with which they recede from each other after collision.*

As for the values of the velocities v_1 and v_2 of the particles after collision, we have from relation (v) above

$$v_2 = v_1 + u_1 - u_2 \quad \text{and} \quad v_1 = v_2 - u_1 + u_2$$

Substituting this value of v_2 in relation (iii) above and solving for v_1 and, similarly, substituting the value of v_1 in (iii) above and solving for v_2 , we have

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 + \left(\frac{2m_2}{m_1 + m_2} \right) u_2 \quad \dots(vi)$$

$$\text{and} \quad v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1 + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) u_2. \quad \dots(vii)$$

Now, the following special cases of interest arise:

(a) **When the colliding particles have the same mass.** In this case, $m_1 = m_2$. So that, from relations (vi) and (vii) we have $v_1 = u_2$ and $v_2 = u_1$, i.e., *in one-dimensional elastic collision of two particles of equal mass, the particles simply interchange their velocities on collision.*

(b) **If one of the particles be also initially at rest.** Let the second particle be at rest. Then, $u_2 = 0$ and, therefore, from relations (vi) and (vii), we clearly have

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1 \quad \dots(viii)$$

$$\text{and} \quad v_2 = \left(\frac{2m_1}{m_1 + m_2} \right) u_1 \quad \dots(ix)$$

and, since $m_1 = m_2$, we have $v_1 = 0$ and $v_2 = u_1$,

i.e., *the first particle comes to rest on collision and the second particle acquires the initial velocity of the first.*

(c) **If the particle at rest be very much more massive than the other.** Let the second particle, at rest, be much more massive than the first, so that m_1 is negligible compared with m_2 . Then, we have, from relations (viii) and (ix),

$$v_1 \approx -u_1 \quad \text{and} \quad v_2 \approx 0,$$

indicating that *when a lighter particle collides against a much more massive particle at rest the latter continues to remain at rest and the velocity of the former gets reversed.*

A common example of this type is the dropping of a steel ball on an equally hard horizontal surface on the ground. The collision thus occurs between the ball and the earth. Since the latter is very much more massive and is at rest, the velocity of the ball gets reversed on impact and it rises to the same height from which it was dropped.

(d) **If the particle at rest be very much lighter than the other.** Again, let the second particle be at rest and very much lighter than the first, so that now m_2 is negligible compared with m_1 . Then, we have, from relations (viii) and (ix) above,

$$v_1 \approx u_1 \text{ and } v_2 \approx 2u_1$$

i.e., the velocity of the massive particle remains practically unaltered on collision with the lighter particle at rest and the lighter particle acquires nearly twice the initial velocity of the massive particle.

We can see from the above why in a reactor, hydrogen is used as a moderator for slowing down neutrons, produced by the fission of Uranium atoms, so as to enable them to produce more fissions. It is obviously because the hydrogen nucleus (*i.e.*, the proton) has nearly the same mass as a neutron and hence in a head-on collision with a hydrogen nucleus at rest, the neutrons are thus almost brought to rest or greatly slowed down. If a massive nucleus like that of lead were to be used as the target, the neutrons will simply bounce back from it with the same velocity with which they impinge on it; and if the target were a lighter one, like an electron, for example, the neutrons will simply continue to move with their original velocity after collision with it. There are, of course, other factors too in the choice of a suitable moderator but from considerations of momentum and energy, a hydrogen nucleus as a stationary target is one of the best bets.

(iii) **Elastic collision in two or three dimensions.** In the case of a two or three-dimensional elastic collision between two particles, the initial velocities of the particles and the conservation laws of momentum and energy alone are not enough for the determination of the velocities of the particles after collision, because the velocity of each particle has three components but we have only four known relations between them, *viz.*, three for the conservation of momentum for each of the three dimensions and one for the conservation of energy. More information is, therefore, needed to be able to calculate the final velocities of the particles and the simplest one is the angle of deflection of one of the particles.

Let us discuss the problem in both the laboratory frame of reference and the centre-of-mass frame of reference.

(a) **In the laboratory frame of reference.** Let a particle of mass m_1 , moving with velocity $\vec{u}_1$ collide with a particle of mass m_2 , initially at rest (*i.e.*, $u_2 = 0$) in the laboratory frame of reference (Fig. 6.7). Let it be deflected or *scattered* at an angle θ_1 with its initial direction, after collision, with velocity $\vec{v}_1$, where $\vec{u}_1$ and $\vec{v}_1$ lie in the X - Y plane. Obviously, then, the velocity $\vec{v}_2$ of m_2 , after collision, making an angle θ_2 with the original direction, will have no component in the z -direction for the simple reason that $m_1\vec{u}_1$ and $m_1\vec{v}_1$ have no z -components. Now, from the law of conservation of momentum, we have

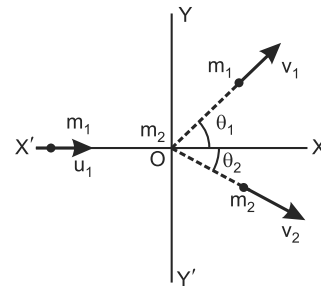


Fig. 6.7

for the x -component of the motion,

$$m_1 u_1 = m_1 v_1 \cos \theta_1 + m_2 v_2 \cos \theta_2 \quad \dots(i)$$

$$\text{and for the } y\text{-component of the motion } 0 = m_1 v_1 \sin \theta_1 - m_2 v_2 \sin \theta_2 \quad \dots(ii)$$

And, from the law of conservation of energy, we have

$$\frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2. \quad \dots(iii)$$

If we know the initial conditions, *viz.*, m_1 , m_2 and u_1 , we have to determine four unknown quantities, *i.e.*, v_1 , v_2 , θ_1 and θ_2 and have only three equations connecting them. We must, therefore, be given the value of one more quantity, such as θ_1 , to be able to determine the motion after collision.

Solving these equations, then, for v_1 , v_2 and θ_2 , in the particular case when $m_1 = m_2$, i.e., when the two masses are equal, we obtain $v_1 = u_1 \cos \theta_1$ and from relation (iii), therefore, $v_2^2 = u_1^2 - v_1^2$, whence v_2 can be obtained.

Finally, $\sin \theta_2 = \frac{v_1}{v_2} \sin \theta_1$. So that, knowing v_1 , v_2 and θ_1 (given), we can also determine θ_2 , where the sum of θ_1 and θ_2 is always 90° , showing that the two equal masses move, after collision, in directions perpendicular to each other.

This method is somewhat lengthy and laborious. If, however, we view the problem in the centre-of-mass frame, its solution becomes very much simpler and perhaps also a little more instructive, as will be seen from the following.

(b) In the centre-of-mass frame of reference. As pointed out earlier under case (i) above, the centre-of-mass frame of reference moves with velocity $\vec{V} = m_1 \vec{u}_1 / (m_1 + m_2)$ relative to the laboratory frame of reference, so that the centre of mass remains at rest in this frame. The initial and final velocities of the two particles (m_1 and m_2) in this frame of reference are thus given by

$$\vec{u}_1 = \vec{u}_1 - \vec{V}, \vec{u}_2 = -\vec{V}, \vec{v}_1' = \vec{v}_1 - \vec{V} \quad \text{and} \quad \vec{v}_2' = \vec{v}_2 - \vec{V}.$$

Since the centre of mass throughout remains at rest in this reference frame, the total momentum is always zero and hence the momentum of the two particles must always be equal and opposite, clearly implying that the velocities $\vec{v}_1'$ and of $\vec{v}_2'$ the two particles, after collision, must be oppositely directed and inclined at the same angle to the initial direction of motion of the particles, as shown in Fig. 6.8. We thus have

$$m_1 \vec{u}_1' = -m_2 \vec{u}_2' \quad \dots (i)$$

$$m_1 \vec{v}_1' = -m_2 \vec{v}_2' \quad \dots (ii)$$

$$\text{Whence, } \vec{v}_2' = \frac{m_1}{m_2} \vec{v}_1' \quad \text{and} \quad \vec{u}_2' = \frac{m_1}{m_2} \vec{u}_1'$$

Since total energy remains conserved on collision, we have

$$\frac{1}{2} m_1 u_1'^2 + \frac{1}{2} m_2 u_2'^2 = \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 v_2'^2. \quad \dots (iii)$$

Substituting the values of $\vec{u}_2'$ and $\vec{v}_2'$, obtained above, we have

$$\frac{1}{2} m_1 u_1'^2 + \frac{1}{2} m_2 \left(\frac{m_1^2 u_1'^2}{m_2^2} \right) = \frac{1}{2} m_1 v_1'^2 + \frac{1}{2} m_2 \left(\frac{m_1^2 v_1'^2}{m_2^2} \right)$$

$$\text{Or, } \frac{1}{2} m_1 u_1'^2 \left(1 + \frac{m_1}{m_2} \right) = \frac{1}{2} m_1 v_1'^2 \left(1 + \frac{m_1}{m_2} \right)$$

$$\text{which gives } \frac{1}{2} m_1 u_1'^2 = \frac{1}{2} m_1 v_1'^2. \quad \text{Or, } u_1' = v_1'.$$

* This may be easily shown by substituting the values of $\vec{u}_1'$ and $\vec{u}_2'$. Thus,

$$m_1 \vec{u}_1' = m_1 (\vec{u}_1 - \vec{V}) = m_1 \left(\vec{u}_1 - \frac{m_1 \vec{u}_1}{m_1 + m_2} \right) = \frac{m_1 m_2 \vec{u}_1}{m_1 + m_2}.$$

$$\text{And } -m_2 \vec{u}_2' = m_2 \vec{V} = m_2 \left(\frac{m_1 \vec{u}_1}{m_1 + m_2} \right) = \frac{m_1 m_2 \vec{u}_1}{m_1 + m_2}$$

Similarly, it can be shown that $m_2 \vec{v}_1' = -m_1 \vec{v}_2'$

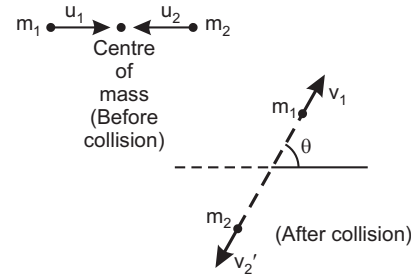


Fig. 6.8

And, hence from energy relation (iii) above, $\vec{u}'_2 = \vec{v}'_2$.

Thus, in the centre-of-mass frame of reference, the magnitudes of the velocities of the particles remain unaltered in an elastic collision.

Advantages of Studying Collision Process. Various types of interactions and forces are operating in nature at microscopic as well as macroscopic levels. A study of collision process *i.e.* relative motion between the two interacting particles helps us to understand the basic nature and characteristics of these interactions and forces. This is done by measuring the initial and final energy and linear as well as angular momentum of the participating particles in accordance with the laws of conservation of energy and momentum.

The deflection of the path of a comet passing near the solar system and the elliptic path of planets within the solar system helps us to understand the nature of gravitational forces.

Collision between molecules of gases according to Kinetic theory leads to understanding of inter-atomic and intermolecular forces.

Scattering of α -particles in Rutherford scattering, scattering of high energy photons in Compton scattering, nuclear reactions involving high energy electrons, protons, neutrons, deuterons and α -particles, scattering of slow neutrons by nuclei and emission of α and β particles from radioactive nuclei help us to understand the nature of nuclear forces, properties of the nucleus and atomic structure.

6.11 VALUE OF THE SCATTERING ANGLE

(i) In the centre-of-mass frame of reference. In this frame of reference, there are absolutely *no limitations* on the value of the scattering angle θ , so that it can have *all* possible values.

(ii) In the laboratory frame of reference. In this frame of reference, there are some restrictions on the value of the scattering angle, θ_1 , (Fig. 6.7), as will be clear from the following:

We have $\tan \theta_1 = v_1 \sin \theta / v_1 \cos \theta_1$.

Now, the y -component of the final velocity of the first particle being the same in the laboratory reference frame as well as the center-of-mass frame, we have $v_1 \sin \theta_1 = v'_1 \sin \theta$. And, since the x -components of this velocity differ by V in the two frames of reference, we have $v_1 \cos \theta_1 = v'_1 \cos \theta + V$. Substituting these values in the relation above, we, therefore, have

$$\tan \theta_1 = \frac{v'_1 \sin \theta}{v'_1 \cos \theta + V} = \frac{\sin \theta}{\cos \theta + (V/v'_1)} \quad \dots(i)$$

As we know,

$$\begin{aligned} V &= \frac{m_1 u_1}{m_1 + m_2} = \frac{m_1 (u'_1 + V)}{m_1 + m_2} \\ &= \frac{m_1 u'_1}{m_1 + m_2} + \frac{m_1 V}{m_1 + m_2} \quad [\because u_1 = u'_1 + V] \end{aligned}$$

$$\text{Or,} \quad V \left(1 - \frac{m_1}{m_1 + m_2} \right) = \frac{m_1 u'_1}{m_1 + m_2}.$$

$$\text{Or,} \quad V \left(\frac{m_2}{m_1 + m_2} \right) = \frac{m_1 u'_1}{m_1 + m_2},$$

$$\text{whence,} \quad V = \frac{m_1}{m_2} u'_1 = \frac{m_1}{m_2} v'_1 \quad [\because u'_1 = v'_1]$$

Substituting this value of V in relation (iv) above, we have

$$\tan \theta_1 = \frac{\sin \theta}{\cos \theta + (m_1/m_2)} \quad \dots(ii)$$

This shows that

- (a) If $m_1 > m_2$, so that m_1/m_2 a little > 1 , the denominator can never be zero and hence $\tan \theta_1$ can never be ∞ . And, therefore, θ_1 must be less than 90° ;
- (b) If $m_1 = m_2$, so that $m_1/m_2 = 1$, the denominator can be zero for $\cos \theta = -1$ and, therefore, $\tan \theta_1$ can be ∞ . Thus, in this case, θ_1 can have any value up to the limiting value of 90° ;
- (c) If $m_1 < m_2$, so that $m_1/m_2 < 1$, the value of $\tan \theta_1$ can also be negative. In this case alone, therefore, **all values of θ_1 can be possible**.

It follows from case (a) that in an elastic collision, *if a massive particle collides against a lighter one at rest, it can never bounce back along its original path*. On the other hand, it follows from case (c) that *if a lighter particle collides against a massive one at rest, it may well bounce back along its original path*. And, it follows from case (b) that in an elastic collision between two particles of equal mass, one of which is initially at rest, the two particles move at right angles to each other after the collision, as mentioned already under 6.10(III) (a).

6.12. SYSTEMS OF VARIABLE MASS—THE ROCKET

In our discussion of the conservation of linear momentum, we have so far dealt with systems whose mass remains constant. We now propose to consider those whose mass is *variable*, i.e., those in which mass enters or leaves the system. A typical case is that of a *rocket* from which hot gases keep on escaping, thereby continuously decreasing its mass.

Principle. Rocket propulsion is based on the principle of *conservation of linear momentum*.

A rocket may use either a liquid or a solid fuel. In the former case the fuel (like liquid hydrogen or liquid paraffin) and a suitable oxidiser (like oxygen, hydrogen peroxide or nitric acid), stored up in separate chambers, are injected into a *combustion chamber* where the fuel is burnt (Fig. 6.9). In the latter case, the fuel itself carries its own oxidiser (as, for example, gun powder) and hence a separate chamber for it is not necessary. In either case, the large quantity of the heat of combustion produced greatly raises the pressure inside the chamber, resulting in the burnt up gases (like CO, steam etc.) issuing out of an orifice at the back or the tail end of the rocket in the form of a high velocity stream, called the *jet*, as indicated in the Figure. In consequence, the rocket is propelled forwards (i.e., opposite to the direction of the jet), since the momentum lost by the jet of fuel gases must be equal to the momentum gained by the rocket.

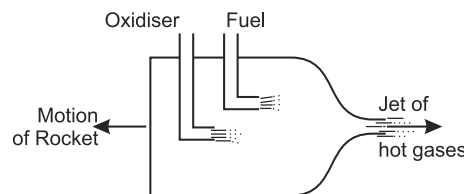


Fig. 6.9

Theory. Let M be the *mass* of the rocket (including that of its fuel and oxidiser) and V , its velocity at any given instant t in the inertial frame of the laboratory. Further, let α be the *rate of change of mass* of the rocket due to the escape of the hot gases through the orifice at its tail end with an *exhaust velocity* $-v$ relative to it. Then, clearly, *rate of change of mass of the rocket* $= dM/dt = -\alpha$, the negative sign indicating that the change of mass is a *decrease* in mass.

Assuming the forward velocity of the rocket to be along the same line with the oppositely directed velocity v of the hot gases or the jet, we have

$$\text{velocity of the jet in the laboratory frame} = -v + V = (V - v)$$

$\therefore$ *rate of change of momentum of the jet issuing out of the rocket,*

$$\text{or} \quad \text{force acting on it} = -\alpha(V - v) = \frac{dM}{dt}(V - v)$$

Hence, in accordance with Newton's third law of motion, this must also be equal to the force acting on the rocket, tending to move it forwards.

So that, *force acting on the rocket, propelling it forwards* $= \frac{dM}{dt}(V - v)$, neglecting the weight (Mg) of the rocket, or the force of attraction on it due to the earth, in comparison with it.

Since M is the mass and V , the *velocity* of the rocket at the instant considered, we have also, (in accordance with Newton's second law), force acting on the rocket given by $d(MV)/dt$. We therefore, have

$$\frac{d}{dt}(MV) = \frac{dM}{dt}(V - v). \quad \dots(i)$$

$$\text{Or,} \quad \frac{dM}{dt}V + M \frac{dV}{dt} = \frac{dM}{dt}V - \frac{dM}{dt}v$$

$$\text{Or,} \quad M \frac{dV}{dt} = -\frac{dM}{dt}v, \quad \dots(ii)$$

which gives the force (*i.e.*, mass $\times$ acceleration) or the **thrust** on the rocket at the instant t , *i.e.*, **force or thrust** $F = -\frac{dM}{dt}v$.

We may re-write equation (ii) above as $dV = -\frac{dM}{M}v$,
when, on integrating it with respect to t , we have

$$\int \frac{dV}{dt} dt = -\frac{1}{M} \int \frac{dM}{dt} v dt. \quad \text{Or, } V = -v \log_e M + C,$$

where C is a constant of integration.

If M_0 be the mass of the rocket and V_0 , its velocity at $t = 0$, (where V_0 would obviously be zero if the rocket just takes off from its launching pad), we have

$$V_0 = -v \log_e M_0 + C, \text{ whence, } C = V_0 + v \log_e M_0$$

$$V = -v \log_e M + V_0 + v \log_e M_0.$$

$$\text{Or,} \quad V = V_0 + v \log_e \frac{M_0}{M}. \quad \dots(iii)$$

This gives the value of the velocity V of the rocket at instant t in terms of the mass of the rocket initially and after time t .

Now, since mass of the rocket (along with the fuel and the oxidiser left in it) is M after time t and since the mass decreases at the rate α , clearly, $M = (M_0 - \alpha t)$, *i.e.*,

$$M = M_0 \left(1 - \frac{1}{M_0} \alpha t \right).$$

Denoting α/M_0 , the *rate of change of mass in terms of the initial mass* M_0 , by β , we have

$$M = M_0 (1 - \beta t).$$

Substituting this value of M in relation (iii) above, we have

$$V = V_0 + v \log_e \left(\frac{M_0}{M_0(1 - \beta t)} \right) = V_0 + v \log_e \left(\frac{1}{1 - \beta t} \right)$$

$$\text{Or,} \quad V = V_0 - v \log_e (1 - \beta t), \quad \dots(iv)$$

which gives the velocity of the rocket at the instant t in terms of β (the rate of change of mass of the rocket in terms of its initial mass).

The distance covered by the rocket in time t can be easily obtained by integrating expression (iii) or (iv) with respect to t .

In case the weight of the rocket is taken into account, the equation of motion (i) for a rocket moving vertically upwards becomes

$$\frac{d}{dt}(MV) = \frac{dM}{dt}(V - v) - Mg.$$

$$\text{Or,} \quad \frac{dM}{dt}V + M \frac{dV}{dt} = \frac{dM}{dt}V - \frac{dM}{dt}v - Mg.$$

Or,
$$M \frac{dV}{dt} = -\frac{dM}{dt}v - Mg, \text{ Whence, } \frac{dV}{dt} = -\frac{dM}{Mdt}v - g,$$

integrating which with respect to t , we have

$$\int \frac{dV}{dt} dt = -v \int \frac{dM}{Mdt} dt - \int g dt. \text{ Or, } V = -v \log_e M - gt + C.$$

Since at $t = 0$, $M = M_0$ and $V = V_0$, we have $V_0 = -v \log_e M_0 + C$. Or, $C = V_0 + v \log_e M_0$. So that,

$$V = -v \log_e M - gt + V_0 + v \log_e M_0.$$

Or,
$$V = V_0 + v \log_e \frac{M_0}{M} - gt. \quad \dots(v)$$

Again, since $M = M_0 (1 - \beta t)$, we have

$$V = V_0 - v \log_e (1 - \beta t) - gt. \quad \dots(vi)$$

It will be noted that both relation (v) and (vi) for the velocity of the rocket are essentially the same as (iii) and (iv) respectively, with the additional term $-gt$ (in view of the downward pull due to gravity), where t is the time of combustion of the fuel.

In case t be small, the addition of the term $-gt$ hardly makes any difference, i.e., there is hardly any appreciable reduction in the value of V .

It t be large, however, i.e., if the fuel is burnt at a slower rate, the term $-gt$ may become appreciably large and may slow down or even stop the ascent of the rocket. The rate of fuel combustion has, in fact, to be judiciously regulated, since both a slower and a more rapid rate of combustion are, for various reasons, equally undesirable.

A glance at the expressions for velocity (V) of the rocket will show that to obtain a higher value of the rocket velocity we must have (i) a higher exhaust velocity v and (ii) a higher value of M_0/M (or of β) i.e., a higher loss of mass, so that the final mass of the rocket may be very much smaller than its initial mass.

Now, the exhaust velocity depends upon the design for the nozzle as also upon the temperature and pressure developed inside the combustion chamber, the highest theoretical value possible being the *root mean square velocity* of the gas molecules at the temperature prevailing in the combustion chamber. Since, however, the temperature can hardly ever exceed 3000°C , the practical limit to the value of the exhaust velocity works out to 2 km/sec.

Then, again, the ratio M_0/M can at best be 10% of the initial value M_0 , (even if the whole of the fuel is burnt, so that M is only the mass of the empty containers), but, its advantage is more than offset by a smaller exhaust velocity in this case.

As can be easily calculated from expression (iii) above, with an exhaust velocity of 2 km/sec, initial velocity (V_0) of the rocket *zero* and the ratio $M_0/M = 10$, the velocity (V) of the rocket works out to $0 + 2 \log_e 10 = 2 \times 2.3026 = 4.6052$ km/sec.

This is less by far than the escape velocity 11.2 km/sec necessary to be able to escape from the gravitational field of the earth, or even the orbiting velocity 8 km/sec, necessary to be able to orbit around the earth close to its surface.

It can be easily seen that, with the exhaust velocity $v = 2$ km/sec, in order that a rocket may attain a velocity V equal to the escape velocity 11.2 km/sec, the mass ratio M_0/M must be 270 (see worked example (23) or even a bit more, actually, in view of the fact that the rocket will encounter, and will have to overcome, the resistance of the air. This means that a little more than 270 kg of the fuel will be consumed for every 1 kg mass of the rocket that leaves the earth. In the present stage of its development no single stage rocket is capable of such a performance and we have, therefore, to use a *multistage*, i.e., a *two or three stage rocket* to achieve the purpose.

A **multistage rocket** is just a combination of rockets, either (i) joined consecutively, i.e., in series, (ii) *one inside the other*, or (iii) with the rear part of one inside the nozzle of the other, as

shown diagrammatically in Fig. 6.10. In all the three types, the first stage is the largest in dimensions as well as in weight, and the last stage the smallest and the lightest.

Naturally, the first stage rocket is used first and when its fuel is all burnt up and it has done its job it gets detached and is discarded, with the second stage taking over the task of producing further acceleration. This too, in its turn, is discarded when its fuel is burnt up the third stage rocket takes over. The velocity thus goes on increasing at each stage by the same amount as it does in a single stage rocket. The fuel consumption and the thrust for the first stage are about 100 times more than for the third stage and the fuel stock carried by it about 60 times that carried by the third stage.

Finally, a word may as well be said here as to the shape of the rocket. As can be readily seen, a rocket is subjected to intense air pressure during its upward flight through the atmosphere and a lot of heat is generated due to the viscous friction of the air. Both these factors are taken into account in designing it. Its frame is made of a heat-resisting material, with its velocity during its flight through the denser regions of the atmosphere kept sufficiently low and its body is given a cylindrical and *tapering, stream line* form to reduce the air pressure on each individual part of it to the very minimum, its overall shape resembling that of a cigar.

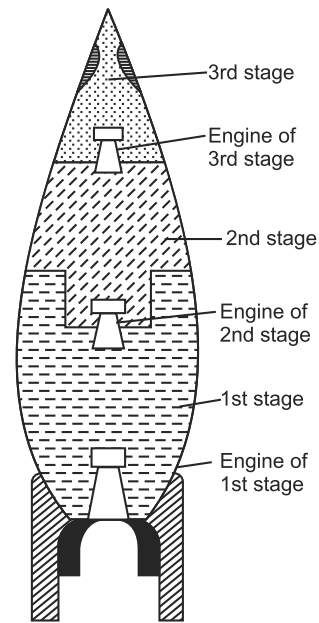


Fig. 6.10

6.13 ANGULAR MOMENTUM — TORQUE

The angular momentum of a particle about a point fixed in an inertial frame is *the moment of its linear momentum about that point*. That is why it is also called **moment of momentum**. It is usually denoted by the symbol $\vec{J}$ and is measured by the *product of the linear momentum* $\vec{p}(=m\vec{v})$ *of the particle and its vector distance* $\vec{r}$ *from the fixed or the reference point in the inertial frame*. Thus,

$$\vec{J} = \vec{r} \times \vec{p} = m(\vec{r} \times \vec{v})$$

$$\dots I [\text{where } \vec{p} = m\vec{v}]$$

It is obviously a *vector quantity* and its direction, perpendicular to both $\vec{r}$ and $\vec{p}$ (or $\vec{v}$), is given by the *right hand screw rule*. So that, in the case of a particle describing a circular motion its value is equal to $m\omega^2 r = mvr$ (where v is the linear velocity of the particle) and is perpendicular to the plane of the circle described, in the same direction with ω (fig. 6.11).

A component of the angular momentum about any axis (*i.e.*, a line) passing through the fixed or the reference point is referred to as the angular momentum of the particle about that axis (or line).

The units of angular momentum in the *C.G.S.* and the *RMKS* (or *SI*) systems are respectively $\text{gm-cm}^2/\text{sec}$ or erg-sec and $\text{kg-m}^2/\text{sec}$ or joule/sec .

Relation I holds good even for a system of particles, each free to move independently of the other or compactly arranged to form a rigid body. Thus, if $\vec{J}_1, \vec{J}_2, \vec{J}_3$ etc. be the angular momenta of the different particles of the system about the fixed or the reference point, the angular momentum of the whole system about that point is given by

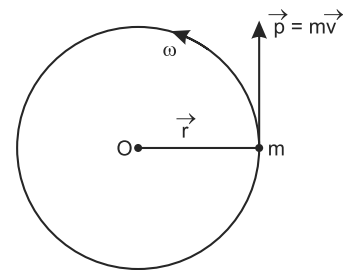


Fig. 6.11

$$\vec{J} = \vec{J}_1 + \vec{J}_2 + \vec{J}_3 + \dots = (\vec{r}_1 \times m \vec{v}_1) + (\vec{r}_2 \times m \vec{v}_2) + (\vec{r}_3 \times m \vec{v}_3) + \dots$$

$$\text{Or,} \quad \vec{J} = \sum (\vec{r} \times m \vec{v}) = \sum (\vec{r} \times \vec{p}),$$

i.e., the angular momentum of the whole system about the point is the vector sum of the angular momenta of the individual particles about that point.

Torque. Differentiating relation I above with respect to t , we have

$$\frac{d\vec{J}}{dt} = \frac{d}{dt}(\vec{r} \times \vec{p}) = \frac{d\vec{r}}{dt} \times \vec{p} + \vec{r} \times \frac{d\vec{p}}{dt}.$$

Now, $\frac{d\vec{r}}{dt} \times \vec{p} = \vec{v} \times m \vec{v} = 0$ and $d\vec{p}/dt = \text{rate of change of linear momentum or the force } \vec{F}$ acting on the particle. So that,

$$\frac{d\vec{J}}{dt} = \vec{r} \times \frac{d\vec{p}}{dt} = \vec{r} \times \vec{F}.$$

The vector product $\vec{r} \times \vec{F}$ is called *torque** (or *moment of the force*) about the fixed or the reference point and is represented by the symbol τ .

Thus, the time rate of change of angular momentum of a particle about a point gives the torque τ about that point, where

$$\vec{\tau} = \vec{r} \times \vec{F} = d\vec{J}/dt. \quad \dots \text{II}$$

In the case of a system of particles, independent of each other or forming a rigid body, we have seen above how $\vec{J} = \sum (\vec{r} \times m \vec{v}) = \sum (\vec{r} \times \vec{p})$. It follows, therefore, that

$$\vec{\tau} = d\vec{J}/dt = \sum \vec{r} \times \frac{d}{dt}(m \vec{v}) = \sum (\vec{r} \times \vec{F}),$$

i.e., the torque or the time rate of change of angular momentum of the system about the fixed or the reference point is the sum of the torques of all the external forces acting on the system. The reason why we explicitly say *external* forces is that the internal forces all form collinear action-and-reaction pairs of equal and opposite forces, having equal and opposite moments about the given point and their sum is, therefore, zero, i.e., they produce no effect.

As will be readily seen, *angular momentum is the rotational analogue of linear momentum*. Just as the rate of change of linear momentum of a particle gives the force acting upon it, so also the rate of change of angular momentum of the particle gives the torque acting upon it. And, as we shall shortly see, just as *linear momentum = (mass) (linear velocity)* or mv , we have, for a rigid body, *angular momentum = (moment of inertia) (angular velocity)* or $I\omega$.

Another expression for angular momentum of a system. Sometimes it is found to be more convenient to express the angular momentum of a system of particles in terms of the velocity of the centre of mass and the velocities of the particles of the system relative to the centre of mass.

Thus, if $\vec{R}$ and $\vec{V}$ be the *position vector* and *velocity* respectively of the centre of mass of a system of particles relative to a fixed or a reference point, and $\vec{r}_c$ and $\vec{v}_c$, the *position vector* and *velocity* of a particle of mass m of the system, *relative to the centre of mass*, the position vector and velocity of the particle *relative to the fixed or reference point* will clearly be $\vec{r} = \vec{R} + \vec{r}_c$ and $\vec{v} = \vec{V} + \vec{v}_c$ respectively. Hence, the *angular momentum of the system about the fixed or the reference point* will, as just explained above, be given by

$$\vec{J} = \sum m(\vec{R} + \vec{r}_c) \times (\vec{V} + \vec{v}_c) = \sum m(\vec{R} \times \vec{V}) + \sum m(\vec{V} \times \vec{r}_c) + \sum m(\vec{R}_c \times \vec{V}) + \sum m(\vec{r}_c \times \vec{v}_c)$$

* From the Latin word 'torquere' meaning 'to twist'.

Now, $\vec{r}_c = (\vec{r} - \vec{R})$ and $\therefore m\vec{r}_c = m\vec{r} - m\vec{R}$

Or, $\sum m\vec{r}_c = \sum m\vec{r} - \sum m\vec{R} = \sum m\vec{r} - M\vec{R}$ [$\because \sum m = M$, mass of the system.]

But, as we know, the inherent property of the centre of mass demands that

$$M\vec{R} = m_1\vec{r}_1 + m_2\vec{r}_2 + \dots = \sum m\vec{r}$$

We, therefore, have $\sum m\vec{r} - M\vec{R} = 0$ or $\sum m\vec{r}_c = 0$.

Similarly, $\sum m\vec{v}_c = 0$.

So that, the above relation for $\vec{J}$ simplifies to

$$\vec{J} = \vec{R} \times M\vec{V} + \sum (\vec{r}_c \times m\vec{v}_c).$$

Here, clearly, $\sum (\vec{r}_c \times m\vec{v}_c)$ is the angular momentum of the system about the centre of mass, say $\vec{J}_{c.m.}$ and $M\vec{V} = \vec{p}$, the momentum of the centre of mass or the total linear momentum of the system.

We, therefore, have

$$\vec{J} = \vec{R} \times \vec{p} + \vec{J}_{c.m.} \quad \dots \text{III}$$

i.e., the total angular momentum of the system about the fixed or the reference point is the vector sum of the angular momentum of the centre of mass about that point and the angular momentum of the system about the centre of mass. The former depends upon the fixed or the reference point chosen and is referred to as the *orbital angular momentum* and the latter, which is independent of the fixed or the reference point, is called the *spin angular momentum*. So that, the *total angular momentum of the system is the vector sum of its orbital and spin angular momenta*.

Differentiating expression III above with respect to t , we have

$$\text{torque acting on the system, } \vec{\tau} = \frac{d\vec{J}}{dt} = \frac{d\vec{J}_{c.m.}}{dt} + \frac{d}{dt}(\vec{R} \times \vec{p}).$$

If the centre of mass be chosen as the origin, so that it remains stationary, we have $\vec{p} = M\vec{V} = 0$ (since $\vec{V} = 0$). The second term in the expression for $\vec{\tau}$ above, therefore, vanishes and we have torque $\vec{\tau} = d\vec{J}_{c.m.}/dt$ and the body is said to be in a *state of spinning*.

Thus, each planet in the solar system possesses both an orbital and a spin angular momentum and its total angular momentum about the sun is the vector sum of the two. In the case of our own planet, *Earth*, the two momenta are inclined to each other at an angle of 23.5° . The sun alone in the solar system (considered in isolation from the rest of the universe) possess *zero orbital angular momentum* (because $\vec{V} = 0$) and it, therefore, has only spin angular momentum.

The total angular momentum of the solar system is thus the vector sum of all these orbital and spin angular momenta of the various planets, the contribution of the spin angular momentum of the sun itself being just 2%.

If, however, we consider the universe, as a whole with the centre of the galaxy as the origin, or the reference point, the sun too has an orbital angular momentum about the origin and thus possesses both types of angular momenta.

6.14 CONSERVATION OF ANGULAR MOMENTUM

We know that the external torque applied to a system of particles is given by the relation $\vec{\tau} = d\vec{J}/dt = \vec{r} \times \vec{F}$. If, therefore, the external torque be *zero*, we have $d\vec{J}/dt = 0$, i.e., $\vec{J} = \text{constant}$. This is the **principle or the law of conservation of angular momentum** and may be formally stated thus:

When the external torque (or the sum of external torques) applied to a system of particles is zero, the total angular momentum of the system remains conserved.

The law is not restricted to only closed orbits but applies equally well to open orbits as well as collisions.

Thus, if $\vec{J}_1, \vec{J}_2, \vec{J}_3$, etc. be the angular momenta of the different particles of a system, their sum total $\vec{J} = \vec{J}_1 + \vec{J}_2 = \vec{J}_3 + \dots$ remains constant in the absence of any external torque applied to the system, though the particles may exchange momenta among themselves.

Similarly, in the case where the centre of mass is taken to be the origin, we have $\vec{\tau} = d\vec{J}_{c.m.}/dt$. If, therefore, the external torque $\vec{\tau}$ be zero, we have $\vec{J}_{c.m.} = \text{constant}$. This may be called the *law of conservation of spin angular momentum*.

It will be seen at once that just as the external force applied determines the motion of the centre of mass of a system; the external torque applied determines the rotation of the system about the centre of mass.

Now, from the relation for τ , mentioned above, viz, $\vec{\tau} = d\vec{J}/dt = \vec{r} \times \vec{F}$, it is abundantly clear that *the torque acting on a particle will be zero, and hence its angular momentum conserved, when (i) its position vector $\vec{r}$ is zero or (ii) the force $\vec{F}$ applied to it is zero or (iii) the direction of both $\vec{r}$ and $\vec{F}$ is the same, i.e., the line of action of force $\vec{F}$ passes through the fixed or the reference point.*

The first two conditions are obvious by themselves and the third condition is satisfied in the case of *central forces*, i.e., forces which are always directed towards (or away from) a fixed reference point. This will be seen from the following:

The magnitude of a central force on a particle depending only on the distance from the fixed or the reference point, we may represent it by $\vec{F} = \hat{r}f(r)$, where $\hat{r}$ is the *unit vector* along the direction of $\vec{r}$ and is equal to $\vec{r}/r$ and $f(r)$, a *scalar function* of distance r .

$\therefore$ *torque acting on a particle under the action of a central force* is given by

$$\vec{\tau} = \frac{d\vec{J}}{dt} = \vec{r} \times \vec{F} = \vec{r} \times \hat{r} f(r) = f(r)(\vec{r} \times \vec{r}/r) = 0, \quad [\because \vec{r} \times \vec{r} = 0]$$

showing that $\vec{J} = \vec{r} \times m \vec{v} = \text{constant}$.

i.e., the angular momentum of a particle moving under the influence of a central force always remains conserved.

It follows, therefore, that $\vec{J}$ being constant, must be perpendicular to the plane containing $\vec{r}$ and $\vec{v}$ i.e., *the path of a particle under the influence of a central force must lie in a plane.*

Thus, since electrostatic forces are central forces, the angular momentum of the electron moving in its orbit around the proton in a hydrogen atom remains conserved.

Again, since gravitational forces too are central forces, the angular momenta of planets orbiting around the sun remain conserved (although they move in elliptical orbits) and so also of the satellites moving around a planet.

Let us examine the case of a planet moving around the sun *in an elliptical orbit* in a little more detail.

6.15 MOTION OF A PLANET IN AN ELLIPTICAL ORBIT AROUND THE SUN

The planets, as we know, move in elliptical orbits around the sun, with the sun at one of one foci. In fact, it is the centre of mass of the sun and the planet which should lie at this focus but, in

view of the enormously greater mass of the sun compared with that of the planet, the centre of mass of the system (*i.e.*, of the sun and the planet) lies close to the centre of the sun itself which may, therefore, be assumed to lie there.

Thus, let O be the centre of the sun lying at one focus of the ellipse and P and P' , the two positions of the planet at instants t and $t + \Delta t$, given by the position vectors $\vec{r}$ and $\vec{r} + \Delta \vec{r}$ respectively, with the gravitational force on it due to the sun always directed towards the latter, (Fig. 6.12).

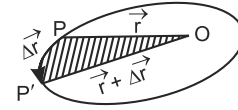


Fig. 6.12

In this interval of time Δt the planet sweeps out, on account of its displacement $\Delta \vec{r}$, a vector areas $\Delta \vec{S}$ (shown shaded) = $\frac{1}{2} \vec{r} \times \Delta \vec{r}$.

$\therefore$ Rate at which the radius vector sweeps out the area = $\Delta \vec{S} / \Delta t = \frac{1}{2} \vec{r} \times \Delta \vec{r} / \Delta t$.

Or, in the limit $\Delta t \rightarrow 0$, we have $d\vec{S}/dt = \frac{1}{2} \vec{r} \times d\vec{r}/dt = \frac{1}{2} \vec{r} \times \vec{v}$

where $d\vec{S}/dt$ may be called the *areal velocity of the radius vector*.

Now, angular momentum $\vec{J} = \vec{r} \times m \vec{v}$; so that, $\vec{r} \times \vec{v} = \vec{J}/m$, and, therefore, *areal velocity of the radius vector*, $d\vec{S}/dt = \vec{J}/2m$.

Since $\vec{J}/2m$ is a constant, it follows that *the areal velocity of the radius vector remains constant under the influence of the central gravitational force*.

This is really **Kepler's second law** which states that *the radius vector from the sun to any planet sweeps out equal areas in equal time*.

A consequence of this is that *a planet moves faster at the point of its closet, than that of its farthest, approach to the sun*. This is obviously so because at these points the vector $\vec{r}$ is perpendicular to $\vec{v}$ (Fig. 6.13) and the angular momentum is thus equal to mvr . For the conservation

of momentum, therefore, the angular momentum mv_2r_2 at the point of nearest approach must be equal to the angular momentum mv_1r_1 at the point farthest approach. We, therefore, have $mv_1r_1 = mv_2r_2$. Or $v_1r_1 = v_2r_2$, showing that *the larger velocity of the planet is associated with its shorter distance from the sun and vice versa*.

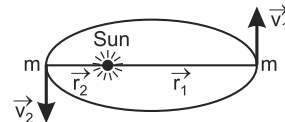


Fig. 6.13

As since the angular momentum ($\vec{J}$) remains constant, $\vec{J}/2m$ or the areal velocity of the radius vector too remains constant.

In the case of the earth, its orbital velocity is 30.26 km/sec when it is nearest the sun and 29.29 km/sec when it is farthest from it.

This result, arrived at by *Kepler* from his long and laborious calculations and most careful observations, is thus just a natural consequence of the law of conservation of angular momentum.

Interestingly enough, *Newton* used the reverse argument, half a century later, to show that the areal velocity $d\vec{S}/dt$ of the radius vector joining a planet and the sun being constant, the gravitational force is a central force.

6.16 CONSERVATION OF ANGULAR MOMENTUM OF A SYSTEM, A CONSEQUENCE OF ROTATIONAL INVARIANCE OF POTENTIAL ENERGY OF THE SYSTEM

Just as the conservation of linear momentum of a system of particles is a consequence of translational invariance of the potential energy of the system (§6.17), so also the conservation of angular momentum of a system of particles is a consequence of rotational invariance of the potential energy of the system, *i.e.*, *the angular momentum of the system remains conserved if its rotation does not bring about any change in its potential energy*.

This follows from the fact that the rotation of a system would normally require work to be done against any external torque that may be acting on the system; thus necessarily resulting in some change in its potential energy. In case, therefore, the rotation of the system brings about no change in its potential energy, the obvious inference is that no work is being done against any external torque or that, in other words, no external torque is acting on the system. And, as we know so well, in the absence of an external torque, the angular momentum of the system must remain conserved.

Analytically, we may proceed in a manner similar to that adopted in § 6.17, in the case of linear momentum.

Thus, let us consider first, for the sake of simplicity, a system consisting of only two particles having position vectors $\vec{r}_1$ and $\vec{r}_2$. Let these be rotated through any arbitrary angle about any arbitrary axis and let the rotated vectors be represented by $\Omega\vec{r}_1$ and $\Omega\vec{r}_2$ (where Ω is an operator and not a mere number), such that their lengths are the same as those of $\vec{r}_1$ and $\vec{r}_2$ respectively, indicating that, on rotation, the vectors $\vec{r}_1$ and $\vec{r}_2$ undergo no change in magnitude but only a change in direction.

Now, the rotational invariance of potential energy demands that the potential energy $U(\vec{r}_1, \vec{r}_2)$ before rotation = the potential energy $U(\Omega\vec{r}_1, \Omega\vec{r}_2)$ after rotation. This can obviously be possible only if $U(\vec{r}_1, \vec{r}_2) = U(\vec{r}_1 - \vec{r}_2)$ and $U(\Omega\vec{r}_1, \Omega\vec{r}_2) = U(\Omega\vec{r}_1 - \Omega\vec{r}_2)$, i.e., if the potential energy depends only on the magnitude of the distance between the two particles, this magnitude being the same before and after the motion. In order that this be so, the force acting on the particles must be directed along the line $(\vec{r}_2 - \vec{r}_1)$, i.e., it must be a *central force*. And since a central force gives rise to no torque on the system, its angular momentum remains conserved.

What is true for a system of two particles is equally true for a system consisting of any number of them, provided the potential energy depends on the magnitude of the distance between the various particles.

The conservation of angular momentum of a system is thus a natural corollary of the rotational invariance of its potential energy.

6.17 SOME ILLUSTRATIVE EXAMPLES OF CONSERVATION OF ANGULAR MOMENTUM

(i) **Motion of a planet or a satellite in its orbit.** This has already been dealt with under § 6.15, where it has been shown (a) how to be able to conserve its angular momentum, a planet must move faster at its point of nearest approach to the sun than at the point of its farthest approach and also (b) how the principle of conservation of angular momentum has led to the deduction of Kepler's second law.

(ii) **Scattering of a positive particle by a massive nucleus.** Suppose a positive particle, like a *proton* or an *alpha particle*, approaches a massive nucleus N of an atom, (Fig. 6.11), and that its velocity is not appreciably affected due to its passage into the atom. Then, clearly, as it approaches the (positive) nucleus of the atom, it experiences an electrostatic force of repulsion, varying inversely as the square of its distance from the nucleus. So that as it gets closer to the nucleus, this repulsive force on it goes on increasing. As a result, it gets deflected or scattered from its straight path (towards the nucleus) into a hyperbolic path, PAP' , as shown, with the nucleus N at its external focus.

Let the particle in question be a *proton*, of mass m and carrying a charge $+e$, and let the atomic number of the nucleus be Z , so that the charge on it is $+Ze$.

Suppose at an infinite distance from the nucleus where the repulsive force on it is zero*, the particle travels with velocity v_0 and enters the atom (with no appreciable change in its velocity)

* The nucleus may be supposed to have infinite mass, so that its velocity of recoil may be negligibly small or zero and hence also its energy of recoil and the angular momentum imparted to it equal to zero.

along the direction PO and that, on reaching the point A , the apse of the hyperbola, it proceeds along AP' such that PA and AP' make equal angles with the straight line from N passing through A .

The perpendicular distance $NB = b$ between the nucleus and the initial direction PO of the particle is called the **impact parameter** or the **collision parameter** and distance $NA = r_s$ is called the **distance of closest approach** of the particle to the nucleus.

Since electrostatic forces (like gravitational ones) are central forces, the angular momentum of the particle must remain conserved and we, therefore, have

$$mv_0 b = mv_A r_A,$$

whence,

$$v_A = v_0 b / r_A$$

where v_A is the velocity of the particle at the point of closest approach, i.e., at A . Also, since the energy of the particle must also remain conserved, we have

$$\begin{aligned} \frac{1}{2}mv_A^2 + Ze^2/r_A &= \frac{1}{2}mv_0^2, \\ &= \frac{1}{2}mv_0^2 + 0, \end{aligned}$$

where $\frac{1}{2}mv_A^2$ is the *kinetic*, and Ze^2/r_A , the *potential* energy of the particle at A and $\frac{1}{2}mv_0^2$, its *initial kinetic energy*, when its *potential energy is zero* (its distance from the nucleus being infinite).

Or, substituting the value of v_A , obtained above, we have

$$Ze^2/r_A = \frac{1}{2}mv_0^2 - \frac{1}{2}m(v_0 b/r_A)^2.$$

$$\text{Or,} \quad Ze^2/r_A = \frac{1}{2}mv_0^2 \left[1 - \left(\frac{b}{r_A} \right)^2 \right].$$

So that, knowing the *impact parameter* b , the *distance of closest approach of the particle* (r_A) can be easily obtained.

(iii) Effect on linear and angular speeds of a particle on contraction of its orbit. Let a particle of mass m be carried at one end of a light string such that it can be rotated in a horizontal circle, i.e., made to move in circular orbit. Let the free end of the string pass through a small hollow tube, as shown in Fig. 6.15, so that the radius of the orbit of the particle can be decreased by simply applying a downward pull F on the free end of the string.

Let the *initial radius* of the circular orbit described by the particle be r_1 , its *linear speed* v_1 and angular speed $\omega_1 = v_1/r_1$, so that its *angular momentum* about the centre O of the orbit $= mv_1 r_1$ and its *kinetic energy* $= \frac{1}{2}mv_1^2$.

And, when the free end of the string is pulled down a little by a force F , so that the radius of the circular orbit of the particle is reduced to r_2 , let the *linear and angular speeds* of the particle be v_2 and ω_2 respectively and hence its *angular momentum* about $O = mv_2 r_2$ and its *kinetic energy* $= \frac{1}{2}mv_2^2$.

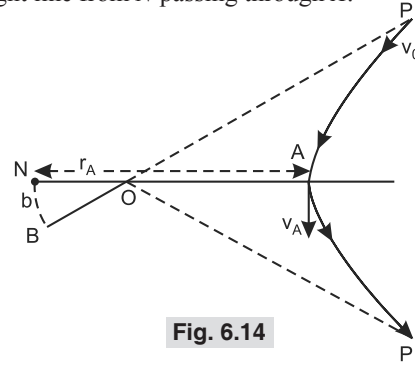


Fig. 6.14

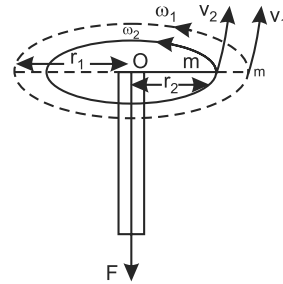


Fig. 6.15

Clearly, the downward pull on the free end of the string is transmitted to the particle as a *radial force* (vis, the centripetal force), directed towards the centre O of its orbit. As a radial force, therefore, it exerts no torque on the particle (its line of action passing through O) and its angular momentum thus remains conserved, so that we have

$$mv_1 r_1 = mv_2 r_2, \text{ whence } v_2 = v_1 r_1 / r_2.$$

Since, $r_2 < r_1$, it follows that $v_2 > v_1$ and hence the *K.E. of the particle in the second case, i.e.*, $\frac{1}{2}mv_2^2 = \frac{1}{2}m(v_1 r_1 / r_2)^2 = \frac{1}{2}mv_1^2 (r_1 / r_2)^2$ is greater than its kinetic energy $\frac{1}{2}mv_1^2$ in the first case.

Thus, by shortening the radius of the orbit of the particle from r_1 to r_2 , its K.E. has increased by

$$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = \frac{1}{2}mv_1^2 \left(\frac{r_1^2}{r_2^2} - 1 \right) \quad \dots I$$

This additional energy must obviously be supplied by the work done on the particle against the centrifugal force in pulling it inwards towards O . Let us see if this is so.

Since angular momentum of the particle remains conserved, we have

$$mv_1 r_1 = mv_2 r_2 \quad \text{Or,} \quad v_1 r_1 = v_2 r_2 = k, \text{ say.}$$

This means that the linear speed of the particle, $v = k/r$. And, therefore, centrifugal force $F = mv^2/r = (m/r)(k^2/r^2) = mk^2/r^3$.

$\therefore$ work done against the centrifugal force in decreasing the orbit radius from r_1 to r_2 is given by

$$W = \int_{r_1}^{r_2} -F dr = \int_{r_1}^{r_2} -\frac{mk^2}{r^3} dr = \left[\frac{mk^2}{2r^2} \right]_{r_1}^{r_2} = \frac{mk^2}{2} \left[\frac{1}{r^2} \right]_{r_1}^{r_2} = mk^2 \left[\frac{1}{r_2^2} - \frac{1}{r_1^2} \right]$$

Or, Substituting $v_1 r_1$ for k , we have

$$W = \frac{mv_1^2 r_1^2}{2} \left(\frac{1}{r_2^2} - \frac{1}{r_1^2} \right) = \frac{1}{2}mv_1^2 \left(\frac{r_1^2}{r_2^2} - 1 \right) \quad \dots II$$

which is the same as expression I for gain in *K.E.*, above, showing that the gain in *K.E.*, of the particle on shortening its orbit, is indeed supplied by the work done against the centrifugal force acting on the particle.

Obviously enough, in the absence of any such supply of energy to the particle, it would not be possible for it to conserve its angular momentum.

This may be easily seen if we simply attach the end of the string to a fixed vertical stick and set the particle in rotation, as before, at the end of the string (6.16). It will be found that as the particle moves in its circular orbit, the radius of the orbit goes on progressively decreasing and the string goes on winding itself around the stick in the form of a spiral.

Clearly, no work is done here against the centripetal force, as in the first case, and hence no energy is supplied to the particle. As a result, despite the continuous decrease in the radius of the orbit, there cannot be any increase in the kinetic energy, and hence, in the linear speed of the particle which therefore continues to remain the same throughout, although the angular speed $\omega = v/r$ goes on increasing with decrease in the value of r . The constancy of the linear speed indicates that the tension in the string is always perpendicular to the direction of v at any point in the spiral path of the particle and is no longer directed towards the centre of its initial circular path. This is so because, in the absence of energy supplied to the particle, its angular momentum mvr cannot possibly remain conserved, but must go on decreasing along with the radius r of the orbit.

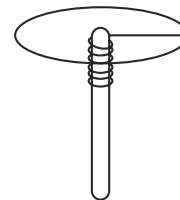


Fig. 6.16

(iv) **The shape of the galaxy.** Galaxies are physical systems, containing anything from 10^9 to 10^{12} stars with a lot of free gas, bound by gravitational attraction, and free to rotate about their own axes of symmetry. One galaxy is separated from another by a huge distance of about 3×10^{24} cm. The galaxy which contains our own sun is referred to as *the galaxy* and has a radius of about 10^{23} cm.

A conjecture as to the formation of galaxies, which some regard to be oversimplified, is due to *Hubble*, according to which there was, to start with, just a gaseous mass spread throughout space, some portion of which acquired angular momentum due to gravitational interaction within itself and the rest, free from angular momentum, condensed in the form of spheres to ultimately form stars.

A galaxy is supposed to have come to acquire its more or less lens-like or pancake shape due to the condensation of a great mass of the gas inside it under gravitational interaction. This may be explained somewhat in the following manner.

Suppose there is a large mass of gas in a more or less spherical form, having an initial angular momentum $\vec{J}$ about any axis, as shown in Fig. 6.17(a). As the gas contracts under gravitational interaction, the forces of interaction being pairs of equal and opposite forces, there is no external torque acting on the system and its angular momentum, therefore, remains conserved. Since however, its volume is greatly reduced, energy has to be supplied to it to enable it to conserve its angular momentum, as we have seen under (iii) above. This energy can obviously come only from the gravitational potential energy of the gas itself.

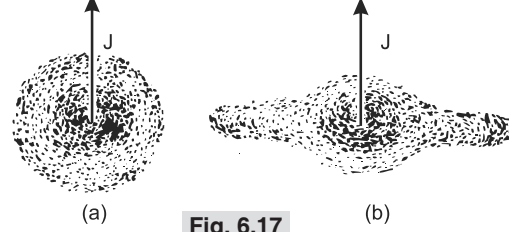


Fig. 6.17

Thus, if m be the mass of a particle of the gas in the outer regions of the galaxy, initially at a distance r_0 from its centre and rotating with velocity v_0 about it, its $K.E. = \frac{1}{2}mv_0^2$.

After gravitational interaction, if the particle lies at a distance r from the centre of the galaxy and if its velocity be v , its $K.E. = \frac{1}{2}mv^2$.

Since the angular momentum remains conserved, we have

$$mv_0r_0 = mvr,$$

whence, $v = v_0r_0/r$ and $\therefore K.E.$ of the particle $= \frac{1}{2}mv_0^2\left(\frac{r_0}{r}\right)^2$.

And its *gravitational P.E.* at this distance r from the centre of the galaxy $= -GMm/r$, where M is the mass of the galaxy.

$$\therefore \text{total energy of the particle} = \frac{1}{2}mv_0^2\left(\frac{r_0}{r}\right)^2 - \frac{GMm}{r}.$$

The contraction of the gas will stop, or equilibrium attained, when the total energy of the gas or the particle is an *extremum* (i.e. either a maximum or a minimum), i.e., when

$$\frac{d}{dr} \left[\frac{1}{2}mv_0^2\left(\frac{r_0}{r}\right)^2 - \frac{GMm}{r} \right] = 0$$

$$\text{or,} \quad -mv_0^2\frac{r_0^2}{r^3} + \frac{GMm}{r^2} = 0$$

$$\text{or,} \quad \frac{GMm}{r^2} = \frac{mv_0^2r_0^2}{r^3} = \frac{mv^2}{r}. \quad \left[\text{Substituting } \frac{vr}{r_0} \text{ for } v_0 \right]$$

Thus, the contraction of the gas continues, with its angular momentum conserved, until *equilibrium is attained when the inward pull due to gravitational force on it is just balanced by the outward pull due to centrifugal force*, the value of r , then being $v_0^2 r^2 / GM$.

Any further reduction in the value of r beyond this is not possible, because the rapidly increasing centrifugal force opposes any further contraction of the gas.

There is, however, no such limitation to the contraction of the gas with its angular momentum conserved *along the direction of the axis of rotation*, any *K.E.* acquired during contraction being dissipated presumably in the form of radiation. As a result, the shape ultimately acquired by the galaxy is that of a *pancake*, with a small spherical, or nearly spherical, protion at the centre, as shown in Fig. 6.17 (b).

(v) **Diving, skating, ballet-dancing etc.** In all these processes, the performer utilises the principle of conservation of angular momentum. For, as mentioned earlier, the angular momentum of a body free to rotate about an axis is equal to $I\omega$ where I is the *moment of inertia* of the body about the axis and ω , its angular velocity. Now, I depends upon the square of the *radius of gyration* of the body about the axis of rotation, *i.e.*, upon the *effective distance* of the particles of the body from the axis of rotation and hence upon the *distribution of mass* about the axis. A large variation in the value of the moment of inertia (I) of the performer and hence a corresponding variation in the value of his angular velocity ω is, therefore, possible on his stretching his limbs out or pulling them in.

Thus, a **diver**, as he springs from the diving board, has a certain angular velocity about a horizontal axis passing through his centre of mass to enable him to spin about the axis or to make a somersault or two before striking water.

Let his initial *M.I.* about the axis be I_0 and his angular velocity about it, ω_0 , so that his initial angular momentum about the axis $I_0\omega_0$. The only external force acting on the diver is the gravitations force due to the attraction of the earth, but since it is a *central force* it produces no torque about his centre of mass and his angular momentum thus remains conserved. He can, therefore, double or triple his angular velocity and thus increase the number of his somersaults before touching water by correspondingly reducing his *M.I.* about the axis of rotation to, say, I , by simply pulling in his hands and feet towards the centre of his body. If ω be his angular velocity corresponding to the value I of his moment of inertia, we have

$$I_0\omega_0 = I\omega$$

Since $I < I_0$, we have $\omega > \omega_0$ and $\therefore \frac{1}{2}I\omega^2 > \frac{1}{2}I_0\omega_0^2$, *i.e.*, the rotational *K.E.* of the diver increases in the process.

The source of this additional K.E. acquired by the diver is obviously the work done by him in pulling in his limbs towards the centre of his body.

In a similar manner, a skater or a ballet dancer, (Fig. 6.18), can increase or decrease his or her angular velocity and thus spin faster or slower about a vertical axis through the centre of mass by pulling in the hands and the feet or by extending them as far out as possible, thereby decreasing or increasing the *M.I.* about the axis of rotation, as desired.



Fig. 6.18

6.18. ANGULAR MOMENTUM OF ELEMENTARY PARTICLES—ITS QUANTISATION

We have seen under § 6.15 how a planet moving in its elliptical orbit around the sun possesses (i) an *orbital angular momentum* due to its motion around the sun and (ii) a *spin angular momentum* due to its rotating or spinning about an axis passing through its own centre of mass. The same is true for almost all the fundamental or elementary particles, including the *photon*, which may possess

one or both these types of angular momenta. Thus, for example, the electron in a hydrogen atom possesses orbital angular momentum on account of its motion in its orbit around the proton and also spin angular momentum on account of spinning about its own axis (passing through its centre of mass).

Now, the student is, in all probability, already aware that in *Bohr's* atomic model, an electron can revolve round the nucleus, without emitting any radiant energy, in only certain specified or permissible orbits for which its orbital angular momentum is an integral multiple of $h/2\pi$, i.e., $nh/2\pi$, where n is an integer, 1, 2, 3, etc. and h , the *Planck's constant* (6.25×10^{-27} erg-sec), these orbits are, for this reason, referred to as stationary orbits or Bohr's orbits.

The orbital angular momentum of the revolving electron is thus restricted to certain fixed or discrete values (*viz.*, integral multiples of $h/2\pi$) and is, therefore, said to be *quantised*.

Bohr, of course, had no theoretical basis for making this assumption except that it could satisfactorily explain the various spectral series of hydrogen.

A mathematical basis for this hypothesis was, however, provided later by *De Broglie* who suggested that every particle has a *wave* associated with it and that the smaller the particle, the longer this wave. Thus, the wave associated with an electron is longer than that associated with a proton. The wavelength of this associated wave, called *de Broglie wavelength*, may be easily obtained. For, in accordance with *Einstein's mass-energy equation* $E = mc^2$ and from *Planck's equation* $E = h\nu$, coupled with the fact that $c = \nu\lambda$ and, therefore, $\nu = c/\lambda$, (where ν is the frequency of the wave), we have $hc/\lambda = mc^2$, whence, $\lambda = h/mc$.

If, therefore, v be the velocity of an electron, the De Broglie wavelength associated with it may be obtained by substituting v for c in the above relation, when we have $\lambda = h/mv$, a fact rested experimentally by *Davisson* and *Germer*.

So that, instead of an electron moving in a stationary orbit, we must now consider a series of waves moving in the orbit and, in order that they may not cancel each other due to interference they must move in such a manner as to produce a stationary or standing wave in the orbit. The necessary condition for this is that the length of the path, i.e., the circumference of the orbit, must be equal to an integral multiple of the wave length. Thus, if r be the radius of a stationary orbit, we must have $2\pi r = n\lambda = nh/mv$, whence, the *orbital angular momentum* of the electron, $mvr = n\lambda/2\pi$, as assumed by *Bohr*.

Now, the *centripetal force* acting on the electron moving with speed v in an orbital of radius r around the nucleus is clearly $-mv^2/r$, the $-ve$ sign indicating that it is directed inwards towards the centre of the orbit or the nucleus.

Since the charge on the nucleus is Ze , where Z is its atomic number and e , a charge equal in magnitude to that on an electron, the force acting on the electron inwards, towards the nucleus, is also $-Ze.e/r^2$, (the charge on the electron being negative).

In the case of hydrogen, $Z = 1$ and, therefore,
 $\therefore$ force acting on the electron, inwards $= e^2/r^2$.

$$\text{We, therefore, have } \frac{e^2}{r^2} = \frac{mv^2}{r}, \text{ whence, } mv^2 = \frac{e^2}{r} \quad \dots(i)$$

And, since angular momentum of the electron, i.e., $mvr = nh/2\pi$, we have

$$m^2v^2r^2 = n^2h^2/4\pi^2 \quad \dots(ii)$$

$$\therefore \text{ dividing relation (ii) by (i), we have } \frac{m^2v^2r^2}{mv^2} = \frac{n^2h^2r}{4\pi^2e^2},$$

$$\text{whence, } r = \frac{h^2}{4\pi^2me^2}n^2. \quad \text{Or, } r \propto n^2.$$

Thus, the radii of the stationary or permissible orbits are proportional to n^2 .

Further, potential at a distance r from the charge Ze on the nucleus $= Ze/r$. We, therefore, have

$$P.E. \text{ of the electron} = (Ze)(-e)/r = -Ze^2/r = -e^2/r \quad [\because Z = 1]$$

And, since $K.E. \text{ of the electron} = \frac{1}{2}mv^2$, we have

$$\begin{aligned} \text{total energy of the electron, } E &= K.E. + P.E. = \frac{1}{2}mv^2 - e^2/r \\ &= \frac{1}{2} \frac{e^2}{r} - \frac{e^2}{r} = -\frac{1}{2} \frac{e^2}{r}. \end{aligned}$$

Or, Substituting the value of r from above, we have

$$E = -\frac{1}{2} \frac{e^2 4\pi^2 m e^2}{n^2 h^2} = -\frac{2\pi^2 m e^4}{n^2 h^2}. \quad \dots(iii)$$

It may be pointed out that *Bohr* had originally taken the electron orbit to be circular. Later, *Sommerfeld* extended the theory to include also elliptical orbits but showed that the value of $E = K.E. + P.E.$ for the electron orbit is still given by relation (iii) above, deduced for circular orbits.

Here, obviously, the motion of the nucleus (or the proton) has been ignored. More correctly, we can imagine the proton (of mass M , say) to be stationary and obtain *relative motion* of the electron about it if we replace the mass m of the electron by the reduced mass $\mu = mM/(m + M)$ of the electron and the proton. In that, case, clearly,

$$r = \frac{h^2}{4\pi^2 \mu e^2} n^2 \text{ and } E = -\frac{2\pi^2 \mu e^4}{n^2 h^2}$$

Correct value of orbital angular momentum of an electron and other fundamental particles. A full fledged *quantum theory of the atom*, based on wave mechanics, was developed in the year 1925, thanks in main to the efforts of *Heisenberg*, *Schrodinger*, *Born* and others, according to which the correct value of the orbital angular momentum of an electron or, in fact, any elementary particle, is given not by $nh/2\pi$ but by $\sqrt{l(l+1)} \frac{h}{2\pi}$, where l is called the **orbital or Azimuthal quantum number** and may have values 0, 1, 2, 3, ... $(n-1)$, where n is the **principal quantum number**.

A significant departure from Bohr's theory is at once noticeable in that here, with $l = 0$, we can have angular momentum $(\vec{J}) = 0$ which is surely not consistent with Bohr's theory which demands that an electron moving in its orbit around the nucleus must necessarily possess angular momentum.

Now, angular momentum is a vector quantity and is represented by a vector along the axis of rotation (or perpendicular to the plane of rotation). If, therefore, we express it in units of $h/2\pi$, i.e., if $h/2\pi$ be taken as *unit angular momentum*, the orbital angular momentum may be represented by a vector $\vec{l}$ whose magnitude is not l but $\sqrt{l(l+1)}$ units, where 1 unit $= h/2\pi = 1.0542 \times 10^{-34}$ erg-sec or $\approx 10^{-34}$ kg-m²/sec, usually denoted by the symbol $\hbar$ (read as h bar).

In a complex atom, where there may be more orbiting electrons than one, the total orbital angular momentum $(\vec{L})$ is the vector sum of the orbital angular momenta of the individual electrons and will always have an integral value.

It may also be mentioned here that any *observable component* of the orbital angular momentum, such, for example, as the one in the direction of a magnetic field, is also *quantised* and is given by $\frac{h}{2\pi} m_l$ where m_l is called the *magnetic orbital quantum number* and may have any of its $(2l+1)$ possible values, $l, (l-1), (l-2), 0, -1, -2, -l$.

Spin angular momentum. In order to explain the *fine structure* of the spectral lines of hydrogen and some other elements (*i.e.*, the splitting up of their spectral lines, when examined by an instrument of high resolving power), as also to explain *anomalous Zeeman effect*, *Uhlenbeck* and *Gondsmit* and also *Dirac* suggested, in the year 1925, that an electron, whether in an orbit or free, spins about an axis just like a top and has, therefore, also a *spin angular momentum*, whose magnitude is $sh/2\pi$, where s is called the *spin quantum number*. The same is true for all other elementary particles except π or K -mesons.

The spin angular momentum too is thus *quantised* and expressed in $h/2\pi$ or $\hbar$ units. Its correct value, however, based on wave mechanics (when measured in terms of $h/2\pi$ as the unit) is represented by the vector $\vec{s}$, whose magnitude is not s but $\sqrt{s(s+1)}$ units.

The value of the spin quantum number is not the same for all elementary particles. Thus, while it is $\frac{1}{2}$ for the electron and for most of the other elementary particles, it is 1 for the *photon* and 0 for π and K -mesons. So that for the electron, proton and neutron etc., the vector $\vec{S}$ (*i.e.*, the *spin angular momentum vector*) has the magnitude $\sqrt{\frac{1}{2}\left(\frac{1}{2}+1\right)} = \sqrt{3/2}$ units, where of course, 1 unit = $h/2\pi$ or $\hbar$.

Since a free and independent electron too spins about its axis, and so do most of the other elementary particles, they all possess spin angular momentum. For this reason, the spin angular momentum of an electron or an elementary particle is also referred to as its *intrinsic angular momentum*, which it preserves under all circumstances.

The vector sum $\vec{S}$ of the spin angular momenta of more than one electron must thus be an odd multiple of $\frac{1}{2}$ if the number of electrons be odd and must be an even integer, or an even multiple of $\frac{1}{2}$, if the number of electrons be even, *i.e.*, being quantised, it must always be a multiple of $\frac{1}{2}\hbar$ or $\frac{1}{2}h/2\pi$. It also follows that the values of spin angular momenta for different directions of spin must differ from each other by a whole unit h or $h/2\pi$. This means, in other words, that the vectors representing the spin angular momentum vectors (or the spin, for short) must always be *either parallel* (*i.e.*, similarly directed), as shown in Fig. 6.19 (i) or *anti parallel* (*i.e.*, oppositely directed), as shown in Fig. 6.19 (ii).

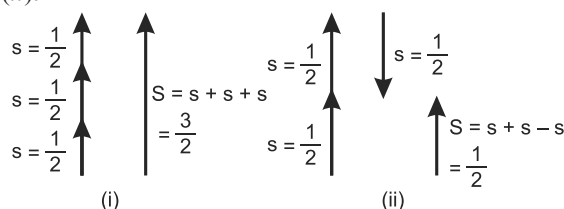


Fig. 6.19

Fundamental particles, like electrons, nucleons (*i.e.*, *protons* and *neutrons*), *neutrinos* μ -mesons etc. for which the spin angular momentum (or spin) is $\frac{1}{2}$ are called **fermions**, after *Fermi*,—the heavier ones among them (like *nucleons*) being called *baryons* and the lighter ones (like *electrons*, μ -mesons or *muons*), *leptons*.

A very important and far-reaching property of fermions is that *no two fermions inhabiting the same region, can possess identical properties*, thus *no two electrons in an atom can exist in the same state*. This was first stated by *Pauli*, in the year 1925, and is known as *Pauli's exclusion principle*. As can be easily understood, it is this principle which accounts for the arrangement of electrons in an atom.

Particles for which the spin is *zero* or an integral multiple of $\hbar$ (or $h/2\pi$) are called **Bosons** (after Bose), as for example, α -particles, *spin zero*, *photons*, *spin 1*, *deuteron* (nucleus of deuterium), *spin 2*.

Example of conservation of spin angular momentum. Such an example is furnished by a positron slowing down in a material and capturing an electron to form an '*atom*' of **positronium**, for a brief period, with the positron as nucleus. Since both positron and electron have the same mass, the two are really distributed about their centre of mass. So that, when they capture each other, the only angular momentum involved is the spin angular momentum of $\frac{1}{2}\hbar$ of each. If their spins be opposed to each other, *i.e.*, if their spin vectors be anti-parallel, we have what is called **para positronium** whose total spin is $\frac{1}{2}\hbar - \frac{1}{2}\hbar = 0$. On decay, therefore, the annihilated mass appears as electromagnetic energy in the form of two *photons*, with their spins of 1 each in opposite directions, so that the spin angular momentum carried off is zero. The spin angular momentum thus remains conserved, being zero both before and after decay of the 'parapositronium atom.' Obviously, we cannot have only one photon produced during annihilation, for it would mean the carrying off of an angular momentum of $1\hbar$ from where it is zero.

If, on the other hand, the spin of the positron and the electron be aligned, *i.e.*, be in the same direction (or their spin vectors parallel), we obtain what is called **ortho positronium** whose total angular momentum is $\frac{1}{2}\hbar + \frac{1}{2}\hbar = 1\hbar$. In order that the angular momentum may remain conserved, therefore, it must decay into three photons, with two of them having opposite spins, so that the total angular momentum carried off is $1\hbar + 1\hbar - 1\hbar = 1\hbar$, the same as that of the 'ortho positronium atom'.

It need hardly be pointed out that we can observe the quantum nature of angular momentum only in the interaction of small entities like the fundamental particles because its unit $\hbar$ or $h/2\pi$ is much too small (equal to 1.0542×10^{-7} erg-sec). If, God forbid, we were living in a world in which $\hbar$ or $h/2\pi$ were equal to 1, large objects of human size would start behaving the way these small particles do and very strange things will happen to us. Thus, for example, a 1 kg mass whirled at the end of a metre long string would revolve only at frequencies 0, $1/2\pi$, $2/2\pi$, $3/2\pi$, etc. and the swing would occur in directions such that the change in angular momentum from one direction to the next is equal to $1\hbar$. And, our bicycles would then pick up speeds which are integral multiples of some basic speed (depending upon their size and weight), their wheels and handles; would turn only through certain discrete angles, and so on.

Total angular momentum. The total angular momentum of a revolving electron would obviously be equal to the sum of its orbital angular momentum and its spin angular momentum and would be given by $j(h/2\pi)$, where j is the *total angular momentum quantum number*. So that, if $\vec{j}$ be the *total angular momentum vector*, we have $\vec{j} = \vec{l} + \vec{s}$, such that the vector sum has always a half odd integral value, *i.e.*, it is always an odd multiple of $\frac{1}{2}$. Since s is always equal to $\frac{1}{2}$, it means that j can have only two values for a given value of l , *viz.*, $l + \frac{1}{2}$ and $l - \frac{1}{2}$, except, of course, when $l = 0$, in which case it will have only the value $\frac{1}{2}$.

Again, on the basis of wave mechanics, the magnitude of $\mathbf{j}$ must be $\sqrt{j(j+1)}$, that of $\mathbf{l}$ must be $\sqrt{l(l+1)}$ and that of $\mathbf{s}$ must be $\sqrt{s(s+1)}$, all in units $\hbar$ or $h/2\pi$.

In the event of there being two or more electrons in an atom, it is found that in an overwhelming number of cases we have what is called an **L—S coupling** (or *Russel-Saunders coupling*) in which all orbital angular momenta combine to give a resultant orbital angular momentum vector $\vec{L}$, irrespective of the spin angular momenta, and, similarly, all spin angular momenta combine to give

a resultant spin angular momentum vector $\vec{S}$, irrespective of the orbital angular momenta. The **total angular momentum of the atom** is thus represented by vector $\vec{J} = \vec{L} + \vec{S}$, such that $\vec{J}$ must be an integer or an odd multiple of according as $\vec{S}$ is an integer or an odd multiple of $\frac{1}{2}$. So that, the possible number of values that $\vec{J}$ can have if $\vec{L} > \vec{S}$ is $(2\vec{S} + 1)$ and the number of values that it can have when $\vec{L} < \vec{S}$ is $(2\vec{L} + 1)$. Of course, in case $\vec{L} = 0$, $\vec{J}$ can have only a single value, viz. $\vec{S}$.

N.B. For worked examples on quantisation of angular momentum, see chapter 7.

WORKED EXAMPLES

I—Conservation of linear momentum

Example 1. A hunter has a rifle that can fire 0.06 kg bullets with a muzzle velocity of 900 m/sec. A 40 kg leopard springs at him at a speed of 10 m/sec. How many bullets must the hunter fire into the leopard in order to stop it in its tracks?

Solution. Here, clearly, *momentum of the leopard* = (mass) (velocity) = (40)(10) kg-m/sec. The momentum of the bullets too that must be fired into the leopard must be at least this much in order to stop it in its tracks. Let this number of bullets be n . Then, *momentum of the bullets (in the opposite direction to that of the leopard)* = $n(0.06)(900)$ kg-m/sec. Equating the two, therefore, we have

$$n(0.06)(900) = (40)(10), \text{ whence, } n = \frac{40 \times 10}{0.06 \times 900} = \frac{400}{54} \approx 8.$$

Thus, the hunter must fire 8 bullets into the leopard in order to stop it.

Example 2. (a) What is the momentum of an electron of a kinetic energy 100 electron volts? (mass of the electron may be taken to be 9×10^{-28} gm).

(b) At what velocity will a 10,000 kg truck have (i) the same momentum, (ii) the same kinetic energy as a 4,000 kg car at 30 m/sec?

Solution. (a) We have K.E. of the electron = $\frac{1}{2}mv^2 = 100 \text{ eV} = 100 \times 1.6 \times 10^{-12} \text{ ergs}$.

$$\begin{aligned} \therefore \text{momentum of the electron} = mv &= \sqrt{m^2 v^2} = \sqrt{2m \times \frac{1}{2}mv^2} \\ &= \sqrt{2 \times 9 \times 10^{-28} \times 100 \times 1.6 \times 10^{-12}} \\ &= \sqrt{18 \times 1.6 \times 10^{-38}} = 5.37 \times 10^{-19} \text{ gm-cm/sec.} \end{aligned}$$

(b) (i) Let the truck have the same momentum as the car at a velocity of v m/sec. Then, clearly, $10,000 v = 4,000 \times 30$, whence, $v = 4,000 \times 30 / 10,000 = 12 \text{ m/sec}$.

(ii) If the truck has the same K.E. as the car at velocity v' , we have

$$\frac{1}{2} \times 10,000 \times v'^2 = \frac{1}{2} \times 4,000 \times 30^2, \text{ whence, } v' = \sqrt{360} = 18.97 \text{ m/sec.}$$

Thus, the truck will have the same momentum as the car at a velocity of 12 m/sec and the same K.E. as the car at a velocity of 18.97 m/sec.

Example 3. A vessel at rest explodes, breaking into three pieces. Two pieces, having equal masses, fly off perpendicular to one another with the same speed of 30 m/sec. The third piece has three times the mass of each other piece. What is the direction and magnitude of its velocity immediately after the explosion?

Solution. Here, the masses of the three pieces of the vessel are given to be as 1 : 1 : 3.

Let one of the smaller pieces move along the axis of x and the other, along the axis of y then clearly *momentum of the first small piece* = $(1)(30\hat{i}) = 30\hat{i}$

and *momentum of the second small pieces* = $(1)(30\hat{j}) = 30\hat{j}$.

If the speed of the third, bigger piece be $\vec{v}$ m/sec, we have

momentum of the bigger piece = $(3)(\vec{v}) = 3\vec{v}$.

Since initially the vessel was at rest and hence its linear momentum, zero, the total momentum of the three pieces, after explosion, should, in accordance with the law of conservation of momentum, be also *zero*, since no external force is acting on the vessel. We, therefore, have

$$30\hat{i} + 30\hat{j} + 3\vec{v} = 0,$$

whence, $3\vec{v} = -30\hat{i} - 30\hat{j}$ or, $\vec{v} = -(10\hat{i} + 10\hat{j})$.

Thus, the magnitude of the velocity of the bigger piece, i.e., $v = \sqrt{10^2 + 10^2} = \sqrt{200} = 10\sqrt{2}$ m/sec. And its direction with the axis of x or the direction of motion of the first piece, say, $\theta = \tan^{-1}(-1/1)$, i.e., $\tan^{-1}(-1) = 135^\circ$, the same as the angle that its direction makes with the axis of y or the direction of motion of the second piece.

Thus, *the velocity of bigger piece immediately after the explosion is $10\sqrt{2}$ m/sec, inclined at an angle of 135° to the direction of motion of either small piece.*

Example 4. Radioactive ${}_8\text{O}^{15}$ decays into ${}_7\text{N}^{15}$ by emitting a positron and neutrino. The positron and the neutrino are observed to move at right angles to each other and carry momenta 2×10^{-22} and 5×10^{-23} kg m sec $^{-1}$ respectively. Find the momentum of the recoiling nucleus.

Solution. Let the positron move along the $+X$ direction and the neutrino along $+Y$ direction with momenta $\vec{p}_1 = 2 \times 10^{-22} = 20 \times 10^{-23}$ kg m sec $^{-1}$ and $\vec{p}_2 = 5 \times 10^{-23}$ kg m sec $^{-1}$ respectively. If $\vec{p}$ is the momentum of the recoiling nucleus, then according to the principle of conservation of linear momentum.

$$\vec{p}_1 + \vec{p}_2 + \vec{P} = 0$$

$$\text{or } \vec{P} = -(\vec{p}_1 + \vec{p}_2)$$

$$\begin{aligned} \text{Now } |\vec{p}_1 + \vec{p}_2| &= \sqrt{(20 \times 10^{-23})^2 + (5 \times 10^{-23})^2} \\ &= 5\sqrt{17} \times 10^{-23} \text{ kg m sec}^{-1}. \end{aligned}$$

and its direction is given by

$$\tan \theta = \frac{5 \times 10^{-23}}{20 \times 10^{-23}} = \frac{1}{4} = 0.25$$

$$\text{or, } \theta = 14^\circ 2'$$

$\therefore \vec{P} = 5\sqrt{17} \times 10^{-23}$ kg m sec $^{-1}$ and it makes an angle $14^\circ 2'$ with $-X$ direction or $(180^\circ + 14^\circ 2')$ = $194^\circ 2'$ with $+X$ direction.

Example 5. Two bodies of different masses are moving with the same kinetic energy of translation. Which has greater momentum?

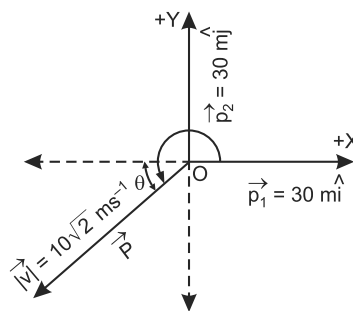


Fig. 6.20

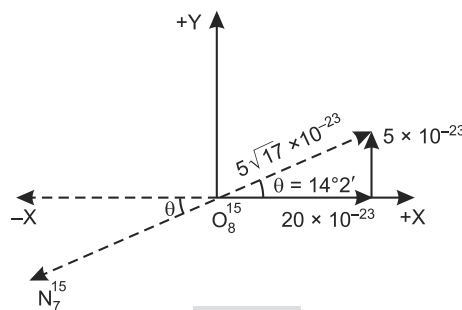


Fig. 6.21

Solution. Let m_a and v_a be the mass and velocity of mass A and m_b , v_b that of mass B with $m_a > m_b$, then

$$\text{K.E. of mass } A = \frac{1}{2} m_a v_a^2 \text{ and K.E. of mass } B = \frac{1}{2} m_b v_b^2$$

$$\text{Now,} \quad \frac{1}{2} m_a v_a^2 = \frac{1}{2} m_b v_b^2 \quad \therefore \frac{m_a}{m_b} = \frac{v_b^2}{v_a^2} \text{ or } \sqrt{\frac{m_a}{m_b}} = \frac{v_b}{v_a}$$

$$\text{Linear momentum of } A, \quad p_a = m_a v_a$$

$$\text{Linear momentum of } B, \quad p_b = m_b v_b$$

$$\therefore \quad \frac{p_a}{p_b} = \frac{m_a v_a}{m_b v_b} = \frac{m_a}{m_b} \sqrt{\frac{m_b}{m_a}} = \sqrt{\frac{m_a}{m_b}} \text{ but } m_a > m_b$$

$$p_a > p_b$$

i.e., the body A with greater mass has greater momentum.

II—Centre of mass

Example 6. (a) The distance between the centres of the carbon and oxygen atoms in the carbon monoxide (CO) molecule is 1.130×10^{-10} metres. Locate the centre of mass of the molecule relative to the carbon atom.

(b) Find the centre of mass of a homogeneous semicircular plate of radius a .

Solution. (a) The centre of mass of the CO molecule will obviously lie on the line joining the C and O atoms. Let it be at a distance x from the carbon atom. Then, since the atomic weights of carbon and oxygen are respectively 12 and 6, we have

$$12x = 6(1.130 \times 10^{-10} - x) = 6 \times 1.130 \times 10^{-10} - 6x.$$

$$\text{Or,} \quad 28x = 6 \times 1.130 \times 10^{-10}, \quad \text{whence, } x = 6 \times 1.130 \times 10^{-10} / 28$$

$$= 6.457 \times 10^{-11} \text{ m.}$$

Hence, the centre of mass of the CO molecule lies at a distance of 6.457×10^{-11} m from the centre of the carbon atom on the line joining the carbon and oxygen atoms.

(b) Here, too, the centre of mass obviously lies on the line of symmetry OA (Fig. 6.22) passing through the mid-point O of the base and perpendicular to it.

Since it is a homogeneous plate, the centre of mass will be at a distance R and O , given by the relation

$$R = \frac{1}{M} \int_0^a r \cdot dm, \text{ where } dm \text{ is the mass of an element of it of}$$

radius r and M , the mass of the whole plate.

Clearly, area of the semicircular plate $= \pi a^2 / 2$. We therefore, have

$$\text{mass per unit area of the plate} = M / \frac{\pi a^2}{2} = 2M / \pi a^2.$$

And, area of the element of the plate, of radius r $=$ (length of shaded strip) $\times$ i

$$(\text{its width } dr) = \pi r dr.$$

$$\therefore \text{ mass of the strip, i.e., } dm = (\pi r dr)(2M / \pi a^2)$$

$\therefore$ distance of centre of mass of the semicircular plate from O , i.e.,

$$R = \frac{1}{M} \int_0^a r \pi r dr 2M / \pi a^2 = \frac{2}{a^2} \int_0^a r^2 dr = \frac{2}{a^2} \left[\frac{r^3}{3} \right]_0^a = \frac{2}{a^2} \left(\frac{a^3}{3} \right) = \frac{2a}{3}.$$

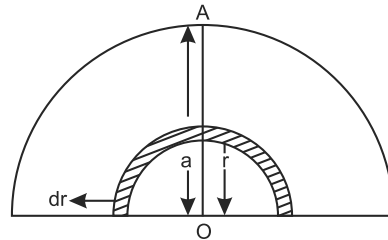


Fig. 6.22

Thus, the centre of mass of the semicircular plate lies on the right bisector OA of its base at a distance of $2a/3$ from its mid-point O .

Example 7. Two particles, of masses 100 and 300 gm, have at a given time positions $2\hat{i} + 5\hat{j} + 13\hat{k}$ and $-6\hat{i} + 4\hat{j} - 2\hat{k}$ cm respectively and velocities $10\hat{i} - 7\hat{j} - 3\hat{k}$ and $7\hat{i} - 9\hat{j} + 6\hat{k}$ cm/sec respectively. Deduce (i) the instantaneous position of the centre of mass, (ii) the velocity of the second particle in a frame of reference travelling with the centre of mass.

(Nagpur 2005)

Solution. (i) We know that the distance of the centre of mass from the chosen origin is given by

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

$\therefore$ Substituting the given values of m_1 , m_2 , r_1 and r_2 , we have

$$\vec{R} = \frac{100(2\hat{i} + 5\hat{j} + 13\hat{k}) + 300(-6\hat{i} + 4\hat{j} - 2\hat{k})}{100 + 300} = \frac{-16\hat{i} + 17\hat{j} + 7\hat{k}}{4} \text{ cm.}$$

Thus, the centre of mass lies at a distance of $(-16\hat{i} + 17\hat{j} + 7\hat{k})/4$ cm from the origin.

(ii) The velocity of the centre of mass is given by

$$\begin{aligned} \vec{V} &= \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2} = \frac{100(10\hat{i} - 7\hat{j} - 3\hat{k}) + 300(7\hat{i} - 9\hat{j} + 6\hat{k})}{100 + 300} \\ &= \frac{31\hat{i} - 34\hat{j} + 15\hat{k}}{4} \text{ cm/sec.} \end{aligned}$$

$\therefore$ velocity of the second particle in the frame of reference travelling with the centre of mass

$$\begin{aligned} &= \vec{v}_2 - \vec{V} = (7\hat{i} - 9\hat{j} + 6\hat{k}) - \frac{31\hat{i} - 34\hat{j} + 15\hat{k}}{4} \\ &= \frac{-3\hat{i} - 2\hat{j} + 9\hat{k}}{4} \text{ cm/sec.} \end{aligned}$$

Example 8. Two particles P and Q are initially at rest, 1.0 metre apart. P has a mass of 0.10 kg and Q , a mass of 0.30 kg. P and Q attract each other with a constant force of 1.0×10^{-2} N. No external forces act on the system. Describe the motion of the centre of mass. At what distance from P 's original position do the particles collide?

Solution. Since no external forces act on the system, the question of motion of the centre of mass does not arise. It, therefore, continues to be at rest.

Let the distance at which the two particles collide be x metres from P 's original position. Then, clearly, the two particles cover distances x metres and $(1 - x)$ metres respectively, in the same interval of time t , say, that they take to come into contact with each other.

Now, acceleration of particle $P = \frac{\text{force}}{\text{mass}} = \frac{1 \times 10^{-2}}{0.1} = 10^{-1} \text{ m/sec}^2$ and that of particle

$$Q = \frac{1 \times 10^{-2}}{0.3} = \frac{1}{3} \times 10^{-1} \text{ m/sec}^2.$$

$\therefore$ distance x covered by particle P in time $t = \frac{1}{2} \times 10^{-1} \times t^2 = \frac{1}{20} t^2$ metres.

and distance $(1 - x)$ covered by particle Q in the same time $t = \frac{1}{2} \times \frac{1}{3} \times 10^{-1} \times t^2 = \frac{1}{60} t^2$ metres.

Clearly, therefore, $\frac{1}{20} t^2 + \frac{1}{60} t^2 = (x) + (1 - x) = 1$, whence, $t^2 = 15$.

Hence
$$x = \frac{1}{20}t^2 = \frac{1}{20} \times 15 = \frac{3}{4} = 0.75 \text{ metres.}$$

Thus, the two particles collide at a distance of 0.75 m from the initial position of particle P.

Example 9. A bomb in flight explodes into two fragments when its velocity is $10\hat{i} + 2\hat{j}$. If the smaller mass M flies with velocity $20\hat{i} + 50\hat{j}$, deduce the velocity of the larger mass $3M$. Deduce also the velocities of the fragments in the centre of mass reference frame and show these in a diagram.

Solution. Here, clearly, initial momentum of the unexploded bomb = (total mass) $\times$ (velocity)

$$= (M + 3M)(10\hat{i} + 2\hat{j}) = 4M(10\hat{i} + 2\hat{j})$$

After explosion, momentum of the smaller fragment = $M\vec{v}_1 = M(20\hat{i} + 50\hat{j})$
 and momentum of the larger fragment = $3M\vec{v}_2$,
 where $\vec{v}_1$ and $\vec{v}_2$ are the velocities of the two fragment respectively.

Since, no external force acts on the system, its momentum must remain conserved. We, therefore, have total momentum after explosion = total momentum before explosion. Or,

$$M(20\hat{i} + 50\hat{j}) + 3M\vec{v}_2 = 4M(10\hat{i} + 2\hat{j}). \text{ Or, } 3M\vec{v}_2 = 4M(10\hat{i} + 2\hat{j}) - M(20\hat{i} + 50\hat{j}),$$

$$\text{whence, } \vec{v}_2 = \frac{1}{3}(20\hat{i} + 42\hat{j}).$$

Thus, the velocity of the larger mass $3M$ is $\frac{1}{3}(20\hat{i} + 42\hat{j})$.

Now, in the absence of any external force, the centre of mass continues to move with the same velocity as that of the unexploded bomb, viz.,

$$\vec{V} = (10\hat{i} + 2\hat{j}).$$

$\therefore$ velocity of the smaller mass in the centre of mass reference frame = $\vec{v}_1 - \vec{V} = (20\hat{i} + 50\hat{j}) - (10\hat{i} + 2\hat{j}) = 10\hat{i} + 48\hat{j}$
 and velocity of the larger mass in the centre of mass reference frame = $\vec{v}_2 - \vec{V} = \frac{1}{3}(20\hat{i} + 42\hat{j}) - (10\hat{i} + 2\hat{j}) = \frac{1}{3}(-10\hat{i} - 48\hat{j})$

$$= -\frac{1}{3}(10\hat{i} + 48\hat{j})$$

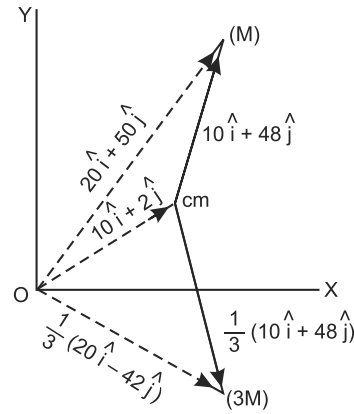


Fig. 6.23

All these velocities are shown in Fig. 6.23, both relative to the origin O and the centre of mass ($c.m.$), the former in dotted, and the latter, in full lines.

Example 10. If the centre of mass of three particles of masses 1, 2 and 3 kg be at the point 3, 3, 3, where should a fourth mass of 4 kg be placed so that the centre of mass of the four particles be at the point 1, 1, 1?

Solution. Let $x_1, y_1, z_1, x_2, y_2, z_2$ and x_3, y_3, z_3 be the positions of the three particles respectively. Then, for the x-coordiante,
$$X = \frac{m_1x_1 + m_2x_2 + m_3x_3}{m_1 + m_2 + m_3}$$

$$\text{Or, } 3 = \frac{x_1 + 2x_2 + 3x_3}{1 + 2 + 3} = \frac{x_1 + 2x_2 + 3x_3}{6}.$$

$$\text{Or, } x_1 + 2x_2 + 3x_3 = 18 \quad \dots(i)$$

Now, suppose the fourth particle of mass 4 kg must be placed at the point x_4, y_4, z_4 so that the centre of mass of the combination of all the four particles lies at 1, 1, 1. Then, again, for the x-coordinate, we have

$$\text{Or, } 1 = \frac{x_1 + 2x_2 + 3x_3 + 4x_4}{1 + 2 + 3 + 4} = \frac{x_1 + 2x_2 + 3x_3 + 4x_4}{10}$$

$$\text{Or, } x_1 + 2x_2 + 3x_3 + 4x_4 = 10 \quad \dots(ii)$$

Subtracting relation (i) from (ii), we have $4x_4 = 10 - 18 = -8$, whence, $x_4 = -2$.

Proceeding in the same manner for the y and z coordinates also, we obtain $y_4 = -2$ and $z_4 = -2$.

Thus, the fourth mass of the 4 kg must be placed at the point $-2, -2, -2$, in order that the centre of mass of the combination of all the four particles may be at the point $1, 1, 1$.

Example 11. Centre of mass is at $P(1, 1, 1)$ when system consists of particles of masses 2, 3, 4 and 5 kg. If the centre of mass shifts to $Q(2, 2, 2)$ on removing 5 kg, what was its position?

Solution. The position vector of centre of mass is given by

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + m_4 \vec{r}_4 + \dots}{m_1 + m_2 + m_3 + m_4 + \dots}$$

$$\text{1st case } \vec{R}_1 = (\hat{i} + \hat{j} + \hat{k}) = \frac{2\vec{r}_1 + 3\vec{r}_2 + 4\vec{r}_3 + 5\vec{r}_4}{2 + 3 + 4 + 5} \quad \dots(i)$$

$$\text{2nd case } \vec{R}_2 = (2\hat{i} + 2\hat{j} + 2\hat{k}) = \frac{2\vec{r}_1 + 3\vec{r}_2 + 4\vec{r}_3}{2 + 3 + 4} \quad \dots(ii)$$

$$\text{From (i) we have } 14\hat{i} + 14\hat{j} + 14\hat{k} = 2\vec{r}_1 + 3\vec{r}_2 + 4\vec{r}_3 + 5\vec{r}_4 \quad \dots(iii)$$

$$\text{From (ii) we have } 18\hat{i} + 18\hat{j} + 18\hat{k} = 2\vec{r}_1 + 3\vec{r}_2 + 4\vec{r}_3$$

$$\text{Subtracting (iv) from (iii), we get } -4\hat{i} - 4\hat{j} - 4\hat{k} = 5\vec{r}_4 \quad \dots(iv)$$

$$\therefore \vec{r}_4 = -\frac{4}{5}\hat{i} - \frac{4}{5}\hat{j} - \frac{4}{5}\hat{k}$$

$$\therefore \text{Co-ordinates of the mass of 5 kg are } \left(-\frac{4}{5}, -\frac{4}{5}, -\frac{4}{5}\right).$$

Example 12. Two bodies of masses 2 g and 10 g have position vectors $(3\hat{i} + 2\hat{j} - \hat{k})$ and $(\hat{i} - \hat{j} + 3\hat{k})$ respectively. Find the position vectors and the distance of centre of mass from the origin.

Solution. The position of centre of mass of two particles is given by $\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$

$$\begin{aligned} \therefore \vec{R} &= \frac{2(3\hat{i} + 2\hat{j} - \hat{k}) + 10(\hat{i} - \hat{j} + 3\hat{k})}{2 + 10} \\ &= \frac{16\hat{i} - 6\hat{j} + 28\hat{k}}{12} = \frac{4}{3}\hat{i} - \frac{1}{2}\hat{j} + \frac{7}{3}\hat{k} \end{aligned}$$

Distance of centre of mass from the origin.

$$|\vec{R}| = \sqrt{\left(\frac{4}{3}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{7}{3}\right)^2} = \sqrt{\frac{16}{9} + \frac{1}{4} + \frac{49}{9}} = 2.73 \text{ units}$$

Example 13. The distance between Carbon and Oxygen atom in 'CO' molecule is $1.12 \text{ \AA}$. Find the centre of mass of 'CO' molecule with respect to 'C' atom.

Solution. Let us take the centre of mass as the origin of the co-ordinate system and axis of 'CO' molecule along X -axis, then

Position of Carbon atom of mass 12

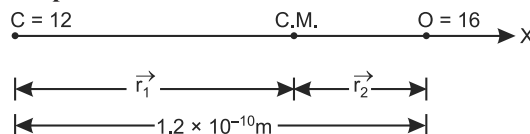


Fig. 6.24

$$= \vec{r}_1 = -r_1 \hat{i}$$

Position of Oxygen atom of mass 16

$$= \vec{r}_2 = +(1.12 \times 10^{-10} - r_1) \hat{i}$$

As the centre of mass is taken as the origin of the co-ordinate system

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} = 0$$

$$\text{or } m_1 \vec{r}_1 + m_2 \vec{r}_2 = 0$$

$$\therefore -12r_1 \hat{i} + 16(1.12 \times 10^{-10} - r_1) \hat{i} = 0$$

$$\text{or } 28r_1 \hat{i} = 16 \times 1.12 \times 10^{-10} \hat{i}$$

$$\therefore r_1 = \frac{16 \times 1.12 \times 10^{-10}}{28} = 0.64 \times 10^{-10} \text{ m} = 0.64 \text{ \AA}$$

As the position of Carbon atom with respect to centre of mass = $-r_1 \hat{i}$, the position of centre of mass with respect to 'C' atom = $+r_1 \hat{i} = +0.64 \times 10^{-10} \text{ m}$.

Example 14. Two masses constrained to move in a horizontal plane collide. Given initially $m_1 = 85 \text{ gms}$, $m_2 = 200 \text{ gms}$; $u_1 = 6.48 \text{ cms/sec}$ and $u_2 = -6.78 \text{ cms/sec}$, find the velocity of centre of mass.

Solution. The velocity of centre of mass is given by $\vec{V}_{cm} = \frac{m_1 \vec{u}_1 + m_2 \vec{u}_2}{m_1 + m_2}$

$$\therefore \vec{V}_{cm} = \frac{85 \times 6.48 + 200 \times (-6.78)}{85 + 200} = 2.82 \text{ cm/sec in the direction of motion of } m_2.$$

Example 15. Two particles each of mass 2 kg are moving with velocities $2\hat{i} + 4\hat{j} \text{ m/s}$ and $5\hat{i} + 6\hat{j} \text{ m/s}$ respectively. Find the kinetic energy of the system relative to centre of mass.

Solution. Given $m_1 = m_2 = 2 \text{ kg}$; $u_1 = 3\hat{i} + 4\hat{j}$; $u_2 = 5\hat{i} + 6\hat{j}$

$$\text{Velocity of centre of mass } \vec{V}_{cm} = \frac{m_1 \vec{u}_1 + m_2 \vec{u}_2}{m_1 + m_2} = \frac{2(3\hat{i} + 4\hat{j}) + 2(5\hat{i} + 6\hat{j})}{2 + 2} = 4\hat{i} + 5\hat{j}$$

$$\text{Velocity of } m_1 \text{ in centre of mass frame } \vec{u}'_1 = \vec{u}_1 - \vec{V}_{cm}$$

$$\text{or } \vec{u}'_1 = 3\hat{i} + 4\hat{j} - 4\hat{i} - 5\hat{j} = -\hat{i} - \hat{j}$$

$$\text{Velocity of } m_2 \text{ in centre of mass } \vec{u}'_2 = \vec{u}_2 - \vec{V}_{cm}$$

$$\text{or } \vec{u}'_2 = 5\hat{i} + 6\hat{j} - 4\hat{i} - 5\hat{j} = \hat{i} + \hat{j}$$

$$\text{Kinetic energy relative to centre of mass before collision} = \frac{1}{2} m_1 u_1'^2 + \frac{1}{2} m_2 u_2'^2$$

$$= \frac{1}{2} m_1 \left| \sqrt{(-1)^2 + (-1)^2} \right|^2 + \frac{1}{2} m_2 \left| \sqrt{(+1)^2 + (+1)^2} \right|^2$$

$$= 2 + 2 = 4 \text{ Joule.}$$

III—Collisions

Example 16. (a) A particle of mass m_1 , moving with a velocity u_1 , collides head-on with a particle of mass m_2 at rest, such that, after collision, they travel with velocities v_1 and v_2 respectively. If the collision be an elastic one, show that $v_2 = 2u_1 \left(1 + \frac{m_2}{m_1}\right)$.

(b) If the collision in (a) be assumed to be an inelastic one, calculate the common velocity of the two particles after collision and the loss of kinetic energy due to collision.

Solution. (a) A 'head-on' collision means that, after the collision, the two particles travel along the same line.

Now, in accordance with the law of conservation of momentum, we have

$$m_1 u_1 + m_2 \times 0 = m_1 v_1 + m_2 v_2. \quad \text{Or, } m_1 u_1 = m_1 v_1 + m_2 v_2, \quad \dots(i)$$

whence,
$$v_1 = (m_1 u_1 - m_2 v_2)/m_1 = u_1 - \frac{m_2}{m_1} v_2. \quad \dots(ii)$$

And, in accordance with the law of conservation of energy, we have

$$\frac{1}{2} m_1 u_1^2 + 0 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2. \quad \text{Or, } \frac{1}{2} m_1 u_1^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \quad \dots(iii)$$

This gives
$$u_1^2 = v_1^2 + \frac{m_2}{m_1} v_2^2 \quad \dots(iv)$$

Or, substituting the value of v_1 from relation (ii) above, we have

$$u_1^2 = u_1^2 + \frac{m_2^2}{m_1^2} v_2^2 - \frac{2m_2 u_1 v_2}{m_1} + \left(\frac{m_2}{m_1} v_2\right)^2$$

Or,
$$0 = \frac{m_2}{m_1} \left(v_2^2 + \frac{m_2}{m_1} v_2^2 - 2u_1 v_2 \right)$$

Or,
$$0 = \frac{m_2}{m_1} \left(v_2 + \frac{m_2}{m_1} v_2 - 2u_1 \right) \quad \text{Or, } v_2 + \frac{m_2}{m_1} v_2 - 2u_1 = 0.$$

Or,
$$v_2 \left(1 + \frac{m_2}{m_1} \right) = 2u_1, \quad \text{whence, } v_2 = \frac{2u_1}{1 + \frac{m_2}{m_1}}$$

(b) Here, the collision **being inelastic**, the two particles stick together on collision and move with a common velocity v , say.

Then, for conservation of momentum, we have

$$m_1 u_1 + 0 = (m_1 + m_2)v, \quad \text{whence, } v = m_1 u_1 / (m_1 + m_2).$$

And, clearly, loss of kinetic energy due to collision $= \frac{1}{2} m_1 u_1^2 - \frac{1}{2} (m_1 + m_2)v^2$.

Or, substituting the value of v , we have

$$\begin{aligned} \text{loss of kinetic energy} &= \frac{1}{2} m_1 u_1^2 - \frac{1}{2} (m_1 + m_2) \left(\frac{m_1 u_1}{m_1 + m_2} \right)^2 = \frac{1}{2} m_1 u_1^2 - \frac{1}{2} \frac{m_1^2 u_1^2}{m_1 + m_2} \\ &= \frac{1}{2} m_1 u_1^2 \left(1 - \frac{m_1}{m_1 + m_2} \right) = \frac{1}{2} m_1 u_1^2 \left(\frac{m_2}{m_1 + m_2} \right). \end{aligned}$$

Example 17. A steel ball weighing 1 lb is fastened to a cord 27 inches long and is released when the cord is horizontal. At the bottom of its path the ball strikes a 5.0 lb block initially

at rest on a frictionless surface (Fig. 6.25). The collision is elastic. Find the speed of the ball and the speed of the block just after collision.

Solution. As the ball comes down and hits the block, it is clearly a head-on collision. If, therefore, m_1 and m_2 be the masses of the ball and the block respectively, u_1 , the velocity acquired by the ball on falling through 27", with which it hits the block, v_1 , the velocity of recoil of the ball and v_2 , the forward velocity of the block, we have, for conservation of momentum,

$$m_1 u_1 + 0 = -m_1 v_1 + m_2 v_2 \quad \dots(i)$$

$K.E.$ acquired by the ball on falling through 27" or 9/4 ft = $\frac{1}{2} m_1 u_1^2 = m_1 g h$,

whence, $u_1^2 = 2gh = 2 \times 32 \times 9/4 = 144$. Or, $u_1 = \sqrt{144} = 12 \text{ ft/sec}$.

Now, proceeding as in example 16 above, we have

$$v_2 = \frac{2u_1}{1 + m_2/m_1} = \frac{2 \times 12}{1 + 5/1} = \frac{24}{6} = 4 \text{ ft/sec}.$$

Substituting $v_2 = 4 \text{ ft/sec}$ in the momentum relation (i) above, we have

$$1 \times 12 = -1 \times v_1 + 5 \times 4, \quad \text{Or, } v_1 = 20 - 12 = 8 \text{ ft/sec}.$$

Thus, the ball recoils with a velocity of 8 ft/sec and the block moves forward with a velocity of 4 ft/sec.

Example 18. A sand bag of mass 10 kg is suspended with a 3 metre long weightless string. A bullet of mass 200 gm is fired with a speed 20 m/sec into the bag and stays in the bag. Calculate (i) the speed acquired by the bag, (ii) the maximum displacement of the bag, (iii) energy converted to heat in the collision.

Solution. (i) This is obviously a case of an inelastic collision between the bullet and the bag, where $m_1 = 200 \text{ gm} = 0.2 \text{ kg}$, $m_2 = 10 \text{ kg}$, $u_1 = 20 \text{ m/sec}$ and v , the common speed of the bullet and the bag is to be determined.

In accordance with the law of conservation of momentum, we have

$$m_1 u_1 = (m_1 + m_2) v, \text{ whence,}$$

$$v = \frac{m_1 u_1}{m_1 + m_2} = \frac{0.2 \times 20}{0.2 + 10} = \frac{4}{10.2} = 0.3922 \text{ m/sec}.$$

(ii) The bag will oscillate like a simple pendulum of length 3 metres about its point of suspension under a restoring force $F = -mg \sin \theta = -Cx$, say (Fig. 6.26), where x is the amplitude or the maximum displacement of the bag and the bullet lodged into it.

If l be the length of the pendulum, we have $-mg \sin \theta = -mgx/l = -Cx$, where $C = mg/l$. Since m (the mass of the bag and the bullet) = 10.2 kg, $g = 9.8 \text{ m/sec}^2$, we have $C = 10.2 \times 9.8/3$.

The total work done in this displacement of the bag and the bullet = $\int_0^x Cx dx = \frac{1}{2} Cx^2$. This work is obviously done at the expense of the $K.E.$ acquired by the bag and the bullet, equal to $\frac{1}{2} mv^2$. We, therefore, have $\frac{1}{2} Cx^2 = \frac{1}{2} mv^2 = \frac{1}{2} (10.2)(0.3922)^2$.

Or, $Cx^2 = (10.2)(0.3922)^2$.

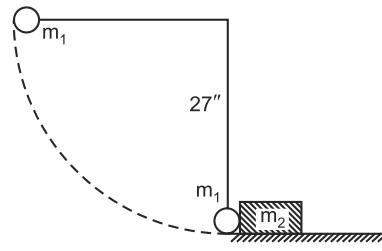


Fig. 6.25

Or,
$$x^2 = \frac{(10.2)(0.3922)^2(3)}{(10.2)(9.8)},$$

whence,
$$x = \sqrt{\frac{(0.3922)^2 \times 3}{9.8}} = 0.217 \text{ metre.}$$

The maximum displacement of the bag is thus 0.217 metre.

(iii) The energy converted into heat is obviously equal to the difference between the initial *K.E.* of the bullet and the final *K.E.* of the bullet and the bag, i.e.,

$$\begin{aligned} &= \frac{1}{2} m_1 u_1^2 - \frac{1}{2} (m_1 + m_2) v^2 \\ &= \frac{1}{2} (0.2)(20)^2 - \frac{1}{2} (10.2)(0.3922)^2 \\ &= 40 - 0.7845 = 39.2 \text{ joules.} \end{aligned}$$

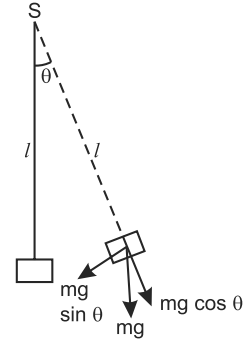


Fig. 6.26

Example 19. Show that in the case of an elastic and glancing collision between two particles of masses m_1 and m_2 respectively, the maximum value of the scattering angle θ_1 in the laboratory frame corresponds to the scattering angle θ in the centre of mass reference frame, where $\theta = \cos^{-1}(-m_2/m_1)$. Also show that this maximum value of the scattering angle

$$\theta_1 = \tan^{-1} \left(\frac{m_1^2}{m_2^2} - 1 \right)^{-\frac{1}{2}}.$$

Solution. We know that the scattering angle θ_1 in the laboratory reference frame is related to the scattering angle θ in the centre of mass reference frame by the relation

$$\tan \theta_1 = \frac{\sin \theta}{\cos \theta + (m_1/m_2)} \quad \dots(i)$$

(§ fig. 6.11(ii))

Differentiating with respect to θ , therefore, we have

$$\begin{aligned} \sec^2 \theta_1 \frac{d\theta_1}{d\theta} &= \frac{\cos \theta \left(\cos \theta + \frac{m_1}{m_2} \right) - \sin \theta (-\sin \theta)}{(\cos \theta + m_1/m_2)^2} \\ &= \frac{\cos^2 \theta + \sin^2 \theta + \frac{m_1}{m_2} \cos \theta}{(\cos \theta + m_1/m_2)^2} = \frac{1 + \frac{m_1}{m_2} \cos \theta}{(\cos \theta + m_1/m_2)^2}. \end{aligned}$$

For maximum value of θ_1 , $d\theta_1/d\theta$ must be equal to zero. This means, in other words, that $1 + \frac{m_1}{m_2} \cos \theta = 0$, or that $\cos \theta = -\frac{m_2}{m_1}$, whence, $\theta = \cos^{-1} \left(-\frac{m_2}{m_1} \right)$

Now, $\sin \theta = (1 - \cos^2 \theta)^{\frac{1}{2}} = (1 - m_2^2/m_1^2)^{\frac{1}{2}}$. So that, substituting the values of $\sin \theta$ and $\cos \theta$ in relation (i) above, we have $\tan \theta_1 = \frac{(1 - m_2^2/m_1^2)^{\frac{1}{2}}}{-\frac{m_2}{m_1} + \frac{m_1}{m_2}}$

$$= \frac{(m_1^2 - m_2^2)^{\frac{1}{2}}/m_1}{(m_1^2 - m_2^2)/m_1 m_2} = \frac{m_2}{(m_1^2 - m_2^2)^{\frac{1}{2}}}$$

$$= \left(\frac{m_2^2}{(m_1^2 - m_2^2)} \right)^{\frac{1}{2}} = \left(\frac{m_1^2 - m_2^2}{m_2^2} \right)^{-\frac{1}{2}}$$

Or,

$$\theta_1 = \tan^{-1} \left(\frac{m_1^2}{m_2^2} - 1 \right)^{-\frac{1}{2}}$$

IV–Rocket

Example 20. A rocket motor consumes 100 kg of fuel per second, exhausting it with a speed of 5×10^3 m/sec. What force is exerted on the rocket? What will be the velocity of the rocket at the instant that its mass is reduced to 1/20th of its initial mass, its initial velocity being zero? (Neglect gravitational and viscous forces etc.)

Solution. We have *force or thrust on the rocket* given by $F = -\frac{dM}{dt}v$. [§ 6.12]

Here, *rate of consumption of fuel*, $dM/dt = 100$ kg/sec and exhaust velocity $v = 5 \times 10^3$ m/sec. So that, *thrust on the rocket* $= -100 \times 5 \times 10^3 = -5 \times 10^5$ newtons.

Now, as we know, (§ 6.12), *velocity of the rocket at an instant t* is given by

$$V = V_0 + v \log_e \left(\frac{M_0}{M} \right),$$

where V_0 is its initial velocity at $t = 0$, M_0 , its initial mass and M , its mass at instant t .

Here, initial velocity V_0 , is given be zero. So that, $V = v \log_e \frac{M_0}{M}$.

Since the mass of the rocket is reduced to 1/20th of its initial mass, $M_0/M = 20$. We therefore, have

$$\begin{aligned} V &= v \log_e 20 = v \times 2.3026 \log_{10} 20 = v \times 2.3026 \times 1.3010 \\ &= 2.995 v \approx 3v, \end{aligned}$$

i.e., very nearly 3 times the exhaust velocity.

Since the exhaust velocity $v = 5 \times 10^3$ m/sec, we have

velocity of the rocket at the instant its mass is reduced to 1/20th of its initial mass
 $= 3 \times 5 \times 10^3 = 15 \times 10^3$ metres per second very nearly.

Example 21. If the maximum possible exhaust velocity of a rocket be 2 km/sec, calculate the ratio M_0/M for it if it is to attain the escape velocity 11.2 km/sec. How long will it take the rocket (starting from rest) to attain this velocity if its rate of change of mass in terms of its initial mass, i.e., $\beta = 1/10$?

Solution. We have velocity of the rocket, starting from rest (i.e., $V_0 = 0$), given by

$$V = v \log_e \frac{M_0}{M} \text{ So that, } \log_e \frac{M_0}{M} = \frac{V}{v} = \frac{11.2}{2} = 5.6.$$

Or,

$$2.3026 \log_{10} \frac{M_0}{M} = 5.6, \quad \text{Or, } \log_{10} \frac{M_0}{M} = \frac{5.6}{2.3026} = 2.432,$$

whence, $M_0/M = 270$.

Thus, the ratio M_0/M for the rocket must be 270.

Now, as we know, $M = M_0(1 - \beta t)$. So that, $M/M_0 = 1 - \beta t$. Or, $\beta t = (1 - M/M_0)$,

whence,

$$t = \frac{1}{\beta} \left(1 - \frac{M}{M_0} \right) = 10 \left(1 - \frac{1}{270} \right) = \frac{10 \times 269}{270} = 9.96 \text{ sec.}$$

Thus, the rocket will, in this case, attain the escape velocity in 9.96 sec.

Example 22. A rocket starts vertically upward with speed v_0 . Show that its speed v at a height h is given by $v_0^2 - v^2 = 2gh/(1 + h/R)$, where R is the radius of the earth and g the acceleration due to gravity at the earth's surface. Hence deduce an expression for the maximum height reached by a rocket fired with a speed 90% of the escape velocity.

Solution. For the velocity of a body going vertically upward with an initial velocity u , we have the relation $v^2 = u^2 - 2gh$, where v is the velocity acquired by it on covering a height h and g , the acceleration due to gravity, assumed to be constant over comparatively small heights above the surface of the earth.

In the case of a rocket, large heights being involved, g cannot be presumed to remain constant throughout. So that, at the surface of the earth g being equal to MG/R^2 (where M is the mass and R , the radius of the earth, and G the gravitational constant), we have at height h above the surface of the earth, the value of acceleration due to gravity given by

$$g' = MG/(R+h)^2 = MG/R^2(1+h/R)^2 = g/(1+h/R)^2 \quad [\because MG/R^2 = g.]$$

So that,

$$\begin{aligned} v^2 &= v_0^2 - 2 \int_0^h \frac{g}{(1+h/R)^2} dh = v_0^2 - 2g \left[-\frac{R}{1+h/R} \right]_0^h \\ &= v_0^2 + 2g \left[\frac{R}{1+h/R} - R \right] \end{aligned}$$

Or,

$$v^2 = v_0^2 - \frac{2gh}{1+h/R}. \quad \text{Or,} \quad v_0^2 - v^2 = \frac{2gh}{1+h/R}.$$

Now, as we know, the *escape velocity* is equal to $\sqrt{2gR}$ and, therefore, 90% of this velocity is $0.9\sqrt{2gR}$. Hence, the *initial velocity of the rocket in the second case* is $v_0 = 0.9\sqrt{2gR}$. And, at the maximum height h_m , obviously, $v = 0$. So that, substituting these values in the expression above, we have

$$(0.9\sqrt{2gR})^2 - 0 = \frac{2gh_m}{1+h_m/R}. \quad \text{Or,} \quad 0.81 \times 2gR = \frac{2gh_m R}{R+h_m}.$$

Or,

$$0.81 = \frac{h_m}{R+h_m}, \text{ whence, } h_m = \frac{0.81}{0.19} R = 4.263 R.$$

Thus, the maximum height attained by the rocket in this case is $4.26 R$.

Example 23. (a) A rocket, set for vertical firing, weighs 50 kg and contains 450 kg of fuel. It can have a maximum exhaust velocity of 2 km/sec. What should be its minimum rate of fuel consumption (i) to just lift it off the launching pad, (ii) to give it an acceleration of 20 m/sec^2 ?

(b) What will be the speed of the rocket when the rate of consumption of fuel is (i) 10 kg/sec (ii) 50 kg/sec?

Solution. (a) We know that, at any instant, the *thrust* on the rocket is given by

$$F = M \frac{dV}{dt} = -\frac{dM}{dt} v - Mg.$$

(i) When the rocket is still on the launching pad, $M = M_0 = 50 + 450 = 500 \text{ kg}$ and to just lift it off the pad with minimum fuel consumption, therefore,

$$-\frac{dM_0}{dt} v = M_0 g.$$

So that,

$$-\frac{dM_0}{dt} = \frac{M_0 g}{v} = \frac{500 \times 9.8}{2 \times 10^3} = 2.45 \text{ kg/sec.}$$

[$\because M_0 = 500 \text{ kg}$ and $v = 2 \text{ km/sec} = 2 \times 10^3 \text{ m/sec.}$]

Or, the *minimum rate of consumption of fuel to just lift the rocket off the launching pad* should be 2.45 kg/sec.

(ii) Again, $F = M dV/dt = Ma = -v dM/dt - Mg$, where a is the *acceleration* of the rocket.

Since the rocket is still on the launching pad, $M = M_0 = 500$ kg. We, therefore, have

$$M_0 a = -\frac{dM_0}{dt} v - M_0 g, \text{ whence, } -\frac{dM_0}{dt} = \frac{M_0(g+a)}{v}.$$

$$\text{Or, } -\frac{dM_0}{dt} = \frac{500(9.8+20)}{2 \times 10^3} = 7.45 \text{ kg/sec.}$$

Or, *the rate of the fuel consumption, in the case, should be 7.45 kg/sec.*

(b) Obviously, the rocket will acquire its maximum speed when its entire fuel has been consumed.

(i) Here, the rate of fuel consumption is 10 kg/sec. So that the time for the consumption of the entire fuel = $450/10 = 45$ sec.

Now, velocity acquired by the rocket, starting from rest, in time t , is given by

$$V = v \log_e(M_0/M) - gt \quad (\S 6.12, \text{ relation } V)$$

Here, $M_0 = 500$ kg and $M = 50$ kg (the entire fuel (450 kg) being burnt up)

$$\begin{aligned} \therefore V &= 2 \times 10^3 \log_e(500/50) - 9.8 \times 45 \\ &= 2 \times 10^3 \times 2.3026 \log_{10} 10 - 9.8 \times 45 \\ &= 2 \times 10^3 \times 2.3026 - 9.8 \times 45 = 4.164 \text{ km/sec.} \end{aligned}$$

Thus, *the speed of the rocket, in the case, will be 4.164 km/sec.*

(ii) Here, the entire fuel is burnt up in $450/50 = 9$ sec. And, therefore, *velocity of the rocket,*

$$\begin{aligned} V &= 2 \times 10^3 \log_e(500/50) - 9.8 \times 9 = 2 \times 10^3 \times 2.3026 - 9.8 \times 9 \\ &= 4517 \text{ m/sec} = 4.517 \text{ km/sec.} \end{aligned}$$

Thus, *the maximum speed of the rocket here will be 4.517 km/sec.*

Example 24. Show that the rocket speed is twice the exhaust speed when $\frac{M_0}{M} = e^2$.

Solution. If V_0 is the speed of the rocket and V_e the exhaust speed, then $\frac{V_0}{V_e} = \log_e \frac{M_0}{M}$.

$$\text{Now, } \frac{M_0}{M} = e^2 \quad \therefore \frac{V_0}{V_e} = \log_e e^2 = 2$$

i.e. the rocket speed is twice the exhaust speed.

Example 25. A 6000 kg rocket is set for vertical firing. If the gas exhaust speed is 1000 ms^{-1} , how much gas must be ejected each second to supply the thrust needed (i) to overcome the weight of the rocket and (ii) to give the rocket an initial upward acceleration of 20 ms^{-2} ?

Solution. (i) Here mass of the rocket and fuel $M = 6000$ kg

Exhaust velocity $V_e = -1000 \text{ ms}^{-1}$. Acceleration due to gravity $g = -9.8 \text{ ms}^{-2}$.

$$\text{Net force on the rocket } M \frac{dV}{dt} = V_e \frac{dM}{dt} + Mg$$

To overcome the weight of the rocket, the resultant force acting on it $M \frac{dV}{dt} = 0$

$$\therefore 0 = -1000 \frac{dM}{dt} - 6000 \times 9.8$$

$$\text{Or } \frac{dM}{dt} = -6 \times 9.8 = -58.8 \text{ kg s}^{-1}$$

(ii) To give the rocket an initial upward acceleration of 20 ms^{-2} , the resultant force

$$M \frac{dV}{dt} = Ma = 20 \times M$$

$$\therefore 20 \times 6000 = -1000 \times \frac{dM}{dt} - 6000 \times 9.8$$

$$\text{or } \frac{dM}{dt} = -\frac{6000(20+9.8)}{1000} = -178.8 \text{ kg s}^{-1}.$$

Example 26. A rocket of mass 20 kg has 180 kg of fuel. The exhaust velocity of fuel is 1.60 km s^{-1} . Calculate the ultimate vertical speed gained by the rocket when the rate of consumption of fuel is 2 kg s^{-1} .

Solution. If V_f is the final velocity of the rocket when the entire fuel has been exhausted, V_e the exhaust velocity, and rocket starts from rest, then $V_f = -V_e \log_e \frac{M_0}{M_f}$

where M_0 is the initial mass of rocket and fuel and M_f the mass of the empty rocket.

$$\text{Now } M_0 = 20 + 180 = 200 \text{ kg}$$

$$\text{Consumption of fuel} = 2 \text{ kg s}^{-1}$$

$$\text{Time to consume whole of fuel} = \frac{180}{2} = 90 \text{ s}$$

$$\text{Exhaust velocity } V_e = -1.6 \text{ km s}^{-1} = -1.6 \times 10^3 \text{ ms}^{-1}$$

$$\therefore V_f = 1.6 \times 10^3 \times 2.3026 \log_{10} \frac{200}{20} = 3.684 \times 10^3 \text{ ms}^{-1}$$

$$= 3.684 \text{ km s}^{-1}$$

V-Conservation of angular momentum

Example 27. A torque of 1 Nm is applied to a wheel of mass 10 kg and radius of gyration 50 cm. What is the resulting acceleration?

Solution. Here mass of the wheel, $M = 10 \text{ kg}$

$$\text{Radius of gyration, } K = 50 \text{ cm} = 0.5 \text{ m}$$

$$\text{Moment of inertia of the wheel, } I = MK^2$$

$$= 10 \times 0.5 \times 0.5 = 2.5 \text{ kg m}^2$$

$$\text{Torque, } \tau = 1 \text{ Nm}$$

Now, $\tau = I\alpha$, where α is the angular acceleration

$$\therefore \alpha = \frac{\tau}{I} = \frac{1}{2.5} = 0.4 \text{ rad s}^{-2}$$

Example 28. A particle of mass m moving in a circular orbit of radius r has angular momentum J about its centre. Calculate the kinetic energy of the particle in terms of J , m and r .

Solution. The angular momentum of a particle of mass m moving in a circular orbit of radius r is given by

$$J = mvr \quad \dots(i)$$

If E is the kinetic energy of the particle, then $E = \frac{1}{2}mv^2$.

$$\text{From equation (i), we get } v = \frac{J}{mr} \quad \text{or} \quad v^2 = \frac{J^2}{m^2 r^2}$$

$$\therefore \text{K.E.} = \frac{1}{2}mv^2 = \frac{1}{2}m \frac{J^2}{m^2 r^2} = \frac{J^2}{2mr^2}$$

Example 29. A particle of mass M travels with velocity $\vec{v} = v\hat{i}$ along the axis of y . At $t = 0$, the particle is at the point $(0, y_1)$. Calculate (a) the angular momentum of the particle about the origin, (b) the total angular momentum about the point $(0, y_2)$ where $y_2 > 0$.

Solution. (a) Here, clearly, the position vector of the particle, i.e., $\vec{r} = y_1\hat{j}$ and its velocity vector $\vec{v} = v\hat{i}$. Therefore, its angular momentum about the origin, i.e., $\vec{J} = \vec{r} \times m\vec{v}$.

$$\text{Or,} \quad \vec{J} = y_1\hat{j} + Mv\hat{i} = -Mvy_1\hat{k}$$

(b) In this case, the position vector of the particle, $\vec{r} = (y_2 - y_1)\hat{j}$ and its velocity vector, the same as before, i.e., $\vec{v} = v\hat{i}$.

$$\begin{aligned} \therefore \text{total angular momentum about the point } 0, y_2, \text{ i.e., } \vec{J} &= \vec{r} \times M\vec{v} \\ &= (y_2 - y_1)\hat{j} \times Mv\hat{i} = -Mv(y_2 - y_1)\hat{k}. \end{aligned}$$

Example 30. (a) A satellite of mass M_S is going round the earth in a circular orbit of radius R_S . Obtain an expression for its angular momentum about the centre of its orbit (i.e., the centre of the earth).

(b) Express the total energy of the satellite in terms of its angular momentum.

Solution. (a) Let $\vec{v}$ be the speed of the satellite in its circular orbit around the earth. Then, its angular momentum about the centre of its orbit, i.e., about the centre of the earth is given by $\vec{J} = \vec{R}_S \times M_S \vec{v}$. And its magnitude is

$$J = R_S M_S v \sin \theta, \quad \dots \text{I}$$

in a direction perpendicular to the plane of its motion.

Since the orbit is a circular one, $\theta = \pi/2$ and, therefore, $\sin \theta = 1$. So that,
angular momentum of the satellite about the centre of its orbit $= R_S M_S v \quad \dots \text{II}$

To obtain the value of v we note that as the satellite goes round the earth, the outward pull on it due to centrifugal force, $M_S v^2 / R_S$ is just balanced by the inward gravitational pull $M_E M_S G / R_S^2$ due to the earth (where M_E is the mass of the earth).

We, therefore, have $M_S v^2 / R_S = M_E M_S G / R_S^2$, whence, $v = (M_E G / R_S)^{1/2}$.

Substituting this value of v in expression II above, we have

$$\text{angular momentum of the satellite about the centre of its orbit, i.e.,} \quad J = (R_S M_S^2 M_E G)^{1/2} \quad \dots \text{III}$$

(b) The kinetic energy of the satellite is given by $T = \frac{1}{2} M_S v^2 = \frac{1}{2} M_S M_E G / R_S$

From relation III above, $M_E = J^2 / R_S M_S^2 G$. So that

$$T = \frac{1}{2} M_S \cdot \frac{J^2}{R_S M_S^2 G} \frac{G}{R_S} = \frac{J^2}{2 R_S^2 M_S}.$$

And, potential energy of the satellite is given by $U = -M_S M_E G / R_S$

Again, substituting the value of M_E , we have $U = -\frac{M_S G}{R_S} \left(\frac{J^2}{R_S M_S^2 G} \right)$

$$= -\frac{J^2}{R_S^2 M_S}.$$

$$\therefore \text{Total energy of the satellite, } E = T + U = \frac{J^2}{2R_S^2 M_S} - \frac{J^2}{R_S^2 M_S} = -\frac{J^2}{2R_S^2 M_S}.$$

Example 31. The moon revolves about the earth so we always see the same face of the moon. (a) How are spin and orbital parts of the angular momentum of the moon with respect to the earth related? (b) By how much would its spin angular momentum have to change if we were to be able to see all the moon's surface during the course of a month? (Mass of the moon, $M_m = 7.35 \times 10^{25}$ gm, radius of the moon, $R_m = 1.74 \times 10^8$ cm, radius of moon's orbit around the earth, $R_E = 3.84 \times 10^{10}$ cm and time of one revolution of the moon round the earth, $T = 2.36 \times 10^6$ sec.

Solution. (a) As we know (see next chapter), the angular momentum of a body rotating about an axis is given by $J = I\omega$, where I , is the *moment of inertia* and ω , the angular velocity of the body about that axis.

For a sphere $M.I.$ about its diameter $= (2/5)MR^2$. Therefore, *spin angular momentum of the moon, due to rotation about its own axis, i.e., about its own diameter, say,*

$$S = I\omega = (2/5) M_m R_m^2 \omega.$$

Now, the moon revolves around the earth so as to always present the same face towards it, showing clearly that its rate of rotation about its own axis is the same as its rate of revolution about the earth, so that $\omega = 2\pi/2.36 \times 10^6$.

$$\begin{aligned} \text{We, therefore, have} \quad S &= \frac{2}{5} M_m R_m^2 \times \frac{2\pi}{2.36 \times 10^6} \\ &= \frac{2}{5} \times 7.35 \times 10^{25} \times (1.74 \times 10^8)^2 \times \frac{2\pi}{2.36 \times 10^6} \\ &= 2.369 \times 10^{36} \text{ erg-sec.} \end{aligned}$$

And, *orbital angular momentum of the moon about the earth, say, $L = R_E M_m v$* , where v is the speed of the moon in its orbit around the earth.

$$\begin{aligned} \text{Since} \quad v &= R_E \omega, \text{ we have } L = R_E M_m \omega. \\ &= (3.84 \times 10^{10})^2 \times 7.35 \times 10^{25} \times 2\pi/2.36 \times 10^6 \\ &= 2.885 \times 10^{41} \text{ erg-sec.} \end{aligned}$$

$$\begin{aligned} \therefore \text{Ratio of the orbital to spin angular momentum of the moon, i.e., } L/S \\ &= 2.885 \times 10^{41} / 2.369 \times 10^{36} = 1.217 \times 10^5. \end{aligned}$$

(b) obviously, the moon must spin about its axis half as fast or half as slow as it actually does now, in order that both faces of it may be visible on the earth during the course of a month. *There should thus be an increase or a decrease in its spin angular momentum of half its present value.*

Example 32. (a) The maximum and minimum distances of a comet from the sun are 1.4×10^{12} m and 7×10^{10} m. If its velocity nearest to the sun is 6×10^4 m/sec, what is its velocity when farthest? Assume in both positions that the comet is moving in a circular orbit.

(b) A wheel is rotating with an angular speed of 500 rev/min on a shaft whose rotational inertia is negligible. A second identical wheel, initially at rest, is suddenly coupled to the same shaft. What is the angular speed of the resultant combination of the shaft and the two wheels?

Solution. (a) Let m be the mass of the comet, r_1 , r_2 , v_1 and v_2 , the radii of its orbits and its velocities when nearest to the sun and farthest from it respectively.

$$\text{The law of conservation of angular momentum demands that } \vec{m v_1 r_1} = \vec{m v_2 r_2}.$$

Since $\mathbf{r}_1$ and $\mathbf{r}_2$ are always perpendicular to the velocity vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively (the orbits being circular), we have $mv_1r_1 = mv_2r_2$.

Or, $v_2 = v_1r_1/r_2 = 6 \times 10^4 \times 7 \times 10^{10}/1.4 \times 10^{12} = 3 \times 10^3 = 3000 \text{ m/sec}$.

Thus, the speed of the comet when farthest from the sun = 3000 m/sec.

(b) If I be the $M.I.$ of the wheel and ω , its angular velocity about the shaft, we have angular momentum of the wheel about the shaft = $I\omega$.

When a second identical wheel is coupled with the first, the $M.I.$ of the combination of the two wheels about the shaft = $2I$.

If ω' be the angular velocity of the combination about the shaft, we have its angular momentum about the shaft = $2I\omega'$.

Since no external torque has been applied to the system, its angular momentum must remain conserved. And we, therefore, have

$$2I\omega' = I\omega, \text{ whence, } \omega' = \omega/2.$$

Since $\omega = 500 \text{ rev/min}$, we have $\omega' = 500/2 = 250 \text{ rev/min}$,

i.e., the angular speed of the resultant combination is 250 rev/min.

Example 33. A 500 gm mass is whirled round in a circle at the end of a string 40 cm. long, other end of which is held in the hand. If the mass makes 5 rev/sec, what is its angular momentum? If the number of revolutions is reduced to just one, after 20 seconds, calculate the mean value of the torque acting on the mass.

Solution. Here, angular momentum of the mass is given by $\vec{J} = \vec{r} \times m\vec{v}$, in a direction perpendicular to the plane of the circle.

$$\text{Or, } J = r \times mv \sin \theta$$

Since $\mathbf{r}$ is at any instant perpendicular to $\mathbf{v}$ (Fig. 6.27),

$$\theta = 90^\circ \text{ and } \therefore \sin \theta = 1.$$

we, therefore, have $J = rmv = rm(r\omega) = mr^2\omega$,

where ω is the angular velocity of the mass = $5 \times 2\pi/\text{sec}$. So that, substituting the values of m , r and ω , we have

$$\text{angular momentum, } J = 500 (40)^2 (5 \times 2\pi) = 8 \times 10^6 \pi = 2.5 \times 10^7 \text{ erg-sec.}$$

$$\text{Now, torque } \tau = dJ/dt = mr^2 d\omega/dt = 500 (40)^2 (2\pi/20) = 2.5 \times 10^5 \text{ dyne-cm.}$$

Example 34. A string of length l , tied to the top of a pole, carries a ball at its other end, as shown in Fig. 6.28. On giving the ball a single hard blow, it acquires an initial velocity v_0 in the horizontal plane and moves in a spiral of decreasing radius, with the string winding itself around the pole whose radius is negligible compared with the length of the string. (i) What is the instantaneous centre of revolution? (ii) Is the angular momentum conserved? (iii) Assuming the kinetic energy to be conserved, calculate the angular velocity of the ball after it has made five complete revolutions.

Solution. (i) The instantaneous centre of the ball is obviously the point of contact of the string with the circumference of the pole at the instant considered.

(ii) Let $\vec{r}$ be the position vector of the ball, measured by the axis of the pole passing through O to the centre B of

the ball (Fig. 6.29) and let $\vec{F}$ be the force applied to the ball, directed towards the point P on the circumference of the pole, which is the centre of revolution at

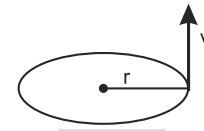


Fig. 6.27

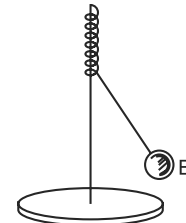


Fig. 6.28

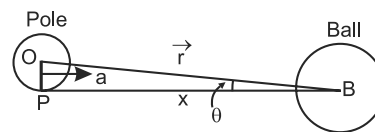


Fig. 6.29

the instant, and not towards O . So that $\vec{F}$ is not parallel to $\vec{r}$ and the torque on the ball is not, therefore, zero. It thus follows as a natural consequence that the angular momentum of the system is not conserved.

(iii) Since the *K.E.* of the system is assumed to be conserved, it is clear that the velocity of the ball remains constant at its initial value v_0 . The component of the velocity perpendicular to the radius vector OB is $v_0 \cos \theta = v_0 x/r$ and the angular velocity of the ball about the axis of the pole is thus $\omega = \frac{v_0 x/r}{r} = v_0 x/r^2$.

After 5 revolutions, the length of the string will decrease to $l - 5(2\pi a) = l - 10\pi a$, where a is the radius of the pole. So that, the length of the string will now be $(l - 10\pi a)$. And, since $r^2 = OP^2 + PB^2 = a^2 + x^2 = a^2 + (l - 10\pi a)^2$, the angular velocity will now be given by $\omega' = v_0 \frac{l - 10\pi a}{a^2 + (l - 10\pi a)^2}$. Or, Since $a \ll l$, we have $\omega' = \frac{v_0}{l - 10\pi a}$.

This will then be the angular velocity of the ball about the point on the circumference of the pole after five complete revolutions of the ball.

Example 35. A neutron of energy 1 MeV passes a proton at such a distance that the angular momentum of the neutron relative to the proton approximately equals 10^{-26} erg-sec. What is the distance of closest approach? (Neglect energy of interaction between the two particles).

Solution. As we know, the angular momentum of the neutron is given by $\vec{J} = \vec{r} \times m \vec{v}$, where m is the mass of the neutron, $\vec{v}$, its velocity and $\vec{r}$, its position vector relative to the proton.

If $\vec{r}$ be equal to the distance of closest approach, r_A , we have $\vec{J} = r_A \times m \vec{v}$.

And, therefore, $J = r_A m v \sin \theta$.

Since at the distance of closest approach, the velocity vector is perpendicular to the position vector, $\theta = 90^\circ$ and, therefore, $\sin \theta = 1$.

Hence, $J = r_A m v$. Or, $r_A = J/mv$.

Now, *K.E. of the neutron*, $T = \frac{1}{2} m v^2$, whence, $v^2 = 2T/m$ and $\therefore v = \sqrt{2T/m}$. Substituting this value of v in the expression for r_A above, we have

$$r_A = \frac{J}{m\sqrt{2T/m}} = \frac{J}{\sqrt{2Tm}}$$

Since $J = 10^{-26}$ erg-sec and $T = 1 \text{ MeV} = 1.6 \times 10^{-12} \times 10^6 = 1.6 \times 10^{-6} \text{ erg}$ and m , the mass of the neutron may be taken to be $1.67 \times 10^{-24} \text{ gm}$, we have

$$\text{distance of closest approach, } r_A = \frac{10^{-26}}{(2 \times 1.6 \times 10^{-6} \times 1.67 \times 10^{-24})^{1/2}} = 4.326 \times 10^{-12} \text{ cm}$$

Example 36. The distance of closest approach of a positive particle to a nucleus of atomic number 80 is $2 \times 10^{-11} \text{ cm}$. If the charge on the particle be equal to e and its initial kinetic energy equal to 2 MeV, obtain the value of the impact parameter for it.

Solution. We have relation $\frac{Ze^2}{r_A} = \frac{1}{2} m v_0^2 \left[1 - \left(\frac{b}{r_A} \right)^2 \right]$, [See § 6.17(ii)]

where r_A is the *distance of closest approach* of the particle, b , the *impact parameter* for it and $\frac{1}{2}mv_0^2$, its *initial K.E.*

Since *initial K.E.* of the particle is given to be 2 MeV , we have

$$\frac{1}{2}Mv_0^2 = 2 \times 1.6 \times 10^{-12} \times 10^{-6} = 3.2 \times 10^{-6} \text{ ergs.}$$

So that,
$$\frac{Ze^2}{r_A} = 3.2 \times 10^{-6} \left[1 - \left(\frac{b}{r_A} \right)^2 \right], \text{ whence, } 1 - \left(\frac{b}{r_A} \right)^2 = \frac{Ze^2}{3.2 \times 10^{-6} r_A}.$$

Or,
$$\left(\frac{b}{r_A} \right)^2 = 1 - \frac{Ze^2}{3.2 \times 10^{-6} \times r_A} = 1 - \frac{80(4.8 \times 10^{-10})^2}{3.2 \times 10^{-6} \times 2 \times 10^{-11}}$$

$$= 1 - 0.288 = 0.712$$

$\therefore b^2 = 0.712 \times (r_A)^2, \text{ Or, } b = r_A \sqrt{0.712} = 2 \times 10^{-11} \times 0.8437$

$$= 1.6874 \times 10^{-11} \approx 1.7 \times 10^{-11} \text{ cm.}$$

Thus, the *impact parameter for the particle* $= 1.7 \times 10^{-11} \text{ cm.}$

Example 37. Given that the orbital velocity of the sun about the centre of our galaxy is $3 \times 10^7 \text{ cm/sec}$ and its distance nearly equal to $3 \times 10^{22} \text{ cm}$ from the axis of the galaxy, estimate the mass of the galaxy ($G = 6.67 \times 10^{-8} \text{ C.G.S. units}$)

Solution. Obviously, the inward pull on the sun due to gravitational attraction between it and the galaxy just balances the outward pull on it due to centrifugal force. If, therefore, M_S and M be the masses of the sun and the galaxy respectively, we have

$$\frac{MM_S}{r^2} G = \frac{M_S v^2}{r}, \text{ whence, } M = \frac{v^2 r}{G}.$$

Substituting the values of v , r and G , therefore, we have

$$\text{mass of the galaxy, } M = \frac{(3 \times 10^7)^2 \times 3 \times 10^{22}}{6.67 \times 10^{-8}} = \frac{27 \times 10^{44}}{6.67} = 4.05 \times 10^{44} \text{ gm.}$$

EXERCISE

I—Conservation of Linear Momentum

- (a) What is linear momentum? State and explain the principle of conservation of linear momentum with examples. (Nagpur U. 2009, 2007)
(b) Derive the law of conservation of linear momentum from Newton's laws of motion.
- (a) State and explain principle of conservation of linear momentum. Show that in the absence of any external force acting on it, the linear momentum of a system of particles remains constant.
(b) A light and a heavy body have equal kinetic energies of translation. Which one has the larger momentum?
- Explain whether (a) it is possible for a body to have energy without having momentum and vice versa, (b) a sail boat can be propelled by air blown at the sails from a fan attached to the boat itself.
- Write a short note on the law of conservation of momentum and its importance in Physics. Does the law also hold good in nuclear and relativistic physics?
- (a) A 27 kg and a 40 kg boy are at rest on frictionless roller skates. The larger boy pushes the other so that the latter rolls away at a speed of 8 km/hr. What is the effect of this action on the larger boy himself? [Ans. He rolls away in the opposite direction with a speed of 5.35 km/hr]

- (b) A 200 kg bomb falling freely explodes and splits into two fragments, a larger one of 160 kg and a smaller one of 40 kg. The former moves away with a speed $\vec{v} = \hat{i} + 2\hat{j} + 3\hat{k}$. What is the speed of the latter? [Ans. $-4(\hat{i} + 2\hat{j} + 3\hat{k})$]
6. Show that when the vector sum of the external forces acting upon a system of particles equals zero, the total linear momentum of the system remains constant.
7. Show that the law of conservation of linear momentum of a system is a direct consequence of the translational invariance of the potential energy of the system.
8. A radioactive nucleus, initially at rest, decays by emitting an electron and a neutrino at right angles to one another. The momentum of the electron is 1.2×10^{-22} kg-m/sec and that of the neutrino is 6.4×10^{-23} kg-m/sec. (a) Find the direction and magnitude of the momentum of the recoiling nucleus. (b) The mass of the residual nucleus is 5.8×10^{-26} kg. What is the kinetic energy of recoil?

[Ans. (a) 1.36×10^{-22} kg-m/sec at $\angle 28^\circ 4'$ with the direction of the electron.

(b) 1.595×10^{-19} joule or 1 eV.]

[Hint. Assuming the direction of motion of the electron to be along the axis of x , momentum of the electron $= 1.2 \times 10^{-22} \hat{i}$ kg-m/sec and momentum of neutrino $= 6.4 \times 10^{-23} \hat{j} = 0.64 \times 10^{-22} \hat{j}$ kg-m/sec.]

And, momentum of the residual nucleus $= m\vec{v}$, where m is its mass and $\vec{v}$, its velocity. Since, initially, the undecayed nucleus was at rest, its momentum was zero. Hence, in accordance with the law of conservation of momentum,

$$1.2 \times 10^{-22} \hat{i} + 0.64 \times 10^{-22} \hat{j} + m\vec{v} = 0, \text{ whence, } m\vec{v} = -(1.2 \times 10^{-22} \hat{i} + 0.64 \times 10^{-22} \hat{j})$$

Hence, magnitude of momentum of recoiling nucleus, mv

$$= \sqrt{(1.2 \times 10^{-22})^2 + (0.64 \times 10^{-22})^2} = 1.36 \times 10^{-22} \text{ kg-m/sec.}$$

If its direction of motion be inclined at θ with the x -axis or direction of motion of the electron, we have $\tan \theta = 0.64/1.2 = 0.5333$, whence, $\theta = \tan^{-1}(0.5333) \approx 28^\circ 4'$.

$$\text{And, velocity of the recoiling nucleus} = \frac{\text{momentum (mv)}}{\text{mass (m)}} = \frac{1.36 \times 10^{-22}}{5.8 \times 10^{-26}} \text{ m/sec.}$$

$$\begin{aligned} \therefore K.E. \text{ of recoil} &= \frac{1}{2}mv^2 = \frac{1}{2} \times 5.8 \times 10^{-26} \times \left(\frac{1.36 \times 10^{-22}}{5.8 \times 10^{-26}} \right)^2 \\ &= 1.595 \times 10^{-19} \text{ joule} \\ &= \frac{1.595 \times 10^{-4}}{1.60 \times 10^{-19}} \approx 1 \text{ eV.} \end{aligned}$$

9. A railroad flat car of weight W can roll without friction along a straight horizontal track. Initially, a man of weight w is standing on the car which is moving to the right with speed v_0 . What is the change in velocity of the car if the man runs to the left so that his speed relative to the car is v just before he jumps off at the left end? [Ans. $wv/(W + w)$]
10. A bomb of mass 40.0 kg explodes in flight at the instant when its velocity is $\vec{v}_0 = 10\hat{i} + 5\hat{j}$ m/sec. It splits in two fragments of mass ratio 1 : 3 and the larger mass is observed to have velocity $\vec{v}_2 = 100\hat{i} + 50\hat{j} + 10\hat{k}$ m/sec just after explosion. Deduce (i) velocity $\vec{v}_1$ of the smallest fragment just after explosion, (ii) kinetic energy produced in the explosion.

[Ans. (i) $-(260\hat{i} + 130\hat{j} + 30\hat{k})$ m/sec. (ii) 6.135×10^2 joules.]

[Hint. If M be the mass of the unexploded bomb, its momentum before explosion $= Mv_0$. And if m_1 and m_2 be the masses of the two fragments after explosion and $\vec{v}_1$ and $\vec{v}_2$, their respective velocities, we have $M\vec{v}_0 = m_1\vec{v}_1 + m_2\vec{v}_2$.

$$\text{Or, } 40(10\hat{i} + 5\hat{j}) = 10\vec{v}_1 + 30(100\hat{i} + 50\hat{j} + 10\hat{k})$$

$$\text{whence, } \vec{v}_1 = -(260\hat{i} + 130\hat{j} + 30\hat{k}) \text{ m/sec.}$$

And $K.E.$ produced in the explosion $= \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 - \frac{1}{2}Mv_0^2 = \frac{1}{2} \times 10(260^2 + 130^2 + 30^2) + \frac{1}{2} \times 30(100^2 + 50^2 + 10^2) - \frac{1}{2} \times 40(10^2 + 5^2) = 6.135 \times 10^2 \text{ joules}$.

11. Three particles of masses 20 gm , 30 gm and 40 gm are initially moving along the positive direction of the three coordinate axes respectively with the same velocity of 20 cm/sec , when due to their mutual interaction, the first particle comes to rest, the second acquires a velocity $10\hat{j} + 20\hat{k}$. What is then the velocity of the third particle? [Ans. $10\hat{i} + 7.5\hat{j} + 5\hat{k}$.]

[Hint. Initial momentum of the system $= 20(20\hat{i}) + 30(20\hat{j}) + 40(20\hat{k})$ and final momentum of the system $= 20(0) + 30(10\hat{j} + 20\hat{k}) + 40\vec{v}$, where $\vec{v}$ is the velocity of the third particle. Equating the two, $\vec{v}$ can be evaluated.]

12. Two blocks A and B of mass m_A and m_B are connected together by means of a spring and are resting on a horizontal frictionless table. The blocks are then pulled apart so as to stretch the spring and then released. Show that the kinetic energies of the blocks are, at any instant, inversely proportional to their respective masses.
13. A projectile is fired from a gun at an angle of 45° with the horizontal and with a muzzle speed of 1500 ft/sec . At the highest point in its flight the projectile explodes into two fragments of equal mass. One fragment, whose initial speed is zero, falls vertically. How far from the gun does the other fragment land, assuming a level terrain? [Ans. $1.1 \times 10^5 \text{ ft}$.]
14. The wire cage of a bird is suspended from a spring balance. How does the reading on the balance differ when the bird flies about from that when it just sits quietly?
15. Define centre of mass. Show that when no external force acts on a body the acceleration of the centre of mass is zero and its velocity is a constant.

(G.N.D.U. 2002; Indore U.2001; Nagpur U.2007; D.A.U. Agra 2006)

16. What is centre of mass? Find the total linear momentum of a system of particles about the centre of mass and show that it is zero. (Nag. U. 2009, 2007; Gharwal U. 2000)
17. (a) What is reduced mass? Reduce two body problem to one body problem and obtain equation of motion for equivalent one body problem for two masses.

(H.P.U. 2001, 2000; Luck. U. 2001; P.U. 2000; Gharwal U. 2000; Purvanchal U. 2005; Nagpur U. 2007)

- (b) Discuss the motion of reduced mass under the influence of inverse square force. (H.P.U. 2000)
- (c) Distinguish between centre of mass and reduced mass. What is their importance in Physics?

II—Centre of mass

18. Show that the position of the centre of mass of a system of particles is quite independent of the frame of reference used.
19. Show that the centre of mass of two particles is on the line joining them at a point whose distance from each particle is inversely proportional to the mass of that particle.
20. (a) What is meant by a centre of mass frame of reference? What is its particular advantage?
- (b) Must there necessarily be any mass at the centre of mass in the case of a solid body and must it always lie within the body?
21. If only an external force can change the state of motion of the centre of mass of a body, how does it happen that the internal force of the brakes can bring a car to rest?
22. (a) Locate the centre of mass of three particles of 2 kg , 3 kg and 4 kg placed at the three corners of an equilateral triangle of 1 metre side. [Ans. $x = 0.56 \text{ m}$, $y = 0.385 \text{ m}$.]

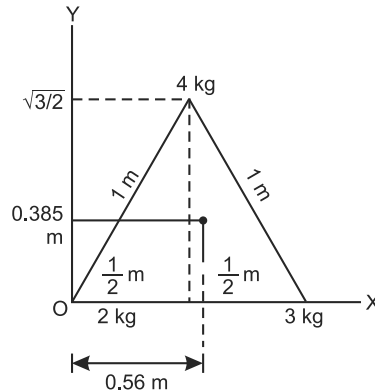


Fig. 6.30

[Hint. Imagine the triangle to lie in X - Y plane, with one side along the axis of x , as shown in Fig. 6.30. Then, we have

$$x = \frac{\sum m_k x_k}{\sum m_k} = \frac{2(0) + 3(1) + 4\left(\frac{1}{2}\right)}{2 + 3 + 4} = \frac{5}{9} = 0.56 \text{ m and}$$

$$y = \frac{\sum m_k y_k}{\sum m_k} = \frac{2(0) + 3(0) + 4(\sqrt{3}/2)}{2 + 3 + 4}$$

$$= \frac{2\sqrt{3}}{9} = 0.385 \text{ m.}]$$

(b) Where does the centre of mass of a triangular plate lie?

[Ans. At the point of intersection of the three medians.]

23. Find the centre of mass (i) a solid hemisphere (ii) a thin hemispherical shell, of radius r .

[Ans. (i) $\frac{3}{8}r$ above the centre of the plane face (ii) $\frac{1}{2}r$ above the mid-point of open face.]

24. The centre of mass of three particles of masses 4 gm, 5 gm and 6 gm respectively lies at the point (2, 2, 2). Where should a fourth particle of mass 12 gm be placed so that the centre of mass of the system of four particles lies at (0, 0, 0)?

[Ans. At the point (-2.5, -2.5, -2.5)]

25. In a given inertial frame, three particles of masses 10 gm, 30 gm and 20 gm are moving with velocities $6\hat{i}, 4\hat{j}$ and $9\hat{k}$ respectively. Calculate the velocity of the centre of mass of the system. What will be the velocities of the three particles in the centre-of-mass frame of reference whose coordinate axes are parallel to those of the inertial frame?

[Ans. $\vec{V} = \hat{i} + 2\hat{j} + 3\hat{k}$, $\vec{v}_1 = 5\hat{i} - 2\hat{j} - 3\hat{k}$, $\vec{v}_2 = -\hat{i} + 2\hat{j} - 3\hat{k}$, $\vec{v}_3 = -\hat{i} - 2\hat{j} + 6\hat{k}$.]

[Hint. $\vec{V} = \frac{\sum m_k \vec{V}_k}{\sum m_k} = \frac{10(6\hat{i}) + 30(4\hat{j}) + 20(9\hat{k})}{10 + 30 + 20} = \hat{i} + 2\hat{j} + 3\hat{k}$.

And, velocities in the centre-of-mass frame of reference, are

$$\vec{v}_1 = (6\hat{i}) - \vec{V}, \vec{v}_2 = (4\hat{j}) - \vec{V} \text{ and } \vec{v}_3 = (9\hat{k}) - \vec{V}.$$

26. The respective positions of three particles A , B and C , of masses 10 kg, 5 kg and 5 kg are given by $(4\hat{i} + 2\hat{j})$, $(-2\hat{i} + 2\hat{j})$ and $(2\hat{i} - 4\hat{j})$ metres. If forces $20\hat{j}$, $-10\hat{i}$ and $25\hat{i}$ newtons act on them respectively, calculate the acceleration of the centre of mass of the system. Illustrate your answer by means of a proper diagram.

[Ans. 1.25 m/sec^2 at an angle 53° with the axis of x .]

[Hint. Here, the x -component of the resultant force $\vec{F}$ acting on the centre of mass is $\vec{F}_x = 25\hat{i} - 10\hat{i} = 15\hat{i}$ and its y -component, $\vec{F}_y = 20\hat{j}$.]

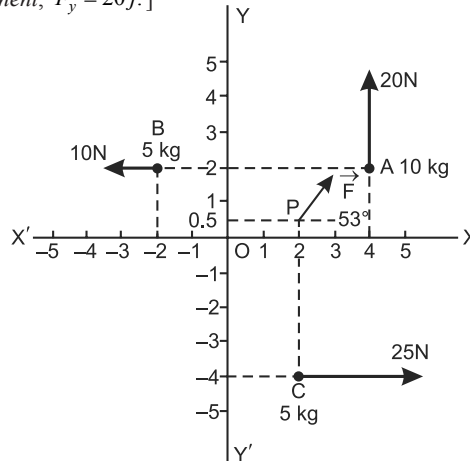


Fig. 6.31

$\therefore$ magnitude of the resultant force acting at the centre of mass $= \sqrt{15^2 + 20^2} = \sqrt{625} = 25N$. If θ be the angle that this force makes with the x -axis, we have $\theta = \tan^{-1}(20/15) = \tan^{-1}(4/3) = 53^\circ$.

Hence, acceleration of the centre of mass $= \frac{\text{force}}{\text{total mass}} = \frac{25}{10+5+5} = \frac{25}{20} = 1.25 \text{ m/sec}^2$ at an angle of 53° with the x -axis.

To illustrate this by means of a diagram, we must first indicate the positions of the particles A , B and C , as shown in Fig. 6.31. Then, the x and y coordinates of the centre of mass are obtained as usual, i.e.,

$$x = \frac{10(4) + 5(-2) + 5(2)}{10 + 5 + 5} = 2m \text{ and}$$

$$y = \frac{10(2) + 5(2) + 5(-4)}{10 + 5 + 5} = 0.5 \text{ m}$$

Or, the position of the centre of mass P is $(2i + 0.5j)$.

Thus, the $c.m.$, lies at the point P , as shown, and the force $F = 25N$ acts at this point P at an angle of 53° with the x -axis.

The acceleration of the centre of mass, $F/M = 25/20 = 1.25 \text{ m/sec}^2$ is thus naturally directed along F , i.e., at an angle of 53° with the x -axis.]

27. Calculate the changes in the values of energy and angular momentum when a two body system interacting through gravitational force is reduced to an equivalent one body problem.
28. (a) (i) what is the position vector of centre of mass?
(ii) What is the velocity and acceleration of centre of mass?
(b) Show that the linear momentum of the system of two particles is equal to the linear momentum of the centre of mass. (Gharwal U. 2000; P.U. 2004; Pbi. U. 2003)
29. A bomb weighing 50 kg explodes into three pieces in flight when its velocity is $20\hat{i} + 22\hat{j} + 10\hat{k} \text{ ms}^{-1}$. Two fragments of the bomb weighing 10 and 20 kg, are found to have velocities $100\hat{i} + 50\hat{j} + 20\hat{k}$ and $30\hat{i} + 20\hat{j} + 10\hat{k} \text{ ms}^{-1}$. Find the third piece.
[Hint. The mass of third piece = 20 kg. Let its velocity be $(x\hat{i} + y\hat{j} + z\hat{k}) \text{ ms}^{-1}$, then $50(20\hat{i} + 22\hat{j} + 10\hat{k}) = 10(100\hat{i} + 50\hat{j} + 20\hat{k}) + 20(30\hat{i} + 20\hat{j} + 10\hat{k}) + 20(x\hat{i} + y\hat{j} + z\hat{k})$] [Ans. $(-30\hat{i} + 50\hat{j} - 25\hat{k}) \text{ ms}^{-1}$]
30. Deduce the equation of motion of the centre of mass. (Nagpur U. 2007)

III—Collision

31. (i) What is collision? Explain elastic and inelastic collision.
(ii) Discuss the phenomenon of collision, in one dimension, between two particles, when collision is perfectly elastic. (Nagpur U. 2009, M.S.U. Tirunelveli 2007)
32. Prove that in centre of mass system, the magnitude of velocities of the particles remain unaltered in elastic collision. (Pbi. U. 2001, P.U. 2000)
33. Explain laboratory and centre of mass systems (or frames of reference). What is the advantage of studying a collision process in centre of mass system?
(G.N.D.U. 2004, 2001, 2000; Meerut U. 2003; P.U. 2004, 2001, 2000; Pbi. U. 2001, 2000; Guwahati U. 2000)
34. Prove that the kinetic energy of the two colliding particles in the centre of mass system are inversely proportional to their masses.
35. Show that in perfectly inelastic collision in Laboratory system there is always loss of kinetic energy. (G.N.D.U. 2004)
36. State the relation between velocities in centre of mass system and laboratory system. (Pbi. U. 2001)
37. A particle of mass m_1 moving with a velocity u collides head on with a particle of mass m_2 at rest such that after the collision they travel with velocity v_1 and v_2 respectively. If the collision is perfectly elastic one, show that $v_2 = \frac{2m_1u_1}{m_1 + m_2}$. (Agra U. 2007; G.N.D.U. 2003)

38. What precisely is meant by the term 'collision'? Show that in an elastic one-dimensional collision, the relative velocity with which the two particles approach each other before collision is equal to the relative velocity with which they recede from each other after collision. (Nagpur U. 2008)
39. Discuss the following special cases in an elastic collision:
 (i) when the colliding particles have the same mass, (ii) when one of the colliding particles is also initially at rest, (iii) when the particle at rest is much more massive than the other, (iv) when the particle at rest is much lighter than the other. (Nagpur U. 2007)
40. Show that in a head-on elastic collision between two particles, the transference of energy is the maximum when their mass ratio is unity.
41. A neutron of mass 1.67×10^{-27} kg and speed 10^5 m/sec collides with a stationary deuteron of mass 3.34×10^{-27} kg. The particles do not stick together and no kinetic energy is lost in collision. What is the subsequent speed of each particle. (Nagpur Uni. 2009, 2006)
42. Obtain an expression for the fractional decrease in kinetic energy of a neutron of mass m_1 when it makes a head-on collision with an atomic nucleus of mass m_2 initially at rest and calculate the percentage decrease when the nucleus concerned is the one of (i) lead, (ii) carbon, (iii) hydrogen.

$$[\text{Ans. } \frac{4m_1m_2}{(m_1+m_2)^2}, (i) 2\%, (ii) 28\%, (iii) 100\%.]$$

[Hint. If u_1 and v_1 be the velocities of the neutron before and after collision respectively, we have K.E.

of the neutron before collision, $T_1 = \frac{1}{2}m_1u_1^2$ and its K.E. after collision, $T_2 = \frac{1}{2}m_1v_1^2$.

$$\therefore \text{fractional loss of energy on collision} = \frac{T_1 - T_2}{T_1} = \frac{\frac{1}{2}m_1u_1^2 - \frac{1}{2}m_1v_1^2}{\frac{1}{2}m_1u_1^2} = \frac{u_1^2 - v_1^2}{u_1^2} = 1 - \frac{v_1^2}{u_1^2} \quad \dots I$$

But, as we have seen in § 6.10 (ii) (b) above, in a collision of this type,

$$v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u_1.$$

Substituting this value of v_1 in relation I above, we have

$$\frac{T_1 - T_2}{T_1} = 1 - \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2 \frac{u_1^2}{u_1^2} = 1 - \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2 = \frac{4m_1m_2}{(m_1 + m_2)^2}.$$

$\therefore$ (i) For lead, $m_2/m_1 = 206$ and $\therefore m_2 = 206 m_1$. So that,

$$\frac{T_1 - T_2}{T_1} = \frac{4 \times 206}{(1 + 206)^2} = 0.01912 \text{ or } 1.91\% \approx 2\%$$

(ii) For carbon, $m_2/m_1 = 12$, or, $m_2 = 12 m_1$, and

$$\therefore \frac{T_1 - T_2}{T_1} = \frac{4 \times 12}{(1 + 12)^2} = 0.2841 \text{ or } 28\%.$$

And (iii) for hydrogen, $m_2/m_1 = 1$ or, $m_2 = m_1$, and

$$\therefore \frac{T_1 - T_2}{T_1} = \frac{4 \times 1}{(1 + 1)^2} = 1 \text{ or } 100\%.]$$

43. Show that in a one-dimensional elastic collision the speed of the centre of mass of two particles, m_1 moving with initial speed u_1 , and m_2 moving with initial speed u_2 is given by $V = \left(\frac{m_1}{m_1 + m_2} \right) u_1 + \left(\frac{m_2}{m_1 + m_2} \right) u_2$ both before and after collision.

[Hint. Before collision, $(m_1 + m_2)V = m_1u_1 + m_2u_2$, whence, $V = \left(\frac{m_1}{m_1 + m_2} \right) u_1 + \left(\frac{m_2}{m_1 + m_2} \right) u_2$. After collision, using the values of v_1 and v_2 , as given under §6.10 and substituting in the relation $(m_1 + m_2)V = m_1v_1 + m_2v_2$ we obtain the same value of V as before collision.]

44. Two objects of masses $m_1 = 2$ gm and $m_2 = 5$ gm possess velocities $\vec{u}_1 = 10\hat{i}$ cm/sec and $\vec{u}_2 = 3\hat{i} + 5\hat{j}$ cm/sec just prior to a collision during which they become permanently attached to each other. Obtain the values of (i) the velocity of the centre of mass of the system, (ii) final momentum of the system in (a) the laboratory (b) the centre-of-mass reference frame, (iii) the fraction of the initial total kinetic energy associated with the motion after collision.

[Ans. (i) $5\hat{i} + \frac{25}{7}\hat{j}$ cm/sec, (ii) (a) $35\hat{i} + 25\hat{j}$ gm-cm/sec, (b) 0, (iii) 0.72.]

[Hint. (i) $(m_1 + m_2)V = m_1u_1 + m_2u_2$. (ii) (a) Since momentum is conserved in both elastic and inelastic collisions, the final momentum is equal to the initial momentum $m_1u_1 + m_2u_2$ or equal to $(m_1 + m_2)V$. (b) since in the centre-of-mass reference frame, the centre of mass remains at rest, $V = 0$ and hence the momentum of the system $= (m_1 + m_2)V = 0$. (iii) Initial K.E., $T_1 = \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2$ and final K.E., $T_2 = \frac{1}{2}(m_1 + m_2)v^2$, where v is the velocity of the combination after collision, equal to $(m_1u_1 + m_2u_2)/(m_1 + m_2)$, i.e., the same as the velocity of the centre of mass. The fraction of the total K.E. associated with the motion after collision is thus T_2/T_1 .]

45. The amount of energy transferred to a stationary body of mass m_2 by a collision with a moving body of mass m_1 can be increased by interposing another body of mass m_3 between them, so that m_1 strikes m_3 which, in turn, strikes m_2 . Show that the energy transferred to m_2 is a maximum when $m_3 = \sqrt{m_1m_2}$.
46. A bullet of mass 20 gm and moving with a speed of 20 m/sec hits the bob of a simple pendulum of mass 0.5 kg and lodges into it. If the length of the pendulum be 100 cm and $g = 980$ cm/sec², find the angular amplitude of the swing of the pendulum.

[Ans. $14^\circ 15'$.]

[Hint. See worked example 12. Angular amplitude $\theta = \sin \theta = x/l$, where l is the length of the pendulum.]

47. A bullet of mass 10 gm strikes a ballistic pendulum of mass 2.0 kg. The centre of masses of the pendulum rises a vertical distance 12 cm. Assuming the bullet remains embedded in the pendulum, calculate its initial speed.

[Ans. 310 m/sec.]

N.B. A ballistic pendulum is just a large wooden block suspended vertically by a pair of cords. The time of impact of the bullet with the pendulum is much smaller than the time-period of the pendulum, so that the cords remain vertical during the collision. No external force thus acts on the system (bullet + pendulum) and the horizontal component of the momentum remains conserved.

48. Calculate the percentage of the initial kinetic energy lost by a neutron colliding head-on with (a) a stationary deuterium nucleus, (b) a stationary carbon nucleus.
49. Show that the relative velocity between the particles after an elastic head-on collision is equal and opposite to the relative velocity before the collision.

[Ans. (a) 67% (b) 28%]

(Agra U. 2003)

50. A body of mass 100 gms moving with a velocity $(8\hat{i} - 6\hat{j} - 10\hat{k})$ cm s⁻¹ collides with a body of mass 200 gm moving with velocity $(-10\hat{i} + 6\hat{j} - 8\hat{k})$ cm s⁻¹ and attain velocities $(3\hat{i} - 4\hat{j} - 5\hat{k})$ and $(-4\hat{i} + 5\hat{j} - 6\hat{k})$ cm s⁻¹ respectively. Find kinetic energy before and after collision in the Lab system. Is the collision elastic?

[Hint. Energy before collision $= \frac{1}{2} \times 100(8\hat{i} - 6\hat{j} - 10\hat{k})^2 + \frac{1}{2} \times 100(-10\hat{i} + 6\hat{j} - 8\hat{k})^2 = 30,000$ erg

Energy after collision $= \frac{1}{2} \times 100(3\hat{i} - 4\hat{j} - 5\hat{k})^2 + \frac{1}{2} \times 100(-4\hat{i} + 5\hat{j} - 6\hat{k})^2 = 10,200$ erg

The collision is inelastic as the kinetic energy is not conserved.]

51. In a head-on collision, a particle with an initial speed V_0 strikes a stationary particle of the same mass. Find the velocities v_1 and v_2 of the two particles respectively after collisions if half of the original K.E. is lost.

(D.A.U. Agra 2006)

52. Two particles of masses 100 and 300 gm have at a given time positions $2\hat{i} + 5\hat{j} + 13\hat{k}$ and $-6\hat{i} + 4\hat{j} - 2\hat{k}$ cm respectively and velocities $10\hat{i} - 7\hat{j} - 3\hat{k}$ and $7\hat{i} - 9\hat{j} + 6\hat{k}$ cm/s respectively. Find (a) the instantaneous position and (b) velocity of centre of mass.

(Nagpur U. 2007)

[Ans. (a) $\frac{-16\hat{i} + 17\hat{j} + 7\hat{k}}{4}$ m and (b) $\frac{31\hat{i} - 34\hat{j} + 15\hat{k}}{4}$ m/s]

IV–Rocket

53. Explain the principle underlying a *rocket*. What is meant by the *thrust* of a rocket and on what factors does it depend?

Establish the relation for the velocity of a rocket both when its weight is ignored and when it is taken into account.

54. What is a two-stage rocket? What is the advantage of a two-stage rocket over a single stage rocket?
55. The final velocity of the final stage of a multistage rocket is much greater than the final velocity of a single stage rocket of the *same* total weight and fuel supply. Why is this so? Explain.
56. Show that the speed of a rocket is (i) equal to the exhaust speed, (ii) twice the exhaust speed, when the ratio M_0/M is very nearly (i) e (about 2.7), (ii) e^2 (about 7.4).
57. A rocket of mass 20 kg has 180 kg fuel. The exhaust velocity of the fuel is 1.6 km/sec. Calculate the minimum rate of consumption of fuel so that the rocket may rise from the ground. Also calculate the ultimate vertical speed gained by the rocket when the rate of consumption of fuel is (i) 2 kg/sec, (ii) 20 kg/sec. [Ans. 1.225 kg/sec (i) 2.8 km/sec, (ii) 3.6 km/sec.]

[Hint. See worked example 16.]

58. If an ion engine with an exhaust velocity of 108 m/sec could be constructed, what percentage of the initial mass must be fuel if the engine is used to power a rocket that can attain the escape velocity of 11.2 km/sec? Neglect gravity and air resistance. Ans. 0.011%.
59. An empty rocket weighs 5000 kg and contains 40,000 kg fuel. If exhaust velocity of the fuel is 2.0 km/sec, calculate the maximum velocity gained by the rocket. Deduce the formula used. Ans. 4.29 km/s

60. What is a rocket? Describe the principle of a rocket? Why multistage rocket is necessary? Establish the following relation for a rocket

$$V = V_0 + v \log_e \left(\frac{M_0}{M} \right)$$

Calculate the burnt out velocity when rocket starts from rest.

(Nagpur U. 2009, 2007, 2003; Cal. U. 2003; Meerut U. 2001, 2000; Agra U. 2007, 2003)

V–Angular momentum

61. Define angular momentum of a particle. Show that the time rate change of angular momentum of a particle is equal to the torque acting on it.

62. (a) Define angular momentum $\vec{J}$ and torque $\vec{\tau}$. What are their units? Show that torque is given by the time rate of change of angular momentum.

(Meerut U. 2002; M.D.U. 2000; Kerala U. 2001; Agra U. 2007, 2005; Nagpur U. 2009)

- (b) If no torque acts on a body will its angular velocity remain conserved? (Nagpur U. 2008)

63. (a) Establish the vector relation between the force and the torque.

- (b) Establish the relation between torque applied and angular acceleration.

64. (a) State and explain the law of conservation of angular momentum. Illustrate with examples. What are the consequences of the law of conservation of angular momentum?

(G.N.D.U. 2001; Luck. U. 2001; Meerut U. 2001; Guwahati U. 2000; Bang. U. 2000; Nagpur U. 2008)

- (b) Explain why an ice skater always utilises the principle of conservation of angular momentum.

(M.D.U. 2000)

65. Discuss the motion of the top. Calculate the precessional velocity of a top and also show that $\vec{\tau} = \vec{\omega}_p \times \vec{L}$ where $\vec{\tau}$ is torque, $\vec{\omega}_p$ is precessional velocity and $\vec{L}$ is angular momentum.

(Meerut 2005)

66. Show that the total angular momentum of a system of particles about a fixed or a reference point is given by the relation $\vec{J} = \vec{R} \times \vec{P} + \vec{J}_{c.m.}$, where $(\vec{R} \times \vec{P})$ is the angular momentum of the centre of mass

about that point and $\vec{J}_{c.m.}$, the angular momentum of the system about the centre of mass.

What are spin and orbital angular momenta?

67. (a) State the law of conservation of angular momentum. Give one example of its application.
 (b) Show that for a central force, the angular momentum is conserved. (Nagpur U. 2005)
68. (a) Define angular momentum $\vec{J}$ and torque $\vec{\tau}$. Using Newton's second law in an inertial frame of reference, show that $d\vec{J}/dt = \vec{\tau}$
 (b) Show that for a particle subjected to a central force, the angular momentum is a constant of motion (G.N.D.U. 2001)
69. (a) A man turns on a rotating table with an angular speed ω . He is holding two equal masses at arm's length. Without moving his arms, he drops the two masses. What change, if any, is there in his angular speed? Is the angular momentum conserved? Explain.
 (b) Show that the conservation of the angular momentum of a system is a consequence of the rotational invariance of its potential energy. (MDU 2003; Pbi U. 2001, P.U. 2000; HPU 2003)
70. Show that the conservation of angular momentum applied to planetary motion leads to the law of constant areal velocity.
71. Why is the velocity of a satellite the *maximum* when it is *closest* to the sun and the *minimum* when it is farthest from it in its orbit around it?
72. State the law of conservation of angular momentum. Show that the sum of all internal torques is zero. Derive an expression for the distance of closest approach of a proton projected into the Coulomb field of a heavy nucleus. [Delhi (Hons), 1971]
73. A proton of speed u travels towards an atomic nucleus such that if it is undeflected, the nearest approach would be b . Show that the actual nearest approach b' is given by $Ze^2/b' = \frac{1}{2}mv^2[1 - (b/b')^2]$, where Z is the atomic number of the nucleus and m is the mass of the proton.
74. (a) For a particle of mass $m = 10.0 \text{ gm}$, position $r = 10\mathbf{i} + 6\mathbf{j} \text{ cm}$ and velocity $v = 5\mathbf{i} \text{ cm/sec}$, calculate the angular momentum about the origin. [Agra (Supp), 1970]
 [Ans. -300 k.]
 (b) Show that the angular momentum of a particle about a fixed point is equal to the product of its mass and double the area $\left(\frac{1}{2}r^2\omega\right)$ described by the rotating line in unit time and has a direction perpendicular to this area.
75. What is the angular momentum of a neutron of kinetic energy 1 MeV about a nucleus if the impact parameter be 1 A.U. (mass of neutron may be taken to be $1.67 \times 10^{-24} \text{ gm}$; $1 \text{ A.U.} = 10^{-8} \text{ cm}$. [Ans. $2.31 \times 10^{-23} \text{ gm-cm}^2/\text{sec.}$]
76. Explain why due to atmospheric friction, the velocity of a satellite in its circular (or nearly circular) orbit around the earth increases. Does it not violate the law of conservation of energy? What happens to its angular momentum?

[Hint. Due to atmospheric friction, the satellite certainly loses some energy in the form of heat. Since, however, its *total energy* $E = \text{its kinetic energy } T + \text{its potential energy } U$ we have

$$E = T + U = \frac{1}{2}mv^2 - MmG/r,$$

where M is the mass of the earth, m , the mass of the satellite, r , the radius of its orbit and v , its velocity in the orbit.

Now, for a circular motion around the earth, $mv^2/r = MmG/r$, whence, $v^2 = MG/r$.

$$\therefore E = \frac{1}{2}m \frac{MG}{r} - \frac{MmG}{r} = -\frac{1}{2} \frac{MmG}{r}.$$

Thus, a reduction in the value of E (due to friction) means that it should become less positive or more negative, i.e., r , the radius of the orbit of the satellite should become smaller.

Since, as we know, $v \propto 1/\sqrt{r}$ [see worked example 33(a)], a reduction in the value of r , the radius of the orbit of the satellite, must result in an increase in the value of v , the velocity of the satellite in the orbit.

Apparently, this increase in the K.E. of the satellite may look like a violation of the law of conservation of energy but a little reflection will show that the P.E. of the satellite decreases to an even greater extent, so that *there is an over all reduction in the energy of the satellite* on account of its conversion into heat due to atmospheric friction just in accordance with the law of conservation of energy.

And, since angular momentum $\mathbf{J}$ is also proportional to $1/\sqrt{r}$ [see worked example 30(a)], *the angular momentum of the satellite decreases with decrease in the radius of its orbit* (due to frictional effects). It may be noted, however, that the atmosphere gains an equal amount of angular momentum, so that *the angular momentum of the system, as a whole, remains conserved.*

77. A flywheel of mass 100 kg and radius of gyration 0.5 m is rotating with a speed of 90 revolutions per minute. Calculate the torque required to bring it to rest in 4 minutes. [Ans. 0.98 Nm]

78. A massive ball comes moving and collides elastically with a comparatively very light stationary ball. Immediately after the collision, what is the approximate ratio of speeds of lighter and massive ball? (Luck. U. 2001).

[Hint. The velocity of heavy particle remains almost the same after collision and the light particle acquires nearly twice the velocity of heavy particle.

$$\frac{\text{Speed of lighter particle}}{\text{Speed of heavy (massive) particle}} = 2]$$

79. The earth is moving around the Sun under gravitational force and its orbit has semi-major axis of 1.495×10^8 km. When the earth passes closest to the Sun (*i.e.*, at its perihelion) its distance is 1.47×10^8 km and its orbital velocity is 0.303 km s^{-1} . Find earth's velocity at the aphelion and also its angular velocities at the two positions.

[Hint. Angular momentum $\vec{J} = \vec{r} \times m\vec{v}$ is conserved.]

$$[\text{Ans. } V_a = 293 \text{ km s}^{-1}; \omega_p = 0.206 \times 10^{-8} \text{ rad s}^{-1}; \omega_a = 0.193 \times 10^{-9} \text{ rad s}^{-1}]$$

80. A grind stone weighing 40 kg has a radius of 1.2 m. Starting from rest it acquires a speed of 150 r.p.m. in 12 sec. Calculate the torque acting on it.

[Hint.

$$I = \frac{1}{2}MR^2 = \frac{1}{2} \times 40 \times 1.2 \times 1.2 = 28.8 \text{ Kg m}^2$$

$$\alpha = \frac{2\pi \times 150}{60 \times 12} = \frac{5\pi}{12} \text{ rad s}^{-2} \quad \tau = I\alpha = \frac{28.8 \times 5\pi}{12} = 37.7 \text{ Nm}]$$

7

Chapter

SIMPLE HARMONIC MOTION

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INTRODUCTION

Every oscillatory motion is simple harmonic in character and continues due to the interaction of *inertia* and *elasticity*. Motion of a clock pendulum, balance wheel of a clock, mass attached to a suspended spring, vibrations of prongs of tuning fork, up and down motion of needle of sewing machine, a bar magnet suspended in uniform magnetic field are some of the familiar examples of simple harmonic motion. When an oscillator is displaced from its position of equilibrium by application of a force and thus doing work on it, a restoring force comes into play tending to bring it back to its equilibrium position. According to Hook's law, this restoring force is proportional to the displacement and depends upon the *elastic properties* or *elasticity* of the system. As soon as the restoring force tries to bring the system back to its equilibrium position, the property of *inertia* opposes this change in velocity. Further, when the system reaches the equilibrium position, it overshoots the mark and moves beyond the mean position again due to inertia of motion. The motion continues till the deforming force due to inertia develops, bring the system to rest. The restoring force again then sets the oscillator into motion back towards the equilibrium position. Throughout the process, the total energy remains conserved, only the conversion of P.E. and K.E. and vice-versa takes place within a limit set by the *potential well*.

7.1 PERIODIC AND HARMONIC MOTION

A motion which repeats itself over and over again after regularly recurring intervals of time, called as *time-period*, is referred to as a periodic motion.

If a particle, undergoing periodic motion, covers the same path *back and forth* about a mean position, it is said to be executing an *oscillatory* (or *vibratory*) motion or an *oscillation* (or a *vibration*). Such a motion is not only periodic but also bounded, *i.e.*, the displacement of the particle on either side of its mean position remains confined within a well defined limit. Such oscillations are termed as *simple harmonic*. It should be noted that all the periodic motions are not simple harmonic motions (S.H.M.). For example: the motion of earth around the sun and the motion of moon around the earth are periodic but not simple harmonic.

Further, since all of the trigonometrical ratios, the *sines* and the *cosines* alone are periodic as well as bounded, the displacement of a particle executing an oscillatory motion is usually expressed in terms of sines or cosines or a combination of both. Thus, coupled with the fact that this type of motion is, in general, associated with *musical instruments*, (particularly wire instruments whose complexity is solved by Fourier's series), is probably the reason why it is also spoken of as *harmonic motion*.

The term oscillatory motion is not restricted only to 'displacement' of a mechanical oscillator, but it may be any physical quantity. For example, in an electrical system an oscillatory vibration of charge, current or voltage may take place.

7.2 HARMONIC OSCILLATOR

We have seen already, under § 5.8*, how the position of minimum potential energy of a particle is its position of stable equilibrium and how its displacement remains confined to a little distance on either side of this equilibrium or mean position, within what is called a *potential well* and also, how a restoring force $F = -dU/dx$ acts upon it at every point of its displacement except at the mean position (where it is *zero*), tending to bring it back to its mean or equilibrium position. It has also been shown there how the energy function $U = F(x)$ may, in general, be expanded about the position

x_0 of stable equilibrium by means of Taylor's theorem, which then gives $U = U_0 + \frac{Cx^2}{2!} + \frac{C_1x^3}{3!} + \dots$

and, therefore, force acting on the particle is given by

$$F = -\frac{dU}{dx} = -Cx - \frac{C_1}{2}x^2 - \frac{C_2}{6}x^3 + \dots$$

So that, if the displacement be small and C_1, C_2 etc are negligible, we have $U = U_0 + \frac{1}{2}Cx^2$ and, therefore, force acting on the particle, $F = -Cx$, where C is a *positive constant*, called the force constant. This relation indicates that a restoring force acts on the particle, tending to bring it back to its mean or equilibrium position and that the potential energy curve is parabolic in form, as shown in Fig. 5.11.

Such an oscillating particle is called a **harmonic oscillator** and in case the limits (x_1 and x_2) are equally spaced about the equilibrium position, it is called a *simple harmonic oscillator* and its motion, a *simple harmonic motion*, (*S.H.M.*, for short).

Thus, in the case of a simple harmonic motion, the *P.E. curve* varies as the square of the displacement and *the force acting on the particle, and hence its acceleration, is proportional to displacement but is directed oppositely to it (towards the mean or equilibrium position), the maximum displacement (or amplitude) of the particle being the same on either side of the mean or equilibrium position.*

In case the displacement is not the same on either side of the equilibrium position, the motion is harmonic all right but not simple harmonic. And, if the values of C_1, C_2 etc. in the relation for U above, cannot be neglected, the P.E. curve would no longer be parabolic but will assume the form shown dotted in Fig. 5.11. Even so, it will be parabolic in form at the bottom of the curve, implying that whatever be the nature of the potential function $U = U(x)$, *small oscillations may always be regarded as simple harmonic.*

Common in nature, the *S.H.M.* is, in fact, the most fundamental type of periodic motion and *all other periodic motions (harmonic as well as non-harmonic)* can be obtained by a suitable combination of two or more simple harmonic motions. Let us, therefore, study this motion in some detail.

7.3 SIMPLE HARMONIC MOTION : DIFFERENTIAL EQUATION OF S.H.M. _____

From what we have just seen in § 7.2 above, we may give a clear-cut definition to S.H.M. as follows:

A particle may be said to execute a simple harmonic motion if its acceleration is proportional to its displacement from its equilibrium position, or any other fixed point in its path, and is always directed towards it.

Thus, if F be the force acting on the particle and x , its displacement from its mean or equilibrium position, we have $F = -Cx$.

Now, in accordance with Newton's second law of motion, $F = ma$. So that, substituting $-Cx$ for F and d^2x/dt^2 for acceleration a , we have

$$-Cx = m \frac{d^2x}{dt^2}. \quad \text{Or,} \quad \frac{d^2x}{dt^2} + \frac{C}{m}x = 0,$$

This equation is called the **differential equation of motion** of a simple harmonic oscillator or a simple harmonic motion, because by solving it we can find out how the displacement of the particle depends upon time and thus know the correct nature of the motion of the particle.

Solution of differential Equation

To solve the equation, we may put it in the form $d^2x/dt^2 = -(C/m)x$, the negative sign, as we know, indicating that the acceleration is directed oppositely to displacement x .

* Students are advised to go through this article again at this stage in order to understand what follows more thoroughly.

Or, putting $(C/m) = \omega^2$, where ω is the *angular velocity* of the particle, the equation takes the form

$$\frac{d^2x}{dt^2} = -\omega^2x = -\mu x, \quad \dots(i)$$

where μ is a constant, equal to ω^2 . Or, since $d^2x/dt^2 = -\mu$ if $x = 1$, we may define μ as the *acceleration per unit displacement of the particle*.

Multiplying both sides of the equation by $2 dx/dt$, we have

$$2 \frac{dx}{dt} \frac{d^2x}{dt^2} = -\omega^2 \cdot 2x \frac{dx}{dt},$$

integrating which with respect to t , we have

$$\left(\frac{dx}{dt}\right)^2 = -\omega^2x^2 + A, \quad \dots(ii)$$

where A is a constant of integration.

Since at the *maximum displacement* (or *amplitude*) a of the oscillator (or the oscillation), the velocity $dx/dt = 0$, we have

$$0 = \omega^2a^2 + A, \text{ whence, } A = -\omega^2a^2.$$

Substituting this value of A in relation (ii), therefore, we have

$$\left(\frac{dx}{dt}\right)^2 = -\omega^2x^2 + \omega^2a^2 = \omega^2(a^2 - x^2),$$

whence, the *velocity of the particle at an instant t* , is given by

$$dx/dt = \omega\sqrt{a^2 - x^2}. \quad \dots(iii)$$

Putting equation (iii) as $dx/\sqrt{a^2 - x^2} = \omega dt$ and integrating again with respect to t , we have

$$\sin^{-1} \frac{x}{a} = \omega t + \phi. \quad \text{Or, } x = a \sin(\omega t + \phi). \quad \dots(iv)$$

This gives the displacement of the particle at an instant t in terms of its amplitude (a) and its total phase ($\omega t + \phi$), made up of the *phase angle* ωt and what is called the **initial phase, phase constant** or the **epoch** ϕ of the particle, usually denoted by the letter e . This *initial phase* or *epoch* arises because of our starting to count time, not from the instant that the particle is in some standard position, like its mean position or one of its extreme positions, but from the instant when it is anywhere else in between.

Other Solution:

Thus, if we start counting time *when the particle is in its mean position, i.e.,* when $x = 0$ at $t = 0$, we have $\phi = 0$ and, therefore,

$$x = a \sin \omega t.$$

And, if we start counting time *when the particle is in one of its extreme positions, i.e.,* when $x = a$ at $t = 0$, we have $a = a \sin(0 + \phi) = a \sin \phi$, i.e., $\sin \phi = a/a = 1$ or $\phi = \pi/2$. So that, $x = a \sin(\omega t + \pi/2) = a \cos \omega t$.

If, on the other hand, we start counting time from an instant t' , *before the particle has passed through its mean position*, we have $x = 0$ when $t = t'$, so that, $0 = a \sin(\omega t' + \phi)$. Or, $\omega t' + \phi = 0$, whence, $\phi = -\omega t' = -e$, say. And, therefore, $x = a \sin(\omega t - e)$.

Similarly, if we start counting time from an instant t' , *after the particle has passed through its mean position*, we have $x = a \sin(\omega t + e)$.

In all these cases, however, the particle executes an oscillation or vibration such that its displacement varies cyclically with time and it is said to execute a simple harmonic motion.

It will be readily seen that if we put $\phi = \phi' + \pi/2$, we shall have

$$x = a \sin(\omega t + \phi' + \pi/2). \quad \text{Or, } x = a \cos(\omega t + \phi').$$

Thus, a simple harmonic motion may be expressed either in terms of a sine or a cosine function. Only, the initial phase or the phase constant will have different values in the two cases.

In fact, the general solution of equation I, viz., $d^2x/dt^2 = -\omega^2 x$, is of the form $x = a \sin \omega t + b \cos \omega t$, which is a combination of both the sine and the cosine terms.

Coming back to our relation $x = a \sin(\omega t + \phi)$, we find that if we increase the time t by $2\pi/\omega$, we have $x = a \sin \left[\omega \left(t + \frac{2\pi}{\omega} \right) + \phi \right]$.

Or, $x = a \sin(\omega t + 2\pi + \phi) = a \sin(\omega t + \phi)$, the same, as before, thus indicating that *the particle repeats its movements after every $2\pi/\omega$ sec*, or that, in other words, the time-period of the particle,

$$T = 2\pi/\omega = 2\pi\sqrt{1/\omega^2} = 2\pi\sqrt{1/\mu}$$

$$\text{Or, } T = 2\pi\sqrt{\frac{1}{\text{acceleration per unit displacement}}}.$$

$$\text{Or, } T = 2\pi\sqrt{\frac{\text{displacement}}{\text{acceleration}}}.$$

$$\text{Or, since } \omega^2 = C/m, \text{ we also have } T = 2\pi\sqrt{m/C}.$$

Since T is quite independent of both a and ϕ (or e), it is clear that the oscillations of the particle are *isochronous*, (i.e., take the same time irrespective of the values of a and e).

The number of oscillations (or vibrations) made by the particle per second is called its *frequency of oscillation* or, simply, its *frequency*, usually denoted by the letter n . Thus, *frequency is the reciprocal of the time-period, i.e.,*

$$n = 1/T = \omega/2\pi = \frac{1}{2\pi}\sqrt{C/m}, \quad \text{whence, } \omega = 2\pi n = 2\pi/T.$$

Since ω is the angle described by the particle per second, it is also referred to as the **angular frequency** of the particle.

All these results are equally true for *angular simple harmonic motion*, if we consider angular velocity, displacement, acceleration etc, instead of the linear ones.

Alternatively, we may obtain the same results, perhaps a trifle more neatly, by solving the differential equation of motion of the particle with the help of *complex numbers*. Thus, taking the solution of the equation to be $x = Ae^{ipt}$, where A and p are arbitrary constants and $i = \sqrt{-1}$, we have

$$dx/dt = ipAe^{ipt} \quad \text{and} \quad d^2x/dt^2 = -p^2Ae^{ipt}.$$

Substituting these values in the differential equation $d^2x/dt^2 = -\omega^2 x$, we have

$$-p^2Ae^{ipt} = -\omega^2Ae^{ipt}, \quad \text{whence, } p^2 = \omega^2 \text{ and, therefore, } p = \pm \omega.$$

There are thus *two* solutions to the equation, viz., $x = A_1e^{i\omega t}$ and $x = A_2e^{-i\omega t}$.

The *most general solution*, therefore, is $x = A_1e^{i\omega t} + A_2e^{-i\omega t}$, where the constants A_1 and A_2 can be determined from the initial conditions.

Now, as we know, $e^{i\theta} = (\cos \theta + i \sin \theta)$. We, therefore, have

$$\begin{aligned} x &= A_1(\cos \omega t + i \sin \omega t) + A_2(\cos \omega t - i \sin \omega t) \\ &= (A_1 + A_2) \cos \omega t + i(A_1 - A_2) \sin \omega t. \end{aligned}$$

Or, putting $(A_1 + A_2) = a \sin \phi$ and $i(A_1 - A_2) = a \cos \phi$, we have

$$x = a \sin \phi \cos \omega t + a \cos \phi \sin \omega t,$$

i.e., $x = a \sin(\omega t + \phi)$, the same relation as obtained above.

Here, obviously, $a = \sqrt{(a \sin \phi)^2 + (a \cos \phi)^2}$

$$= \sqrt{(A_1 + A_2)^2 + [i(A_1 - A_2)]^2} = \sqrt{(A_1 + A_2)^2 - (A_1 - A_2)^2}$$

and

$$\phi = \tan^{-1} \frac{A_1 + A_2}{i(A_1 - A_2)}.$$

7.4 PHASE RELATIONSHIP BETWEEN DISPLACEMENT, VELOCITY AND ACCELERATION OF S.H. OSCILLATOR

Displacement. The displacement of a simple harmonic oscillator at any instant of time t is given by $x = a \sin(\omega t + \phi)$ (i)

Velocity. The velocity is defined as the time rate of change of displacement.

$$\begin{aligned} \therefore \text{Velocity } v &= \frac{dx}{dt} = \dot{x} = a \omega \cos(\omega t + \phi) \\ &= a \omega \sin\left(\omega t + \phi + \frac{\pi}{2}\right) \end{aligned} \quad \dots (ii)$$

$$\text{As } \sin(\omega t + \phi) = \frac{x}{a}, \cos(\omega t + \phi) = \sqrt{1 - \frac{x^2}{a^2}} = \sqrt{\frac{a^2 - x^2}{a^2}}$$

$$\text{Or } v = \omega \sqrt{a^2 - x^2}$$

Maximum velocity. The velocity of the oscillator is maximum when $\sin\left(\omega t + \phi + \frac{\pi}{2}\right) = 1$.

Hence, $V_{\max} = a\omega$

The value of $v = V_{\max}$ when $x = 0$ i.e., the particle executing S.H.M. is in its mean position.

Comparing equations (i) and (ii), we find that the velocity of simple harmonic oscillator at any instant of time t leads the displacement by a phase difference $\frac{\pi}{2}$ radian (or 90°) i.e., the two are in quadrature. The velocity varies harmonically with the same frequency ω .

Acceleration. Acceleration is defined as the time rate of change of velocity. Now velocity

$$\begin{aligned} V &= a \omega \cos(\omega t + \phi) = a \omega \sin\left(\omega t + \phi + \frac{\pi}{2}\right) \\ \therefore \text{Acceleration} &= \frac{dv}{dt} = \frac{d^2x}{dt^2} = \ddot{x} = -a \omega^2 \sin(\omega t + \phi) \\ &= a \omega^2 \sin(\omega t + \phi + \pi) \end{aligned} \quad \dots (iii)$$

Maximum acceleration. The acceleration of the oscillator is maximum when $\sin(\omega t + \phi + \pi) = 1$

and is given by $\left(\frac{d^2x}{dt^2}\right)_{\max} = a\omega^2$.

Phase relationship between displacement, velocity and acceleration. The displacement of a particle executing S.H.M. is $x = a \sin \omega t$, the velocity $v = a \omega \cos(\omega t + \phi) = a \omega \sin\left(\omega t + \phi + \frac{\pi}{2}\right)$

and the acceleration $= a\omega^2 \sin(\omega t + \phi + \pi)$. From this we conclude that the acceleration of a simple harmonic oscillator leads the displacement by π radian (or 180°) in phase *i.e.*, the acceleration and displacement are *out of phase i.e., antiphase*.

Graphical representation. The displacement of a particle executing S.H.M. is

$$x = a \sin(\omega t + \phi)$$

$$\therefore \text{Velocity, } V = \frac{dx}{dt} = a\omega \cos(\omega t + \phi)$$

$$\text{and} \quad \text{acceleration} = \frac{d^2x}{dt^2} = -a\omega^2 \sin(\omega t + \phi)$$

Hence, for	$y = 0,$	$v = +a\omega,$	$a = 0$
	$y = +a,$	$v = 0,$	$a = -\omega^2 a$
	$y = -a,$	$v = 0,$	$a = +\omega^2 a$

The same is shown graphically in Fig. 7.1

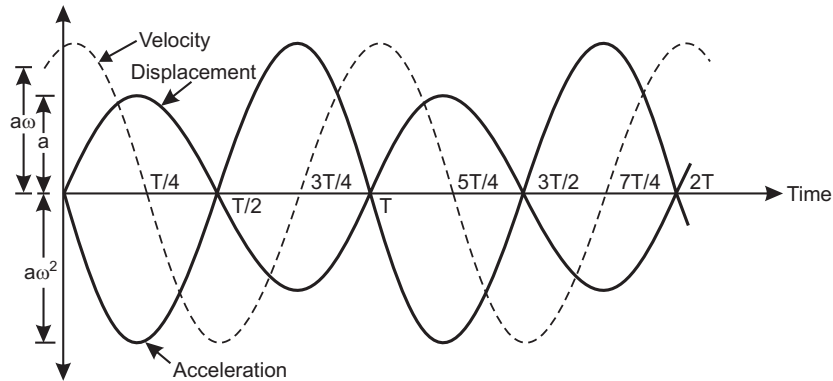


Fig. 7.1

7.5 ENERGY OF A HARMONIC OSCILLATOR

The acceleration of a harmonic oscillator, *i.e.*, of a particle executing a simple harmonic motion, is, as we know, directed towards its mean or equilibrium position, *i.e.*, opposite to the direction in which its displacement x increases. Work is, therefore, done during the displacement of the particle and it thus possesses *potential energy*. Also, the particle has velocity and, therefore, it possesses kinetic energy. Thus, a particle executing simple harmonic motion has, in general, both kinetic and potential energies. And if there be no dissipative forces (like friction etc.) at work, the sum total of the two remains constant, *i.e.*, the *mechanical energy of the oscillator (or the particle) remains conserved* though, as the displacement increases, the *P.E.* increases and the *K.E.* decreases. Thus, at its maximum displacement (a), when it is momentarily at rest, the whole of its energy is present in the potential form, the kinetic energy being *zero*, and at its mean position, where its velocity is the maximum, the whole of the energy is present in the kinetic form, its potential energy now being *zero*. In between these two extreme positions, the energy of the oscillator or the particle is partly kinetic and partly potential, the sum total of the two remaining constant throughout. This may be seen from the following:

Since the acceleration of the particle, $d^2x/dt^2 = -\omega^2 x$, the force F required to maintain the displacement x is $m\omega^2 x$, where m is the mass of the particle.

$$\therefore \text{work done for a small displacement } dx \text{ of the particle} \\ = Fdx = m\omega^2 x dx.$$

This work is obviously a measure of the *P.E.* of the particle at this displacement, so that *P.E. of the particle at displacement $dx = m\omega^2 x dx$.*

$\therefore$ work done for the whole displacement x of the particle or its *P.E. at displacement x* , say,

$$U = \int_0^x m\omega^2 x dx = \frac{1}{2} m\omega^2 x^2 = \frac{1}{2} m \left(\frac{C}{m} \right) x^2 = \frac{1}{2} Cx^2, \quad [\because \omega^2 = c/m]$$

i.e., *P.E. of the particle $\propto x^2$.*

The maximum value of the potential energy is thus at $x = a$ and is, obviously, $\frac{1}{2} m\omega^2 a^2 = \frac{1}{2} C a^2$.

Now, *velocity of the particle at displacement x* is $v = \frac{dx}{dt} = \frac{d}{dt} a \sin(\omega t + \phi) = \omega \sqrt{a^2 - x^2}$.
(see equation (iii) under § 7.3 above).

$$\begin{aligned} \therefore \text{K.E. of the particle at displacement } x &= \frac{1}{2} m\omega^2 (a^2 - x^2) \\ &= \frac{1}{2} C (a^2 - x^2) \end{aligned}$$

The maximum value of *K.E.* is obviously at $x = 0$ and is also equal to $\frac{1}{2} m\omega^2 a^2 = \frac{1}{2} C a^2$.

Hence, *total energy of the particle at displacement x* , i.e., $E = \text{K.E.} + \text{P.E.} = \frac{1}{2} m\omega^2 (a^2 - x^2) + \frac{1}{2} m\omega^2 x^2 = \frac{1}{2} m\omega^2 a^2 = \frac{1}{2} C a^2$.

As will be readily seen, *the total energy of the particle is quite independent of its displacement x* and thus remains the *same* throughout.

Since $\omega = 2\pi/T$, where T is the *time-period* of the particle, we may also express *total energy of the particle* as

$$E = \frac{1}{2} m (2\pi/T)^2 a^2 = 2\pi^2 m a^2 / T^2.$$

Or, since $1/T = n$, the *frequency* of the particle, we also have

$$E = 2\pi^2 n^2 m a^2.$$

Now, since in a conservative system, such as this, the sum total of the kinetic and potential energies of the system remains a constant, it follows that any one of them can increase only at the expense of the other and thus attains its maximum value when that of the other is the minimum or zero.

This means, in other words, that *the maximum value of any one of the two forms of energy is the same as the total energy of the system.*

Thus, *maximum value of K.E. = maximum value of P.E. = total energy $E = \frac{1}{2} m\omega^2 a^2 = \frac{1}{2} C a^2$.*

And, since during motion of the particle, the kinetic energy varies from 0 to the maximum, the graph showing the relation between *K.E.* and time (t) is as shown in full line in Fig. 7.2.

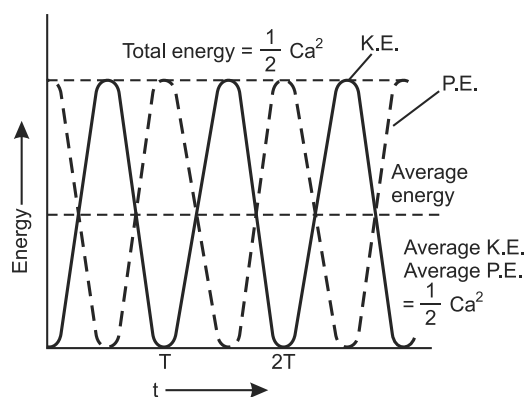


Fig. 7.2

Similarly, during motion of the particle, the *P.E.* of the particle varies between 0 and the same maximum $\frac{1}{2}m\omega^2 a^2$ or $\frac{1}{2}Ca^2$, but reciprocally with *K.E.*, i.e., when *K.E.* is the maximum, *P.E.* is zero and vice versa. Hence the graph between *P.E.* and *t*, shown 7.2, is the reciprocal of that for *K.E.*

The total energy is represented by the upper horizontal line parallel to the time axis and touching the two curves at points representing the maximum values of kinetic and potential energies respectively.

If, on the other hand, we plot the potential energy (*U*) of the particle against its displacement (*x*), the relation between which is given by $U = \frac{1}{2}Cx^2$, we obtain a *parabola* with its vertex at $x = 0$, as shown in full line in Fig. 7.3.

The maximum displacement of the particle on either side of its mean position is the *same*, $+a$, and $-a$ respectively. The value of the *P.E.* is, therefore, the maximum at these points and equal to the total energy of the particle. The upper horizontal line passing through these points of maximum potential energy and parallel to the displacement axis thus represents the total energy curve of the particle.

A graph between *K.E.* and displacement (*x*) of the particle will obviously be the reciprocal of the *P.E.* curve, as shown dotted in the figure, with the vertex of the parabola touching the upper horizontal line or the total energy curve at $x = 0$.

Since the total energy curve is a horizontal line in either case (Fig. 7.2 and 7.3), being parallel to the time-axis in the former and to the displacement axis in the latter case, it follows that *the total energy of the particle remains constant throughout and is independent of both time and displacement*, (being equal to the sum of the *K.E.* and the *P.E.* of the particle at any given instant).

It is also clear from Fig. 7.3 that at $x = a$ or $-a$, the total energy is wholly potential in form and at $x = 0$, it is entirely kinetic in form, being partly kinetic and partly potential at all other points.

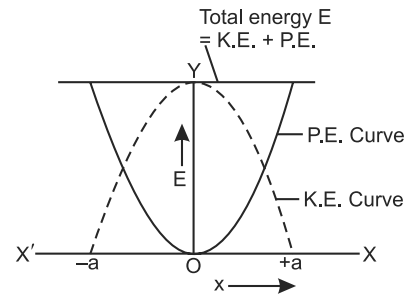


Fig. 7.3

7.6 AVERAGE VALUES OF KINETIC AND POTENTIAL ENERGIES OF A HARMONIC OSCILLATOR

It can be seen at a glance from Fig. 7.2 that the *average value of K.E. of the particle is equal to the average value of its potential energy* $= \frac{1}{4}m\omega^2 a^2 = \frac{1}{4}Ca^2$. The same result may, however, be obtained directly as follows:

We have *P.E. of the particle at displacement* $x = \frac{1}{2}m\omega^2 x^2$

$$= \frac{1}{2}m\omega^2 a^2 \sin^2(\omega t + \phi)$$

$\therefore$ *Average P.E. of the particle over a complete cycle or a whole time-period*

$$\begin{aligned} T &= \frac{1}{T} \int_0^T \frac{1}{2} m\omega^2 a^2 \sin^2(\omega t + \phi) dt. \\ &= \frac{m\omega^2 a^2}{4T} \int_0^T 2 \sin^2(\omega t + \phi) dt = \frac{m\omega^2 a^2}{4T} \int_0^T 1 - \cos 2(\omega t + \phi) dt. \end{aligned}$$

Since the average value of both a sine and a cosine function for a complete cycle or a whole time-period T is 0, we have *average P.E. of the particle* $= \frac{1}{4T} m\omega^2 a^2 [t]_0^T - 0 = \frac{1}{4T} m\omega^2 a^2 T$

$$= \frac{1}{4} m\omega^2 a^2 = \frac{1}{4} Ca^2.$$

$$\text{And, since K.E. of the particle at displacement } x = \frac{1}{2} m \left(\frac{dx}{dt} \right)^2 = \frac{1}{2} m \left[\frac{d}{dt} (a \sin(\omega t + \phi)) \right]^2$$

$$= \frac{1}{2} m\omega^2 a^2 \cos^2(\omega t + \phi),$$

we have *average K.E. of the particle over a complete cycle or a whole time-period*

$$T = \frac{1}{T} \int_0^T \frac{1}{2} m\omega^2 a^2 \cos^2(\omega t + \phi) dt$$

$$= \frac{m\omega^2 a^2}{4T} \int_0^T [1 + \cos 2((\omega t + \phi))] dt.$$

Again, the average value of a sine or a cosine function over a complete cycle or a whole time-period T being zero, we have *average K.E. of the particle* $= \frac{m\omega^2 a^2}{4T} [t]_0^T = \frac{m\omega^2 a^2}{4T} T.$

$$= \frac{1}{4} m\omega^2 a^2 = \frac{1}{4} Ca^2.$$

Thus, **Average P.E. of the particle = Average K.E. of the particle** $= \frac{1}{4} m\omega^2 a^2 = \frac{1}{4} Ca^2 = \text{Half the total energy.}$

7.7 SOME EXAMPLES OF S.H.M.

Let us now proceed to examine some important examples of simple harmonic motion.

7.7.1 The simple pendulum

A simple (or a mathematical) pendulum is just a heavy particle (ideally, a *point-mass*) suspended from one end of an *inextensible, weightless* string whose other end is fixed in a rigid support, —this point being referred to as the *point of suspension* of the pendulum.

Obviously, it is simply impossible to obtain such an idealised simple pendulum. In actual practice, therefore, we take a small and heavy spherical bob tied to a long and fine silk thread, the other end of which passes through a split cork securely clamped in a suitable stand, the length (l) of the pendulum being measured from the point of suspension to the centre of mass of the bob.

In Fig. 7.4, let S be the point of suspension of the pendulum and O , the mean or equilibrium position of the bob. On taking the bob a little to one side and then gently releasing it, the pendulum starts oscillating about its mean position, as indicated by the dotted lines. At any given instant, let the displacement of the pendulum from its mean position SO into the position SA be θ . Then, the weight mg of the bob, acting vertically downwards, exerts a torque or a moment $-mg l \sin\theta$ about the point of suspension, tending to bring it back to its mean position, the negative sign of the torque indicating that it is oppositely directed to the displacement (θ).

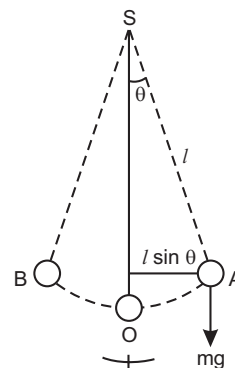


Fig 7.4

If $d^2\theta/dt^2$ be the acceleration of the bob, towards O , and I , its *M.I.* about the point of suspension (S), the moment of the force or the torque acting on the bob is also equal to $I.d^2\theta/dt^2$. We, therefore, have

$$I \frac{d^2\theta}{dt^2} = mg l \sin \theta.$$

Now, expanding $\sin \theta$ into a power series, in accordance with *Maclaurin's theorem*, we have $\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots$. If, therefore, θ be small, *i.e.* if the amplitude of oscillation be small, we may neglect all other terms except the first and take $\sin \theta = \theta$. So that,

$$I \frac{d^2\theta}{dt^2} = -mg l \theta, \text{ whence, } \frac{d^2\theta}{dt^2} = -\frac{mg l}{I} \theta$$

Or, since *M.I.* of the bob (or the point mass) about the point of suspension (S) is ml^2 , we have $\frac{d^2\theta}{dt^2} = \frac{mg l}{ml^2} \theta = \frac{g}{l} \theta = \mu \theta$

where $g/l = \mu$, the *acceleration per unit displacement*.

The acceleration of the bob is thus proportional to its angular displacement θ and is directed towards its mean position O . The pendulum thus executes a simple harmonic motion and its time-period is, therefore, given by

$$T = 2\pi \sqrt{\frac{1}{\mu}} = 2\pi \sqrt{\frac{1}{g/l}} = 2\pi \sqrt{\frac{l}{g}}$$

it being clearly understood that the amplitude of the pendulum is small.*

The displacement here being angular, instead of linear, it is obviously an example of an angular simple harmonic motion.

Drawbacks of a simple pendulum. Although one of the simplest methods for determining the value of g at a place a simple pendulum suffers from a number of drawbacks, the more important of which are the following:

(i) It is just an ideal conception, not realisable in actual practice, since it is impossible to have both a point-mass and a weightless string. So that, the string too has a moment of inertia about the axis of suspension.

(ii) The resistance and the buoyancy of the air appreciably affect the motion of the bob.

(iii) The expression for the time-period ($T = 2\pi\sqrt{l/g}$) is true only for oscillations of infinitely small amplitude.

(iv) The motion of the bob is not strictly linear. It has also rotatory motion about the axis of suspension.

(v) The bob also has a relative motion with respect to the string at the extremities of its amplitude on either side.

For all these reasons, a *compound pendulum* is preferred to a simple pendulum for the determination of the value of g .

7.7.2 The Compound Pendulum

Also called a *physical pendulum* or a *rigid pendulum*, a *compound pendulum* is just a rigid body, of whatever shape capable of oscillating about a horizontal axis passing through it.

* For correction for the finite amplitude of the pendulum, see §7.12

The point in which the vertical plane passing through the *c.g.* of the pendulum meets the axis of rotation is called *its point or centre of suspension* and the distance between the point of suspension and the *c.g.* of the pendulum measures the *length* of the pendulum.

Thus, Fig. 7.5 shows a vertical section of a rigid body or a compound pendulum, free to rotate about a horizontal axis passing through the point or centre of suspension *S*. In its normal position of rest, its *c.g.*, *G*, naturally lies vertically below *S*, the distance between *S* and *G* giving the length *l* of the pendulum.

Let the pendulum be given a small angular displacement θ into the dotted position shown, so that its *c.g.* takes up the new position *G'* where, of course, $SG' = l$. The weight of the pendulum, mg , acting vertically downwards at *G'* and its reaction at the point of suspension *S* constitute a couple (or a torque), tending to bring the pendulum back into its original position.

Obviously, *moment of this restoring couple* = $-mg l \sin \theta$, the negative sign indicating that the couple is oppositely directed to the displacement θ .

If *I* be the *moment of inertia* of the pendulum about the axis of suspension (through *S*) and $d^2\theta/dt^2$, its angular acceleration, the couple is also equal to $I \cdot d^2\theta/dt^2$. So that, we have

$$I \cdot d^2\theta/dt^2 = -mg l \sin \theta.$$

Again, $\sin \theta = \theta - \theta^3/3! + \theta^5/5! \dots$, so that, if θ be small, $\sin \theta \approx \theta$ and, therefore, $I \cdot d^2\theta/dt^2 = -mg l \theta$, whence, $d^2\theta/dt^2 = -(mg l/I)\theta = -\mu\theta$, whence $mg l/I = \mu$, the *acceleration per unit displacement*.

The pendulum thus executes a simple harmonic motion and its *time-period* is given by

$$T = 2\pi\sqrt{\frac{1}{\mu}} = 2\pi\sqrt{\frac{1}{mg l/I}} = 2\pi\sqrt{\frac{I}{mg l}}$$

Now, if I_0 be the moment of inertia of the pendulum about an axis through its *c.g.*, *G*, parallel to the axis through *S*, we have, from the theorem of parallel axes, $I = I_0 + ml^2$. And if *k* be the radius of gyration of the pendulum about this axis through *G*, we have $I_0 = mk^2$. So that, $I = mk^2 + ml^2 = m(k^2 + l^2)$.

$$\therefore T = 2\pi\sqrt{\frac{m(k^2 + l^2)}{mg l}} = 2\pi\sqrt{\frac{k^2 + l^2}{gl}} = 2\pi\sqrt{\frac{k^2/I + l}{g}}$$

Thus, *the time-period* of the pendulum is the same as that of a simple pendulum of length $L = (k^2/l + l)$ or $(k^2 + l^2)/l$* . This length *L* is, therefore, called the **length of an equivalent simple pendulum** or the **reduced length** of the compound pendulum. Since k^2 is always greater than zero, *the length of the equivalent simple pendulum (*L*) is always greater than *l*, the length of the compound pendulum*.

Centre of oscillation. A point *O* on the other side of the *c.g.* (*G*) of the pendulum in a line with *SG* and at a distance k^2/l from *G* is called the *centre of oscillation* of the pendulum (Fig. 7.6) and a horizontal axis passing through it, parallel to the axis of suspension (through *S*) is called the axis of oscillation of the pendulum.

Now, clearly, $GO = k^2/l$ and $SG = l$. So that, $SO = SG + GO = l + k^2/l = L$, the length of the equivalent simple pendulum, *i.e.*, *the distance between the centers*

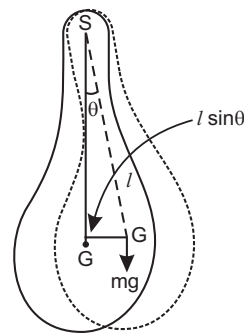


Fig. 7.5

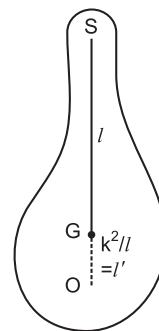


Fig. 7.6

* Here, the amplitude of the pendulum has been supposed to be infinitely small. Since it is almost always finite, a proper correction has to be applied, as explained under § 7.12.

of suspension and oscillation is equal to the length of the equivalent simple pendulum or the reduced length (L) of the pendulum and we, therefore, have

$$T = 2\pi\sqrt{L/g}$$

Interchangeability of centers of suspension and oscillation. If we put $k^2/l = l'$, we have $L = l + k^2/l = l + l'$ and therefore,

$$T = 2\pi\sqrt{(l + l')/g}$$

If now we invert the pendulum, so that it oscillates about the axis of oscillation through O , its time period, T' , say is given by

$$T' = 2\pi\sqrt{(k^2 + l'^2)/l'g}$$

Since $k^2/l = l'$, we have $k^2 = ll'$,

Substituting ll' for k^2 in the expression for T' , therefore, we have

$$T' = 2\pi\sqrt{(ll' + l'^2)/l'g} = 2\pi\sqrt{(l + l')/g} = T$$

i.e., the same as the time-period about the axis of suspension

Thus, the centers of suspension and oscillation are interchangeable or reciprocal to each other; i.e. the time-period of the pendulum is the same about either.

In fact, there are two other points on either side of G , about which the time-period of the pendulum is the same as about S and O . For, if with G as centre and radii equal to l and k^2/l respectively, we draw two circles, so as to cut SG produced in S and O' above, and at O and S' below G , as shown in Fig. 7.7 we have $SG = GS' = l$ and $GO = GO = k^2/l = l'$

$\therefore$

$$O'S' = GS' + GO = l + k^2/l = l + l' = SO.$$

Thus, there are four points in all, viz., S , O , S' and O' , collinear with the c.g. of the pendulum (G) about which its time-period is the same.

Maximum and minimum time-periods of a compound pendulum. For the time-period of a compound pendulum, we have the relation.

$$T = 2\pi\sqrt{(k^2 + l^2)/lg}$$

squaring which, we have

$$T^2 = \frac{4\pi^2(k^2 + l^2)}{lg} = \frac{4\pi^2}{g} \left(\frac{k^2 + l^2}{l} \right) = \frac{4\pi^2}{g} \left(\frac{k^2}{l} + l \right)$$

Differentiating with respect to l , we have

$$2T \frac{dT}{dl} = \frac{4\pi^2}{g} \left(-\frac{k^2}{l^2} + 1 \right)$$

a relation showing the variation of T with length (l) of the pendulum.

Clearly, T will be a maximum or a minimum when $dT/dl = 0$, i.e., when $l^2 = k^2$ or $l = \pm k$ or when $l = k$, because the negative value of k is simply meaningless.

Since d^2T/dl^2 comes out to be positive it is clear that T is a minimum when $l = k$, i.e., the time-period of compound pendulum is the minimum when its length is equal to its radius of gyration about the axis through its c.g. And the value of this minimum time-period will obviously be

$$T_{min} = 2\pi\sqrt{(k^2 + k^2)/kg} = 2\pi\sqrt{2k/g}$$

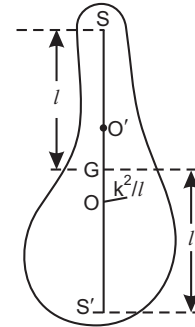


Fig. 7.7

On the other hand, we see from the expression for T above that if $l = 0$ or ∞ , $T = \infty$ or a maximum.⁷

Ignoring $l = \infty$ as absurd, we thus find that *the time-period of a compound pendulum is the maximum when its length is zero, i.e., when the axis of suspension passes through its c.g. or the c.g. itself is the point of suspension.*

This is obviously so because the pendulum is then in a state of neutral equilibrium, with no restoring action due to gravity on it.

Determination of the value of g . From the interchangeability of the points of suspension and oscillation it would appear that the easiest method of determining the value of g at a place would be to locate two points on either side of the c.g. of the pendulum about which the time-period of the pendulum is the same. These points would then correspond to the centres of suspension and oscillation of the pendulum and the distance between them would give L , the length of the equivalent simple pendulum. So that, if T be the time-period of the pendulum about either of these, we shall have $T = 2\pi\sqrt{L/g}$ and, therefore, $g = 4\pi^2 L/T^2$.

This is however, easier said than done, for it is extremely difficult if not impossible, to locate two such points in the pendulum. We, therefore, take recourse to one of the following two methods, using special forms of the pendulum.

(i) **By means of a bar pendulum.** A bar pendulum is the simplest form of a compound pendulum and consists of a uniform metal bar AB (7.8) having equally spaced holes drilled along its length on either side of its c.g., G , so that any one of them may be slipped on to a horizontal knife-edge K and the bar made to oscillate about it in the vertical plane.

First, the time-periods of different lengths of the pendulum are determined by slipping on to the knife-edge one hole after another from A to G and each length (i.e., the distance between the point of suspension and the c.g.) carefully noted.

A graph ABD is then plotted between distances of the holes from the c.g. (or the lengths of the pendulum) along the x -axis and the time-periods (T) along the y -axis, and the time periods (T) along the y -axis, as shown in Fig. 7.9.

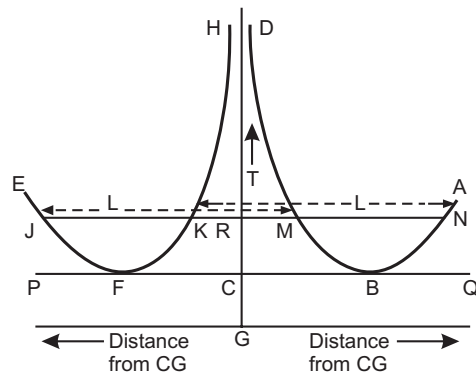


Fig. 7.9

Let a horizontal line JN be drawn, parallel to the x -axis so as to cut the two curves in points J , K , M and N . Then, clearly, the time period of the pendulum for lengths corresponding to all these points is the same. They thus correspond to the four points S , O' , O and S' , collinear with the c.g. G

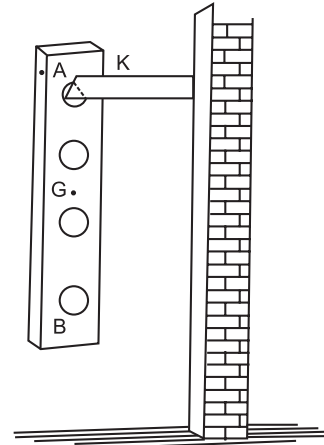


Fig. 7.8

The experiment is next repeated with the holes on the other side of the c.g. of the pendulum and a similar graph EFH drawn alongside the first on the same graph paper, using the same scale, or, this graph may be drawn symmetrically with the first without actually repeating the experiment with the holes on the other side of the c.g. This graph (EFH) will be a mirror image of the first (ABD), as is clear from the figure.

It will be seen at once that as the c.g. of the bar (G) is approached (i.e., as the centre of suspension comes closer to the c.g.), the time-period first decreases, acquires minimum value and then increases until it becomes infinite at the c.g. itself.

in Fig 7.7 above about which the period of the pendulum is the same. Clearly, therefore, $JM = KN = L$, the length of the equivalent simple pendulum, for JR here corresponds to l and RM , to k^2/l ; so that $JM = JR + RM = l + k^2/l = L$. And, similarly, $KN = KR + RN = k^2/l + l = L$.

Thus, knowing the value of T corresponding to these points (J , K , M and N) from the graph, we easily obtain the value of g at the place from the relation $T = 2\pi\sqrt{L/g}$, which gives $g = 4\pi^2 L/T^2$.

It will also be seen that if we draw a tangential line PQ , touching the two curves at B and F , then B and F represent the points where the centers of suspension and oscillation coincide with each other in the two cases respectively. The time-period of the pendulum corresponding to these points is therefore, the minimum. In other words, at these points $k^2/l = l$ or $k^2 = l^2$ and, therefore, $k = l$. So that, each of the distance (or lengths of the pendulum) CB and CF is equal to k , or $BF = 2k$, whence, the value of the radius of gyration (k) of the pendulum about the axis through its *c.g.* can also be easily obtained.

Now, instead of calculating the value of g as above, a better method, suggested by *Ferguson* in the year 1928 is to plot lT^2 along the axis of x and l^2 along the axis of y , which from the relation $l^2 + k^2 = (lT^2/4\pi^2)g$ must give a straight line graph, as shown in Fig. 7.10. The slope of the curve is $g/4\pi^2$, whence the value of g may be easily obtained.

Further, the intercept of the curve on the axis of y gives $-k^2$ and thus the values of both g and k can be obtained at once.

A drawback of the method is that it being well nigh (*i.e.* near) impossible to pin-point the position of the *c.g.* of the bar or the pendulum (as, infect, of any other body), the distances measured from it are not really accurate. Any error due to this is, however, eliminated automatically as the graph is smoothed out into the form of a straight line.

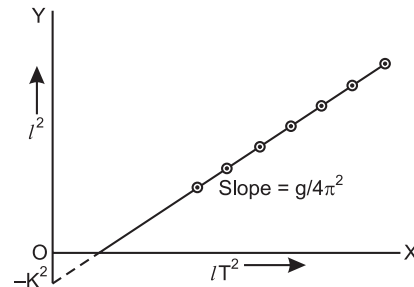


Fig. 7.10

(ii) **By means of a Kater's pendulum.** A Kater's pendulum is a rigid bar AB of brass or steel, fitted with adjustable and mutually facing knife-edges K_1 , and K_2 , near its two ends, as shown in Fig. 7.11, so that it may be made to oscillate about any one of them, as desired-hence the name *reversible pendulum* given to it.

Two cylindrical weights w and W , sliding along the bar, can be clamped in any desired position, on it such that the *c.g.* of the pendulum lies in between the two knife-edges and nearer to one of them than to the other. For this purpose, the heavier weight (W) is clamped near one end of the bar and the lighter weight (w) in between the knife-edges, its position being adjusted by means of a micrometer screw M .

In case the weights be so arranged that the time-period of the pendulum is *exactly the same* about either knife-edge, clearly, their positions corresponds to those of the centers of suspension and oscillation respectively and the distance between them directly gives the length L of the equivalent simple pendulum. We therefore have $T = 2\pi\sqrt{L/g}$ whence, $g = 4\pi^2 L/T^2$.

As mentioned earlier, however, it is extremely difficult and tiring to thus adjust the positions of the two weights and as *Bessel* pointed out, not really necessary. It is quite enough to adjust their positions such that the time-periods of the pendulum about the two knife edges are only very nearly equal. For, then, if l_1 and l_2 be the lengths of the pendulum (*i.e.* distances of knife-edges K_1 and K_2 respectively from the *c.g.*) and T_1 and T_2 (very nearly equal), their respective time periods, we have



Fig. 7.11

$$T_1 = 2\pi\sqrt{\frac{k^2 + l_1^2}{l_1 g}} \quad \text{and} \quad T_2 = 2\pi\sqrt{\frac{k^2 + l_2^2}{l_2 g}}$$

Squaring and rearranging, we have $T_1^2 l_1 g = 4\pi^2 (k^2 + l_1^2)$ and $T_2^2 l_2 g = 4\pi^2 (k^2 + l_2^2)$. Subtracting the second relation from the first we have

$$\begin{aligned} g(T_1^2 l_2 - T_2^2 l_1) &= 4\pi^2(k^2 + l_1^2) - 4\pi^2(k^2 + l_2^2) \\ &= 4\pi^2(l_1^2 + l_2^2) = 4\pi^2(l_1 + l_2)(l_1 - l_2). \end{aligned}$$

$$\begin{aligned} \text{Or,} \quad \frac{4\pi^2}{g}(l_1 + l_2) &= \frac{T_1^2 l_1 - T_2^2 l_2}{l_1 - l_2} = \frac{2(T_1^2 l_1 - T_2^2 l_2)}{2(l_1 - l_2)} \\ &= \frac{(T_1^2 + T_2^2)(l_1 - l_2) + (T_1^2 - T_2^2)(l_1 - l_2)}{2(l_1 - l_2)} \end{aligned}$$

$$\text{Whence,} \quad \frac{4\pi^2}{g}(l_1 + l_2) = \left(\frac{T_1^2 + T_2^2}{2} \right) + \left(\frac{T_1^2 - T_2^2}{2} \right) \left(\frac{l_1 + l_1}{l_1 - l_2} \right)$$

Here $(l_1 + l_2)$ is clearly the distance between the two-knife-edges and can, therefore, be accurately measured, $(l_1 - l_2)$ being the difference between the distances of the two knife-edges from the *c.g.* of the between the distance of the two knife-edges from the *c.g.* of the pendulum can not, of course, be measured accurately on account of the difficulty in locating the exact position of the *c.g.* However, since T_1 and T_2 are very nearly equal $(T_1^2 - T_2^2)$ is much too small and the second term on the right hand side, involving $(l_1 - l_2)$ becomes negligible compared with the first term. We, therefore, have

$$\frac{4\pi^2}{g}(l_1 + l_2) = \frac{T_1^2 + T_2^2}{2} \quad \dots(ii)$$

$$\text{Whence,} \quad g = \frac{8\pi^2(l_1 + l_2)}{T_1^2 + T_2^2}$$

The accurate value of g at the place can thus be obtained.

N.B. Comparing expression (ii) above with that for a simple pendulum, viz., $4\pi^2 g/l = T^2$, we find that if T be the time period for a length $(l_1 + l_2)$ of the pendulum, we have

$$T^2 = \left(\frac{T_1^2 + T_2^2}{2} \right) + \left(\frac{T_1^2 - T_2^2}{2} \right) \left(\frac{l_1 + l_1}{l_1 - l_2} \right)$$

Here T is called the **computed time** of the pendulum.

Superiority of a compound pendulum over a simple pendulum. The main points of the superiority of a compound pendulum over a simple pendulum are the following:

(i) Unlike the ideal simple pendulum, a compound pendulum is easily realisable in actual practice.

(ii) It oscillates as a whole and there is no lag like that between the bob and the string in the case of a simple pendulum.

(iii) The length to be measured is clearly defined (viz., the distance between the two knife-edges in a Kater's pendulum). In the case of a simple pendulum, the point of suspension and the *c.g.* of the bob, the distance between which gives the length to the pendulum, are both more or less indefinite points, so that the distance between them, i.e., l , cannot be measured accurately.

(vi) On account of its large mass, and hence a large moment of inertia, it continues to oscillate for a longer time, thus enabling the time for a large number of oscillations to be noted and its time-period calculated more accurately. The time-period must, however, be corrected in the manner explained under § 7.12.

7.7.3 Loaded Spring

We have seen earlier under § 5.9 how when a particle is displaced from its mean or equilibrium position, a *linear restoring force* acts upon it, tending to bring it back into its original position.

Thus, suppose we have a mass m attached to the free end of a massless flat spiral spring*, with its other end fixed to a rigid support, like a wall etc. (Fig. 7.12). If the mass be displaced through a distance x , as shown, a *linear restoring force* $F = -Cx$ at once starts acting on the spring, tending to bring it back into its original condition, where C is the *force constant* of the spring. The $-ve$ sign of the force simply indicates that it is directed oppositely to the displacement of the mass.

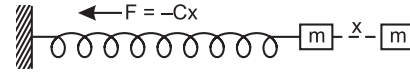


Fig. 7.12

Imagining the system to lie on a smooth horizontal surface, if the mass be released, it starts oscillating back and forth due to the spring getting alternately compressed and extended under the action of this force.

If d^2x/dt^2 be the acceleration set up in the spring, the force acting on the mass is also equal to $m.d^2x/dt^2$. We therefore, have

$$m \frac{d^2x}{dt^2} = -Cx \quad \text{or,} \quad \frac{d^2x}{dt^2} = -\frac{C}{m}x = -\mu x, \quad \dots(i)$$

Where $C/m = \mu$, the acceleration per unit displacement.

Thus, $d^2x/dt^2 \propto x$ and is directed oppositely to it.

The mass m thus executes a *simple harmonic motion* and its *time-period* is given by

$$T = 2\pi\sqrt{\frac{1}{\mu}} = 2\pi\sqrt{\frac{1}{C/m}} = 2\pi\sqrt{\frac{m}{C}} \quad \dots(ii)$$

The solution of equation of motion I above is also of the form $x = a \sin(\omega t + \phi)$, where $\sqrt{C/m} = \omega$.

Or, if as is usually the case, the time is counted from the moment that the mass just crosses its mean or equilibrium position towards the positive direction of x , i.e., if $x = 0$, at $t = 0$ we have $\phi = 0$ and

$$\therefore x = a \sin \omega t.$$

In actual practice, the spiral spring is arranged vertically with either its lower end fixed and the mass m placed on its upper end or with its upper end fixed and the mass suspended from its lower end (particularly if the wire of the spring be thin), as shown in figs. 7.13 (a) and (b) respectively.

Taking the more usual case (b), if the spring be extended through a distance l , say, due to the weight mg of the mass, as linear restoring force Cl at once comes into play in the opposite direction, so that the equilibrium position is attained when the two forces just balance each other, i.e., when $mg = Cl$. Whence, $C = mg/l$.

If the mass be now pulled down through a distance x from this equilibrium position, the linear restoring force $F = mg - C(l+x) = mg - (mg/l)(l+x) = mg - mg - (mg/l)x = -(mg/l)x = -Cx$, i.e., we have $m d^2x/dt^2 = -Cx$, as before, clearly showing that *there is no effect of gravity on the force constant C and hence on the period of oscillation of the mass*, given by $T = 2\pi\sqrt{m/C}$.

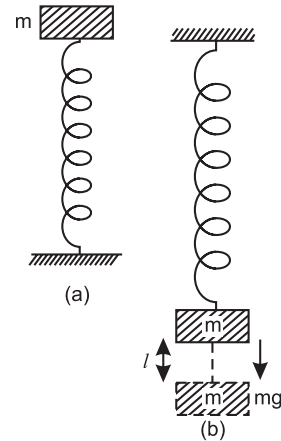


Fig. 7.13

* Spring is a uniform wire or ribbon which, in its normal unstrained condition, has the form of a helix such as the one obtained by closely and uniformly winding a wire around a cylinder of a radius much larger than that of the wire. If the plane of each coil of the spiral be/perpendicular to the axis of the cylinder, it is called a *flat spiral*.

7.7.4 Torsion Pendulum

A heavy body, like a cylinder or a cylinder or a disc, fastened at its mid-point to a fairly long and thin wire, suspended from a rigid support, constitutes a *torsional pendulum*, (Fig. 7.14). It is so called because, if the cylinder or the disc be turned in its own (*i.e.*, the horizontal) plane to twist the wire a little and then released, it executes torsional vibrations or oscillations about the wire as axis.

Thus, if the disc (or cylinder) be turned through an angle θ , say, the suspension wire too gets twisted through the same angle θ and this gives rise to a restoring torsional couple $C\theta$ in it, tending to bring it back into its original condition. Here, C is the *torsional couple per unit twist* of the wire, equal to $\pi n R^4 / 2L$, where R is the radius and L , the length of the wire and n , the modulus of rigidity of its material (see chapter on Elasticity).

If I be the *moment of inertia* of the disc (or cylinder) about the wire as axis, passing through its centre, and $d^2\theta/dt^2$, its angular acceleration, the couple acting on it is also equal to $I \cdot d^2\theta/dt^2$. We, therefore, have $I d^2\theta/dt^2 = -C\theta$, the $-ve$ sign indicating that the restoring couple or torque is oppositely directed to the angular displacement. So that,

$$d^2\theta/dt^2 = -(C/I)\theta = \mu\theta \quad \dots(i)$$

where $C/I = \mu$, the acceleration per unit angular displacement.

Or, $d^2\theta/dt^2 \propto \theta$ and is directed oppositely to it.

The disc (or the cylinder) thus executes an *angular simple harmonic motion* and its time-period is given by

$$T = 2\pi\sqrt{\frac{1}{\mu}} = 2\pi\sqrt{\frac{1}{C/I}} = 2\pi\sqrt{\frac{I}{C}} \quad \dots(ii)$$

It may be noted that no approximations whatever have been used in arriving at this relation for T , unlike in the case of a simple or a compound pendulum. *The time-period of a torsional pendulum, therefore, remain unaffected (i.e. the oscillations remain isochronous) even if the amplitude be large, provided, of course, the elastic limit of the suspension wire is not exceeded.*

Further, the solution of relation I above is also of the form

$$\theta = \theta_m(\omega t + \phi),$$

where θ_m is the *maximum angular displacement* or *amplitude* of the disc (or the cylinder), ω , its *angular velocity* equal to $\sqrt{C/I}$ and ϕ , the *initial phase* or *phase constant*.

It will be readily seen that the angular *S.H.M.* is analogous to linear *S.H.M.* Only, we take angular in place of linear displacement, moment of inertia instead of mass and torsional constant instead of force constant.

N.B. It need hardly be pointed out that a torsion pendulum entirely different from simple or a compound pendulum. For, here, the *c.g.* of the suspended body, Instead of moving in an arc, remains fixed in its position, the body merely rotating about axis through it and, unlike there, the time-period of the pendulum is quite independent of the value of g .

Inertia table. A simple application of the torsional pendulum is what is called the Inertia table which is just a device to determine the moments of inertia of bodies of both regular geometrical shapes and irregular shapes.

It is merely a large disc D of aluminum, about 15 cm in diameter, fitted with two small vertical pillars P, P at the extremity of one diameter and connected at the top by a cross bar as shown in Fig. 7.15.

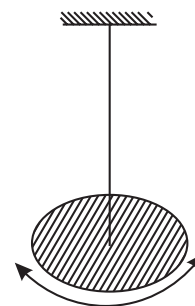


Fig. 7.14

The disc, along with the pillars and the cross bar, is suspended by means of a long and thin wire from the top of a large frame-work F , mounted on a heavy circular base, provided with levelling screws. A small mirror is fixed right at the mid-point of the cross bar enables the oscillations to be observed by a lamp and scale arrangement.

To ensure that the disc remains horizontal, a concentric circular groove is cut into it, such that three *balancing weights* can be made to slide into the desired positions inside it. Further, a number of concentric circles are usually drawn on the disc or the table to enable the body, whose moment of inertia is to be determined, to be so placed on it that its centre of mass lies on the suspension wire, which thus becomes the axis of rotation of the body through its centre of mass and perpendicular to its plane. The whole apparatus is enclosed in a glass envelope as a safeguard against disturbance due to air draughts.

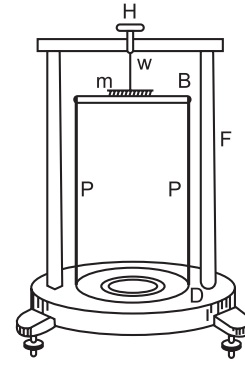


Fig. 7.15

Procedure. First, the inertia table is made perfectly horizontal by means of the levelling screws and the sliding balancing weights, as indicated. It is then set into torsional vibration all by itself alone and its time-period T_0 determined. If I_0 is the moment of inertia of the *Inertia table* about the suspension wire as axis, we have

$$T_0 = 2\pi\sqrt{I_0/C} \quad \dots(i)$$

where C is the torsional couple per unit twist of the wire.

The body whose moment of inertia I is to be determined is now placed centrally on the Inertia table in the manner explained (*i.e.*, with its centre of mass lying on the suspension wire), taking care to see that the table remains horizontal, so that its moment of inertia (I_0) about the suspension wire does not get altered. If T_1 be the time-period of the Inertia table, thus loaded, we have

$$T_1 = 2\pi\sqrt{(I_0 + I)/C} \quad \dots(ii)$$

Finally, the given body is replaced by another of a known moment of inertia I_1 (about the axis passing through its centre of mass and perpendicular to its plane) and, again taking care to keep the table horizontal (by changing the positions of the balancing weights, if necessary), the time-period of the loaded table determined. Let it be T_2 . Then, we have

$$T_2 = 2\pi\sqrt{(I_0 + I_1)/C} \quad \dots(iii)$$

Now, squaring and dividing relation (ii) by (i) we have

$$T_1^2/T_0^2 = (I_0 + I)/I_0. \text{ Or, } I/I_0 = (T_1^2 - T_0^2)/T_0^2. \quad \dots(iv)$$

Similarly, squaring and dividing relation (iii) by (i), we have

$$T_2^2/T_0^2 = (I_0 + I_1)/I_0. \text{ Or, } I_1/I_0 = (T_2^2 - T_0^2)/T_0^2. \quad \dots(v)$$

So that, dividing relation (iv) by (v), we have $I/I_1 = (T_1^2 - T_0^2)/(T_2^2 - T_0^2)$.

Or,

$$I = \left(\frac{T_1^2 - T_0^2}{T_2^2 - T_0^2} \right) I_1^*$$

whence, the moment of inertia (I) of the given body about the wire as axis can be easily obtained,

The amplitude (θ) here need not be small since the restoring couple, as pointed out above, is found to be proportional to θ even if θ be large. The assumption, however, that even with different loads suspended from the wire and with different longitudinal tensions in it, the value of C remains unaffected is not strictly true.

* Although we could the value of I from expression (i) and (ii), we adopt this procedure to eliminate I_0 , which may not always be accurately known.

An obvious disadvantage of the method is that it can be used to determine the moment of inertia of a body only about one particular axis, *viz.*, the axis perpendicular to its plane and passing through its centre of mass.

7.7.5 Helmholtz Resonator

A resonator is a device to analyse a complex note of sound *i.e.*, to find out what particular frequencies are present in the given note. For this purpose it is necessary that the resonator should exhibit a sharp resonance, *i.e.* it should resound with a note of *only one particular frequency*, namely, its own natural frequency.

Thus, resonator of different natural frequencies are used to detect the different frequencies comprising the given note.

Helmholtz showed that sharpness of resonance in what is called a *volume resonator* is ensured if the resonator is a large vessel, spherical or cylindrical, of glass or metal, containing air, with a narrow neck *N*, through which it communicates with the outside air and receives the complex note to be analysed, and a narrow aperture *O* at the opposite end, to be plugged into the ear, [Figs. 7.16 (a) and (b)]. Whereas the *spherical type of resonator* (a) can be used to detect *just one frequency equal to its own*, the *cylindrical type* (b), being made up of two parts, one sliding over the other, can have its frequency altered at will and can thus be used to detect the different frequencies of the given note.

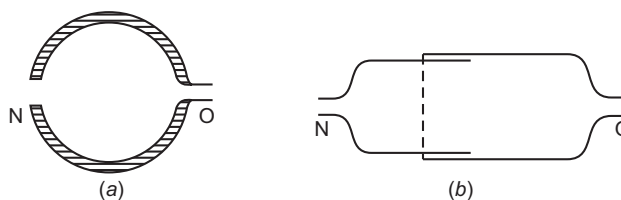


Fig. 7.16

The principle underlying the working of the resonator is that the *air in its neck serves as an air plug, as it were, and performs an oscillatory motion like a piston in a cylinder containing air, or a mass suspended from a spring.*

Thus, if l be the *length of the neck* and α , its *area of cross section*, the *mass of the air plug* or the *air piston in the neck* = $l\alpha\rho$, where ρ is the *density of the air* in the vessel.

If, therefore, this air plug be forced *inwards* through a small distance x , the decrease in the volume of air inside the vessel is, say, $-\delta V = -\alpha x$ (the $-ve$ sign indicating a decrease), resulting in a slight increase in pressure δp on the air inside the vessel. So that, if K be the *volume elasticity of the air* and V , the total volume of the vessel (or the initial volume of the air in it), we have

$$K = -\frac{\delta p}{\delta V/V} = -\delta p \frac{V}{\delta V}, \quad \text{whence, } \delta p = -K \frac{\delta V}{V}$$

$$\therefore \text{Force acting on the air plug outwards} = \delta p \alpha = -K \frac{\delta V}{V} \alpha = \frac{K \alpha^2 x}{V}$$

If d^2x/dt^2 be the acceleration of the air plug, the force acting on it is also = $(l\alpha\rho) d^2x/dt^2$. We, therefore, have

$$l\alpha\rho \frac{d^2x}{dt^2} = -\frac{K \alpha^2 x}{V}, \quad \text{whence, } \frac{d^2x}{dt^2} = -\frac{K \alpha^2 x}{V} \times \frac{1}{l\alpha\rho} = \frac{K \alpha}{V l \rho} x = -\mu x$$

where $K\alpha/Vl\rho = \mu$ = acceleration per unit displacement.

Thus, $d^2x/dt^2 \propto -x$ and is directed oppositely to it.

The *air plug* or the *air piston*, therefore, executes a *S.H.M.* and its *time-period* is given by

$$T = 2\pi\sqrt{\frac{1}{\mu}} = 2\pi\sqrt{\frac{1}{Ka/Vl\rho}} = 2\pi\sqrt{\frac{Vl\rho}{K\alpha}}$$

Now, the velocity of sound in air is given by the relation $v = \sqrt{K/\rho}$. So that, substituting v for $\sqrt{K/\rho}$ in the expression for T , we have

$$T = \frac{2\pi}{v}\sqrt{\frac{Vl}{\alpha}} \text{ and the frequency of oscillation } n = 1/T = \frac{v}{2\pi}\sqrt{\frac{\alpha}{Vl}}$$

Thus, the frequency of the resonator depends upon the total volume of the vessel and the length and area of cross section of its neck. By suitably adjusting these therefore, resonators of any desired frequencies may be constructed.

7.7.6 Inductance-Capacitance or L-C Circuit

Just as we have harmonic oscillators in mechanical systems performing *S.H.M.*, so also we come across harmonic oscillators in electrical systems where charge, current or voltage executes *S.H.M.* under suitable conditions. As an example we shall take up here an inductance-capacitance circuit, written for short as an *L-C* circuit.

Let a capacitor of capacitance C , an inductance coil of inductance L , (with negligible resistance) be connected up with a battery through a Morse key, as shown in Fig. 7.17.

On pressing the knob K of the Morse key to connect stud a , the capacitor gets directly connected to the battery and thus gets charged. On releasing the knob, it gets disconnected from the battery but acts connected to the inductance coil, as shown, through which therefore it discharges itself.

Now, as we know, the inductance in an electric circuit plays the same part as mass and moment of inertia do in translatory and rotatory motion respectively and thus *opposes both growth and decay of current in the circuit*. So that, as the capacitor discharges itself through the inductance coil and the current in the latter grows, the increasing magnetic flux due to it gives rise to an induced *emf* in the circuit opposing the growth of the current in it. Thus, if I be the instantaneous value of the current in the coil (or the circuit) at any given instant, the opposing *emf* set up across the coil is $-LdI/dt$, where dI/dt is the rate of change of current through the coil or the circuit.

And, if Q be the charge on the capacitor at the instant considered, the voltage across it tending to drive the current through the coil (or the circuit) is Q/C . Since the external *emf* is now zero (the battery being cut off), we have net *emf* in the circuit $= Q/C + L dI/dt = 0$... (i)

A current being just the *rate of flow of charge*, we have $I = dQ/dt$.

The above relation thus takes the form $Q/C + Ld^2Q/dt^2 = 0$.

$$\text{Or, } Q/LC + d^2Q/dt^2 = 0.$$

$$\text{Or, } \frac{d^2Q}{dt^2} = -\left(\frac{1}{LC}\right)Q = -\mu Q, \quad \dots (ii)$$

where $1/LC = \mu$, a constant for the circuit.

This is clearly an equation similar to the one for simple harmonic motion of a mechanical system, with x replaced by Q , mass (m) by inductance L and the force constant by reciprocal capacitance $1/C$.

Thus, the charge on the capacitor oscillates simple harmonically with time, i.e., the discharge of the capacitor is oscillatory in character, its time-period being

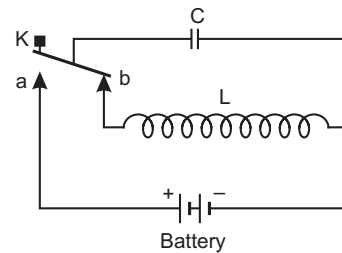


Fig. 7.17

$$T = 2\pi\sqrt{1/\mu} = 2\pi\sqrt{LC}$$

And, therefore, its frequency $n = 1/T = 1/2\pi\sqrt{LC}$

As will be readily seen, the solution of relation (ii) above is

$$Q = Q_0 \sin(\omega t + \phi), \quad \dots(iii)$$

where Q_0 is the *maximum value* or the *amplitude* of the charge, $\omega = 1/\sqrt{LC}$, the angular frequency (of the variation of charge) and ϕ , the *phase constant* which depends, as usual, on the initial conditions.

The charge in the circuit thus oscillates between $+Q_0$ and $-Q_0$, with a frequency $n = 1/2\pi\sqrt{LC}$.

Differentiating equation (iii) with respect to t , we have

$$\text{Instantaneous value of the current, } I = dQ/dt = Q_0\omega \cos(\omega t + \phi),$$

where the maximum value or the amplitude of the current $= Q_0\omega$, i.e., when $\cos(\omega t + \phi) = 1$.

Denoting this by I_0 , we have

$$I = I_0 \cos(\omega t + \phi)$$

Showing that the current in the circuit too is oscillatory in character and has same frequency as the charge, viz.,

$$n = 1/2\pi \cdot 1/2\pi\sqrt{LC}$$

7.7.7 Vibration of Magnet

Let NS be magnet of *pole-strength* m and *magnetic moment* M , suspended at an angle θ with the earth's field H , as shown in Fig. 7.18.

Clearly, forces mH and mH acting on the two poles of the magnet, being *equal, opposite and parallel*, constitute a *couple of moment* $mH \times ST = mHNS \sin \theta = MH \sin \theta$, tending to rotate the magnet back so as to lie parallel to the earth's field (H).

Since θ is small, we may take $\sin \theta = \theta$ and, therefore

couple acting on the magnet $= MH\theta$.

If I be the *moment of inertia* of the magnet about the suspension thread and $d^2\theta/dt^2$, its *angular acceleration*, the couple on it is also equal to $Id^2\theta/dt^2$.

So that, we have $Id^2\theta/dt^2 = MH\theta$, whence $d^2\theta/dt^2 = -(MH/I)\theta = -\mu\theta$, the $-ve$ sign indicating that the acceleration is directed oppositely to the angular displacement θ from its equilibrium position parallel to the field.

The angular acceleration of the magnet is thus proportional to its angular displacement from its mean of equilibrium position and is directed oppositely to displacement. It, therefore, executes a S.H.M. and its time period is given by

$$T = 2\pi\sqrt{\frac{1}{\mu}} = 2\pi\sqrt{\frac{1}{MH/I}}$$

$$T = 2\pi\sqrt{\frac{I}{MH}}$$

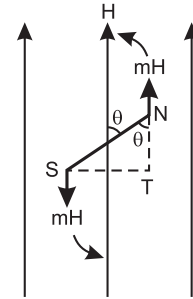


Fig. 7.18

7.7.8 Bifilar Oscillations

Bifilar Oscillations (with parallel threads): If a heavy and uniform bar or cylinder (or any rigid body), be suspended horizontally by means of two equal, vertical, flexible and inelastic thread, equidistant from its center of gravity, the arrangement constitutes bifilar suspension.

On being displaced a little in its own plane i.e., horizontal plane and then released, the bar or cylinder executes S.H.M. about the vertical axis through its centre of gravity.

Let AB [fig. 7.19(a)] represents the equilibrium position of a cylinder of mass m with its c.g. at O where its weight mg acts vertically downwards. Let the two suspension threads PA and QB be parallel to each other, and distance $2d$ apart and let the length of each thread be l .

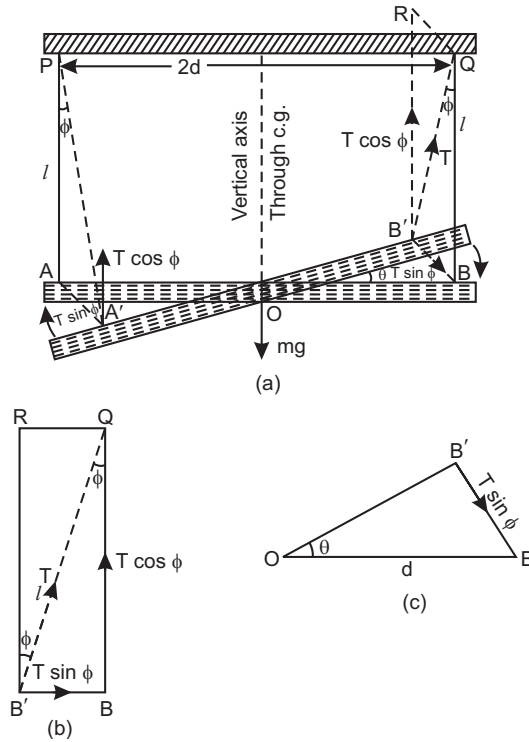


Fig. 7.19

Now, If the cylinder is displaced a little into the position $A'B'$ through a small angle ϕ , the suspension threads take up the positions PA' and QB' at an angle θ with their original positions, where ϕ is small.

Let T be the tension in each thread acting upwards along it. Then, resolving it into its two rectangular components, we have (i) component $T \cos \phi$, acting vertically upwards and (ii) component $T \sin \phi$, acting horizontally along $B'B$ or $A'A$.

Vertical components support the weight of the cylinder. Hence,

$$2 T \cos \phi = mg$$

$$T \cos \phi = mg/2$$

And since ϕ is small, $\cos \phi = 1$

So that
$$T = \frac{mg}{2}$$

The components $T \sin \phi$ acting at A' and B' being equal, opposite and parallel constitute a couple, tending to bring the cylinder back to its original position. Since $A'A$ and $B'B$ are practically at right angles to AB , we have

$$\begin{aligned} \text{Moment of the restoring couple} &= T \sin \phi \cdot 2d \\ &= T \phi \cdot 2d \end{aligned}$$

Now,
$$\phi = \frac{BB'}{OB} = \frac{BB'}{d}$$

$\therefore$
$$BB' = \theta d$$

And
$$= \frac{BB'}{QB'} = \frac{BB'}{l} = \frac{\theta d}{l}$$

$$\begin{aligned}
 \therefore \text{Moment of the restoring couple} &= T \cdot \frac{\theta d}{l} \cdot 2d \\
 &= \frac{mg}{2} \cdot \frac{\theta d}{l} \cdot 2d \\
 &= \frac{mgd^2}{2} \cdot \theta
 \end{aligned}$$

$$\text{But, the moment of the restoring couple} = I \cdot \frac{d^2\theta}{dt^2}$$

where I is M.I. of the cylinder about the vertical axis through its *c.g.* and $\frac{d^2\theta}{dt^2}$ is its angular acceleration

$$\begin{aligned}
 \therefore I \cdot \frac{d^2\theta}{dt^2} &= -\frac{mgd^2}{l} \cdot \theta \\
 \therefore \frac{d^2\theta}{dt^2} &= -\frac{mgd^2}{Il} \cdot \theta = -\mu\theta
 \end{aligned}$$

where $\frac{mgd^2}{I \cdot l} = \mu$, angular acceleration per unit angular displacement.

$$\therefore \frac{d^2\theta}{dt^2} \propto -\theta$$

Hence, the bifilar oscillations are S.H.M. and its time period is given by

$$T = 2\pi \sqrt{\frac{1}{\mu}} = 2\pi \sqrt{\frac{I \cdot l}{mgd^2}}$$

$$\therefore T = 2\pi \frac{1}{d} \cdot \sqrt{\frac{Il}{mg}}$$

Or, if we put $I = mk^2$, where k is radius of gyration of the cylinder about the vertical axis through O , we have

$$\begin{aligned}
 \therefore T &= 2\pi \frac{1}{d} \cdot \sqrt{\frac{mk^2 \cdot l}{mg}} = 2\pi \frac{1}{d} \cdot \sqrt{\frac{k^2 \cdot l}{mg}} \\
 T &= 2\pi \frac{k}{d} \cdot \sqrt{\frac{l}{g}}
 \end{aligned}$$

This equation gives the time period of the bifilar oscillator having parallel threads.

7.8 TWO-BODY HARMONIC OSCILLATOR

A system of two bodies connected by a spring so that both are free to oscillate simple harmonically along the length of the spring constitutes a *two-body harmonic oscillator* or a *coupled oscillator*.

Now, we have seen under §7.7.3 that a mass m , attached to the free end of a spring whose other end is fixed to a rigid support like a wall, constitutes a harmonic oscillator. This too is in fact a two-body oscillator, only one of the bodies, *viz.* the wall, is rigidly connected to the earth and has thus effectively an infinite mass and is there, immovable. Since the end of the spring connected to the wall does not move, the change in the length of the spring is given by the displacement of mass m itself. In short, the extension of the spring is determined by the motion of mass m alone, the infinite mass at its other end remaining fixed in its position.

In general, however, we seldom come across such infinite and immovable masses and have, therefore, to consider the motion of both the masses connected to the two ends of the spring. Among example of such two-body oscillators may be cited the diatomic molecules like those of hydrogen, carbon monoxide, HCl etc which can oscillate along their respective axes of symmetry. Although, of course, the two atoms in their molecules are coupled by electromagnetic forces, we may, for all practical purposes, imagine them to be connected together by means of tiny, massless springs.

Let us, however, consider the general case of a two-body oscillator consisting of two masses m_1 and m_2 (Fig. 7.20), connected by a horizontal massless spring of force constant C , so as to be free to oscillate along length of the spring on a frictionless horizontal surface.

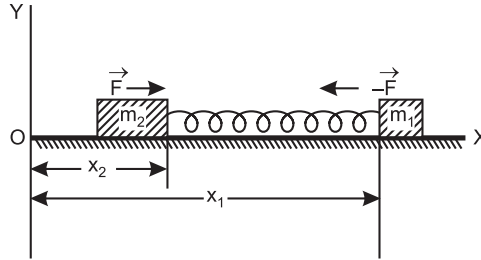


Fig. 7.20

Let the normal length of the spring be l and let, at any given instant, the coordinates of the two ends of the spring be x_1 and x_2 , as shown. Then, clearly,

$$\text{Extension of the spring, } x = (x_1 - x_2) - l,$$

where x is positive if the spring is stretched, zero, if the spring has its *normal length* and *negative* if it be *compressed*. Here, we assume it to be positive. The forces (F) exerted by the spring on the two masses are obviously equal in magnitude but opposite in sign, as indicate in the figure, the magnitude of each being Cx .

If, therefore, d^2x_1/dt^2 be the acceleration of mass m_1 and d^2x_2/dt^2 , that of mass m_2 , we have

$$m_1 d^2x_1/dt^2 = -Cx \dots (i) \quad \text{and} \quad m_2 d^2x_2/dt^2 = Cx. \dots (ii)$$

Multiplying relation (i) by m_2 and relation (ii) by m_1 and subtracting the latter from the former, we have

$$m_1 m_2 \frac{d^2x_1}{dt^2} - m_1 m_2 \frac{d^2x_2}{dt^2} = -m_2 Cx - m_1 Cx$$

$$\text{Or,} \quad m_1 m_2 \frac{d^2}{dt^2}(x_1 - x_2) = -Cx(m_1 + m_2)$$

$$\text{Or,} \quad \left(\frac{m_1 m_2}{m_1 + m_2} \right) \frac{d^2}{dt^2}(x_1 - x_2) = -Cx$$

Now, l being a constant, $d^2(x_1 - x_2)/dt^2 = d^2x/dt^2$. So that,

$$\left(\frac{m_1 m_2}{m_1 + m_2} \right) \frac{d^2x}{dt^2} = -Cx$$

Or, putting $m_1 m_2 / (m_1 + m_2) = \mu$, called the *reduced mass* of the system (being smaller than either of the two masses m_1 and m_2), we have

$$\frac{dx^2}{dt^2} = -\frac{C}{\mu}x$$

This is identical in form with the equation of motion obtained for a singly-body oscillator under §7.7.3, with the difference that μ here is the *reduced mass of the system instead of mass m* of the

single-body there and x here is the *relative displacement of the two masses from their equilibrium positions instead of the displacement of mass m alone from its equilibrium position there.*

The system thus oscillates simple harmonically with a time-period

$$T = 2\pi\sqrt{\frac{1}{C/\mu}} = 2\pi\sqrt{\frac{\mu}{C}} \quad \text{and} \quad \text{frequency, } n = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{C}{\mu}} \quad \dots(iv)$$

This means that *the two-body system oscillates along the axis of the spring with the same time-period and frequency as a one-body oscillator of mass μ and force constant C .*

Each of the two masses oscillates relatively to the other as though the latter were fixed and its own mass were reduced to μ but, since their directions of motions are opposite to each other (both moving inwards during compression and outwards during extension of the spring), they differ in phase by π .

Also, since no *external* force acts on the system, its centre of mass remains stationary and the amplitudes of masses m_1 and m_2 , therefore, are respectively $m_2/(m_1 + m_2)$ and $m_1/(m_1 + m_2)$ times the extension x of the spring.

Further, as in the case of a single-body oscillator, so also here, the *P.E. of the system* is given by $U = \frac{1}{2}Cx^2$ and hence the *P.E. curve* here too is *parabolic* in form. Here, x , however, depends upon

the relative positions of the two masses and is equal to $(r - r_0)$ where r_0 is the distance between the two masses when they are at rest and r , the distance between them when the spring has been extended by a length x . So that,

$$U = \frac{1}{2}C(r - r_0)^2$$

Obviously, therefore, the *curve between U and r (i.e. the *P.E. curve*) is parabolic in form as in form as in the case of a single-body system. *The potential energy here, however, is a characteristic of the system as a whole and not of the individual masses constituting it.**

7.9 OSCILLATION OF A DIATOMIC MOLECULE

As mentioned earlier, under §7.8 above, a diatomic molecule behaves as a two-body oscillator, with its two atoms connected by a small, massless spring, as it were.

The potential energy U of the system or the molecule changes with distance r between the atoms or rather between their nuclei. Thus, putting $r - r_0 = x$, where r_0 is the distance between the atoms, when they are in their normal positions, or stationary, (and hence a constant), we have

force acting on the atoms, i.e., $F = dU/dr = -dU/dx$,

whence

$$dU = -Fdx.$$

$\therefore$

$$U = \int Fdx = \int Cxdx = \frac{1}{2}Cx^2 + A$$

where A is a constant of integration.

Now, assuming the *P.E.* to be *zero* at infinite separating of the atoms, it will be negative for ordinary finite distance (as measured on the atomic scale) and will have its *minimum value*, $-U_0$, say, at $x = 0$, i.e., when $r = r_0$ or the atoms are in their normal, stationary state.

We, therefore, have $-U_0 = 0 + A$, Or, $A = -U_0$.

Substituting the value of A in the expression for U above, we have

$$U = -U_0 + \frac{1}{2}Cx^2 = -U_0 + \frac{1}{2}C(r - r_0)^2$$

indicating that $U - r$ or the potential energy curve for the diatomic molecule should also be parabolic in form like that of a two-body or a one-body harmonic oscillator.

The actual *P.E.* curve of a diatomic molecule, however, comes out to be of the form shown in Fig. 7.21, which departs somewhat from the true parabolic form (shown dotted) near the top of the potential well, where it more or less spreads out. This departure from the true parabolic form is due to two reasons: (i) *a rather large amplitude of the oscillations* and (ii) *the rotation of the molecule about its centre of mass*. Nevertheless, the bottom of the potential well is almost truly parabolic in form, indicating that for amplitudes about the equilibrium position (r_0), the system oscillates like a two-body harmonic oscillator, with its natural frequency $\nu_0 = \frac{1}{2\pi} \sqrt{\frac{C}{\mu}}$, where μ is its *reduced mass*.

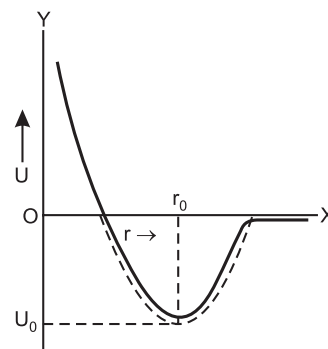


Fig. 7.21

Energy-level diagram of a diatomic molecule. Classical mechanics, as we know, puts no limit to the energy of vibration possessed by a harmonic oscillator which must go on increasing continuously with its amplitude. But quantum—and this assertion is borne out by experiment—that the oscillator can possess only certain discrete values of energy and no others. These discrete or permissible values of vibrational energy (E_V) are given by the relation,

$$E_V = \left(n + \frac{1}{2}\right) h\nu_0$$

where n can be either zero or a positive integer and is called the *vibrational quantum number*; h is the familiar *planck's constant* (6.62×10^{-27} erg-sec.) and ν_0 , the *natural frequency of oscillation* of the molecule, equal to $\frac{1}{2\pi} \sqrt{C/\mu}$

This vibrational energy (E_V) of the molecule is of course relative to the vibrational energy *zero* that it possesses when at rest in its position of equilibrium, (*i.e.*, when $r = r_0$).

Now, the *total energy of molecule*, which, as we have seen above, is normally always negative, is given by $E = (-U_0 + E_V)$.

Or,

$$E = -U_0 + \left(n + \frac{1}{2}\right) h\nu_0$$

which gives a various permissible discrete values of energy the molecule may possess if n be given the values 0, 1, 2, 3, etc.

The vibrational energy E_{V0} , corresponding to $n = 0$, being obviously the lowest, is referred to as the **zero-point energy** and is clearly equal to $\left(0 + \frac{1}{2}\right) h\nu_0 = \frac{1}{2} h\nu_0$. The *minimum total energy* of the molecule (corresponding to $n = 0$) is thus $E_0 = \left(-U_0 + \frac{1}{2} h\nu_0\right)$. And the **dissociation energy** (*i.e.*, the energy needed to completely separate the atoms) and hence also, the **binding energy** (or the energy necessary to bind the atoms together) of the molecule is thus $-\left(-U_0 + \frac{1}{2} h\nu_0\right) = U_0 - \frac{1}{2} h\nu_0$.

It will also be easily seen that the vibrational energy of the molecule, and hence also its total energy, will go on increasing in equal steps of $h\nu_0$ as n takes on the successive value 1, 2, 3 etc., for

$$E_{Vn} - E_{V(n-1)} = \left(n + \frac{1}{2}\right) h\nu_0 - \left(n - 1 + \frac{1}{2}\right) h\nu_0 = h\nu_0.$$

Thus, if we take $E = 0$, in the equilibrium position of the molecule, as its *upper limit* of energy (the energy in all other positions being energy, we can build up an *energy-level diagram*, consisting of a set of parallel straight lines representing energies of the molecule corresponding to $n = 0, 1, 2, 3$ etc, as shown in Fig. 7.22, such that the distance of any particular line from the reference line $E = 0$ is proportional to the energy represented by that line.

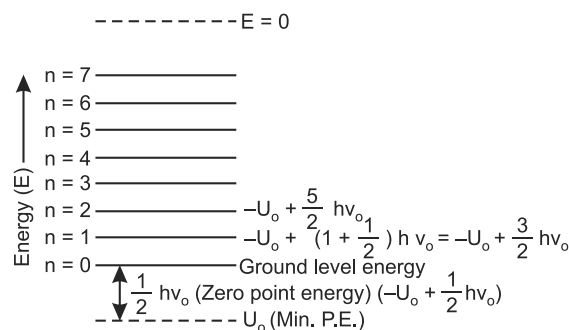


Fig. 7.22

Thus, proceeding from the ground level upward, for *quite some distance the lines are all equally spaced* since an upper level differs from the one immediately below it by $h\nu_0$. Only at the levels higher up, which correspond to the top of the potential well of fig. 7.21, above, where the *P.E.* curve departs from the parabolic form and spreads out sideways and where, therefore, the frequency is less than its value $\frac{1}{2\pi}\sqrt{C/\mu}$ in the lower parts, do the lines get closer together, as shown.

Now, it may as well be mentioned here that just as a molecule, when it rises from a lower to a higher energy level, absorbs energy which is an integral multiple of $h\nu_0$, so also when it falls from a higher to a lower energy level, it parts with energy which too, naturally, is an integral multiple of $h\nu_0$. This energy is usually given out in the form of a *photon* or electromagnetic radiations, which are a direct consequence of the oscillations of electric charges in the molecule. Such radiations, however, are given out by what are called *polar molecules* (*i.e.*, molecules in which there are free electric poles or charges that can oscillate) like those of *HCl*, for example, but not by symmetrical molecules like those of hydrogen (because of the absence of electrical poles or charges). Here too, there are permissible and forbidden transitions, the permissible ones that occur most frequently are those in which the value of n change by 1. So that, if ν be the frequency of the radiation emitted, we have *energy given out*, *i.e.*,

$$h\nu = E\nu_n - E\nu_{(n-1)} = \left(n + \frac{1}{2}\right)h\nu_0 - \left(n - 1 + \frac{1}{2}\right)h\nu_0 = h\nu_0$$

i.e., the frequency ν of the radiation emitted is equal to the classical frequency $\nu_0 = \frac{1}{2\pi}\sqrt{C/\mu}$

And, therefore, the corresponding *wave number* ν (*i.e.*, number of wave contained in a length of

$$\begin{aligned} 1 \text{ cm}) &= \frac{1}{\lambda} = \frac{\nu}{c} = \frac{\nu_0}{c} \\ &= \frac{1}{2\pi c} \sqrt{\frac{C}{\mu}} \end{aligned}$$

where c is the velocity of light in free space (equal to 3×10^{10} cm/sec).

Some of the *forbidden transitions*, *i.e.*, those corresponding to the changes of more than 1 in the value of n , also occur quite often, giving electromagnetic radiations in the form of faint spectral lines.

It may also happen sometimes that the molecule oscillates and rotates simultaneously. In such cases, obviously, many more of the spectral lines are emitted.

7.10 SIMILARITY BETWEEN MECHANICAL AND ELECTRICAL OSCILLATIONS

There is a similarity between mechanical and electrical oscillations as discussed below:

The equation of motion of a simple harmonic mechanical oscillator is given by

$$\frac{d^2x}{dt^2} + \frac{s}{m}x = 0 \quad \dots(i)$$

where x is the displacement, s the force constant of proportionality or stiffness and m the mass of oscillator.

The angular frequency is given by

$$\omega^2 = \frac{s}{m} \quad \text{or} \quad \omega = \sqrt{\frac{s}{m}}$$

or frequency

$$n = \frac{1}{2\pi} \sqrt{\frac{s}{m}}$$

The equation for displacement is given by

$$x = a \sin(\omega t + \phi)$$

where a and ϕ are constants, known as amplitude and initial phase angle.

The total energy of the mechanical oscillator

$$E = \frac{1}{2}mv^2 + \frac{1}{2}sx^2 = \frac{1}{2}m\dot{x}^2 + \frac{1}{2}sx^2$$

where $\frac{1}{2}m\dot{x}^2$ is the *K.E.* and $\frac{1}{2}sx^2$ the *P.E.*

The equation of motion of a simple harmonic electrical oscillator is given by

$$\frac{d^2Q}{dt^2} + \frac{1}{LC}Q = 0$$

where Q is the charge, L the inductance and C the capacitance of the electrical circuit.

The angular frequency is given by

$$\omega = \sqrt{\frac{1}{LC}}$$

or frequency

$$n = \frac{1}{2\pi\sqrt{LC}}$$

The charge on the capacitance varies harmonically and is represented by an equation similar to displacement equation *i.e.*

$$Q = Q_0 \sin(\omega t + \phi)$$

where Q_0 is the amplitude of charge and ϕ the phase difference.

The current $I = \frac{dQ}{dt}$ corresponds to velocity $v = \frac{dx}{dt}$ and is given by

$$I = \omega Q_0 \cos(\omega t + \phi).$$

The voltage across the capacitor $V = \frac{Q}{C} = \frac{Q_0}{C} \sin(\omega t + \phi)$

Both I and V , therefore vary harmonically with the same angular velocity ω .

The total energy of an electrical oscillator is the sum of the magnetic energy and electric energy.

The magnetic energy can be calculated from the current I and potential $V = L \frac{dI}{dt}$ across the inductance and is given by

$$\int VI \, dt = \int L \frac{dI}{dt} I \, dt = \int LI \, dI = \frac{1}{2} LI^2 = \frac{1}{2} L \dot{Q}^2$$

Compare it with kinetic energy in a mechanical oscillator given by $\frac{1}{2}mx^2$. Thus mass in a mechanical circuit corresponds to inductance in an electrical circuit and velocity to electric current.

The electrostatic energy can be calculated from the voltage across the capacitor and is given by

$$\frac{1}{2}CV^2 = \frac{1}{2}C\left(\frac{Q}{C}\right)^2 = \frac{1}{2}\frac{Q^2}{C}$$

Compare it with potential energy in a mechanical oscillator given $\frac{1}{2}sx^2$. Thus stiffness s in a mechanical circuit corresponds to $\frac{1}{C}$ in an electrical circuit.

7.11 ROLE OF L AND C IN ELECTRICAL OSCILLATOR

An oscillating electrical circuit consists of an inductance L and a capacitance C . In the electrical oscillator it is the charge on the capacitance that oscillates. In other words, *the charge on the capacitor is the harmonically varying quantity which gives rise to electrical oscillations*. In an electrical oscillator charge corresponds to displacement in a mechanical oscillator. The inductance L is the electrical counterparts of mass m (inertia) and the reciprocal of capacitance ($1/C$) is the counter part of stiffness s . The frequency is given by

$$n = \frac{1}{2\pi\sqrt{LC}}$$

We can realise an LC circuit in practice and obtain simple harmonic oscillations if the circuit has zero (ohmic) resistance and there is no loss of energy. In practice, however, it is not possible to have a circuit with zero resistance. Hence energy is supplied to the LC circuit to make up for this small inevitable loss by electronic devices like thermionic valves or transistors.

7.12 AN HARMONIC OSCILLATOR

An harmonic oscillator is one in which the relationship between *force* and *displacement* is *linear*. The harmonic oscillator is a highly idealized system that oscillates with a single frequency, irrespective of the amount of pumping or energy injected into the system. Consequently, the fundamental frequency of vibration of an harmonic oscillator is *independent of the amplitudes* of the vibrations. Most commonly studied is the mass-spring system obeying Hook's law, as seen in detail.

However, in a mechanical anharmonic oscillator, the relationship between force and *displacement* is *not linear*, but *depends upon the amplitude* to the displacement. The non-linearity arises from the fact that the spring is not capable of exerting a restoring force that is proportional to its displacement because of, for example, stretching in the material comprising the spring. As a result of non-linearity, the vibration frequency can change, depending upon the system's displacement.

These changes in the vibration frequency result in energy being coupled from the fundamental vibration frequency to other frequencies through a process known as *parametric coupling*.

There are many systems throughout the physical world exhibiting an anharmonic oscillations in addition to the non-linear mass-spring system. For example, an atom, which consists of a positively charged nucleus surrounded by a negatively charged electronic cloud, experiences a displacement between the centre of mass of the nucleus and the electronic cloud when an electric field is present. The amount of that displacement, called the electric dipole moment, is related linearly to the applied field for small fields, but as the magnitude of the field is increased, the field-dipole moment relationship becomes non-linear, just as in the mechanical system.

Further example of anharmonic oscillator include the large-angle pendulum, which exhibits chaotic behaviour as a result of its anharmonicity non-equilibrium semiconductor, that possess

a large not carrier population, which exhibit nonlinear behaviour of various types related to the effective mass, of the carriers; and ionosphere plasmas, which also exhibit nonlinear behaviour based on the anharmonicity of the plasma. Anharmonicity played a role in *lattice and molecular vibrations*, in *quantum oscillations* and in acoustics.

In fact, virtually all oscillators becomes anharmonic when their pump amplitude increases beyond some threshold and as a result, it is necessary to use non-linear equations of motion to describe their behaviour.

Analytical Treatment

We have seen under §5.9 how that *P.E.* of a particle at a distance x from its position of stable equilibrium is given by

$$U = U_0 + \frac{Cx^2}{2!} + \frac{C_1x^3}{3!} + \frac{C_2x^4}{4!} + \dots \quad \dots(i)$$

and the force acting on the particle by $F = -Cx - \frac{C_1x^2}{2!} - \frac{C_2x^3}{3!} \dots \quad \dots(ii)$

It will also be recalled that in the case of a harmonic oscillator, the terms containing C_1 , C_2 , etc., viz., $C_1x^3/3!$, $C_2x^4/4!$ etc., are all equal to zero, so that $U = \frac{1}{2}Cx^2$ and hence the *P.E.* curve is a *parabola*, And the force $F = -Cx$ and hence proportional to displacement of the particle or the oscillator.

In case any of the terms containing C_1 , C_2 etc is not equal to zero, the motion of the particle or the system no longer remain harmonic and it is, therefore, called an *anharmonic oscillator*. And since it is the presence of the terms containing C_1 , C_2 etc that really makes it so, they are referred to as *anharmonic terms*.

Let us study the anharmonic oscillator in some detail.

With the forces acting on an anharmonic oscillator given by relation (ii) above, its equation of motion is

$$m \frac{d^2x}{dt^2} = -Cx - \frac{C_1x^2}{2!} - \frac{C_2x^3}{3!}$$

which may be put as $\frac{d^2x}{dt^2} + \frac{C}{m}x = -\frac{C_1x^2}{(2 \times 1)m} - \frac{C_2x^3}{(3 \times 2 \times 1)m} - \dots$

Or, putting $C/m = \omega_0^2$, $C_1/2m = \alpha$, $C_2/6m = \beta \dots$, we have

$$\frac{d^2x}{dt^2} + \omega_0^2x = -\alpha x^2 - \beta x^3 - \dots \quad \dots(iii)$$

There is really no general solution to this equation. Approximate solution may however be obtained in specific cases if, as is usually the case, the anharmonic terms are small and, in particular, if the terms are small and, in particular, if the terms containing the 4th and the higher powers of X are negligible.

Now, as mentioned earlier (under §7.2), any complex periodic motion is really made up of a number of simple harmonic motions (called *harmonics*) whose frequencies are integral multiples of the lowest or the fundamental frequency. This means, in other words, that the displacement x is a *periodic function of t* and can as such, be expressed as a *Fourier series* as follows:

$$x = A_0 + A_1 \sin \omega t + A_2 \sin 2\omega t + \dots A_n \sin (n \omega t) + B_1 \cos \omega t + B_2 \cos 2\omega t + \dots B_n \cos (n \omega t). \quad \dots(iv)$$

As will be easily seen, many of the coefficients in this equation will be equal to zero in view of the sheer symmetry of the motion. Thus, for example, if we start counting time from an instant

such that the motion is an *even* functions of time, *i.e.*, when x has equal values at times t and $-t$ and, therefore, its maximum or minimum value at $t = 0$, all sine terms will be absent from the equation because the general *sine term*, $\sin(n\omega t)$, being an odd function of time t , its value changes sign with that of t . We, therefore, take as a possible solution the expression.

$$x = A_0 + B_1 \cos \omega t + B_2 \cos 2\omega t + B_3 \cos 3\omega t + \dots, \quad \dots(v)$$

where the values of ω and the coefficients A_0, B_1, B_3 , etc. are to be determined.

As we know, in the event of α and β etc being each equal to zero, we have (from relation (iii) above) a harmonic oscillator, with the solution $x = B_1 \cos \omega_0 t$.

Here, although α and β etc are not equal to *zero*, they are nevertheless supposed to be small, so that all coefficients other than B_1 , *viz.*, A_0, B_2, B_3 etc. must also be small and hence the products and squares of $\alpha, \beta, A_0, B_2, B_3$ etc must be negligibly small. Keeping this in mind and taking value of x as given by relation (iv) above, we have

$$\begin{aligned} \frac{d^2 x}{dt^2} + \omega_0^2 x &= -\omega^2 (B_1 \cos \omega t + 4B_2 \cos 2\omega t + 9B_3 \cos 3\omega t + \dots) \\ &\quad + \omega_0^2 (A_0 + B_1 \cos \omega t + B_2 \cos 2\omega t + B_3 \cos 3\omega t + \dots) \\ &= \omega_0^2 A_0 + (\omega_0^2 - \omega^2) B_1 \cos \omega t + (\omega_0^2 - 4\omega^2) B_2 \cos 2\omega t \\ &\quad + (\omega_0^2 - 9\omega^2) B_3 \cos 3\omega t + \dots \end{aligned}$$

$$\begin{aligned} \text{and} \quad -\alpha x^2 - \beta x^2 &= \alpha (A_0 + B_1 \cos \omega t + B_2 \cos 2\omega t + B_3 \cos 3\omega t + \dots)^2 \\ &\quad - \beta (A_0 + B_1 \cos \omega t + B_2 \cos 2\omega t + B_3 \cos 3\omega t + \dots)^2 \\ &= -\alpha B_1^2 \cos^2 \omega t - \beta B_1^3 \cos^3 \omega t - \alpha \frac{L_1^2}{2} (1 + \cos 2\omega t) \\ &\quad - \beta \frac{B_1^3}{4} (3 \cos \omega t + \cos 3\omega t) \\ &= -\alpha \frac{B_1^2}{2} - 3\beta \frac{B_1^3}{4} \cos \omega t - \alpha \frac{B_1^2}{2} \cos 2\omega t - \beta \frac{B_1^3}{4} \cos 3\omega t \end{aligned}$$

Substituting these values in expression (iii) above, we have

$$\begin{aligned} \omega_0^2 A_0 - (\omega_0^2 - \omega^2) B_1 \cos \omega t + (\omega_0^2 + 4\omega^2) B_2 \cos 2\omega t + (\omega_0^2 + 9\omega^2) B_3 \cos 3\omega t + \dots \\ = -\alpha \frac{B_1^2}{2} - 3\beta \frac{B_1^3}{4} \cos \omega t - \alpha \frac{B_1^2}{2} \cos 2\omega t - \beta \frac{B_1^3}{4} \cos 3\omega t \quad \dots(vi) \end{aligned}$$

Clearly, in order that relation (iii) may hold good, relation (v) must be satisfied for all values of t . This will obviously happen when the coefficients of the respective cosine functions on both sides of equation (vi) are equal, *i.e.*, when

$$\begin{aligned} \omega_0^2 A_0 &= -\alpha B_1^2/2 \quad \dots(a), & (\omega_0^2 - \omega^2) B_1 &= 3\beta B_1^2/4 \quad \dots(b), \\ (\omega_0^2 - 4\omega^2) B_2 &= -\alpha B_1^2/2 \quad \dots(c) & (\omega_0^2 - 9\omega^2) B_3 &= -\alpha B_1^2/4 \quad \dots(d) \end{aligned}$$

The terms containing the 4th and higher powers of x being assumed to be negligible, the coefficients B_4, B_5, B_6 , etc. are all equal to zero.

$$\text{Now, relation (b) above gives } \omega^2 = \omega_0^2 + \frac{3}{4} \beta B_1^2$$

$$\omega = \left(\omega_0^2 + \frac{3}{4} \beta B_1^2 \right)^{1/2}$$

$$\text{Or,} \quad \omega = \omega_0 \left(1 + \frac{3}{4} \frac{\beta B_1^2}{\omega_0^2} \right)^{1/2} = \omega_0 \left(1 + \frac{1}{2} \frac{3}{4} \frac{\beta B_1^2}{\omega_0^2} \right)$$

Or,
$$\omega = \omega_0 \frac{3}{8} \frac{\beta B_1^2}{\omega_0^2} \quad \dots(vii)$$

Assuming ω to be very nearly equal to ω_0 , we have from relation (c),

$$-3 \omega_0^2 B_2 = -\alpha B_1^2/2, \text{ whence, } B_2 = \alpha B_1^2/6\omega_0^2.$$

Similarly, from relation (d), we have $B_3 = \beta B_1^2/32\omega_0^2$,

and from relation (a), we have $A_0 = \alpha B_1^2/2\omega_0^2$.

Substituting these values of the coefficients in relation (v) above, we have

$$x = -\frac{\alpha B_1^2}{2 \omega_0^2} + B_1 \cos \omega t + \frac{\alpha B_1^2}{6 \omega_0^2} \cos 2 \omega t + \frac{\beta B_1^2}{32 \omega_0^2} \cos 3 \omega t + \dots \quad \dots(viii)$$

where the value of ω is as given by relation (vii) above.

Here, the term $B_1 \cos \omega t$ is referred to as the **fundamental component of oscillation** of the particle or the system and B_1 as its *amplitude* and the terms in $\cos 2\omega t$ and $\cos 3\omega t$ as the second and third harmonics, with frequencies *twice* and *thrice* respectively of the fundamental.

As mentioned earlier also, if $\alpha = \beta = 0$, we have the case of a *harmonic* oscillator, with $x = B_1 \cos \omega_0 t$, for then $\omega = \omega_0$ [from relation (vii) above].

The same happens when B_1 is small so that terms containing B_1^2 may be neglected. Relation (viii) then reduces again to $x = B_1 \cos \omega_0 t$ [because $\omega = \omega_0$ from relation (vii)].

This shows that *provided the amplitude be small, the oscillations even in the case of an anharmonic potential well are simple harmonic in character**, as already mentioned in §7.2.

If, however, the amplitude increases, the following changes occur:

(i) As can be seen from relation (vii) above, if the amplitude (B_1) be large, ω will no longer be equal to ω_0 and hence the *time-period* and, therefore, also the frequency, of the oscillation, will change. Thus, if T be the *time-period* of the oscillations, we shall have

$$\begin{aligned} T &= \frac{2\pi}{\omega} = \frac{2\pi}{\omega_0 (1 + 3\beta B_1^2 / 8\omega_0^2)} = \frac{2\pi}{\omega_0} \left(1 + \frac{3\beta B_1^2}{8\omega_0^2} \right)^{-1} \\ &= \frac{2\pi}{\omega_0} \left(1 - \frac{3\beta B_1^2}{8\omega_0^2} \right) \end{aligned}$$

Since $2\pi/\omega_0 = T_0$, the time-period of the small amplitude (or simple harmonic) oscillations, we have

$$T = T_0 \left(1 - \frac{3\beta B_1^2}{8\omega_0^2} \right)$$

(ii) The amplitudes of the second, third and higher harmonics, having twice, thrice etc frequencies of the fundamental, will increase still more rapidly, being proportional to B_1^2 , B_1^3 etc.

(iii) The amplitude of the particle will clearly not be equal to B_1 , the amplitude of the fundamental component of the oscillation, nor will move equally on either side of its equilibrium position, (see (iv) below).

(iv) Since the average values of the cosine functions ($\cos \omega t$, $\cos 2 \omega t$ etc) over a time-period $T = 2\pi/\omega$ are zero, the average value of the displacement x will become

This is actually *not so universally true*. For, there are cases where, despite small amplitudes, the oscillations are not simple harmonic; for example, (i) when the potential well, instead being parabolic in its lower parts, is *V-shaped* with a well-marked cusp at the bottom. For, in such a case, du/dx and other derivatives simply, do not exist at the bottom of the well and U can not, therefore, be expanded in a Taylor's series (ii) Or, again, if the value of U , be given by some such relation as $U = -U_0 e^{-ax^4}$, with $F = -dU/dx = -4 a U_0 x^3$, so that the equation of motion turns out to be $m d^2x/dt^2 + 4aU_0 x^3 = 0$, clearly showing that the oscillations, although periodic in the range $-U_0 \ll E < 0$ are by no means simple harmonic despite a small amplitude.

$$\langle x \rangle = -\alpha B_1^2/2\omega_0^2, \text{ with } \alpha, \text{ of course, not being zero.}$$

Thus, there will be a shift in the mean position of the particle, proportional to B_1^2 , indicating a greater restoring force acting on one side of its equilibrium position than on the other. This means, in other words, that *the potential energy curve or the potential well no longer remain symmetrical and the particle, in consequence, will move more to one side of its equilibrium position than to the other. Its oscillation will thus no longer be simple harmonic.*

An interesting example of the shift in the mean position of a particle is provided by the oscillating atoms in a solid material. The atoms experience a considerable repulsive force when they are closer together, so that they can move apart more easily than they can come nearer to each other. When the solid is heated, therefore, the amplitude of their oscillation increases and their mean position shifts *outwards*, resulting in the linear expansion of the solid, —the greater the shift in the mean position of the atoms, the greater the linear expansion of the solid. Thus, *linear expansion of the solid $\propto$ shift in the mean position of the atoms.*

Now, as we have just seen above, this shift in the mean position is proportional to the square of the amplitude of the oscillating atom which, in its turn, is proportional to the energy of the atom. And, since the average energy of the atom depends upon the temperature of the solid, it follows that *the linear expansion of a solid is proportional to its temperature.*

7.13 TIME-PERIOD OF A PENDULUM FOR LARGE AMPLITUDE OSCILLATION

A simple or a compound pendulum oscillating with a large amplitude is a familiar example of an anharmonic oscillator. The oscillation is simple harmonic only if the angular amplitude θ be infinitely small, *i.e.*, when in the expansion of $\sin\theta$ into a power series (*viz.*, $\sin\theta = \theta - \theta^3/3! + \theta^5/5! \dots$) all other terms except the first are negligibly small. So that, as shown in §7.71, the equation of motion of a *simple pendulum* comes out to be $d^2\theta/dt^2 = -(g/l)\theta$ and of a *compound pendulum*, $d^2\theta/dt^2 = -(mgl/I)\theta = -[mgl/(I_0 + ml^2)]\theta$

$$= -[mgl/(mk^2 + ml^2)]\theta = [mgl/m(k^2 + l^2)]\theta = \left[g \left(\frac{k^2}{l^2} + 1 \right) \right] \theta.$$

Or, putting $(k^2/l^2) + 1 = L$, the length of the equivalent simple pendulum, we have

$$d^2\theta/dt^2 = -(g/L)\theta,$$

the two cases being thus identical except that we have L in place of l in the case of a compound pendulum.

If θ be appreciable large, so that the second term ($\theta^3/3!$) in the power series can not be neglected*, we have (taking the case of a simple pendulum),

$$\frac{d^2\theta}{dt^2} = -\frac{g}{l}\theta + \frac{g}{(3 \times 2 \times 1)l}\theta^3$$

Now, referring back to equation (iii) of §7.12 above, and taking θ in place of x we see that $\omega_0^2 = g/l$, $\alpha=0$, $\beta = -g/6l = -\omega_0^2/6$.

$\therefore$ From relation (viii) of §7.9, taking amplitude θ_1 in place of B_1 , we have

$$\begin{aligned} \theta &= \theta_1 \cos \omega t + \frac{\theta_1^3 \beta}{32 \omega_0^2} \cos 3\omega t = \theta_1 \cos \omega t - \frac{\theta_1^3 \omega_0^2}{6 \times 32 \omega_0^2} \cos 3\omega t \\ &= \theta_1 \cos \omega t - \frac{\theta_1^3}{192} \cos 3\omega t, \end{aligned}$$

where (as given by relation (viii) of §7.12) $\omega = \omega_0 \left(1 - \frac{3}{8} \frac{\omega_0^2}{6} \cdot \frac{\theta_1^2}{\omega_0^2} \right)$

Thus, for example, if $\theta = 60^\circ$ or $\pi/3$ radian, we have $\theta^3/3! = \theta^3/3 \times 2 \times 1 = 0.19$ which cannot be neglected, although $\theta^5/5! = 0.001$ may easily be ignored.

$$= \omega_0 \left(1 - \frac{\theta_1^2}{16} \right) = \sqrt{\frac{g}{l}} \left(1 - \frac{\theta_1^2}{16} \right)$$

Hence, *time-period of the simple pendulum, i.e.,*

$$T = \frac{2\pi}{\omega} = 2\pi / \sqrt{g/l} \left(1 - \frac{\theta_1^2}{16} \right) = 2\pi \sqrt{\frac{l}{g}} \left(1 + \frac{\theta_1^2}{16} \right)$$

Putting $2\pi\sqrt{l/g}$, the time-period when the amplitude small, equal to T_0 , we have

$$T = T_0(1 + \theta_1^2/16).$$

Similarly, for a compound pendulum, we shall obtain

$$T = 2\pi \sqrt{\frac{L}{g}} \left(1 + \frac{\theta_1^2}{16} \right)$$

Or

$$T = 2\pi \sqrt{\frac{(k^2/l^2) + l}{g}} \left(1 + \frac{\theta_1^2}{16} \right)$$

Since $2\pi\sqrt{\frac{(k^2/l^2) + l}{g}}$ is the time-period T_0 for oscillations of small amplitude, we have

$$T = T_0(1 + \theta_1^2/16),$$

Indicating that the *time-period increases with amplitude*.

Since the amplitude of the pendulum in both cases does not remain constant but goes on progressively decreasing from θ_1 in the beginning to, say, θ_2 , at the end, we may take $\theta_1\theta_2$ in place of θ_1^2 . So that, in either case, $T = T_0(1 + \theta_1\theta_2/16)$

Now, what we actually observe is T . The correct value of the time-period (*i.e.*, if the amplitude be small) is, therefore, given by

$$T_0 = T / \left(1 + \frac{\theta_1\theta_2}{16} \right) = T \left(1 - \frac{\theta_1\theta_2}{16} \right)$$

WORKED EXAMPLE

I-General-simple harmonic Motion

Example 7.1. A particle of mass 100 gm is placed in a field of potential $U = 5x^2 + 10$ ergs/gm. Find the frequency. (Meerut U. 2001)

Solution. Here potential energy $U = 5x^2 + 10$ ergs/gm; $m = 100$ gm

$$\text{Now} \quad F = -\frac{dU}{dx} \quad \therefore \quad F = -\frac{d}{dx}(5x^2 + 10) = -10x$$

Taking x as the displacement, we have $F = m \frac{d^2x}{dt^2}$

$$\therefore \quad m \frac{d^2x}{dt^2} = -10x \quad \text{or} \quad \frac{d^2x}{dt^2} = -\frac{10}{m}x$$

This is the equation of a simple harmonic motion with force constant $s = 10$ dyne/cm.

$$\therefore \text{ Frequency} \quad n = \frac{1}{2\pi} \sqrt{\frac{s}{m}} = \frac{1}{2\pi} \sqrt{\frac{10}{100}} = \frac{1}{2\pi\sqrt{10}} = 0.05 \text{ Hz}$$

Example 7.2. (a) An oscillatory motion of a body is represented by $y = ae^{i\omega t}$ where symbols have usual meaning. Show that the motion is simple harmonic.

(b) The displacement of a moving particle at any time t is given by $y = a \cos \omega t + b \sin \omega t$ show that the motion is simple harmonic.

Solution. (a) Given $y = ae^{i\omega t}$

Differentiating with respect to 't' we get

$$\frac{dy}{dt} = ae^{i\omega t} \times i\omega = ia\omega e^{i\omega t}$$

Differentiating again, we get

$$\frac{d^2y}{dt^2} = ia\omega e^{i\omega t} \times i\omega = -a\omega^2 e^{i\omega t} = -\omega^2 y$$

Or
$$\frac{d^2y}{dt^2} + \omega^2 y = 0$$

This is differential equation of S.H.M. Hence $y = ae^{i\omega t}$ represents a S.H.M.

(b) $y = a \cos \omega t + b \sin \omega t$

$\therefore \frac{dy}{dt} = -a\omega \sin \omega t + b\omega \cos \omega t$

and
$$\begin{aligned} \frac{dy}{dt} &= -a\omega^2 \cos \omega t - b\omega^2 \sin \omega t \\ &= -\omega^2 (a \cos \omega t + b \sin \omega t) \\ &= -\omega^2 y \end{aligned}$$

Or
$$\frac{d^2y}{dt^2} + \omega^2 y = 0$$

Hence $y = a \cos \omega t + b \sin \omega t$ is the equation of simple harmonic motion.

Example 7.3. A particle executes S.H.M. of period 10 sec. and amplitude 5 cm. calculate the maximum amplitude of velocity. (Guhati U. 2002)

Solution. Here displacement amplitude $a = 5$ cm; Time period $T = 10$ sec.

$\therefore$ Angular frequency $\omega = \frac{2\pi}{T} = \frac{2\pi}{10} \text{ s}^{-1}$

Maximum velocity $= a\omega = \frac{5 \times 2\pi}{10} \text{ cm/s}$

$= \pi \text{ cm/sec or } 3.14 \text{ cm/sec.}$

Example 7.4. Calculate the displacement to amplitude ratio for a S.H.M when K.E. is 90% of total energy.

Solution. If m is the mass of the particle executing S.H.M. a the amplitude and ω the angular velocity, then

$$\text{Total energy} = \frac{1}{2} \omega^2 a^2$$

Let y be the displacement when K.E. = 90% of total energy.

As K.E. is 90% of total energy.

$$\text{Potential energy} = \text{Total energy} - \text{Kinetic energy} = 10\% \text{ of total energy}$$

Now potential energy $= \frac{1}{2} m\omega^2 y^2$

$$\therefore \frac{\frac{1}{2} m \omega^2 y^2}{\frac{1}{2} m \omega^2 a^2} = \frac{y^2}{a^2} = \frac{10}{100} = 0.1$$

$$\text{Or } \frac{\text{Displacement}}{\text{Amplitude}} = \frac{y}{a} = \sqrt{0.1} = 0.316$$

Example 7.5. What is the ratio of kinetic energy at displacement one fourth to one third of the amplitude in case of simple harmonic motion?

Solution. If m is the mass of the particle executing S.H.M., a the amplitude and ω the angular velocity, then

$$\text{Total energy} = \frac{1}{2} m \omega^2 y^2$$

$$\text{When the displacement is } y, \text{ potential energy} = \frac{1}{2} m \omega^2 y^2$$

$$\therefore \text{Kinetic energy} = \text{Total energy} - \text{Potential energy} = \frac{1}{2} m \omega^2 (a^2 - y^2)$$

$$\text{At displacement } y = \frac{a}{4}, \text{ kinetic energy } E_1 = \frac{1}{2} m \omega^2 \left(a^2 - \frac{a^2}{16} \right) = \frac{15}{32} m \omega^2 a^2$$

$$\text{At displacement } y = \frac{a}{3}, \text{ kinetic energy } E_2 = \frac{1}{2} m \omega^2 \left(a^2 - \frac{a^2}{9} \right) = \frac{8}{18} m \omega^2 a^2$$

$$\therefore \frac{E_1}{E_2} = \frac{15}{32} \times \frac{18}{8} \times \frac{135}{128} = 1.055$$

Example 7.6. A simple harmonic oscillator is characterised by $y = a \cos \omega t$. calculate the displacement at which kinetic energy is equal to its potential energy.

(Nagpur U. 2003; Pbi. U, 2000)

Or

At what displacement from the mean position the total energy of a simple harmonic oscillator is half kinetic and half potential. (G.N.D.U. 2001)

Solution. Let y be the displacement at which kinetic energy of the simple harmonic oscillator is equal to its potential energy.

$$\text{Now } y = a \cos \omega t \quad \therefore \frac{dy}{dt} = -a \omega \sin \omega t$$

The kinetic energy of a simple harmonic oscillator is given by

$$\begin{aligned} \text{K.E.} &= \frac{1}{2} m v^2 = \frac{1}{2} m \left(\frac{dy}{dt} \right)^2 = \frac{1}{2} m (-a \omega \sin \omega t)^2 \\ &= \frac{1}{2} m a^2 \omega^2 \sin^2 \omega t = \frac{1}{2} m a^2 \omega^2 (1 - \cos^2 \omega t) \\ &= \frac{1}{2} m \omega^2 (a^2 - a^2 \cos^2 \omega t) = \frac{1}{2} m \omega^2 (a^2 - y^2) \end{aligned}$$

$$\text{P.E.} = \frac{1}{2} s y^2 \text{ where } s = \text{stiffness and } \frac{s}{m} = \omega^2 \text{ or } s = m \omega^2$$

$$\therefore \text{P.E.} = \frac{1}{2} m \omega^2 y^2$$

$$\text{when } \text{K.E.} = \text{P.E.}$$

$$\frac{1}{2}m\omega^2(a^2 - y^2) = \frac{1}{2}m\omega^2 y^2$$

or $a^2 - y^2 = y^2$ or $a^2 = 2y^2$

$$\therefore \text{Displacement } y = \pm \frac{a}{\sqrt{2}}$$

In other words, for a displacement $y = \pm \frac{a}{\sqrt{2}}$ from the mean position P.E. = $\frac{1}{4}m^2\omega^2 a^2$, K.E. = $\frac{1}{4}ma^2\omega^2$ and energy = $\frac{1}{2}m\omega^2 a^2$

i.e., half of the total energy is kinetic and half potential.

Example 7.7. The amplitude of a simple harmonic oscillator is doubled. How does this effect the time period, total energy and maximum velocity of the oscillator.

(P.U., 2003; Pbi, U. 2000)

Solution. When the displacement y of a simple harmonic oscillator is given by $y = A \sin(\omega t + \phi)$

$$\text{velocity } v = \frac{dy}{dt} = A\omega \cos(\omega t + \phi)$$

$$\text{and acceleration } a = \frac{d^2y}{dt^2} = -A\omega^2 \sin(\omega t + \phi) = -\omega^2 y$$

$$\text{Now time period } T = 2\pi \sqrt{\frac{\text{Displacement}}{\text{Acceleration}}} = 2\pi \sqrt{\frac{y}{\omega^2 y}} = \frac{2\pi}{\omega} \text{ (ignoring the negative sign)}$$

As $\frac{2\pi}{\omega}$ is a constant, the time period does not depend upon amplitude but remains constant.

Maximum velocity $v_{\max} = A\omega$ when $\cos(\omega t + \phi) = 1$

When amplitude A is doubled, the maximum velocity is also doubled.

$$\text{Total energy} = \frac{1}{2}mA^2\omega^2$$

When the amplitude A is doubled, the total energy becomes four times, because energy $\propto A^2$.

Example 7.8. Show that for a particle executing S.H.M. average value of kinetic and potential energy is the same and each is equal to half the total energy.

(Nagpur U., 2009; Pbi. U., 2003; Luck. U., 2002)

Solution. Average kinetic energy. Kinetic energy at any instant

$$= \frac{1}{2}mv^2 = \frac{1}{2}m\dot{y}^2 = \frac{1}{2}m\left(\frac{dy}{dt}\right)^2$$

$$\text{For a S.H.M. } y = a \sin(\omega t + \phi)$$

$$\text{and } \frac{dy}{dt} = a\omega \cos(\omega t + \phi)$$

$$\therefore \text{Instantaneous } K.E. = \frac{1}{2}ma^2\omega^2 \cos^2(\omega t + \theta)$$

If T is the time period, then

$$\begin{aligned} \text{Average } K.E. &= \langle K.E. \rangle = \frac{1}{T} \int_0^T \frac{1}{2}ma^2\omega^2 \cos^2(\omega t + \phi) dt \\ &= \frac{ma^2\omega^2}{2T} \int_0^T \cos^2(\omega t + \phi) dt \\ &= \frac{ma^2\omega^2}{2T} \int_0^T \frac{1}{2} [1 + \cos 2(\omega t + \phi)] dt \end{aligned}$$

$$\begin{aligned}
&= \frac{ma^2\omega^2}{4T} \left[\int_0^T dt + \int_0^T \cos 2(\omega t + \phi) dt \right] \\
&= \frac{ma^2\omega^2}{4T} T \left[\because \int_0^T \cos (2\omega t + \phi) dt = 0 \right] \\
&= \frac{1}{4} ma^2\omega^2
\end{aligned}$$

Average potential energy. Instantaneous potential energy is given by

$$\begin{aligned}
P.E. &= \frac{1}{2} sy^2 = \frac{1}{2} sa^2 \sin^2 (\omega t + \phi) \\
\therefore \text{Average } P.E. &= \frac{1}{T} \int_0^T \frac{1}{2} sa^2 \sin^2 (\omega t + \phi) dt \\
&= \frac{sa^2}{2T} \int_0^T \frac{1}{2} [1 - \cos 2(\omega t + \phi)] dt \\
&= \frac{sa^2}{4T} \left[\int_0^T dt - \int_0^T \cos 2(\omega t + \phi) dt \right] \\
&= \frac{sa^2}{4T} T = \frac{1}{4} sa^2 \left[\because \int_0^T \cos 2(\omega t + \phi) dt = 0 \right]
\end{aligned}$$

But $\omega^2 = \frac{s}{m} \quad \therefore s = m\omega^2$

$\therefore$ Average $P.E. = \frac{1}{4} ma^2\omega^2$

Total Energy. The total energy = $\frac{1}{2} ma^2\omega^2$

Thus it is clear from equations (i), (ii), and (iii) that the average kinetic energy of a harmonic oscillator is equal to the average potential energy and is equal to half the total energy *i.e.*,

$$\langle K.E. \rangle = \langle P.E. \rangle = \frac{1}{2} E_{\text{total}}$$

Example 7.9. The displacement of a simple harmonic oscillator is given by

$$x = a \sin (\omega t + \phi)$$

If the oscillations started at time $t = 0$ from a position x_0 with velocity $x = v_0$ show that

$$\tan \phi = \frac{\omega x_0}{v_0} \text{ and } a = \left[x_0^2 + \frac{v_0^2}{\omega^2} \right]^{\frac{1}{2}}$$

Solution. Given $x = a \sin (\omega t + \phi)$; at $t = 0, x = x_0$

$\therefore x_0 = a \sin \phi$

Also $\dot{x} = a \omega \cos (\omega t + \phi)$; at $t = 0, \dot{x} = v_0$

$\therefore v_0 = a \omega \cos \phi$

Hence $\tan \phi = \frac{\sin \phi}{\cos \phi} = \frac{x_0}{a} \times \frac{a\omega}{v_0} = \frac{\omega x_0}{v_0}$

and $a^2 \sin^2 \phi + a^2 \cos^2 \phi = x_0^2 + \frac{v_0^2}{\omega^2}$

or
$$a = \left[x_0^2 + \frac{v_0^2}{\omega^2} \right]^{\frac{1}{2}}$$

Example 7.10. A man stands on a platform which vibrates simple harmonically in a vertical direction at a frequency of 5 Hertz. Show that the mass loses contact with the platform when the displacement 10^{-2} metres.

Solution. The mass contact with the platform when the upward force acting on it exceeds its weight mg .

The mass vibrates simple harmonically given by

$$x = a \sin(\omega t + \phi)$$

$$\therefore \text{Acceleration} \quad \ddot{x} = a \omega^2 \sin(\omega t + \phi) = -\omega^2 x$$

$$\text{Upward force} = -m\omega^2 x$$

$$\text{In the limiting case } mg = m\omega^2 x$$

or
$$x = \frac{g}{\omega^2} = \frac{9.81}{4\pi^2 5^2} = 0.1 \text{ m} = 10^{-2} \text{ metre}$$

Example 7.11. A body executing S.H.M. has velocities 80 cm/s and 60 cm/s when displacements are 3 cm and 4 cm respectively. Calculate the amplitude of vibration and the time taken to travel 2.5 cm from positive extremity of the oscillation.

Solution. Velocity
$$v = \omega \sqrt{a^2 - y^2}$$

where ω = angular velocity; a = amplitude and y = displacement from the mean position

$$80 \text{ cm s}^{-1} = \omega \sqrt{a^2 - 3^2} \quad \dots(i)$$

$$60 \text{ cm s}^{-1} = \omega \sqrt{a^2 - 4^2} \quad \dots(ii)$$

$$\therefore \frac{80}{60} = \frac{4}{3} = \frac{\sqrt{a^2 - 3^2}}{\sqrt{a^2 - 4^2}} \quad \text{or} \quad \frac{16}{9} = \frac{a^2 - 9}{a^2 - 16}$$

or
$$a = 5 \text{ cm}$$

Substituting $a = 5 \text{ cm}$ in (i) we get

$$80 = \omega \sqrt{5^2 - 3^2}$$

$$\omega = 20 \text{ rad. s}^{-1}$$

or
$$y = a \sin \omega t$$

$\therefore$ Time taken to reach the positive extremity is given by

$$5 = 5 \sin \omega t$$

or
$$\sin \omega t = 1$$

or
$$\omega t = \pi/2$$

$\therefore$
$$t = \frac{\pi}{2\omega} = \frac{\pi}{40} \text{ s}$$

Distance of the point 2.5 cm from positive extremity, from the mean position = $5 - 2.5 = 2.5 \text{ cm}$
Time taken to reach a point 2.5 cm from the mean position is given by

$$2.5 = 5 \sin \omega t$$

or
$$\sin \omega t = \frac{1}{2}$$

or
$$\omega t = \pi/6$$

$$\therefore t = \frac{\pi}{6\omega} = \frac{\pi}{120} \text{ sec}$$

Time taken to travel from a point 2.5 cm from the positive extremity to the positive extremity

$$= \frac{\pi}{40} - \frac{\pi}{120} = \frac{\pi}{60} = 0.052 \text{ sec}$$

Hence time taken to travel from positive extremity a point 2.5 cm away = 0.052 sec.

Example 7.12. If a particle moves in a potential energy field $U = U_0 - ax + bx^2$, where a and b are positive constants, obtain an expression for the force acting on it as a function of position. At what point does the force vanish? Is this a point of stable equilibrium?

Calculate the force constant, time-period and frequency of the particle.

Solution. As we know, force acting on the particle is given by

$$F = -\frac{dU}{dx} = -\frac{d}{dx}(U_0 - ax + bx^2), \quad \text{whence, } F = a - 2bx$$

(ii) The force vanishes at the points where $dU/dx = 0$, i.e., where $a - 2bx = 0$. So that, $2bx = a$

Or, $x = a/2b$,

which gives the position of the point where the force vanishes.

(iii) We have $d^2U/dx^2 = 2b$ So that, b being positive, the point $x = a/2b$ represents the points of *minimum potential energy* on the energy curve of the particle. It is, therefore, a point of *stable equilibrium*.

(iv) From the expression for force F in (i) above, it is clear that it is a *linear restoring force* and the force constant C is therefore equal to $2b$.

(v) The time-period of the particle, $T = 2\pi\sqrt{1/\mu}$
where, as we know, $\mu = C/m = 2b/m$.

$$\text{So that, } T = 2\pi\sqrt{1/\frac{2b}{m}} = 2\pi\sqrt{\frac{m}{2b}}$$

$$\text{And, therefore, frequency of the particle, } n = 1/T = \frac{1}{2\pi}\sqrt{2b/m}$$

Example 7.13. (a) The potential energy of a harmonic oscillator of mass 2 kg in its resting position is 5 joules, its energy is 9 joules and its amplitude, 1 cm. Calculate its time-period.

(b) A particle of mass 5 gm lies in a potential field $V = (8x^2 + 200)$ ergs/gm. Calculate its time-period.

Solution. (a) Clearly, the energy gained by the oscillator in moving from its resting position to its extreme position, i.e., in moving through a distance equal to its amplitude a , is equal to $(9 - 5) = 4$ joules.

Since, energy gained $= \frac{1}{2}Cx^2$, where x is the displacement of the particle and C , the force constant, we have $\frac{1}{2}C(1)^2$, whence, $C = 4 \times 2 = 8 \text{ joules/gm}$.

$$\therefore \text{Time-period of the oscillator, } T = 2\pi\sqrt{m/C} = 2\pi\sqrt{2/8} = 2\pi\sqrt{1/4} = 2\pi \times 1/2 = \pi = 3.142 \text{ sec.}$$

(b) Here, P.E. of the particle, i.e., $U = mV = 5(8x^2 + 200)$ ergs.

$$\therefore \text{Force acting on the particle, } F = -dU/dx = -5 \times 16x = 80x \text{ dynes.}$$

So that, its equation of motion is $m d^2x/dt^2 = -80x$.

whence, $d^2x/dt^2 = -80x/m = -80x/5 = -16x = -\mu x$. Or, $d^2x/dt^2 \propto x$.

The particle thus executes a simple harmonic motion and its time-period is given by

$$T = 2\pi\sqrt{1/\mu} = 2\pi\sqrt{1/16} = 2\pi\sqrt{1/4} = \pi/2 = 1.571 \text{ sec.}$$

Example 7.14. A particle of mass 10 gm moves under a potential V_x given by $V(x) = 8 \times 10^5 x^2$ ergs/gm, where x is in centimeters. Deduce the time-displacement relation when the total energy is 8×10^5 ergs. (Agra (supp), 2001)

Solution. Clearly, maximum energy acquired by the particle $= mV(x) = 10 \times 8 \times 10^5 x^2 = 8 \times 10^6 x^2$ ergs. The whole of this must be present in the form of potential energy (U) at the end of its maximum displacement and must be equal to the total energy of the particle, given to be 8×10^5 ergs. Thus, $8 \times 10^6 x^2 = 8 \times 10^5$, where x is the maximum displacement.

Since the maximum displacement is the amplitude a of the particle, we may replace x by a . So that, $8 \times 10^6 a^2 = 8 \times 10^5$, whence, $a^2 = 8 \times 10^5 / 8 \times 10^6 = 1/10$.

$$\text{Or,} \quad a = 1/\sqrt{10}$$

Now, force acting on the particle, $F = -dU/dx = -d(8 \times 10^6 x^2)/dx = -2 \times 8 \times 10^6 x = -16 \times 10^6 x$.

$\therefore$ the equation of motion of the particle is $md^2x/dt^2 = -16 \times 10^6 x$,

whence, $d^2x/dt^2 = -16 \times 10^6 x/m = -16 \times 10^6 x/10 = -16 \times 10^5 x = -\omega^2 x$ (A)

$$\text{So that,} \quad \omega^2 = 16 \times 10^5 \quad \text{and} \quad \therefore \omega = \sqrt{16 \times 10^5} = 400\sqrt{10}$$

If, therefore, the solution of the differential equation of motion of the particle, (which is clearly a simple harmonic motion) be $x = a \sin(\omega t + \phi)$, we have

$$x = \frac{1}{\sqrt{10}} \sin(400\sqrt{10}t + \phi)$$

Example 7.15. Show that if the displacement of a moving point at any time is given by an equation of the form $x = a \cos \omega t + b \sin \omega t$, the motion is simple harmonic. If $a = 3$, $b = 4$ and $\omega = 2$, determine the period, amplitude, maximum velocity and maximum acceleration of the motion. (Madras 2004)

Solution. Since the displacement of the particle or the point is given by $x = a \cos \omega t + b \sin \omega t$, its velocity, $v = dx/dt = -a \omega \sin \omega t + b \omega \cos \omega t$ and its acceleration, $a = d^2x/dt^2 = \omega^2 (a \cos \omega t + b \sin \omega t) = -\omega^2 x$.

The particle, therefore, executes a simple harmonic motion of amplitude $\sqrt{a^2 + b^2} = \sqrt{3^2 + 4^2} = 5$ cm

The time-period of the particle, $T = 2\pi/\omega = 2\pi/2 = \pi = 3.142$ sec.,

Maximum velocity of the particle $= \omega a = 2 \times 5 = 10$ cm/sec.

and its maximum acceleration $= \omega^2 a = (2)^2 \times 5 = 20$ cm/sec².

Example 7.16. A vertical U-tube of uniform cross section contains water to a height of 30 cm. Show that if the water on one side is depressed and then released, its motion up and down the two sides of the tube is simple harmonic, and calculate its time-period.

Solution. Let AA' be the initial level of water in the U-tube and let the column on the left be depressed through distance y to B , (Fig. 7.22). Then, obviously, the column on the right will rise up through the same distance y (liquids being incompressible) to the level C , so that the difference of level between the two columns $= B'C = 2y$, where B' is in a level with B .

The weight of this column of water $= 2y \times a \times \rho \times g$, where a is internal area of cross section of the tube (or of the water column), ρ , the density of water and g , the acceleration due to gravity at the place.

This is then the force acting on the total mass of water, m , say, $= 2hap = 2 \times 30 \times a \times \rho = 60a\rho$ gm in the two limbs of the U-tube.

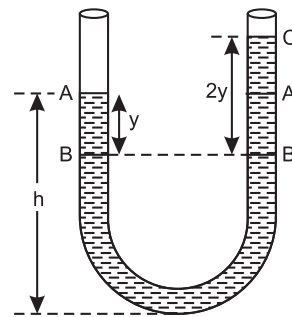


Fig. 7.22

$\therefore$ acceleration of the mass of water = force/mass = $-2ya\rho g/60a\rho = -(g/30)y = \mu y$, the negative sign indicating that it is directed opposite to the direction of displacement. Thus, the acceleration of the mass of water is proportional to its displacement and is directed opposite to it. It, therefore, executes a S.H.M. and its time-period is given by

$$T = 2\pi\sqrt{1/\mu} = 2\pi\sqrt{1/\frac{g}{30}} = 2\pi\sqrt{30/g} = 2\pi\sqrt{30/981} = 1.098 \text{ sec.}$$

Example 7.17. A particle is moving with S.H.M. in a straight line. When the distance of the particle from the equilibrium position has the values x_1 and x_2 , the corresponding values of the velocity are u_1 and u_2 . Show that the period is $2\pi[(x_2^2 - x_1^2)/(u_1^2 - u_2^2)]^{1/2}$.

Find also the maximum velocity and amplitude of the particle.

Solution. As we know, the velocity of a particle, executing S.H.M., at a distance x from its equilibrium position is given by $u = \omega\sqrt{a^2 - x^2}$, where ω is its angular velocity and a , its amplitude (§7.3). So that, we have here

$$u_1 = \omega\sqrt{a^2 - x_1^2} \quad \text{and} \quad u_2 = \omega\sqrt{a^2 - x_2^2}$$

$$\therefore u_1^2 - u_2^2 = \omega^2(x_2^2 - x_1^2). \quad \text{Or} \quad \omega = (u_1^2 - u_2^2)/(x_2^2 - x_1^2)^{1/2} \quad \dots(i)$$

And, therefore, time-period of the particle, $T = 2\pi/\omega$

$$= 2\pi[(x_2^2 - x_1^2)/(u_1^2 - u_2^2)]^{1/2}.$$

$$\text{Now, we have, from above, } u_1^2 = \omega^2(a^2 - x_1^2), \text{ Or, } \omega^2 = u_1^2/(a^2 - x_1^2) \quad \dots(ii)$$

From relations (i) and (ii), therefore, we have

$$\frac{u_1^2 - u_2^2}{x_2^2 - x_1^2} = \frac{u_1^2}{(a^2 - x_1^2)}. \quad \text{Or} \quad a^2 - x_1^2 = u_1^2 \left(\frac{x_2^2 - x_1^2}{u_1^2 - u_2^2} \right)$$

$$\begin{aligned} \text{And } \therefore a^2 &= \frac{u_1^2 x_2^2 - u_1^2 x_1^2}{u_1^2 - u_2^2} + x_1^2 = \frac{u_1^2 x_2^2 - u_1^2 x_1^2 + u_1^2 x_1^2 - u_2^2 x_1^2}{u_1^2 - u_2^2} \\ &= \frac{u_1^2 x_2^2 - u_2^2 x_1^2}{u_1^2 - u_2^2} \end{aligned}$$

$$\text{whence, amplitude of the particle, } a = \left(\frac{u_1^2 x_2^2 - u_2^2 x_1^2}{u_1^2 - u_2^2} \right)^{1/2}$$

Finally, we known that the maximum velocity of a particle executing S.H.M. is given by $v_{\max} = a\omega$. So that here,

$$\begin{aligned} \text{Maximum velocity of the particle, } v_{\max} &= \left(\frac{u_1^2 x_2^2 - u_2^2 x_1^2}{u_1^2 - u_2^2} \right)^{1/2} \left(\frac{u_1^2 - u_2^2}{x_2^2 - x_1^2} \right)^{1/2} \\ &= \left(\frac{u_1^2 x_2^2 - u_2^2 x_1^2}{x_2^2 - x_1^2} \right)^{1/2} \end{aligned}$$

Example 7.18. Two cities on the surface of the earth are joined by a straight, smooth underground tunnel of length 640 km. A body is released into the tunnel from one city. How much time will it take to reach the other city? Derive the formula used $G = 6.67 \times 10^{-8}$ c.g.s units and $\rho = 5.52$ gm/c.c. Calculate the velocity when the body would be nearest to the centre of the earth in its journey.

(Bombay, 2002)

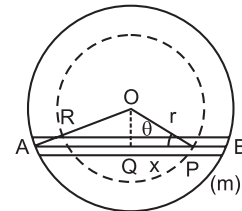


Fig. 7.23

Solution. Taking the earth to be a *solid and homogeneous sphere* of centre O , radius R and density ρ , let A and B be the two cities on its surface, joined by a straight, smooth underground tunnel AB , 640 km long, as shown in Fig. 7.23. Then, a perpendicular dropped from O on to AB meets it in Q where Q is the midpoint of AB .

Let the body of mass m , released at city A be at a point P inside the tunnel at a given instant such that it is at a distance x from Q (the mid-point of the tunnel) and $OP = r$.

Then, clearly, the body at P lies on the surface of a solid sphere, concentric with the earth, of radius r and density ρ , as shown dotted in the Figure It, therefore experiences a gravitational force of attraction towards the centre O , given by $F = (\text{mass of the sphere}) \times m \times G/r^2$. Since the mass of the sphere $= 4\pi r^3 \rho / 3$, we have

$$F = \frac{(4/3)\pi r^3 \rho \times m}{r^2} \cdot G = \frac{4}{3} \pi r \rho m G$$

Multiplying and dividing by R^3 , we have $F = (4/3)\pi r \rho m G R^3/R^3$.

Clearly $(4/3)\pi R^3 \rho = M$, the mass of the earth. So that,

force acting on the mass, $F = MmrG/R^3$, directed along PO .

The component of this force along the tunnel $= \frac{Mmr}{R^3} G \cos \theta$, where $\theta = \angle OPQ$

Or, since $\cos \theta = x/r$, we have

$$\text{force acting on the mass, along the tunnel, towards } Q = \frac{-MmrG}{R^3} \frac{x}{r}$$

the $-ve$ sign indicating that its direction is opposite to that of displacement x .

$$\therefore \text{Acceleration of the body towards } Q = \frac{MmG}{R^3 m} x = \frac{-MG}{R^3} x = -\omega^2 x = -\mu x$$

where

$$MG/R^3 = \omega^2 = \mu, \text{ a constant.}$$

The acceleration of the body is thus proportional to its displacement and is directed oppositely to it. It, therefore, executes a simple harmonic motion along AB about the point Q and its time-period is given by

$$T = 2\pi \sqrt{\frac{1}{\mu}} = 2\pi \sqrt{\frac{1}{MG/R^3}} = 2\pi \sqrt{\frac{R^3}{MG}}$$

Since the body is released at the end A , it will take time T to reach B and come back to A , completing one full oscillation.

$$\therefore \text{time taken in reaching from } A \text{ to } B = T/2 = \pi \sqrt{R^3/MG} = \sqrt{R^3 \frac{4}{3} \pi R^3 \rho G}$$

$$= \sqrt{\pi^2 R^3 \frac{4}{3} \pi R^3 \rho G} = \sqrt{3\pi / 4 \rho G} = \sqrt{3\pi / 4 \times 5.52 \times 6.67 \times 10^{-8}}$$

$$= 2529 \text{ sec.}$$

$$= 42 \text{ min } 29 \text{ sec.}$$

Now, it is obvious that the body would be nearest to the centre of the earth (O) in its Journey at the point Q , OQ being the perpendicular from O on to AB . Its velocity here will clearly be the maximum, this being its mean its mean position or its position of equilibrium.

So that, velocity of the body at Q (nearest to the of the earth), say, $v = \omega a$, where a is its amplitude equal to $OA = OB = AB/2$.

Now, $\mu = \omega^2 = MG/R^3$ and, therefore, $\omega = \sqrt{MG/R^3}$.

So that, *velocity of the body at Q, nearest to the centre of the earth, i.e.,*

$$\begin{aligned} v &= \sqrt{MG/R^3} \cdot a = \sqrt{\frac{4\pi R^3 \rho g}{3R^3}} \cdot a = \sqrt{\frac{4}{3} \pi \rho G} \times \frac{AB}{2} \\ &= \sqrt{\frac{4}{3} \pi \times 5.52 \times 6.67 \times 10^{-8}} \times \frac{640}{2} \times 10^5 = 3.973 \times 10^4 \text{ cm/sec.} \end{aligned}$$

Example 7.19. (a) A hydrogen atom has a mass of 1.68×10^{-24} gm. When attached to a certain massive molecule, it oscillates as a classical oscillator with a frequency of 10^{14} cycles per second and with an amplitude of 10^{-9} cm. Calculate the force acting on the hydrogen atom.

(Delhi, 2002)

(b) A test tube of weight 6 gm and of external diameter 2 cm is floated vertical in water by placing 10 gm of mercury at the bottom of the tube. The tube is depressed by a small amount and then released. Find the time of oscillation.

Solution. (a) We know that the frequency of a particle executing S.H.M. is given by $n = \omega/2\pi$. Here, $n = 10^{14}$ /sec. So that, $10^{14} = \omega/2\pi$, whence, $\omega = 2\pi \times 10^{14}$.

Now, acceleration of the particle $= -\omega^2 x$, where x is its displacement from its mean or equilibrium position, the $-ve$ sign indicating that it is directed towards that position, i.e. opposite to displacement.

Here, since displacement = amplitude a , we have acceleration of the particle, or the hydrogen atom $= -\omega^2 a = -(2\pi \times 10^{14})^2 \times 10^{-9}$ cm/sec²

$\therefore$ force acting on the hydrogen atom = mass $\times$ acceleration

$$= 1.68 \times 10^{-24} \times (2\pi \times 10^{14})^2 \times 10^{-9} = 6.63 \times 10^{-4} \text{ dynes.}$$

(b) Here mass of the tube + mercury = $6 + 10 = 16$ gm, external radius of the tube = $2/2 = 1$ cm and, therefore, its area of cross section $= \pi r^2 = \pi (1)^2 = \pi$ sq cm.

If the tube be depressed into water through a distance y cm, clearly, volume of water displaced $= y \times \pi$ c.c. Hence upthrust on the tube due to displaced water

$$= y \times \pi \times \rho \times g = y\pi g \quad (\because \rho \text{ for water} = 1 \text{ gm/c.c.}).$$

$$\therefore \text{Acceleration of the tube} = \frac{\text{Force}}{\text{mass}} = -\frac{\pi g}{16} y = \mu y \quad \text{where } \pi g/16 = \mu.$$

The $-ve$ sign, as usual, indicates that the acceleration is directed oppositely to displacement. Thus, the acceleration $\propto$ displacement and is directed oppositely to it. The tube, therefore, executes a S.H.M. and its time-period is given by

$$\begin{aligned} T &= 2\pi\sqrt{1/\mu} = 2\pi\sqrt{1/\frac{\pi g}{16}} = 2\pi\sqrt{16/\pi g} \\ &= \sqrt{4\pi \times 16/g} = 0.4527 \text{ sec.} \end{aligned}$$

Example 7.20. If the earth were a homogeneous sphere of radius R and a straight hole were bored in it through its centre, show that a particle dropped into the hole will execute a simple harmonic motion and find its time-period, (Jodhpur)

Solution. We know that acceleration due to gravity on the surface of the earth is given by $g = MG/R^2$ where M is the mass of the earth, R , its radius and G , the gravitational constant.

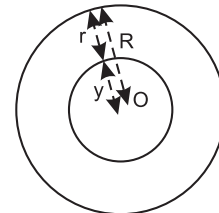


Fig. 7.24

If ρ be the density of the earth (assumed to be a homogeneous solid sphere), we have

$$M = \frac{4}{3} \pi R^3 \rho$$

So that,
$$g = \frac{4}{3} \pi R^3 \rho G / R^2 = \frac{4}{3} \pi R \rho G$$

At a depth r below surface of the earth, (Fig. 7.24), similarly, the acceleration due to gravity will be, say, $g' = \frac{4}{3} \pi [(R-r)\rho]G$

$$\therefore g'/g = (R-r)/R.$$

Or,
$$g' = [(R-r)/R]g = (g/R)(R-r)$$

If we put $(R-r) = y =$ distance from the centre of the earth, we have

$$g' = -\frac{g}{R} y = -\mu y,$$

the $-ve$ sign indicating its opposite direction to y .

Thus, the *acceleration $\propto$ displacement* from the centre of the earth and is directed towards it, i.e., opposite to displacement.

The particle, therefore, executes a S.H.M. about the centre of the earth.

And, its time-period is given by $T = 2\pi \sqrt{\frac{1}{\mu}} = 2\pi \sqrt{\frac{1}{g/R}} = 2\pi \sqrt{\frac{R}{g}}$

Example 7.21 The total energy of a particle executing a simple harmonic motion of period 2π sec is 10240 ergs. $\pi/4$ sec after the particle passes mid-point of the swing, its displacement is $8\sqrt{2}$ cm. Calculate the amplitude of the motion and the mass of the particle.

Solution. We know that the *total energy of a particle executing S.H.M.* $= 2\pi^2 m a^2 / T^2$, where the symbols have their usual meanings.

Here, $T = 2\pi$ sec and, therefore, *total energy of the particle* $= 2\pi^2 m a^2 / (2\pi)^2$
 $= 10240,$

whence,
$$\frac{1}{2} m a^2 = 10240. \text{ Or, } m a^2 = 20480. \quad \dots(i)$$

Since the time here is counted from the instant the particle passes through its mean or equilibrium position, we have $\phi = 0$. Hence its *displacement* is given by the relation $x = a \sin \omega t$.

Now, $T = 2\pi/\omega = 2\pi$ (given). So that, $\omega = 1$ radian/sec, $x = 8\sqrt{2}$ cm and $t = \pi/4$.

$$8\sqrt{2} = a \sin (1 \times \pi/4) = a \sin \pi/4 = a \times 1/\sqrt{2},$$

whence
$$a = 8\sqrt{2} \times \sqrt{2} = 8 \times 2 = 16 \text{ cm.}$$

i.e., the *amplitude of the particle* is 16 cm.

Substituting this value of a in expression (i) above, we have

$$m \times (16)^2 = 20480. \text{ whence, } m = 20480/16 \times 16 = 80 \text{ gm.}$$

Thus, the *mass of the particle* is 80 gm.

Example 7.22. The particle performing S.H.M. has a mass 2.5 gm and frequency of vibration 10 Hz. It is oscillation with an amplitude of 2 cm. Calculate the total energy of the particle. (Nagpur U. 2009, s/2007)

Solution. Total energy,
$$E_T = \frac{1}{2} m \omega^2 a^2$$

$$\omega = 2\pi n = 2\pi \times 10 = 20 \pi \text{ s}^{-1}$$

$$\begin{aligned}
 a &= 2 \text{ cm} = 0.02 \text{ m} \\
 m &= 2.5 \text{ gm} = 2.5 \times 10^{-3} \text{ kg} \\
 E_T &= \frac{1}{2} \times (20\pi)^2 \times (2.5 \times 10^{-3}) \times (0.02)^2 \\
 &= \frac{1}{2} 400 (3.14)^2 \times 6.25 \times 10^{-6} \\
 &= 200 \times 9.86 \times 6.25 \times 10^{-6} \\
 &= 12324.5 \times 10^{-6} \\
 &= 12.3245 \times 10^{-3} \text{ J}
 \end{aligned}$$

II – PENDULUMS

Type-I Simple Pendulum

Example 7.23. (a) what is the frequency of a simple pendulum 2.0 metres long?

(b) Assuming small amplitude, what would its frequency be in an elevator accelerating upwards at a rate of 2.0 metres/sec²?

(c) What would its frequency be in free fall?

Solution. (a) We know that the time-period of a simple pendulum is given by

$$T = 2\pi\sqrt{l/g}. \text{ Hence, its frequency } n = 1/T = \frac{1}{2\pi}\sqrt{g/l}$$

Here, $l = 2.0$ metres. Therefore, taking $g = 9.8 \text{ m/sec}^2$, we have frequency of the pendulum,

$$n = \frac{1}{2\pi}\sqrt{9.8/2} = \frac{1}{2\pi}\sqrt{4.9} = 0.3524 / \text{sec}$$

(b) In the elevator going up with an acceleration a , the effective weight of the pendulum is $mg(1 + a/g) = m(g + a)$. So that, here the effective value of g is, say, $g' = (g + a) = (9.8 + 2.0) = 11.8 \text{ m/sec}^2$.

$$\begin{aligned}
 \therefore \text{Frequency of the pendulum in the elevator, say, } n' &= \frac{1}{2\pi}\sqrt{g'/l} \\
 &= \frac{1}{2\pi}\sqrt{11.8/2} = \frac{1}{2\pi}\sqrt{5.9} = 0.3867 / \text{sec}
 \end{aligned}$$

(c) In free fall of the pendulum, the effective weight of the pendulum is $mg(1 - g/g) = 0$, i.e., the effective value of g , i.e., $g' = 0$.

$\therefore$ time-period of the pendulum $= 2\pi\sqrt{l/g'}$ = infinite, i.e. there is no oscillation of the pendulum at all. Its frequency (n) is, therefore, zero.

Example 7.24. A simple pendulum of length l and mass m is suspended in a car that is travelling with a constant speed v around a circle of radius R . If the pendulum undergoes small oscillations about its equilibrium position, what will its frequency of oscillation be?

Solution. Here, in addition to g , the acceleration due to gravity, the pendulum will also be subject to the centripetal acceleration v^2/R in a direction perpendicular to that of g on account of the car going round with speed v in a circle of radius R . Thus, resultant acceleration to which the pendulum is subjected $= [g^2 + (v^2/R)^2]^{1/2}$

$$= (g^2 + v^4/R^2)^{1/2}.$$

On a small angular displacement θ from its mean or equilibrium position, therefore, the restoring couple acting on the pendulum $= m(g^2 + v^2/R^2)^{1/2} l \sin \theta$.

$$= m(g^2 + v^2/R^2)^{1/2} l \theta \quad [\therefore \theta \text{ is small}].$$

If $d^2\theta/dt^2$ be the acceleration of the pendulum and I , its moment of inertia about the suspension thread, we have the couple on the pendulum also equal to $I \cdot d^2\theta/dt^2$, or equal to $ml^2 d^2\theta/dt^2$, because $I = ml^2$.

We, therefore, have $ml^2 d^2\theta/dt^2 = -m(g^2 + v^4/R^2)^{1/2} l^0$,

whence,
$$d^2\theta/dt^2 = -\frac{(g^2 + v^4/R^2)^{1/2} l^0}{l} = -\mu\theta,$$

where
$$\mu = \frac{(g^2 + v^4/R^2)^{1/2}}{l}$$

The pendulum thus executes a *S.H.M.* and therefore,

$$\text{frequency of the pendulum} = n = 1/T = \frac{1}{2\pi} \sqrt{\mu} = \frac{1}{2\pi} \sqrt{\frac{(g^2 + v^4/R^2)^{1/2}}{l}}$$

Example 7.25. (a) Show that the maximum tension in the string of a simple pendulum, when the amplitude θ_m is small, is $mg(1 + \theta_m^2)$. At what position of the pendulum is the tension a maximum?

(b) Using conservation of energy, show that the angular speed of a simple pendulum is given by $d\theta/dt = \left[\frac{2}{ml^2} \{E - mgl(1 - \cos\theta)\} \right]^{1/2}$ where E is the total energy of oscillations, l and

m are length and mass of the pendulum and θ is angular displacement from the vertical.

Solution. (a) Let SO be a simple pendulum of length l , with S , as its point of suspension and O , its mean or equilibrium position, (Fig. 7.25). If its angular amplitude be θ_m so that it is in the position SP , the whole of its energy is present in the potential form and is equal to $mg \times OP'$ where OP' is the vertical distance through which the bob rises with respect to its mean position O .

Or, since $OP' = SO - SP' = l - l \cos \theta_m = l(1 - \cos \theta_m)$, we have total energy of the pendulum $= mgl(1 - \cos \theta_m)$.

Or, because θ_m is small, we may take $\cos \theta_m = 1 - \frac{1}{2}\theta_m^2$.

So that, total energy of the pendulum

$$= mgl \left[1 - \left(1 - \frac{1}{2}\theta_m^2 \right) \right] = \frac{1}{2} mgl \theta_m^2$$

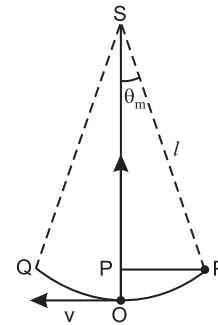


Fig. 7.25

When the bob arrives at O , the whole of the energy of the pendulum is converted into the kinetic form, so that if v be its velocity at O tangential to the arc PO , its $K.E. = \frac{1}{2}mv^2$. Thus, $\frac{1}{2}mv^2 = \frac{1}{2}mgl\theta_m^2$ whence, $v^2 = gl\theta_m^2$, Or, $v = (gl\theta_m^2)^{1/2}$.

Now, the *centripetal force necessary* to keep the bob moving in an arc of radii $l = mv^2/l$ towards S and it must obviously be equal to $T - mg$, where T is the tension in the string along OS and mg , the weight of the bob, i.e., $T - mg = mv^2/l$.

Or, substituting the value of v from above, we have

$$T - mg = mgl\theta_m^2/l = mg\theta_m^2, \text{ whence, } T = mg + mg\theta_m^2 = mg(1 + \theta_m^2).$$

The tension is clearly the maximum when the bob passes through its mean or equilibrium position, because only in this position is the velocity of the bob, and hence the centripetal force acting upon it, the maximum, as also the entire weight of the bob effective.

(b) If the angular displacement of the pendulum bob be θ , its *potential energy in its displaced position* = $mg l (1 - \cos \theta)$, where, as we have just seen in part (a) above, $l(1 - \cos \theta)$ is the vertical distance through which the bob rises relative to its mean position.

Since θ is less than the amplitude of the pendulum, the bob in this position possesses also kinetic energy = $\frac{1}{2} mv^2$, where v is the velocity of the bob in the displaced position. Therefore,

$$\text{Total energy of the pendulum, } E = \frac{1}{2} mv^2 + mg l (1 - \cos \theta)$$

which through remains constant in accordance with the *law of conservation of energy*.

$$\text{We thus, have } \frac{1}{2} mv^2 = E - mg l (1 - \cos \theta). \text{ Or, } mv^2 = 2E - 2 mg l (1 - \cos \theta)$$

Now, $v = l\omega = l d\theta/dt$, where $\omega = d\theta/dt$ is the angular velocity of the bob.

$$\text{So that, } m l^2 (d\theta/dt)^2 = 2E - 2 mg l (1 - \cos \theta)$$

$$\text{Or, } (d\theta/dt)^2 = (2/m l^2) [E - mg l (1 - \cos \theta)].$$

$$\text{And, therefore, } \frac{d\theta}{dt} = \left[\frac{2}{m l^2} \{E - mg l (1 - \cos \theta)\} \right]^{1/2}$$

Example 7.26. A simple pendulum of length 100 cm has an energy equal to 2×10^6 ergs when its, amplitude is 4 cm. Calculate its energy when (i) its length is doubled (ii) its amplitude is doubled.

Solution: We have *energy of a simple pendulum* = its max. *P.E.* = its max.

$$K.E. = \frac{1}{2} mv^2 = \frac{1}{2} m \omega^2 a^2$$

$$\text{Now, } \omega = 2\pi/T, \text{ where } T \text{ is the time-period of the pendulum} = 2\pi\sqrt{l/g}, \text{ Or, } \omega = \sqrt{g/l}.$$

Initially $l = 100$ cm. $\therefore \omega = \sqrt{g/100}$. So that, initially, with amplitude 4 cm, the energy of the pendulum = $E = \frac{1}{2} m(g/100)(4)^2 = 8 mg/100 = 100 \times 10^6$ ergs.

(i) If the length of the pendulum is doubled, we have $l = 200$ cm and

$$\therefore \omega = \sqrt{g/200}$$

$$\text{Hence, energy of the pendulum, } E' = \frac{1}{2} m (g/200)(4)^2 = mg/200$$

$$\text{So that, } E'/E = \frac{8mg/200}{8mg/100} = \frac{1}{2}, \text{ Or } E' = \frac{1}{2} E$$

Since the initial energy of the pendulum, $E = 2 \times 10^6$ ergs (given), we have

$$\text{energy of the pendulum now} = \frac{1}{2} \times 2 \times 10^6 = 10^6 = 10^6 \text{ ergs}$$

(ii) If the amplitude of the pendulum be doubled, $a = 8$ cm, its length remaining 100 cm; we have

$$\text{energy of the pendulum, } E'' = \frac{1}{2} m (g/100) (8)^2 = 32 mg/100.$$

$$\therefore E''/E = \frac{32mg/100}{8mg/100} = 4 \quad \text{Or, } E'' = 4E.$$

$$\text{Or, } E'' = 4 \times 2 \times 10^6 \text{ ergs.} = 8 \times 10^6 \text{ ergs.}$$

Thus, the energy of the pendulum is now 8×10^6 ergs.

Example 7.27. A hollow sphere is filled with water, used as pendulum bob. If water trickles out slowly through a hole made at the bottom, how will the time period be effected?

Solution. The time period of a simple pendulum given by

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of the pendulum and g the acceleration due to gravity.

When the water trickles out slowly through a hole made at the bottom of the hollow sphere used as bob of the pendulum, the mass of the bob goes on slowly decreasing. As time period does not depend upon the mass of the bob, there is no change in time period.

Example 7.28. A lift is ascending at acceleration of 3 m/s^2 . What is the period of oscillation of a simple pendulum of length one metre suspended in the lift?

Solution. The time period of a simple pendulum is given by

$$T = 2\pi\sqrt{\frac{l}{g}}$$

where l is the length of a simple pendulum and g acceleration due to gravity.

As the lift is ascending with an acceleration of 3 ms^{-2} , acceleration due to gravity

$$g' = 9.8 + 3 = 12.8 \text{ ms}^{-2}.$$

Given: length of pendulum, $l = 1 \text{ m}$.

$\therefore$ Time period of the simple pendulum suspended in the lift

$$T' = 2\pi\sqrt{\frac{l}{g'}} = 2\pi\sqrt{\frac{1}{12.8}} = 1.756 \text{ sec.}$$

Example 7.29: A uniform circular disc of radius R oscillates in a vertical plane about a horizontal axis. Find the distance of the axis of rotation from the centre for which the period is minimum. What is the value of this period? (Bombay, 2002; sagar, 2004)

Solution. The circular disc here oscillates as a compound or a physical pendulum of length l ,

whose time-period is given by $T = 2\pi\sqrt{\frac{(k^2/l) + l}{g}}$

where k its radius of gyration about an axis through its c.g., parallel to the axis of suspension.

Now, as we know, the time-period of compound pendulum is the minimum, when its length is equal to its radius of gyration about its c.g., i.e., when $l = k$.

So that,

$$T_{\min.} = 2\pi\sqrt{\frac{(k^2/k) + k}{g}} = 2\pi\sqrt{\frac{2k}{g}}$$

Since the moment of inertia of a disc about an axis perpendicular to its plane and passing through its centre is equal to $I = Mk^2 = \frac{1}{2}MR^2$, where M is the mass of the disc and R , its radius, we have $k^2 = R^2/2$ and $\therefore k = R/\sqrt{2}$.

Thus, the disc will oscillate with the minimum time-period when the distance of the axis of rotation from the centre is $R/\sqrt{2}$.

The axis of rotation from the centre is $R/\sqrt{2}$.

And the value of this minimum time-period will be

$$T_{\min} = 2\pi\sqrt{\frac{2R/\sqrt{2}}{g}} = 2\pi\sqrt{\frac{\sqrt{2}R}{g}} = 2\pi\sqrt{\frac{1.414R}{g}}.$$

Example 7.30. Three particles of the same mass m are fixed to a uniform circular hoop of mass M and radius a at the corners of an equilateral triangle. The hoop is free to move in a vertical plane about the point on the circumference opposite to one of the masses m . prove that the equivalent simple pendulum is equal in length to the diameter of the hoop.

Solution. Let the three equal masses (m) be fixed to the hoop of mass M and radius a , as shown in Fig 7.26, so as to lie at the corners of an equilateral triangle. Since all of them are equidistance from the centre, the c.g., of the triangle lies at O , the centre of the hoop. The whole arrangement is thus equivalent to a hoop of mass $(M + 3m)$ and radius a , with its centre of gravity at its centre O .

$\therefore$ $M.I.$ of this loaded hoop about an axis passing through its centre and perpendicular to its plane $= (M + 3m)a^2$ and hence, by the principle of parallel axes, its $M.I.$ about a parallel axis through its point of suspension $S = (M + 3m)a^2 + (M + 3m)a^2 = 2(M + 3m)a^2$.

$\therefore$ Its time-period about S , i.e.,

$$T = 2\pi\sqrt{\frac{1}{(M + 3m)ga}}$$

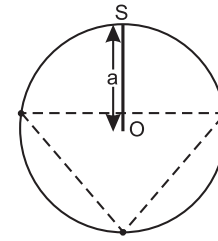


Fig. 7.26

Because, here mass of the pendulum $= (M + 3m)$ and its length $l = a$.

(See §7.7.2) where it is shown that $T = 2\pi\sqrt{I/mgl}$

Or, $T = 2\pi\sqrt{\frac{2(M + 3m)a^2}{(M + 3m)ga}} = \sqrt{\frac{2a}{g}}$, the same as that of a simple pendulum of length $2a$.

So that, the length of the equivalent simple pendulum is equal to $2a$ or the diameter of the hoop.

III–Torsional pendulum

Example 7.31 A solid sphere of radius 0.3 metre executes torsional oscillations of time-period $2\pi\sqrt{12}$ sec at the end of a suspension wire whose upper end is fixed in a rigid support. If the torque constant of the wire be 6×10^{-3} N-m/radian, calculate the mass of the sphere.

Solution. We know the $M.I.$ of a solid sphere about any diameter, i.e., about any axis passing through its centre is $\frac{2}{5}MR^2$, where M is its mass and R , its radius. We, therefore, have

$$I = \frac{2}{5}MR^2, \text{ torque constant or torque per unit twist of the suspension wire. i.e.,}$$

$$C = 6 \times 10^{-3} \text{ N-m/radian and } T = 2\pi\sqrt{12} \text{ sec.}$$

Substituting these values in the relation $T = 2\pi\sqrt{I/C}$ for the time-period of a torsional oscillation, we have

$$2\pi\sqrt{12} = 2\pi\sqrt{I/C}, \text{ whence } \sqrt{12} = \sqrt{I/C} \text{ or, } I/C = 12. \text{ Or, } I = 12C,$$

$$\text{i.e., } \frac{2}{5}MR^2 = 12 \times 6 \times 10^{-3} = 72 \times 10^{-3}$$

$$\text{Or, } MR^2 = 72 \times 10^{-3} \times 5/2 = 180 \times 10^{-3} = 18 \times 10^{-2}.$$

$$\therefore M = \frac{18 \times 10^{-2}}{R^2} = \frac{18 \times 10^{-2}}{(0.3)^2} = \frac{18 \times 10^{-2}}{9 \times 10^{-2}} = 2 \text{ kg}$$

Thus, the mass of the sphere is 2 kg.

Example 7.32. The time-period of a rod of mass 120 gm and length 10 cm, executing torsional vibration about a suspension wire passing through its centre and perpendicular to its length is 2 sec. The period of a flat plate of the shape of an equilateral triangle suspended from the suspension wire, passing through its centre of mass and perpendicular to its plane is found to be 4 sec. What is the moment of inertia of the triangular plate about the suspension wire as axis?

Solution. Let I_1 be the *M.I.* of the rod about the suspension wire as axis and I_2 , that of the triangular plate about the *same* axis. Then, if T_1 and T_2 be the respective time-periods of the two torsional oscillations, we have

$$T_1 = 2\pi\sqrt{I_1/C} \text{ and } T_2 = 2\pi\sqrt{I_2/C} \text{ So that, } T_2/T_1 = \sqrt{I_2/I_1} \text{ Or } I_2/I_1 = T_2^2/T_1^2$$

whence,

$$I_2 = (T_2^2/T_1^2) I_1 = (T_2/T_1)^2 I_1.$$

Now, *M.I.* of the rod about the suspension wire *i.e.*, $I_1 = ML^2/12 = 120 \times (10)^2/12 = 10^3 \text{ gm-cm}^2$, $T_1 = 2 \text{ sec}$ and $T_2 = 4 \text{ sec}$.

$\therefore$ *M.I.* of the triangular plate about the suspension wire, $I_2 = (4/2)^2 \times 10^3 = 4 \times 10^3 = \text{gm-cm}^2$.

Example 7.33. The balance wheel of a watch vibrates with an angular amplitude of π radians and a period of 0.5 sec. Find (a) the maximum angular speed of the wheel, (b) the angular speed of the wheel when its displacement is $\pi/2$ radians and (c) the angular acceleration of the wheel when its displacement is $\pi/4$ radians.

Solution (a) Just as in linear *S.H.M.*, so also, here in torsional vibration, we have

$$\theta = \theta_0 \sin(\omega t + \phi),$$

where θ is the angular displacement at the given instant and θ_0 , the *maximum angular displacement* or *amplitude*.

$\therefore$ *angular speed*, $d\theta/dt = \theta_0 \omega \cos(\omega t + \phi)$, the maximum value of which will clearly be $\theta_0 \omega$, *i.e.*, when $\cos(\omega t + \phi) = 1$.

Here, $\theta_0 = \pi$ radians and $\omega = 2\pi n = 2\pi \times 1/T = 2\pi \times 1/0.5 = 4\pi$.

$\therefore$ *maximum angular speed of the balance wheel* $= \theta_0 \omega = \pi \times 4\pi = 4\pi^2 = 39.48 \text{ radians/sec}$.

(b) Again, as in linear *S.H.M.*, so also here, angular speed at angular displacement $\theta = \omega\sqrt{\theta_0^2 - \theta^2}$. So that,

$$\begin{aligned} \text{Angular speed of the wheel at displacement } \pi/2 \text{ radian} &= 4\pi\sqrt{\pi^2 - (\pi/2)^2} = 4\pi\sqrt{3\pi^2/4} \\ &= 2\pi^2\sqrt{3} = 34.18 \text{ radians/sec.} \end{aligned}$$

(c) Again, acceleration of the balance wheel at displacement θ is given by $d^2\theta/dt^2 = -\omega^2\theta$. Therefore,

acceleration of the balance wheel at displacement $\theta = \pi/4$ radian is clearly equal to $-(4\pi)^2 (\pi/4) = -4\pi^3 = -124 \text{ radians/sec}^2$.

Example 7.34. A mass of 1 kg is attached to a spring of stiffness constant 16 Nm^{-1} . Find its natural frequency. (Nagpur U. 2008; H.P.U. 2001)

Solution. Here stiffness constant $e = 16 \text{ Nm}^{-1}$, mass $m = 1 \text{ kg}$

$$\therefore \text{Frequency } n = \frac{1}{2\pi} \sqrt{\frac{e}{m}} = \frac{1}{2\pi} \sqrt{\frac{16}{1}} = \frac{2}{\pi} = 0.64 \text{ Hz.}$$

Example. 7.35 A spring whose force constant is 80 N/m. hangs vertically supporting a 1 kg mass at rest. Find the distance by which the mass should be pulled down so that on being released it may pass the equilibrium position with a velocity of 1 m/s.

Solution. For a loaded spring.

$$\omega = \sqrt{\frac{c}{m}}$$

Here, $c = 80 \text{ N/m}$, $m = 1 \text{ kg}$

$$\therefore \omega = \sqrt{\frac{80}{1}} = 8.944$$

Let a be the distance by which the mass has to be pulled down (amplitude of vibration)

$$v = a\omega$$

or
$$a = \frac{v}{\omega}$$

Here $v = 1 \text{ m/s}$ and $\omega = 8.944$

$$\therefore a = \frac{1}{8.944} = 0.118 \text{ m}$$

Example 7.36. A uniform spring of force constant S is cut into two pieces whose lengths are in the ratio of 1:3. Calculate the force constant of each piece.

Solution. Here force $= F$

Increase in length $= l$

$$s = \frac{F}{l}$$

When the springs are cut in the ratio 1 : 3, increase in length for the first piece for force will be $\frac{l}{4}$, and increase in length for the second piece will be $\frac{3l}{4}$.

$$s_1 = \frac{F}{l/4} = 4 \left[\frac{F}{l} \right]$$

$$s_1 = 4s \quad \dots(i)$$

$$s_2 = \frac{F}{3l/4} = \frac{4}{3} \left[\frac{F}{l} \right]$$

$$s_2 = \frac{4}{3}s \quad \dots(ii)$$

Example 7.37. A spring when compressed by 10 cm developed a restoring force of 10 N. A body of mass 4 kg is attached to it. Calculate the compression of the spring due to the weight of the body and calculate the period of oscillation.

Solution. Here restoring force $F = 10 \text{ N}$; displacement $y = 10 \text{ cm} = 0.1 \text{ m}$

$$\therefore \text{Force constant} \quad s = \frac{F}{y} = \frac{10}{0.1} = 100 \text{ Nm}^{-1}$$

Mass attached $m = 4 \text{ kg}$

Force applied $F = 4 \text{ kg } \omega t = 4 \times 9.8 = 39.2 \text{ N}$

$$\text{Displacement} \quad y' = \frac{F'}{s} = \frac{39.2}{100} \text{ m} = 39.2 \times 10^{-2} = 39.2 \text{ cm}$$

$$\text{Time period} \quad T = 2\pi\sqrt{\frac{m}{s}} = 2\pi\sqrt{\frac{4}{100}} = \frac{2}{5}\pi = 1.26 \text{ sec.}$$

Example 7.38. A spring of force constant 1200 N/m is mounted on a horizontal table. A mass of 3 kg is attached to the free end of the spring which is pulled sideways to a distance of 2 cm and released. What is (a) the frequency of oscillation of the mass, and (b) the max. acceleration of the mass.

Solution. (a) period $T = 2\pi\sqrt{\frac{m}{s}} = 2\pi\sqrt{\frac{3}{1200}} = 2\pi\sqrt{\frac{1}{400}}$

$\therefore$ Frequency $n = \frac{1}{T} = \frac{1}{2\pi} \sqrt{400} = \frac{20}{2 \times 3.14} = 3.185 \text{ Hz.}$

(b) Acceleration $\frac{d^2x}{dt^2} = -\frac{c}{m}x = -\frac{1200}{3} \times 0.02 \quad (\because x = 0.02 \text{ m})$
 $= -8 \text{ m/s}^2$

Example 7.39. The scale of a spring balance reading from 0 to 14.5 kg is 10.25 cm long. A package suspended from the balance is found to oscillate vertically with a frequency of 2.0 oscillations per second. How much does the package weigh?

Solution. Here, clearly, a mass of 14.5 kg suspended from the spring balance extends the spring by 10.25 cm = 0.1025 m.

$\therefore$ Force constant of the spring, i.e., $C = \frac{\text{force applied}}{\text{extension}} = \frac{14.5 \times 9.81}{0.1025} = 1387.75 \text{ Nm}^{-1}.$

Now, time-period of oscillation of the spring, $T = 2\pi\sqrt{m/C}$

$\therefore T^2 = 4\pi^2 m/C$, whence, $m = CT^2/4\pi^2$.

Since $C = 1387.75 \text{ Nm}^{-1}$ and $T = 1/2 \text{ sec}$, we have

$$m = \frac{1387.75 \times 1/4}{4\pi^2} = \frac{86.73}{\pi^2}$$

The package thus weighs 8.796 kg.

Example 7.40. Two springs S_1 and S_2 are connected to a mass M , as shown in Figs. 7.27 (a) and (b), with the mass lying on a frictionless surface. If the respective force constants of the springs be C_1 and C_2 , obtain expression for the time-periods of the mass in the two cases. What are their electrical analogues?

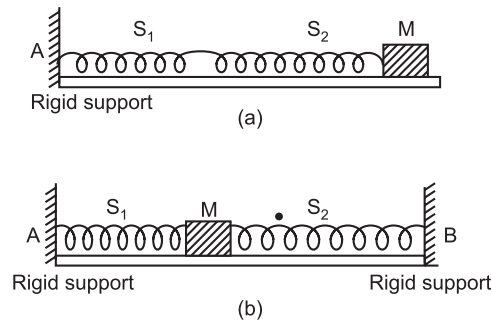


Fig. 7.27

Solution. (a) If x_1 and x_2 the extensions produced in the two springs respectively as the mass (M) is displaced outwards, the total extension $x = x_1 + x_2$. Since the restoring force due to each spring is the same, F , say, we have $F = -C_1x_1 = -C_2x_2$; so that, $x_1 = -F/C_1$ and $x_2 = -F/C_2$.

$$\therefore x = x_1 + x_2 = -F \left(\frac{1}{C_1} + \frac{1}{C_2} \right) = -F \left(\frac{C_1 + C_2}{C_1 C_2} \right) \text{ whence } F = - \left(\frac{C_1 C_2}{C_1 + C_2} \right) x$$

indicating that the effective force constant of the combination of the two springs, say, $C = \left(\frac{C_1 C_2}{C_1 + C_2} \right)$.

$$\begin{aligned} \therefore \text{Time-period of the oscillating mass, } T &= 2\pi \sqrt{\frac{M}{C}} = 2\pi \sqrt{\frac{M}{(C_1 C_2)(C_1 + C_2)}} \\ &= 2\pi \sqrt{\frac{(C_1 + C_2)M}{C_1 C_2}} \end{aligned}$$

(b) In this case, if the mass (M) be displaced to one side or the other, through a distance x , say, one springs gets extended and the other compressed, both exerting a restoring force on the mass in the same direction, tending to bring it back to its original position. Thus, if F_1 and F_2 be the restoring forces due to the two springs respectively, we have resulting restoring force $F = F_1 + F_2 = -C_1 x - C_2 x = -(C_1 + C_2)x = Cx$, indicating that the effective force constant C of the combination of the two springs is now equal to $(C_1 + C_2)$, the sum of their individual force constants.

The time-period of the oscillating mass is, therefore, $T = 2\pi \sqrt{M/C} = 2\pi \sqrt{M/(C_1 + C_2)}$.

The electrical analogues of the two systems are a parallel and a series arrangement of two capacitors respectively.

Example 7.41. A uniform spring of normal length l has a force constant C . It is cut into two pieces of lengths l_1 and l_2 such that $l_1 = nl_2$ where n is an integer. What are the force constants C_1 and C_2 of the two pieces respectively in terms of n and C ?

Solution. Here, clearly, $l = l_1 + l_2$. Or, since $l_1 = nl_2$ and, therefore, $l_2 = l_1/n$, we have

$$l = l_1 + l_1/n = l_1 (1 + 1/n) = l_1 (n + 1)/n \dots (i) \text{ Also, } l = nl_2 + l_2 = l_2(n + 1) \dots (ii)$$

Now, for a given spring, its force constant is inversely proportional to its length, i.e., $C \propto 1/l$ or $Cl = a$ constant. We, therefore, have

$$C_1 l_1 = Cl = C l_1 (n + 1)/n, \text{ whence, } C_1 = C(n + 1)/n,$$

$$\text{Similarly } C_2 l_2 = Cl = C l_2 (n + 1), \text{ whence, } C_2 = C(n + 1).$$

Thus, the force constants of the two pieces of the spring are respectively

$$C_1 = C(n + 1)/n \quad \text{and} \quad C_2 = C(n + 1).$$

Example 7.42. Show that if a given body oscillates at the end of a spring whose mass is not negligible, its time-period is the same as though its mass were increased by one-third of that of the spring.

Solution. Let a uniform spring of length l and mass m be suspended from a rigid support and carry a mass M at its lower free end, (Fig. 7.28). Then, clearly, mass per unit length of the spring $= m/l$, and therefore mass of an element of length ds of it at a distance s from the upper fixed end $= (m/l) ds$.

If at a given instant, the velocity of the lower end of the spring (and therefore also of mass M) be v , the velocity of the elements ds at a distance s from the fixed end $= (v/l)s$.

$$\therefore \text{K.E. of the element} = \frac{1}{2} \left(\frac{m}{l} ds \right) \left(\frac{vs}{l} \right)^2 = \frac{1}{2} \frac{mv^2}{l^3} s^2 ds$$

Hence K.E. of the whole spring at the given instant.

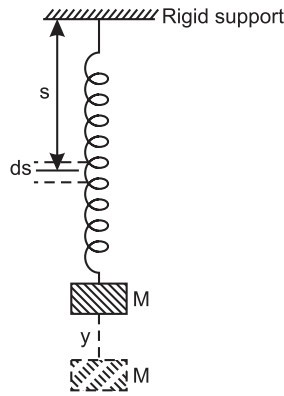


Fig. 7.28

$$= \int_0^l \frac{1}{2} \frac{mv^2}{l^3} s^2 ds = \frac{1}{2} \frac{mv^2}{l^3} \cdot \frac{l^3}{3} = \frac{mv^2}{6}$$

And *K.E.* of the suspended mass (*M*) = $\frac{1}{2} Mv^2$.

$$\therefore \text{K.E. of the system, as a whole} = \frac{1}{2} Mv^2 + \frac{mv^2}{6} = \frac{1}{2} \left(M + \frac{m}{3} \right) v^2$$

If at the instant considered, *y* be the displacement of mass *M* from its mean or equilibrium position, we have, substituting *dy/dt* for *v*.

$$\text{K.E. of the system, as a whole} = \frac{1}{2} \left(M + \frac{m}{3} \right) \left(\frac{dy}{dt} \right)^2 \quad \dots(i)$$

If *C* be the force constant of the spring, we have restoring force set up in the spring = $-Cy$ and, therefore,

$$\text{P.E. of the system} = \int_0^y Cy \, dy = \frac{1}{2} Cy^2 \quad \dots(ii)$$

$$\therefore \text{Total energy of the system, } E = \text{K.E.} + \text{P.E.} = \frac{1}{2} \left(M + \frac{m}{3} \right) \left(\frac{dy}{dt} \right)^2 + \frac{1}{2} Cy^2.$$

Since the total energy of the system must remain conserved, we have

$$\frac{dE}{dy} = 0, \text{ i.e., } \frac{d}{dy} \left[\frac{1}{2} \left(M + \frac{m}{3} \right) \left(\frac{dy}{dt} \right)^2 + \frac{1}{2} Cy^2 \right] = 0$$

$$\text{i.e., } \left(M + \frac{m}{3} \right) \frac{d^2y}{dt^2} + Cy = 0. \quad \text{Or, } \frac{d^2y}{dt^2} = -\frac{C}{(M + m/3)} y,$$

Indicating that the spring executes a *S.H.M.* of time-period

$$T = 2\pi \sqrt{1 / \frac{C}{M + m/3}} = 2\pi \sqrt{\frac{M + m/3}{C}}$$

which is clearly the same as though the value of the suspended mass has been increased by one-third of the mass of the spring.

Example 7.43. An elastic cord of length l is suspended from a rigid support, A mass m , attached to its lower end, stretches it by a length a . An additional mass m' , attached to it, stretches it further by a length b . When the system has come to rest, mass m' gets detached. Show that the distance of mass m from the upper end of the cord is, at any instant t , given by $l + a + b \cos \sqrt{g/a} \cdot t$.

Solution. In Fig. 7.29 let SA be the elastic cord of length l , suspended from a rigid support at S . When a mass m is attached to its lower end, let it extend by a length $AB = a$ and, on attaching another mass m' to it, further extend by a length $BC = b$.

When the system has come to rest at the point C , mass m' is suddenly detached, so that mass m starts oscillating up and down about its mean position B , the equation of motion of the mass being $d^2y/dt^2 = -(C/m)y$, where C is the force constant of the cord.

Clearly, before the mass m was attached to it, the position of the lower end of the cord was at A . The addition of mass m , i.e., the application of a force mg to it, brought it down to B or produced in it an extension $AB = a$. So that, $mg = Ca$, whence, $C = mg/a$. And, therefore, $d^2y/dt^2 = -(g/a)y$, i.e. mass m executes a *S.H.M.* about B as its mean or equilibrium position, its displacement at any instant t being given by $y = a \sin (\omega t + \phi)$, where a is the amplitude of the oscillation and $\omega^2 = g/a$ or $\omega = \sqrt{g/a}$.

$\therefore$ velocity of the mass at the given instant $= dy/dt = \omega \sqrt{a^2 - y^2}$.

At $t = 0$, i.e., at the instant that the mass m' just gets detached, $y = b$ and $dy/dt = 0$. So that,

$$0 = \omega \sqrt{a^2 - b^2} \text{ . Or, } a^2 - b^2 = 0 \text{ Or, } a = b$$

Substituting these values in the expression for y above we, have

$$b = b \sin \phi, \text{ whence, } \sin \phi = 1 \text{ or } \phi = \pi/2.$$

We therefore have $y = b \sin (\omega t + \pi/2) = b \cos \omega t = b \cos \sqrt{g/a} t$

$\therefore$ Distance of mass m from the point of suspension (or the upper end of the cord) at instant

$$t = l + a + y = l + a + b \cos \sqrt{g/a} t$$

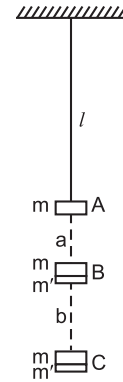


Fig. 7.29

IV—Hemholtz resonator—L-C Circuit

Example 7.44. A spherical Helmholtz resonator of capacity 10 litres has a narrow neck of length 1 cm and radius 1 cm. If the velocity of sound in air be 340 m/sec, calculate the frequency and the wavelength of the note to which it will sharply respond.

Solution. We know that the frequency of the note to which a spherical resonator responds is given by $n = (v/2\pi) \sqrt{\alpha/Vl}$ [(§ 7.6 (5))], where v is the velocity of sound in air, l , the length and α , the area of cross-section of the neck and V , the volume (or capacity) of the resonator.

Here, $v = 340 \text{ m/sec} = 34000 \text{ cm/sec}$, $\alpha = \pi (1)^2 = \pi \text{ sq cm}$, $l = 1 \text{ cm}$ and $V = 10 \text{ litres} = 10000 \text{ c.c.}$

$$\therefore \text{ frequency of the note, } n = \frac{34000}{2\pi} \sqrt{\frac{\pi}{10000 \times 1}} = 170 / \sqrt{\pi} = 95.92/\text{sec}$$

And, wavelength of the note, $\lambda = v/n = 34000/95.92 = 345.5 \text{ cm}$.

Example 7.45. Calculate the frequency of an L-C circuit in which the inductance and capacitance are respectively 5 millihenry and $2\mu\text{F}$. If the maximum potential difference across the capacitor be 10 volts, what is the energy of the system?

Solution. For the frequency of the oscillations in an L - C circuits, we have the relation

$$n = \frac{1}{2\pi\sqrt{LC}}$$

where L and C are the inductance and capacitance in the circuit respectively.

Here, $L = 5$ milli henry $= 5 \times 10^{-3}$ henry and $C = 2\mu F = 2 \times 10^{-6}$ farad.

$$\therefore \text{Frequency of the } L\text{-}C \text{ circuit, } n = \frac{1}{2\pi\sqrt{5 \times 10^{-3} \times 2 \times 10^{-6}}} = \frac{10^4}{2\pi} = 1592/\text{sec}$$

$$\text{And, with } V = 10 \text{ volts, energy of the system} = \frac{1}{2} CV^2 = \frac{1}{2} \times 2 \times 10^{-6} \times (10)^2 = 10^{-4} \text{ Joule}$$

V-Two-body harmonic oscillator

Example 7. 46. (a) What is the reduced mass of each of the following diatomic molecules: O_2 , HCl , CO ? Express your answers in atomic mass units, the mass of a hydrogen atom being approximately 1.00 amu.

(b) An HCl molecule is known to vibrate at a fundamental frequency of 8.7×10^{13} cycles/sec. What is the effective 'force constant' C for the coupling forces between the atoms?

Solution. (a) We know that a diatomic molecule behaves like a system of two masses connected by a small massless spring. So that, its reduced mass is given by

$$m = \frac{m_1 \times m_2}{m_1 + m_2}$$

Now, in the case of O_2 , $m_1 = 16$ amu and $m_2 = 16$ amu

$$\therefore \mu = \frac{16 \times 16}{16 + 16} = \frac{256}{32} = 8 \text{ amu}$$

In the case of HCl , $m_1 = 1$ amu and $m_2 = 35.5$ amu.

$$\mu = \frac{1 \times 35.5}{1 + 35.5} = \frac{35.5}{36.5} = 0.97 \text{ amu.}$$

And, in the case of CO , $m_1 = 12$ amu and $m_2 = 16$ amu

$$\therefore \mu = \frac{12 \times 16}{12 + 16} = \frac{192}{28} = 6.857 \text{ amu}$$

(b) The frequency of a two-body harmonic oscillator, as we know, is given by $n = \frac{1}{2\pi} \sqrt{\frac{C}{\mu}}$ where C is the 'force constant' for the coupling forces and μ , the reduced mass of the system.

Here, the reduced mass μ for the HCl molecule $= 0.97$ amu [see above under (a)], i.e., equal to $0.97 \times \text{mass of a hydrogen atom} = 0.97 \times 1.67 \times 10^{-24}$ gm and $n = 8.7 \times 10^{13}$ cycles/sec So that,

$$8.7 \times 10^{13} = \frac{1}{2\pi} \sqrt{\frac{C}{0.97 \times 1.67 \times 10^{-24}}}, \text{ whence, } C = 4\pi^2 (8.7 \times 10^{13})^2 \times 0.97 \times 1.67 \times 10^{-24}$$

$$= 485200 \text{ dyne/cm} = 485 \text{ N/m.}$$

Example 7.47. Two discs D_1 and D_2 of masses 1 kg and 2 kg respectively are connected to the two ends of a massless spring and placed on a table, as shown in Fig. 7.30, with the spring vertical and the heavier disc (D_2) resting on the table. Calculate the frequency of oscillation of the system, if the force constant of the spring be 10^6 dynes/cm.

(b) After the system has come to rest, the table is suddenly removed from under disc D_2 . Calculate (i) the frequency of the system, (ii) the instantaneous acceleration of the two discs of the centre of mass of the system.

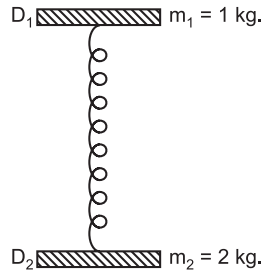


Fig. 7.30

Solution. (a) Obviously, here, the system functions as an ordinary one-body harmonic oscillator, with the lower end of the spring fixed and the upper end carrying disc D_1 of mass $m_1 = 1$ kg.

If, therefore, n_1 be the frequency of the system, we have

$$n_1 = \frac{1}{2\pi} \sqrt{\frac{C}{m_1}} = \frac{1}{2\pi} \sqrt{\frac{10^6}{1000}} = 5.025 \text{ or } 5.0/\text{sec}.$$

(b) (i) When the table is suddenly removed from under disc D_2 , the system oscillates as a two-body harmonic oscillator, with both the discs oscillating simultaneously,

If n_2 be the frequency of the system, we have $n_2 = \frac{1}{2\pi} \sqrt{\frac{C}{\mu}}$

Where μ is the reduced mass of the system, equal to $\frac{m_1 m_2}{m_1 + m_2} = \frac{1 \times 2}{1 + 2} = \frac{2}{3}$ kg

$$\therefore n_2 = \frac{1}{2\pi} \sqrt{\frac{10^6}{(2/3) \times 1000}} = \frac{1}{2\pi} \sqrt{\frac{3 \times 10^6}{2000}} = \frac{1}{2\pi} \sqrt{\frac{3 \times 10^3}{2}} = 6.165 \text{ sec}.$$

(ii) Clearly, with the sudden removal of the table from under disc D_2 , the latter is subjected to an instantaneous force equal to $(1 \text{ kg} + 2 \text{ kg})g = 3 \text{ kg wt} = 3 \times 10^3 \times g$ dynes.

If its acceleration be a , the force on it is clearly $m_2 a = 2 \times 10^3 a = 3 \times 10^3 \times g$, whence, $a = 3g/2$.

Thus, the instantaneous acceleration of disc $D_2 = \frac{3}{2} g \text{ cm/sec}^2$

As to acceleration of disc D_1 , we note that the system was at rest before removal of the table from under disc D_2 . Since disc D_1 was at rest on top of the spring, its weight was being supported by the elastic reaction of the spring and it continues to be so supported at the instant the table is just removed. So that, the instantaneous acceleration of disc D_1 is zero.

The acceleration of the centre of mass of the system will naturally be the same as the of a body falling freely under the action of gravity, viz., g . This may be shown as follows:

If y be the instantaneous displacement of the centre of mass (taking y -direction along the vertical) and y_1 and y_2 , the corresponding displacements of the two discs of masses m_1 and m_2 respectively, we have $(m_1 + m_2)y = m_1 y_1 + m_2 y_2$. Differentiating with respect to t , we have $(m_1 + m_2) \frac{d^2 y}{dt^2} = m_1 \frac{d^2 y_1}{dt^2} + m_2 \frac{d^2 y_2}{dt^2}$, where, as we know, $\frac{d^2 y_1}{dt^2} = 0$ and $\frac{d^2 y_2}{dt^2} = 3g/2$.

If, therefore, the acceleration of the centre of mass, i.e., $\frac{d^2 y}{dt^2}$ be $a' \text{ cm/sec}^2$, we have

$$(1 + 2) a' = 0 + 2 \times \frac{3}{2} g. \text{ Or, } 3a' = 3g, \text{ whence, } a' = g,$$

i.e., the acceleration of the centre of mass is g .

Example. 7.48. The angular vibration frequency of the carbon monoxide molecule (CO) is $0.6 \times 10^{15}/\text{sec}$. Calculate (i) the reduced mass of the molecule in gms, (ii) the reduced mass of the molecule in gms, (ii) the force constant for stretching the molecule and (iii) the work done in stretching the molecule by 0.5 Angstrom unit.

Solution. (i) Clearly, *reduced mass of the CO molecule, i.e.,* $\mu = \frac{m_1 m_2}{m_1 + m_2} = \frac{12 \times 16}{12 + 16}$
 $= \frac{192}{28} \text{ amu} = \frac{192}{28} \times \text{mass of hydrogen atom} = \frac{192}{28} \times 1.67 \times 10^{-24}$
 $= 1.145 \times 10^{-23} \text{ gm.}$

(ii) As we know, the vibrational frequency of the molecule is given by

$$n = (1/2\pi) \sqrt{C/\mu} \quad \text{Or} \quad 2\pi n = \sqrt{C/\mu}$$

Now, $\sqrt{C/\mu} = \omega$, whence, $C = \mu \omega^2$, where, $\mu = 1.145 \times 10^{-23} \text{ gm}$ and $\omega = 0.6 \times 10^{15} \text{ (given)}$.

$$\therefore \text{Force constant } C = (1.145 \times 10^{-23})(0.6 \times 10^{15})^2 = 1.145 \times 0.36 \times 10^7 = 4.122 \times 10^6 \text{ dynes/cm.}$$

(iii) Work done in stretching the molecule by $(r - r_0)$ is given by $\frac{1}{2} C(r - r_0)^2$ [§7.9]

Here, $C = 4.122 \times 10^6 \text{ dynes/cm}$ and $(r - r_0) = 0.5 \text{ \AA} = 0.5 \times 10^{-8} \text{ cm}$.

$$\therefore \text{work done in stretching the molecule through } 0.5 \text{ \AA} = \frac{1}{2} \times 4.122 \times 10^6 (0.5 \times 10^{-8})^2 = 2.061 \times 10^6 \times 0.25 \times 10^{-16} = 5.15 \times 10^{-11} \text{ erg.}$$

Example 7.49. HCl gas absorbs light of wave length λ , thereby causing the transition in the vibrational energy level from $n = 0$ to $n = 1$. Calculate the force constant of the HCl molecule and the wavelength of the light absorbed if the vibrational energy levels be separated by 0.36 eV.

Solution. Here transition in the energy level occurring from $n = 0$ to $n = 1$, i.e., with n changing by 1, the energy absorbed $= h\nu = h\nu_0$, where h is the planck's constant $= 6.6 \times 10^{-27} \text{ erg-sec}$ and ν = frequency of the light absorbed.

Since $h\nu = 0.36 \text{ eV} = 0.36 \times 1.6 \times 10^{-12} \text{ ergs}$, we have

$$\nu = \nu_0 = 0.36 \times 1.6 \times 10^{-12} / 6.6 \times 10^{-27} = 8.728 \times 10^{13} / \text{sec.}$$

Now, $\nu = \nu_0 = (1/2\pi) \sqrt{C/\mu}$, where C is the *force constant* and μ , the *reduced mass of the molecule*.

$$\text{Clearly, } \mu = \frac{1 \times 35.5}{1 + 35.5} \times 1.67 \times 10^{-24} = 0.97 \times 1.67 \times 10^{-24} \text{ gm. We, therefore, have}$$

$$8.728 \times 10^{13} = \frac{1}{2\pi} \sqrt{\frac{C}{0.97 \times 1.67 \times 10^{-24}}}$$

whence,

$$C = (8.728 \times 10^{13})^2 \times 4\pi^2 \times 0.97 \times 1.67 \times 10^{-24}.$$

Or, force constant $C = (8.728)^2 \times 0.97 \times 1.67 \times 4\pi^2 \times 10^2 \times 4.87 \times 10^5 \text{ dynes/cm.}$

$$\text{And, wavelength of light absorbed, } \lambda = \frac{\text{velocity of light } (c)}{\text{frequency } (\nu)} = \frac{3 \times 10^{10}}{8.728 \times 10^{13}}$$

$$= 3.438 \times 10^{-4} \text{ cm} = 34380 \text{ \AA.}$$

Example 7.50. (a) If the transition from one energy level to an adjacent one involves an energy change of 0.5 eV in the case of hydrogen, calculate the energy change involved in a similar transition in the case of HCl.

(b) While comparing the potential energy curve of two diatomic molecules A and B, it is observed that the potential well of the latter is narrower. What conclusions can be drawn from this? Can an estimate of the binding energies of the two molecules be made from these curves?

Solution. (a) The energy, change involved in transition between adjacent energy levels is, as we know, $h\nu$, where $h\nu = h\nu_0 = (h/2\pi) \sqrt{C/\mu}$, with C as the force constant of the molecule and μ , its reduced mass.

Now, in the case of hydrogen, $h\nu = 0.5$ eV, and, taking the mass of an atom of hydrogen to be m amu, $\mu = m \times m/(m + m) = m/2$ amu. And, therefore,

$$\therefore 0.5 = \frac{h}{2\pi} \sqrt{\frac{C}{m/2}} = \frac{h}{2\pi} \sqrt{\frac{2C}{m}}$$

In the case of HCl ,
$$\mu = \frac{m \times 35.5}{m + 35.5} = \frac{35.5m}{36.5} = 0.97 \text{ amu.}$$

If, therefore the energy change involved in transition between adjacent energy levels be E , we have

$$E = \frac{h}{2\pi} \sqrt{\frac{C}{0.97m}}$$

$\therefore$ dividing relation (ii) by (i), we have

$$\frac{E}{0.5} = \sqrt{\frac{C}{0.97m} \times \frac{m}{2C}} = \sqrt{\frac{1}{2 \times 0.97}} \text{ or, } E = \sqrt{\frac{1}{2 \times 0.97}} \times 0.5 = 0.359$$

Thus, the energy change involved in transition between adjacent energy levels in the case of HCl is 0.36 electron volts.

(b) A narrower potential well in the case of molecule B signifies a higher value of d^2U/dr^2 and, in consequence, a larger value of the force constant C . So that, for the same value of the reduced mass (μ) in the two cases the vibrational frequency n [equal to $(1/2\pi)\sqrt{C/\mu}$] will be higher in the case of molecule B .

An estimate of the binding energies of the two molecules can be made by observing the minima of the potential energies in the two cases. For, if we ignore the zero-point energy, the numerical values of the minimum potential energy and the binding energy of a molecule are the same.

Example 7.51. The vibrational frequency of H^1Cl^{35} molecule is 8990×10^{10} cycles/sec. Deduce the vibrational frequency of H^2Cl^{35} molecule, assuming the force constant to be the same in the two cases. (Agru (supp), 2002)

Solution. Since the vibrational frequency of a diatomic molecule is given by $(1/2\pi)\sqrt{C/\mu}$, it is clear that for the same value of the force constant (C), $n \propto \sqrt{1/\mu}$.

If therefore, μ_1 and μ_2 , be the reduced masses of the HCl molecule in the two cases, and n_1 and n_2 , their vibrational frequencies respectively, we have

$$\frac{n_2}{n_1} = \sqrt{\frac{\mu_1}{\mu_2}}, \text{ whence, } n_2 = \sqrt{\frac{\mu_1}{\mu_2}} n_1 \quad [C \text{ being the same for both.}]$$

Now, $\mu_1 = 1 \times 35/(1 + 35) = 35/36$, $\mu_2 = 2 \times 35/(2 + 35) = 70/37$ and

$$n_1 = 8990 \times 10^{10} \text{ c.p.s. (given)}$$

$$\therefore n_2 = \sqrt{\frac{35}{36} \times \frac{37}{70}} \times 8990 \times 10^{10} = \sqrt{\frac{37}{72}} \times 8990 \times 10^{10} = 6445 \times 10 \text{ c.p.s.}$$

Thus, the vibrational frequency of the H^2C^{35} molecule = 6445×10^{10} cycles/sec.

VI—Anharmonic Oscillator

Example 7.52. Calculate the percentage change in the time-period of a simple pendulum if the angular amplitude of the pendulum be (i) 30° , (ii) 60° .

Solution. Let the length of the simple pendulum be l . Then, its true time-period (for infinitely small amplitudes) is $T_0 = 2\pi\sqrt{l/g}$

If its amplitude be θ_1 , the time-period $T = T_0 (1 + \theta_1^2/16)$...[see 7.13]

$$\therefore \text{Percentage change in the time-period} = \left(\frac{T - T_0}{T} \right) \times 100 = \frac{\theta_1^2}{16} \times 100$$

Now, in case (i) $\theta_1 = 30^\circ = \pi/6$ radian. So that, we have

$$\text{percentage change in the time-period} = \frac{(\pi/6)^2}{16} \times 100 = \frac{25\pi^2}{144} = 1.7$$

And, in case (ii) $\theta_2 = 60^\circ = \pi/3$ radian and we, therefore, have

$$\text{Percentage change in the time-period} = \frac{(\pi/3)^2}{16} \times 100 = \frac{25\pi^2}{36} = 6.85$$

Example 7.53. An oscillating particle of mass 1 gm has potential to $500x^2 + 200x^4$ ergs what is its frequency for small oscillations? What will be its frequency if the amplitude of its fundamental component be 3 cm? Also calculate the frequency and the amplitude of the first overtone in the latter case.

We have here $U = 5000x^2 + 200x^4$ ergs $\therefore F = -dU/dx = -10,000x - 800x^3$ dynes.

For small oscillations, the second term in x^3 may be neglected. So that,

$$F = -10,000x \text{ dynes.}$$

And $\therefore$ acceleration, $d^2x/dt^2 = -10,000x/1 = -10,000x = -(C/m)x = -\omega_0^2x$,

$$\text{i.e.,} \quad \omega_0^2 = 10,000 \text{ and therefore, } \omega_0 = \sqrt{10,000} = 100$$

$$\therefore \text{ frequency for small oscillations, } n_o = \omega_o/2\pi = 100/2\pi = 15.92/\text{sec}$$

Now, $\omega = \omega_0 + \frac{3}{8} \beta \frac{B_1^2}{\omega_0^2}$ (Equation VII, §7.12). And here, $\beta = 800$, $B_1 = 3$ cm.

So that,

$$\omega_0 = 100 + \frac{3}{8} \times 800 \times \frac{(3)^2}{100} = 127$$

$$\therefore \text{ frequency, when amplitude of fundamental component is 3 cm is } n = \omega/2\pi,$$

$$\text{Whence,} \quad n = 127/2\pi = 20.21/\text{sec.}$$

Clearly, the first overtone here has thrice the frequency of the fundamental component, i.e., a frequency n' , say, $= 3\omega/2\pi = 20.21 \times 3 = 60.63/\text{sec.}$

$$\text{and its amplitude is } \frac{\beta B_1^3}{32 \omega_0^2} = \frac{800 \times (3)^2}{32 \times (100)^2} = \frac{27}{400} = 0.0675 \text{ cm.}$$

EXERCISE

1. Differentiate between simple harmonic motion and oscillatory motion. Define simple harmonic motion. Discuss potential well. (Nagpur U. 2009, H.P.U. 2001, Bang. U. 2000, Gauhati. U. 2000)
2. Explain how interaction of inertia and elasticity account for simple harmonic motion (P.U. 2002)
3. Enumerate the characteristics of simple harmonic motion. (Purvanchal U. 2005)
4. Derive a general differential equation of motion of a simple harmonic oscillator and obtain its various solutions. (Nagpur U. 2006)
5. (a) Find an expression for the velocity of a simple harmonic oscillator. (Nagpur U. 2008, 2003, Bang. U. 2000)
(b) Velocity of simple harmonic oscillator at any time t leads the displacement by an angle $\pi/2$ radian. Explain why? (Nagpur U. 2003)
6. (a) Find an expression for the acceleration of a simple harmonic oscillator. (Nagpur U. 2008)
(b) Show that for the body executing simple harmonic motion, the acceleration leads the velocity by $\pi/2$ and displacement by π .

7. Show that the phase relation between the displacement, velocity and acceleration diagrammatically, given displacement $y = A \sin (\omega t + \phi)$.
8. What is meant by a harmonic oscillator? Obtain expressions for (i) its displacement and velocity at a given instant, (ii) time-period and frequency.
9. What are the important properties of a simple harmonic oscillator? Why is a simple harmonic motion considered to be a fundamental periodic motion? Show that the time-period of a simple harmonic oscillation is given by

$$T = 2\pi \sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

10. Explain the vibration of the kinetic and potential energies of a simple harmonic oscillator. Illustrate your answer with suitable graphs.
11. Show that for a simple harmonic oscillator, mechanical energy remains conserved and that its energy is, on an average, half kinetic and half potential in form. At what particular displacement is this exactly so? And what is the ratio between its kinetic and potential energies at a displacement equal to half its amplitude?

[Ans. At displacement $a\sqrt{2}$, where a is the amplitude. 3:1.]

12. It is said that all oscillations, if their amplitudes be infinitely small, may be regarded as simple harmonic. Do you agree? If so, explain why. Is it possible to have an oscillator for which even with small amplitudes, the oscillations are not simple harmonic?
13. Solve the differential equation $d^2x/dt^2 + \omega^2x = 0$ to obtain the expression $x = a \sin (\omega t + \phi)$ for the displacement of a particle executing S.H.M.
14. Show that a simple harmonic motion may be expressed as either a sine or a cosine function, there being only a difference of initial phase in the two cases.
15. (a) A particle of mass 10 gm lies in a potential field $V = 32x^2 + 200$ ergs/gm. What is its frequency of oscillation. [Ans. 1.27 sec^{-1} .]
 (b) A block executes a S.H.M. on a horizontal surface. If its frequency be 2 oscillations per second and the coefficient of static friction between the block and the surface, 0.50, how great can the amplitude be if the block does not slip on the surface?
16. A quantity of gas is enclosed in a cylinder, fitted with a smooth, heavy piston. The axis of the cylinder is vertical. The piston is thrust downwards to compress the gas and then let go. Is the ensuing motion of the piston a simple harmonic motion? If so, what is its time-period?

Ans. Yes; $T = \frac{2\pi}{a} \sqrt{\frac{Vm}{P}}$, where V and P are the initial volume and pressure of the gas respectively, a , the area of cross-section of the piston and m , its mass.

17. Prove that in a simple harmonic motion, the average potential energy equals the average kinetic energy $= \frac{1}{4} Ca^2$ when the average is taken with respect to time over one period of the motion, and that, if the average is taken with respect to position over one cycle, it is equal to $\frac{1}{2} Ca^2$. Explain why.
18. (a). A particle executing a S.H.M. has a maximum displacement of 4 cm and its acceleration at a distance of 1 cm from its mean position is 3 cm/sec^2 . What will its velocity be when it is at a distance of 2 cm from its mean position? [Ans. 6 cm/sec .]
 (b) A particle, moving in a straight line with S.H.M. of period $2\pi/\omega$ about a fixed point O , has a velocity $\sqrt{3b\omega}$ when at a distance b from O . shown that its amplitude is $2b$ and that it will cover that rest of its distance in time $\pi/3\omega$.
19. A particle executes linear harmonic motion about the point $x = 0$; at $t = 0$, it has displacement $x = 0.37 \text{ cm}$ and zero velocity. If the frequency of the motion is $0.25/\text{sec}$, determine (a) the period, (b) the amplitude, (c) the maximum speed and (d) the maximum acceleration.

[Ans. (a) 4.0 sec, (b) 0.37 cm, (c) 0.58 cm/sec, (d) 0.91 cm/sec².]

20. A light string is suspended vertically from a point and carries a heavy mass at its lower free end which stretches it through distance l cm. Show that the vertical oscillations of the system are simple harmonic in nature and of a time-period equal to that of a simple pendulum of length l cm.
21. Two particles execute *S.H.M.* of the same amplitude and frequency along adjacent straight lines. They pass one another when going in opposite directions each time their displacement is half their amplitude. What is the phase difference between them? [Ans. $2\pi/3$]
22. Are all periodic motions simple harmonic? Is the reverse true? Explain. (P.U. 2004 pbi. 2003)
23. A particle is dropped down in a deep hole which extends to the centre of the earth. Calculate its velocity at a depth of 1 kilometer from the surface of the earth. Assume that $g = 1000 \text{ cm/sec}^2$ and radius of the earth = 6400 kilometers. (Gorakhpur, 2005)
[Ans. $1.414 \times 10^4 \text{ cm/sec.}$]
24. The displacement of an oscillating particle at an instant t is given by $x = a \cos \omega t + b \sin \omega t$. Show that it is executing a *S.H.M.*
If $a = 5 \text{ cm}$, $b = 12 \text{ cm}$ and $\omega = 4 \text{ radians/sec}$, calculate (i) the amplitude, (ii) the time-period, (iii) the maximum velocity and (iv) the maximum acceleration of the particle.
[Ans. (i) 13 cm, (ii) 1.57 cm (iii) 52 cm/sec and (iv) 208 cm/sec².]
25. A smooth, straight tunnel is bored through the moon along any line takes at random. Show that if a smooth ball is dropped inside the tunnel, it executes *S.H.M.* In how much time will it reach one end of the tunnel from the other end? Given: mass of the moon = $7.35 \times 10^{25} \text{ gm}$, radius of the moon = $1.75 \times 10^8 \text{ cm}$ and $G = 6.67 \times 10^{-8} \text{ dynes-cm}^2 \text{ gm}^{-2}$. [Ans. 54 min. 44 sec.]
[Hint. See worked example 7.]
26. The position of a particle executing *S.H.M.* along the x axis are $x = A$ and $x = B$ at time t and time $2t$ respectively. Show that its period of oscillation is given by $T = 2\pi/\cos^{-1}(B/2A)$.
[Hint. Here, $A = a \sin \omega t$ and $B = a \sin 2\omega t = 2a \sin \omega t \cos \omega t$. If at $t = 0$, $x = 0$, we have $\cos \omega t = B/2A$, i.e., $\omega = \cos^{-1}(B/2A)/t$. $\therefore T = 2\pi/\omega = 2\pi t/\cos^{-1}(B/2A)$.]
27. If the potential energy curve is $U = -A/x + B/x^2$, where A and B are positive constants, obtain the position of stable equilibrium and the force constant for small oscillations. What is the time-period for such oscillations?
[Ans. $2B/A$; $A^4/8B^3$; $4\pi B\sqrt{2mB/A^2}$]
28. The ends of one of the prongs of a tuning fork, which executes *S.H.M.* of frequency 1000/sec, has an amplitude of 0.40 nm. Calculate (a) its maximum acceleration and maximum speed (b) its acceleration and its speed when it has a displacement equal to half its amplitude.
[Ans. (a) $1.6 \times 10^4 \text{ m/sec}^2$; 2.5 m/sec. (b) $7.9 \times 10^3 \text{ m/sec}^2$; 2.2 m/sec.]

II—Pendulums—(I)—Simple pendulum

29. Show that a simple pendulum executes simple harmonic motion find its period.
30. Is it really possible to construct a truly simple pendulum? show that as the angular amplitude of the pendulum approaches 180° , its time-period approaches infinity.
31. What are the drawbacks of a simple pendulum? The time-period of a simple pendulum for infinitely small amplitudes is given by $T = 2\pi\sqrt{l/g}$. What will it be for a infinite amplitude?
32. A simple pendulum of length 100 cm has energy equal to 0.3 joule when its amplitude is 2 cm. what will be its energy if (i) its length is increased to 150 cm, (ii) its amplitude is increased to 3 cm?
[Ans. (i) 0.2 joule, (ii) 0.675 joule]
33. A hollow sphere is filled with water through a small hole in it. It is hung by a long thread and, as the water slowly flows out of the hole at the bottom, the period of oscillations first increases and then decreases. Why is it so?
34. Show that the time-period of a simple pendulum at depth h below the earth's surface is inversely proportional to $\sqrt{R-h}$, where R is the radius of the earth. Calculate its time-period when $h = R/2$.
[Ans. 28 sec.]

35. What is a compound pendulum? How does a compound pendulum differ from a simple pendulum? Obtain an expression for its time-period and mention its points of superiority over a simple pendulum.
(Nagpur U. 2009, 2005, 2006; P.U. 2004, 2003)
36. How is the time period of a pendulum affected when its points of suspension is (a) moved horizontally with acceleration a , (b) moved vertically upwards with acceleration a , (c) moved vertically downward with acceleration $a < g$? which case, if any, applies to a pendulum mounted on a cart rolling down an inclined plane?
37. Define centres of suspension and oscillation of a compound pendulum and show that they are interchangeable. What length of the pendulum has its minimum time-period?
38. Show that if a uniform stick of length l is mounted so to rotate about a horizontal axis perpendicular to the stick and at a distance d from its centre of mass, the period has the minimum value when $d = l\sqrt{12} = 0.289l$.
39. Give the theory of kater's pendulum and find an expression for acceleration due to gravity terms of two nearly equal periods of oscillation about the two parallel knife-edges.
(Bombay; punjab)
40. A thin uniform bar of length 120 cm is made to oscillate about an axis through its end. Find the period of oscillation and other points about which it can oscillate with the same time-period.
(Punjab; Poona Raipur)
[Ans. 1.795 sec; at 40 cm, 80 cm and 120 cm from either end.]
41. In a reversible pendulum, the periods about the two knife-edges are t and $(t + T)$, where T is a small quantity. The knife-edges are distance l and l' from the centre of gravity of the pendulum. Prove that
$$l + l' = \frac{gl}{4\pi^2} \left(t + \frac{2l'}{l' - l} T \right).$$

(Madras; Agra)
42. Obtain an expression for the time-period of a compound pendulum and show that (i) there are four points collinear with the c.g. about which the time-period is the same, (iii) its time-period remains unaffected by the fixing of a small additional mass to it at its centre of suspension.
43. What is meant by an equivalent simple pendulum? If the periods of a kater's pendulum in the erect and inverted positions are equal, prove that the distance between the knife-edges is equal to the length of the equivalent simple pendulum.
44. A body of mass 200 gm oscillates about a horizontal axis at a distance of 20 cm from its centre of gravity. If the length of the equivalent simple pendulum be 35 cm, find its moment of inertia about the axis of suspension.
(Patna)
[Ans. 1.4×10^6 gm/cm²]
45. A uniform rod AB of mass 100 gm and length 120 cm can swing in a vertical plane about A as a pendulum. A particle of mass 200 gm is attached to the rod at a distance x from A . Find x such that the period of vibration is a minimum.
(Gorakhpur, Madras)
[Ans. $x = 2.748$ cm.]
46. A uniform circular disc of radius 25 cm oscillates in a vertical plane about a horizontal axis. Find the distance of the axis of rotation from the centre for which the period is the minimum. What is the value of this period?
(Nagpur U. 2009)
[Ans. $25\sqrt{2}$ cm; $2\pi\sqrt{1.44 \times 25/g}$]
47. A uniform cube is free to turn about one edge which is horizontal. Find in terms of a seconds pendulum the length of the edge, so that it may execute a complete oscillation in 2 sec.
[Ans. $3\sqrt{2}l$]
48. At a certain place the value of g is 980 cm/sec² and the length of the pendulum is so adjusted that period is 1 sec. When the period of the pendulum is measured on an elevator, undergoing uniform acceleration, it is found to be 1.025 sec. (i) For an acceleration $a \ll g$ of the elevator, show that the period measured on the elevator is given by $T = T_0 (1 - a/2g)$, where T_0 is the period measured on the elevator is given by $T = T_0 (1 - a/2g)$, where T_0 is the period of the unaccelerated pendulum and upward acceleration is assumed to be positive. (ii) What is the acceleration of the elevator?
[Ans. 49 cm/sec² downward.]

49. An annular ring of internal and outer radii r and R respectively oscillates in the vertical plane about a horizontal axis perpendicular to its plane and passing through a point on its outer edge. Calculate its time-period and show that the length of an equivalent simple pendulum is $2R$ as $r \rightarrow 0$ and $3R$ as $r \rightarrow R$.

$$[\text{Ans. } T = 2\pi\sqrt{r^2 + 2R^2/Rg}]$$

(iii) Torsional Pendulum

50. what is torsion pendulum? Show that oscillations of torsion pendulum are S.H.M. find its period. What are the merits of torsion pendulum. (Nagpur U. 2009, 2006, 2007)
51. What is a torsional pendulum? Obtain an expression for its time-period. Explain why, unlike a simple or a compound pendulum, the time-period in this case remains unaffected even if the amplitude be large.
52. A disc of 10 cm radius and mass 1 kg is suspended in a horizontal plane by a vertical wire attached to its centre. If the diameter of the wire is 1 mm and its length is 1.5 metres and period of torsional vibration of the disc is 5 sec, find the rigidity (n) of material of the wire. (Madras)
- [Ans. 1.206×10^{12} dynes-cm².]
53. A solid cylinder of 2 cm radius weighing 200 gm is rigidly connected, with its axis vertical, to the lower end of a fine wire. The period of oscillation of the cylinder under the influence of the torsion wire is 2 seconds. Calculate the couple necessary to twist it through four complete turns. (Alahebed)
- [Ans. 9.9×10^4 dyne km⁻¹]
54. A body suspended by means of a wire, executes torsional vibrations about the wire as axis, of time-period 5.0 sec. If the moment of inertia of the body about the suspension wire as axis be 7.5×10^3 gm-cm², calculate the maximum angular velocity of the body if its amplitude be $\pi/2$ radian. What would then be its average kinetic energy? [Ans. 1.973 radian/sec. 7.30×10^3 ergs.]

[Hint. Since $\theta = \theta_0 \sin(\omega t + \phi)$, we have angular velocity $d\theta/dt = \omega\theta_0 \cos(\omega t + \phi)$,

Where θ_0 is the amplitude. $\therefore$ maximum value of $d\theta/dt = \omega\theta_0 = 2\pi n\theta_0$ ($\therefore \omega = 2\pi n$).

Here, $n = 1/T = 1/5$ per sec and $\theta_0 = \pi/2$ radian. Hence maximum angular velocity of the body
 $= 2\pi \times (1/5) \times (\pi/2) = \pi^2/5 = 1.973$ radian/sec.

$$\text{And, average K.E.} = \frac{1}{2} \text{ total energy} \quad \text{total energy} = \frac{1}{2} \left(\frac{1}{2} C\theta_0^2 \right) = \frac{1}{2} C\theta_0^2$$

Since $T = 2\pi\sqrt{I/C}$ and $\therefore C = 4\pi^2 I/T^2$, we have

$$\text{average K.E.} = \frac{1}{4} \times \frac{4\pi^2 I}{T^2} \left(\frac{\pi}{2} \right)^2$$

III—Loaded spring (Mass on spring)

55. An unstressed spring has a force constant C . It is stretched by a weight hung from it to an equilibrium length well within the elastic limit. Does the spring have the same force constant C for displacements from this new equilibrium position?
56. Show that an oscillating loaded exhibits S.H.M. find its time period (Nagpur U. 2006)
57. Show that the time period of oscillations of a loaded spring is

$$T = 2\pi\sqrt{\frac{Mx}{mg}}$$

58. Derive an expression for time period of oscillations of a loaded spring. (Nagpur U. 2009)
59. What are the dimensions of force constant of vibrating spring? (Pbi. U. 2002, H.P.U. 2000)
60. Explain the terms: restoring force and force constant.
61. Explain how we can determine the period of oscillation of a spring-mass system from the extension produced in the spring on suspending the mass from it, even though we neither know the magnitude of the mass nor the force constant of the spring.

62. A spring has a force constant C . It is cut into two pieces whose lengths are in the ratio 1:2, will the force constant continue to be the same (C) for each piece? If not, obtain the value of the force constant of each piece of C . [Ans. $3C$, $3C/2$]

[Hint. See worked example 7.41]

63. A massless spring, as we know, is purely a myth. Any real spring must have some mass. If this fact be taken into account, explain what change, if any, will it bring about in the expression for the time-period of a spring-mass system.

[Hint. See worked example 7.42]

64. A spring of force constant 19.6 N/m hangs vertically. A body of mass 0.20 kg is attached to its free end and then released. Assume that the spring was unstretched before the body was released and find how far below the initial position the body descends. Find also the frequency and amplitude of the resulting S.H.M. [Ans. 0.1 m ; $1.576/\text{sec}$; 0.1 m .]

65. A 2 kg mass hangs from a spring. A 300 gm body hung below the mass stretches the spring 2 cm further. If the 3000 gm body is removed and the mass set into oscillation, find the period of motion. [Ans. 0.73 sec .]

66. (a) show that 'gravity' has no effect upon the force constant or the period of oscillation of a spring-mass system.

- (b) Calculate the ratio of $K.E.$ to $P.E.$ when a mass oscillating at the lower end of a spring has covered half its amplitude. [Ans. $3:1$]

67. An 8 lb block is suspended from a spring with a force constant of 3.0 lb/in . A bullet weighing 0.10 lb is fired into the block from below with a velocity of the 500 ft/sec and comes to rest in the block. (a) Find the amplitude of the resulting S.H.M. (b) What fraction of the original $K.E.$ of the bullet is stored in the harmonic oscillator? [Ans. (a) 6.2 in. , (b) 1.2%]

68. Fig. (7.27) show a mass M resting on a smooth table between two firm supports A and B and controlled by two massless springs. If the mass M is 30 gm and the force constants of the two springs are 2000 and 3000 dynes/cm , deduce (i) the frequency of small oscillations of M , (ii) the energy of oscillations for amplitude 0.5 cm . (Agra, 1969)

[Ans. (i) $1.592/\text{sec.}$, (ii) 375 ergs .]

[Hint. See worked example 7.40 (b): $n = 1/T$; Energy of oscillations $= \frac{1}{2} Ca^2$]

69. (a) Calculate the frequency for small oscillations of mass M in problem 53 above, if it were connected to the two spring as shown in Fig 7.27 (a). [Ans. $0.75/\text{sec}$.]

[Hint. See worked example 7.40 (a); $n = 1/T$]

- (b) A car may be considered to be mounted on a stout vertical spring so far as its vertical oscillations are concerned. Thus, if the frequency of such vertical oscillations of a car weighing 1700 kg be 4 when unoccupied, what would be the frequency if it were occupied by five persons, each weighing 60 kg on an average? [Ans. $3.4/\text{sec}$.]

70. The scale of a spring balance reading $0-5 \text{ kg}$ is 12.5 cm . A body suspended from the balance oscillates with a frequency $5/\pi$ Hertz. Calculate the mass of the body attached to the spring. [Ans. $m = 3.92 \text{ kg}$]

71. A spring hung vertically and loaded with a mass of 400 gm made to oscillate. Calculate (i) the time period, and (ii) the frequency of oscillation. When the spring is loaded with 100 gm , it extends by 5 cm . [Ans. (i) 0.8985 (ii) 1.11 Hz]

72. A spherical vessel of capacity 500 c.c. is fitted with a tube of 5 cm length and $\sqrt{\pi} \text{ cm}$ radius. Calculate the wavelength of sound for which it will serve as a resonator. If the velocity of sound be 340 m/sec , what will be the frequency of the sound to which it will respond? [Ans. $\lambda = 100 \text{ cm}$; $n = 340 \text{ Hz}$.]

$$\left[\text{Hint. } n = \frac{v}{2\pi} \sqrt{\frac{\alpha}{Vl}} \therefore \lambda = \frac{v}{n} = 2\pi \sqrt{\frac{Vl}{\alpha}} \right]$$

73. A spherical resonator with a neck of length 4 cm and area of cross section $4\pi^2 \text{ sq cm}$ responds sharply to a note of frequency 112 . If the velocity of sound in air be 336 m/sec , what is the capacity of the resonator? [Ans. 10 litres .]

74. What is helmholtz resonator? Show that the oscillations are simple harmonic. Find its frequency.
(Nagpur U. 2007, 2008, meerut 2005)
75. A spherical Helmholtz resonator of capacity 10 liters has a narrow neck of length 0.01 m and radius 0.01 m. If the velocity of sound in air is 340 m/sec, calculate the frequency of the note to which sharply responds.
(Nagpur U. 2008) (Ans. 95.92 Hz)
76. Discuss the LC circuit and show that the charge on capacitor oscillates simple harmonically. Find the frequency of oscillations. Can it be realized in practice?
(Agra U. 2007, 2004)
77. What oscillates in a simple harmonic electrical oscillator? Can we realize in practice?
78. Derive the differential equation of S.H.M. for an electrical circuit. Can it be realized in practice?
79. It is desired to have an $L-C$ circuit of frequency 10 kilocycles/sec. Calculate the capacitance that must be put in series with an inductance of 1 millihenry to achieve the purpose. [Ans. 0.2533 μF .]
80. An $L-C$ circuit oscillates with a frequency of 20 cps. The capacitance in the circuit is 5 μF . Calculate the value of inductance in the circuit. [Ans. 12.67 henry.]
81. Show that the time period of the swing of a magnet in the earth's field is given by $t = 2\pi\sqrt{I/MH}$, where M is the magnetic moment, I its moment of inertia about the axis of suspension and H , the earth's field.
(Sagar 2005)

V-Two-body harmonic oscillator

82. What is meant by a two-body harmonic oscillator? Obtain an expression for the time-period of its oscillation and point out its similarities with a one-body oscillator. What is meant by reduced mass? Why is it so called.
(Nagpur U. 2006, 2007, 2008)
83. Show that the $K.E.$ of a two-body oscillator (*i.e.*, two bodies connected by a spring) is given by $\frac{1}{2}\mu v^2$, where μ is the reduced mass and $v(=v_1 - v_2)$ the relative velocity.
[Hint. Linear momentum remains conserved during oscillations of the system.]
84. Is there any connection between the F versus x relation at the molecular level and the macroscopic relation between F and x in a spring? Explain briefly.
85. Draw a typical potential energy curve for a diatomic molecule, clearly pointing out and explaining its peculiarities, if any show that, for small amplitudes, its oscillations are simple harmonic and deduce their time-period.
86. (a) Two masses m_1 and m_2 of 1.0 kg and 3.0 kg respectively are connected by a spring of force constant 250 N/m and are placed on a smooth table. They are slightly pulled apart and released. What is the frequency of the two-body system? [Ans. 2.90/sec.]
(b) What is the ratio between the kinetic energies of the two masses of the system? [Ans. 3:1.]
[Hint. Since linear momentum remains conserved during the oscillations, we have $m_1v_1 = m_2v_2$, or $m_1/m_2 = v_2/v_1$] Hence $K.E.$ of mass $m_1/K.E.$ of mass $m_2 = \frac{1}{2}m_1v_1^2 / \frac{1}{2}m_2v_2^2 = (m_1/m_2)(v_1^2/v_2^2) = (m_1/m_2), (m_2^2/m_1^2) = m_2/m_1 = 3/1.$
87. Draw characteristic potential energy diagram of a diatomic molecule and mark on it (i) the binding energy, (ii) equilibrium internuclear distance.
88. Draw the energy-level diagram of a diatomic molecule and explain its salient points. What is meant by the terms: vibrational quantum and zero-point energy?
89. In an HCl molecule, the force required to alter the distance between the atoms from its equilibrium value is 5.4×10^5 dynes per cm. what is the fundamental frequency of the vibration of the molecule, assuming the vibration to be simple harmonic and the mass of the Cl atom to be infinite compared to that of the H atom, which is 1.66×10^{-24} gm? [Ans. 9.078×10^{13}]
90. The separation of energy levels in HCl molecules made from H^1 and Cl^{35} isotopes of atomic weights 1 and 35 is 0.36 eV. What will be the value of the separation levels in HCl molecules made from D^2 and Cl^{37} isotopes. [Ans. 0.258 eV.]

VI—Bifilar oscillations

91. Show that bifilar oscillations are *S.H.M*. Derive the expression for periodic time of a bifilar oscillator with parallel threads. (Nagpur U. s/2003, 2007, 2008).
92. Describe bifilar oscillations and bifilar suspension. (Nagpur U. 2009)

VII—Anharmonic oscillator

93. Discuss a large amplitude pendulum as an anharmonic oscillator.
94. Assuming the solution of the differential equation $m d^2x/dt^2 = -Cx + C_1x^2$ for an anharmonic oscillator to be $x = A \cos \omega t + B \cos^2 \omega t + x_1$, and neglecting all terms involving products of A and B and the second and higher powers of B, show that the average value of x is $x_1 = \alpha A^2 / \omega_0^2$, where $\omega^2 = \omega_0^2 = C/m$ and $\alpha = C_1/m$.
95. A particle oscillates in a potential field $U = \frac{1}{2}Cx - \frac{1}{2}C_1x^2$, where C_1 is positive and very much smaller than C . Will its potential well be symmetrical? If not, how not, how will it affect the mean position of the particle?

8

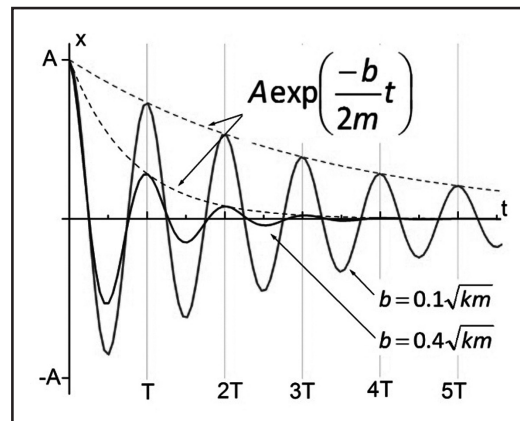
Chapter

DAMPED AND DRIVEN HARMONIC OSCILLATORS

Brief Contents

Introduction

- 8.1 Damping (frictional effects)
- 8.2 Damped Harmonic Oscillator
- 8.3 Power Dissipation
- 8.4 Quality factor Q
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- 8.6 Driven Harmonic Oscillator
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- 8.13 Driven L-C-R Circuit
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- 8.16 R.M.S. or Effective Values of Alternating Current and emf
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INTRODUCTION

In an ideal simple harmonic motion (S.H.M.), the displacement follows a sinusoidal curve as shown in Fig. 8.1(a). The amplitude (a) of oscillations remain constant for an *infinite time*. This is because there is no loss of energy and total energy remains constant. Such oscillations are called '**Free oscillations**'. However, in actual practice, the situation is different. In our discussion in the last chapter, on harmonic oscillator, we completely ignored the effect of frictional forces on it. Since, however, an oscillator, in actual practice, always experiences *frictional* or *resistive medium*, like air, oil etc., part of its energy is dissipated in overcoming the opposing frictional or viscous forces and its amplitude, therefore goes on decreasing progressively as shown in Fig. 8.1(b). Due to the presence of opposing forces, the energy of free oscillator is continuously lost and consequently the amplitude of vibration decreases gradually and ultimately the body comes to rest. Hence, decay of amplitude with time is called **damping**. These opposing forces, being non-conservative nature, produces this **damping effect** are also referred as *damping, resistive* or *dissipative forces*. Such oscillations are called **damped harmonic oscillations**.

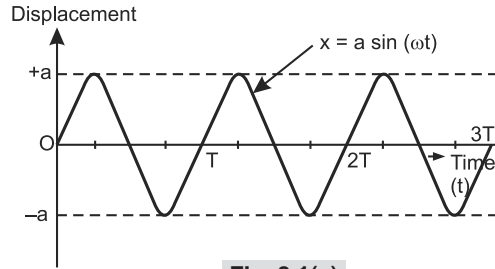


Fig. 8.1(a)

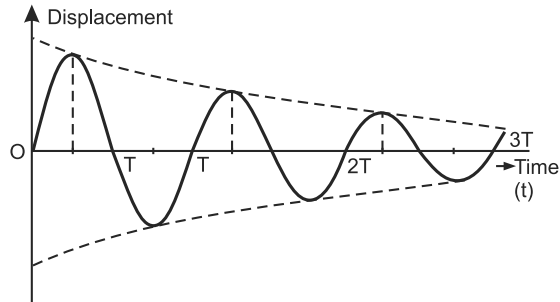


Fig. 8.1(b)

In order to maintain the amplitude constant, an *external periodic force* is applied. These forced vibrations initially gains the frequency equal to its natural frequency and then after short time, the oscillator acquires the frequency of the impressed periodic force. The externally applied periodic force is, hence, known as *driven force* and the oscillator is named as *driven harmonic oscillator* or *forced harmonic oscillator*. In this chapter, we will discuss amplitude resonance, quality factor and energy considerations of both damped and driven harmonic oscillators in detail.

8.1 DAMPING (FRICTIONAL EFFECTS)

Let us discuss the damping effect on the oscillations of a harmonic oscillator in detail. It has been shown by Mayevski that at ordinary velocities—and most cases of interest to us fall in this category—the opposing, resistive or damping force is, to a first approximation, proportional to velocity and may thus be represented by

$$F = -\gamma v = -\gamma dx/dt, \quad \dots(i)$$

where γ is a *positive constant*, called **damping coefficient** of the medium and may be looked upon as the *resistive force per unit velocity*.

So that, if there is no other force, other than this resistive or damping force, acting on the oscillating body or particle, of mass m , we have, in accordance with Newton's second law of motion,

$$F = m \frac{d^2x}{dt^2} = m \frac{dv}{dt} = -\gamma v, \quad \text{Or,} \quad \frac{dv}{dt} + \frac{\gamma}{m} v = 0. \quad \dots(ii)$$

Here, m/γ is usually denoted by τ , a constant, having the dimensions of time and called **relaxation time**,

We, therefore, have $\frac{dv}{dt} + \frac{1}{\tau}v = 0$... (iii)

The constant $1/\tau = \gamma/m$, or the *resistive force per unit mass per unit velocity*, is often denoted by $2k$, where k is called the **damping constant** of the medium.

Now, rewriting differential equation (iii) in the form $dv/v = -dt/\tau$, we have

$$\int \frac{dv}{v} = -\frac{1}{\tau} \int dt, \text{ which gives } \log_e v = -\frac{t}{\tau} + C,$$

where C is a constant of integration to be determined from the initial conditions.

Thus, if at $t = 0$, $v = v_0$, we have $\log_e v_0 = C$. And, therefore,

$$\log_e v - \log_e v_0 = -t/\tau, \quad \text{Or,} \quad v = v_0 e^{-t/\tau}, \quad \dots (iv)$$

clearly showing that *the velocity decreases exponentially with time*, as shown by the curve in Fig. 8.2 between the function $e^{-t/\tau}$ and t . We express this by saying that *the velocity is damped, with time constant τ* .

As will be readily seen, at $t = \tau$, $v = v_0 e^{-1} = v_0/e = v_0/2.718 = 0.368 v_0$.

This enables us to define the *time constant* (or the *relaxation time*) τ as the time in which the velocity of the oscillating particle falls to $1/e$ th (i.e., 0.368 or, roughly, one-third) of its initial value.

And, since the kinetic energy of the oscillating particle is given by $T = \frac{1}{2}mv^2$, we have, on substituting the value

of v from relation (iv) above, $T = \frac{1}{2}mv_0^2 e^{-2t/\tau}$. Or,

representing the initial kinetic energy $\frac{1}{2}mv_0^2$ by T_0 , we have $T = T_0 e^{-2t/\tau}$ indicating that *the kinetic*

energy of the oscillating particle too falls exponentially with time, with a relaxation time half that for velocity, i.e., $\tau/2$, which is only to be expected since $K.E. \propto (\text{velocity})^2$.

Putting dx/dt for v in relation (iv) above, we have $dx/dt = v_0 e^{-t/\tau}/\tau$, which, on integration, gives $x = -v_0 \tau e^{-t/\tau} + C$, where C is a constant of integration.

At $t = 0$, $x = 0$, so that $C = v_0 \tau$. And, therefore, $x = v_0 \tau (1 - e^{-t/\tau})$.

Now, as $t \rightarrow \infty$, $e^{-t/\tau} \rightarrow 0$ and, therefore, $x \rightarrow v_0 \tau$.

Thus, *the maximum value of x is the distance that would be covered by the particle in time τ if its velocity remained constant at its initial value v_0 .*

Examples. (i) As an example of the resistive or damping force of the type represented by relation (i) above may be cited the force experienced by a flat disc moving normally to its plane through a gas (or air), at very low pressure, at a speed very much smaller than that of its molecules.

(ii) Or, perhaps a more familiar example is that furnished by the ohmic resistance in an electrical circuit containing an inductance, such as the one shown in Fig. 8.3. On suddenly breaking the circuit, an induced *emf*, $-L di/dt$, is set up across the inductance L , where I is the value of the current flowing through the circuit at the instant it is broken. Since there is now no external *emf* operative, we have

$$RI = -L di/dt. \quad \text{Or,} \quad L di/dt + RI = 0 \quad \dots (v)$$

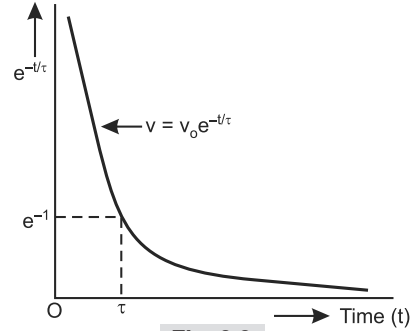


Fig. 8.2

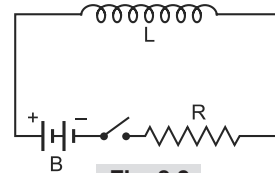


Fig. 8.3

where RI is the potential drop across the resistance R .

This is a relation, identical in form with relations (ii) and (iii) above, with the time constant $\tau = L/R$, clearly indicating that the ohmic resistance plays the same part here as the damping force there, so that *the current in the circuit falls exponentially with time*. For, putting relation (v) in the form $dI/I = -(R/L) dt$ and integrating, we have

$$\log_e I = -\frac{R}{L}t + C, \text{ where } C \text{ is a constant of integration.}$$

Since at $t = 0$, $I = I_0$, its initial (or maximum) value, we have $C = \log_e I_0$. And, therefore,

$$\log_e I = -\frac{R}{L}t + \log_e I_0.$$

Or,
$$\log_e \frac{I}{I_0} = -\frac{R}{L}t, \text{ whence, } I = I_0 e^{-(R/L)t}.$$

It will be seen at once that if $t = \text{time constant } L/R$, we have

$$I = I_0 e^{-1} = I_0/e.$$

Thus, *the time constant here too is the time in which the value of the current in the circuit falls, to $1/e$, or roughly, one-third, of its initial or maximum value.*

8.2 DAMPED HARMONIC OSCILLATOR

A harmonic oscillator in which the oscillations are damped on account of resistive forces, as explained, above under § 8.1, with its amplitude progressively decreasing to zero, is called a *damped harmonic oscillator*.

Obviously, in the case of such an oscillator, in addition to the restoring force $-Cx$, a resistive or damping force $\gamma dx/dt$ also acts upon it, where dx/dt is its velocity at displacement x . We, therefore, have

$$m \frac{d^2 x}{dt^2} = -\gamma \frac{dx}{dt} - Cx. \quad \text{Or,} \quad \frac{d^2 x}{dt^2} + \frac{\gamma dx}{mdt} + \frac{C}{m}x = 0.$$

Or, since $\gamma/m = 2k$ (where k is the *damping constant* of the resistive medium) and $\sqrt{C/m} = \omega_0$, the *natural angular frequency of the oscillating particle, i.e.*, its frequency in the absence of damping, we have

$$\frac{d^2 x}{dt^2} + 2k \frac{dx}{dt} + \omega_0^2 x = 0, \quad \dots(vi)$$

which is called the **differential equation of a damped harmonic oscillator**.

Solution of the equation. The differential equation, which is a *homogeneous linear type, of the second order*, has at least one solution of the form $x = Ae^{\alpha t}$, where A and α are both *arbitrary constants*. We shall, therefore, use this as a trial solution.

Differentiating with respect to t , we have $dx/dt = \alpha Ae^{\alpha t}$ and $d^2x/dt^2 = \alpha^2 Ae^{\alpha t}$.

Substituting these values in the differential equation (vi) above, we have

$$\alpha^2 Ae^{\alpha t} + 2k\alpha Ae^{\alpha t} + \omega_0^2 Ae^{\alpha t} = 0.$$

Or, dividing throughout by $Ae^{\alpha t}$, we have $\alpha^2 + 2k\alpha + \omega_0^2 = 0. \quad \dots(vii)$

This is clearly a *quadratic equation* in α and, therefore,

$$\alpha = -k \pm \sqrt{k^2 - \omega_0^2}$$

The differential equation (vi) is thus satisfied by two values of x , viz.,

$$x = Ae^{(-k + \sqrt{k^2 - \omega_0^2})t} \quad \text{and} \quad x = Ae^{(-k - \sqrt{k^2 - \omega_0^2})t}.$$

The equation being a linear one, the linear sum of the two linearly independent solutions of the equation is also a — and, indeed, the most — general solution. Thus, the general solution is

$$x = A_1 e^{(-k + \sqrt{k^2 - \omega_0^2})t} + A_2 e^{(-k - \sqrt{k^2 - \omega_0^2})t} \quad \dots(viii)$$

Now, $k = 1/2\tau$ and if we put $\sqrt{k^2 - \omega_0^2} = \beta$, we may also put the solution in the form

$$x = A_1 e^{-\frac{t}{2\tau} + \beta t} + A_2 e^{-\frac{t}{2\tau} - \beta t}$$

where the values of the *arbitrary constants* A_1 and A_2 may be determined as follows:

Differentiating expression (viii) with respect to t , we have

$$\begin{aligned} \frac{dx}{dt} &= (-k + \sqrt{k^2 - \omega_0^2}) A_1 e^{(-k + \sqrt{k^2 - \omega_0^2})t} \\ &\quad + (-k - \sqrt{k^2 - \omega_0^2}) A_2 e^{(-k - \sqrt{k^2 - \omega_0^2})t} \quad \dots(ix) \end{aligned}$$

So that, if the displacement x be the *maximum*, equal to $x_{max} = a_0$, say, at $t = 0$ and, therefore, the velocity $dx/dt = 0$, we have from relation (viii) above,

$$x_{max} = a_0 = (A_1 + A_2) \quad \dots(A)$$

and from relation (ix).

$$(-k + \sqrt{k^2 - \omega_0^2}) A_1 + (-k - \sqrt{k^2 - \omega_0^2}) A_2 = 0.$$

$$\text{Or,} \quad -k(A_1 + A_2) + \sqrt{k^2 - \omega_0^2} (A_1 - A_2) = 0.$$

$$\text{Or,} \quad \sqrt{k^2 - \omega_0^2} (A_1 - A_2) = k(A_1 + A_2) = k a_0,$$

$$\text{whence,} \quad (A_1 - A_2) = k a_0 / \sqrt{k^2 - \omega_0^2} \quad \dots(B)$$

Adding relations A and B , therefore, we have

$$2A_1 = a_0 + k a_0 / \sqrt{k^2 - \omega_0^2}$$

$$\begin{aligned} \text{And } \therefore A_1 &= \frac{1}{2} \left(a_0 + \frac{k a_0}{\sqrt{k^2 - \omega_0^2}} \right) = \frac{1}{2} a_0 \left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}} \right) \\ &= \frac{1}{2} a_0 \left(1 + \frac{1}{2\beta\tau} \right) \end{aligned}$$

$$\begin{aligned} \text{and } A_2 &= (A_1 + A_2) - A_1 = a_0 - \frac{1}{2} a_0 \left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}} \right) \\ &= \frac{1}{2} a_0 \left(1 - \frac{k}{\sqrt{k^2 - \omega_0^2}} \right) = \frac{1}{2} a_0 \left(1 - \frac{1}{2\beta\tau} \right) \end{aligned}$$

Substituting these values in expression (viii) above, we have

$$\begin{aligned} x &= \frac{1}{2} a_0 e^{-kt} \left[\left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}} \right) e^{\sqrt{(k^2 - \omega_0^2)}t} \right. \\ &\quad \left. + \left(1 - \frac{k}{\sqrt{k^2 - \omega_0^2}} \right) e^{-\sqrt{(k^2 - \omega_0^2)}t} \right] \quad \dots(x) \end{aligned}$$

Or, since $k = 1/2\tau$ and $\sqrt{k^2 - \omega_0^2} = \beta$, we have

$$x = \frac{1}{2}a_0 e^{-t/2\tau} \tau \left[\left(1 + \frac{1}{2\beta\tau}\right) e^{\beta t} + \left(1 - \frac{1}{2\beta\tau}\right) e^{-\beta t} \right] \quad \dots(xi)$$

Now, three important cases arise:

1. When k (or $1/2\tau$) $> \omega_0$ — (Case of overdamping). In case of high damping such as this, clearly, $\sqrt{k^2 - \omega_0^2}$ (or $\sqrt{1/4\tau^2 - \omega_0^2}$) is a *real quantity* with a *positive value, less than k* . So that, each of the two terms on the right hand side of equation (x) or (xi) has an exponential term with a negative power. The displacement, after attaining its maximum value, therefore, dies off exponentially with time, without changing direction. There is thus no oscillation and the motion is, therefore, called **overdamped, aperiodic** or **dead beat**, as we have in the case of a *dead beat galvanometer* (§ 8.7, I) or that of a pendulum oscillating in a viscous fluid like oil.

2. When k (or $1/2\tau$) $= \omega_0$ — (Case of critical damping). In this case, obviously, $\sqrt{k^2 - \omega_0^2}$ (or $\sqrt{1/4\tau^2 - \omega_0^2}$) $= 0$, so that each of the two terms on the right hand side of equation (x) or (xi) above, becomes infinite and the solution breaks down.

Let us, however, consider the case when $\sqrt{k^2 - \omega_0^2} = h$, a *very small quantity but not zero*. We shall then have, from relation (viii) above,

$$\begin{aligned} x &= A_1 e^{(-k+h)t} + A_2 e^{(-k-h)t} = e^{-kt} (A_1 e^{ht} + A_2 e^{-ht}). \\ \text{Or,} \quad x &= e^{-kt} \left[A_1 \left(1 + ht + \frac{h^2 t^2}{2!} + \frac{h^3 t^3}{3!} + \dots \right) \right. \\ &\quad \left. + A_2 \left(1 - ht + \frac{h^2 t^2}{2!} - \frac{h^3 t^3}{3!} + \dots \right) \right] \end{aligned}$$

Or, neglecting terms containing the second and higher powers of h , we have

$$x = e^{-kt} [A_1(1 + ht) + A_2(1 - ht)] = e^{-kt} [(A_1 + A_2) + (A_1 - A_2)ht]$$

Or, putting $(A_1 + A_2) = M$ and $(A_1 - A_2)h = N$, we have

$$x = e^{-kt} (M + Nt).$$

Now, taking $x = x_{\max} = a_0$ and $dy/dt = 0$ at $t = 0$, we have

$$M = x_{\max} = a_0 \quad \text{and} \quad N = ka_0.$$

So that,

$$x = e^{-kt} (a_0 + ka_0 t) = a_0 e^{-kt} (1 + kt) = a_0 e^{-t/2\tau} \left(1 + \frac{t}{2\tau} \right).$$

Here, the second term $a_0 + kte^{-kt}$ [or $(a_0/2\pi)e^{-t/2\tau}$] decays less rapidly than the first term $a_0 e^{-kt}$ (or $a_0 e^{-t/2\tau}$) and the displacement of the oscillator first increases but it then returns back quickly to its equilibrium position. The motion of the oscillator thus becomes *just aperiodic or non-oscillatory, i.e., it just ceases to oscillate*. This is called the case of **critical damping**, the necessary condition for which, as we have just seen, is the $k \rightarrow \omega_0$. It finds an application in many pointer-type instruments like galvanometers where the pointer moves at once to, and stays at, the correct position, without any annoying oscillations.

3. When k (or $1/2\tau$) $< \omega_0$ — (Case of underdamping). Here, clearly, the quantity $\sqrt{k^2 - \omega_0^2}$ will be an *imaginary* one, say, equal to $i\omega$, where $i = \sqrt{-1}$ and $\omega = \sqrt{\omega_0^2 - k^2}$, a *real quantity*. Expression (viii) above thus becomes $x = A_1 e^{(-k+i\omega)t} + A_2 e^{(-k-i\omega)t}$

Or,

$$\begin{aligned} x &= e^{-kt} [A_1 (\cos \omega t + i \sin \omega t) + A_2 (\cos \omega t - i \sin \omega t)] \\ &= e^{-kt} [(A_1 + A_2) \cos \omega t + i (A_1 - A_2) \sin \omega t] \end{aligned}$$

Or, putting $(A_1 + A_2) = A$ and $i(A_1 - A_2) = B$, we have

$$x = e^{-kt} (A \cos \omega t + B \sin \omega t)$$

If A, B and a_0 be related as shown in Fig. 8.4, we have

$$\begin{aligned} x &= e^{-kt} \left(a_0 \cos \omega t \frac{A}{a_0} + a \sin \omega t \frac{B}{a_0} \right) \\ &= e^{-kt} (a_0 \cos \omega t \sin \phi + a_0 \sin \omega t \cos \phi) \end{aligned}$$

Or,

$$\begin{aligned} x &= a_0 e^{-kt} \sin(\omega t + \phi) \\ &= a_0 e^{-t/2\tau} \sin(\omega t + \phi) \end{aligned} \quad \dots(xii)$$

This is the equation of a damped harmonic oscillator, with amplitude $a_0 e^{-kt}$ (or $a_0 e^{-t/2\tau}$) and frequency $\omega/2\pi = \sqrt{\omega_0^2 - k^2}/2\pi$. It is so called because the sine term in the equation suggests the oscillatory character of the motion and the exponential term, the gradual damping out of the oscillations.

Damping thus clearly produces two effects:

(i) *The frequency of the damped harmonic oscillator, $\omega/2\pi$ is smaller than its undamped or natural frequency $\omega_0/2\pi$, i.e., damping somewhat decreases the frequency or increases the time-period of the oscillator. In actual practice, in a majority of cases, particularly in the case of musical instruments, the damping is small and its effect on the frequency or the time-period of the oscillator, therefore, quite negligible.*

(ii) *The amplitude of the oscillator does not remain constant at a_0 , which represents the amplitude in the absence of damping, but decays exponentially with time, to zero, in accordance with the term e^{-kt} , called the damping factor.*

This is illustrated by the time-displacement curve of the damped harmonic oscillator, shown in Fig. 8.5.

Since the maximum values of $\sin(\omega t + \phi)$ are $+1$ and -1 alternately, the time-displacement graph of the oscillating particle is bounded by the dotted curves

$$a = a_0 e^{-kt} \text{ (or } a = a_0 e^{-t/2\tau} \text{) and } a = -a_0 e^{-kt} \text{ (or } a = -a_0 e^{-t/2\tau} \text{)}.$$

Thus, although its amplitude decreases exponentially with time, the underdamped harmonic oscillator does perform a sort of oscillatory motion. *The motion does not, of course, repeat itself and is thus not periodic in the usual sense of the term.* However, it has still a time-period $= 2\pi/\omega = 2\pi/\sqrt{\omega_0^2 - k^2}$, which is the time-interval between its successive passages in the same direction past the equilibrium point. It is obviously also the time-interval between successive maximum displacements on the same side of the equilibrium point.

N.B. Actually, as we shall see later, a damped oscillation may, in terms of *Fourier's theorem*, be imagined to result from the superposition of a very large number (i.e., an infinite series) of undamped oscillations, with their frequencies varying continuously on either side of the main or principal frequency $\omega/2\pi$ and with their amplitudes diminishing in proportion to the departure of their frequencies from the principal one.

Logarithmic decrement. As we have just seen above, the amplitude in the case of a damped harmonic motion goes on decreasing progressively. So that, if a_n and $a_{(n+1)}$ be the successive amplitudes of the oscillating particle on the two sides of the equilibrium position respectively, the

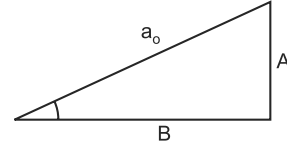


Fig. 8.4

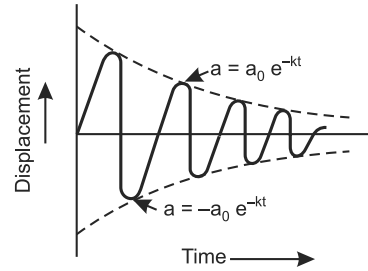


Fig. 8.5

time-interval between them is clearly half the time-period (T) of oscillation and is thus $T/2$. We thus have

$$a_n = a_0 e^{-kt} \quad \text{and} \quad a_{(n+1)} = a_0 e^{-k(t+T/2)}$$

and therefore,
$$\frac{a_n}{a_{(n+1)}} = \frac{e^{-kt}}{e^{-k(t+T/2)}} = e^{kT/2} = d, \text{ a constant.}$$

This constant d between two successive amplitudes of a given damped harmonic motion is called the **decrement** for that motion.

The same naturally also applies to angular amplitudes, where we have

$$\frac{\theta_n}{\theta_{(n+1)}} = d.$$

The logarithm of the decrement, i.e., $\log_e d = kT/2 = \lambda$. Or, $d = e^\lambda$.

This constant, λ , which is obviously the *natural logarithm of the decrement or the ratio between two successive amplitudes of the oscillation* is referred to as the **logarithmic decrement** for that motion or oscillation.

Use is made of this in applying the necessary correction for damping to the *first deflection or the first 'throw'* θ_1 of a ballistic galvanometer as follows:

Each half oscillation, as we know, comprises one swing from θ_1 to θ_2 or from θ_2 to θ_3 , i.e., from the extreme position on one-side to the extreme position on the other, as shown in Fig. 8.6.

So that,
$$\theta_1/\theta_2 = \theta_2/\theta_3 = d = e^\lambda.$$

Now, the first throw of a ballistic galvanometer, from the mean or equilibrium position to either extreme, constitutes only a *quarter oscillation on a half swing*. If, therefore, θ be the *true* value of this first throw, i.e., its value in the absence of damping, and θ_1 , its observed value,

we have
$$\frac{\theta}{\theta_1} = d^{1/2} = e^{\lambda/2} = \left(1 + \frac{\lambda}{2} + \frac{(\lambda/2)^2}{2!} + \dots\right) = 1 + \frac{\lambda}{2} \text{ very nearly.}$$

Hence,
$$\theta = \theta_1 (1 + \lambda/2).$$

Thus, knowing the logarithmic decrement (λ) for the given galvanometer, we can easily correct its first throw (θ_1) for damping.

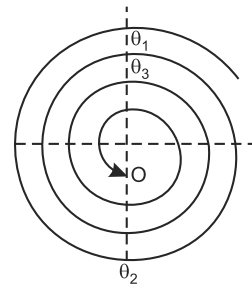


Fig. 8.6

8.3 POWER DISSIPATION

Since the amplitude of a damped harmonic oscillator goes on falling exponentially with time, on account of the resistive or damping forces it has to overcome, it is clear that its energy gets continuously dissipated during its oscillation. Let us calculate its *rate of dissipation of energy* or its *power dissipation*, as it is more commonly called (because *energy or work/time = power*).

Since the displacement of the oscillator is given by

$$x = a_0 e^{-kt} \sin(\omega t + \phi),$$

we have, *velocity of the particle at a given instant t , i.e.,*

$$dx/dt = a_0 e^{-kt} [-k \sin(\omega t + \phi) + \omega \cos(\omega t + \phi)]$$

$\therefore$ *K.E. of oscillation of the particle at the given instant t*

$$\begin{aligned} &= \frac{1}{2} m \left(\frac{dx}{dt} \right)^2 = \frac{1}{2} m a_0^2 e^{-2kt} [-k \sin(\omega t + \phi) + \omega \cos(\omega t + \phi)]^2 \\ &= \frac{1}{2} m a_0^2 e^{-2kt} [k^2 \sin^2(\omega t + \phi) + \omega^2 \cos^2(\omega t + \phi) \\ &\quad - 2k\omega \sin(\omega t + \phi) \cos(\omega t + \phi)] \end{aligned}$$

If the damping be small (as it usually is), the amplitude does not change appreciably over a time-period, so that the factor e^{-2kt} may be taken to be constant over the period. Further, since the average value over a time-period of both $\sin^2(\omega t + \phi)$ and $\cos^2(\omega t + \phi)$ is $1/2$ and that of $2\omega \sin(\omega t + \phi) \cos(\omega t + \phi) = \sin 2(\omega t + \phi) = 0$, we have *average K.E. of the particle or the oscillator over one cycle at the given instant* $t = \frac{1}{2} m a_0^2 e^{-2kt} \left(\frac{1}{2} k^2 + \frac{1}{2} \omega^2 - 0 \right) = \frac{1}{4} m a_0^2 \omega^2 e^{-2kt}$,

neglecting k^2 in comparison with ω^2 , in view of its very small value.

And, *P.E. of the particle or the oscillator at the given instant* t (when the displacement is x) = $\frac{1}{2} C x^2 = \frac{1}{2} m \omega_0^2 x^2$, where ω_0 is the natural (undamped) angular frequency of the particle.

$$\left[\because P.E. = \int_0^x Cx dx = \frac{1}{2} Cx^2 \text{ and } \frac{C}{m} = \omega_0^2 \right]$$

Or, substituting the value of x , we have

$$\begin{aligned} P.E. \text{ of the oscillating particle} &= \frac{1}{2} m \omega_0^2 [a_0 e^{-kt} \sin(\omega t + \phi)]^2 \\ &= \frac{1}{2} m a_0^2 e^{-2kt} [\omega_0^2 \sin^2(\omega t + \phi)] \end{aligned}$$

Again, since the average value of $\sin^2(\omega t + \phi)$ over a time-period is $1/2$, we have

$$\begin{aligned} \text{average P.E. of the particle or the oscillator over one cycle, at the given instant } t &= \frac{1}{4} m a_0^2 \omega_0^2 e^{-2kt} \\ &= \frac{1}{4} m a_0^2 \omega^2 e^{-2kt}, \text{ because, damping being small, } \omega_0 \approx \omega. \end{aligned}$$

Hence, *average total energy of the oscillator at the given instant* = (average K.E. + average P.E.) at the instant = $\frac{1}{4} m a_0^2 \omega^2 e^{-2kt} + \frac{1}{4} m a_0^2 \omega^2 e^{-2kt} = \frac{1}{2} m a_0^2 \omega^2 e^{-2kt} = E = E_0 e^{-2kt} = E_0 e^{-t/\tau}$, where $\frac{1}{2} m a_0^2 \omega^2 = E_0$, the average total energy of the undamped oscillator.

$\therefore$ average power dissipation (P) = rate of loss of energy

$$= -dE/dt = m a_0^2 \omega^2 k e^{-2kt} = 2kE = E/\tau. \quad (\because 2k = 1/\tau)$$

This loss of energy is obviously due to the work done against the damping or dissipative force $-\gamma dx/dt$ and usually appears in the form of heat.

Alternatively, we could obtain the same result as follows:

As in the case of a harmonic oscillator, so also here, the energy of an oscillator must be proportional to the square of its amplitude, i.e.,

$$E \propto (a_0 e^{-kt})^2, \quad \text{Or, } E = C a_0^2 e^{-2kt},$$

where C is a constant of proportionality.

After one full cycle (i.e., one full time-period T), the energy is, say,

$$E_T = C a_0^2 e^{-2k(t+T)} = E e^{-2kT}$$

$$\therefore \text{ loss of energy in one time-period } T = E - E e^{-2kT} = E - E \left(1 - 2kT + \frac{4k^2 T^2}{2!} \dots \right)$$

$$= E - E(1 - 2kT) = 2kET,$$

neglecting higher terms in k , since k is small.

And $\therefore$ rate of loss of energy over a time-period, i.e., $P = 2kET/T = 2kE = E/\tau$.

8.4 QUALITY FACTOR Q

As its very name indicates, it is a factor which measures the quality of a harmonic oscillator in so far as damping is concerned. *The less the damping, the better the quality of the harmonic oscillator as an oscillator and, therefore, the higher its quality factor Q .*

It is also referred to as the figure of merit of a harmonic oscillator and is defined as 2π times the ratio between the energy stored and the energy lost per period. Being thus a mere number, it is a dimensionless quantity.

$$\text{Thus,} \quad Q = 2\pi \frac{\text{Energy stored}}{\text{Energy lost per period}} = \frac{2\pi E}{PT},$$

where P is the average loss of energy over a period $= E/\tau$.

(See § 8.3 above)

$$\text{And, since } 2\pi/T = \omega, \text{ we have } Q = \frac{E_\omega}{P} = \frac{E_\omega}{E/\tau} = \omega\tau.$$

In the case of low damping, $\omega = \omega_0$ and, therefore, $Q = \omega_0\tau$.

$$\text{But, as we know,} \quad \omega_0 = \sqrt{\frac{C}{m}} \quad \text{and} \quad \tau = \frac{m}{\gamma}. \text{ So that,}$$

$$Q = \sqrt{\frac{C}{m}} \cdot \frac{m}{\gamma} = \frac{\sqrt{Cm}}{\gamma}.$$

clearly indicating that the lower the value of γ , i.e., the lower the damping, the higher the value of Q , i.e., as $\gamma \rightarrow 0$, $Q \rightarrow \infty$.

Again; we know that the energy of a damped harmonic oscillator is given by $E = E_0 e^{-t/\tau}$. Hence, if $t = \tau$, we have $E = E_0 e^{-1} = E_0/e$, i.e., the energy of the oscillator falls to $1/eth$ of its initial value in time $t = \tau$. In this interval of time, the oscillator executes $(\omega_0/2\pi)\tau$ or $Q/2\pi$ oscillations, so that *the phase of the oscillator changes by Q .*

This gives us another method of defining Q , viz. *as the phase change brought about in the oscillator in the time taken by its energy to fall to $1/eth$ of its initial value.*

8.5 IMPORTANT EXAMPLES OF DAMPED HARMONIC OSCILLATORS

I. Dead beat and ballistic galvanometers. In the case of a moving coil galvanometer, the turning of the coil under the *deflecting couple* is opposed, as we know, by a *restoring couple* $C\theta$, where C is the torsional couple per unit twist in the suspension fibre (equal to $\pi n r^4/2l$) and θ , the deflection of the coil at the given instant t . In addition to this, there are two other damping couples acting on the coil, viz., (i) a couple $-\gamma d\theta/dt$ due to the *mechanical damping* on account of viscosity of air and the elastic hysteresis of the suspension fibre, where γ is the damping coefficient for this type of damping and (ii) a couple $-(\Lambda/R) d\theta/dt$ due to *electromagnetic damping*, i.e., damping on account of the induced current set up in the coil, which apart from being directly proportional to its velocity $d\theta/dt$, is also inversely proportional to its resistance and directly proportional to the magnetic flux generated through it and its area etc., all included in the constant Λ .

If, therefore, I be the moment of inertia of the coil or the suspended system about the suspension fibre, the *equation of motion* of the system is

$$I \frac{d^2\theta}{dt^2} = -\gamma \frac{d\theta}{dt} - \frac{\Lambda}{R} \frac{d\theta}{dt} - C\theta.$$

$$\text{Or,} \quad \frac{d^2\theta}{dt^2} + \left(\frac{\gamma + \Lambda/R}{I} \right) \frac{d\theta}{dt} + \frac{C\theta}{I} = 0.$$

Now, $(\gamma + \Lambda/R)I = 2k = 1/\tau$ and $C/I = \omega_0^2$, where ω_0 is the undamped angular frequency of the system.

So that,
$$\frac{d^2\theta}{dt^2} + 2k \frac{d\theta}{dt} + \omega_0^2\theta = 0.$$

on,
$$\frac{d^2\theta}{dt^2} + \frac{1}{\tau} \frac{d\theta}{dt} + \omega_0^2\theta = 0.$$

This differential equation of motion of the coil, it will be readily seen, is the same as that of a damped harmonic oscillator, discussed under § 8.2, except that we have here an angular, instead of a linear, displacement. And, therefore, its solution, as explained there, is given by

$$\theta = A_1 e^{\left(-k + \sqrt{k^2 - \omega_0^2}\right)t} + A_2 e^{\left(-k - \sqrt{k^2 - \omega_0^2}\right)t}$$

where the values of the arbitrary constants A_1 and A_2 , obtained in the same manner, are

$$\frac{1}{2}\theta_0 \left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}}\right) \text{ and } \frac{1}{2}\theta_0 \left(1 - \frac{k}{\sqrt{k^2 - \omega_0^2}}\right)$$

respectively, with θ_0 as the initial or maximum deflection.

So that,
$$\theta = \frac{1}{2}\theta_0 e^{-kt} \left[\left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}}\right) e^{\left(\sqrt{k^2 - \omega_0^2}\right)t} + \left(1 - \frac{k}{\sqrt{k^2 - \omega_0^2}}\right) e^{\left(-\sqrt{k^2 - \omega_0^2}\right)t} \right]$$

As before, *three cases* arise:

(i) **When $k > \omega_0$.** This is the case of overdamping or heavy damping and $k^2 > \omega_0^2$ or $\left(\frac{\gamma + \Lambda/R}{4I^2}\right)^2 > \frac{C}{I}$. So that, both the terms in the expression for θ above have an exponential term

with a negative power and θ , therefore, decays exponentially without changing sign. *The motion is thus non-oscillatory, aperiodic or dead beat.*

Hence, **for a galvanometer to be dead beat**, $k = 1/2\tau = (\gamma + \Lambda/R)/2I$ must be large, i.e., (a) I , the M.I. of the coil or the suspended system must be small, (b) R , the resistance of the coil should be small, (c) Λ should be large, indicating that the induced currents set up in the coil and hence the magnetic flux through it should be large, i.e., the coil must be wound on a conducting frame and (d) the damping coefficient γ for mechanical damping must be large.

(ii) When $k = \omega_0$, or $(\gamma + \Lambda/R)^2/4I^2 = C/I$. In this case, the damping is critical (See § 8.2, above) and the coil comes back to its original position quickly.

(iii) When $k < \omega_0$ or $(\gamma + \Lambda/R)^2/4I^2 < C/I$. This is the case of underdamping when, as explained under § 8.2, (3), the solution of the differential equation of motion of the coil is of the form $\theta = \theta_0 e^{-kt} \sin(\omega t + \phi)$, where $\omega = \sqrt{\omega_0^2 - k^2} = \sqrt{C/I - (\gamma + \Lambda/R)^2/4I^2}$ and, therefore, frequency $n = \omega/2\pi$.

The amplitude here goes on decreasing exponentially with time.

A galvanometer used for measuring charges must have the minimum damping and is called a **ballistic galvanometer**. Thus, in a ballistic galvanometer, $k = 1/2\tau = (\gamma + \Lambda/R)/2I$ must be as small as possible and, therefore, the conditions for it are just the opposite of those for a dead beat galvanometer, i.e., (a) I must be large, (b) R must be large, (c) the mechanical damping coefficient γ must be small and (d) Λ should be small, i.e., the magnetic flux through the coil should be small and

it should, therefore, be wound on a non-conducting frame of ebonite, ivory, bamboo etc.—all these conditions being the very opposite of those required for a dead beat type of galvanometer.

The first throw of the ballistic galvanometer is corrected for any damping that may still be present, with the help of the logarithmic decrement of the instrument.

II—The L-C-R circuit. Any electrical circuit containing inductance (L), capacitance (C) and resistance (R) is an excellent example of a damped harmonic oscillation, with the resistance, by itself alone, playing the part of a resistive or dissipative force analogous to that of friction or viscosity in the case of mechanical oscillations.

Thus, suppose we have such an electrical circuit, connected up as shown in Fig. 8.7, such that on pressing the knob of the Morse Key, $M.K.$, the capacitor gets charged by the battery and on releasing it, the battery is thrown out of the circuit and the capacitor (C) connected in series with the inductance (L) and the resistance (R) through which it then discharges itself.

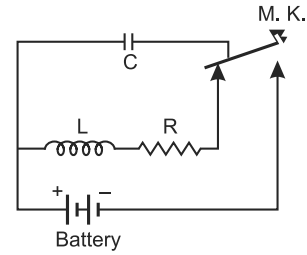


Fig. 8.7

Suppose further that at a given instant t , the charge on the capacitor is Q , the current flowing through the circuit (consisting of capacitor, inductance and resistance, with the battery thrown out) is I , so that the induced *emf* across the inductance is LdI/dt and the potential difference across the resistance is RI . Then, since there is now no external *emf* in the circuit (the battery being out), we have, in accordance with Kirchoff's law,

$$RI + \frac{Q}{C} = -L \frac{dI}{dt}. \quad \text{Or,} \quad L \frac{dI}{dt} + RI + \frac{Q}{C} = 0,$$

$$\text{But } I = dQ/dt \text{ and, therefore, } L \frac{d^2Q}{dt^2} + R \frac{dQ}{dt} + \frac{Q}{C} = 0.$$

$$\text{Or,} \quad \frac{d^2Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{Q}{LC} = 0. \quad \dots I$$

This equation is of the same form as that for a damped harmonic oscillator (equation (vi), page 378), with the difference that displacement x there is replaced by charge Q here, $2k$ (or $1/\tau$) by R/L and ω_0^2 by $1/LC$.

Proceeding in the same manner, therefore, we find the solution of the equation to be

$$Q = A_1 e^{(-k + \sqrt{k^2 - \omega_0^2})t} + A_2 e^{(-k - \sqrt{k^2 - \omega_0^2})t},$$

where the values of the arbitrary constants A_1 and A_2 respectively are

$$\frac{1}{2}Q_0 \left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}} \right) \text{ and } \frac{1}{2}Q_0 \left(1 - \frac{k}{\sqrt{k^2 - \omega_0^2}} \right).$$

Thus,

$$Q = \frac{1}{2}Q_0 e^{-kt} \left[\left(1 + \frac{k}{\sqrt{k^2 - \omega_0^2}} \right) e^{(\sqrt{k^2 - \omega_0^2})t} + \left(1 - \frac{k}{\sqrt{k^2 - \omega_0^2}} \right) e^{(-\sqrt{k^2 - \omega_0^2})t} \right] \quad \dots II$$

Again, the following three cases arise:

(i) **When the damping is large**, i.e., when $k^2 > \omega_0^2$ or $(R^2/4L^2) > 1/LC$ or $R^2 > 4L/C$. In this case, both the terms in expression II for Q above have exponential terms with a negative power, so that the charge decays exponentially with time, without changing sign, as shown by curve (i) in

Fig. 8.8, and there is thus no oscillation, i.e., the discharge of the capacitor is non-oscillatory or aperiodic.

(ii) **When the damping is critical**, i.e., when $k^2 = \omega_0^2$ or $R^2/4L^2 = 1/LC$ or $R^2 = 4L/C$. Here, the discharge becomes just aperiodic and dies down in the shortest possible time, as shown by curve (ii) in Fig. 8.8.

(iii) **With the damping is small**, i.e., when $k^2 < \omega_0^2$ or $R^2/4L^2 < 1/LC$ or $R^2 < 4L/C$. This is the case of interest to us, since here the solution of the equation of motion of the charge takes the form $Q = Q_0 e^{-kt} \sin(\omega t + \phi)$, viz, that of a *damped harmonic oscillator*. The charge on the capacitor thus repeatedly acquires positive and negative values before ultimately decaying to zero. The discharge curve of the capacitor is of the form shown by curve (iii) in Fig. 8.8 above, which it will be recalled, is identical in form with the displacement curve of a damped harmonic oscillator, shown in Fig. 8.5 and is similarly bound by the curves $Q = Q_0 e^{-kt}$ and $Q = -Q_0 e^{-kt}$.

Thus, the discharge of the capacitor is oscillatory in character, the oscillations being damped as in the case of a damped harmonic oscillator.

Differentiating the expression for Q with respect to t , we have

$$\frac{dQ}{dt} = \frac{dQ_0}{dt} e^{-kt} \sin(\omega t + \phi).$$

Or, since $dQ/dt = I$ and $dQ_0/dt = I_0$, the maximum value or the amplitude of the current, we have

$$I = I_0 e^{-kt} \sin(\omega t + \phi)$$

$$\begin{aligned} \text{The frequency of the oscillations is } n &= \frac{\omega}{2\pi} = \frac{\sqrt{\omega_0^2 - k^2}}{2\pi} \\ &= \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}. \end{aligned}$$

That the resistance R alone is responsible for the damping of the oscillations in this case (the energy dissipated appearing as heat) is clear from the fact that in the absence of R , i.e., if $R = 0$, we shall have $n = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$, the same as that of an L - C circuit, with no damping of the oscillations.

The *quality factor of the circuit*, $Q = \omega\tau = \omega/2k = L\omega/R$, where $\omega = \sqrt{\omega_0^2 - k^2} = \sqrt{\omega_0^2 - R/4L^2}$.

Since R is small, $\omega \approx \omega_0$ and, therefore, quality factor $Q = L\omega_0/R$.

It will be readily seen that in a purely inductive circuit, i.e., with $R = 0$, the quality factor Q will be infinite.

8.6 DRIVEN HARMONIC OSCILLATOR

When a harmonic oscillator oscillates in a medium like air, its oscillations, as we know, get damped, i.e., its amplitude falls exponentially with time to zero.

If, however, we apply an *external periodic force* to the oscillator, of a frequency *not necessarily the same as its own natural frequency*, a sort of tussle ensues between the damping force tending to retard its motion and the applied force tending to continue it. So that, after some initial erratic movements, it ultimately succumbs to the applied or the driving force and *settles down to oscillating with the forcing or the driving frequency* (i.e., the frequency of the applied or the driving force) and a constant amplitude and phase so long as the applied force remains operative.

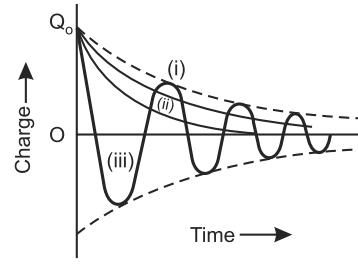


Fig. 8.8

An oscillator, thus compelled to oscillate with a frequency other than its own natural frequency, is called a *driven harmonic oscillator* and the oscillations executed by it are, therefore, called *driven or forced oscillations*.

In the particular case of forced oscillations *when the frequency of applied force is the same as the natural frequency of the oscillator itself, we have the phenomenon of resonance or resonant oscillations*.

Let us first deal with the *theory of a forced or driven oscillator*, in general.

Suppose the periodic force to which a damped harmonic oscillator is subjected is $F = F_0 \sin pt$ which is obviously a sinusoidal force, of *amplitude* (or *maximum value*). F_0 and frequency $p/2\pi$.

Since the damping and the restoring forces acting on the oscillator are respectively $-\gamma dx/dt$ and $-Cx$, its *equation of motion* becomes

$$m \frac{d^2x}{dt^2} = -\gamma \frac{dx}{dt} - Cx + F. \text{ Or, } \frac{d^2x}{dt^2} + \frac{\gamma}{m} \frac{dx}{dt} + \frac{C}{m} x = \frac{F_0}{m} \sin pt.$$

Now, $\gamma/m = 2k$ and $C/m = \omega_0^2$, where ω_0 is the *natural angular frequency* of the oscillator. So that, putting $\frac{F_0}{m} \sin pt = f_0 \sin pt$, representing the *applied force per unit mass*, we have

$$\frac{d^2x}{dt^2} + 2k \frac{dx}{dt} + \omega_0^2 x = f_0 \sin pt. \quad \dots I$$

When the *steady state* has been attained, *i.e.*, when, after the tussle between the damping and the applied forces, the oscillator has settled down to oscillate with the forcing frequency $p/2\pi$ and a constant amplitude, let us try $x = A \sin (pt - \theta)$ as a particular solution of its equation of motion (*I*, above), where θ is the possible phase difference between the applied force and the displacement of the oscillator.

Then, clearly, $dx/dt = A p \cos (pt - \theta)$ and $\frac{d^2x}{dt^2} = -A p^2 \sin (pt - \theta)$.

Substituting these values in equation *I*, we have

$$\begin{aligned} -A p^2 \sin (pt - \theta) + 2k A p \cos (pt - \theta) + \omega_0^2 A \sin (pt - \theta) \\ = f_0 \sin [(pt - \theta) + \theta] \\ = f_0 \sin (pt - \theta) \cos \theta + f_0 \cos (pt - \theta) \sin \theta \end{aligned}$$

$$\begin{aligned} \text{Or, } A(\omega_0^2 - p^2) \sin (pt - \theta) + 2k A p \cos (pt - \theta) \\ = f_0 \cos \theta \sin (pt - \theta) + f_0 \sin \theta \cos (pt - \theta) \quad \dots II \end{aligned}$$

If this solution is to hold good for all values of t , the coefficients of $\sin (pt - \theta)$ on either side of equation *II* must respectively be equal, *i.e.*, we must have

$$A (\omega_0^2 - p^2) = f_0 \cos \theta \quad \dots (i)$$

$$\text{and} \quad 2k A p = f_0 \sin \theta. \quad \dots (ii)$$

Squaring and adding relations *(i)* and *(ii)*, we have

$$A^2(\omega_0^2 - p^2)^2 + 4k^2 p^2 = f_0^2 \cos^2 \theta + f_0^2 \sin^2 \theta.$$

$$\text{Or, } A^2[(\omega_0^2 - p^2)^2 + 4k^2 p^2] = f_0^2,$$

whence,

$$A^2 = \frac{f_0^2}{(\omega_0^2 - p^2)^2 + 4k^2 p^2}$$

$\therefore$ *amplitude of the driven or forced oscillator,*

$$A = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}}, \quad \dots (III)$$

taking only the positive value of the square root. Its negative value will mean opposite phase but then θ too will change by π and there would, therefore, be no effect on the value of A .

And, $\tan \theta = \frac{f_0 \sin \theta}{f_0 \cos \theta} = \frac{2kp}{(\omega_0^2 - p^2)}$. Or, the *phase difference* between the driven or forced oscillator and the applied force is

$$\theta = \tan^{-1} \frac{2kp}{(\omega_0^2 - p^2)} \quad \dots (IV)$$

Since $\sin \theta$ is positive, it follows that θ lies within the range 0 to π . Substituting these values in the relation $x = A \sin (pt - \theta)$, we have

$$x = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}} \sin \left(pt - \tan^{-1} \frac{2kp}{\omega_0^2 - p^2} \right) \quad \dots (V)$$

which represents a *S.H.M.* of frequency $p/2\pi$, i.e., the same as that of the driving force, but lagging behind it in phase by $\theta = \tan^{-1} [2kp/(\omega_0^2 - p^2)]$, where θ lies between 0 and π ($\therefore \sin \theta$ is positive).

Now, $x = A \sin (pt - \theta)$ is not really the complete solution of equation (I), which is an *inhomogeneous differential equation* because of the presence of the term $f_0 \sin pt$ which contains neither the variable (x) nor its derivative. The solution will, therefore, be complete only if we add to it a complementary function which is a solution of the related homogeneous equation $d^2x/dt^2 + 2k dx/dt + \omega_0^2 x = 0$.

One such solution, as we know, is $x = a_0 e^{-kt} \sin (\omega t + \phi)$, representing a damped harmonic oscillation, where $\omega = \sqrt{\omega_0^2 - k^2}$. The addition of this term [$x = a_0 e^{-kt} \sin (\omega t + \phi)$] to the solution $x = A \sin (pt - \theta)$ does not in any way impair the validity of the latter, since the term, taken by itself, reduces the left hand side of equation (I) to zero.

The *complete solution of equation (I)* is thus $x = a_0 e^{-kt} \sin (\omega t + \phi) + A \sin (pt - \theta)$, where, obviously, the first term on the right hand side represents an initial damped oscillation of frequency $\omega/2\pi$, with its amplitude decaying *exponentially* to zero, and the second represents a forced or driven oscillation of the forcing (or driving) frequency $p/2\pi$ and a *constant amplitude* A . The former oscillation dies out quickly and the latter alone then remains effective, so that we are left with $x = A \sin (pt - \theta)$ as the *equation of motion of the forced or driven oscillation*, with its *amplitude* A and *phase angle* θ given by relations (III) and (IV) above respectively.

Now, when k has a finite value, greater than zero, the value of A will obviously be the maximum when the denominator in expression (III) (for A) has its minimum value,

$$\text{i.e., when } \frac{d}{dp} [(\omega_0^2 - p^2)^2 + 4k^2 p^2] = 0,$$

$$\text{Or, } -4(\omega_0^2 - p^2)p + 8k^2 p = 0,$$

$$\text{i.e., when } p^2 = \omega_0^2 - 2k^2 \text{ and, therefore, } p = \sqrt{\omega_0^2 - 2k^2}.$$

Alternatively, we can obtain the same result by putting expression (III) for A in the form

$$A = \frac{f_0}{\sqrt{(p^2 + 2k^2 - \omega_0^2)^2 + 4k^2 \omega_0^2 - 4k^4}},$$

from which also, A will be maximum when $p^2 + 2k^2 - \omega_0^2 = 0$, or when

$$p^2 = \omega_0^2 - 2k^2, \text{ or } p = \sqrt{\omega_0^2 - 2k^2}.$$

Thus, the amplitude of the driven oscillator will be the maximum when the driving frequency is $\sqrt{\omega_0^2 - 2k^2}/2\pi$, which we may denote by PR/π .

This phenomenon where for a particular driving frequency, the response or the amplitude of the driven oscillator is the maximum is called **amplitude resonance** and the particular driving frequency is referred to as the **resonant frequency**.

It may be noted that *this resonant frequency* ($p_R/2\pi = \sqrt{\omega_0^2 - 2k^2}/2\pi$) is smaller than both the natural, undamped frequency $\omega_0/2\pi$ and the natural damped frequency $\omega/2\pi = \sqrt{(\omega_0^2 - k^2)}/2\pi$ of the oscillator.

Substituting p_R for P in the relation for A above therefore, we have

$$\text{maximum amplitude, } A_{\max} = \frac{f_0}{2k(\omega_0^2 - k^2)^{1/2}} \quad \dots (i)$$

Or, because $\omega_0^2 - 2k^2 = p^2$ or, $\omega_0^2 = p^2 + 2k^2$, we have

$$A_{\max} = \frac{f_0}{2k(p^2 + k^2)^{1/2}}, \quad \dots (ii)$$

showing that the smaller the value of k , the greater the value of $A_{\max}$.

(i) It follows, therefore, that *in case the damping be low*, $p \approx \omega_0$ and the maximum amplitude in that case is $A_{\max} = f_0/2k \omega_0 = f_0\tau/\omega_0$, ($\because \tau/2k = \tau$). Obviously then, *in the absence of damping* (i.e., $k = 0$), the amplitude should become *infinite*. Since, however, damping is never actually zero, this never happens.

(ii) Again, if *the driving frequency be negligibly small or zero and the damping low*, we have

$$A = f_0/\omega_0^2.$$

Thus, the ratio of the response (or amplitude) of the oscillator when the driving frequency is zero or negligible is clearly

$$\frac{f_0\tau/\omega_0}{f_0/\omega_0^2} = \omega_0\tau = Q, \text{ the quality factor of the oscillator;}$$

which thus actually controls its response or amplitude at this frequency.

It will also be easily seen that $f_0/\omega_0^2 = \frac{F_0/m}{C/m} = F_0/C$, showing that if *the driving frequency be negligible or zero and the damping low*, the amplitude or the response of the oscillator is controlled neither by its mass nor any damping but simply by the force constant C .

(iii) Finally, if *the damping be low and the driving frequency high*, so that ω_0 is negligible in comparison, the amplitude of the driven oscillator is given by

$$A = f_0/p^2,$$

indicating that *the response in this case falls as p increases*.

It is thus clear from the above discussion, as indeed from the expression for A itself, that the magnitude of A depends upon the relative values of p and ω_0 and that it is, infact, controlled by the factor $(\omega_0^2 - p^2)$. The natural consequence is that its value diminishes both when $p < \omega_0$ and when $p > \omega_0$. This will be amply clear from Fig. 8.9, where a number of curves are drawn showing the relation between p and the amplitude (A) of the driven oscillator for different cases of damping. It will be noted that

(a) *The peak value of amplitude, $A_{\max} = f_0/2k \omega_0 = f_0/2kp$ occurs when $(\omega_0^2 - p^2) = 0$, i.e., when $P = \omega_0$, which represents the condition for amplitude resonance.*

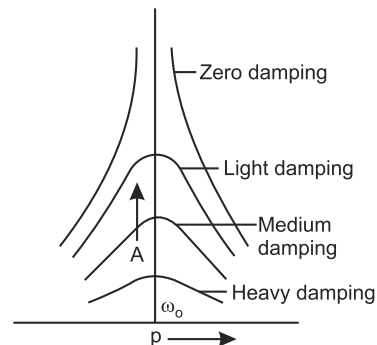


Fig. 8.9

And, if $k = 0$, i.e., if there be no damping, A_{max} becomes infinite for this value of p , the curve being asymptotic to the y or the amplitude axis. Since damping is never zero, we do not obtain such a curve in actual practice but only those of the type shown for light and medium damping.

(b) The peak value of A (i.e., A_{max}) is naturally different for different cases of damping, becoming less and less as **damping increases**, but always occurs at, or very nearly at, $(\omega_0^2 - p^2) = 0$ provided the damping is **not too** large, i.e., the resonant frequency (p_R) in all these cases is equal to, or very nearly equal to, ω_0 .

Only in the case of heavy damping does the peak value occur at a frequency less than ω_0 (being equal to $\sqrt{\omega_0^2 - 2k^2}$).

(c) The fall in the curves on either side of $(\omega_0^2 - p^2) = 0$ is steeper in the case of smaller than in the case of heavier damping.

8.7 SHARPNESS OF RESONANCE

We have just seen in Fig. 8.9 above how the curves between p and A fall more steeply on either side of the respective peak values of A when the damping is low than when it is high. Thus, in the case of high damping, the amplitude remains more or less at its peak value over an appreciable range on either side of $(\omega_0^2 - p^2) = 0$, i.e., even when $p \neq \omega_0$. The oscillator thus responds to a number of frequencies near about ω_0 on either side of it. The resonance in this case is, therefore, said to be **flat**.

On the other hand, if the damping be low, the steep fall of curve on either side of the peak value of A shows that the oscillator responds only to the frequency exactly equal to its natural frequency ω_0 and to none others. The resonance here is, therefore, said to be **sharp**.

Thus, sharpness of resonance is, in a way, a measure of the rate of fall of amplitude from its maximum value at resonant frequency, on either side of it. The sharper the fall in amplitude, the sharper the resonance.

Familiar examples of *flat* and *sharp* resonance are the case of air column and the sonometer wire respectively. On account of the large value of k for air, the curve between p and A is comparatively flat at the peak value of A . The air column thus continues to respond to frequencies over an appreciable range on either side of its natural frequency ω_0 , so much so that it becomes difficult to determine the correct length of the air column that *exactly* responds to the oscillator. This is, therefore, a case of *flat resonance*.

A sonometer wire, on the other hand, with little or no damping, responds only to one particular frequency, viz., its own natural frequency ω_0 and to none others. The resonance here is, therefore, *sharp*.

Thus, the smaller the damping, the sharper the resonance. In fact, it can be shown that sharpness of resonance is inversely proportional to the square of the damping constant k .

8.8 PHASE OF THE DRIVEN OSCILLATOR

We have seen above (§ 8.6) how a forced or driven oscillation is represented by the equation $x = A \sin(pt - 0)$, indicating that it always lags a phase angle $\theta = \tan^{-1} [2kp/(\omega_0^2 - p^2)]$ behind the driving force $f_0 \sin pt$.

Obviously, this phase angle (θ) depends upon the damping and the relative values of ω_0 and p . If the damping be negligible, the following three cases arise:

(i) **When $p < \omega_0$** , i.e., when the forcing or driving frequency is smaller than the natural, undamped frequency of the oscillator. In this case, clearly, $\tan \theta = 2kp/(\omega_0^2 - p^2)$ will be a small positive quantity and the driven oscillator will, therefore, very nearly be in phase with the driving force.

(ii) **When $p > \omega_0$, i.e., when the driving frequency is greater than the natural, undamped frequency of the oscillator:** Here, obviously, $\tan \theta$ will be a small negative quantity and the driven oscillator will thus be out of phase with the driving force, i.e., the phase difference between the two will be π (or $T/2$).

(ii) **When $p = \omega_0$, i.e., when the driving frequency is equal to the natural, undamped frequency of the oscillator:** In this case, $\tan \theta = \infty$ and, therefore, $\theta = \pi/2$, i.e., the driven or forced oscillator will differ in phase from the driving force by $\pi/2$ (or $T/4$). Thus, its displacement will be the maximum when the driving force is zero and vice versa.

It is clear from the above that the phase angle θ changes from 0 to π but remains positive throughout.

The general nature of the phase lag of a driven oscillator as the driving frequency increases, passing through the value ω_0 (the natural or undamped frequency of the oscillator) will be clear from the curves in Fig. 8.10, drawn for different cases of damping. The following points emerge:

(i) When the damping is zero, the curve runs along the frequency axis from 0 to ω_0 and again parallel to it from ω_0 to $2\omega_0$ but removed from the first part by π . This means, in other words, that for $p \ll \omega_0$, $\theta = 0$ but when $p \gg \omega_0$, θ suddenly jumps to π .

(ii) With damping present, the phase lag (θ) increases from 0 to $\pi/2$ as the driving frequency p increases from 0 to ω_0 and then approaches π as p continues to increase beyond ω_0 .

(iii) The rate of change of the phase angle is more rapid when the damping is low than when it is high.

(iv) Except at very high damping, all curves pass through $\pi/2$ when $p = \omega_0$, i.e., amplitude resonance occurs when the driving frequency is equal to the natural, undamped frequency of the oscillator, or the resonance frequency in all these cases remains ω_0 .

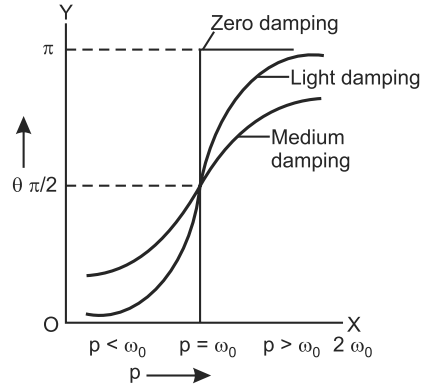


Fig. 8.10

8.9 VELOCITY RESONANCE

The displacement of a driven oscillator at a given instant t , is, as we know, given by

$$x = A \sin(pt - \theta) = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}} \sin(pt - \theta) \quad \dots \text{I}[\S 8.6]$$

$\therefore$ the velocity of the driven oscillator at that instant is $v = dx/dt = pA \cos(pt - \theta) = v_0 \cos(pt - \theta)$, because

$$pA = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}} = v_0.$$

Or,

$$v = v_0 \sin(pt - \theta + \pi/2),$$

showing that the velocity leads the displacement in phase by $\pi/2$.

Clearly, the velocity amplitude $v_0 = pA = f_0 p / \sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}$ and 0, as we know, equal to $\tan^{-1}[2kp/(\omega_0^2 - p^2)]$.

Thus, the velocity amplitude (v_0) varies with p , being zero when $p = 0$ and the maximum when $p = p_R = \omega_0$, this maximum value being $f_0/2k = f_0\tau$.

This acquiring of its maximum velocity by a driven oscillator is spoken of as velocity resonance and obviously occurs when the driving frequency (p) is equal to the resonant frequency (p_R), equal to the natural, undamped frequency of the oscillator, $\omega_0 = C/m$.

For all other values of p , greater or smaller than $p_R = \omega_0$, the velocity amplitude is naturally smaller, because the displacement amplitude (A) is then much smaller.

Now, as we know, the displacement at resonance lags in phase by $\pi/2$ behind the driving force and, as we have just seen, the velocity, then, leads the displacement in phase by $\pi/2$, i.e., the displacement also lags in phase by $\pi/2$ behind the velocity. Obviously, therefore, *at resonance, the velocity of the driven oscillator is in phase with the driving force*. This is, therefore, the most favourable circumstance for the transference of energy from the driving force (F) to the driven oscillator, for both F and v being in the same phase, with v at its maximum, the rate of transference of energy, Fv , has its highest positive value.

8.10 HALF WIDTH OF RESONANCE CURVE

The frequency—amplitude curve of a driven oscillator, (Fig. 8.9), is, as we know, symmetrical (for small damping) about the resonant frequency $p_R = \omega_0$, indicating that its amplitude, which is the maximum (A_{max}) at p_R , falls on either side of it. There must, therefore, be a frequency p_H , say, for which the amplitude will be half of the maximum, i.e., equal to $A_{max}/2$, as shown in Fig. 8.11.

The change Δp , (from p_R to p_H), in the value of the driving frequency p for which the amplitude of the driven harmonic oscillator falls from its maximum value A_{max} to half this maximum value $A_{max}/2$ is called the half width of the resonance curve.

Thus, half width of the resonance curve, $\Delta p = |p_H - p_R|$.

Now, as we know, (§ 8.6, Expression (i)),

$$A_{max} = \frac{f_0}{2k(\omega_0^2 - k^2)^{1/2}} = \frac{f_0}{\sqrt{4k^2\omega_0^2 - 4k^4}}.$$

$$\therefore A_{max}/2 = \frac{f_0}{2\sqrt{4k^2\omega_0^2 - 4k^4}} \quad \dots(i)$$

Now,

$$\begin{aligned} A &= \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2p^2}} \\ &= \frac{f_0}{\sqrt{(\omega_0^2 - 2k^2 - p^2)^2 + 4k^2\omega_0^2 - 4k^4}} \quad [\S 8.6, \text{Exp. III.}] \end{aligned}$$

$$\text{So that, } A_{max}/2 \text{ also} = \frac{f_0}{\sqrt{(\omega_0^2 - 2k^2 - p_H^2)^2 + 4k^2\omega_0^2 - 4k^4}} \quad [\because p \neq p_H \text{ when } A = A_{max}/2.] \dots(ii)$$

From the two values of $A_{max}/2$, as given by relations (i) and (ii), we, therefore, have

$$(\omega_0^2 - 2k^2 - p_H^2)^2 + 4k^2\omega_0^2 - 4k^4 = 4(4k^2\omega_0^2 - 4k^4)$$

But, as we know, $(\omega_0^2 - 2k^2) = p_R^2$. And, therefore,

$$(p_R^2 - p_H^2)^2 = 3(4k^2\omega_0^2 - 4k^4) = 3(4k^2p_R^2 - 4k^4) \quad [\because \omega_0 = p_R.]$$

Since with low damping, k is small and, therefore, $4k^4$ quite negligible, we have

$$(p_R^2 - p_H^2)^2 = 3(4k^2p_R^2), \text{ whence, } p_R^2 - p_H^2 = \pm\sqrt{3}(2kp_R).$$

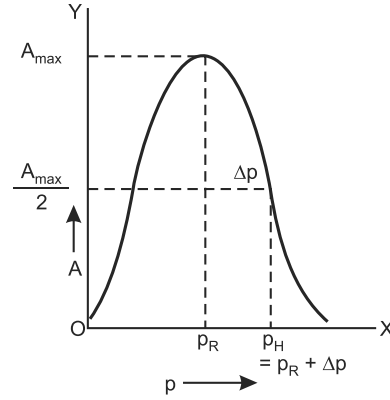


Fig. 8.11

Or,
$$p_H^2 = p_R^2 \mp \sqrt{3}(2kp_R) = p_R^2 \left(1 \mp \sqrt{3} \frac{2k}{p_R}\right).$$

Or,
$$p_H = p_R \left(1 \mp \sqrt{3} \frac{2k}{p_R}\right)^{1/2} \approx p_R \left(1 \mp \frac{\sqrt{3}k}{p_R}\right) = p_R \mp \sqrt{3}k.$$

$\therefore$ for low damping, half width of the resonance curve, $\Delta p = (p_H - p_R) \approx \sqrt{3}k$.

It is thus possible to estimate the value of the damping constant, as also, of course, that of ω_0 , from the frequency-amplitude curve of a driven harmonic oscillator.

8.11 POWER ABSORPTION

In many an important problem on driven harmonic oscillators, as for example, in discussing the driven oscillations of an electron about a point of stable equilibrium, we are interested not so much in its displacement from the equilibrium, position—which, in the case of an electron, at any rate, we cannot even hope to observe directly—as in the energy absorbed by the oscillator to keep itself in motion.

We shall, therefore, calculate the average power absorbed per cycle by the driven oscillator to offset its loss of power in overcoming the frictional or resistive forces and thus to maintain its oscillations. *When the oscillator has settled down to a steady state of oscillation, the average power absorbed will obviously be just equal to the average power dissipated.*

If $F = F_0 \sin pt$ be the applied or the driving force and dx , the displacement of the oscillator in time dt , we have

work done or energy supplied by the applied force to the oscillator, is given by

$$dE = Fdx = F \frac{dx}{dt} dt, \text{ whence, } \frac{dE}{dt} = F \frac{dx}{dt}. \quad \dots(i)$$

Now, $F = F_0 \sin pt = mf_0 \sin pt$ and

$$\frac{dx}{dt} = \frac{pf_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}} \cos(pt - \theta) \quad \dots(ii)$$

[From relation I, § 8.9]

Substituting these values in relation (i) above, we have

$$\begin{aligned} \text{power absorbed by the oscillator, } P &= \frac{dE}{dt} = F \frac{dx}{dt} \\ &= mf_0 \sin pt \frac{pf_0 \cos(pt - \theta)}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}}. \end{aligned}$$

Or,
$$P = F \frac{dx}{dt} = \frac{mf_0^2 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}} \sin pt \cos(pt - \theta).$$

To obtain the average power, $P_{(av)}$, over a full cycle or time-period, we note that the average value of $\sin pt \cos(pt - \theta)$

$$= \frac{1}{2} [\sin(2pt - \theta) - \sin \theta] \text{ over a cycle or time-period } T \text{ is}$$

$$\frac{1}{T} \int_0^T \frac{1}{2} [\sin(2pt - \theta) - \sin \theta] dt = \frac{1}{2} \sin \theta.$$

And, since $2kAp = f_0 \sin \theta$ and $A = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}}$, [See § 8.6 (ii) and III]

we have
$$\sin \theta = \frac{2kAp}{f_0} = \frac{2kp}{f_0} \cdot \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}}$$

$$= \frac{2kp}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}}.$$

So that, average value of $\sin pt \cos (pt - \theta)$ over a full cycle or time-period

$$= \frac{1}{2} \sin \theta = \frac{1}{2} \frac{2kp}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}}.$$

$$= \frac{kp}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}}$$

$\therefore$ average power,
$$P_{(av)} = \frac{mf_0^2 p}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}} \cdot \frac{kp}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}}$$

$$= \frac{mkf_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4k^2 p^2} \quad \dots(iii)$$

But $\frac{f_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4k^2 p^2} = v_0^2$, where v_0 is the velocity amplitude (See § 8.9).

Therefore, average power absorbed by the driven oscillator per cycle or time-period,

$$P_{(av)} = mkv_0^2 = mv_0^2/2\tau = \gamma v_0^2/2 \quad [\because k = 1/2\tau \text{ and } m/\tau = \gamma.] \quad \dots(iv)$$

It is clear from relation (iii) above that the average power absorbed will have its *maximum value*, P_{max} , when $p = \omega_0$. So that,

$$\text{maximum power absorbed, } P_{max} = \frac{1}{4} \frac{mf_0^2}{k} = \frac{1}{2} mf_0^2 \tau. \quad \dots(v)$$

Thus, $\omega_0 = p_R$, the resonant frequency for velocity also turns out to be the resonant frequency for the average power absorbed.

It may be noted that the maximum power is absorbed at the frequency of velocity resonance, $p_R = \omega_0$, and not at the frequency of amplitude resonance, $\sqrt{\omega_0^2 - 2k^2}$.

Half width of average power absorbed versus frequency curve. As we have just seen above (expression v), the maximum value of average power absorbed is $P_{max} = mf_0^2/4k$ at resonance frequency $p_R = \omega_0$. So that, half its maximum value of average power absorbed $= P_{max}/2 = mf_0^2/8k$. Let us calculate the frequency at which this is the average power of absorbed.

We have relation (iii) above for average power absorbed ($P_{(av)}$). So that, if $P_{(av)} = P_{max}/2 = mf_0^2/8k$, we have

$$P_{(av)} = \frac{mkf_0^2 p^2}{(\omega_0^2 - p^2)^2 + 4k^2 p^2} = \frac{mf_0^2}{8k},$$

whence,
$$\frac{kp^2}{(\omega_0^2 - p^2)^2 + 4k^2 p^2} = \frac{1}{8k}.$$

Or,
$$(\omega_0^2 - p^2)^2 + 4k^2 p^2 = 8k^2 p^2.$$

Or,
$$(\omega_0^2 - p^2)^2 = 4k^2 p^2,$$

whence,
$$\omega_0^2 - p^2 = \pm 2kp$$

Or,
$$\omega_0 - p = \pm \frac{2kp}{\omega_0 + p} = \frac{2k}{\omega_0/p + 1}$$

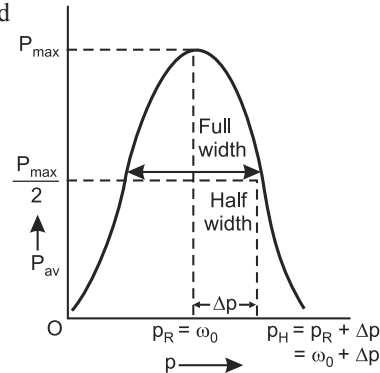


Fig. 8.12

$$= \pm \frac{2k}{2} = +k. \quad [\because \omega_0 \approx p.]$$

Thus, $p = \omega_0 \mp k = p_H$, say.

Or, since $\omega_0 = p_R$, we have $|p_H - p_R| = \Delta p = k$.

This change Δp , (from p_R to p_H), in the value of the driving frequency p for which the average power absorbed by the driven oscillator falls from its maximum value P_{max} to half this value $P_{max}/2$ is called the half width of the average power absorbed versus driving frequency curve, shown in Fig. 8.12.

So that, *half width of the average power absorbed versus driving frequency curve*, $\Delta p = k = 1/2\tau$.

Quality factor Q of the driven oscillator. We know that $Q = \omega_0\tau$. (§8.4).

Here, $\Delta p = k = 1/2\tau$ and $\therefore \tau = 1/2 \Delta p$.

$$\text{So that, } Q = \omega_0\tau = \omega_0/2\Delta p = \frac{\text{Resonance frequency}}{\text{Full width of power-frequency curve at half maximum power}}$$

It may, however, also be obtained directly as follows:

$$\begin{aligned} \text{By definition, } Q &= 2\pi \frac{\text{average energy stored}}{\text{energy dissipated per cycle or per time-period}} \\ &= \frac{E_{(av)}}{P_{(av)} \times T} \end{aligned}$$

Now, at any instant,

$$\begin{aligned} E &= K.E. + P.E. \\ &= \frac{1}{2}mp^2A^2\cos^2(pt - \theta) + \frac{1}{2}m\omega_0^2A^2\sin^2(pt - \theta). \end{aligned}$$

Since the average value of $\cos^2(pt - \theta)$ and $\sin^2(pt - \theta)$ over a whole time-period is $\frac{1}{2}$ each and the average power absorbed, $P_{(av)} = mv_0^2/2\tau = mA^2p^2/2\tau$, we have

$$Q = 2\pi \frac{\frac{1}{4}mp^2A^2 + \frac{1}{4}m\omega_0^2A^2}{(mA^2p^2/2\tau) \times T} = 2\pi \frac{\frac{1}{4}mA^2(p^2 + \omega_0^2)}{(mA^2p^2/2\tau) \times (2\pi/6)} = \frac{1}{2} \frac{(p^2 + \omega_0^2)}{p} \tau \quad [\because T = 2\pi/p.]$$

$$\text{Or, } Q = \frac{1}{2}(p + \omega_0^2/p)\tau = \frac{1}{2}(1 + \omega_0^2/p^2)p\tau.$$

Since at, or near, resonance, $p = \omega_0$, we have

$$Q = \frac{1}{2}(2)p\tau = p\tau = \omega_0\tau = \omega_0/2 \Delta p, \text{ as obtained above.}$$

At low damping, obviously $k = 1/2\tau$ will be small or τ large, so that $Q = \omega_0\tau$ will also be large, making for sharpness of resonance. Thus, the *quality factor Q is a measure of sharpness of resonance in the case of a driven harmonic oscillator.*

8.12 SUPERPOSITION PRINCIPLE

This principle is a general one, applicable to all cases where the equations governing motion or displacement are linear. It merely states that *solutions in all such cases are additive*, i.e., if x_1 be the displacement due to a force F_1 and x_2 due to another force F_2 , the displacement due to the combined force $(F_1 + F_2)$ is equal to $(x_1 + x_2)$. This process of vector addition of displacements is termed *superposition*.

It is thus possible to obtain the displacement under the combined force $(F_1 + F_2)$ if the displacement under each of the two forces individually be known.

The principle is also applicable to electromagnetic waves because the relationship between electrical and magnetic fields is linear. It, however, fails to apply the moment the displacements or disturbances are large and the ordinary linear laws of mechanical action cease to hold. Thus, in the case of sound, it certainly applies to ordinary sound waves but not to shock waves caused by violent explosions.

In our present case of harmonic oscillators it applies because of the linearity of their equations of motion. It ceases to apply when this linearity is lost. Thus, it does not apply in the case of an harmonic oscillators or even in the case of harmonic oscillators with large displacements when the linear relationship given by Hooke's law no longer holds.

In view of this principle, therefore, we can look upon the response of a harmonic oscillator to a sinusoidal force as being the sum of its responses to a number of components of various frequencies into which the given force may be imagined to be resolved.

8.13 DRIVEN L-C-R CIRCUIT

An LCR circuit, *i.e.*, a circuit containing an inductance (L), a capacitance (C) and a resistance (R) in series, (Fig. 8.13), functions as a driven oscillator if it be connected to an external source of alternating *emf* to supply the necessary energy to maintain the oscillations.

If I be the current in the circuit at a given instant, with Q as the charge on the capacitor at that instant, we have *potential difference across the inductance* $= -L dI/dt$, *potential difference across the capacitor* $= Q/C$ and *potential difference across the resistance* $= RI$.

If, therefore, $E = E_0 \sin pt$ be the external *emf* applied to the circuit, we have

$$E - L(dI/dt) = (Q/C) + RI.$$

$$\text{Or, } L(dI/dt) + RI + (Q/C) = E = E_0 \sin pt \quad \dots(i)$$

$$\text{Or, } dI/dt + (R/L) I + Q/LC = (E_0/L) \sin pt.$$

Or, since $I = dQ/dt$, we have

$$d^2Q/dt^2 + (R/L)(dQ/dt) + Q/LC = (E_0/L) \sin pt.$$

This is an equation of an identical form with that for a driven harmonic oscillator, discussed earlier (Equation I, §8.6), with Q replacing x and R/L , $1/LC$ and E_0/L replacing $2k$, ω_0^2 and f_0 respectively.

If steady state solution too is similarly

$$Q = \frac{E_0/L}{\sqrt{(1/LC - p^2)^2 + (pR/L)^2}} \sin(pt - \theta), \quad \dots(ii)$$

where

$$\theta = \tan^{-1} \frac{pR/L}{1/LC - p^2}.$$

We can easily obtain the value of the current flowing through the circuit by differentiating this expression for Q with respect to t , just as we can obtain the value of the velocity v of the oscillator by differentiating the earlier expression for x with respect to t .

Alternatively, we can proceed directly with a trial solution $I = I_0 \sin(pt - \phi)$

So that, $dI/dt = I_0 p \cos(pt - \phi)$

$$\text{and } Q = \int I dt = \int I_0 \sin(pt - \phi) dt = \frac{I_0}{p} \cos(pt - \phi).$$

Substituting these values in equation (i) above, we have

$$LpI_0(\cos pt - \phi) + RI_0 \sin(pt - \phi) - \frac{I_0}{Cp} \cos(pt - \phi) = E_0 \sin pt.$$

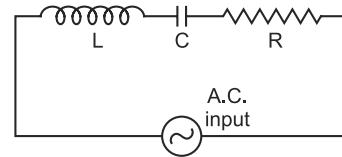


Fig. 8.13

$$\text{Or, } I_0[R \sin(pt - \phi) + (Lp - 1/Cp) \cos(pt - \phi)] = E_0 \sin pt \quad \dots(iii)$$

Now, putting $R = a \cos \phi$ and $(Lp - 1/Cp) = a \sin \phi$, we have

$$a = \sqrt{R^2 + (Lp - 1/Cp)^2} \text{ and } \tan \phi = (Lp - 1/Cp)/R.$$

Substituting these values of R and $(Lp - 1/Cp)$ in relation (iii) above, we have
 $I_0 a [\cos \phi \sin(pt - \phi) + \sin \phi \cos(pt - \phi)] = E_0 \sin pt.$

$$\text{Or, } I_0 a \sin[(pt - \phi) + \phi] = I_0 a \sin pt = E_0 \sin pt.$$

$$\text{Or, } I_0 \sqrt{R^2 + (Lp - 1/Cp)^2} \sin pt = E_0 \sin pt,$$

$$\text{whence, } I_0 \sqrt{R^2 + (Lp - 1/Cp)^2} = E_0.$$

$$\text{Or, } I_0 = \frac{E_0}{\sqrt{R^2 + (Lp - 1/Cp)^2}} \quad \dots(iv)$$

$$\therefore I = I_0 \sin(pt - \phi) = \frac{E_0}{\sqrt{R^2 + (Lp - 1/Cp)^2}} \sin(pt - \phi), \quad \dots(v)$$

$$\text{where } \phi = \tan^{-1} (Lp - 1/Cp)/R.$$

It will be readily seen that in expression (iv) for I_0 , the denominator $\sqrt{R^2 + (Lp - 1/Cp)^2}$ functions as the *effective resistance* in the circuit. It is called **impedance** of the circuit, denoted by the letter Z and measured in *ohms*.

Clearly, the impedance is made up of two parts, the *ohmic* resistance R and the quantity $(Lp - 1/Cp)$, called **reactance**, usually denoted by X and also measured in *ohms* (though L and C are individually measured in *henry* and *farad* respectively).

The relation between *resistance*, *reactance*, *impedance* and ϕ is best illustrated as in Fig. 8.14.

The *reactance* too, as can readily be seen, is made up of two parts: Lp , the *reactance due to inductance*, usually denoted by X_L and $1/Cp$, the *reactance due to capacitance*, denoted by X_C ; so that,

$$X = X_L + X_C.$$

$$\text{We thus have } Z = \sqrt{R^2 + X^2}.$$

$$\text{Or, } \text{Impedance} = \sqrt{\text{Resistance}^2 + \text{Reactance}^2}.$$

And, clearly, *peak value or amplitude* of the current, i.e.,

$$I_0 = E_0/Z = E_0/\sqrt{R^2 + X^2}.$$

$$\text{And, therefore, } I = \frac{E_0}{Z} \sin(pt - \phi) = I_0 \sin(pt - \phi).$$

Obviously, the *current* and *emf* in the circuit differ in phase by ϕ ,

$$\begin{aligned} \text{where } \phi &= \tan^{-1} \frac{Lp - 1/Cp}{R} = \tan^{-1} \left(\frac{\text{Reactance}}{\text{Resistance}} \right) \\ &= \tan^{-1} \left(\frac{X_L - X_C}{R} \right). \end{aligned}$$

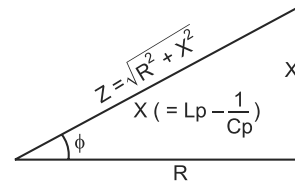


Fig. 8.14

Now, whereas the *ohmic resistance* (R) has no effect on the phase angle (because it is quite independent of the frequency), the inductance in a circuit makes the emf lead the current (in phase), and the capacitance in a circuit makes the current lead the emf. When both are present in a circuit, therefore, the lead or lag of the emf or the current depends upon the relative values of X_L or Lp and X_C or $1/Cp$ and the following **three cases** arise:

(i) **When the value of p is such that $X_L = X_C$.** In this case, obviously, $Lp = 1/Cp$, i.e., the reactance in the circuit, $X = X_L - X_C = 0$. So that,

impedance in the circuit = resistance R , and, therefore, the least.

$\therefore$ amplitude or peak value of the current, $I_0 = E_0/R$, the maximum, which approaches ∞ as $R \rightarrow 0$.

Obviously, here, $\tan \phi = \frac{X_L - X_C}{R}$ and $\therefore \phi = 0$, i.e., the current in the circuit is in phase with the applied emf.

This is, therefore, the case of resonance, with $p^2 = 1/LC$ and $\therefore p = 1/\sqrt{LC} = \omega_0$.

Thus, resonance frequency = $p/2\pi = \omega_0/2\pi = 1/2\pi\sqrt{LC}$.

The circuit is, therefore, called a **series resonance circuit**.

It may be noted that current resonance here is analogous to velocity resonance in the case of mechanical oscillators.

(ii) **When p has a value such that $X_L > X_C$.** Here, since $Lp > 1/Cp$, the net reactance in the circuit is inductive, $\tan \phi$, a positive quantity and, therefore, $\phi = \pi/2$. The applied emf thus leads the current by $\pi/2$ and the value of p is greater than ω_0 .

(iii) **When the value of p is such that $X_L < X_C$.** In this case, Lp being less than $1/Cp$, the net reactance in the circuit is capacitive, $\tan \phi$, a negative quantity and, therefore, $\phi = -\pi/2$. The current thus leads the applied emf by $\pi/2$ and the value of p is less than ω_0 .

These changes in the peak value of the current I_0 and in the value of the phase angle ϕ with p are shown in Figs. 8.15(a) and (b) respectively, for different values of R , which corresponds to k in the case of mechanical oscillators.

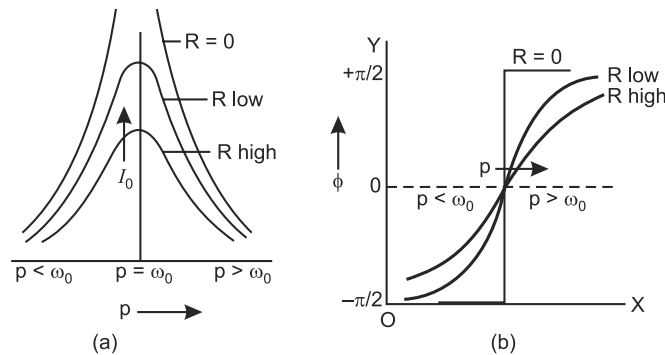


Fig. 8.15

It will be seen at once that the lower the value of R , the sharper the resonance and that the current lags or leads the emf according as p is greater than or less than ω_0 .

8.15 PARALLEL RESONANCE CIRCUIT

When the inductance (L) and the capacitance (C) in a circuit are placed in parallel (instead of in series), as shown in Fig. 8.16, the applied voltage is the same across both inductance and capacitance.

Now, as we know, the current I_L through the inductance *lags*, and the current I_C through the capacitance *leads*, the *emf* the by $\pi/2$, i.e.,

$$I_L = \frac{E_0}{Lp} \sin(pt - \pi/2)$$

and

$$I_C = \frac{E_0}{1/Cp} \sin(pt + \pi/2)$$

$\therefore$ total current in the circuit, $I = I_L + I_C$

$$= \frac{E_0}{Lp} \sin(pt - \pi/2)$$

$$+ \frac{E_0}{1/Cp} \sin(pt + \pi/2)$$

Or, $I = E_0(Cp - 1/Lp) \cos pt.$

The current thus leads or lags the *emf* by a phase angle π .

For a particular value of p , we have $Cp = 1/Lp$, when $p^2 = 1/LC$ or $p = 1/\sqrt{LC}$. The current through the inductance is equal in magnitude to that through the capacitance, the two differing in phase by π . The net current through the circuit is thus *zero*.

This frequency p is called the *resonant frequency* of the circuit and the circuit itself is referred to as a *parallel resonance circuit*.

As in the series resonant circuit, therefore, we have $p = 2\pi n = 1/\sqrt{LC}$ or $n = 1/2\pi\sqrt{LC}$.

The circuit, under this condition, oscillates with its natural frequency, without requiring any supply of energy from an outside source. In actual practice, however, the inductance coil does possess some resistance and the oscillations of the circuit can be maintained only by some supply of energy from an external source.

It will be seen at once that *whereas in a series circuit, the current at resonance is the maximum and the impedance minimum (or zero), in a parallel circuit, the current at resonance is zero and the impedance maximum*.

The parallel resonance circuit is thus used in wireless transmitting circuits to filter or to cut out the current of resonant frequency, allowing currents of other frequencies to pass through the circuit. It is also, therefore, called a *rejector circuit*.

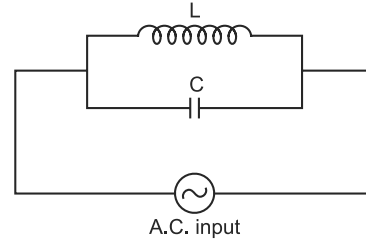


Fig. 8.16

8.16 R.M.S. OR EFFECTIVE VALUES OF ALTERNATING CURRENT AND EMF

Since an alternating current and *emf* vary continuously from a maximum in one direction, through zero, to a maximum in the opposite direction, their average values over one complete cycle or time-period are zero each. We, therefore, obtain their mean or average value $I_{(av)}$ and $E_{(av)}$ over half their time-periods and these can be shown to be equal to $2I_0/\pi$ and $2E_0/\pi$ respectively.

For most of our purposes, however, we are interested in knowing the *effective value* of an alternating current or an alternating *emf* in terms of its equivalent direct current or *emf*.

This effective or virtual value of an alternating current is defined as that steady (or direct) current which would produce the same heating effect as the alternating current in a given resistance in a given time.

Thus, if I_v be the effective or virtual value of an alternating current $I = I_0 \sin \omega t$, clearly, the heating effect produced by this steady or direct current in a resistance R in time $t = I_v^2 Rt$.

And the heating effect produced by the alternating current in the same resistance in the same time $t = (\text{average value of } I^2 = I_0^2 \sin^2 \omega t \text{ over a cycle or time-period } T) \times R \times t = (I_0^2/2) \times R \times t$, (because the average value of $\sin^2 \omega t$ over a whole cycle or time-period is $1/2$).

We, therefore, have $I_v^2 R t = (I_0^2/2) R t$. Or, $I_v^2 = I_0^2/2$, whence, $I_v = I_0/\sqrt{2}$, i.e., **effective or virtual value of alternating current** $= I_0/\sqrt{2}$.

Since this is the square root of the mean square value ($I_0^2/2$) of the current, it is also called the *root mean square* or the *r.m.s. value* of the alternating current.

Similarly, we have **effective, virtual or r.m.s value of alternating emf** equal to $E_0/\sqrt{2}$.

8.17 POWER IN AN ALTERNATING CURRENT CIRCUIT

The power (P) in an electrical circuit is the amount of energy delivered per second by the source to the circuit, and in a direct current (or D.C) circuit is measured by the product EI , where E is the emf in the circuit and I , the current.

Since in a D.C. circuit, we have only resistance and no inductance or capacitance, $E = IR$ and, therefore, power $P = I^2 R$, appearing in the form of heat.

In an *alternating current* (or A.C.) circuit, say in an LCR circuit, the emf and current both vary continuously, sinusoidally, and are given by $E = E_0 \sin pt$ and $I = I_0 \sin (pt - \phi)$ respectively. We, therefore, have

$$\text{power } P = EI = (E_0 \sin pt) [I_0 \sin (pt - \phi)] = \frac{E_0 I_0}{2} [\cos \phi - \cos (2pt - \phi)]$$

Since the average value of $\cos (2pt - \phi)$ over a whole cycle or time-period is zero, we have

$$P = \frac{E_0 I_0}{2} \cos \phi = \frac{E_0}{\sqrt{2}} \frac{I_0}{\sqrt{2}} \cos \phi = E_v I_v \cos \phi, \quad [\because E_0/\sqrt{2} = E_v \text{ and } I_0/\sqrt{2} = I_v]$$

where $\cos \phi = R/Z = R/\sqrt{R^2 + (Lp - 1/Cp)^2}$. [See Fig. 8.14.]

The unit of power is the **watt** = 1 joule/sec.

Obviously, the maximum power will be $E_v I_v$ (when $\cos \phi = 1$) and, therefore,
$$\frac{\text{True power expended}}{\text{Maximum power}} = \frac{E_v I_v \cos \phi}{E_v I_v} = \cos \phi.$$

Thus, $\text{true power expended} = \text{maximum power} \times \cos \phi$.

This is why $\cos \phi = R/Z$ is called the **power factor** of the A.C. circuit.

As we know, $\phi = 0$ and, therefore, $\cos \phi = 1$ only in the case of resonance, for reactance is then zero and impedance $Z = R$. So that, in this case, $\text{true power} = \text{maximum power} = E_v I_v = I_v^2 R$.

[$\because E_v = I_v R$ here.]

It may as well be pointed out that, even otherwise, the power in an A.C. circuit, like that in a D.C. circuit, appears in the form of heat in the resistance alone, because the power expended in a pure inductance or a pure capacitance is always zero. This may also be seen from the following:

We have $P = E_v I_v \cos \phi$. Or, since $E_v = I_v Z$, we have

$$P = (I_v Z) I_v \cos \phi.$$

Or, $P = I_v^2 Z(R/Z) = I_v^2 R,$

showing that the power appears in the form of heat in the resistance R .

Finally, as in the case of a mechanical oscillator, so also here, the *quality factor of the circuit*, $Q = Lp/R = L\omega_0/R$ ($\because p \approx \omega_0$ at or near resonance) $= \omega_0 \tau = \omega_0/2 \Delta p = \text{frequency at resonance}/\text{full width of power frequency curve at half maximum power}$, and is a measure of the sharpness of resonance of the LCR circuit. See § 8.11).

WORKED EXAMPLES

I—Damping force—Damped harmonic oscillator

Example 8.1. A particle of mass 5 kg lies in a potential field $V = 8x^2 + 200$ Joules/kg. Calculate its time period. (Nagpur U.2004)

Solution. Potential energy (U) of a particle

$$\begin{aligned} U &= mV \\ &= m(8x^2 + 200) \end{aligned}$$

Restoring force (F) acting is given by

$$\begin{aligned} F &= -\frac{dU}{dx} \\ &= -\frac{d}{dx}(8mx^2 + 200m) = -16mx \\ &= -kx \end{aligned} \quad (\text{where } k = 16m, \text{ a constant})$$

As $f \propto -x$, the motion is S.H.M.

$\therefore$ Angular frequency,

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{16m}{m}} = 4$$

and Time period,

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = 1.57 \text{ sec.}$$

Example 8.2. A mass of 25×10^{-2} kg is suspended from the lower end of a vertical spring having a force constant 25 Nm^{-1} . What should be the damping constant of the system so that the motion is critically damped? (Nagpur U. 2004, 2001)

Solution. Here $m = 25 \times 10^{-2}$ kg; Force (spring) constant, $s = 25 \text{ Nm}^{-1}$

Damping constant $r = ?$

For critical damping $\frac{r^2}{4m^2} - \frac{s}{m} = 0$

or

$$\begin{aligned} r^2 &= 4ms \text{ or } r = 2\sqrt{ms} = 2\sqrt{25 \times 10^{-2} \times 25} \\ &= 2 \times 25 \times 10^{-1} = 5 \text{ kg s}^{-1} \end{aligned}$$

Example 8.3. A mass 25×10^{-3} kg is suspended from the lower end of a vertical spring having a force constant 25 N/m . The mechanical resistance of the system is 1.5 Ns/m . The mass is displaced vertically and released. Find whether the motion is oscillatory? If so, calculate its period of oscillation.

Solution. Given : $m = 25 \times 10^{-3}$ kg, $s = 25 \text{ N/m}$, $r = 1.5 \text{ Ns/m}$

$$b^2 = \frac{r^2}{4m^2} = \frac{(1.5)^2}{4 \times (25 \times 10^{-3})^2} = \frac{2.25}{25 \times 10^{-4}} = 900$$

and

$$\omega^2 = \frac{s}{m} = \frac{25}{25 \times 10^{-3}} = 10^3 = 1000$$

Since, $b^2 < \omega^2$, the motion is oscillatory.

$\therefore$ Period,

$$T = \frac{2\pi}{\sqrt{\frac{s}{m} - \frac{r^2}{4m^2}}} = \frac{2 \times 3.14}{\sqrt{1000 - 900}} = \frac{6.28}{\sqrt{100}} = \frac{6.28}{10} = 0.628 \text{ s.}$$

Example 8.4. A mass of 1 kg is suspended from a spring of stiffness constant 25 N m^{-1} . If the undamped (or natural) frequency is $\frac{2}{\sqrt{3}}$ times the damped frequency, calculate the damping factor (or constant).

Solution. Damped frequency $f' = \frac{1}{2\pi} \sqrt{\frac{s}{m} - \frac{r^2}{4m^2}}$

and undamped frequency $f_0 = \frac{1}{2\pi} \sqrt{\frac{s}{m}}$

Now $\frac{f_0}{f'} = \frac{2}{\sqrt{3}}$

or $\frac{\sqrt{\frac{s}{m}}}{\sqrt{\frac{s}{m} - \frac{r^2}{4m^2}}} = \frac{2}{\sqrt{3}}$ or $\frac{\frac{s}{m}}{\frac{s}{m} - \frac{r^2}{4m^2}} = \frac{4}{3}$

or $\frac{3}{4} \frac{s}{m} = \frac{s}{m} - \frac{r^2}{4m^2}$

or $\frac{r^2}{4m^2} = \frac{1}{4} \frac{s}{m}$

$\therefore r^2 = sm$

Given $s = 25 \text{ Nm}^{-1}$ and $m = 1 \text{ kg}$

$r^2 = 25 \text{ Nm}^{-1} \text{ kg} = 25 \text{ kg}^2 \text{ s}^{-2}$ [$\because N = \text{kg ms}^{-2}$]

$\therefore r = 5 \text{ kg s}^{-1}$

Example 8.5. Show that the unit of damping term (or damping coefficient) b is s^{-1} .

(G.N.D.U. 2001; H.P.U. 2002)

Solution. For a damped harmonic oscillator the equation of motion is

$$m \frac{d^2 y}{dt^2} + r \frac{dy}{dt} + sy = 0$$

or $\frac{d^2 y}{dt^2} + \frac{r}{m} \frac{dy}{dt} + \frac{s}{m} y = 0$

Substituting $\frac{r}{m} = 2b$ and $\frac{s}{m} = \omega^2$, we get

$$\frac{d^2 y}{dt^2} + 2b \frac{dy}{dt} + \omega^2 y = 0$$

$\therefore$ The unit of $r \frac{dy}{dt}$ is that of force, i.e., Newton

$\therefore$ Unit of damping constant r is $\frac{\text{Force}}{\text{velocity}} = \text{Nm}^{-1} \text{ s} = \text{kg ms}^{-2} \text{ m}^{-1} \text{ s} = \text{kgs}^{-1}$

Hence unit of damping (coefficient) or term $b = \frac{r}{m} = \frac{\text{kg s}^{-1}}{\text{kg}} = \text{s}^{-1}$ i.e., b has the dimensions of frequency.

Example 8.6. In an oscillatory circuit $L = 0.5 \text{ H}$, $C = 1.8 \text{ } \mu\text{fd}$ what is the maximum value of resistance to be connected so that the circuit may produce oscillations. (Nagpur U., 2006)

Solution. $L = 0.5 \text{ H}$ $C = 1.8 \text{ } \mu\text{fd} = 1.8 \times 10^{-6} \text{ F}$.

Let R be the maximum resistance for which the discharge is oscillatory. For the circuit to produce oscillations,

$$\frac{R^2}{4L^2} = \frac{1}{LC}$$

or
$$R = 2\sqrt{\frac{L}{C}} = 2 \times \sqrt{\frac{0.5}{1.8 \times 10^{-6}}} = 1054 \text{ ohm}$$

Example. 8.7. The frequency of an underdamped harmonic oscillator is adjusted to be equal to half the frequency experienced by the oscillator without damping. Calculate the logarithmic decrement of the system.

Solution. Let the frequency of the harmonic oscillator without damping be ω_0 and the frequency of the underdamped harmonic oscillator be ω' , then,

$$\omega' = \frac{1}{2} \omega_0$$

Now
$$\omega' = \sqrt{\omega_0^2 - b^2} \text{ where } b \text{ is the damping coefficient.}$$

$$\therefore \frac{1}{2} \omega_0 = \sqrt{\omega_0^2 - b^2} \text{ or } \frac{\omega_0^2}{4} = \omega_0^2 - b^2$$

or
$$b^2 = \omega_0^2 - \frac{\omega_0^2}{4} = \frac{3}{4} \omega_0^2$$

$$b = \frac{\sqrt{3}}{2} \omega_0 = \sqrt{3} \omega'$$

Now logarithmic decrement $\lambda = bT$ and $T = \frac{2\pi}{\omega'}$

$$\therefore \lambda = \sqrt{3} \omega' \cdot \frac{2\pi}{\omega'} = 2\pi\sqrt{3}$$

Example 8.8. A damped vibrating system, starting from rest, reaches a first amplitude of 500 mm which reduces to 50 mm in that direction after 100 oscillations each of period 2.3 sec. Find the damping constant, relaxation time and the correction for the first displacement for damping [$\log_e^{10} = 2.3$]. (Indore U. 2001)

Solution. The damping constant b is given by the relation

$$\log_e \frac{A_1}{A_2} = \log_e \frac{A_2}{A_3} = \dots = b$$

where A_1, A_2, A_3 etc. are the successive amplitudes and T the time period. The amplitude n vibrations after the first i.e., $n + 1$ vibration is A_{n+1} , then

$$\log_e \frac{A_1}{A_{n+1}} = \log_e \frac{A_1}{A_2} \times \log_e \frac{A_2}{A_3} \times \dots \times \log_e \frac{A_n}{A_{n+1}} = nkT$$

Now $A_1 = 500 \text{ mm}$; $A_{101} = 50 \text{ mm}$; $n = 100$; $T = 2.3 \text{ sec}$

$$\therefore \log_e \frac{500}{50} = 100 b \times 2.3$$

or
$$2.3 \log_{10} 10 = 2.3 \times 100 b$$

or
$$100 b = 1$$

$$\therefore \text{Damping constant } b = \frac{1}{100} = 0.01$$

Relaxation time
$$t_r = \frac{1}{b} = \frac{1}{0.01} = 100 \text{ Sec.}$$

Logarithmic decrement $\lambda = bT = 0.01 \times 2.3 = 0.023$

The amplitude A_0 which could be obtained if the damping force were absent, is given by

$$A_0 = A_1 \left(1 + \frac{\lambda}{4} \right) = 500 \left(1 + \frac{1}{4} \times 0.01 \times 2.3 \right) = 502.875$$

$\therefore$ Correction for the first displacement = 2.875 mm.

Example 8.9. Deduce the frequency and quality factor of an LCR circuit with $L = 2\text{mH}$, $C = 5\mu\text{F}$ and $R = 0.2\text{ ohm}$. (G.N.D.U. 2003; P.U. 2004, 2003)

Solution. Here $L = 2\text{mH} = 20 \times 10^{-3}\text{ H}$; $C = 5\mu\text{F} = 5 \times 10^{-6}\text{ F}$; $R = 0.2\ \Omega = 2 \times 10^{-1}\ \Omega$

$$\begin{aligned} \text{Angular frequency } \omega' &= \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \sqrt{\frac{1}{2 \times 10^{-3} \times 5 \times 10^{-6}} - \frac{2 \times 2 \times 10^{-2}}{4 \times 4 \times 10^{-6}}} \\ &= \sqrt{10^8 - 25 \times 10^2} = 10^4 \text{ rad s}^{-1} \quad (\text{approx}) \end{aligned}$$

$$\text{Frequency } \nu = \frac{\omega'}{2\pi} = \frac{10^4}{2\pi} = \frac{10}{2\pi} \times 10^3 = 1.59 \times 10^3 = 1590 \text{ Hz}$$

$$\text{and Quality factor } Q = \frac{L\omega'}{R} = \frac{2 \times 10^{-3} \times 10^4}{2 \times 10^{-1}} = 10^2 = 100$$

Example 8.10. A simple pendulum has a period of 1 sec. and an amplitude of 10° . After 10 complete oscillations its amplitude is reduced to 5° . What is the relaxation time of the pendulum and quality factor?

Solution. For a damped simple harmonic oscillator, the amplitude is given by

$$A = A_0 e^{-bt} = A_0 e^{-\frac{r}{2m}t} \quad \dots(i)$$

Here

$$A_0 = 10^\circ; A = 5^\circ$$

after 10 complete oscillations i.e., after a time = $10 \times \text{Time period} = 10 \times 1 = 10 \text{ sec}$. Substituting in (i)

$$5 = 10e^{-10b}$$

$$\therefore e^{-10b} = \frac{1}{2} \quad \therefore e^{10b} = 2$$

$$\text{or } b = \frac{\log_e 2}{10} = \frac{0.6931}{10} = 0.06931$$

$$\text{The relaxation time } t_r = \frac{1}{b} = \frac{1}{0.06931} = 14.428 \text{ sec}$$

$$\text{Quality factor } Q = \omega \frac{m}{r} = \frac{\omega}{2b} \text{ where } \omega = \frac{2\pi}{T} = 2\pi \quad [\because T = 1\text{s}]$$

$$\therefore Q = \frac{2\pi}{T \times 2b} = \frac{\pi}{bT} = \frac{\pi}{b} = \frac{\pi}{0.06931} = 45.33$$

Example 8.11. A condenser of capacity $1\mu\text{F}$, an inductance of 0.2 Henry and a resistance of 800 ohm are connected in series. Is the circuit oscillatory? If yes, calculate the frequency and quality factor of the circuit. (G.N.D.U. 2002; H.P.U 2002)

Solution. Here $C = 1\mu\text{F} = 10^{-6}\text{ F}$; $L = 0.2\text{ H}$; $R = 800\ \Omega$.

$$\text{For the circuit to be oscillatory } \frac{R^2}{4L^2} < \frac{1}{LC}$$

$$\text{Now } \frac{R^2}{4L^2} = \frac{800 \times 800}{4 \times 0.2 \times 0.2} = 4 \times 10^6$$

$$\text{and } \frac{1}{LC} = \frac{1}{0.2 \times 10^{-6}} = 5 \times 10^6$$

As $\frac{R^2}{4L^2} < \frac{1}{LC}$, the circuit is oscillatory.

The angular frequency $\omega' = \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} = \sqrt{5 \times 10^6 - 4 \times 10^6} = 10^3 \text{ rad s}^{-1}$

Hence frequency $f' = \frac{\omega'}{2\pi} = \frac{10^3}{2\pi} = 159 \text{ s}^{-1}$

and quality factor $Q = \frac{L\omega'}{R} = \frac{0.2 \times 10^3}{800} = 0.25$

Example 8.12. A damped oscillator consists of a mass 200 gm attached to a spring of constant 100 Nm^{-1} and damping constant $5 \text{ Nm}^{-1} \text{ s}$. It is driven by a Force $F = 6 \cos \omega t$ Newton, where $\omega = 30 \text{ s}^{-1}$. If displacement in steady state is $x = A \sin (\omega t - \phi)$ metre, find A and ϕ . Also calculate the power supplied to the oscillator. (Pbi. U., 2003; P.U., 2001, 2000)

Solution. Amplitude of the driving force $F_0 = 6 \text{ N}$

Frequency of the driving force $\omega = 30 \text{ s}^{-1}$

Mass of damped oscillator $m = 200 \text{ gm} = 0.2 \text{ kg}$

Spring constant $s = 100 \text{ Nm}^{-1}$

Damping constant $r = 5 \text{ Nm}^{-1} \text{ s}$

When an external force $F = F_0 \cos \omega t$ acts on a damped oscillator the steady state is given by

$$x = A \sin (\omega t - \phi)$$

where $A = \frac{F_0}{\omega Z_m}$, Z_m being the 'impedance' of the mechanical system given by

$$Z_m = \sqrt{r^2 + \left(\omega m - \frac{s}{\omega} \right)^2}$$

and

$$\tan \phi = \frac{\omega m - \frac{s}{\omega}}{r}$$

$\therefore$

$$\tan \phi = \frac{30 \times 0.2 - \frac{100}{30}}{5} = 0.534$$

$\therefore$

$$\phi = 28^\circ 6' \text{ and } \cos \phi = 0.8821$$

$$Z_m = \sqrt{(5 \times 5) + \left(30 \times 0.2 - \frac{100}{30} \right)^2} = 5.67 \text{ Nm}^{-1} \text{ s}$$

$$A = \frac{F_0}{\omega Z_m} = \frac{6}{30 \times 5.67} = 0.0352 \text{ m} = 3.52 \text{ mm}$$

$$\text{Average power supplied} = \frac{F_0^2}{2Z_m} \cos \phi = \frac{6 \cdot 6}{2 \times 5.67} \times 0.8821 = 2.8 \text{ watt}$$

Example 8.13. If the resonant (angular) frequency of acoustic system is 280 Hz and half power frequencies are 200 Hz and 360 Hz respectively, calculate the quality factor.

(Nagpur U., 2002)

Solution. Here $\omega_0 = 280 \text{ Hz}$, $\omega_1 = 200 \text{ Hz}$, $\omega_2 = 360 \text{ Hz}$

$\therefore$ Quality factor $Q = \frac{\omega_0}{\omega_2 - \omega_1} = \frac{280}{360 - 200} = \frac{280}{160} = 1.75$

Example 8.14. Light of wavelength 6×10^{-5} cm is emitted by an electron in an atom (a damped simple harmonic oscillator) with a quality factor 3×10^6 . Find the width of the spectral line from such an atom from resonance bandwidth.

Solution. The quality factor Q is given by

$$Q = \frac{\omega_0}{\omega_2 - \omega_1}$$

where ω_0 is the angular frequency of maximum velocity response (resonance) and $\omega_2 - \omega_1$ is the bandwidth in terms of angular frequency at half maximum power.

Quality factor in terms of wavelength. The above relation can be put in terms of wavelength as under. Let ν_1 , ν_2 and ν_0 be the frequencies and λ_1 , λ_2 and λ_0 the wavelengths corresponding to angular frequencies ω_1 , ω_2 and ω_0 , then

$$\begin{aligned} \frac{\omega_0}{\omega_2 - \omega_1} &= \frac{2\pi\nu_0}{2\pi\nu_2 - 2\pi\nu_1} = \frac{\nu_0}{\nu_2 - \nu_1} \\ &= \frac{\frac{c}{\lambda_0}}{\frac{c}{\lambda_2} - \frac{c}{\lambda_1}} = \frac{\frac{1}{\lambda_0}}{\frac{1}{\lambda_2} - \frac{1}{\lambda_1}} \quad \left[\because v = \frac{c}{\lambda} \right] \\ &= \frac{1}{\lambda_0} \frac{\lambda_1 \lambda_2}{\lambda_1 - \lambda_2} = \frac{\lambda_0}{\lambda_1 - \lambda_2} \end{aligned}$$

since $\lambda_1 \lambda_2 = \lambda_0^2$ (approx.). Hence $Q = \frac{\lambda_0}{\lambda_1 - \lambda_2}$

$$\therefore \text{Bandwidth} \quad \lambda_1 - \lambda_2 = \frac{\lambda_0}{Q}$$

$$\text{Now} \quad \lambda_0 = 6 \times 10^{-5} \text{ cm} = 6 \times 10^{-7} \text{ m and } Q = 3 \times 10^6$$

$$\therefore \text{Bandwidth} = \frac{\lambda_0}{Q} = \frac{6 \times 10^{-7}}{3 \times 10^6} = 2 \times 10^{-13} \text{ m} = 2 \times 10^{-3} \text{ \AA}$$

Example 8.15. A particle of mass 50 gm, moving with an initial velocity of 100 cm/sec, is acted upon by a damping force which brings it to rest in a distance of 10 metres. Assuming the damping force to be proportional to velocity, calculate (i) its relaxation time, (ii) the time in which (a) its velocity is halved (b) its kinetic energy is halved, (iii) the damping force on it when its velocity is 20 cm/sec. (Take $\log_e 10 = 2.30$).

Solution. We have the relation $x = v_0 \tau (1 - e^{-t/\tau})$ for the distance covered by a particle in time t , where v_0 is its initial velocity and τ , its relaxation time. (See § 8.1)

(i) Clearly, x will have its maximum value as $t \rightarrow \infty$ and, therefore, the quantity $e^{-t/\tau} \rightarrow 0$. So that,

$$x_{\max} = v_0 \tau \text{ and } \therefore \tau = x_{\max} / v_0.$$

Here, $x_{\max} = 10 \text{ m} = 1000 \text{ cm}$ and $v_0 = 100 \text{ cm/sec}$. We, therefore, have

$$\text{relaxation time } \tau = 1000/100 = 10 \text{ sec.}$$

(ii) (a) The velocity of the particle after time t from the start is given by the relation $v = v_0 e^{-t/\tau}$ (§ 8.1), whence $\log_e (v/v_0) = -t/\tau$. Or, $t = \tau \log_e (v_0/v)$.

$\therefore$ The velocity will be reduced to half, i.e., 50 cm/sec in time t given by

$$t = 10 \times 2.30 \log_{10} (100/50) = 10 \times 2.30 \times \log_{10} 2 = 10 \times 2.30 \times 0.3010 = 6.92 \text{ sec.}$$

(b) We know that if T be the kinetic energy of the particle after time t and T_0 , its initial kinetic energy, we have $T = T_0 e^{-2t/\tau}$. [See § 8.1]

$\therefore \log_e(T/T_0) = -2t/\tau$, whence, $2t = t \log_e(T_0/T)$.

Or, $2t = 10 \times 2.30 \log_{10}(2/1)$. Or $t = 5 \times 2.30 \log_{10} 2 = 5 \times 2.30 \times 0.3010 = 3.46 \text{ sec}$.

Or, we could have obtained this result directly from (a) above, since, as we know, relaxation time for $K.E.$ is half that for velocity. (See § 8.1).

Thus, the $K.E.$ of the particle will be reduced to half its initial value in 3.46 sec.

(iii) The damping force $= \gamma v$, where $\gamma = m/\tau$.

$\therefore$ damping force at velocity 20 cm/sec $= mv/\tau = 50 \times 20/10 = 100 \text{ dynes}$.

Example 8.16. A charged oil drop falls through air under the combined action of an electric field E and gravity g , both acting along the same direction. If m be the mass of the drop and Q , the charge on it, show that the terminal velocity attained by it (i.e., when $t \rightarrow \infty$) is given by $v_{\text{terminal}} = (qE/m)\tau + g\tau$, assuming a solution of the form $v = A + Be^{-\alpha t}$. (Assume damping force to be proportional to velocity).

Solution. Since the electric field E and g are both similarly directed, they both act downwards. So that, the force acting on the drop is given by

$$F = m(dv/dt) = mg + qE - \gamma v,$$

where γv is the damping force.

Or, $m(dv/dt) + \gamma v = mg + qE$. Or, $(dv/dt) + (\gamma/m)v = g + (q/m)E$.

Or, Since $\gamma/m = 1/\tau$, where τ is the relaxation time, the equation of motion of the drop is

$$\frac{dv}{dt} + \frac{v}{\tau} = g + \frac{qE}{m}. \quad \dots \text{I}$$

Now, let us assume a solution of this equation to be $v = A + Be^{-\alpha t}$, where A and B are arbitrary constants to be determined from the initial conditions.

Since at $t = 0$, i.e., when the drop just starts from rest, $v_0 = 0$, we have

$0 = A + B$, whence, $B = -A$. And, therefore,

$$v = A + Ae^{-\alpha t} = A(1 - e^{-\alpha t}). \quad \dots \text{II}$$

Hence, $\frac{dv}{dt} = A\alpha e^{-\alpha t}$

Substituting this value of dv/dt in the equation of motion (I) above, we have

$$A\alpha e^{-\alpha t} + \frac{1}{\tau}A(1 - e^{-\alpha t}) = g + \frac{qE}{m}.$$

Clearly, at $t \rightarrow \infty$, we have $A/\tau = g + qE/m$.

Or, $A = \tau(g + qE/m)$...III

And, at $t = 0$, we have $A\alpha = g + qE/m$IV

From relations III and IV, we thus have $\alpha = 1/\tau$.

Hence from equation II, we have

$$v = \tau(g + qE/m)(1 - e^{-\alpha t}) = \tau(g + qE/m)(1 - e^{-t/\tau}) \quad \dots \text{V}$$

Clearly, therefore, when $t \rightarrow \infty$, we have

$$v_{\text{terminal}} = \tau(g + qE/m) = (qE/m)\tau + g\tau.$$

Example 8.17. A small ball bearing is falling freely through a viscous medium, of coefficient of viscosity η , under the action of gravity alone. The viscous drag on it due to the medium is $-6\pi\eta av$, where a is its radius and v , its velocity. Obtain (i) the relaxation time, (ii) the terminal velocity of the ball bearing. (iii) If the ball bearing has an initial horizontal velocity u , what will be the maximum horizontal distance covered by it? (iv) If the radius of the ball bearing be doubled, how will it affect its terminal velocity?

Solution. (i) Here, obviously, the *damping force* $= 6\pi\eta av = \gamma v$, where $\gamma = 6\pi\eta a$, and is thus *proportional to velocity*.

$\therefore$ *relaxation time* $\tau = m/\gamma = m/6\pi\eta a$.

(ii) Proceeding as in worked example 8.16 above and remembering that, here, the ball bearing (a tiny sphere) is falling only under the action of gravity, we have terminal velocity of the ball bearing given by

$$v_{\text{terminal}} = g\tau = g(m/6\pi\eta a) = mg/6\pi\eta a.$$

(iii) The *maximum horizontal distance covered by the ball bearing* is, as we know, equal to its *initial velocity* (u) $\times$ *relaxation time* (τ) $= u(m/6\pi\eta a) = mu/6\pi\eta a$.

(iv) We have seen under (ii) above that the *terminal velocity of the ball bearing* $= mg/6\pi\eta a = \frac{4}{3}\pi a^3 \rho g/6\pi\eta a = 2a^2 \rho g/9\eta$, where $\frac{4}{3}\pi a^3$ is the volume and, therefore, $\frac{4}{3}\pi a^3 \rho$, the mass (m) of the ball bearing.

Thus, *the terminal velocity* $\propto a^2$, (ρ and g being constants).

Hence, *if the radius of the ball bearing be doubled, its terminal velocity will be four times its previous value, i.e., equal to* $4mg/6\pi\eta a = 2mg/3\pi\eta a$.

Example 8.18. An underdamped harmonic oscillator has its amplitude reduced to 1/10th of its initial value after 100 oscillations. If its time-period be 1.15 sec, calculate (i) the damping constant, (ii) the relaxation time.

If the observed value of the first amplitude of the oscillator be 2 cm. What would be its value in the absence of damping?

Solution. (i) The amplitude (a) of an underdamped oscillator is, as we know, given by the relation $a = a_0 e^{-kt}$, where a_0 is its amplitude in the absence of damping and k , the *damping constant*.

Now, for the first amplitude (a_1), obviously, $t = T/4$ where T is the *time-period* of the oscillator. So that, $a_1 = a_0 e^{-kT/4}$.

Since successive amplitudes occur at intervals of $T/2$ sec each, the 201th amplitude after 100 oscillations occurs after $(200T/2 + T/4) = (100T + T/4)$ sec. And, therefore,

$$a_{201} = a_0 e^{-k(100T + T/4)}.$$

$$\text{Hence, } \frac{a_{201}}{a_1} = \frac{a_0 e^{-k(100T + T/4)}}{a_0 e^{-kT/4}} = e^{-100kT}.$$

But we are given that $a_{201}/a_1 = 1/10$ or 10^{-1} and $T = 1.15$ sec.

$$\therefore 10^{-1} = e^{-100k(1.15)} = e^{-115k}. \text{ Or, } \log_e 10 = 115k.$$

$$\text{Or, } 2.30 \log_{10} 10 = 115k, \text{ i.e., } 2.30 = 115k, \text{ whence, } k = 2.30/115 = 0.02.$$

Thus, the *damping constant* $= 0.02$.

(ii) And, therefore, *relaxation time* $\tau = 1/2k = 1/2 \times 0.02 = 25$ sec.

Now, since the *first amplitude* $a_1 = a_0 e^{-kT/4} = 2$ cm, we have

$$2 = a_0 e^{(0.02 \times 1.15/4)} \text{ or, } a_0 = 2e^{0.023/4} \text{ or, } \log_{10} a_0 = \log_{10} 2 + \frac{0.023}{4} \times \frac{1}{2.30}$$

$$\text{Or, } \log_{10} a_0 = 0.3010 + 0.0025 = 0.3025, \text{ whence, } a_0 = 2.011 \text{ cm.}$$

Thus, *the undamped amplitude of the oscillator would be 2.011 cm.*

Example 8.19. Write down the equation of motion of the damped harmonic oscillator of example (8.18) above and calculate the time in which (i) its amplitude (ii) its energy falls to $1/e^{\text{th}}$ of its undamped value.

If the mass of oscillator be 1.127 gm, calculate its average rate of loss of energy.

Solution. The differential equation of a damped harmonic oscillator, as we know (§ See 8.2), is

$$\frac{d^2 x}{dt^2} + 2k \frac{dx}{dt} + \omega_0^2 x = 0,$$

which, when the oscillator is underdamped, i.e., when $k \ll \omega_0$, gives

$$x = a_0 e^{-kt} \sin(\omega t + \phi).$$

So that its amplitude is given by $a = a_0 e^{-kt} = a_0 e^{-t/2\tau}$, where a_0 is its amplitude in the absence of damping.

(i) Now, as we have seen under example 8.18 above, the damping constant k for the given oscillator works out to 0.02. If, therefore, its amplitude is reduced to $1/e^{\text{th}}$ of its undamped value, we have

$$a = a_0/e = a_0 e^{-1} \text{ and, therefore, } a_0 e^{-1} = a_0 e^{-kt}. \text{ Or, } kt = 1,$$

whence,

$$t = 1/k = 1/0.02 = 50 \text{ sec.}$$

Thus, the amplitude of the oscillator falls to $1/e^{\text{th}}$ of its undamped value in 50 sec.

(ii) Again, as we have seen in § 8.3, the energy of a damped harmonic oscillator at a given instant t is given by $E = E_0 e^{-2kt}$. So that, when its energy falls to $1/e^{\text{th}}$ of its value E_0 in the absence of damping, we have

$$E = E_0/e = E_0 e^{-1}. \text{ And, therefore, } E_0 e^{-1} = E_0 e^{-2kt}. \text{ Or, } 2kt = 1,$$

whence,

$$t = 1/2k = 1/2 \times 0.02 = 25 \text{ sec.}$$

Thus, the energy of the oscillator falls to $1/e^{\text{th}}$ of its value in the absence of damping in 25 sec.

And, as shown in § 8.3, the rate of loss of energy $= E/\tau = 2kE$.

Now, $E = \frac{1}{2} m a_0^2 \omega_0^2 e^{-2kt}$, where $m = 1.127 \text{ gm}$, $a_0 = 2.011 \text{ cm}$, $\omega_0 = 2\pi/T = 2\pi/1.15$, $k = 0.02$ and $t = 25 \text{ sec}$. So that,

$$\begin{aligned} \text{rate of loss of energy} &= 2 \times 0.02 \times \frac{1}{2} \times 1.127 \times (2.011)^2 \times (2\pi/1.15)^2 e^{-2(0.02 \times 25)} \\ &= \frac{0.02 \times 1.127 \times (2.011)^2 \times 4\pi^2}{(1.15)^2 \times e} = \frac{0.02 \times 1.127 \times (2.011)^2 \times 4\pi^2}{(1.15)^2 \times 2.7183} \\ &= 1 \text{ erg/sec.} \end{aligned}$$

Example 8.20. If the amplitude of a seconds pendulum, with a bob of mass 200 gm, is reduced to half its undamped value in 200 seconds, what is its quality factor Q ? What should be the mass of the bob, its size and shape remaining unchanged, in order that the damping of the pendulum may become critical? (Take $\log_e 10 = 2.30$)

Solution. We know that the amplitude of the pendulum (a damped harmonic oscillator) at a given instant t is given by $a = a_0 e^{-kt} = a_0 e^{-t/2\tau}$.

Here, $a = a_0/2$ and $t = 200 \text{ sec}$. So that, $a_0/2 = a_0 e^{-200/2\tau} = a_0 e^{-100/\tau}$.

Or,

$$100/\tau = \log_e 2 = 2.30 \log_{10} 2 = 2.30 \times 0.3010$$

whence,

$$\tau = 100/2.30 \times 0.3010 = 100/0.6923.$$

And, since $\omega_0 = 2\pi/T = 2\pi/2 = \pi$ ($\because T = 2 \text{ sec}$), we have

Quantity factor of the pendulum, $Q = \omega_0 \tau = \pi) 100/0.6923 = 453.7$.

For critical damping, as we know, the condition is that $k = \omega_0$. So that, if τ' be the value of the relaxation time in this case, we have $1/2\tau' = \omega_0$

Or, since $\omega_0 = \pi$, we have $1/2\tau' = \pi$ and, therefore, $\tau' = 1/2\pi$.

If m and m' be the masses of the bob in the two cases respectively, we have $\gamma/m = 1/\tau$ and $\gamma/m' = 1/\tau'$ ($\because \gamma$ remains the same in either case).

$\therefore$

$$\tau'/\tau = (\gamma/m)(\gamma/m') = m'/m, \text{ whence, } m' = m \times \tau'/\tau.$$

Substituting the values of m , τ' and τ , therefore, we have

$$m' = 200 \times \frac{1}{2\pi} \times \frac{0.6923}{100} = \frac{0.6923}{\pi} = 0.2204 \text{ gm.}$$

Thus, for the damping to be critical, the mass of the bob must be 0.2204 gm.

Example 8.21. If the quality factor of an underdamped harmonic oscillator of frequency 512 be 8×10^4 , calculate the time in which its energy is reduced to $1/e^{\text{th}}$ of its energy in the absence of damping. How many oscillations does the oscillator make in this time?

Calculate the percentage reduction in the frequency of the oscillator due to damping. What inference do you draw from it?

Solution. The energy of a damped harmonic oscillator at an instant t , as we know, is given by $E = E_0 e^{-2kt} = E_0 e^{-t/\tau}$, where E_0 is its energy in the absence of damping.

Since here $E = E_0/e = E_0 e^{-1}$, we have $E_0 e^{-1} = E_0 e^{-t/\tau}$, whence, $1 = t/\tau$ or, $t = \tau$. And since $Q = \omega_0 \tau$, we have

$$\tau = \frac{Q}{\omega_0} = \frac{Q}{2\pi n_0} = \frac{8 \times 10^4}{2\pi \times 512} = \frac{10^4}{2\pi \times 64} = \frac{10^4}{128\pi}.$$

And, therefore, the time in which the energy of the oscillator falls to $1/e$ th of its energy in the on line absence of damping, i.e., $t = \tau = 10^4/128\pi = 24.87$ sec.

And, clearly, number of oscillations made by the oscillator in this time

$$= nt = (\omega_0/2\pi)\tau = Q/2\pi = 8 \times 10^4/2\pi = 12740.$$

Now, in the case of an underdamped harmonic oscillator, we have

$$\omega = \sqrt{\omega_0^2 - k^2}. \text{ Or, } \omega^2 = \omega_0^2 - k^2 = \omega_0^2 - (1/2\tau)^2 = \omega_0^2 - 1/4\tau^2.$$

So that, dividing by ω_0 throughout, we have

$$\frac{\omega^2}{\omega_0^2} = 1 - \frac{1}{4\omega_0^2\tau^2} = 1 - \frac{1}{4Q^2} \quad [\because \omega_0\tau = Q]$$

And, therefore,

$$\frac{\omega}{\omega_0} = \left(1 - \frac{1}{4Q^2}\right)^{1/2} = \left(1 - \frac{1}{2} \times \frac{1}{4Q^2} + \dots\right) = 1 - \frac{1}{8Q^2}.$$

Or,

$$\frac{2\pi n}{2\pi n_0} = \frac{n}{n_0} = 1 - \frac{1}{8Q^2},$$

where n_0 is the frequency of the undamped oscillator.

$$\begin{aligned} \therefore \text{Percentage reduction in the frequency of the oscillator} &= \frac{n_0 - n}{n_0} \times 100 \\ &= \left(1 - \frac{n}{n_0}\right) \times 100 = \left[1 - \left(1 - \frac{1}{8Q^2}\right)\right] \times 100 = \frac{100}{8Q^2} = \frac{12.5}{Q^2} \\ &= \frac{12.5}{(8 \times 10^4)^2} = 1.953 \times 10^{-9}. \end{aligned}$$

The obvious inference from the above result is that for oscillators with large values of the quality factor Q (which is usually the case), the reduction in frequency due to damping is negligibly small.

Example 8.22. A massless spring, suspended from a rigid support, carries a mass of 500 gm at its lower end and the system oscillates with a frequency of 5/sec, with the amplitude reduced to half its undamped value in 20 sec. Calculate (i) the force constant of the spring. (ii) the relaxation time of the system and (iii) its quality factor. (iv) What would be the quality factor of the system if the suspended mass be reduced to 150 gm?

Solution. (i) We have the relation $\omega_0 = \sqrt{C/m}$ or $C = m\omega_0^2$, where C is the force constant of the spring and m , the mass suspended from it.

Here, $\omega_0 = 2\pi n = 2\pi \times 5 = 10\pi$ radian/sec and $m = 500$ gm.

So that, *force constant of the spring*, $C = 500 \times (10\pi)^2 = 5\pi^2 \times 10^4$
 $= 4.934 \times 10^5$ dynes/cm.

(ii) The amplitude at an instant t is given by $a = a_0 e^{-kt}$.

Since $a = a_0/2$ and $t = 20$ sec, we have $a_0/2 = a_0 e^{-20k}$. Or, $\log_e 2 = 20k$.

Or, $2.30/\log_{10} 2 = 20k$. Or, $2.30 \times 0.3010 = 20k$, whence, $k = 2.30 \times 0.3010/20$.

$\therefore$ *relaxation time of the system*, $\tau = 1/2k = \frac{1 \times 20}{2 \times 2.30 \times 0.3010} = 14.44$ sec.

(iii) The *quality factor of the system*, $Q\tau = \omega_0 t = 10\pi \times 14.44 = 453.7$.

(iv) Since the force constant (C) of the spring remains the same irrespective of the mass suspended from it, and so does the damping factor $\gamma = 2km = 2 \times 2.30 \times 0.010 \times 500/20 = 115 \times 0.3010$, we have

$$\omega'_0 = \sqrt{C/m'} = \sqrt{5\pi^2 \times 10^4 / 150} = 18.26\pi, \text{ since } m' = 150 \text{ gm.}$$

$\therefore$ τ' , the relaxation time now $= 1/2k = m'/\gamma = 150/115 \times 0.3010$.

And, therefore, the *quality factor now*, $Q = \omega'_0 \tau'$

$$= 18.26\pi \times 150/115 \times 0.3010 = 248.5.$$

Example 8.23. The restoring torque exerted by the suspension fibre of a ballistic galvanometer is 10^3 dyne-cm/radian. If the period of vibration of the coil be 2π seconds and its amplitude be reduced to $1/e^{\text{th}}$ of its undamped value in 100 seconds, calculate the values of (i) the moment of inertia of the suspended system, (ii) the damping factor γ and (iii) the quality factor Q of the system.

Solution. (i) The oscillating system in a ballistic galvanometer, as we know, is an underdamped one (§ 8.5 I (ii)) and, therefore, the angular frequency of the system is given by $\omega = \sqrt{\omega_0^2 - k^2}$,

where ω_0 is the undamped angular frequency of the system and k , the damping constant.

$\therefore$ *Time-period of the suspended system or the coil*, $T = 2\pi / \sqrt{\omega_0^2 - k^2}$.

Or, since $T = 2\pi$ sec, we have $2\pi = 2\pi / \sqrt{\omega_0^2 - k^2}$, whence, $\sqrt{\omega_0^2 - k^2} = 1$.

Or, $\omega_0^2 - k^2 = 1$.

Now, the amplitude of the coil at an instant t is given by $\theta = \theta_0 e^{-kt}$, where θ_0 is its undamped amplitude. And, since the amplitude is reduced to $\theta_0/e = \theta_0 e^{-1}$ in time $t = 100$ sec, we have

$$\theta_0 e^{-1} = \theta_0 e^{-100k} \quad \text{or,} \quad 100k = 1 \quad \text{or} \quad k = 1/100 = 0.01.$$

$\therefore$ $\omega_0^2 - k^2 = \omega_0^2 - (0.01)^2 = 1$. Or, $\omega_0^2 \approx 1$ and $\therefore \omega \approx 1$ rad/sec.

Hence $C/I = \omega_0^2 = 1$, where C is the torque exerted by the suspension fibre per radian twist of the fibre.

Since $C = 10^3$ dyne-cm/radian (given), we have $10^3/I = 1$. Or, $I = 10^3$ dyne-cm².

Thus, the *moment of inertia of the suspended system about the suspension fibre*,

i.e., $I = 10^3$ or 1000 dyne-cm².

(ii) Now $(\gamma \times \wedge/R)/I = 2k$. Or, $\gamma/I = 2k$, (the electromagnetic damping being obviously ignored in the problem). So that,

$$\begin{aligned} \text{damping factor } \gamma &= I \times 2k \\ &= 1000 \times 2 \times 0.01 \\ &= 20 \text{ dyne-cm/sec.} \end{aligned}$$

(iii) The *quality factor of the system*, $Q = \omega_0 \tau = \omega_0/2k = 1/2 \times 0.01 = 50$.

Example 8.24. Give the theory of oscillations in a series LCR circuit with small damping. Deduce the frequency and quality factor for a circuit with $L = 2\text{mH}$, $C = 5\mu\text{F}$ and $R = 0.2\text{ ohm}$.

(Agra, 2001)

Solution. For answer to the first part of the question, see § 8.5 II. The circuit being equivalent to an underdamped harmonic oscillator, we have $\omega = \sqrt{\omega_0^2 - k^2}$ and, therefore, frequency of the oscillations, $n = \omega/2\pi = \sqrt{\omega_0^2 - k^2}/2\pi$.

$$\text{Or, } n = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}}. \quad [\text{§ 8.5 II, (iii)}]$$

Here, $L = 2\text{mH} = 0.002\text{ henry}$, $C = 5\mu\text{F} = 5 \times 10^{-6}\text{ farad}$ and $R = 0.2\text{ Ohm}$.

We, therefore, have frequency of the circuit.

$$\begin{aligned} n &= \frac{1}{2\pi} \sqrt{\frac{1}{0.002 \times 5 \times 10^{-6}} - \frac{(0.02)^2}{4(0.002)^2}} = \frac{1}{2\pi} \sqrt{\frac{4 \times 10^8 - 10^4}{4}} \\ &= \frac{10^2}{2\pi} \sqrt{10^4 - 1} = 10^4/2\pi = 1.592 \times 10^3 \text{ cps.} \end{aligned}$$

Now, the resistance being small, quality factor $Q = L\omega_0/R = L\omega/R$, where $\omega = 2\pi n = 2\pi \times 10^4/2\pi = 10^4\text{ rad/sec}$.

$\therefore$ Quality factor for the circuit, $Q = 0.002 \times 10^4/0.2 = 0.01 \times 10^4 = 100$.

Example 8.25. The line width of orange line of Kr^{86} at $\lambda = 6058\text{ \AA}$ is found to be $5.50 \times 10^{-3}\text{ \AA}$. Calculate (a) line frequency (ii) line width in Hertz.

Solution. Line wavelength $\lambda_0 = 6058\text{ \AA} = 6058 \times 10^{-10}\text{ m}$

$$\therefore \text{Line frequency } \nu_0 = \frac{c}{\lambda_0} = \frac{3 \times 10^8}{6058 \times 10^{-10}} = 4.95 \times 10^{14} \text{ Hertz}$$

Line width in wavelength $\lambda_1 - \lambda_2 = 5.50 \times 10^{-3}\text{ \AA} = 5.50 \times 10^{-13}\text{ m}$

$$\therefore \text{Line width in terms of frequency} = \frac{c}{\lambda_1 - \lambda_2} = \frac{3 \times 10^8}{5.50 \times 10^{-13}} = 5.45 \times 10^{20} \text{ Hz}$$

Example 8.26. A root mean square voltage of 100 volts is applied to a series LCR circuit having $R = 10\text{ ohm}$, $L = 10\text{ mH}$ and $C = 1\mu\text{F}$. Calculate

(i) The natural frequency

(ii) Current at resonance

(iii) Q value of the circuit at resonance

(iv) Bandwidth of the circuit.

Solution.

$$(i) \text{ Natural frequency } \nu = \frac{1}{2\pi\sqrt{LC}} = 1592 \text{ Hz}$$

$$(ii) \text{ Current at Resonance } I_{\max} = \frac{V_{\max}}{R} = \frac{\sqrt{2}V_{\text{rms}}}{R} = \frac{\sqrt{2} \times 100}{10} = 14.14 \text{ A.}$$

$$(iii) Q \text{ value} = \omega_0 \frac{L}{R} = \frac{1}{\sqrt{LC}} \frac{L}{R} = \sqrt{\frac{L}{C}} \frac{1}{R} = \sqrt{\frac{10 \times 10^{-3}}{1 \times 10^{-6}}} \times \frac{1}{10} = 10$$

(iv) Bandwidth $\omega_2 - \omega_1$

$$Q = \frac{\omega_0}{\omega_2 - \omega_1}$$

$$\therefore \omega_2 - \omega_1 = \frac{\omega_0}{Q} = \frac{1}{\sqrt{LC}} \cdot \frac{1}{Q} = \sqrt{\frac{1}{10 \times 10^{-3} \times 10^{-6}}} \times \frac{1}{10} = 1000 \text{ rad s}^{-1}$$

Example 8.27. The equation of motion is

$$2 \times 10^{-4} \frac{d^2x}{dt^2} + 4 \times 10^{-2} \frac{dx}{dt} + 5x = 0.124 \sin 100t, \text{ where all quantities are in S.I. units}$$

Find (i) Natural frequency of undamped oscillation.

(ii) Mechanical impedance.

(Nagpur U. 2009, 2004)

Solution. Comparing with

$$m \frac{d^2x}{dt^2} + r \frac{dx}{dt} + sx = F_0 \sin \omega_0 t$$

we get, $m = 2 \times 10^{-4}$ kg, $r = 4 \times 10^{-2}$ Nsm⁻¹, $s = 5$ Nm⁻¹, $F_0 = 0.124$ N, and $\omega_0 = 100$ rad s⁻¹

(i) Natural frequency of undamped oscillation is

$$\begin{aligned} F_0 &= \frac{\omega_0}{2\pi} = \frac{1}{2\pi} \sqrt{\frac{s}{m}} \\ &= \frac{1}{2 \times 3.14} \sqrt{\frac{5}{2 \times 10^{-4}}} = \frac{10^2}{6.28} \sqrt{\frac{5}{2}} = 25.16 \text{ Hz.} \end{aligned}$$

(ii) Mechanical impedance (z):

$$\text{Amplitude, } a = \frac{f_0}{\sqrt{4b^2\omega_0^2 + (\omega^2 - \omega_0^2)^2}} = \frac{F_0}{m\sqrt{4b^2\omega_0^2 + (\omega^2 - \omega_0^2)^2}} = \frac{F_0}{z}$$

$$\therefore \text{ Mechanical impedance, } z = m\sqrt{4b^2\omega_0^2 + (\omega^2 - \omega_0^2)^2}$$

Here

$$\begin{aligned} \omega^2 - \omega_0^2 &= \frac{s}{m} - \omega_0^2 \\ &= \frac{5}{2 \times 10^{-4}} - (10^2)^2 = 1.5 \times 10^4 \end{aligned}$$

and

$$4b^2\omega_0^2 = (2b)^2\omega_0^2 = \left(\frac{r}{m}\right)^2\omega_0^2 = \left(\frac{4 \times 10^{-2}}{2 \times 10^{-4}}\right)^2 \times (10^2)^2 = 4 \times 10^8$$

$\therefore$

$$\begin{aligned} z &= 2 \times 10^{-4} \sqrt{4 \times 10^8 + 2.25 \times 10^8} = 2 \times 10^{-4} \sqrt{10^8 (4 + 2.25)} \\ &= 2 \times 10^{-4} \times 10^4 \times \sqrt{6.25} = 2 \times 2.25 = 5 \text{ Nsm}^{-1} \end{aligned}$$

Example 8.28. A harmonic oscillator consisting of 50 gm mass attached to a mass less spring has a quality factor 200. If it oscillates with an amplitude of 2 cm in resonance with a periodic force of frequency 20 Hz. Calculate (i) the average energy stores in it and (ii) the rate of dissipation of energy. (Nagpur U. 2007)

Solution. Here, the average energy stored in the oscillator is equal to its maximum potential energy = $\frac{1}{2}cx^2 = \frac{1}{2}m\omega_0^2x_0^2$

Here, $m = 50$ gm, $\omega_0 = 2\pi n = 2\pi \times 20 = 40\pi$, and $x_0 = 2$ cm.

(i) The average energy stored or absorbed in the oscillator

$$= \frac{1}{2} \times 50 \times (40\pi)^2 \times (2)^2 = 1.58 \times 10^6 \text{ ergs.}$$

(ii) By definition, the quality factor

$$Q = 2\pi \times \frac{\text{Average energy stored}}{\text{Energy dissipated per cycle}}$$

$$\therefore \text{ Energy dissipated per cycle} = 2\pi \times \frac{\text{Average energy stored}}{Q}$$

$$= 2\pi \frac{1.58 \times 10^6}{200} = \pi \times 1.58 \times 10^4 \text{ ergs.}$$

Since there are 20 cycles per cycle, we have energy dissipated per second or rate of dissipated, of energy

$$= 20 \times \pi \times 1.58 \times 10^4 = 9.926 \times 10^5 \cong 10^6 \text{ erg/sec.}$$

II–Driven harmonic oscillator

Example 8.29. In an experiment on forced oscillations, the frequency of a sinusoidal driving force is changed while its amplitude is kept constant. It is found that the amplitude of vibrations is 0.01 mm at very low frequency of the driver and goes upto a maximum of 5.0 mm at driving frequency 200 sec^{-1} . Calculate (i) Q of the system, (ii) relaxation time τ and (iii) half width of resonance curve. (Agra (Supp.) 2005)

Solution. We know that if the driving frequency be negligibly small and the damping be low, the amplitude of the driven oscillator, $A = f_0/\omega_0^2$ and that at low damping, the maximum amplitude (or amplitude at resonance) is $A_{\max} = f_0\tau/\omega_0$, where ω_0 is the natural frequency of the oscillator (or the system) and τ , its relaxation time, (See § 8.6)

$$(i) \text{ We, therefore, have } \frac{A_{\max}}{A} = \frac{f_0\tau/\omega_0}{f_0/\omega_0^2} = \omega_0\tau = Q.$$

Since $A_{\max} = 5.0 \text{ mm}$ and $A = 0.01 \text{ mm}$, we have
quality factor of the system, $Q = 5/0.01 = 500$.

(ii) Since $Q = \omega_0\tau$, we have $\tau = Q/\omega_0$.

Here, $Q = 500$ and $\omega_0 = 2\pi n = 2\pi \times 200 = 400\pi$

$\therefore$ relaxation time of the system, $\tau = 500/400\pi = 0.3971 \approx 0.40 \text{ sec.}$

(iii) For low damping, half width of the resonance curve, $\Delta p = \sqrt{3}$ [§ 8.10]

Since $k = 1/2\tau = 1/2 \times 0.40 = 1.25$, we have

$$\begin{aligned} \text{half width of the resonance curve, } \Delta p &= \sqrt{3} \times 1.25 = 2.165 \text{ radian/sec} \\ &= 2.165/2\pi = 0.3445 \text{ c.p.s.} \end{aligned}$$

Example 8.30. A harmonic oscillator of quality factor 10 is subjected to a sinusoidal applied force of frequency one and a half times the natural frequency of the oscillator. If the damping be small, obtain (i) the amplitude of the forced oscillation in terms of its maximum amplitude and (ii) the angle θ by which it will be out of phase with the driving force.

Solution. As we know, the amplitude of the forced oscillation is given by

$$A = \frac{f_0}{\sqrt{(\omega_0^2 - p^2)^2 + 4k^2 p^2}} = \frac{f_0}{\omega_0^2 \sqrt{\left(1 - \frac{p^2}{\omega_0^2}\right)^2 + 4 \frac{k^2 p^2}{\omega_0^4}}},$$

where ω_0 is the natural angular frequency of the oscillator and p , the angular frequency of the driving force.

Now, the quality factor $Q = \omega_0\tau = \omega_0/2k = 10$. So that, $2k = \omega_0/10$ and $p/\omega_0 = 3/2$ (given). We therefore, have

$$A = \frac{f_0}{\omega_0^2 \sqrt{\left(1 - \frac{p^2}{\omega_0^2}\right)^2 + \frac{1}{100} \times \frac{p^2}{\omega_0^2}}} = \frac{f_0}{\omega_0^2 \left(\frac{25}{16} + \frac{1}{100} \times \frac{9}{4}\right)^{1/2}}$$

$$= \frac{f_0}{\omega_0^2 \left(\frac{634}{400} \right)^{1/2}} = \frac{20f_0}{\omega_0^2 \sqrt{634}}.$$

And, for low damping, amplitude is maximum when $P = \omega_0$. So that,

$$A_{\max} = \frac{f_0}{\omega_0^2 \sqrt{(1-1)^2 + \frac{1}{100} \times 1}} = \frac{f_0}{\omega_0^2 \left(\frac{1}{100} \right)^{1/2}} = \frac{f_0}{\omega_0^2 (1/10)} = \frac{10f_0}{\omega_0^2}.$$

$$\therefore \frac{A}{A_{\max}} = \frac{20f_0}{\omega_0^2 \sqrt{634}} \times \frac{\omega_0^2}{10f_0} = \frac{2}{\sqrt{634}} = 0.07943 \approx 0.08. \text{ Or, } A = 0.08 A_{\max}.$$

Thus, the amplitude of the forced oscillation will be 0.08 of the maximum amplitude.

$$\begin{aligned} \text{Now, } \tan \theta &= \frac{2kp}{\omega_0^2 - p^2} = \frac{p\omega_0/10}{\omega_0^2 - p^2} = \frac{(3\omega_0/2)(\omega_0/10)}{\omega_0^2 - 9\omega_0^2/4} \\ &= \frac{3\omega_0^2}{20} \times \frac{4}{5\omega_0^2} = \frac{3}{25} = 0.12. \end{aligned}$$

$$\text{Or, } \theta = \tan^{-1}(0.12) = 6^\circ 51'$$

Therefore, the forced oscillation is $(180^\circ - 6^\circ 51') = 173^\circ 9'$ out of phase with the driving force.

Example 8.31. A torsion pendulum of moment of inertia 10^5 gm-cm^2 suffers an angular displacement of 4° under a constant torque of 2000 dynes-cm. What should be the frequency of the periodic torque that would set the pendulum in resonant vibration?

Solution. Here, angular displacement of the pendulum $= 4^\circ = 4 \times \pi/180$ radian and the torque applied $= 2000$ dyne-cm.

$$\begin{aligned} \therefore \text{Torque per unit angular displacement (C)} &= \frac{\text{Torque applied}}{\text{Angular displacement}} = \frac{2000}{4\pi/180} \\ &= \frac{2000 \times 180}{4\pi} \text{ dyne-cm/rad.} \end{aligned}$$

Now, for setting the pendulum into resonant vibration, the frequency of the applied torque must be the same, or very nearly the same, as the natural frequency of the pendulum, $n = \frac{1}{2\pi} \sqrt{C/I}$.

Substituting the values of C and I , therefore, we have

$$\text{required frequency of the periodic torque, } n = \frac{1}{2\pi} \sqrt{\frac{2000 \times 180}{4\pi \times 10^5}} = 0.1704 \text{ or } 0.17 \text{ sec}^{-1}.$$

Example 8.32. A harmonic oscillator consisting of a 50 gm mass attached to a massless spring has a quality factor 200. If it oscillates with an amplitude of 2 cm in resonance with a periodic force of frequency 20 c.p.s., calculate (i) the average energy stored in it, (ii) the rate of dissipation of energy.

Solution. In the case of resonant vibration, the average energy stored in the oscillator is equal to its maximum potential energy

$$= \frac{1}{2} Cx_0^2 = \frac{1}{2} m\omega_0^2 x_0^2. \quad [\because C = m\omega_0^2]$$

Here, $m = 50 \text{ gm}$, $\omega_0 = 2\pi n = 2\pi \times 20 = 40\pi$ and $x_0 = 2 \text{ cm}$. So that,

$$\begin{aligned} \text{average energy absorbed or stored up in the oscillator} &= \frac{1}{2} \times 50 \times (40\pi)^2 \times (2)^2 \\ &= 16 \times 10^4 \pi^2 = 1.58 \times 10^6 \text{ ergs.} \end{aligned}$$

Now, by its very definition, the quality factor

$$Q = 2\pi \frac{\text{average energy stored}}{\text{energy dissipated per cycle}}$$

$$\begin{aligned} \therefore \text{energy dissipated per cycle} &= 2\pi \frac{\text{average energy stored}}{Q} \\ &= 2\pi \frac{1.58 \times 10^6}{200} = \pi \times 1.58 \times 10^4 \text{ ergs.} \end{aligned}$$

Since there are 20 cycles to the second, we have

$$\begin{aligned} \text{energy dissipated per second or rate of dissipation of energy} &= 20\pi \times 1.58 \times 10^4 \\ &= 9.926 \times 10^5 \approx 10^6 \text{ erg/sec.} \end{aligned}$$

Example 8.33. If in problem 8.32 above, the rate of dissipation of energy is desired to be halved, what should be (i) the amplitude of the applied periodic force, its frequency remaining unchanged, (ii) the frequency of the applied periodic force, its amplitude remaining unchanged?

Solution. (i) We know that the rate of dissipation of energy $\propto (\text{amplitude of the force})^2$.

Thus, if R_1 and R_2 be the rate of dissipation of energy and f_{01} and f_{02} the amplitudes of the force in the two cases respectively, the frequency of the force remaining the same in either case, we have

$$R_1/R_2 = (f_{01}/f_{02})^2$$

Since, $R_2 = \frac{1}{2} R_1$, we have $R_1/\frac{1}{2} R_1 = f_{01}^2/f_{02}^2$, whence, $f_{01}^2/f_{02}^2 = 2$.

So that, $f_{02}^2 = f_{01}^2/2$ and, therefore, $f_{02} = f_{01}/\sqrt{2} = 0.7071 f_{01}$.

Thus, the amplitude of the applied force in the second case must be 0.7071 of its amplitude in the first case.

(ii) We have quality factor of the oscillator, $Q = \omega_0 \tau = p_0 \tau$, because at resonance, $\omega_0 = p_0$.

Since $\Delta p = k = 1/2\tau$, we have $\tau = 1/2 \Delta p$.

$\therefore Q = p_0/2 \Delta p$, whence, $\Delta p = p_0/2Q = 2\pi \times 20/2 \times 200 = \pi/10 = 0.3142$.

Or, angular frequency of the applied force must be $p = p_0 + \Delta p$.

$$\begin{aligned} \text{And, therefore, frequency of the applied force} &= \frac{p}{2\pi} = \frac{p_0}{2\pi} \pm \frac{\Delta p}{2\pi} \\ &= 20 \pm \frac{0.3142}{2\pi} = 20 \pm 0.05. \end{aligned}$$

Example 8.34. A resistance of 10 ohms is joined in series with an inductance of 0.5 henry. What capacitance should be put in series with the combination to obtain maximum current? What will be the potential difference across the resistance, inductance and capacitance? The current is being supplied by a 200 volts and 50 cycles A.C. mains. (Agra, 2004)

Solution. The impedance in an LCR circuit is, as we know, given by

$$Z = \sqrt{R^2 + (Lp - 1/Cp)^2}$$

For resonance or maximum current, $Z = R$, i.e., $(Lp - 1/Cp) = 0$. Or, $Lp = 1/Cp$, and, therefore, $C = 1/Lp^2$.

Here, $L = 0.5 \text{ H}$ and $p = 2\pi n = 2\pi \times 50 = 100\pi$. So that,

$$\text{the required capacitance } C = \frac{1}{0.5 \times (100\pi)^2} = 20.27 \times 10^6 \text{ farad} = 20.27 \text{ } \mu\text{F.}$$

The maximum current will obviously be $I_0 = E_0/Z = 200/10 = 20 \text{ amp}$ [$\because Z = R = 10 \text{ ohms.}$]

$\therefore$ Potential difference across resistance $R = I_0 \times R = 20 \times 10 = 200$ volts.

Since here reactance $X_L = Lp$ due to inductance = reactance $X_C = 1/Cp$ due to capacitance, we have $p.d.$ across $X_L = p.d.$ across $X_C = I_0 Lp = I_0/Cp = 20 \times 0.5 \times 2\pi \times 50 = 3142$ volts.

Example 8.35. An A.C. potential of $1 \times 10^5 \text{ sec}^{-1}$ frequency and 1.0 volt amplitude is applied to a series LCR circuit. If $R = 2.0$ ohm, $L = 0.50$ millihenry, calculate the value of C to secure resonance. Also calculate the rms value of current and peak potential difference across the condenser.

Solution. As in problem 8.34 above, so also here, for resonance, $Z = R$, i.e., $Lp = 1/Cp$, whence,

$$C = 1/Lp^2 = 1/L \times (2\pi n)^2 = 1/0.5 \times 10^{-3} \times (2\pi \times 10^5)^2$$

$$= 5.067 \times 10^{-9} \text{ F} = 5.067 \times 10^{-3} \mu\text{F}.$$

Now, rms or virtual value of current, $I_v = E_v/Z = E_v/R$ [$\because$ at resonance $Z = R$.]

Since $E_v = E_0/\sqrt{2}$ and $R = 2$ ohms, we have

rms value of current, $I_v = E_0/2\sqrt{2} = 1/2\sqrt{2} = 0.3536$ amp. [$\because E_0 = 1$ volt.]

And, peak p.d. across the condenser or capacitor = peak value of current $\times$ reactance (X_C) due to capacitor = $(E_0/R) \times (1/Cp) = (1/2) (2 \times 10^7 \pi^2 / 2\pi \times 10^5) = 157$ volts.

Example 8.36. A circuit contains a resistance of 4 ohms and an inductance of 0.68 henry, and an alternating effective emf of 500 volts at a frequency of 120 cps is applied to it. Find the value of the effective current in the circuit and the power factor.

Solution. We know that the effective current in an LCR circuit

$$= \frac{\text{Effective emf}}{\text{Impedance of the circuit}} = \frac{E}{Z}.$$

Here, effective emf $E = 500$ volts and impedance $Z = \sqrt{R^2 + L^2 p^2} = \sqrt{R^2 + L^2 (2\pi n)^2}$.

Or, $Z = \sqrt{(4)^2 + (0.68)^2 (2\pi \times 120)^2} = \sqrt{16 + 262800} = 512.6$ ohms.

$\therefore$ Effective current in the circuit, $I = \frac{500}{512.6} = 0.9754$ amp.

And, power factor of the circuit = $R/Z = 4/512.6 = 0.0078$.

Example 8.37. An electric lamp which runs at 40 volts D.C. and consumes 10 amperes current is connected to A.C. mains at 100 volts, 50 cycles. Calculate the inductance of the choke.

Solution. Clearly, resistance of the lamp, $R = \frac{\text{voltage}}{\text{current}} = \frac{40}{10} = 4$ ohms.

And, impedance of A.C. circuit, $Z = \sqrt{R^2 + (Lp)^2}$, where L is the inductance of the choke.

Here, $R = 4$ ohms, $P = 2\pi n = 2\pi \times 50 = 100\pi$ and $I = 10$.

$$\therefore I = \frac{E}{Z} = \frac{E}{\sqrt{R^2 + (Lp)^2}} = \frac{100}{\sqrt{4^2 + (L + 100\pi)^2}} = 10.$$

$$\text{Or, } \frac{10^4}{16 + 10^4 \pi^2 L^2} = 100. \text{ Or, } 10^6 \pi^2 L^2 = 10^4 - 1600 = 8400.$$

So that, $L^2 = 8400/10^6 \pi^2$. Or, $L = \sqrt{84/10^4 \pi^2} = \sqrt{84}/100\pi = 0.02917$.

Thus, the inductance of the choke = 0.02917 henry.

Example 8.38. An inductance coil of inductance 0.05 henry and resistance 12.4 ohms is connected to an A.C. supply at 200 volts and 50 cycles. Find (a) power factor, and (b) angle of lag. (c) Show that the power consumed is just equal to the rate of production of heat.

Solution. (a) We know that the power factor, $\cos \phi = R/Z = R/\sqrt{R^2 + (Lp)^2}$.

Here, $R = 12.4 \text{ ohms}$, $L = 0.05 \text{ henry}$ and $p = 2\pi n = 2\pi \times 50 = 100 \pi$.

$$\therefore \text{power factor, } \cos \phi = \frac{12.4}{\sqrt{(12.4)^2 + (0.05 \times 100\pi)^2}} = \frac{12.4}{\sqrt{153.8 + 246.7}} = \frac{12.4}{20} = 0.62.$$

(b) And, therefore, angle of lag, $\phi = \cos^{-1}(0.62) = 51^\circ 41'$.

(c) Clearly, power consumed, $E_v I_v \cos \phi$.

$$\begin{aligned} \text{Now, } E_v &= 100 \text{ volts, } I_v = E_v/Z = 100/\sqrt{(12.4)^2 + (0.05 \times 100\pi)^2} = 100/20 \\ &= 5 \text{ ampere and } \cos \phi = 0.62. \end{aligned}$$

$$\therefore \text{power consumed} = 100 \times 5 \times 0.62 = 310 \text{ watts.}$$

And, rate of production of heat $= I^2 R = 25 \times 12.4 = 310 \text{ watts.}$

Thus, power consumed = rate of production of heat.

EXERCISE

I–Damping force–Damped harmonic oscillator

- What is meant by a damping or a dissipative force? What is its effect on a harmonic oscillator? Illustrate your answer by one or two homely examples. *(Purvanchal 2004)*
- Prove that the damping force is independent of acceleration or displacement and is proportional to velocity. *(H.P.U. 2003, 2000; GNDU 2000)*
- Obtain an expression for the velocity of a damped harmonic oscillator and deduce from it the definition of *relaxation time* (τ) of the oscillator.
- Show that both velocity and kinetic energy of a damped harmonic oscillator fall exponentially with time, the relaxation time in the latter case being half that in the former. *(Guhati 2004)*
- (a) What are damped vibrations? Establish the differential equation of motion for a damped harmonic oscillator and obtain an expression for displacement. Discuss the case of heavy damping, and light damping. *(P.U. 2003; Nagpur U. 2002, 2001; Kerala U. 2001; Pbi. U. 2001; H.P.U. 2000; Purvanchal U. 2007, 2006, 2004)*
(b) Using the general solution of equation of damped simple harmonic motion discuss the case of critical damping. *(Nagpur U. 2001; P.U.)*
- Write down the equation of damped simple harmonic oscillator. Find the expression for displacement and discuss when we get oscillatory damped simple harmonic motion. *(P.U. 2003; Luck. U. 2002; Nagpur U. 2001; Indore U. 2001; Purvanchal U. 2004; H.P.U. 2000)*
- Deduce the differential equation of a damped harmonic oscillator and discuss in detail the case of critical and underdamping.
- Write the differential equations for the following cases of damped harmonic oscillator: (i) mechanical, (ii) electrical. *(Kanpur 2004)*
- What is meant by the terms (i) *logarithmic decrement* and (ii) *quality factor* (Q) of a damped harmonic oscillator? Obtain the usual expressions for them. *(Bangalore 2006)*
- Obtain the differential equation for the motion of the coil of a moving coil galvanometer and discuss the conditions under which its motion is (i) *dead beat*, (ii) *critical* and (iii) *ballistic*. Enumerate the essential requirements of a ballistic galvanometer. *(Patna, 2006)*

11. Show that $x = (A + Bt)e^{\frac{-r}{2m}t}$ is the solution of the differential equation $\frac{d^2x}{dt^2} + \frac{r}{m} \frac{dx}{dt} + \frac{s}{m}x = 0$ for the critically damped oscillations.
12. A charged capacitor discharges through an inductance and resistance in series. Discuss possible solutions and represent these by graphs. Deduce the condition under which the discharge is (i) critically damped and (ii) oscillatory. Obtain expression for the frequency. Does the presence of resistance effect the amplitude and frequency of damped oscillations?
(P.U., 2003; H.P.U., 2002, 2000)
13. Define logarithmic decrement and derive a relation for it for a mechanical oscillator and an electrical oscillator. How can you determine experimentally the value of logarithmic decrement and damping co-efficient?
(P.U. 2003, 2002; Meerut U. 2001; G.N.D.U. 2004, 2000; Pbi. U., 2000)
14. (a) In a ballistic galvanometer the coil is wound over a non-metallic frame, comment. (Pbi. U. 2002)
(b) Why the coil of a moving coil galvanometer, ammeter and voltmeter is wound over a metallic frame?
(Pbi. U. 2001)
15. Define relaxation time of damped oscillatory system. Show that it varies inversely as damping constant. Derive an expression for relaxation period of (i) a mechanical oscillator (ii) electrical oscillator. How is it useful in determining the logarithmic decrement of a system?
(P.U., 2004; H.P.U., 2001; Pbi. U., 2000; G.N.D.U., 2001)
16. (a) Define quality factor of a damped oscillator. Deduce an expression for it for a mechanical oscillator and an electrical oscillator.
(b) Show that (i) lower the damping higher will be the quality factor. (ii) For large quality factor damping has little or no effect on the frequency. (G.N.D.U. 2002; H.P.U. 2001; Pbi. U. 2000)
17. (a) What is damping? On what factors the damping depends?
(b) What is the effect of damping on the natural frequency of an oscillator? (Pbi. U. 2003)
(c) Does viscous damping remain proportional to velocity under all conditions? (Pbi. U. 2000)
(d) Write unit of damping constant and damping coefficient for mechanical and electrical oscillator.
(H.P.U. 2002)
18. Distinguish between transient and steady state in a forced oscillator. Explain the transient and steady state behaviour of a mechanical oscillator driven by a force.
$$F = F_0 e^{i\omega t}$$

Discuss the case when $F = F_0 \cos \omega t$ and when $F = F_0 \sin \omega t$ and show that the driven oscillator is always behind the driving force in phase.
(Nagpur U. 2003; H.P.U., 2003, 2002, 2000; P.U., 2004, 2003, 2002, 2000; G.N.D.U., 2003; Luck. U., 2001; Pbi. U., 2002, 2000; Meerut U., 2002)
19. Discuss the behaviour of displacement versus driving force frequency in case of a forced oscillator. Show that (i) The displacement at low frequency is independent of frequency (ii) The resonant frequency of driving force is slightly less than the natural frequency of the oscillator (iii) Maximum amplitude $A_{max} = \frac{F_0}{r\omega'}$ where r is damping constant and $\omega' = \sqrt{\frac{s}{m} - \frac{r^2}{4m^2}}$
(H.P.U., 2003, 2001; P.U., 2002, 2000; G.N.D.U. 2001; Meerut U., 2000)
20. For a mechanical oscillator driven by a force $F = F_0 \cos \omega t$, discuss the variation of phase difference between displacement and driving force with driving force frequency. What is the phase difference between displacement and force at very high frequency? (G.N.D.U., 2002; H.P.U., 2001, 2000)
21. (a) Draw a graph between velocity amplitude and driving force frequency of a forced oscillator.
(H.P.U., 2001)
(b) Show that whereas at resonance displacement lags behind the driving force by $\pi/2$, the velocity is in phase with driving force.
(G.N.D.U., 2002)
22. Discuss the variation of acceleration amplitude with driving force frequency of a forced mechanical oscillator and show that

- (i) Acceleration amplitude at high frequency is frequency independent and
(ii) When damping constant is very small the acceleration resonance frequency equals the natural frequency of the forced oscillator. (P.U., 2003; G.N.D.U., 2003)
23. (a) What is absorption resonance curve? Draw it for a forced oscillator. Draw it so called? (P.U., 2001, 2000)
(b) What is meant by resonance absorption band width? Define it and show it on a $P - \omega$ graph. Explain how the damping of the medium effects the band width? (Pbi. U., 2003; Nagpur U., 2001; H.P.U., 2000)
24. Derive an expression for the Q -value of a forced oscillator in terms of resonance absorption band width. If the Q -value of an oscillator is increased how is band width of the absorption curve affected. Prove that band width $= \frac{\omega_0}{Q}$ where the symbols have their usual meaning. (Pbi. U. 2001; P.U. 2004, 2001; Nagpur U. 2003, 2001; G.N.D.U., 2000)
25. (a) Find the Q -value of an oscillator in terms of the energy decay.
(b) What is figure of merit of a vibration and how is it defined in terms of bandwidth? The quality factor Q is a measure of sharpness of resonance in case of a forced oscillator. Explain. (P.U., 2000)
26. An electric oscillator is driven by an *e.m.f.* $E = E_0 e^{i\omega t}$. Derive an expression for the current in the electric circuit in the steady state and find its value when the applied *e.m.f.* is
(i) $E = E_0 \cos \omega t$. (ii) $E_0 \sin \omega t$ (P.U. 2001, 2000)
27. (i) What are damped oscillations?
(ii) Obtain differential equation of motion of a damped harmonic oscillator.
(iii) Obtain general solution of differential equation of a damped harmonic oscillator.
(iv) In an oscillatory circuit, $L = 0.2$ Henry, $C = 0.0012 \mu\text{F}$. What is the maximum value of resistance for the circuit to be oscillatory? (Nagpur U. 2006)
28. Obtain an expression for the power dissipated in damped harmonic oscillator. (Nagpur U. 2007)
29. Discuss an LCR circuit as an example of a damped harmonic oscillator and show that it is the resistance alone which is responsible for the damping of the oscillations. Discuss the conditions under which the discharge of the capacitor is *aperiodic*, *critical* and *oscillatory*. What is the quality factor or such a circuit?
30. A body is propelled through a viscous medium with a speed of 300 cm/sec . On the propelling force being suddenly removed, it comes to rest after covering a distance of 45 metres. What is its relaxation time? How long would it be before its speed falls to 30 cm/sec ? [Ans. 15 sec; 34.5 sec.]
31. Show that whereas the amplitude of a pendulum oscillating in a viscous medium like air is affected, its time-period remains practically unaffected.
[Hint. Damping force due to a viscous medium $= -6\pi\eta r v = -6\pi\eta r l d\theta/dt$, where r is the radius of the bob, v , its velocity and l , the length of the pendulum. The equation of motion of the pendulum is thus $m l d^2\theta/dt^2 + 6\pi\eta r l d\theta/dt + mg\theta = 0$, where m is the mass of the bob. This reduces to $d^2\theta/dt^2 + (6\pi\eta r/m) d\theta/dt + (g/l)\theta = 0$ which gives $k = 3\pi\eta r/m = 3\pi\eta r/ \frac{4}{3} \pi r^3 \rho = 9\eta/4r^2 \rho$, (where ρ is the density of the material of the bob) and ω_0^2 is g/l .
Now, time-period $T = 2\pi/\omega = 2\pi/\sqrt{\omega_0^2 - k^2}$. Since k is small compared with ω_0 , we have $T = 2\pi/\sqrt{\omega_0^2} = 2\pi/\omega_0 = 2\pi/\sqrt{g/l} = 2\pi\sqrt{l/g}$, the same as in the absence of damping.
And the amplitude is of course given by $a = a_0 e^{-kt}$, where a_0 is the amplitude in the absence of damping.]
32. If the damping in the case of a damped harmonic oscillator be very small, obtain expression for (i) average total energy of the oscillator and (ii) average rate of energy dissipation. (iii) Show that the latter is equal to the work done by the damping force per second.
[Ans. (i) $E = \frac{1}{2} m \omega^2 a_0^2 e^{-2kt}$, (ii) E/τ , (iii) $-\gamma dt/dt = P$.]

33. Show that the relaxation time of a damped harmonic oscillator has the dimensions of time. Also show that (i) in this time (τ) the amplitude of the oscillator falls to 0.6084 of its undamped value and that (ii) it is reduced to half its undamped value in time 1.3846τ . ($e = 2.718$ and $\log_e 10 = 2.30$).
34. (a) Show that damping has little or no effect on the frequency of a harmonic oscillator if its quality factor (Q) be large.
[Hint. See worked example 7]
(b) If the quality factor of a sonometer wire of frequency 300 cps be 2×10^3 , in what time will its energy be reduced to $1/e^{\text{th}}$ of its energy in the absence of damping. [Ans. In 1.061 sec.]
35. The amplitude of a simple pendulum, with a bob of mass 400 gm and oscillating in air, falls to half its undamped value on completion of 50 complete oscillations. (i) If its time-period be 2 sec, calculate its quality factor. (ii) What should be the mass of the bob, its size and shape remaining unchanged, so that the damping may become critical? [Ans. (i) 453.7, (ii) 0.44 gm.]
36. A capacitor of capacitance 1 μF is discharged through a resistance of 2 ohms and an inductance of 1 henry. Is the discharge of the capacitor oscillatory? If so, obtain its frequency and calculate the time in which the amplitude of the oscillations falls to 10% of its undamped value.
[Ans. Discharge oscillatory; $n = 159$; In 2.3 sec.]
37. (a) In an LCR circuit, $L = 0.2$ henry and $C = 2\mu\text{F}$. What should be the maximum resistance in the circuit for the discharge of the capacitor to be oscillatory in character?
(b) The quality factor of an oscillatory circuit is 100 and its frequency 8 kilo-cycles per second. How will its frequency change if the resistance in the circuit be reduced to zero?
[Ans. (a) Less than 20 ohms, (b) The frequency will increase by 1 cycle per sec.]
38. Find whether the discharge of a condenser for the following inductive circuit is oscillatory: $C = 0.1 \mu\text{F}$, $L = 10 \text{ mH}$, $R = 200 \text{ ohms}$. If the circuit is oscillatory, calculate its frequency. (Agra, 1963)
[Ans. Discharge oscillatory; $n = 4772 \text{ cps}$]
39. (a) What is physical significance of logarithmic decrement of a damped oscillatory system?
(b) Explain how the conditions of critical damping are used in designing of electrical instruments.
40. Express amplitude, energy, logarithmic decrement and relaxation time in terms of Q the quality factor. (Pbi. U. 2002; H.P.U. 2000)
41. Discuss the methods (logarithmic decrement, relaxation time and quality factor) for quantitative measurement of damping effect in a damped simple harmonic oscillator.
42. Which is greater the natural frequency of damped oscillations or frequency of displacement resonance?
43. Why is there large amplitude, when frequency of external periodic force is same as natural frequency of the body.
44. Show that the displacement resonance occurs at a frequency slightly less than the frequency of velocity resonance. (P.U., 2002; H.P.U., 2001)
45. What is sharpness of resonance? Explain the effect of damping on sharpness of resonance.
(Meerut U., 2005, 2003, 2001; Nagpur U., 2003, 2002; G.N.D.U., 2003; Luck., 2002)

II—Driven harmonic oscillator

46. Write down and solve the differential equation of a damped harmonic oscillator subjected to a sinusoidal force and obtain expressions for its maximum amplitude and quality factor.
47. For a driven harmonic oscillator of natural frequency $\omega/2\pi$, described by $m\ddot{x} + r\dot{x} - kx = F \sin pt$, deduce an expression for velocity amplitude x_0 . Show that for the cases $p \gg \omega$, $p = \omega$ and $p \ll \omega$, this amplitude is governed by the inertia factor, the resistance factor and the spring factor respectively.
48. Show that in the case of a system undergoing a forced oscillation, the response is (i) independent of its mass if $p \ll \omega_0$ and (ii) independent of the spring constant if $p \gg \omega_0$.
49. Show that in a driven oscillator, the maximum power is absorbed at the frequency of velocity resonance and not at the frequency of amplitude resonance.

50. Assuming the results of forced oscillations, discuss the sharpness of resonance.
Explain the statement '*the quality factor (Q) is a measure of the sharpness of resonance in the case of a driven oscillator*'.
51. Show that the half width of the velocity versus driving frequency curve and that of the power absorbed versus driving frequency curve are $\sqrt{3k}$ and k respectively.
52. Why should two tuning forks be very accurately in unison to show resonance while a tuning fork and an air column require to be approximately tuned. (Nagpur U. 2002)
53. Derive the expression for the average power supplied to a forced (*i.e.*, driven) oscillator by an external driving force
- $$F = F_0 \cos \omega t \quad (P.U. 2000)$$
54. What is driven harmonic oscillator? How does it differ from simple and damped harmonic oscillator? (Nagpur U. 2009)
55. (a) Explain the terms *reactance*, *impedance* and *power factor*. (Patna, 2001)
(b) Define *virtual current* and *virtual potential* in an *A.C.* circuit. (Delhi (Subs.), 2000)
56. Show that a driven oscillator is always behind the driving force in phase and that its displacement is maximum when the driving force is zero and vice versa.
Illustrate for zero, low and medium damping, the general nature of the phase lag of a driven oscillator as the driving frequency gradually increases and passes through the natural frequency of the oscillator.
57. Give the theory of oscillations in a *LCR* circuit with small resistance, deducing expressions for (i) frequency of oscillations, (ii) power dissipation and (iii) quality factor of the circuit. (Agra (Supp), 2001)
58. Write explanatory notes on the terms: *reactance*, *impedance*, *rms value of current*, *rms value of emf* and *power factor* in a driven *LCR* circuit.
Show that the power in such a circuit, like that in a D.C. circuit, appears in the form of heat in the resistance alone.
59. Obtain an expression for the current in a driven *LCR* circuit and discuss how the current lags or leads the applied emf in phase (a) when the net reactance in the circuit is (i) capacitive, (ii) inductive and (b) when reactance in the circuit is equal to resistance. Illustrate your answer with the help of suitable graphs.
60. A damped harmonic oscillator is subjected to a sinusoidal driving force whose frequency is altered but amplitude kept constant. It is found that the amplitude of the oscillator increases from 0.02 mm at very low driving frequency to 8.0 mm at a frequency of 100 cps. Obtain the values of (i) the quality factor, (ii) the relaxation time, (iii) the damping factor and (iv) the half-width of the resonance curve.
[Ans. (i) $Q \cong 400$, (ii) $\tau = 0.4245 \text{ sec}$, (iii) $k = 1.178$, (iv) 2.04 rad/sec .]
61. A damped harmonic oscillator of quality factor 20 is subjected to a sinusoidal driving force of frequency twice the natural frequency of the oscillator. If the damping be small, what fraction will the amplitude of the oscillator be of its maximum value and by what angle will it differ in phase from the driving force? [Ans. $a = 0.017 a_{\max}$; 178.5° out of phase with the driving force.]
62. Obtain the frequency of the periodic torque which, when applied to a torsion pendulum of moment of inertia 900 gm-cm^2 , sets it into resonant oscillation. The torsion constant of the suspension fibre may be taken to be $4 \times 10^4 \text{ dyne-cm/radian}$. It is found that a small increase of 0.2% in the frequency of the periodic torque results in a fall of 80% in the amplitude of the pendulum. Calculate the quality factor of the pendulum. [Ans. $p = 1.06 \text{ cps}$; $Q = 187$.]
63. An alternating sinusoidal emf is applied to a circuit containing an inductance L and a resistance R . Calculate the current at any instant in the circuit.
64. Find the natural frequency of a circuit containing inductance of 50 micro-henry and a capacitance of 0.0005 microfarad. Find the wavelength to which it corresponds. (Punjab, 2002)

[Ans. 1.007 cps ; 298.1 metres .]

[Hint. $c = n\lambda$, where c is the velocity of light (or electromagnetic radiation)]

65. An alternating potential of 110 volts and 50 cycles frequency is applied to a circuit containing a resistance of 200 *ohms*, inductance of 5 henry and a capacitance of 2 μF . Calculate the maximum current in the circuit. **(Delhi (Subs.), 2003)**
[Ans. peak value of current = 0.78 amp.]
66. To a circuit containing an inductance of 10 millihenry, a capacitance of 1 μF and a resistance of 10 *ohms* in series, an *rms* voltage of 100 V is applied. Calculate the frequency at which the circuit will be in resonance with the current of the same frequency and find the value of the current.
[Ans. $n = 1592$ and $I_v = 10$ amp.]
67. Find the capacitance of the capacitor which must be placed in series with a resistance of 5 *ohms* and inductance of 200 millihenry to bring the current in phase with the voltage if the frequency be 60 cycles per sec. What current will flow if 100 volts be imposed on the circuit? **(Delhi (Hons.), 1957)**
[Ans. 35.20 μF ; 20 amp.]
68. Find the value of the inductance which should be connected in series with a capacitance of 0.5 μF , a resistance of 10 *ohms* and an A.C. source of frequency 50 *cps*, so that the power factor of the circuit be unity.
[Ans. 20.29 henry.]
69. An alternating voltage of 4 volts is applied at the resonant frequency to a coil of inductance 0.004 H and resistance 20 *ohms* in series with a 0.001 μF capacitor. Find (i) the resonant frequency (ii) the current flowing, (iii) the voltage across the capacitor. [Ans. (i) 79600 *cps*; 0.2 amp; 400 volts.]
70. An alternating voltage of 100 *rms* is applied to a circuit of resistance 0.5 *ohm* and an inductance of 0.01 henry, the frequency being 50 *cycles/sec*. What is the current and the time lag?
(Patna (Supp.), 1963) Ans. 31 amp; 0.0045 sec.
71. (a) Find whether the discharge of a condenser through the following inductive circuit is oscillatory: $C = 0.1 \mu F$, $L = 10$ milli henry, $R = 200$ *ohms*. If the circuit is oscillatory, calculate its frequency.
(Agra, 1963) Ans. Oscillatory; 4772 *cps*.
(b) In an oscillatory circuit, $L = 0.2$ henry, $C = 0.0012 \mu F$. What is the maximum value of resistance for the circuit to be oscillatory? **(Punjab, 1959)** [Ans. Less than 2.582×10^4 *ohms*.]
72. A coil connected to a sinusoidal alternating supply of *emf* 240 volts (*rms*) and of frequency 50 *cps* takes a current of 5.0 amp (*rms*). A wattmeter in the circuit reads 1000 watts. Find (a) the power factor of the circuit, (b) the resistance of the coil, (c) the reactance of the coil, (d) the maximum stored energy. [Ans. (a) 0.8; (b) 40 *ohms*; (c) 27 *ohms* very nearly, (d) 21 joules very nearly.]
73. Calculate the band width of an acoustic system having $Q = 1.75$ and resonant frequency 280 Hz.
[Ans. 160 Hz] **(Nagpur U., 2001, 2006)**
74. Find the resonant frequency of an acoustic system having Q equal to 1.60 and half power frequencies equal to 180 Hz and 380 Hz respectively. [Ans. 320 Hz] **(Nagpur U., 2008)**
75. Find the frequency of a circuit containing inductance of 1000 μH and capacity of 0.01 μF . To which wavelength (of radio wave) its response will be maximum? [Ans. 1890 m] **(P.U. 2001, 2000)**
76. A torsion pendulum of moment of inertia 10^5 *gm-cm²* suffers an angular displacement of 4° under a constant torque of 2000 *dynes-cm*. What should be the frequency of the periodic torque that would set the pendulum in resonant vibration? [Ans. 9.926×10^5 *erg/sec*]

9

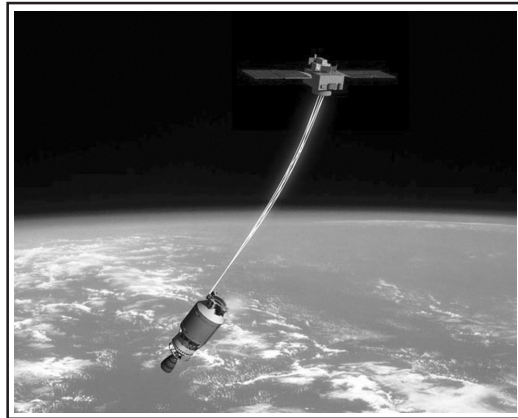
Chapter

COUPLED OSCILLATORS

Brief Contents

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INTRODUCTION

In the previous chapter, we have studied damped and driven oscillators. The harmonic oscillator and the system it models have a *single degree of freedom*. More complicated systems have more degrees of freedom, for example two masses and three spring system. In such cases, the behaviour of each variable influences that of the others. This leads to a *coupling* of oscillations of the individual degrees of freedom. When the motion of one oscillating system influences another, the two are said to be *coupled*.

The two or more oscillators linked together in such a way that an exchange of energy transfer takes place between them are called coupled oscillators.

In a forced oscillator, it is assumed that the external driving force is practically not affected by the oscillations of the driven system, *i.e.*, the flow of energy between the driving agency and the driven system is only in one direction—from the driving agency to the driven system. However, in actual practice, there is always some feed back of energy, although small. In this chapter, let us study two coupled oscillators and their normal modes of vibration and extend the idea to N coupled oscillators.

9.1 SOME DEFINITIONS

Normal Coordinates: Normal co-ordinates are those co-ordinates which help us to express the equations of motion of the harmonic oscillators of a coupled system in the form of a set of *linear differential equations with constant co-efficients* and in which each equation contains only *one variable*.

Normal Modes of Vibration. *The manner in which a coupled system oscillates is called a mode.* The mode of a coupled system may be harmonic or non-harmonic.

The harmonic modes of a coupled system are called normal modes.

Normal modes have definite characteristics and are represented by linear differential equations with constant coefficients and only one dependent variable or normal co-ordinates. A normal mode has its own frequency known as *normal frequency*. In each normal mode all the components of the system vibrate with the same normal frequency. The normal modes of vibration are *entirely independent of each other*, since the energy associated with a normal mode is never exchanged with the energy associated with another normal mode. The total energy of the oscillator is equal to the sum of the energies of all the normal modes. If at any time only one mode is excited and vibrates the other modes will always be at rest and unexcited and these will acquire no energy from the vibrating mode.

Degrees of Freedom. A degree of freedom of a system is the *independent way* by which the system may *acquire energy*. A degree of freedom is assigned its own particular normal co-ordinates. The *number* of degrees of freedom and the number of normal co-ordinates of a system is the number of different ways in which the system can acquire energy. Each harmonic oscillator has two degrees of freedom as it may have *both kinetic* as well as *potential energy*.

The kinetic energy of a simple harmonic oscillator of mass m and having displacement co-ordinate x is given by $\frac{1}{2}m\dot{x}^2 = a\dot{x}^2$ where $a = \frac{1}{2}m$. The potential energy is given by $\frac{1}{2}sx^2 = bx^2$ where s is the stiffness constant.

If the normal modes of a harmonic oscillator are represented by normal co-ordinates X and Y , then the total energy corresponding to the two modes will be

$$E_X = a \dot{X}^2 + bX^2$$

and

$$E_Y = c \dot{Y}^2 + dY^2$$

where a , b , c and d are constants, $a \dot{X}^2$ and $c \dot{Y}^2$ give the kinetic energy and bX^2 and dY^2 the potential energy.

9.2 EQUATION OF MOTION OF STIFFNESS COUPLED SYSTEM OF TWO PENDULUMS

Consider a coupled system of two identical pendulums each having a pendulum bob of mass m suspended by a light weightless, rigid rod of length l . The bobs are connected by a light spring of stiffness s , whose normal length is equal to the distance between the bobs when none of the two pendulums is displaced from its equilibrium position. Such pendulums are known as **stiffness coupled**.

Let the two pendulums A and B be set into vibrations with a *small* amplitude in the plane of the paper and let x and y be the displacements (in the same direction) of the pendulums marked A and B respectively as shown in Fig. 9.1 and Fig. 9.2, then the spring is elongated by a length $(x - y)$ and the corresponding force comes into play is $s(x - y)$. The component of the force due to gravity tending to bring the bobs of the pendulums back to its mean position is

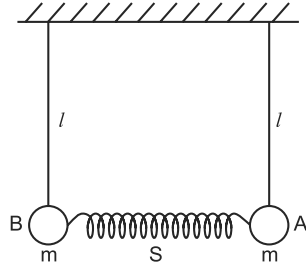


Fig. 9.1

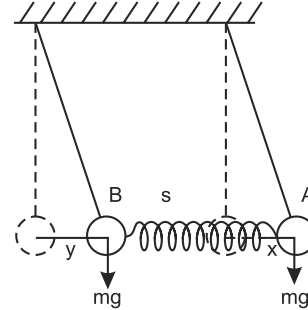


Fig. 9.2

$$= -\frac{mgx}{l} \text{ for the pendulum } A$$

and

$$= -\frac{mgy}{l} \text{ for the pendulum } B.$$

∴ The equations of motion for the pendulums A and B respectively, are

$$m\ddot{x} = -mg\frac{x}{l} - s(x - y) \quad \dots(i)$$

and

$$m\ddot{y} = -mg\frac{y}{l} + s(x - y) \quad \dots(ii)$$

The first term in each equation is the normal simple harmonic motion term and the second term is due to the *coupling of the spring*. If $x > y$ the spring is extended beyond its normal length and will, therefore, apply a force against the acceleration of x but in favour of the acceleration of y . Dividing equations (i) and (ii) by m and substituting $\frac{g}{l} = \omega_0^2$, we have

$$\ddot{x} + \frac{g}{l}x = \ddot{x} + \omega_0^2x = -\frac{s}{m}(x - y) \quad \dots(iii)$$

$$\ddot{Y} + \frac{g}{l}y = \ddot{Y} + \omega_0^2 y = -\frac{s}{m}(m-x) \quad \dots(iv)$$

Adding (iii) and (iv), we get

$$\ddot{x} + \ddot{Y} + \frac{g}{l}(x+y) = \ddot{x} + \ddot{Y} + \omega_0^2(x+y) = 0 \quad \dots(v)$$

Subtracting (iv) from (iii), we get

$$\ddot{x} - \ddot{Y} + \frac{g}{l}(x-y) = \ddot{x} - \ddot{Y} + \left(\omega_0^2 + \frac{2s}{m}\right)(x-y) = 0 \quad \dots(vi)$$

Let us choose two new co-ordinates X and Y so that

$$X = x + y \quad \text{and} \quad Y = x - y$$

$$\text{then} \quad \dot{X} = \dot{x} + \dot{y} \quad \text{and} \quad \dot{Y} = \dot{x} - \dot{y}$$

$$\text{and} \quad \ddot{X} = \ddot{x} + \ddot{y} \quad \text{and} \quad \ddot{Y} = \ddot{x} - \ddot{y}$$

Substituting the values in (v) and (vi), we get

$$\ddot{X} + \frac{g}{l}X = \ddot{X} + \omega_0^2 X = 0 \quad \dots(vii)$$

$$\text{and} \quad \ddot{Y} + \left(\frac{g}{l} + \frac{2s}{m}\right)Y = \ddot{Y} + \left(\omega_0^2 + \frac{2s}{m}\right)Y = 0 \quad \dots(viii)$$

Normal co-ordinates. It is seen that whereas equations (v) and (vi) each have two variables, we find that equations (vii) and (viii) have only one variable. The motion of the coupled system is thus described in terms of two co-ordinates X and Y . *Each equation of motion is a linear differential equation of a simple harmonic oscillator with constant coefficients with only one variable.* The co-ordinates X and Y are therefore *normal co-ordinates* of the coupled system.

Thus the coupled system of two simple pendulums has *two normal modes*, one described by *normal co-ordinate* X and the other by the *normal co-ordinate* Y .

In phase mode. $Y = 0$. When normal co-ordinate $Y = 0$, $x - y = 0$ i.e., $x = y$ for all times. The motion is completely described by

$$\ddot{X} + \frac{g}{l}X = \ddot{X} + \omega_0^2 X = 0$$

As the relative displacement of the two pendulum always remain the same ($\because x = y$), *the stiffness of the coupling has no effect*. The spring always remains at its normal length. The frequency of oscillations is the same as that of either pendulum in isolation i.e., $\omega_0^2 = \frac{g}{l}$. Both pendulums are always swinging in the same phase. Such vibrations are called *in-phase vibrations* and the corresponding mode is known as *in-phase mode*.

Out of phase mode: $X = 0$. When normal co-ordinate $X = 0$, $x + y = 0$ or $x = -y$ for all times. The motion is completely described by

$$\ddot{Y} + \left(\frac{g}{l} + \frac{2s}{m}\right)Y = \ddot{Y} + \left(\omega_0^2 + \frac{2s}{m}\right)Y = 0$$

When the displacement (x) of the pendulum A is *positive* that of $B(y)$ is *negative* but equal to that of $A(x = -y)$. At zero displacement the two pendulums will be moving in opposite directions. Thus the periodic motion of the pendulums will be 180° out of phase. Such vibrations are called *out of phase vibrations* and the corresponding mode is known as *out of phase mode*.

9.3 TOTAL ENERGY OF TWO IDENTICAL STIFFNESS COUPLED PENDULUMS

The equation of motion of a stiffness coupled system of two identical pendulums A and B is given by

$$\ddot{X} + \omega_0^2 X = 0 \quad \dots(i)$$

$$\ddot{Y} + \left(\omega_0^2 + \frac{2s}{m} \right) Y = 0 \quad \dots(ii)$$

where $X = x + y$ and $Y = x - y$ are the *normal co-ordinates of the system* x and y are the displacement (in the same direction) of the pendulums A and B respectively as shown in Fig. 9.2 and $\omega_0^2 = \frac{g}{l}$ where l is the length of the pendulum and s is the stiffness constant. Solving equations (i) and (ii), we get

$$X = x + y = X_0 \cos(\omega_1 t - \phi_1) \quad \dots(iii)$$

$$\text{and} \quad Y = x - y = Y_0 \cos(\omega_2 t - \phi_2) \quad \dots(iv)$$

X_0 and Y_0 are the *normal mode amplitudes* and $\omega_1 = \omega_0 = \left(\frac{g}{l} \right)^{\frac{1}{2}}$ and

$$\omega_2 = \left(\omega_0^2 + \frac{2s}{m} \right)^{\frac{1}{2}} = \left(\frac{g}{l} + \frac{2s}{m} \right)^{\frac{1}{2}} \text{ are normal mode frequencies.}$$

To simplify further discussions, let

$$X_0 = Y_0 = 2a$$

$$\text{and} \quad \phi_1 = \phi_2 = 0$$

$\therefore$ Equations (iii) and (iv) give

$$X = x + y = 2a \cos \omega_1 t$$

$$\text{and} \quad Y = x - y = 2a \cos \omega_2 t$$

Adding (v) and (vi), we get

$$2x = X + Y = 2a \cos \omega_1 t + 2a \cos \omega_2 t$$

$$\therefore \quad x = \frac{1}{2} (X + Y) = a \cos \omega_1 t + a \cos \omega_2 t$$

Subtracting (vi) from (v), we get

$$y = \frac{1}{2} (X - Y) = a \cos \omega_1 t - a \cos \omega_2 t$$

The corresponding velocities are given by

$$\dot{x} = -a\omega_1 \sin \omega_1 t - a\omega_2 \sin \omega_2 t$$

$$\text{and} \quad \dot{y} = -a\omega_1 \sin \omega_1 t + a\omega_2 \sin \omega_2 t$$

Now let the system be set in motion by displacing the bob x to the right by a distance $x = 2a$ keeping $y = 0$ and both the bobs be released from rest so that $\dot{x} = \dot{y} = 0$ at a time $t = 0$. The motion of pendulum A is given by

$$x = a \cos \omega_1 t + a \cos \omega_2 t = 2a \cos \frac{(\omega_2 - \omega_1)t}{2} \cos \frac{(\omega_1 + \omega_2)t}{2}$$

and the motion of pendulum B is given by

$$y = a \cos \omega_1 t - a \cos \omega_2 t = -2a \sin \frac{(\omega_2 - \omega_1)t}{2} \sin \frac{(\omega_1 + \omega_2)t}{2}$$

$\left(\frac{\omega_2 - \omega_1}{2}\right) = \omega_m$ is called the *modulated* or beat frequency and $\frac{\omega_1 + \omega_2}{2} = \omega_a$ is called the *average frequency*.

Hence the modulated amplitude of pendulum $A = A = 2a \cos \omega_m t$

and $x = A \cos \omega_a t$

The modulated amplitude of pendulum $B = B = 2a \sin \omega_m t$

and $y = -B \sin \omega_a t$

If we assume that the spring is very weak and does not store any energy we can consider the modulated amplitude $2a \cos \omega_m t$ and $2a \sin \omega_m t$ to remain constant over one cycle of average frequency.

$$\begin{aligned} \therefore \text{Energy of pendulum } A; \quad E_A &= \frac{1}{2}mv^2 = \frac{1}{2}m(A\omega_a)^2 \\ &= \frac{1}{2}m(2a \cos \omega_m t)^2 \omega_a^2 = 2ma^2 \omega_a^2 \cos^2 \omega_m t \end{aligned}$$

$$\begin{aligned} \text{Energy of pendulum } B; \quad E_B &= \frac{1}{2}mv^2 = \frac{1}{2}m(B\omega_a)^2 \\ &= \frac{1}{2}m(2a \sin \omega_m t)^2 \omega_a^2 = 2ma^2 \omega_a^2 \sin^2 \omega_m t \end{aligned}$$

$$\text{Hence total energy} \quad E = E_A + E_B = 2ma^2 \omega_a^2$$

Evidently, the total energy $E = E_A + E_B$ is constant as m , a and ω_a are constants.

$$\begin{aligned} \text{Also} \quad E_A - E_B &= 2ma^2 \omega_a^2 [\cos^2 \omega_m t - \sin^2 \omega_m t] \\ &= E \cos 2\omega_m t = E \cos(\omega_2 - \omega_1)t \end{aligned}$$

$$\therefore E_A = \frac{1}{2}E[1 + \cos(\omega_2 - \omega_1)t]$$

$$\text{and} \quad E_B = \frac{1}{2}E[1 - \cos(\omega_2 - \omega_1)t]$$

This shows that total energy is *constant* but it *flows back and forth between the pendulums at the modulated (or beat) frequency*.

Thus we see that after drawing aside the bob of the first pendulum by a distance $2a$ and releasing in the pendulum shows the cosine behaviour at a frequency ω_a which is the average of the two normal mode frequencies and its amplitude also varies according to the cosine law at a low frequency of half the difference between the normal mode frequencies. On the other hand pendulum 2 starts from zero and vibrates according to the sine law with the same average frequency and its amplitude builds up to $2a$ and then decays according to sine law at the same low frequency of half the difference between the normal mode frequencies.

The initial configuration $x = 2a, y = 0$ can be decomposed into X and Y modes as shown in Fig 9.3. The X mode *i.e.*, $x = y = a$ so that $X_0 = x + y = 2a$ is known as the '*in phase*' mode and the Y mode *i.e.*, $x = a, y = -a$ so that $Y_0 = x - y = 2a$ is known as the '*out of phase*' mode.

As the Y -mode has a higher frequency it will gain half a vibration (phase difference π radian) on the X -mode after a number of vibrations and the combination of X and Y mode then will give $x = 0$ and

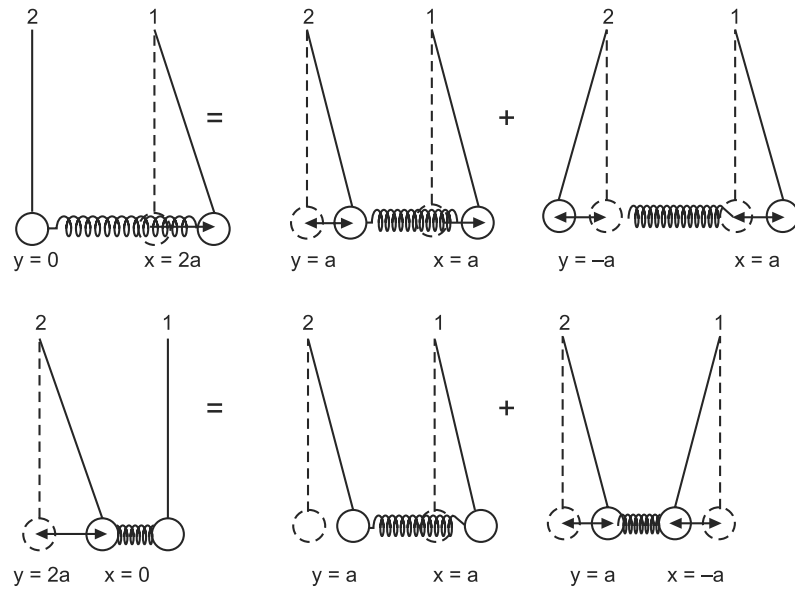


Fig. 9.3

$y = 2a$ as shown in the lower Fig. 9.3. After some time when Y gains another half a vibration $x = 2a$ and $y = 0$. Thus the pendulums only exchange energy the normal modes do not. The 'in phase' mode remains 'in phase' mode and 'out of phase mode' remains 'out of phase' mode.

9.4 TYPES OF COUPLING

Two oscillators are coupled together to bring about an exchange of energy between them. The common coupling components for mechanical oscillators may be 'stiffness' or 'mass' and in case of electrical oscillators it may be 'capacitance' or 'inductance'. Capacitance and inductance are energy storing electrical components and coupling through them consumes no power, thereby making it possible for the energy transfer to take place over a number of oscillations. Coupling may also be done through the 'resistance' component but this causes an inevitable loss of energy that the amplitude goes on rapidly falling.

9.5 FREQUENCY OF OSCILLATION IN-PHASE AND OUT-OF-PHASE MODE

The differential equations of motion of coupled system of two identical (mechanical) oscillators say two simple pendulums in terms of normal co-ordinates X and Y are given by equations (vii) and (viii) of article 9.2 as

$$\ddot{X} + \frac{g}{l}X = \ddot{X} + \omega_0^2 X = 0$$

$$\text{and} \quad \ddot{Y} + \left(\frac{g}{l} + \frac{2s}{m} \right) Y = \ddot{Y} + \left(\omega_0^2 + \frac{2s}{m} \right) Y = 0$$

where $X = x + y$ and $Y = x - y$; x and y being the displacements of the two (mechanical) oscillations, s the stiffness and m the mass.

(i) **In Phase Mode:** In the *in-phase* mode $x = y$ so that $Y = x - y = 0$. In such a case the motion is completely represented by the equation.

$$\ddot{X} + \frac{g}{l}X = \ddot{X} + \omega_0^2 X = 0$$

The frequency of oscillations of the coupled system is, therefore, $\omega_0 = \sqrt{\frac{g}{l}}$ which is the same as that of uncoupled oscillators.

Thus, in the *in-phase mode the frequency of oscillations of the coupled system is the same as that of the uncoupled oscillators or of either pendulum in isolation*. This is due to the fact that the two pendulums are always oscillating in phase as shown in Fig. 9.2 so that the light spring always has its

(ii) Out of Phase Mode: In the *out of phase mode* $x = -y$ so that $X = x + y = 0$. In such a case the motion is completely represented by the equation

$$\ddot{Y} + \left(\omega_0^2 + \frac{2s}{m} \right) Y = 0$$

The frequency of oscillations of the coupled system is, therefore, given by

$$\omega' = \sqrt{\omega_0^2 + \frac{2s}{m}}$$

which is greater than ω_0 , the frequency of the uncoupled oscillators.

Thus in the out of phase mode the frequency of oscillations of the coupled system gets raised.

This is due to the fact that the two pendulums are always out of phase either in the *extended* or in the *compressed* position as shown in Fig. 9.3 i.e., *extreme positions* and hence the *coupling becomes*

9.6 CHARACTERISTICS OF IN-PHASE AND OUT-OF-PHASE MODES OF VIBRATION

Two oscillating, identical coupled simple pendulums have as we know, two normal modes of vibration.

(i) In-phase mode and (ii) Out of phase mode, each having its own characteristics.

(a) Characteristics of in phase mode. (i) The in-phase mode can be excited by displacing the bobs of the two pendulums to the same side by the same amount and then letting them oscillate purely on their own.

(ii) The equation of the in-phase mode is

$$\ddot{X} + \omega_0^2 X = 0 \quad \dots(i)$$

It describes the oscillatory behaviour of the system when

$$Y = (x - y) = 0 \text{ or } x = y$$

(iii) The displacement of both the (bobs) or masses is the *same in magnitude* as well as in *direction* and both the masses continuously oscillate in the *same phase*. It means that both the pendulums pass through their mean position or through either of the extreme positions simultaneously.

(iv) The amplitude of the two simple pendulums is the same i.e., a .

(v) The shape or configuration of the mode is $\frac{x}{y} = +1$

(vi) Each pendulum executes simple harmonic oscillation at frequency $\omega_0 = \sqrt{g/l}$ which is the natural frequency of free oscillations of either pendulum in isolation.

(vii) In the in-phase mode there is no effect of the stiffness term s of the coupling on the motion of the two masses because the spring always has its natural length — it is neither stretched nor compressed.

(viii) A solution of Eq. (i) gives

$$X = X_0 \cos(\omega_0 t - \phi).$$

If the maximum value of x and y be a , then $X_0 = 2a$ is the *normal mode amplitude*.

(b) Characteristics of out of phase mode. (i) The out of phase mode can be excited by displacing the bobs of the two pendulums in opposite directions by the same amount and then let them oscillate freely on their own.

(ii) The equation of the out of phase mode is

$$\ddot{Y} + \left(\omega_0^2 + \frac{2s}{m} \right) Y = 0 \quad \dots(ii)$$

It describes the oscillatory behaviour of the system when

$$X = (x + y) = 0 \text{ or } x = -y$$

(iii) The displacement of either mass (bob) is equal in magnitude but opposite in direction to the other mass and the two masses continuously oscillate *out of phase* by an angle of 180° or π -radian with respect to each other.

It means that both the pendulums pass through their position (in opposite directions) or are at their (opposite) extreme positions simultaneously.

(iv) The amplitude of the two simple pendulums is the same *i.e.*, ' a '.

(v) The shape or configuration of the mode is $\frac{x}{y} = -1$.

(vi) Each pendulum executes simple harmonic oscillations at a frequency

$$\omega' = \sqrt{\frac{g}{l} + \frac{2s}{m}} = \sqrt{\omega_0^2 + \frac{2s}{m}}$$

Which is higher than the natural frequency of the free oscillations of either pendulum.

(vii) In the out of phase mode the coupling term s dominates the motion of the two masses and raises the frequency of oscillation. It is because the coupling spring is either in the stretched or in the compressed state. Only once in one time period of oscillation the spring acquires its normal length when the masses pass through their mean positions.

(viii) A solution of Eq. (ii) gives

$$Y = Y_0 \cos(\omega' t - \phi')$$

If the maximum value of x and y be a , then $Y_0 = 2a$ is the normal mode amplitude.

Note. X_0 and Y_0 are purely mathematical quantities and do not represent the amplitude of the oscillation of either pendulum which has a value $a = \frac{X_0}{2}$ for the in-phase mode and $a = \frac{Y_0}{2}$ for the out of phase mode.

9.7 GENERAL METHOD OF FINDING NORMAL MODE FREQUENCIES

Consider a system of two coupled simple pendulums of the same length coupled by a spring of stiffness s , then their equations of motion are

$$m\ddot{x} + \frac{mg}{l}x + s(x - y) = 0 \quad \dots(i)$$

$$\text{and} \quad m\ddot{y} + \frac{mg}{l}y - s(x - y) = 0 \quad \dots(ii)$$

where x and y are the displacements of the bobs of the two pendulums respectively from their mean positions, l is the length of each pendulum, m the mass of each pendulum bob as proved in Eq.(i) and (ii) of article 9.2.

Now, when a coupled system oscillates in a *single normal mode* each component of the system vibrates with the frequency of that mode.

Therefore, supposing that the system of the coupled pendulums vibrate only in one of its normal modes, let the frequency be ω , then the solutions of the above equations are

$$x = A \cos \omega t$$

and

$$y = B \cos \omega t$$

where A and B are the displacement amplitudes of x and y at the frequency ω .

$$\ddot{x} = -A\omega^2 \cos \omega t \text{ and } \ddot{y} = -B\omega^2 \cos \omega t$$

Substituting in (i) and (ii), we have

$$\left[-m\omega^2 A + \frac{mg}{l} A + s(A - B) \right] \cos \omega t = 0 \quad \dots(iii)$$

and

$$\left[-m\omega^2 B + \frac{mg}{l} B - s(A - B) \right] \cos \omega t = 0 \quad \dots(iv)$$

First normal mode frequency. Adding (iii) and (iv), we get

$$(A + B) \left(-m\omega^2 + \frac{mg}{l} \right) = 0 \quad \dots(v)$$

This equation is satisfied when $\omega^2 = \frac{g}{l}$.

Thus this gives the *first normal mode frequency (or frequency of in-phase mode)* $\omega = \sqrt{\frac{g}{l}}$.

Second normal mode frequency. Subtracting (iv) from (iii), we get

$$(A - B) \left(-m\omega^2 + \frac{mg}{l} + 2s \right) = 0 \quad \dots(vi)$$

This equation is satisfied when

$$\omega^2 = \frac{g}{l} + \frac{2s}{m}$$

This gives the *second normal mode. Frequency for frequency of out of phases mode.*

$$\omega = \sqrt{\frac{g}{l} + \frac{2s}{m}}$$

In phase conditions. Substituting $\omega^2 = \frac{g}{l}$ in (vi), we have

$$(A - B) 2s = 0$$

on

$$A - B = 0$$

$\therefore$

$$A = B$$

which gives the *in phase* condition.

Out of phase condition. Substituting $\omega^2 = \frac{g}{l} + \frac{2s}{m}$ in (v), we have

$$(A + B) 2s = 0$$

or

$$A + B = 0$$

$\therefore$

$$A = -B$$

which gives the *out of phase* or (antiphase) condition.

9.8 EXCHANGE OF ENERGY BETWEEN TWO NORMAL MODES

The equation of motion of a stiffness coupled system of two identical simple pendulums A and B is given by

$$\ddot{X} + \omega_0^2 X = 0 \quad \dots(i)$$

and
$$\ddot{Y} + \left(\omega_0^2 + \frac{2s}{m} \right) Y = 0 \quad \dots(ii)$$

where $X = x + y$ and $Y = x - y$ are the normal co-ordinates of the system. x and y are the displacements (in the same direction) of the pendulums A and B respectively, $\omega_0^2 = \frac{g}{l}$ where l is the length of the pendulum and s the stiffness constant as given by eq. (vii) and (viii) of article 9.2.

Solving equation (i) and (ii), we get

and
$$\begin{aligned} X &= x + y = X_0 \cos(\omega_1 t - \phi_1) \\ Y &= x - y = Y_0 \cos(\omega_2 t - \phi_2) \end{aligned}$$

where X_0 and Y_0 are the normal mode amplitudes, $\omega_1 = \omega_0 = \left(\frac{g}{l} \right)^{\frac{1}{2}}$ and $\omega_2 = \left(\omega_0^2 + \frac{2s}{m} \right)^{\frac{1}{2}} = \left(\frac{g}{l} + \frac{2s}{m} \right)^{\frac{1}{2}}$ are normal mode frequencies.

Putting $X_0 = Y_0 = 2a$, we have

$$X = 2a \cos(\omega_1 t - \phi_1)$$

and

$$Y = 2a \cos(\omega_2 t - \phi_2)$$

As $X = x + y$, X mode is known as *in-phase* mode. The total energy of the system in X -mode is given by

$$E_x = \frac{1}{2} m \omega_1^2 (2a)^2 = 2ma^2 \omega_1^2$$

As $Y = x - y$, Y -mode is known as *out of phase* mode. The total energy of the system in Y -mode is given by

$$E_y = \frac{1}{2} m \omega_2^2 (2a)^2 = 2ma^2 \omega_2^2$$

Both E_x and E_y do not vary with time. They are constant quantities. Therefore, *we conclude that no exchange of energy takes place from one normal mode to another. In other words, normal modes are independent of each other.*

When we say that normal co-ordinates are independent of each other, we mean that between the in phase mode represented by the normal co-ordinate X and the out of phase mode represented by the normal co-ordinate Y there is no exchange of energy and hence the two are independent of each other.

9.9 FREQUENCY OF A TWO BODY COUPLED OSCILLATOR

Consider two masses m_1 and m_2 connected by a spring of stiffness or force constant s . When the two masses are displaced from their equilibrium positions, the spring either contracts or extends, depending upon the displacement of the masses. This causes a linear restoring force to be produced in the spring and both masses begin to vibrate harmonically about their equilibrium position. Such a system is known as a *two body oscillator* or *coupled oscillator*. A familiar example is a diatomic molecule in which the two atoms are connected by some internal force known as 'bond'.

As no external force acts on the system, the centre of mass either remains stationary or moves with constant velocity. The motion executed is simple harmonic. The system will oscillate with a

frequency $\frac{1}{2\pi} \sqrt{\frac{s}{\mu}}$ where $\mu = \frac{m_1 m_2}{m_1 + m_2}$ [For proof refer 7.8 (iv)]

9.10 APPLICATIONS

Inductively coupled circuits. Two electrical circuits are said to be inductively coupled when the magnetic flux due to the current flowing in one circuit threads the second circuit. The two circuits are then said to have a *mutual inductance*. According to Faraday's law of electromagnetic induction, whenever the magnetic flux in one circuit changes an induced *e.m.f.* is set up in the other which is proportional to the time rate of change of magnetic flux and lasts only for the time the change is taking place. The most familiar example is that of a transformer whose working depends on the mutual induction between its primary and secondary coils.

The power source is connected to the primary and the secondary is wound over the primary in the *same sense*.

Two inductively coupled electrical circuits are shown in Fig. 9.4.

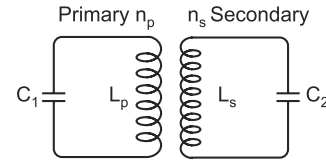


Fig. 9.4

Let n_p represent the number of turns in the primary coil. If ϕ is the magnetic flux set up in the *primary* when a unit current flows through a single turn of the primary coil, then assuming that there is no leakage of flux outside the coil,

Flux linked with *each* primary turn = $n_p \phi$

$\therefore$ Total flux linked with primary coil = $n_p \cdot n_p \phi$

or $L_p = n_p^2 \phi$

where L_p is the co-efficient of self-induction of the primary. Similarly, if a unit current flowing through a single turn of the secondary coil also produces a magnetic flux ϕ , then

Flux linked with each secondary turn = $n_s \phi$

or $L_s = n_s^2 \phi$

where L_s is the co-efficient of self-induction of the secondary.

If we suppose that all the lines of magnetic flux due to a unit current in the primary, thread all the turns of the secondary, then

Total flux lines linking the secondary = $n_s(n_p \phi)$

or $M = n_s n_p \phi = \sqrt{L_p L_s}$

where M is the co-efficient of mutual induction between the two coils.

Co-efficient of coupling. The above result is true only when there is no leakage of magnetic flux. In practice, however, some leakage of flux does take place and $M < \sqrt{L_p L_s}$.

The ratio $\frac{M}{\sqrt{L_p L_s}} = k$ is called the *co-efficient of coupling*.

For *small* value of k the two circuits have a *loose coupling* and are said to be *lightly coupled*.

9.11 ENERGY TRANSFER BETWEEN TWO ELECTRICALLY COUPLED CIRCUITS

To consider the *energy transfer* between the two inductively coupled circuits as shown in Fig. 9.4. let the two circuits be made to oscillate with a frequency ω and let $I_p = I_1 e^{i\omega t}$ and $I_s = I_2 e^{i\omega t}$ be the currents in the two circuits respectively, then

$$\begin{aligned} \text{Rate of change of primary current} &= \frac{dI_p}{dt} \\ &= \frac{d}{dt} I_1 e^{i\omega t} = i\omega I_1 e^{i\omega t} = i\omega I_p \end{aligned}$$

$$\therefore \text{Induced } e.m.f. \text{ in the primary coil} = -L_p \frac{dI_p}{dt} = -i\omega L_p I_p$$

$$\begin{aligned} \text{and Rate of change of secondary current} &= \frac{dI_s}{dt} \\ &= \frac{d}{dt} I_2 e^{i\omega t} = i\omega I_2 e^{i\omega t} = i\omega I_s \end{aligned}$$

$$\therefore \text{Induced } e.m.f. \text{ in the secondary coil} = -L_s \frac{dI_s}{dt} = -i\omega L_s I_s$$

$$\begin{aligned} \text{and Potential difference across the capacitor } C_1 \text{ (Fig. 9.4) in the primary circuit} &= \frac{\int I_p dt}{C_1} \\ &= \frac{\int I_1 e^{i\omega t} dt}{C_1} = \frac{I_1 e^{i\omega t}}{i\omega C_1} = -\frac{iI_p}{\omega C_1} \end{aligned}$$

$$\begin{aligned} \text{Potential difference across the capacitor } C_2 \text{ (Fig. 9.4) in the secondary circuit} &= \frac{\int I_s dt}{C_2} \\ &= \frac{\int I_2 e^{i\omega t} dt}{C_2} = \frac{I_2 e^{i\omega t}}{i\omega C_2} = \frac{iI_s}{\omega C_2} \end{aligned}$$

If the two circuits are considered free from resistance and have inductance and capacitance only, then the *e.m.f.* equations for the primary and the secondary respectively are

$$-L_p \frac{dI_p}{dt} + \frac{\int I_p dt}{C_1} - M \frac{dI_s}{dt} = 0$$

$$\text{and} \quad -L_s \frac{dI_s}{dt} + \frac{\int I_s dt}{C_2} - M \frac{dI_p}{dt} = 0$$

Substituting the values of various quantities, we have

$$i\omega L_p I_p - \frac{i}{\omega C_1} I_p + i\omega M I_s = 0 \quad \dots(i)$$

$$\text{and} \quad i\omega L_s I_s - \frac{i}{\omega C_2} I_s + i\omega M I_p = 0 \quad \dots(ii)$$

Multiplying (i) by $\frac{\omega}{iL_p}$ and (ii) by $\frac{\omega}{iL_s}$, we have

$$\omega^2 I_p - \frac{I_p}{L_p C_1} + \frac{M}{L_p} \omega^2 I_s = 0$$

$$\text{and} \quad \omega^2 I_s - \frac{I_s}{L_s C_2} + \frac{M}{L_s} \omega^2 I_p = 0$$

Now $\frac{1}{L_p C_1} = \omega_1^2$ where ω_1 is the *natural frequency* of the *primary circuit* containing inductance L_p and capacitance C_1 and $\frac{1}{L_s C_2} = \omega_2^2$ where ω_2 is the *natural frequency* of the *secondary circuit* containing inductance L_s and capacitance C_2 .

Substituting $\frac{1}{L_p C_1} = \omega_1^2$ and $\frac{1}{L_s C_2} = \omega_2^2$, we get

$$I_p (\omega_1^2 - \omega^2) = \frac{M}{L_p} \omega^2 I_s$$

and
$$I_p (\omega_2^2 - \omega^2) = \frac{M}{L_s} \omega^2 I_p$$

Multiplying we have

$$I_p I_s (\omega_1^2 - \omega^2) (\omega_2^2 - \omega^2) = \frac{M^2}{L_p L_s} \omega^4 I_p I_s$$

or
$$(\omega_1^2 - \omega^2) (\omega_2^2 - \omega^2) = \frac{M^2}{L_p L_s} \omega^4 = k^2 \omega^4$$

where $k = \frac{M}{\sqrt{L_p L_s}}$ is the co-efficient of coupling.

To simplify put $\omega_1 = \omega_2 = \omega_0$, then

$$(\omega_0^2 - \omega^2)^2 = k^2 \omega^4$$

or
$$\omega_0^2 - \omega^2 = \pm k \omega^2$$

$\therefore \omega = \pm \frac{\omega_0}{\sqrt{1 \pm k}}$

The negative sign on the right hand side has no meaning as ω cannot be negative.

The positive sign gives two frequencies

$$\omega_a = \frac{\omega_0}{\sqrt{1+k}} \text{ and } \omega_b = \frac{\omega_0}{\sqrt{1-k}}.$$

These are the *normal mode frequencies*.

In phase and out of phase normal modes. From the relation

$$I_p (\omega_1^2 - \omega^2) = \frac{M}{L_p} \omega^2 I_s$$

We have
$$\frac{I_s}{I_p} = \frac{\omega_1^2 - \omega^2}{\omega^2} \frac{L_p}{M} = \frac{\omega_0^2 - \omega^2}{\omega^2} \frac{L_p}{M} \quad [\because \omega_1 = \omega_0]$$

$$= \left(\frac{\omega_0^2}{\omega^2} - 1 \right) \frac{L_p}{M} \quad \dots(iii)$$

Selecting the value of $\omega = \frac{\omega_0}{\sqrt{1+k}}$, we have

$$\frac{\omega_0^2}{\omega^2} = 1 + k \text{ or } \frac{\omega_0^2}{\omega^2} - 1 = +k$$

Substituting in Eq. (iii), we get

$$\frac{I_s}{I_p} = +k \frac{L_p}{M} \quad \dots(iv)$$

It is clear that the right hand side of the above equation is positive *i.e.*, I_s and I_p are *in phase*.

Hence $\omega = \frac{\omega_0}{\sqrt{1+k}} = \omega_a$ represents the frequency of the *in-phase mode*.

Similarly selecting $\omega = \frac{\omega_0}{\sqrt{1-k}}$, we find that

$$\frac{I_s}{I_p} = -k \frac{L_s}{M} \quad \dots(v)$$

i.e., I_s and I_p are *out of phase*. Hence $\omega = \frac{\omega_0}{\sqrt{1-k}} = \omega_s$ represents the frequency of *out of phase mode*.

Loose and tight coupling. In loose coupling k is small ($\ll 1$) and both the systems behave almost independently. In this case ω_a and ω_b are very nearly equal to ω_0 . In tight coupling k is large (very nearly equal to unity) so that ω_a and ω_b differ from ω_0 by a large quantity, the peak values of current are displaced and the dip between the peaks is more pronounced. The variation of current amplitude with driving force frequency ω for different values of k is shown in Fig. 9.5. The three cases shown are

- (1) for k small (2) for k having intermediate value and (3) for k large.

It may be noted that for k small (loose coupling) bandwidth is negligible *i.e.*, resonance occurs at one frequency. For large k (tight coupling) bandwidth is large *i.e.*, resonance occurs at two widely apart frequencies.

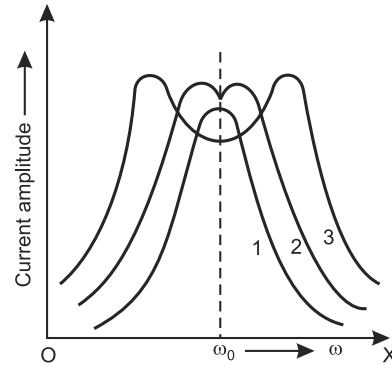


Fig. 9.5.

9.12 N-COUPLED OSCILLATORS

We have seen so far a system in which two atoms coupled together by a massless spring, with an analogy of diatomic molecules like O_2 , H_2 , N_2 etc. However, in a solid, say in a crystal large number of molecules (*viz.* mono-, di-, poly-atomic) arranged in a particular regular manner depending upon its structure *i.e.*, simple cubic (s.c.), body-centred cubic (b.c.c.), or face-centred cubic (f.c.c). A lattice may be regarded as a regular arrangement of atoms which are joined together by elastic springs in three-dimension. At room temperature (or above), the connected atoms start oscillating about their mean position and thereby forms a case of N -coupled oscillator, in which N number of atoms coupled together by massless springs. The lattice may vibrate freely in its normal modes due to its internal energy or may experience forced vibrations under the action of external forces, which may be mechanical or electromagnetic in nature.

Simplest Case: One dimensional monoatomic lattice.

Consider a one-dimensional chain of atoms, each of mass m attached by massless springs as shown in Fig. 9.6. In figure both, the state of equilibrium and state of displacement (disturbed state) are shown.

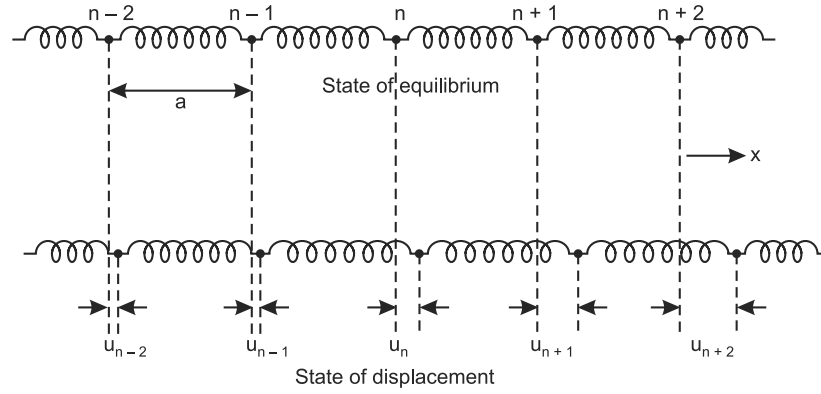


Fig. 9.6

Consider an equilibrium state of the atoms in which the atoms are equally spaced by sites, $\dots, n-2, n-1, n, n+1, n+2, \dots$ as shown in figure. Let a be the lattice parameter, (the distance between two atoms in equilibrium state), the x -coordinates of the corresponding atoms are given by $\dots, (n-2)a, (n-1)a, na, (n+1)a, (n+2)a, \dots$. In the state of vibratory motion along x -axis, the atoms execute simple periodic motion (S.H.M.) about their mean positions and become source of elastic waves which propagate through the medium. Let the displacements, at any instant of time t , of $\dots, (n-2)$ th, $(n-1)$ th, n th, $(n+1)$ th, $(n+2)$ th $\dots$ from their mean position be $\dots, u_{n-2}, u_{n-1}, u_n, u_{n+1}, u_{n+2}, \dots$ respectively.

Assuming the springs to be ideally elastic, force between any two atoms is directly proportional to its linear displacement. If u is the displacement of spring with spring constant β , the force exerted by the spring on the atom is given by $F = \beta u$.

Since n th atom is connected to its neighbours $(n-1)$ th and $(n+1)$ th atoms by two springs, it experiences two opposite forces. The net force on the n th atom will be

$$\begin{aligned} F &= \beta(u_{n+1} - u_n) - \beta(u_n - u_{n-1}) \\ &= \beta(u_{n+1} + u_{n-1} - 2u_n) \end{aligned} \quad \dots(i)$$

Using Newton's second law of motion, we write

$$m \frac{d^2 u_n}{dt^2} = \beta(u_{n+1} + u_{n-1} - 2u_n) \quad \dots(ii)$$

where $\frac{d^2 u_n}{dt^2}$ represents the acceleration of the n th atom. Let the solution for this wave function be

$$u_n = u_0 e^{i(\omega t - Kna)} \quad \dots(iii)$$

where na represents the x -coordinate of n th atom in equilibrium state, $K = 2\pi/\lambda$ is the wave vector or propagation vector and ω is the angular frequency of the wave. Similar quantities for $(n-1)$ th and $(n+1)$ th atoms are

$$u_{n-1} = u_0 e^{i[\omega t - K(n-1)a]} \quad \dots(iv)$$

$$u_{n+1} = u_0 e^{i[\omega t - K(n+1)a]} \quad \dots(v)$$

From eq. (iii), (iv) and (v), we obtain

$$-m\omega^2 = \beta[e^{iKa/2} - e^{-iKa/2}]^2 \quad \dots(vi)$$

$$\text{Since, } \sin x = \frac{e^{ix} - e^{-ix}}{2i} \text{ or } \sin^2 x = -\frac{1}{4}(e^{ix} - e^{-ix})^2$$

Substituting in eq. (vi),

$$-m\omega^2 = -4\beta \sin^2\left(\frac{Ka}{2}\right)$$

$$\text{or} \quad \omega = \pm \sqrt{\frac{4\beta}{m}} \sin\left(\frac{Ka}{2}\right) \quad \dots(vii)$$

If c and ρ denotes the longitudinal stiffness and the mass per unit length of the line respectively, then

$$c = \beta a \text{ and } \rho = \frac{m}{a}$$

$$\therefore \quad \omega = \pm \frac{2}{a} \sqrt{\frac{c}{\rho}} \sin\left(\frac{Ka}{2}\right)$$

$$\text{or} \quad \omega = \pm \frac{2}{a} v_s \sin\left(\frac{Ka}{2}\right) \quad \dots(viii)$$

where $v_s = \sqrt{c/\rho}$ is a constant for a given lattice.

Eq. (viii) gives the angular frequency in terms of v_s , generally referred as velocity of sound waves in solids.

WORKED EXAMPLES

Example 9.1. One of the pendulums of a coupled oscillator is clamped while the other is free to oscillate. Show that the frequency of the single pendulum is given by

$$\nu = \frac{1}{2\pi} \sqrt{\frac{g}{l} + \frac{s}{m}}$$

Solution. Suppose the two pendulums A and B of length l and having bobs of mass m are coupled together by a light spring of stiffness constant s as shown in Fig. 8.1. Then, as proved in (article 9.2) the equations of motion of the pendulums A and B respectively, when the coupled system is set into oscillation are

$$m \frac{d^2x}{dt^2} = -mg \frac{x}{l} - s(x - y)$$

$$\text{and} \quad m \frac{d^2y}{dt^2} = -mg \frac{y}{l} + s(x - y)$$

When the pendulum B is clamped, $y = 0$ and the equation of motion of A is given by

$$m \frac{d^2x}{dt^2} = -mg \frac{x}{l} - sx$$

$$\text{or} \quad \frac{d^2x}{dt^2} = -\left(\frac{g}{l} + \frac{s}{m}\right)x$$

This is the equation of motion of a simple harmonic oscillator and the angular frequency of oscillation ω_1 is given by

$$\omega_1 = \sqrt{\frac{g}{l} + \frac{s}{m}}$$

$$\text{and frequency} \quad \nu_1 = \frac{1}{2\pi} \sqrt{\frac{g}{l} + \frac{s}{m}}$$

When the pendulum A is clamped $x = 0$ and the equation of motion of B is given by

$$m \frac{d^2 y}{dt^2} = -mg \frac{y}{l} + s(-y)$$

or
$$\frac{d^2 y}{dt^2} = -\left(\frac{g}{l} + \frac{s}{m}\right)y$$

This is again the equation of motion of a simple harmonic oscillator and the angular frequency of oscillation ω_2 is given by $\omega_2 = \sqrt{\frac{g}{l} + \frac{s}{m}}$

and frequency
$$\nu_2 = \frac{1}{2\pi} \sqrt{\frac{g}{l} + \frac{s}{m}}$$

Thus whether the pendulum A is clamped or the pendulum B , the frequency of oscillation of the unclamped single pendulum is given by

$$\nu = \frac{1}{2\pi} \sqrt{\frac{g}{l} + \frac{s}{m}}$$

or
$$\omega = \sqrt{\frac{g}{l} + \frac{s}{m}}$$

Example 9.2. Two identical pendulums are connected by a light spring attached to their bobs. The mass of each bob is 10 gm and the stiffness constant of the spring is $8 \times 10^{-3} \text{ Nm}^{-1}$. When one pendulum is clamped the period of the other is found to be 1.20 sec. Find the periods of the normal modes. (H.P.U., 2001)

Solution. When one of the pendulums is clamped the angular frequency ω of the other is given by

$$\omega = \sqrt{\frac{g}{l} + \frac{s}{m}}$$

But $\frac{g}{l} = \omega_0^2$ where ω_0 is the normal mode frequency of the in-phase mode.

$\therefore \omega = \sqrt{\omega_0^2 + \frac{s}{m}}$

or
$$\omega^2 = \omega_0^2 + \frac{s}{m} \quad \dots(i)$$

Here $T = 1.20 \text{ sec.} \therefore \omega = \frac{2\pi}{T} = \frac{2\pi}{1.20} = 5.237 \text{ s}^{-1}$

$$s = 8 \times 10^{-3} \text{ Nm}^{-1}; m = 109 = 10^{-2} \text{ kg}$$

Substituting in Eq. (i), we have

$$(5.237)^2 = \omega_0^2 + \frac{8 \times 10^{-3}}{10^{-2}}$$

or
$$\omega_0^2 = (5.237)^2 - 0.8 = 26.626$$

$\therefore \omega_0 = 5.16 \text{ sec}^{-1}$

and $T_0 = \frac{2\pi}{\omega_0} = 1.218 = 1.22 \text{ sec.}$

This is the time period of the in-phase mode.

The normal mode frequency of the out of phase mode is given by

$$\omega_2 = \sqrt{\omega_0^2 + \frac{2s}{m}}$$

$$\therefore \omega_2^2 = 26.626 + \frac{2 \times 8 \times 10^{-3}}{10^{-2}} = 26.626 + 1.6$$

$$\text{or} \quad \omega_2 = 28.226$$

$$\text{Hence } T_2 = \frac{2\pi}{\omega_2} = 1.18 \text{ sec.} \quad \omega_2 = 5.312$$

This is the time period of the out of phase mode.

Example 9.3. The angular vibrational frequency of CO molecule is $0.6 \times 10^{15} \text{ s}^{-1}$. Calculate the amount of work required for stretching it by $0.5 \text{ \AA}$ from the equilibrium position.

Solution. Reduced mass of CO molecule

$$\mu = \frac{m_1 \times m_2}{m_1 + m_2} = \frac{12 \times 16}{12 + 16} \text{ a.m.u.} = 6.85 \times 1.67 \times 10^{-27} \text{ kg}$$

(where $1 \text{ a.m.u.} = 1.67 \times 10^{-27} \text{ kg}$)

Angular vibration frequency $\omega = 2\pi n = 0.6 \times 10^{15} \text{ s}^{-1}$

$$\text{Now} \quad \omega = \sqrt{\frac{s}{\mu}}$$

$$\therefore \text{Inter-atomic force constant} \quad s = \omega^2 \mu$$

$$= (0.6 \times 10^{15})^2 \times 6.85 \times 1.67 \times 10^{-27}$$

$$= 4.118 \times 10^3 \text{ Nm}^{-1}$$

Work done for stretching by $x = 0.5 \text{ \AA} = 0.5 \times 10^{-10} \text{ m}$

$$W = \frac{1}{2} s x^2 = \frac{1}{2} \times 4.118 \times 10^3 \times 0.5 \times 0.5 \times 10^{-20} = 5.15 \times 10^{-18} \text{ J}$$

Example 9.4. Sodium chloride molecule vibrates with natural frequency $1.14 \times 10^{13} \text{ Hz}$. Calculate the interatomic force constant for the molecule. Given mass of sodium atom = 23 a.m.u. and that of chlorine atom is 35 a.m.u. ($1 \text{ a.m.u.} = 1.67 \times 10^{-27} \text{ kg}$).

Solution. The vibrations of sodium chloride molecule are similar to the vibrations of a mechanical system of two masses coupled by a spring of force constant s , the frequency of which is given by

$$n = \frac{1}{2\pi} \sqrt{\frac{s}{\mu}}$$

where μ is the reduced mass equal to $\frac{m_1 m_2}{m_1 + m_2}$.

$\therefore$ Reduced mass of sodium chloride molecule

$$\mu = \frac{23 \times 35}{23 + 35} = 13.88 \text{ a.m.u.} = 13.88 \times 1.67 \times 10^{-27} = 23.18 \times 10^{-27} \text{ kg}$$

$$\text{Now frequency } n = \frac{1}{2\pi} \sqrt{\frac{s}{\mu}}$$

$$\therefore n^2 = \frac{1}{4\pi^2} \frac{s}{\mu}$$

$$\text{or} \quad s = 4\pi^2 n^2 \mu = 4 \times \left(\frac{22}{7}\right)^2 \times (1.14 \times 10^{13})^2 \times 23.18 \times 10^{-27}$$

$$= 118.9 \text{ Nm}^{-1}.$$

Example 9.5. In a transformer the mutual inductance of two coils is 0.3 H where as the self inductance of primary and secondary are 0.28 H and 0.36 H respectively. Is the transformer loose or tight coupled. (G.N.D.U. 2004, P.U., 2002, 2000; H.P.U., 2003)

Solution. Co-efficient of coupling, $k = \frac{M}{\sqrt{L_p L_s}} = \frac{0.3}{\sqrt{0.28 \times 0.36}} = 0.95$

As the co-efficient of coupling k is nearly equal to unity, the transformer is almost *tight coupled*.

EXERCISE

- (a) Explain the meaning of couple oscillator.
(b) Define and explain normal co-ordinates, degrees of freedom and normal modes of vibration of an oscillatory system. (Pbi.U., 2002; H.P.U., 2000; P.U., 2000; G.N.D.U., 2000)
- What is ment by coupling of two oscillators? Discuss completely the oscillations of two identical stiffness coupled pendulums and write the equation of motion of the system in different cases in terms of normal co-ordinates, X and Y . (H.P.U., 2002, 2001; P.U., 2002)
- Derive an expression for the total energy of a stiffness coupled system of identical pendulums and show that the total energy of the system remains constant. (H.P.U., 2003, 2002)
- Which different types of coupling are used for coupling two oscillators? Show that in the in phase mode, the frequency of oscillations is the same as of uncoupled oscillators whereas in the out of phase mode, the frequency of oscillations gets raised.
- Give a general method of finding normal mode frequencies and obtain expression for the normal mode frequencies of stiffness coupled pendulums. (G.N.D.U., 2002, P.U., 2000)
- (a) Do normal modes exchange energy with each other?
(b) Show that normal modes are independent of each other and there is no exchange of energy between two coupled pendulums. What do you mean by the statement that normal co-ordinates are independent of each other? (P.U., 2003, 2000; Pbi., 2002, 2000; H.P.U., 2003, 2002, 2000)
- Explain transfer of energy between two resistance free electric circuits which are inductively coupled. When is the coupling loose or tight? Obtain an expression for the normal mode frequencies and show that they are almost equal for loose coupling. (P.U., 2004, 2003, 2002, 2001, 2000; Pbi. U., 2003, 2002; H.P.U., 2001, 2000)
- Give the characteristics of the in-phase and out of phase mode of vibration of two identical coupled simple pendulums (oscillators). (G.N.D.U., 2003, 2002, 2001; Pbi.U., 2003; H.P.U., 2000; P.U., 2000,)
- Explain the inductance coupling of two electrical oscillators and define co-efficient of coupling. (G. N. D.U., 2001; P.U., 2004, 2001; H.P.U., 2001)
- Discuss coupled oscillations using coupled system of two identical bodies. (Nagpur U. 2007)
- Obtain an expression for the time period of a two body harmonic oscillator. (Nagpur U. 2008, 2006)

12. Define and explain Normal co-ordinates, normal modes and degree of vibration of an oscillatory system.
13. Discuss exchange of energy between two normal modes of frequencies of two coupled pendulums.
14. What do you understand by degenerate (normal) modes of vibration? Writing down the equations of motion of two identical simple pendulums connected by light spring, find out the frequency of faster normal modes of vibration. **(Purvanchal U. 2007)**
15. Explain the oscillations of two coupled oscillators. What are different modes of vibration? **(Nagpur U. 2009)**
16. Two masses m_1 and m_2 are coupled by a spring of stiffness s . The masses are pulled apart and released. Show that their motion is simple harmonic. Prove that the system will oscillate with a frequency $\sqrt{\frac{s}{\mu}}$ where $\mu = \frac{m_1 m_2}{m_1 + m_2}$. **(P.U., 2001, 2000; Nagpur U. 2007, 2008)**
- [Hint: For proof Pl. refer article 5.9.8]
17. Find self inductance of secondary coil of a transformer having tight coupling with co-efficient of coupling 0.96. The self inductance of primary coil is 0.3 H and mutual inductance 0.2 H between the two coils. **[Ans. $L_s = 0.1477$ H]**

10

Chapter

WAVE MOTION

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- 10.2** Transverse and longitudinal wave motion
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- 10.5** Displacement, velocity and pressure curves
- 10.6** Expression for a plane progressive harmonic wave
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- 10.8** Differential equation of a wave motion
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- 10.37** Group velocity in a conducting medium



INTRODUCTION

More complicated systems possess more degrees of freedom. These compound oscillations (coupled) typically appears very complicated but can be simplified by *resolving the motion* into *normal modes*. As the number of degrees of freedom becomes arbitrarily large, the system approaches continuity. For example, a string or the surface of a body of water. In the classical limit, such systems have an infinite number of *normal modes* and their oscillations occur in the form of *waves* that can characteristically propagate.

10.1 WAVE MOTION - TYPES OF WAVES

One of the commonest modes of transfer of energy from one point to the another is in the form of *waves*. The waves propagates from the point of disturbance, as which they are generated. Thus, for example, we known that *water waves*, *sound waves*, *light waves* and *radio waves* all carry energy of one form or the other from one place to another, without any bulk motion of the intervening medium of the first two types of waves. For light and radio waves propagation, no medium is required at all. There is just propagation of energy takes place. Such a mode of transfer of energy is called **Wave motion**, and may be divided into two broad categories: (i) *mechanical wave motion* and (ii) *non-mechanical or electromagnetic wave motion*.

(i) **Mechanical wave motion** is possible only in material media (solids, liquids and gases) which possess inertia as well as elasticity; so that, water waves and sound waves are both example of this type of wave motion and are, therefore referred to as *mechanical waves*.

(ii) **Non mechanical or electromagnetic wave motion**, on the other hand, requires no material medium for its propagation; so that, light and radio waves, which can travel through empty space, belong to this category and are, therefore referred to as *non mechanical or electromagnetic waves*.

We are concerned here with only the mechanical type of wave motion and shall, for simplicity, refer to it merely as wave motion.

Since all material media possess elasticity as well as inertia, it is easy to see how a wave motion is produced and propagated through them. For, not particle of an elastic medium can be disturbed without affecting its immediate neighbour and, tending to recover its original position, it first stores up potential energy and then converts it back into kinetic energy. The neighbouring particle which has thus been disturbed then performs a similar motion, so that each successive particle repeats, in turn, the movements of its predecessor a little later than it and hands the same on to its successor, resulting in the transference of energy from particle to particle all along the line. One complete oscillation of a particle of the medium obviously produces *one single wave* or a *pulse* and its repeated periodic motion, a succession of waves or a *wave train*.

We may thus define *a wave motion as a disturbance (or a condition) that travels onwards through a medium due to the repeated periodic motion of its particles about their mean or equilibrium position, each particles repeating the movements of its predecessor a little later than it and handing it on to its successor; so that there is a regular phase difference between one particle and the next.*

The simplest type of periodic motion performed by a particle is, of course, the simple harmonic motion and the corresponding wave motion is therefore called a *simple harmonic or a sinusoidal wave motion*, which alone is the most general type of wave motion and the one we shall deal with in the following discussion.

It may be emphasised again that, but for the properties of elasticity and inertia, no wave motion could be produced in, or propagated through, a medium. These two properties in fact determine the velocity of propagation of the wave motion through the medium, as we shall presently see. And, in order that a wave may travel through a medium over fairly large distances without attenuation

(i.e., without any decrease in its amplitude), a third property is also necessary, viz., that the medium should offer the least frictional resistance so as not to unduly damp the periodic motion of the particles.

A wave motion which thus progresses onwards through a medium, with energy transferred across every section of it, is called a *travelling* or a *progressive wave motion* to distinguish it from what is called a *standing* or a *stationary wave motion*, which we shall study a little later and in which there is no onward movement of the wave motion through the medium and hence no transference of energy across any section of it.

10.2 TRANSVERSE AND LONGITUDINAL WAVE MOTION

There are two distinct class of wave motion: (i) *transverse* and (ii) *longitudinal*

(i) In a *transverse wave motion*, the particles of the medium oscillate up and down about their mean or equilibrium position, at *right angles to the direction of propagation of the waves itself*. This form of wave motion therefore travels in the form of *crests and troughs* (Fig. 10.1), as, for example, waves travelling along a stretched string. *A crest and an adjoining trough constitute one wave or pulse* and a succession of them, a *wave train*.

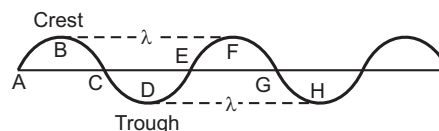


Fig. 10.1

This type of waves motion is possible in media which possess elasticity of shape or rigidity, i.e. in *solids*. But, as we know, they are also possible in water and liquids, in general, even though they do not possess the property of rigidity. This is because they possess another equally effective property of resisting any vertical displacement of their particles (or keeping their level). Gases, however, possess neither rigidity nor do they resist any vertical displacement of their particles (or keep their level). *A transverse wave motion is therefore, not possible in a gaseous medium.*

N.B. It may just be mentioned for the information of the student that an electromagnetic wave is necessarily a transverse wave because of the electric and magnetic fields being perpendicular to its direction of propagation.

(ii) In a *longitudinal wave motion*, the particles of the medium oscillate to and fro about their mean or equilibrium position, *along the direction of propagation of the wave motion itself*. This type of wave motion, therefore, travels in the form of *compressions* (or *condensations*) and *rarefactions*, i.e. in the particles of the medium getting closer together and further apart alternately, (Fig. 10.2), and is possible in media possessing elasticity of volume, i.e., in solids, liquids as well as gases. As examples may be mentioned sound waves in air [Fig. 10.2 (a)] and waves in a spring or helix when one end of it is suddenly compressed or pulled out and then released [Fig. 10.2 (b)].

Here, *one compression and the adjoining rarefaction constitute one wave or pulse* and, as in case (i) a succession of them, a *wave train*.

In some cases, the waves are neither purely transverse nor purely longitudinal as, for example, *ripples* or *surface waves* on water (produced by dropping a stone in water), in which the particles of the medium (here, water) oscillate across as well as along the direction of propagation of the wave motion, describing elliptical paths.

We need not, however, bother ourselves with any such types of waves here.

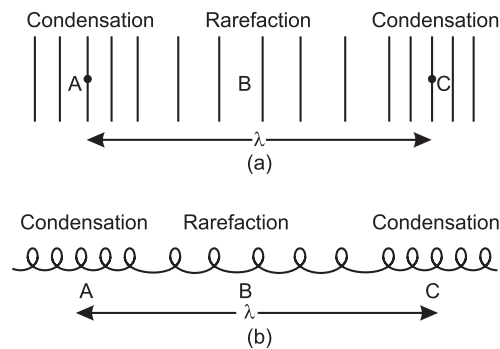


Fig. 10.2

Again, waves may be *one-dimensional*, *two dimensional* or *three dimensional* according as they propagate energy in just one, two or three dimensions. Thus, transverse wave along a string or longitudinal wave along a spring are one-dimensional, surface waves or ripples on water are two-dimensional and sound wave proceeding radially from a point-source are three dimensional.

Before proceeding further, we had better summarise here the important characteristics of a wave motion, in general, whether transverse or longitudinal, viz., that

(i) It is simply a disturbance, a condition or a state of motion that travels through the medium and nothing material, *i.e.*, *there is no transference of any part of the medium (or matter) from one point to another*. For, as we have seen, the particles of the medium simply oscillate up and down or to and fro about their mean position and do not move onwards with the wave motion itself.

(ii) Each particle of the medium receives the disturbance a little later than its predecessor, repeats its movements and hands it on to the next succeeding particle, *i.e.*, *there is a regular phase lag between one particle and the next*.

(iii) *The velocity of the particles of the medium or the particles velocity, as it is referred to, is entirely different from the velocity of the wave motion or the wave velocity.*

Characteristics of wave motion. A progressive simple harmonic wave is that whose amplitude remains constant with time. Its characteristics are:

- (i) A wave is a form of disturbance which propagates in a medium.
- (ii) When the wave propagates the particles of the medium simply oscillate about their mean position.
- (iii) There is a definite phase different between every two consecutive particles.
- (iv) The velocity of the wave is different from the velocity of the particles. The velocity of the wave is a constant for a given medium but the velocity of the particle goes on changing, being maximum in the mean position and minimum in the extreme position.
- (v) Wave motion is possible only in media which have the property of inertia and elasticity.
- (vi) The energy of the vibrating particle at the extreme position is wholly potential and in the mean position wholly kinetic.
- (vii) When a wave travels in a medium, there is a flow of energy without any transfer of matter.

10.3 WAVE FRONT

A plane or a surface on which all particles of the medium are in an identical state of motion at a given instant, *i.e.* in the same phase, is called a *wave front*.

In a homogeneous and isotropic medium, the wave front is always perpendicular to the direction of propagation of the wave motion and the position of a given wave front shifts onwards with the

wave. In other words, it moves onwards through the medium with the velocity of the wave motion itself. A line normal to the wave front thus gives the direction of propagation of the wave and is called a *ray*.

Although wave fronts may have various shapes, the two most important types are plane and the spherical wave fronts. In case the waves are one dimensional, such as those travelling along a stretched string or a spring or produced by the motion of a plane surface back and forth in a gaseous medium, we obtain a plane wave front and such waves are, therefore, also referred to as **plane waves**.

On the other hand, in the case of **spherical waves**, or *waves proceeding from a point-source*, we obtain a *spherical wave front*. Since the radius of a spherical wave front goes on increasing as the wave moves onwards, its curvature goes on progressively decreasing until at an infinite or a large distance from the source it becomes almost a plane surface or a plane wave front, implying thereby that *at large distances from the source, spherical waves—and, in fact, waves of all types—may be treated as plane waves*.

We shall therefore, restrict our discussion here to only plane or one-dimensional waves. For, this, without in any way detracting from the applicability of our result to waves in general, will greatly facilitate our task by doing away with undue mathematical complexities.

10.4 WAVE LENGTH, FREQUENCY AND WAVE NUMBER

It may be recalled that in § 10.2 above, a crest and an adjoining trough, in the case of a transverse wave, and a compression and an adjoining rarefaction, in the case of a longitudinal wave, have been said to constitute one wave. The distance between the particles at the two ends of the wave or the pulse, therefore, constitutes a *wave length*, usually denoted by the symbol λ . Since these two particles are in an identical state of motion or *in the same phase*, a more general definition of wavelength would be *the distance between any two nearest particles of the medium in the same phase*. Thus, in the case of a transverse wave (Fig. 10.1) the *distance between the centres of two nearest crests or two nearest troughs* (i.e. distance BF or DH) also constitutes one wave length. Similarly, in the case of longitudinal wave (Fig. 10.2) the *distance between the centres of two nearest compressions or two nearest rarefactions* (e.g. the distance AC) also constitutes one wavelength.

Again, since a wave or a pulse is produced in the time taken by a particle of the medium to complete its one full oscillation about its mean position, a *wavelength may also be defined as the distance covered by the disturbance during one full time-period (T) of a particle of the medium*.

The *number of waves produced per second* (which is obviously the same as the number of complete oscillations made by a particle of the medium per second) is called the **frequency (n)** of the wave motion or the wave, and is obviously the *reciprocal of the time-period T* , i.e., $n = 1/T$.

The *time rate of change of phase* (i.e., its rate of change with time) is called the **angular frequency (ω)** of the wave motion and is equal to $2\pi/T$ because a complete cycle of phase change 2π occurs in one full time-period T of a particle of the medium.

On the same analogy, we have a quantity, called the **wave number k** , representing the *space rate of change of phase* (i.e. its rate of change with distance) and since one complete cycle of phase change 2π occurs in a distance λ (the wave length), we have $k = 2\pi/\lambda$.

Quite often, however, the wave number is defined as the *number of waves is unit distance*, i.e., equal to $1/\lambda$ and is denoted by the symbol ν . So that, $k = 2\pi/\lambda$ is, in this case, referred to as the propagation constant.

It will also be seen from the above that if the number of waves produced per second, i.e. the frequency of the wave motion, be n and the length of each wave (or wavelength) be λ , we have

$$\nu = n\lambda \text{ or, since } n = 1/T, \text{ we also have } \nu = \lambda/T$$

where ν is the *phase velocity*, usually referred to simply as the *velocity of the wave motion or of the wave*.

10.5 DISPLACEMENT, VELOCITY AND PRESSURE CURVES

(i) **Displacement curve.** In the case of a transverse wave, as we know, the particles of the medium oscillate up and down about their mean position, at right angles to the direction of propagation of the wave. So that, if we plot the displacements of the various particles at a given instant against their respective distances from the origin or, in fact any fixed point, we obtain the *displacement curve* of the wave, which, for a sinusoidal wave, will be a series of sine curves, (Fig. 10.1), with one complete sine curve representing one full wave, as shown in Fig. 10.3. The displacement curve thus gives the form of the wave or the wave profile, as it were.

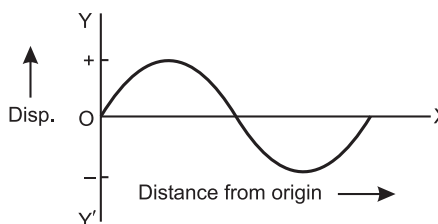


Fig. 10.3

In the case of a longitudinal wave, on the other hand, the particles oscillate to and fro about their mean position along the direction of the wave motion, so that the displacement curve would, in this case, be a straight line along the direction of wave propagation itself. For our convenience, however, we *conventionally* represent the displacement curve of a longitudinal wave motion too by a sine curve of the type shown in Fig. 10.1, the curve here being referred to as the *associated displacement curve* in and it is clearly understood that the displacements of the particles shown perpendicular to the direction of wave propagation are in fact along it.

(ii) **Velocity curve.** The graph between the velocity (U) of an oscillating particle of the medium and its distance from the origin or a fixed point is called the velocity curve of the wave motion. Using the same convention as in case (i) above, the velocity curve in the case of both a transverse and a longitudinal wave is of the form shown in Fig. 10.4, which is obviously also a *sine curve*, a *phase angle* $\pi/2$ or a *quarter period* $T/4$ ahead of the displacement curve.

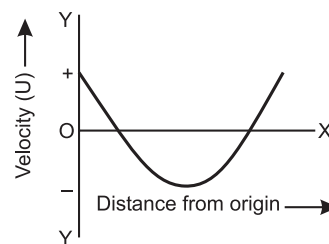


Fig. 10.4

(iii) **Pressure curve.** Let $ABCDE$ (Fig. 10.5) be the associated displacement curve of a longitudinal wave travelling towards the right along the axis ACE of a cylinder of unit area of cross section (shown dotted).

Consider two particles P and Q of the medium (say, air), a small distance δx apart. Before the wave is setup in the cylinder, i.e. when the air inside it is yet undisturbed, the volume of the air enclosed between P and Q is obviously $PQ \times 1 = \delta x \times 1 = \delta x$ (the cross section of the cylinder being unity). After the wave is set up and takes the form shown in the figure, displacement at P is PP' , and that at Q is QQ' and the change in displacement is, therefore, $PP' - QQ' = P'R = \delta y$, say, where $Q'R$ is the perpendicular from Q' on the PP' . So that, remembering that the displacements are in fact longitudinal, along the direction of wave propagation, the change in volume between P and $Q = (PP' - QQ') \times 1 = P'R \times 1 = \delta y \times 1 = \delta y$.

$\therefore$ volume strain set up in the medium = change in volume/original volume = $\delta y/\delta x = dy/dx$ if PQ be really small.

But dy/dx is the slope of the displacement curve at P and represents a compression or a rarefaction according as it has a negative or a positive sign. If, therefore, we plot dy/dx for the particles along the axis of y and their displacements from the origin along the axis of x , we obtain

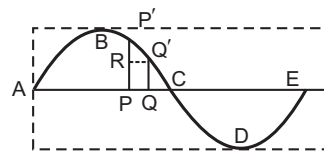


Fig. 10.5

the *pressure curve* of the wave (Fig. 10.6) which is of the same form as the velocity curve (Fig. 10.4). Thus, both the velocity and the pressure curves of a wave are $p/2$ or $T/4$ ahead of the displacement curve in phase. Incidentally, we also see that volume strain at a point = slope of the displacement curve at that point.

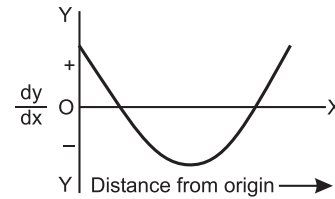


Fig. 10.6

10.6 EXPRESSION FOR A PLANE PROGRESSIVE HARMONIC WAVE

We know that a plane progressive wave is one which travels onwards through the medium in a given direction without attenuation, *i.e.*, with its amplitude constant.

Now, a progressive wave may be transverse or longitudinal. In either case, however, there exists a regular phase difference between any two successive particles of the medium.

Thus, suppose a wave, originating at O , travels to the right along the axis of x (Fig. 10.7). Then, if we start counting time when the particle at O just passes through its mean position in the positive direction (*i.e.*, upwards in the case of a transverse wave and forwards in the case of a longitudinal wave) the equation of motion of this particle (at O) is obviously $y = a \sin \omega t$, where y is its displacement at time t , a , its amplitude and ω , its angular velocity.

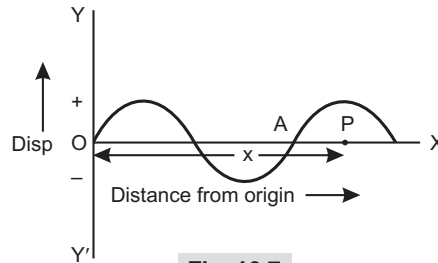


Fig. 10.7

Since the successive particles to the right of O receive and repeat its movements after definite intervals of time, the phase lag goes on increasing as we proceed away from O towards the right. So that, for a particle at P , distant x from O , it is ϕ , say. The equation of motion of the particle at P is thus $y = a \sin (\omega t - \phi)$.

Now, as we know, in a distance λ , the phase lag increases by 2π . In distance x , therefore, the phase lag increase by $2\pi x/\lambda$, *i.e.*, $\phi = 2\pi x/\lambda$. Substituting this value of ϕ in the expression for y above, we have

$$y = a \sin (\omega t - 2\pi x/\lambda). \text{ Or, } y = a \sin (\omega t - kx) \quad \dots(i)$$

where $2\pi/\lambda = k$, the *propagation constant*.

Again, $\omega = 2\pi/T$, where T is the time-period of each particle of the medium. And, since $v = \lambda/T$, where, v is the *phase velocity* or the *wave velocity* and n , the frequency of the oscillating particles or of the wave, we have $1/T = v/\lambda$ and, therefore, $\omega = 2\pi v/\lambda$. We thus have from relation (i) above,

$$y = a \sin \left(\frac{2\pi v t}{\lambda} - \frac{2\pi x}{\lambda} \right).$$

$$\text{Or, } y = a \sin \frac{2\pi}{\lambda} (vt - x). \quad \text{Or, } y = a \sin k(vt - x). \quad \dots(ii)$$

Any one of these expressions or any one of their variations, such as

$$y = a \sin \frac{2\pi v}{\lambda} \left(t - \frac{x}{v} \right) \quad \dots (iii) \quad y = a \sin 2\pi n \left(t - \frac{x}{v} \right) \quad \dots(iv)$$

$$\text{Or } y = a \sin \frac{2\pi}{T} \left(t - \frac{x}{v} \right) \quad \dots (v) \text{ [because } n = 1/T]$$

$$\text{Or } y = a \sin 2\pi \left(\frac{t}{T} - \frac{x}{\lambda} \right) \quad \dots (vi) \text{ [because } v = n\lambda = \lambda]$$

is referred to as the *equation of a plane, progressive harmonic or sinusoidal wave* – the one most commonly used being equation (ii).

Since $\lambda = v/T$, $\omega = 2\pi/T$ and $k = 2\pi/\lambda$, it will be easily seen that

$$v = \lambda/T = \lambda / \frac{2\pi}{\omega} = \lambda\omega/2\pi = \omega / \frac{2\pi}{\lambda} = \omega/k$$

For a wave travelling *towards the left*, since x will be negative, we shall have

$$y = a \sin \frac{2\pi}{\lambda} (vt + x) \quad \text{or,} \quad y = a \sin (\omega t + kx)$$

Note. In deducing these equations, we have assumed that the particle at O just passes through its mean position, in the positive direction, at $t = 0$, i.e., $y = 0$ at $t = 0$. If this be not so, the particle is said to have an *initial phase* θ , say, and the equation of the wave then becomes

$$y = a \sin \frac{2\pi}{\lambda} (\omega t - x + \theta) \quad \text{Or,} \quad y = a \sin (\omega t - kx + \theta),$$

where θ is referred to as the *phase constant*.

Thus, if $\theta = 90^\circ$, we have $y = a$ at $x = 0$ and $t = 0$

The following points emerge from the above discussion may be carefully noted.

- (i) *At a given point x (i.e., for a constant value of x) the displacement (y) varies simple harmonically with time, completing one full cycle in time λ/v which, therefore, gives the periodic time T of the wave and hence also its frequency $n = 1/T = v/\lambda$.*
- (ii) *At a given instant t (i.e., for a constant value of t), the displacement (y) varies simple harmonically with distance from the origin and for $x = \lambda$, y is restored to its original value, so that the wave length $= \lambda$.*
- (iii) *The phase lag for a distance x is $2\pi x/\lambda$ and hence for distance λ it is $2\pi\lambda/\lambda = 2\pi$ which is, in effect, the same thing as zero. So that, particles separated by a distance λ or an integral multiple of λ are in the same phase.*
- (iv) *If we increase time t by δt , and distance x by $v\delta t$, the value of y remains the same, showing that a disturbance at one point is repeated an interval δt later at a point $v\delta t$ further away. This means, in other words, that the disturbance or the wave travels, without attention, with velocity v .*

10.7 PARTICLE VELOCITY

We have the equation of plane progressive wave motion,

$$y = a \sin \frac{2\pi}{\lambda} (vt - x),$$

where y is the displacement of a particle of the medium at distance x from the origin at instant t , a its amplitude and v , the wave velocity (i.e., the *phase velocity*).

$\therefore$ Differentiating the expression for y with respect to t , we have

$$\text{particle velocity} \quad U = \frac{dy}{dt} = \frac{2\pi v}{\lambda} a \cos \frac{2\pi}{\lambda} (vt - x) \quad \dots(i)$$

Relation between particle velocity and wave velocity:

Differentiating the same expression for y with respect to x , we have

Slope of the displacement curve (or strain or compression)

$$\frac{dy}{dx} = -\frac{2\pi}{\lambda} a \cos \frac{2\pi}{\lambda} (vt - x) \quad \dots(ii)$$

$$\text{From relations (i) and (ii) we thus obtain } U = \frac{dy}{dt} = -v \frac{dy}{dx} \quad \dots(iii)$$

i.e. **Particle velocity at a point = – (Wave velocity) $\times$ Slope of displacement curve at that point).**

10.8 DIFFERENTIAL EQUATION OF A WAVE MOTION

Differentiating equation (i), under § 10.7 above, for the particle velocity again with respect to t , we have

$$\begin{aligned} \text{acceleration of the particle, } \frac{d^2 y}{dt^2} &= -\frac{4\pi^2 v^2}{\lambda^2} a \sin \frac{2\pi}{\lambda} (vt - x) \\ &= -\frac{4\pi^2 v^2}{\lambda} y. \end{aligned} \quad \dots(iv)$$

And, similarly, differentiating equation (ii) under the same article, for strain or compression, again with respect to x , we have

rate of change of compression with distance,

$$\frac{d^2 y}{dx^2} = -\frac{4\pi^2}{\lambda^2} a \sin \frac{2\pi}{\lambda} (vt - x) = -\frac{4\pi^2}{\lambda^2} y \quad \dots(v)$$

$$\text{From relations (iv) and (v) we thus obtain } \frac{d^2 y}{dt^2} = v^2 \frac{d^2 y}{dx^2} \quad \dots(vi)$$

This is referred to as the differential equation of a plane or one dimensional progressive wave, in which the coefficient of $d^2 y/dx^2$ straightaway gives square of the wave velocity.

Any equation of this form can unphesitatingly be declared to represent a plane progressive harmonic wave, the velocity of which is given by the square root of the coefficient of $d^2 y/dx^2$.

Since $d^2 y/dx^2$ also gives the curvature of the displacement curve, we may interpret the differential equation to mean that

Particle acceleration at a point

$$= (\text{Wave velocity})^2 \times (\text{Curvature of displacement curve at that point})$$

It may once again be pointed out here that the essential characteristics of a wave motion, in general, is the onward progression of a disturbance with a velocity depending upon factors pertaining to the particular situation in question. It does not by any means imply that y in the wave equation must necessarily represent a physical displacement of particles as we have assumed here. In fact, for a wave, in general, we replace the symbol y in the differential equation of the wave by ψ (psi), referred to as the wave function. So that, for a one dimensional wave the differential equation takes the form

$$\frac{d^2 \psi}{dt^2} = v^2 \frac{d^2 \psi}{dx^2}$$

where ψ may represent physical quantities as disparate as, for example, (i) pressure in a medium in the case of a sound wave travelling through it, (ii) the magnitude of the electric vector of a plane-polarised electromagnetic wave or (iii) even the probability amplitude of what is called a matter wave (from which the probability of finding a particle within a given volume of space may be determined).

10.9 DIFFERENTIAL EQUATION OF A THREE-DIMENSIONAL WAVE

A three-dimensional wave, as have seen, is one that travels out in all direction. For such a wave, the differential equation takes the form

$$\frac{d^2 \psi}{dt^2} = v^2 \nabla^2 \psi$$

$$\text{where } \nabla^2 \psi = \frac{\delta^2 \psi}{\delta x^2} + \frac{\delta^2 \psi}{\delta y^2} + \frac{\delta^2 \psi}{\delta z^2}.$$

One possible solution of the equation is $\psi = a \sin(\omega t - \mathbf{k} \mathbf{r} + \theta)$

where ψ is the displacement at a point $\mathbf{r}$ at an instant t and $\mathbf{k}$, the *propagation vector* whose magnitude is equal to $k = 2\pi/\lambda$ (the propagation constant) and direction the same as that of the wave propagation.

10.10 ENERGY DENSITY OF A PLANE PROGRESSIVE WAVE

We have already seen how in a progressive wave motion, the energy derived from the source is passed on from particle to particle, so that *there is a regular transmission of energy across every section of the medium*.

By the energy density of a plane progressive wave we mean *the total energy (kinetic + potential) per unit volume of the medium through which the wave passing*. It is usually denoted by the letter E .

Let us proceed to obtain an expression for it

As we know, the displacement of a particle of the medium distant x from the origin or the source, at an instant t , is given by

$$y = a \sin \frac{2\pi}{\lambda} (vt - x)$$

where the symbols have their usual meanings.

$$\therefore \text{the velocity of the particle, } U = \frac{dy}{dt} = \frac{2\pi v}{\lambda} a \cos \frac{2\pi}{\lambda} (vt - x)$$

$$\text{and acceleration of the particle, } \frac{dU}{dt} = \frac{d^2 y}{dt^2}$$

$$= -\frac{4\pi^2 v^2}{\lambda^2} a \sin \frac{2\pi}{\lambda} (vt - x) = -\frac{4\pi^2 v^2}{\lambda^2} y$$

Now, if we consider *unit volume* of the medium in the form of an extremely thin element of the medium parallel to the wave front, we have

mass of the element = ρ , its density (because density is mass per unit volume).

And since the layer is very thin, the velocity of all the particles in it may be assumed to be the same. We therefore have

$$K.E. \text{ per unit volume of the medium} = \frac{1}{2} (\text{mass}) (\text{velocity})^2 = \frac{1}{2} \rho U^2$$

$$= \frac{1}{2} \rho \frac{4\pi^2 v^2}{\lambda^2} a^2 \cos^2 \frac{2\pi}{\lambda} (vt - x) = \frac{2\pi^2 v^2 \rho}{\lambda^2} a^2 \cos^2 \frac{2\pi}{\lambda} (vt - x) \quad \dots(i)$$

Clearly, *force acting on unit volume* = *mass* $\times$ *acceleration*

$$= \rho \frac{d^2 y}{dt^2} = \frac{4\pi^2 v^2 \rho}{\lambda^2} y,$$

ignoring the $-ve$ sign which merely indicates that the acceleration is directed oppositely to the displacement.

$\therefore$ *work done in a small displacement dy of the layer*

$$= \text{force} \times \text{displacement} = \frac{4\pi^2 v^2 \rho}{\lambda^2} y dy$$

Hence, *work done during the whole displacement from 0 to y*

$$\begin{aligned} &= \int_0^y \frac{4\pi^2 v^2 \rho}{\lambda^2} y dy = \frac{4\pi^2 v^2 \rho}{\lambda^2} \int_0^y y dy \\ &= \frac{4\pi^2 v^2 \rho}{\lambda^2} \cdot \frac{y^2}{2} = \frac{2\pi^2 v^2 \rho}{\lambda^2} y^2 = \frac{2\pi^2 v^2 \rho}{\lambda^2} a^2 \sin^2 \frac{2\pi}{\lambda} (vt - x) \end{aligned}$$

This work done must obviously be stored up in the medium in the form of potential energy. So that,

P.E. per unit volume of the medium $= \frac{2\pi^2 v^2 \rho}{\lambda^2} a^2 \sin^2 \frac{2\pi}{\lambda} (vt - x)$.
 $\therefore$ total energy per unit volume of the medium or energy density of the plane progressive wave,
i.e., $E = \text{K.E.} + \text{P.E.}$

$$\begin{aligned} \text{or, } E &= \frac{2\pi^2 v^2 \rho}{\lambda^2} a^2 \cos^2 \frac{2\pi}{\lambda} (vt - x) + \frac{2\pi^2 v^2 \rho}{\lambda^2} a^2 \sin^2 \frac{2\pi}{\lambda} (vt - x) \\ &= \frac{2\pi^2 v^2 \rho}{\lambda^2} a^2 \quad \dots(ii) \\ &= 2\pi^2 \left(\frac{v}{\lambda} \right)^2 \rho a^2 = 2\pi^2 n^2 a^2 \rho \quad [\because v/\lambda = n] \dots(iii) \end{aligned}$$

where n is the frequency of the wave

In case the cross section of the beam be unity, expressions (ii) and (iii) give the *total energy of the beam or the wave per unit length*.

A point of interest that must clearly be noted here is that *although both kinetic and potential energies of the wave depend upon the values of x and t , its total energy or the energy density is quite independent of either*.

10.11 DISTRIBUTION OF ENERGY IN A PLANE PROGRESSIVE WAVE

Having calculated the total energy of a plane progressive wave in § 9.10 above, let us see how it is distributed over a complete wave length.

Clearly, average K.E. over a complete wavelength

$$\begin{aligned} &= \frac{1}{\lambda} \int_0^\lambda \frac{2\pi^2 v^2 \rho}{\lambda^2} \cos^2 \frac{2\pi}{\lambda} (vt - x) dx \\ &= \frac{2\pi^2 n^2 a^2 \rho}{\lambda} \int_0^\lambda \frac{1}{2} \left[1 + \cos \frac{4\pi}{\lambda} (vt - x) \right] dx \quad [\because v^2/\lambda^2 = n^2] \end{aligned}$$

$$\begin{aligned} \text{Now, } \int_0^\lambda \cos \frac{4\pi}{\lambda} (vt - x) dx &= -\frac{\lambda}{4\pi} \left[\sin \frac{4\pi}{\lambda} (vt - x) \right]_0^\lambda \\ &= -\frac{\lambda}{4\pi} \sin \frac{4\pi}{\lambda} \left[(vt - x) - \sin \frac{4\pi}{\lambda} vt \right] \\ &= -\frac{\lambda}{4\pi} \left[\sin \left(\frac{4\pi}{\lambda} vt - 4\pi \right) - \sin \frac{4\pi}{\lambda} vt \right] = 0 \end{aligned}$$

So that, average K.E. of the wave over a complete wavelength $= 2\pi^2 n^2 a^2 \rho \lambda / 2\lambda$

$$= \pi^2 n^2 a^2 \rho = \frac{1}{2} \text{ total energy}$$

Similarly, average P.E. of the wave over a complete wavelength

$$= \frac{1}{\lambda} \int_0^\lambda \frac{2\pi^2 v^2 a^2 \rho}{\lambda^2} \sin^2 \frac{2\pi}{\lambda} (vt - x) dx = \pi^2 n^2 a^2 \rho = \frac{1}{2} \text{ total energy}$$

Thus, at any given instant, the energy of a plane progressive harmonic wave is, on an average, half kinetic and half potential form.

10.12 ENERGY CURRENT—INTENSITY OF A WAVE

It has been pointed out earlier under § 10.10 above that if the cross section of the beam or the wave be taken to be unity, we may regard the total energy per unit volume or the energy density $E = 2\pi^2 n^2 a^2 \rho$ as the *total energy per unit length of the wave*. And, since in progressive wave train, a